Bayesian Clinical Trials

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2026-11-11

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Preface

Companion resources

- **R package bayestrials**, which computes the worked calculations and figures in the book (numbers quoted from cited publications are taken from those sources): https://github.com/VadimSokolov/bayestrials (install with `remotes::install_github("VadimSokolov/bayestrials")`).
- **Interactive lab**, fifteen short browser modules that let you run the trials yourself, from "are these two groups different?" to a Bayesian trial simulator: https://vsokolov.org/bayesian_trials_lab/.
- **Encyclopedia**, an online companion with entries for the methods, trials, regulatory documents and people in this book, each linked to the section where the book treats it, with its equations explained in words: https://vsokolov.org/bayesian_trials_lab/encyclopedia/.

This book is for the people who run clinical trials: clinical investigators, physicians, biostatisticians at sponsors and contract research organizations, members of data monitoring committees, and the regulatory reviewers who decide whether the evidence a trial produces is enough to approve a drug. Most of them were trained on the traditional trial: a sample size fixed by a power calculation, equal randomization, a significance level of 2.5 percent one-sided, interim looks paid for with an alpha-spending function, and a p-value and confidence interval in the clinical study report. That is where the book starts.

The book's argument is that a Bayesian adaptive design is the same trial with a different set of decision rules, and that the rules can be written into a protocol, simulated, and defended before a regulator in the same terms the traditional design is defended. The Bayesian design can decide sooner, exposes fewer patients to inferior arms, and uses information the traditional design discards. Against a single unadjusted analysis it does not lower the bar without an informative prior. The saving comes from how the trial is monitored, how several arms are handled, and how prior information is used, and each of those gains has a price that we itemize.

Regulation is the spine of the book. Drugs are approved on substantial evidence from adequate and well-controlled investigations, and every method here is tied to the statute, regulation or guidance that governs it, with its status (final or draft) and date as of November 2026. Chapter 3 is the map; each later chapter opens with a "Regulatory anchor" box naming the documents that apply.

How to read this book

Part I sets the question and the rules. Part II rebuilds the basics from the ground up: how to tell whether two groups differ, why and how we randomize, and how a Bayesian

updates beliefs. Part III revisits the standard apparatus (power, multiplicity, interim looks, anytime-valid inference) and shows what each piece costs. Part IV covers the Bayesian adaptive designs in use today, and Part V works through real trials and the protocol a sponsor submits.

Boxes marked "Technical note" contain derivations that a clinical reader can skip. Boxes marked "In the protocol" give language of the kind that goes into a protocol or statistical analysis plan.

The companion laboratory

An interactive tutorial at https://vsokolov.org/bayesian_trials_lab/ accompanies the book. Its modules let the reader predict an outcome, run the trial many times, and check the prediction, and it includes a Bayesian trial simulator for designing and calibrating a two-arm adaptive trial in the browser. Chapters point to the matching module in a "Try it in the lab" box.

Code

The worked calculations and figures are produced by short R snippets that call the companion package `bayestrials`, which implements the methods used in the book and is tested against published values where they exist (numbers quoted from cited publications are taken from those sources, not computed); see Appendix C.

The figures are in color in the digital editions (the PDF linked from the book's DOI and the HTML edition). In the printed book they appear in grayscale, and where a caption names a color, the digital edition shows it.

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Digital Object Identifier (DOI): 10.5281/zenodo.24147174

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Part I.

Trials, Decisions, and Regulation

1. A Trial Is a Decision

1.1. One trial, from the start

SELECT randomized 17,604 adults with overweight or obesity and established cardiovascular disease, without diabetes, to weekly semaglutide 2.4 mg or placebo, at 804 sites in 41 countries, enrolling from 2018 to 2021 (Lincoff et al. 2023). The primary endpoint, a composite of cardiovascular death, nonfatal myocardial infarction and nonfatal stroke, occurred in 569 of 8,803 patients on semaglutide and 701 of 8,801 on placebo: 1,270 events in all, a hazard ratio of 0.80 with a 95 percent confidence interval from 0.72 to 0.90. Mean follow-up was 39.8 months. A result of this kind changes practice, and it was earned honestly: the trial was designed for a hazard ratio of 0.83 with 90 percent power at one-sided 0.025, 1,225 planned events, and one superiority interim analysis at two thirds of them (D. H. Ryan et al. 2020). At that interim (831 events) the data monitoring committee recommended continuing (FDA clinical review), so the trial ran to its end.

! Regulatory anchor

- **Section 505(d) of the Federal Food, Drug, and Cosmetic Act** (United States Congress 1962). The statutory standard: a drug is approved on "substantial evidence" of effectiveness, which the Act defines as evidence consisting of adequate and well-controlled investigations. Statute, binding.
- **21 CFR 314.126** (U.S. Food and Drug Administration 2024b). The regulation that says what "adequate and well-controlled" means: a clear objective, a valid comparison with a control, measures against bias, and an analysis adequate to assess the effect. Regulation, binding.
- **ICH E8(R1), General Considerations for Clinical Studies** (International Council for Harmonisation 2021) (Step 4, October 2021). Asks that quality be designed in before the study starts ("quality by design"), by identifying what is critical to the decision the trial will support. Final guideline.
- **ICH E9, Statistical Principles for Clinical Trials** (International Council for Harmonisation 1998) (Step 4, 1998). The statistical standard that most protocols still follow: pre-specified analysis, control of type I error, and justification of sample size. Final guideline, with the E9(R1) addendum on estimands (International Council for Harmonisation 2019) (Step 4, November 2019).
- **FDA draft guidance on Bayesian methodology** (U.S. Food and Drug Administration 2026g) (January 2026). Draft, not binding. It is the first FDA guidance on Bayesian methods for drugs and biological products (devices have had one since 2010), and it allows a success criterion to be calibrated to type I error, read from the posterior, or based on a decision-theoretic rule or a benefit-risk threshold.

The full map, with status and dates, is in Chapter 3.

The question this book asks is the one the trial raises. Hold the error rate fixed at the level the design promised. When could the answer have been known, and what did waiting cost? Every month before a result is known, half the participants are on placebo, and a drug that works is withheld from them while the trial runs.

Here is the answer in headline form, as a counterfactual reconstruction of ours and not a statement about what the sponsor or the committee did. Suppose the true hazard ratio is 0.80 and the trial can run to at most 1,270 events, and simulate 40,000 such trials (details in Chapter 21). A Bayes rule monitored on the prespecified event grid (every second event), with its posterior threshold calibrated so that the type I error is 2.5 percent, stops at a median of 464 events among the simulated trials that declare efficacy. SELECT's planned design, with its one interim, stops at a median of 817 events when it succeeds, and a four-look O'Brien-Fleming design at 952. Counting every simulated trial, where noncrossing trials end at each rule's maximum (1,270 events for the hypothetical rules and 1,225 for SELECT's planned two-look design), the medians are 486, 817 and 952. The powers are 95, 97 and 98 percent, and the realized type I errors are 2.5, 2.5 and 2.5 percent, so the comparison is between rules held to the same 2.5 percent ceiling. The Bayesian figure is fixed at 2.5 by construction, because the threshold is calibrated on the same simulated null paths; an independent check on fresh paths, with its Monte Carlo uncertainty, is in Chapter 21. An anytime-valid e-value rule, also in the reconstruction, stops at a median of 612 events among successes but acts in only 83 percent of trials, because it is conservative (a realized type I error of 0.6 percent).

Under a modeled accrual (calendar time is a counterfactual, not an observation), a Bayes decision at the conditional median events would fall about 16 months before the four-look design's and 27 months before the read-out, and about 18,500 modeled placebo patient-years would accrue after that point. The gain is earlier stopping at a calibrated bar, not a lower bar: the same Bayes rule at the conventional 0.975, checked on the same grid, would have a type I error of 11.8 percent.

What waiting cost, in three currencies. Every figure in this paragraph and in Table 1.1 is a modeled counterfactual of ours, computed from the all-trial distribution of simulated stopping points, in which a trial that never crosses is counted at its maximum and so saves nothing. Averaged over all simulated trials the Bayes rule had accrued 44 percent of SELECT's events when it stopped, so on average 56 percent of the events and of the person time would have come afterwards, and the all-trial median decision falls 2.2 years before the read-out under the accrual model above. The three currencies are the trial cost avoided, the cardiovascular label reached earlier inside a fixed patent life (so that label-driven revenue is brought forward), and the health value of the events and quality-adjusted life years (QALYs) the placebo arm would have been spared.

Table 1.1.: What waiting cost in SELECT, in three currencies, all modeled counterfactuals from the all-trial stopping distribution. Dollar figures are not profits and not forecasts.

currency	SELECT, modeled	assumption that carries it
(a) trial cost avoided	\$44 million to \$237 million	a variable share of cost of 50 to 70 percent, on a total of \$157 million, the cardiovascular mean of published pivotal-trial cost estimates (T. J. Moore et al. 2018), up to \$600 million (a scenario for a larger trial)
(b) label revenue brought forward	\$1.8 billion to \$4.6 billion of Wegovy revenue, not profit	2024 global sales of DKK 58.2 billion (Novo Nordisk A/S 2025) at an assumed 6.9 DKK per dollar, 10 to 25 percent of sales attributable to the cardiovascular label (an assumption, not a public figure), brought forward by 2.2 years; an upper bound
(c) health value	in the trial about 75 excess placebo-arm primary events averted and 112 to 224 QALYs; outside it, as a bound, about 10,400 events per million patients treated	events uniform in person time, placebo participants switched at the decision, 1.5 to 3 QALYs per event; the bound adds a placebo event rate of 2.4 percent a year, a 20 percent relative reduction and instant uptake over the whole window

These are not savings the sponsor or the committee forgone: they say what the evidence would have permitted at a calibrated bar, and each currency stacks assumptions on the one before, so the trial cost is the firmest and the population bound the loosest. Two cautions apply to all of them. Among trials stopping early for efficacy, naive estimates tend to exaggerate benefit. Skeptical priors may reduce this bias; bias and interval coverage should be evaluated for the specified estimator and stopping rule. And a regulator wants safety follow-up that would continue after an efficacy decision, which recovers only part of the cost saving. The same reconstruction for REWIND crosses in only 49 percent of the simulated trials, so the all-trial median saving there is zero, although the unconditional expected saving can still be positive (Section 21.6); the counterweight below stands. The full accounting, with the REWIND figures stated conditionally, is in Section 21.6.

The counterweight is a trial with a smaller effect. REWIND, the dulaglutide cardiovascular trial, reported a hazard ratio of 0.88 on 1,257 events (Gerstein et al. 2019). Reconstructed in the same way, the four-look O'Brien-Fleming design acts by the end with probability 0.61 and the Bayes rule with probability 0.49, so the scheduled design acts more often there, even though the Bayes rule decides earlier when it decides (median 680 events against 942). The

book is about when each of these holds and how to show it by simulation before the trial starts. SELECT returns in Chapter 21 with the full reconstruction.

Drug development usually fails, a pivotal trial is expensive and takes years, and a day of delay at the end is worth more than most trials cost per week. A design that reaches the same legal standard of evidence with fewer patients and less time is therefore not a refinement. It is the main thing a sponsor can still change. This chapter makes that case with published numbers first, and only then turns to the machinery. Every figure quoted here is cited; where a figure is a simulation or a registry search, we say so.

1.2. A ledger before the trial

The decision view of medical evidence is older than the randomized trial, and its first great practitioner was a nurse. Polson and Scott tell the story in *AIQ* as the beginning of the idea that the numbers a hospital already collects can change what the hospital does (Polson and Scott 2018). Florence Nightingale worked at the British army's Barrack Hospital at Scutari, across the Bosphorus from Constantinople, from the winter of 1854 to 1855. Her figures are the founding data set of evidence based hospital care. The death rate among soldiers admitted was worst in February 1855, at 42.7 percent, and after sanitary reforms that began on 17 March 1855 it had fallen to 2.2 percent by June 1855 (Neuhauser 2003; Nightingale 1858). These are secondary-source figures: percentages of soldiers admitted who died in the month, as quoted by Neuhauser, and we have not rechecked them against the 1858 report.

```
# Uniform priors: rate ~ Beta(k + 1, n - k + 1); ratio of June to February
 ↪  rate.
ratio_ci <- function(k_feb, k_jun, n, B = 2e6) {
  set.seed(1855)
  r <- rbeta(B, k_jun + 1, n - k_jun + 1) / rbeta(B, k_feb + 1, n - k_feb +
 ↪  1)
  quantile(r, c(0.025, 0.5, 0.975))
}
rat1 <- unname(ratio_ci(427, 22, 1000))  # (lower, median, upper)
rat2 <- unname(ratio_ci(43, 2, 100))
```

Reading the ledger as a Bayesian. Suppose each of the two months had 1000 admissions, so that 427 and 22 deaths are recorded (illustrative counts, since the sources give percentages). With uniform priors on the two death rates, the posterior for the ratio of the June rate to the February rate has median 0.053 and 95 percent interval 0.034 to 0.078. With only 100 admissions a month (43 and 2 deaths, rounded to whole patients), the interval is 0.014 to 0.17. The size of the change is not in doubt at any plausible sample size, and no p-value is needed to see it.

What is in doubt is the cause. The comparison is before and after, not randomized: winter gave way to summer, and other things changed in the same months as the sewers were flushed. Historians dispute how much credit the sanitary reforms deserve, and Nightingale herself noted the arrival of the Sanitary Commission in explaining the decline. A Bayesian reading separates the two questions. The posterior for the change in mortality is concentrated. The posterior for the effect of sanitation on mortality depends on a model of what else changed, and the ledger alone cannot supply it.

Why acting was defensible. The decision framework does not require a trial here. The intervention was cheap, the harm of trying it small, and the loss from inaction was measured in hundreds of deaths a month, so the expected utility of reform plausibly exceeded that of waiting under a wide range of priors, while a randomized comparison that left some wards unreformed would have been costly. This is a judgment about inputs (the size of the loss, the cost of acting, the strength of the prior, the value of further information), and the same inputs govern every decision in this book. The ledger is also a reminder that the first step is to count.

1.3. Why trialists should want this

1.3.1. The price of a trial, of being wrong, and of waiting

Developing one approved drug is estimated to cost between roughly one and two and a half billion dollars once failures and the cost of capital are included: $2,558 million (2013 dollars) in the survey-based estimate of DiMasi and colleagues (DiMasi, Grabowski, and Hansen 2016), and a median of $985 million (mean $1,336 million, 2018 dollars) for 63 products with public R&D data in Wouters and colleagues (Wouters, McKee, and Luyten 2020). The pivotal trials themselves are a smaller but still large share. For the 138 pivotal trials behind 59 novel agents approved by the FDA in 2015 and 2016, the median estimated cost was $19.0 million per trial (T. J. Moore et al. 2018). Counted per approved drug, a median of $48 million went on pivotal trials, at $41,413 per patient (T. J. Moore et al. 2020). Phase 3 studies in the United States averaged between $11.5 million (dermatology) and $52.9 million (pain and anesthesia) (Sertkaya et al. 2016). A patient enrolled is a cost of tens of thousands of dollars, so a design that needs fewer patients saves money in proportion.

Most of the spending buys nothing that reaches a patient. Over 2011 to 2020 the likelihood that a drug entering phase I was eventually approved was 7.9 percent, and only 28.9 percent of candidates cleared phase II (BIO, Informa Pharma Intelligence, and QLS Advisors 2021). In oncology the overall probability of success was 3.4 percent (Wong, Siah, and Lo 2019). Phase II promise does not transfer easily: of 43 phase III studies that repeated the regimen of a preceding phase II study, only 12 (28 percent) were positive (Zia et al. 2005). A design that learns earlier that a drug does not work, and stops, converts a large part of that waste into patients and money for the next candidate.

Time is the third price. From phase I to approval took on average 10.5 years (2.3, 3.6, 3.3 and 1.3 years for phase I, phase II, phase III and regulatory review) (BIO, Informa Pharma Intelligence, and QLS Advisors 2021). A single day is worth about $500,000 in lost sales of a launched drug, plus about $40,000 of direct trial cost per day in phases II and III (Z. P. Smith, DiMasi, and Getz 2024). Those are revenue, not profit, and they are averages over 645 launched drugs, but the order of magnitude is clear: a month saved is of the order of $15 million of sales (our arithmetic, 30 days at the quoted daily figure) and, more important to the patient, a month sooner for an effective drug and a month less on an ineffective one.

1.3.2. What efficiency looks like in real trials

The numbers that follow come from five kinds of evidence, and they should not be mixed up. One is a simulation, three are trials or platforms (one of them frequentist) with their regulatory outcomes, and one is a reconstruction of a real trial.

A simulated drug-development program. Saville and Berry simulated 5,000 development programs in which one in ten treatments works. A traditional sequence of two-arm trials needed 1,966 patients on average and 12.0 years to find an effective treatment, and found one with probability 0.78. An open adaptive platform with response-adaptive randomization (ten active treatments) needed 640 patients and 3.2 years, and found one with probability 0.87 (Benjamin R. Saville and Berry 2016). That platform compares each arm with all control patients, including those enrolled before the arm entered; their concurrent-control response-adaptive design 3d, which also used five active treatments and a 100-patient cap per arm, needed 912 patients and 4.6 years, with probability 0.83. These designs differ in several respects, so the comparison does not isolate the contribution of nonconcurrent controls (the bias that non-concurrent controls can bring is discussed in Chapter 20). That is a 67 percent reduction in patients and a 73 percent reduction in time. It is a simulation under stated assumptions, not an empirical result, and Chapter 20 says which assumptions carry it.

A platform that reached a decision in months. REMAP-CAP is a Bayesian adaptive platform with predefined criteria for superiority, efficacy, equivalence and futility. Its first patient in the immune-modulation domain was randomized on 19 April 2020. On the data to 28 October 2020 the data monitoring board reported that tocilizumab had met the efficacy criterion, a posterior probability of 99.75 percent, an odds ratio of 1.87 for organ-support-free days (95% credible interval 1.20 to 2.76), about six months after the domain opened (REMAP-CAP Investigators, Gordon, et al. 2021). The RECOVERY platform was also fast: the dexamethasone result was announced on 16 June 2020, nearly 100 days after the protocol was first drafted, and was in UK practice that day (RECOVERY Collaborative Group et al. 2021). RECOVERY's analysis was frequentist, so it shows what the platform structure buys (a shared control, no setup time for the next question) and not what a Bayesian analysis buys.

A trial that graduated a drug on 115 evaluable patients. I-SPY 2 randomizes adaptively in neoadjuvant breast cancer, and an arm graduates when its Bayesian predictive probability of success in a 300-patient phase 3 trial reaches 85 percent for some biomarker signature. Neratinib randomized 127 patients against 84 concurrent controls, of whom 115 and 78 were evaluable. In the HER2-positive, hormone-receptor-negative signature the estimated pathological complete response rate was 56 percent (95% probability interval 37 to 73) against 33 percent (11 to 54), and the final predictive probability of phase 3 success was 79 percent. The design had anticipated that value would fall once all pathology data were in, which is why the bar was set at 85 percent (J. W. Park et al. 2016).

A decision that the regulator accepted on a Bayesian criterion. The pivotal trial of the BNT162b2 vaccine used a Bayesian success criterion: the posterior probability that vaccine efficacy exceeds 30 percent had to be above 0.995 at an interim analysis and above 0.986 at the final one, with the thresholds calibrated so that the overall one-sided type I error was 2.5 percent (Pfizer Inc. 2022). It reported 8 cases on vaccine against 162 on placebo (Polack et al. 2020). The REBYOTA product was approved in 2022 on a phase 3 trial analyzed with a Bayesian hierarchical model that borrowed a phase 2 trial (U.S. Food and Drug Administration, Center for Biologics Evaluation and Research 2022), and the FDA's January 2026 draft guidance names REBYOTA, the GBM AGILE and Precision Promise platform trials, and the pediatric diabetes supplements for empagliflozin and linagliptin as examples of Bayesian methods in use (U.S. Food and Drug Administration 2026g). Details are in Section 3.11.

A real trial, reconstructed. SELECT, the semaglutide cardiovascular outcomes trial that opens this chapter (Section 1.1), is the fifth kind of evidence: a published trial rebuilt as a

counterfactual, with a true hazard ratio of 0.80 and at most 1,270 events, and with calendar time and exposure labeled as modeled. Its headline numbers are given there, and Chapter 21 carries the full reconstruction.

1.3.3. Is it spreading?

The registry suggests so, with large caveats. Figure 1.1 plots the share of interventional trials starting each year (phase 2 or 3 and combined phases) whose ClinicalTrials.gov record mentions Bayesian methods, adaptive designs or a platform or master protocol. In 2010, 8 of 4,912 phase 2 or 3 trials mentioned Bayesian (0.16 percent); in 2025 it was 29 of 6,534 (0.44 percent), nearly a tripling of the share. Platform or master-protocol trials went from 0 in 2010 to 50 in 2025 (0.77 percent of phase 2 or 3 starts).

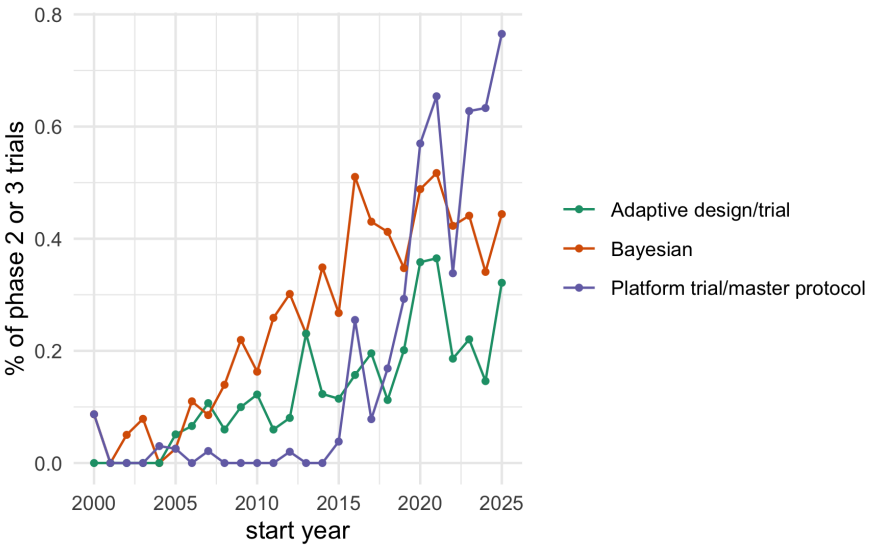


Figure 1.1.: Share of phase 2 or 3 interventional trials (including combined phases) starting each year whose ClinicalTrials.gov record mentions the term, 2000 to 2025. These are registry mention rates, not adoption rates: the search is free text, about 10 percent of Bayesian hits in a hand check of 20 records were false positives, many sponsors do not describe the design in the registry, terminology and registration practice change over time, and 2025 is incomplete. Downloaded 8 November 2026 (package dataset ctgov_counts).

The query is a text search, and in a hand check of 20 Bayesian records 18 truly used a Bayesian design or analysis, though most of those were Bayesian tools inside early-phase designs (BOIN, CRM, Bayesian safety monitoring) and not fully Bayesian confirmatory trials. Registries also under-report adaptive designs: in a review of registered trials, only 3 percent of a sample of adaptive trials known from the published literature were found by the registry search (Hatfield et al. 2016). Mentions increased between these cohorts, led by early phases and oncology, but changes in actual use cannot be separated from changes in reporting.

1.3.4. The counterweight

None of this says that a Bayesian design is automatically smaller or faster, and the book is built on saying where it is not. In the SELECT reconstruction the earlier decisions come from monitoring on the event grid at a bar that controls type I error, not from a lenient one. Against a single unadjusted look at a fixed level, the Bayesian bar is never lower without an informative prior, and the Bayes-factor bar is higher once $I_n\tau^2$ exceeds about 42 (derived below; here I_n is the total information accumulated and τ is the standard deviation of the prior on the effect under the alternative). And in a trial where the effect is modest, the scheduled design can win: in the REWIND reconstruction of Section 1.1 the Bayes rule decides earlier when it decides, but it decides less often. A sponsor should prefer a Bayesian design when the efficiency is real in that sponsor's setting, and should be able to show it by simulation. The rest of the book is how.

1.4. A trial is a measurement and a decision

D. R. Cox, in *Planning of Experiments* (Cox 1958), described the setting of the clinical trial as one of "large uncontrolled variations." In physics, once the apparatus is working, closely reproducible results emerge: the uncontrolled variation is small next to the effect being measured. A clinical trial lives in a different world. A patient's outcome swings with the natural history of the disease, with the environment, with the psychological response to being treated, and with the supportive care received. The effect of a new drug, often only a modest improvement over existing therapy, has to be picked out from underneath all of that.

Everything in this book follows from that sentence. Because the variation between patients is large and cannot be removed, a single trial can never prove anything. It can only change what we believe, and then someone has to act on the changed belief: approve or not, continue or stop, give the next patient drug A or drug B. A trial is therefore two things at once. It is a measurement, and it is a decision. The traditional apparatus that most readers use every day was built to make the measurement honest and to put a fixed ceiling on one kind of wrong decision. It does that well. This book asks what changes when the decision is put in the foreground, and what the regulator will want to see when it is.

1.5. The traditional trial, fairly stated

Most readers can recite the traditional design, and it deserves a careful statement before anything is set against it, because it is a coherent and successful answer to a real problem.

The sponsor picks a primary endpoint and a treatment effect θ that the trial should be able to detect. For a time-to-event endpoint, $\theta = -\log \mathrm{HR}$, so that benefit is positive. The null hypothesis is that the new treatment is no better than control. The sponsor fixes a one-sided type I error rate α, conventionally 0.025, which is the same as the familiar two-sided 0.05 when only benefit counts. It also fixes power $1-\beta$, say ninety percent, against the design alternative θ. A short calculation then gives the sample size,

$$I_n = \frac{(z_\alpha + z_\beta)^2}{\theta^2}, \qquad \text{which is} \qquad I_n = \frac{10.51}{\theta^2} \text{ at } \alpha = 0.025,\ 1-\beta = 0.90,$$

where I_n is the total information required (for a time-to-event endpoint with one-to-one randomization each event carries information 1/4, so the number of events is $4I_n$, as Chapter 9 shows) and z_α is the upper α quantile of the standard normal distribution. Patients are randomized one to one, because equal arms minimize the variance of the comparison for a given total. The protocol states in advance when the data will be analyzed, and the clinical study report (CSR) reports a point estimate, a confidence interval, and a p-value. The trial is "positive" if the p-value is below α.

Three features of this design are easy to underrate.

First, the type I error rate is a promise about the procedure, made before any data exist. If the drug does nothing, then over many trials run this way, at most a fraction α will declare success. That promise is checkable without anyone's opinion about how likely the drug is to work, and it is the reason the design is trusted by regulators who must deal with sponsors who have every incentive to see a benefit.

Second, the promise survives repeated use only if the analysis plan is fixed in advance. If the trial may be examined before it ends, the right to look is paid for. A group sequential design spends part of α at each interim analysis according to a pre-specified function (Lan and DeMets 1983), and the boundaries of O'Brien and Fleming (O'Brien and Fleming 1979) are very strict early and nearly the fixed-sample value at the end. If the trial has several arms, doses, endpoints or subgroups, the level is divided among the contrasts (Bonferroni, Holm) or passed along a pre-specified graph, so that the chance of any false claim stays at α. The apparatus is complete and well understood, and ICH E9 describes it at length.

Third, the regulatory machinery has grown around it. Statisticians at sponsors and at the regulator share a vocabulary, a set of software tools, and decades of precedent. A reviewer who sees "one-sided $\alpha = 0.025$, two interim looks with Lan-DeMets O'Brien-Fleming spending, graphical multiplicity procedure" knows what is meant without reading further. Nothing in this book is a reason to abandon that shared language. The question is which parts of the design can be improved, at what price, and in what terms a regulator will accept the improvement.

1.5.1. What the apparatus costs

Each component has a price that is rarely itemized, and the Bayesian case rests on stating those prices exactly. Most of the numbers below are computed in the code that follows; the combined four-look, six-contrast boundary of about 5.3 and the interval of 0.01 to 0.51 are quoted, not computed.

A sample size fixed in advance. The formula $I_n = 10.51/\theta^2$ commits the trial to a guess θ. If the true effect is larger than the guess, the trial runs longer than it needed to and patients stay on the inferior arm for no reason. If it is smaller, power is below target, although the estimate and confidence interval may remain informative. The cost of the guess is paid in one direction or the other, and the formula has no way to adjust.

An error budget spent in advance. For four equally spaced analyses, the O'Brien-Fleming boundaries in Figure 1.2 run from about 4.05 at the first look down to about 2.02 at the last. A trial that stops at the first interim look must have produced a result four standard errors from zero. That is a very high bar for a treatment effect of the size a confirmatory trial usually studies. What the sponsor purchases is the right to stop early *if the drug is spectacular*, and the price is paid by every trial where the drug is merely good. Testing

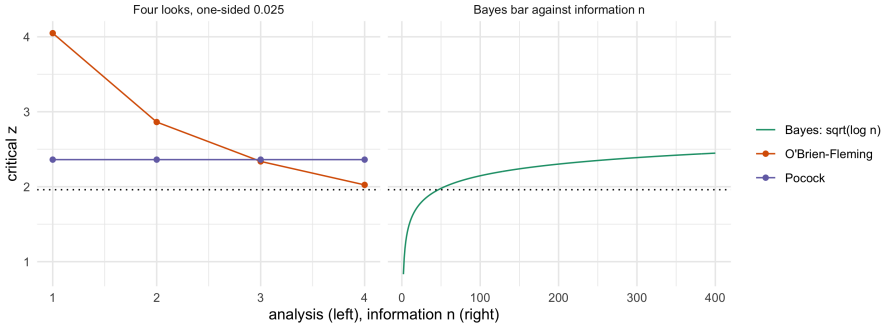


Figure 1.2.: Critical values on the standardized (z) scale. Left: four equally spaced looks at one-sided level 0.025. The fixed-sample value is 1.96. A naive repeated test at 1.96 has an overall one-sided error of about 0.063, so the boundaries must be raised: O'Brien-Fleming is very high early, Pocock is flat at 2.36. Right: the value of the Bayes factor rule, sqrt(log n), against the fixed value 1.96. In the logarithmic approximation the two cross at n = exp(1.96^2), about 46.6, where n is information times prior variance (a unit-information prior); the exact mixture boundary crosses at about 41.6, and at the prior scale used later in the book the crossing needs far more information (Chapter 11).

naively at 1.96 at each of four looks does not work either: the overall one-sided error is about 0.063, more than twice the nominal 0.025. That is the problem the spending function solves.

A multiplicity correction applied at the end. With four arms there are six pairwise contrasts. A Bonferroni correction takes the level to $0.025/6 = 0.00417$, with a critical value of 2.64 in place of 1.96. Combine that with four looks and the first interim boundary moves to about 5.3. The correction is conservative because it must be valid whatever the correlation between contrasts and whatever the truth, and it treats a contrast the same whether or not the other arms resemble it.

In small trials, an appearance of decisiveness. In dose-finding, the standard design enrolls twelve to thirty patients. Observing one toxicity in six patients at the recommended dose, with a Beta$(0.25, 0.75)$ prior, leaves a ninety-five percent posterior interval for the toxicity probability running from about 0.01 to 0.51 (Polson and Sokolov 2026b). A procedure that returns an interval that wide has not identified a dose, though the protocol will name one. We return to this in Chapter 17.

None of these costs is a flaw in the logic of the traditional design. They are what it costs to deliver the one promise, that of a fixed ceiling on false positives, under every circumstance in which the design might be used.

1.6. From a test to a decision

At any moment in a trial, someone with authority to act (a sponsor, a data monitoring committee, a regulator) has a small set of options. Continue enrolling. Stop for futility. Stop because the drug has clearly worked. Take the drug to the regulator. Each option has consequences for patients inside the trial, for patients outside it who will or will not

receive the drug later, and for the sponsor. A hypothesis test answers a narrower question, "is the data surprising if the drug does nothing?", and leaves the rest to judgment outside the statistics. The decision framework tries to put that judgment inside the model, where it can be inspected.

It needs three ingredients.

1. **A prior.** A probability distribution $\pi(\theta)$ for the treatment effect before the trial, summarizing what is known from earlier phases, from related drugs, from mechanism, and from the history of how often drugs entering a confirmatory trial turn out to work.
2. **A model for the data.** The likelihood $p(y \mid \theta)$ is the same as in the traditional analysis. Nothing about the Bayesian approach changes how the outcome data are modeled. After the data arrive, Bayes' rule turns the prior into a posterior $\pi(\theta \mid y)$.
3. **A utility (or loss).** A statement of what the possible outcomes are worth. Approving a drug that works gains something, approving one that does not harms the patients who take it and wastes resources, and failing to approve a good drug loses what those patients would have gained. Every additional month of the trial and every additional patient randomized to the inferior arm carries a cost.

The decision rule follows. At a terminal analysis, suppose the only choices are to approve or to reject. Act when the posterior expected loss of approving is smaller than that of rejecting. Suppose that approving a drug that does not work costs L_{I}, that failing to approve a drug that does work costs L_{II}, and that correct decisions cost nothing. If p is the posterior probability that the drug works, the expected loss of approving is $(1-p)L_{\mathrm{I}}$ and the expected loss of not approving is $p\,L_{\mathrm{II}}$. Approval is better when

$$p > \frac{L_{\mathrm{I}}}{L_{\mathrm{I}} + L_{\mathrm{II}}}. \tag{1.1}$$

The posterior probability must exceed a threshold that depends on the ratio of the two costs, and nothing else. If a false approval is thirty-nine times as costly as a missed drug, the threshold is $39/40 = 0.975$. The one-sided $\alpha = 0.025$ of the traditional design is, in this reading, an implicit statement of that ratio. An interim decision is harder: it must also compare both terminal losses with the sampling cost and the expected value of continuing, so the displayed threshold alone is not an interim stopping rule (Chapter 15).

```
# Flat prior on the effect, normal likelihood: posterior is N(est, se^2),
 ↪  so
# Pr(theta > 0 | data) = pnorm(est / se). Threshold = L_I / (L_I + L_II).
r <- c(1, 9, 39, 99); t <- r / (r + 1)
knitr::kable(data.frame(r, t, qnorm(t)), digits = c(0, 3, 2), col.names =
 ↪  c("Cost ratio (false approval : missed drug)", "Approve when posterior
 ↪  probability exceeds", "Equivalent z"), align = "rrr")
```

Cost ratio (false approval : missed drug)	Approve when posterior probability exceeds	Equivalent z
1	0.500	0.00
9	0.900	1.28
39	0.975	1.96
99	0.990	2.33

With a flat prior and a normal likelihood the posterior probability that the effect is positive equals one minus the one-sided p-value, so approving when it exceeds 0.975 is the same as rejecting when $p < 0.025$. The two designs agree numerically on that analysis, and that is the point of departure. A traditional analysis at the end of a fixed trial with no interim looks and one primary contrast is *already* a Bayesian decision with a flat prior and a particular cost ratio. The disagreements begin when there is more than that: looks, multiple arms, historical information, a cost per patient.

1.6.1. The three things a protocol fixes

Seen this way, every protocol, traditional or Bayesian, fixes three things before the first patient is randomized.

- **What is known going in.** In the traditional protocol this is implicit: the design alternative θ and a variance. In a Bayesian protocol it is explicit, as a prior, and the protocol must justify it (Chapter 8).
- **What will count as success.** In the traditional protocol, a p-value below α with adjustments. In a Bayesian protocol, a posterior criterion such as $\Pr(\delta > a \mid \text{data}) > c$, where δ is the treatment effect, a is the smallest effect that matters, and c is the posterior probability threshold, the form that appears in the FDA draft guidance (U.S. Food and Drug Administration 2026g).
- **What the data-generating process allows the sponsor to do along the way.** The sample size, the interim analyses, the permitted adaptations, the allocation ratio. These are the protocol's operating rules, and they determine the operating characteristics: the probability of declaring success under each possible truth.

The third item is where the regulator's interest is concentrated, and it is why the book returns so often to simulation. A regulator does not need to share a sponsor's prior or utility to evaluate a design. It can ask what the design *does*: under the null, how often does it declare success? Under a plausible alternative, how often does it succeed and how many patients does it use? A Bayesian design is judged, in practice, by the same standard as a traditional one, with the trial simulated many times under pre-specified truths (Chapter 14). The Bayesian approach changes how the rule is built, not the court in which it is judged.

1.6.2. Two kinds of "decision maker"

There is a second reason to put the decision in front. Two different people are deciding, and they are not asking the same question. In a study of the prostate cancer choices of two high-profile patients (Polson and Sokolov 2026d), we found that the physician and the patient reasoned about different quantities. The physician, trained to think in terms of the population and the most likely disease state, reasons from the modal case: the cancer is probably low risk, the patient can probably be cured by something not too aggressive, so why be aggressive for a small gain in cure probability? The patient, who has one life and one decision, reasons about the whole posterior, including the thin tail in which the disease has spread. A colleague of the first patient wrote that this is exactly the distinction: "It's different to think of things in population terms (and for other people) than to think about them as an individual patient with one life." Both patients recognized that the published risk estimate was a point estimate with known calibration error, and that it should propagate into the decision rather than be treated as exact.

The same asymmetry shows up in trial design. The regulator and the sponsor think at the level of the population: will the drug meet the standard? The patient in the trial, and the patient who will be prescribed the drug afterwards, bears the consequences one at a time. A design that lets the trial stop earlier when the posterior is decisive, or that allocates fewer patients to a losing arm, is a design that takes the second perspective seriously. Using the same decision-theoretic language for both is the attraction of the framework, and a utility function is where the disagreement between them can be written down and argued about instead of left implicit.

1.7. Preview: the thesis and four mechanisms

The thesis of this book can be stated in two sentences. A regulator approves a drug on evidence that meets a legal standard, and a Bayesian adaptive design can meet that standard sooner and with fewer patients exposed to inferior arms. The saving comes from monitoring, from handling multiple arms jointly, and from using prior information, and it does not come from a lower bar: against a single unadjusted analysis, the Bayesian bar is never lower without an informative prior, and the Bayes-factor bar is higher once $I_n\tau^2$ exceeds about 42.

That last claim is the one a skeptical reader should test first, so here is the arithmetic. A fixed-sample test at one-sided level 0.025 demands a standardized score of $z_\alpha = 1.96$ whatever the amount of information. The Bayes-factor rule, which with even prior odds and equal costs acts when the posterior probability of the alternative exceeds that of the null, demands, approximately, a score whose square exceeds the logarithm of the accumulated information n. This logarithmic penalty is sometimes called the Ockham penalty: a hypothesis that could explain any data is rewarded less than one that made a sharp prediction, and the reward shrinks as the amount of data grows (Chapter 7). For the stated mixture prior and $b = 1$ the exact boundary is $Z_n^2 \geq (1 + 1/q)\log(1 + q)$ with $Z_n > 0$ and $q = I_n\tau^2$ (information times prior variance); on the score scale the logarithmic approximation of the bar is $\sqrt{\log n}$. The exact boundary crosses 1.96^2 at $q \approx 41.6$, and the approximation $n = e^{1.96^2} \approx 46.6$ shown in the right-hand panel of Figure 1.2 overstates the crossing (Chapter 11). Beyond the crossing the Bayes-factor bar is *higher* than the fixed-sample bar (Polson and Sokolov 2026b). SELECT at the prior scale $\tau = 0.25$ has $q = 1270/64 \approx 19.8$, below the crossing. Holding the Bayes-factor threshold fixed can increase the required information, whereas calibrating a single-analysis rule to the same type I error can instead reproduce the conventional test, and Chapter 9 does the calculation carefully. Three rules are called Bayesian in this book and they should not be confused: the flat-prior posterior threshold of equation Equation 1.1 (numerically a one-sided test), the Bayes-factor rule just described, and the continuously monitored rule calibrated to a type I error of 0.025 (a posterior probability threshold of about $p^* = 0.995$ in the SELECT reconstruction of Chapter 21). The saving claimed in this book belongs to the third.

So where does the saving come from? Claims that Bayesian designs need fewer patients draw on four mechanisms that are usually run together. They differ in how much they assume, and the book keeps them apart (Polson and Sokolov 2026b).

M1. The deviation scale. The deviation scale is the size of the standardized score that a rule must see before it acts, as a function of the amount of data. A decision rule that must be valid at every possible stopping time, as an anytime-valid e-value rule is (a rule whose false-positive guarantee holds however often and whenever the data are examined),

pays a surcharge on top of the logarithmic term, which for the symmetric normal mixture is approximately $2\log(1/\alpha)$ in squared standardized units when information is large (exactly $(1 + 1/q)\,2\log(1/\alpha)$, with $q = I_n\tau^2$). In plain terms the surcharge is the extra evidence demanded in return for the right to look at any time without planning the looks. The uncalibrated Bayes-factor rule does not pay it. Asymptotically the surcharge is about seventy percent of the boundary term 10.51 of a conventional trial, a comparison of boundary terms and not an exact sample-size identity; calibration narrows the realized gap (Chapter 11 and Chapter 12). The surcharge is a property of this normal-mixture construction, not a universal cost of anytime validity, and it is not free.

M2. One calibrated threshold for monitoring, and shrinkage for estimation. A group sequential design pays for looking by raising the early boundaries, and Bonferroni pays for several contrasts by raising them again. A Bayesian analysis can monitor continuously with a single calibrated threshold (about 0.995 rather than 0.975 in SELECT), which is another monitoring boundary and not an escape from monitoring costs. For several arms it can shrink the estimates toward a common mean (pulling each arm's estimate toward the average of all the arms, more strongly when the arm has little data). Shrinkage improves estimation, but it does not by itself control the familywise error rate, particularly under partial null configurations, so the complete confirmatory decision rule still needs its error evaluated and, where required, calibrated. Any saving must be shown between designs with comparable error and power (Chapter 10, Chapter 11).

M3. Informative priors. If effects reaching a confirmatory trial are known to be small (A. Sokolova, Sokolov, and Polson 2026a), a prior concentrated on small effects carries real information, formally equivalent to a number of pseudo-observations whose size can be reported as a prior effective sample size (Morita, Thall, and Müller 2008). It is genuine when the prior is right and a bias when it is wrong, so it must come with a sensitivity analysis (Chapter 8, Chapter 19).

M4. Optimal stopping under a sampling cost. When each patient carries a cost, the best Bayes rule is found by working backward from the end of the trial (a dynamic program), and the continuation region (the set of interim results for which the trial keeps going) narrows as the horizon approaches, so its expected sample size is typically below that of a fixed design with comparable error rates (Chapter 15).

> **i** Technical note: the arithmetic of M1 and M2
>
> With $q = I_n\tau^2$, the exact symmetric normal-mixture bar is $Z_n^2 \geq (1 + 1/q)(\log(1 + q) + 2\log b)$ with $Z_n > 0$ for a one-sided claim, and the e-value bar is the same with $b = 1/\alpha$. The exact surcharge over the Bayes-factor bar at $b = 1$ is therefore exactly $(1 + 1/q)\,2\log(1/\alpha)$, whose large-information approximation is $2\log(1/\alpha) = 7.38$ at $\alpha = 0.025$. The seventy percent quoted above compares 7.38 with $(z_\alpha + z_\beta)^2 = 10.51$ for a conventional trial at 90% power; it compares boundary terms asymptotically and is not a predicted sample-size reduction (subtracting the sample-size equations adds a positive log term). We do not quote information ratios between the first O'Brien-Fleming boundary and the uncalibrated Bayes-factor bar, because that compares a calibrated rule with an uncalibrated one.

Two of the four mechanisms (M2 and M4) rest on structure that a regulator can verify by simulation. One (M3) rests on a judgment that must be defended. One (M1) is a mathematical fact, under stated approximations, that comes with a trade-off. Each is quantified in the chapters named, and the case studies of Chapter 21 apply them to real

trials, among them SELECT, the semaglutide cardiovascular outcomes trial of 17,604 patients that read out on 1,270 primary events (Lincoff et al. 2023), and the four-arm ulcer-surgery trial in which Novick and Grizzle (Novick and Grizzle 1965) made the original Bayesian case for treating several treatments together. The ulcer trial, taken up in Chapter 10, is worth previewing because it shows multiplicity and repeated analysis together defeating the frequentist reading of what looked like a real effect.

1.7.1. What we claim and what we do not

We claim that a Bayesian design can decide sooner when it is monitored, that the reason is the structure of the monitoring and of the multiple comparisons rather than a lenient threshold, and that the magnitude can matter in patient-years. We do not claim that it dominates. The surcharge a Bayesian rule declines to pay buys a real guarantee: a sponsor who cannot commit to a monitoring plan should pay it. A sponsor who runs a single well-powered look has no reason to pay an anytime-valid surcharge at all, and should use the fixed-sample test. The REWIND trial of dulaglutide is an example in which the scheduled design wins, and Chapter 21 says why. Equally, every Bayesian gain has a price (a prior that must be defended, simulations that must be run, a design that is more work to specify), and the price must be itemized in the currency the regulator accepts, which is simulated operating characteristics under a pre-specified design.

1.8. What the regulator needs, and what it does not

The regulatory standard is easy to misread in either direction. The statute says that a drug may be approved if there is substantial evidence that it has the effect claimed, and defines substantial evidence by reference to adequate and well-controlled investigations (United States Congress 1962). The regulation 21 CFR 314.126 lists what makes an investigation adequate and well-controlled (objectives, a valid control, bias-minimizing assignment and assessment, reliable response measures, and an adequate analysis; the full list is in Chapter 3) (U.S. Food and Drug Administration 2024b). Nowhere does the statute or the regulation say “p-value” or “type I error.” A test of significance is the method that ICH E9 describes and that decades of practice have settled on, but the legal standard is evidence, not a procedure.

This has two consequences that run through the book. First, a Bayesian design is not exotic in law. It is a different way of showing that the evidence is substantial, and the FDA’s 2026 draft says so by listing three routes to a success criterion: calibration to type I error, direct reading of the posterior, and a benefit-risk decision rule (U.S. Food and Drug Administration 2026g). Only the first is tied to type I error. Because the draft is a draft, it describes what the agency is prepared to discuss, not what it requires. Second, the standard is about the *whole design*: a valid comparison with a control, measures against bias, and a pre-specified analysis, usually achieved through randomization and blinding (the regulation also recognizes historical controls and settings where blinding is not feasible). A Bayesian posterior computed from a trial without a valid comparison and bias control does not meet 314.126 however sophisticated the model. Bayesian methods change the analysis and the adaptation rules, and they do not change the need for a fair comparison. Chapter 5 explains why a Bayesian still randomizes.

ICH E8(R1) adds a point that fits this book's emphasis on design. It asks sponsors to build quality into the trial from the start, by identifying the factors that are critical to the decision the trial will support and by concentrating effort on those (International Council for Harmonisation 2021). That is the decision framing in regulatory language: if you cannot say which decision the trial supports and which features of the trial bear on it, you cannot design it well. A Bayesian adaptive protocol makes those statements explicit, since the prior, the success criterion and the operating rules are all written down, and in that respect the Bayesian approach has an affinity with quality by design. ICH E9 and its addendum E9(R1) (International Council for Harmonisation 1998, 2019) remain the statistical baseline, and the FDA adaptive-design guidance of November 2019 (U.S. Food and Drug Administration 2019a) is the final guidance that allows pre-planned adaptations provided they preserve the integrity of the trial and control the chance of a false conclusion. All of these are described in Chapter 3.

The flat-prior equivalence of the Bayes rule and the one-sided p-value is derived for the normal model in Chapter 7, where it is also contrasted with a point null.

1.9. A short history

The decision-theoretic view of trials is as old as the randomized trial. Thompson asked in 1933 for the probability that one treatment is better than another and proposed allocating patients accordingly (Thompson 1933). Wald's sequential analysis came from the wartime problem of sampling only as much as needed (Wald 1947). Grundy, Healy and Rees posed the economic choice of the amount of experimentation in 1956 (Grundy, Healy, and Rees 1956), and Raiffa and Schlaifer's 1961 decision theory supplied the preposterior analysis (Raiffa and Schlaifer 1961). Anscombe and Colton observed in 1963 that a trial is run to inform the treatment of a much larger population of future patients, and derived trial sizes from that population rather than from an error rate (Anscombe 1963a; Colton 1963a). Novick and Grizzle gave a Bayesian analysis of a four-arm surgical trial in 1965 (Novick and Grizzle 1965). Cornfield gave a Bayesian sequential test for clinical trials in 1966, and set out the Bayesian outlook for trials in 1969, with a prior probability on the null hypothesis and relative betting odds as the summary of the evidence (Cornfield 1966a, 1969), and with Halperin and Greenhouse an adaptive allocation procedure (Cornfield, Halperin, and Greenhouse 1969); Lindley's decision theory (Dennis V. Lindley 1985) gives the general framework. Berry restated the case for Bayesian interim analysis in 1985 and spent the following decades building Bayesian adaptive designs (D. A. Berry 1985), including decision-theoretic stopping boundaries with Ho (D. A. Berry and Ho 1988b). Lewis and Berry evaluated Bayesian decision-theoretic designs by their classical operating characteristics (Lewis and Berry 1994a), the idea behind the calibration route that the FDA draft describes (Chapter 3). Whitehead's "case for frequentism" is the standard statement of the opposing view (Whitehead 1993). Inside the pharmaceutical industry, the Basel group of Racine, Grieve, Flühler and Smith showed that Bayesian methods could be used routinely in drug development (Racine et al. 1986). In January 2026 the FDA issued draft guidance on Bayesian methodology for drugs and biologics (U.S. Food and Drug Administration 2026g), and the EMA released a concept paper on Bayesian methods for consultation (European Medicines Agency, Methodology Working Party 2026); the regulatory history and the response to the FDA draft are in Section 3.5.2 and Section 3.5.4.

The objections deserve their strongest form. The prior can decide the answer. Prior choice can change the conclusion, and the FDA draft recommends prespecified sensitivity analyses

across reasonable alternative priors, with the influence of the primary prior quantified and justified (answered by sensitivity analysis and the effective sample size of Chapter 8). The type I error rate of a Bayesian rule is not controlled by the Bayes calculation itself (answered by calibration through simulation, Chapter 14). Interim peeking at accumulating data can bias the trial and the people who run it (answered by firewalls and prespecification, Chapter 11). And a Bayesian design is harder to review than a fixed-sample trial (answered, so far as it can be, by the simulation report and by the protocol content of Chapter 23). Berry, a leading Bayesian designer, wrote of the promise and the caution of adaptive trials (D. A. Berry 2011), Bauer and colleagues reviewed the opportunities and pitfalls of confirmatory adaptive designs (Peter Bauer et al. 2016), and Lee and Chu reviewed Bayesian trials in action (J. J. Lee and Chu 2012). The response to the 2026 draft guidance repeats these objections in current form (Section 3.5.4).

1.10. A map of the book

The book is built in five parts, and each starts from what the reader already does.

Part I, Trials, Decisions, and Regulation. This chapter frames the problem. Chapter 2 follows a drug through phases I to IV and shows where each kind of evidence, and each kind of error, enters. Chapter 3 is the map of the regulatory texts that every later chapter cites: statute and regulation, the ICH guidelines, the FDA guidances on adaptive designs, Bayesian methods, master protocols and substantial evidence, and the programs through which sponsors talk to the agency.

Part II, Basics. These chapters start from first principles, so that a reader who has forgotten testing can follow. Chapter 4 rebuilds the two-group comparison, the p-value, error rates and power from simulation. Chapter 5 explains why we allocate by chance, how it is done, and why a Bayesian still does it. Chapter 6 introduces prior, likelihood and posterior and repeats the two-group comparison the Bayesian way. Chapter 7 compares the evidence from p-values and Bayes factors and shows how the posterior probability criterion of the FDA draft reads. Chapter 8 is about choosing and defending a prior, which is where most regulatory discussion is concentrated.

Part III, The Standard Apparatus Reconsidered. Each chapter takes one component of the traditional design and asks what the decision framework says about it. Chapter 9 treats sample size and power, with the Bayesian replacement for the textbook formula. Chapter 10 treats multiple comparisons and shrinkage, including the ulcer trial. Chapter 11 treats interim looks and the three deviation scales that give the surcharge. Chapter 12 treats anytime-valid inference, which is a frequentist answer to the problem of looking at any time. Chapter 13 applies the same machinery to safety signals and safety monitoring, with the rofecoxib cumulative record as the running example.

Part IV, Bayesian Adaptive Designs. These chapters build the designs. Chapter 14 shows how to calibrate a Bayesian rule to type I error and how to write the simulation report the FDA draft and ICH E20 (draft, Step 2, June 2025) ask for. Chapter 15 treats predictive probability, futility and stopping. Chapter 16 treats response-adaptive randomization and bandits, including the controversies. Chapter 17 treats dose finding and dose optimization. Chapter 18 treats time-to-event endpoints, where events rather than patients carry the information. Chapter 19 treats borrowing of information from other trials, external controls and pediatric extrapolation. Chapter 20 treats master protocols and platform trials.

Part V, From Design to Approval. Chapter 21 applies the machinery to real trials, with the semaglutide cardiovascular outcomes trial SELECT, the dulaglutide trial REWIND, and TRAILBLAZER-ALZ 2 in Alzheimer's disease. It also covers the BNT162b2 vaccine trial with its Bayesian success criterion, and it values the saving in patient-years and money. Chapter 22 reanalyzes four landmark trials (GUSTO, EOLIA, ANDROMEDA-SHOCK and RECOVERY) and says what a reanalysis can and cannot claim. Chapter 23 turns the machinery into the document: the protocol table, the statistical analysis plan, the simulation report, the data monitoring committee charter, and what to say to the agency and when.

The appendices give the notation (Appendix A), a list of regulatory documents and programs with status and links (Appendix B), and the software and code used (Appendix C).

Three conventions hold throughout. The traditional version comes first, and the Bayesian version of the same decision follows on the same numbers, followed by a side-by-side comparison of what changes in the protocol, what stays the same, and what the regulator will ask. One-sided $\alpha = 0.025$ is the default level, and two-sided is stated whenever it is used. And whenever a Bayesian design is said to be better, the book says what it costs and where the traditional design is the better choice.

> In the protocol
>
> Whatever the design, the protocol and statistical analysis plan should answer five questions in so many words. What decision does this trial support, and who makes it? What prior information is used, if any, and why? What is the success criterion, in a form that can be evaluated by computer? What are the permitted adaptations and the stopping rules? What are the operating characteristics (type I error, power or probability of success, expected sample size) under pre-specified scenarios, and how were they computed? A traditional protocol answers most of these implicitly. A Bayesian protocol must answer them aloud, and that is as much of a benefit as it is a burden.

> Try it in the lab
>
> Open Module 12, "Decisions: what Type I control costs" (https://vsokolov.org/bayesian_trials_lab/12-decisions.html). Before you touch a control, predict what happens to the power of a fixed trial when the number of looks goes from one to four and the number of Bonferroni tests from one to six. Then set the sliders in section B and compare. In section A, move the cost ratio from 39:1 to 9:1 and see how the threshold $L_{\mathrm{I}}/(L_{\mathrm{I}} + L_{\mathrm{II}})$ in Equation 1.1 changes.

1.11. In practice

Three regulatory episodes show the decision framing at work. The detailed accounts, with thresholds and sources, are in Section 3.11, and the BNT162b2 calculation is worked in Section 6.12.

- **A Bayesian criterion and a frequentist guarantee are not alternatives.** The BNT162b2 posterior bar was tuned by simulation to a one-sided type I error of 2.5 percent (Chapter 14).

- **The number of trials needed is itself a threshold.** REBYOTA's two posterior thresholds stood for one trial and for two, and the stricter one, for two trials, was not met (Chapter 19).
- **A decision rule on a predictive probability stops arms early.** I-SPY 2 graduated neratinib on 115 evaluable patients (127 randomized), and the final predictive probability fell below the bar because the last outcomes arrive after the decision (Chapter 15).

Key points

- A clinical trial works against large uncontrolled variation between patients (Cox). It measures a treatment effect and, at the same time, supports a decision to approve, continue or stop.
- The traditional design (fixed $I_n = 10.51/\theta^2$ for $\alpha = 0.025$ and ninety percent power, one-to-one randomization, spending for looks, multiplicity correction, a p-value in the CSR) is coherent. Its central strength is a fixed ceiling on the false-positive rate, whatever the prior.
- A Bayesian trial adds a prior, a utility and a rule: at a terminal analysis, approve when the posterior expected loss of approving is smaller than that of rejecting (an interim decision must also weigh the value of continuing). With a flat prior and one look this is the same decision as a one-sided test, with a cost ratio of 39 to 1 at $\alpha = 0.025$.
- The law asks for substantial evidence from adequate and well-controlled investigations (FD&C Act 505(d), 21 CFR 314.126), not for a statistical philosophy. Bayesian methods change the analysis and the adaptation rules and leave the need for a fair, randomized, blinded comparison where it was.
- The saving from a Bayesian design comes from four mechanisms: the deviation scale, a single calibrated threshold for monitoring (with shrinkage for estimation, not a substitute for error control), informative priors, and optimal stopping. It does not come from a lower bar. Once information times prior variance exceeds about 42 ($I_n\tau^2 \gtrsim 42$ for the exact mixture boundary; the logarithmic approximation gives 46.6) the Bayes-factor bar is higher than a single unadjusted look.
- Each gain has a price, and the price is stated in simulated operating characteristics under a pre-specified design.

i Discussion questions

1. Pick a trial you have worked on. Write down the three things its protocol fixed: what was known going in, what counted as success, and what the sponsor could do along the way. Which were stated explicitly and which were left implicit?
2. A one-sided $\alpha = 0.025$ corresponds to a cost ratio of 39 to 1 between a false approval and a missed drug. In what disease settings would you argue for a different ratio, and who should decide it?
3. Explain in your own words why, with a flat prior and a single look, a posterior probability above 0.975 and a one-sided p-value below 0.025 are the same rule. What has to be added to the trial for the two to disagree?
4. The Bayes-factor bar is higher than 1.96 once $I_n\tau^2$ exceeds about 42, a crossing that SELECT at $\tau = 0.25$ does not reach. Under what circumstances would a sponsor still be right to choose a Bayesian design? Under what circumstances

would a single fixed-sample look be the better choice?

5. The physician in the prostate cancer example reasons from the modal state and the patient from the posterior. Where in a clinical trial does the same difference arise between the sponsor, the regulator and the patient?

2. The Life of a Drug: Phases I to IV

A new medicine reaches patients through a sequence of experiments, each larger, longer and more expensive than the last, and each built to answer a different question. The conventional names, Phase I to Phase IV, describe this sequence, and every later chapter of this book sits inside one of them. This chapter is the map. It says what each phase is for, what can go wrong in it, how the traditional design handles it, and where a Bayesian decision maker would act differently. Throughout, a phase is treated as an experiment whose value lies in the decision it informs: whether to proceed to the next phase, at what dose, in which patients.

! Regulatory anchor

The phases have regulatory definitions but need not form a rigid sequence. Chapter 3 describes each document in full and gives its status. Here only the roles matter.

- **21 CFR 312.21.** Describes Phases 1 to 3 of an investigation under an IND and expressly allows them to overlap, so a program is not obliged to complete one phase before starting the next.
- **ICH E8(R1), General Considerations for Clinical Studies (Step 4, October 2021).** Classifies studies by objective (human pharmacology, exploratory, confirmatory, postmarketing) and says that the classic phase labels do not fit every program. A program is a set of studies, each with a stated question, organized by objective and not necessarily in a fixed sequence.
- **ICH E9, Statistical Principles for Clinical Trials (1998).** Distinguishes exploratory from confirmatory trials. The type I error discipline of this book applies to the confirmatory stage.
- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft guidance, January 2026).** A draft, not binding. It is the document that most directly invites Bayesian methods at every stage, including the use of earlier-phase data to inform later ones.

2.1. Before Phase I

A compound enters human testing after discovery, lead optimization, and preclinical studies in cells and animals that establish pharmacology and a safe starting dose. In the United States the sponsor then files an Investigational New Drug application. Some programs run a Phase 0 or microdosing study, giving subtherapeutic doses to a handful of volunteers to check that the drug reaches its target and behaves pharmacokinetically as predicted. The prior for everything that follows is built here, from animal data, from mechanistic models and from the history of similar compounds.

2.2. Phase I: is it safe, and at what dose?

Phase I is the first exposure of humans to the drug. In most therapeutic areas it enrolls 20 to 100 healthy volunteers. In oncology, where the drugs are toxic, it enrolls patients for whom standard therapy has failed. The questions are tolerability, pharmacokinetics (what the body does to the drug), pharmacodynamics (what the drug does to the body), and, in oncology, the maximum tolerated dose.

The traditional oncology design is the 3+3 rule: treat three patients at a dose, escalate if none has a dose-limiting toxicity, add three more if one does, and stop if two or more do. The rule uses toxicity counts at the current dose, including any expansion cohort, without fitting a dose-toxicity curve, so it ignores the shape of that curve and identifies a dose with little precision. Model-based Bayesian designs replace it with a dose-toxicity model, for example

$$\operatorname{logit} \Pr(\text{toxicity} \mid d) = \log a + b \log(d/d^{\star}), \qquad a, b > 0,$$

the Bayesian logistic regression model of Neuenschwander, Branson, and Gsponer (2008). After each cohort the posterior over (a, b) is updated, and the next dose is the one with the highest posterior probability of a toxicity rate in the target interval, subject to a cap on the posterior probability of overdosing. The continual reassessment method (O'Quigley, Pepe, and Fisher 1990) and escalation with overdose control (Babb, Rogatko, and Zacks 1998) belong to the same family. Section 17.1 simulates the 3+3 and sets it beside these designs on the same scenarios, and Chapter 17 develops them.

i TGN1412: a design that could not learn between patients

In March 2006 six healthy volunteers in London received the first human doses of TGN1412, a monoclonal antibody against CD28, and two others received placebo (Suntharalingam et al. 2006). Within hours all six who received the drug developed a cytokine storm with multiple organ failure. The doses were given within minutes of one another, so the first volunteer's reaction could not inform the decision to dose the others. In the language of this book, the design gave the first observation no chance to update the action for the rest of the cohort. Staggered dosing, in which the first participant is dosed alone and observed before the next, is now standard practice. It is the cheapest possible way to let the posterior after one participant inform the next decision, and a reminder that a sequential design is a safety device before it is an efficiency device.

2.3. Phase II: does it work, and at what dose?

Phase II is where most drugs fail, and where most of the value of a development program is created or destroyed. Phase IIa is proof of concept: a small randomized or single-arm study asking whether the drug moves a clinical or biomarker endpoint at all. Phase IIb is dose ranging: several doses against placebo, to estimate the dose-response curve and choose the dose for Phase III.

Phase II is where Bayesian methods have the strongest claim, because its purpose is learning rather than confirming, a distinction made central to clinical pharmacology by Sheiner

(1997). The natural output of Phase II is not a p-value but a posterior over the dose-response curve and a predictive probability that Phase III will succeed. Chapter 15 builds that predictive probability, beginning from the traditional conditional power of Section 15.1, and Section 9.6.3 shows why a Phase III designed around the Phase II point estimate is usually underpowered. Chapter 17 treats dose finding and dose optimization, and Chapter 16 the allocation rules that adaptive Phase II designs borrow from bandit theory.

ASTIN: an early and well-founded no

The ASTIN trial tested a neutrophil inhibitory factor (UK-279,276) in acute ischemic stroke (Krams et al. 2003). Patients were allocated adaptively among many doses and placebo, with a Bayesian dose-response model updated as outcomes accrued and allocation chosen to learn most about the dose giving 95 percent of the maximum effect. The design included rules to stop for futility or efficacy. The trial found no treatment effect and stopped early for futility, at 966 patients according to the published report. A fixed design without interim analyses would have run to completion before learning the same thing, although a conventional group sequential design with a futility rule could also have stopped early. ASTIN is the standard example of a Bayesian adaptive design delivering its most valuable output, a fast and well-founded negative result.

2.4. Phase III: confirming, and the regulatory decision

Phase III trials are large, randomized and usually blinded, designed to confirm efficacy and characterize safety in the population for which the drug will be licensed. They enroll hundreds to tens of thousands of patients and cost from tens to hundreds of millions of dollars. The FDA's traditional evidentiary standard is two adequate and well-controlled trials, or one adequate and well-controlled trial with confirmatory evidence. This is the stage at which the type I error rate of Chapter 9 is the contract with the regulator, and at which interim analyses (Chapter 11, Chapter 15) are governed by the strictest rules. Chapter 3 describes the documents that apply.

Torcetrapib: a signal that belonged in the prior

In December 2006 Pfizer stopped ILLUMINATE, a Phase III outcomes trial of torcetrapib, a drug that raised HDL cholesterol, after the data monitoring committee found an excess of deaths on the drug (Barter et al. 2007). The program had cost hundreds of millions of dollars. Torcetrapib had been associated with higher blood pressure in earlier studies, a signal visible before the outcomes trial began. The episode illustrates two themes of this book: information from earlier phases about off-target effects belongs in the prior for Phase III, and the value of an interim analysis lies as much in stopping for harm as for benefit. Chapter 13 develops safety monitoring.

2.5. Phase IV: after approval

Approval is not the end of learning. Phase IV comprises postmarketing studies required or agreed at approval, observational surveillance of adverse events, and confirmatory trials for

drugs granted accelerated approval on a surrogate endpoint. Premarketing trials are often too small or too short to characterize rare harms reliably, and such harms become clear only when many more patients are exposed.

> **Rofecoxib: surveillance as a sequential decision**
>
> Rofecoxib (Vioxx) was approved in 1999 and withdrawn on 30 September 2004, after the APPROVe trial, a randomized comparison with placebo in 2,586 patients with a history of colorectal adenoma, showed a higher rate of confirmed thrombotic events on the drug (46 events against 26, rate ratio 1.92, 95% CI 1.19 to 3.11) (Bresalier et al. 2005). Earlier, the VIGOR trial in rheumatoid arthritis had shown fewer gastrointestinal events than with naproxen but more myocardial infarctions (Bombardier et al. 2000). A cumulative meta-analysis found that a statistically clear excess of myocardial infarction was already present at the end of 2000, years before withdrawal (Jüni et al. 2004). Postmarketing surveillance is a sequential decision problem in which the action is to warn, restrict or withdraw, and the cost of waiting is counted in patients harmed. Chapter 13 returns to the case.

Postmarketing surveillance is also where empirical Bayes has had its greatest regulatory success. Spontaneous reporting systems collect millions of reports of drug and adverse-event pairs. The observed count n_{ij} for drug i and event j is compared with the count E_{ij} expected under independence, and the crude ratio $\hat{\lambda}_{ij} = n_{ij}/E_{ij}$ is unstable when counts are small. The gamma-Poisson shrinker of DuMouchel (1999) instead defines a latent relative reporting rate λ_{ij} through the model $n_{ij} \sim \text{Poisson}(\lambda_{ij} E_{ij})$, with a mixture of gammas for λ_{ij}, and reports its posterior estimate, so that an estimate based on three reports is shrunk heavily toward the common distribution and one based on three hundred is not. This is the same kind of partial pooling that Chapter 20 describes for platform trials, applied at the scale of a national database.

2.6. The phases as a sequence of options

Let p_{12}, p_{23} and p_{3A} denote the probabilities of moving from Phase I to II, from II to III, and from III to approval, and let c_1, c_2, c_3 be the costs of the phases. Phase k is paid for only if all earlier phases succeed, so the expected cost per drug started is $c_1 + p_{12}c_2 + p_{12}p_{23}c_3$. The expected number of approvals per start is $p_{12}p_{23}p_{3A}$, and the ratio of the two expectations is the long-run cost per approval:

$$C_{\text{approved}} = \frac{c_1 + p_{12}c_2 + p_{12}p_{23}c_3}{p_{12}\, p_{23}\, p_{3A}}. \tag{2.1}$$

The formula explains why attrition dominates the economics of drug development. Wong, Siah, and Lo (2019) used over 400,000 trial records for more than 21,000 compounds, and estimated that about 13.8 percent of programs entering Phase I eventually reach approval, and 3.4 percent in oncology. Programs that use biomarkers to select patients have higher success probabilities. The sensitivity is easy to see with a stylized example. The probabilities and costs below are illustrative, not estimates from any source: they are chosen only so that the overall success probability is near the published range.

Table 2.1.: Stylized cost per approved drug from Equation 2.1, in millions of dollars. The baseline has p12 = 0.5, p23 = 0.3, p3A = 0.6 and phase costs of 25, 60 and 250 (illustrative values, not data). The rows change one input at a time.

Scenario	Cost per approval	Change (%)
Baseline	1028	0.0
Phase III 20% cheaper	944	-8.1
Phase II success 0.30 to 0.40	875	-14.9
Phase I success 0.50 to 0.60	981	-4.5

Raising the Phase II to III transition probability from 0.30 to 0.40 lowers the cost per approval by 15 percent, and making Phase III a fifth cheaper lowers it by 8 percent. The ranking of these two savings depends on the perturbations chosen: the first changes Phase II success by a third, the second reduces one cost by a fifth, so the comparison is not a general ranking of inputs. Nor is advancing more compounds in itself better selection; the case for the predictive methods of Chapter 15 is that they advance the right ones.

2.7. Bioequivalence: TOST and a posterior summary

A generic or reformulated drug must be shown to deliver the same exposure as the reference product. Exposure is measured by the area under the concentration-time curve, and the parameter of interest is the ratio $\rho = \mu_T/\mu_R$ of geometric means for test and reference. Bioequivalence is declared when ρ lies in an equivalence interval, conventionally $(0.80, 1.25)$. The classical analysis is the two one-sided tests (TOST) procedure of Schuirmann (1987), which gives the confidence-interval criterion a test form, on $\eta = \log \rho$, equivalent to checking that the 90 percent confidence interval lies inside $(\log 0.80, \log 1.25)$; Westlake had proposed a symmetric confidence interval for the same purpose (Westlake 1976), and Hauck and Anderson a different test of equivalence for two-group bioavailability trials (Hauck and Anderson 1984).

The Bayesian version asks the question directly; it goes back to Rodda and Davis (Rodda and Davis 1980), Mandallaz and Mau (Mandallaz and Mau 1981) and Selwyn, Dempster and Hall (Selwyn, Dempster, and Hall 1981), with the Basel worked example of Flühler et al. (1983), a two-stage procedure by Racine-Poon and colleagues (Racine-Poon et al. 1987), and a decision-analytic treatment by Lindley (Dennis V. Lindley 1998). Take a simplified model in which the n within-subject log differences d_i are independent and normal with mean η, sample mean $\bar{d}$ and standard deviation s, and use a reference prior. (A standard two-sequence, two-period crossover analysis also models period and sequence effects and generally has different residual degrees of freedom, so this is a simplification, not the regulatory analysis.) Then the posterior of η is a scaled t distribution, $\eta \mid d \sim \bar{d} + (s/\sqrt{n})\, t_{n-1}$, and the reported quantity is

$$\Pr(\log 0.80 < \eta < \log 1.25 \mid d) = F_{t_{n-1}}\left(\frac{\log 1.25 - \bar{d}}{s/\sqrt{n}}\right) - F_{t_{n-1}}\left(\frac{\log 0.80 - \bar{d}}{s/\sqrt{n}}\right). \tag{2.2}$$

A rule that concludes bioequivalence when this probability exceeds a threshold such as 0.90 is an alternative to TOST, not the conventional criterion: TOST is the standard in FDA's final guidance *Statistical Approaches to Establishing Bioequivalence* (U.S. Food and Drug

Administration 2026h) (May 2026, Section II.C.1.a), and a posterior-probability rule would need agreement with the regulator and an evaluation of its operating characteristics. The relation to TOST is exact under the reference prior. Each one-sided posterior tail probability equals the corresponding one-sided TOST p-value, so the posterior probability of equivalence is $1 - p_L - p_U$. If both p-values are below 0.05, the posterior probability exceeds 0.90. The two procedures can therefore disagree only in one direction: TOST at level 0.05 implies a posterior probability above 0.90, and a posterior probability above 0.90 does not imply TOST success. Choosing the threshold 0.95 (and, to match, TOST at 0.025 on each side) changes the comparison. The posterior probability is therefore a supplementary summary of the same data that is easy to read, and it is a weaker rule at 0.90: some data sets pass it and fail TOST (the third row of Table 2.2, with posterior probability 0.9291 and TOST p-value 0.0709). Racine-Poon et al. (1987) made the procedure sequential, using the posterior and its predictive distribution after a first stage to decide whether to stop or how many further subjects to recruit, the template for the interim decision rules of Chapter 15. Average bioequivalence is also not the only approach regulators use: for highly variable drugs some product-specific recommendations use reference-scaled approaches, so the Bayesian rule is one design choice within a family and not a replacement for it.

Table 2.2.: Bayesian and TOST bioequivalence for four illustrative crossover scenarios (made-up values): within-subject SD of the log differences 0.30 throughout, and mean log difference 0.03 (n = 24 and n = 12) or 0.13 and 0.15 (n = 24). Posterior probability from the formula above under the reference prior; TOST p-value is the larger of the two one-sided p-values.

	Posterior prob. of equivalence	TOST p-value
Centered, n = 24	0.9976	0.0022
Centered, n = 12	0.9693	0.0238
Shifted to 0.13, n = 24	0.9291	0.0709
Shifted to 0.15, n = 24	0.8778	0.1222

2.8. Recruitment and the value of speed

Most trials are slow because of accrual, not analysis, and the recruitment rate is the quantity a sponsor can least control. One episode shows both how to estimate a rate and what a recruit is worth. In November 1995 Michael Milken, who had founded CaP CURE (later the Prostate Cancer Foundation) after his own prostate cancer diagnosis, appeared on CNN's Larry King Live with General Norman Schwarzkopf and the molecular biologist Leroy (Lee) Hood of the University of Washington, and asked families with several men affected by prostate cancer to volunteer for a genetic study (Prostate Cancer Foundation 2025; Daniels 2004). More than 3,000 people called. Within three weeks Hood's group had a sample of about 300 qualified families, where Johns Hopkins had recruited 90 families over ten years (Daniels 2004); a University of Washington release of 1999 reports 291 families enrolled in the resulting PROGRESS study, many of whom came after seeing the broadcast (University of Washington 1999). The coauthors' working paper analyzes the episode (A. Sokolova, Sokolov, and Polson 2026b), and the calculations below follow it.

Model enrollment of eligible families as a Poisson process with rate λ per week. With the reference prior Gamma$(1/2, 0)$ and n families observed in t weeks, the posterior is

Gamma$(n + 1/2, t)$, the conjugate update of Chapter 6. The conventional pace is not known. "Several years" for 300 families is an assumption, and we bracket it at three to five years, which we treat as fixed scenario rates of 300/156 to 300/260 families a week, not as data.

The posterior for the broadcast rate has mean 100 families a week and standard deviation 5.8, so the counting error is small. The ratio of the broadcast rate to an assumed fixed conventional rate is 52 (95 percent interval 46 to 58) if the conventional pace was three years, and 87 (77 to 97) if it was five. These are conditional comparisons against assumed conventional rates, rather than estimates from conventional recruitment data. The real uncertainty is the counterfactual pace, not the count. The reported three-week identification of eligible families was much faster than the assumed conventional scenarios, and completed enrollment and specimen-collection rates are not established by this account. This is the Bayesian answer to the planning question of how long accrual will take, and it is the same calculation that, with a prior borrowed from earlier sites, turns a protocol's accrual assumption into a predictive distribution of the end date.

A rate is not the whole story, because a recruit is worth the evidence it contributes. Linkage studies report a LOD score, the base ten logarithm of the likelihood ratio for linkage against no linkage (Morton 1955). At a fixed recombination fraction and genetic model, a LOD score is a log likelihood ratio. A posterior probability of linkage requires a marginal Bayes factor B in the sense of Chapter 7, integrating over uncertain linkage parameters, and with prior probability π that a disease gene lies in the region tested it equals $\pi B/(\pi B + 1 - \pi)$. A maximized LOD of 3 does not automatically imply $B = 1000$ (Génin, Martinez, and Clerget-Darpoux 1995). If it did, with a prior of one in fifty the posterior probability would be 0.95. Because log likelihood ratios add across independent families, the value of a family is the expected LOD it contributes. A broad appeal brings in families whose clustering reflects screening rather than inheritance (phenocopies) and families whose disease is not linked to the locus under study, and both dilute that expectation. Under a crude working approximation, expected LOD per family is $e_1\alpha(1 - \phi)$, with e_1 the expected LOD of a fully informative linked family, α the fraction of families linked to the locus and ϕ the phenocopy dilution. These are illustrative inputs, not estimates: with $e_1 = 0.3$, $\alpha = 0.2$ and $\phi = 0.3$, a family is worth 0.042 LOD, and under this toy assumption about expected information approximately 71 families would give an expected cumulative LOD of 3, against 29 if $\alpha = 0.34$ and $\phi = 0$. These are mean-path calculations, not sample-size requirements: enrollment planning needs a specified probability of reaching the threshold and a model for family heterogeneity, and unlinked families need not contribute zero expected log likelihood ratio. Volume rose by a factor of fifty or more while quality per family may have fallen (a hypothesis, since comparative pedigree and informativeness data would be needed to establish it), so the sensible design takes the volume and then weights families by their posterior probability of carrying the signal, which is what a hierarchical model does automatically (Chapter 19). The same logic applies to a trial: count the information a patient contributes, not the patients.

The value of the acceleration is the part that depends on assumptions, and every one is labeled. Suppose the resulting family resource has probability p of leading to a clinically useful genetic test (assumed), that the test reduces mortality among hereditary cases by a fraction r (assumed), and that the broadcast brings that benefit forward by y years (assumed, and smaller than the years of accrual saved, because genotyping and replication take their own time). Take 37,000 United States prostate cancer deaths a year (late 1990s, as the working paper reports), a hereditary share of 9 percent (an assumption), ten life years lost per death (assumed) and \$100,000 per life year (assumed). The expected value is $V = 37{,}000 \times 0.09 \times r \times 10 \times \$100{,}000 \times y \times p$, which is \$67 million at $p = 0.1$, $r = 0.1$,

$y = 2$ and \$500 million at $p = 0.25$, $r = 0.2$, $y = 3$. If the appeal replaced an assumed \$2,000 to \$5,000 of conventional outreach expenditure per family and had negligible marginal cost, the modeled gross saving would be \$0.6 million to \$1.5 million. Under the chosen assumptions, the larger modeled value comes from earlier clinical benefit. The range is wide because the inputs are, and it is an ex ante expectation: the first locus reported, HPC1 on 1q24-25 from 91 high-risk families (J. R. Smith et al. 1996), did not become a clinical test, and the actionable variant, HOXB13 G84E, was reported in 2012 (Ewing et al. 2012). An expectation can be the right basis for a decision even when one draw is disappointing. The full treatment of the value of information is in the companion book *Personalized Medicine*.

2.9. A Bayesian reading of the phases

Each phase updates a posterior and hands it to the next. Phase I produces a posterior over the dose-toxicity curve and the pharmacokinetic parameters, which becomes the prior for dose selection in Phase II. Phase II produces a posterior over the dose-response curve, and its predictive distribution gives the probability that Phase III will succeed. Phase III produces the evidence on which approval is decided. Phase IV keeps updating after approval. Traditional development also passes information forward: estimates and their uncertainty, dose-response and pharmacokinetic models, and safety findings all inform the next phase, and the confirmatory phase is where formal hypothesis tests carry the decision. The distinction is not that the traditional path passes verdicts and the Bayesian path passes distributions. It is that a Bayesian development program specifies one probability model across the phases, so that what one phase learns propagates into the predictive probabilities and decisions of the next, whereas the traditional path links phases through estimates and judgment. The rest of the book is about what this buys, and about the conditions (calibration, pre-specification, simulation) under which a regulator will accept the alternative. Table 2.3 is the map from phase to chapter.

Table 2.3.: Phases, the questions they answer, and where this book treats each.

Phase	Question	Traditional design	Bayesian method	Chapter
I	Safe? Which dose?	3+3	Model-based escalation (BLRM, CRM, EWOC)	Chapter 17
II	Works? Which dose?	Fixed-dose arms, single-arm response test	Dose-response models, predictive probability, adaptive allocation	Chapter 17, Chapter 15, Chapter 16
III	Confirmed?	Fixed or group sequential, type I error 0.025	Posterior success criteria calibrated to error control, borrowing	Chapter 11, Chapter 9, Chapter 19

Phase	Question	Traditional design	Bayesian method	Chapter
IV	Harm after approval?	Spontaneous reports, observational studies	Empirical Bayes shrinkage, sequential monitoring	Chapter 13

Key points

- The phases are experiments in which each hands the next a decision. Traditional development passes estimates, models and safety findings forward; a Bayesian program additionally propagates one probability model into predictive decisions.
- Phase I is a safety problem first: TGN1412 shows the cost of a design that cannot update between patients. Model-based escalation fits a dose-toxicity curve to all accumulated toxicity data, where the 3+3 uses only the counts at the current dose.
- Phase II is where most programs fail and where Bayesian learning pays most: a posterior over the dose-response curve and a predictive probability of Phase III success replace a p-value.
- Phase III is confirmatory, and a Bayesian criterion there is judged by its operating characteristics under a pre-specified design.
- Phase IV is a sequential decision problem. The rofecoxib record shows that evidence of harm can be in hand years before action.
- Cost per approval, Equation 2.1, falls with each transition probability and each phase cost. In the stylized example a one-third rise in Phase II success saves more than a one-fifth cut in Phase III cost, but that ranking depends on the perturbations chosen, and advancing more compounds is not itself better selection.
- Bayesian bioequivalence is a familiar posterior probability. Under a reference prior, TOST success at 0.05 implies a posterior probability above 0.90, but not conversely, so the posterior rule is a supplementary summary or an alternative needing regulatory agreement, while TOST remains the conventional criterion.
- Recruitment is usually the slowest step, and a recruit is worth the evidence it adds. A gamma-Poisson posterior turns an enrollment count into a rate with an interval, and a LOD score, a log likelihood ratio at a fixed model, shows why more participants can mean less information per participant when a broad appeal dilutes the signal.

Discussion questions

1. For each of Phases I to IV, state in one sentence the decision the phase informs and the kind of error that matters most in it. Where does a type I error rate become the contract with the regulator?
2. The stylized cost-per-approval example changes one input at a time. What does it establish, and what does it not? Why can a ranking of two savings depend on the size of the perturbations chosen?
3. Why might advancing more compounds from Phase II to Phase III be a worse decision, even though it raises the Phase II transition probability?
4. Under a reference prior, TOST success at level 0.05 implies a posterior probability of equivalence above 0.90, but not the reverse. Explain why, and say what a

sponsor would have to show a regulator before using the posterior rule as the primary criterion.

5. TGN1412 and ASTIN are both sequential designs. What did each show about the value of letting early data change later actions?
6. In the Larry King episode the enrollment rate rose fiftyfold or more while the expected LOD per family probably fell. How should a trial team decide whether a faster, broader recruitment channel is worth using, and what would it need to measure?

3. The Regulatory Framework

A drug is approved or not approved on a legal finding, and everything in this book is aimed at that finding. A trial can be elegant, efficient and ethically admirable, and it still fails its sponsor if the evidence it produces does not meet the standard the law sets. The standard does not say how the evidence must be produced, and that gap is where design choices, including Bayesian and adaptive ones, live. This chapter is the map. It says what the law requires, what international and national guidance recommends on top of it, which meetings a sponsor can ask for, and where the European and British regulators agree or differ.

Two cautions before starting. First, the regulatory text changes quickly. Several of the documents below were issued as drafts in 2025 and 2026, and a draft can be revised or withdrawn. The statuses in this chapter are as of November 2026, and the dates are those printed on the documents. Second, regulatory documents differ in force. A statute binds. A regulation binds. A guidance "contains nonbinding recommendations". This chapter keeps those categories apart.

! Regulatory anchor

This chapter is about the anchors themselves. The documents that carry the legal and procedural weight are:

- **FD&C Act section 505(d)** (United States Congress 1962), as amended in 1997 (United States Congress 1997): the drug cannot be approved without "substantial evidence" that it has the effect it claims, defined as adequate and well-controlled investigations. This is law.
- **21 CFR 314.126** (U.S. Food and Drug Administration 2024b): the regulation that lists the characteristics of an adequate and well-controlled study. This is regulation.
- **FDA draft guidance on substantial evidence, June 2026** (U.S. Food and Drug Administration 2026a): how FDA reads the standard, including when one trial plus confirmatory evidence can suffice. Draft, nonbinding.
- **ICH E8(R1), E9, E9(R1), E20 (draft), M15** (International Council for Harmonisation 2021, 1998, 2019, 2025a, 2026a): the harmonized statistical and design principles that FDA, EMA and other regulators adopt as guidance.
- **FDA Adaptive Designs (final, November 2019)** (U.S. Food and Drug Administration 2019a) and **Bayesian Methodology (draft, January 2026)** (U.S. Food and Drug Administration 2026g): the two FDA documents that govern the designs in this book.
- **FDA Formal Meetings guidance (August 2026)** (U.S. Food and Drug Administration 2026c) and the **Complex Innovative Trial Design guidance (December 2020)** (U.S. Food and Drug Administration 2020a): how a sponsor gets answers from FDA before the trial starts.

3.1. The legal standard

3.1.1. What the statute says

The central sentence of US drug regulation comes from the Kefauver-Harris Amendments of 1962, passed after the thalidomide tragedy, which for the first time required manufacturers to show that a drug works and not only that it is reasonably safe. Section 505(d) of the Federal Food, Drug, and Cosmetic Act directs FDA to refuse approval of a new drug application if there is a lack of substantial evidence that the drug will have the effect it purports or is represented to have under the conditions of use in the labeling. The Act then defines substantial evidence as "evidence consisting of adequate and well-controlled investigations", including clinical investigations, by experts qualified by training and experience, on the basis of which it could fairly and responsibly be concluded by such experts that the drug will have the claimed effect (United States Congress 1962).

Two features of this definition matter for everything that follows. The first is the plural: the statute speaks of investigations, and for decades FDA read that as two independent adequate and well-controlled trials, each with a positive result. The second is that the standard is one of evidence and not of method: the statute names no test, level, sample size, prior or design, and asks only whether qualified experts could fairly and responsibly conclude that the drug works. Frequentist hypothesis testing at a one-sided 0.025 level became the customary way to show this, and the customary way hardened into expectation, but the law does not mention it. The Bayesian methodology draft says so in effect when it allows that its default calibration to 0.025 "may not be applicable or appropriate" in some circumstances (U.S. Food and Drug Administration 2026g).

3.1.2. The 1997 amendment: one trial and confirmatory evidence

The Food and Drug Administration Modernization Act of 1997 (United States Congress 1997) added a sentence to section 505(d): if the Secretary determines, based on relevant science, that "data from one adequate and well-controlled clinical investigation and confirmatory evidence" are sufficient to establish effectiveness, the Secretary may consider those data to constitute substantial evidence. The confirmatory evidence may come from before or after the trial. The power is discretionary: it is FDA's determination, based on relevant science, not a right of the sponsor.

The amendment has been used more often over time, especially in oncology, rare diseases and pediatric settings where a second trial is impractical or unethical. FDA's draft guidance on the topic was first issued in December 2019 (U.S. Food and Drug Administration 2019c) and was revised in June 2026 (U.S. Food and Drug Administration 2026a), which, when finalized, will replace the 1998 guidance *Providing Clinical Evidence of Effectiveness for Human Drug and Biological Products*. The revision was announced in the Federal Register on 24 June 2026 under docket FDA-2019-D-4964.

3.1.3. The regulation: what "adequate and well-controlled" means

The statute leaves "adequate and well-controlled" undefined. FDA's regulation at 21 CFR 314.126 defines it, and it is the single most useful list for a trialist to carry in the head (U.S. Food and Drug Administration 2024b). The regulation says the purpose of conducting

clinical investigations of a drug is to distinguish the effect of the drug from other influences, such as spontaneous change in the course of the disease, placebo effect, or biased observation. It then lists the characteristics of an adequate and well-controlled study. In paraphrase, the study:

1. has a clear statement of its objectives and of the method of analysis to be used;
2. uses a design that permits a valid comparison with a control, so that the effect of the drug can be quantified (the regulation names placebo concurrent control, dose-comparison, no treatment, active treatment and historical control as recognized types);
3. selects subjects by an adequate method, with reasonable assurance that they have the disease or condition being studied;
4. assigns subjects to treatment and control groups in a way that minimizes bias and assures comparability on baseline factors such as age, sex, severity and duration of disease (randomization is the usual way);
5. takes adequate measures to minimize bias on the part of subjects, observers and analysts of the data (blinding is the usual way);
6. uses well-defined and reliable methods of assessing response;
7. analyses the results adequately to assess the effects of the drug.

Two observations. Item 7 asks for an adequate analysis, and item 1 expects the protocol to state the proposed methods of analysis, and where it did not, the report to explain how the methods were chosen. Items 1 and 7 together are the regulatory root of the pre-specification that every guidance in this chapter repeats, and it applies to Bayesian analyses as it applies to frequentist ones. Item 2 admits historical control as a recognized type, but the 2026 draft cautions that external controls have a weakness: differences between groups arise "because of the lack of randomized treatment assignment" (U.S. Food and Drug Administration 2026a). The hierarchy of controls will come back in Chapter 19.

3.1.4. The 2026 substantial-evidence draft

The June 2026 revised draft guidance, *Demonstrating Substantial Evidence of Effectiveness for Human Drug and Biological Products* (U.S. Food and Drug Administration 2026a), restates how FDA reads the statute. It is a draft, so it is not for implementation, and it revises the draft issued in December 2019 (U.S. Food and Drug Administration 2019c). What it says that matters here, in the order the document gives it:

- It describes the conventional route (two trials) and a single-trial route. The single-trial route "will involve a highly persuasive trial or a source of strong confirmatory evidence" (Section IV.A of the draft).
- It lists the factors FDA weighs in deciding whether one trial and confirmatory evidence suffice: the seriousness of the disease and the unmet need, the strength of the single trial, the strength and source of the confirmatory evidence, and "whether it is ethical and practicable to conduct more than one adequate and well-controlled trial".
- It allows confirmatory evidence of types that are not trials. For instance, "mechanistic evidence of the drug's treatment effect in a particular disease may be appropriate to use as confirmatory evidence", within the regulatory flexibility the draft describes in its Section V for serious disease and unmet need.
- It says that the plan for one trial "should be discussed with FDA prior to initiating the single trial".

The draft explicitly recognizes Bayesian analyses, and it matters for them: a Bayesian analysis that uses an informative prior built from earlier data is a close cousin of "one trial plus confirmatory evidence", and the borrowing discussion in Chapter 19 returns to the connection. Prespecified borrowing formalizes how earlier evidence enters the analysis, but FDA still evaluates its relevance, weight and contribution to substantial evidence, including in discussions before the trial.

Try it in the lab

Open Module 11, *What counts as proof* (https://vsokolov.org/bayesian_trials_lab/11-substantial-evidence.html). Before you touch a control, predict: a first trial just clears the win line with p = 0.04 (two-sided, counting luck in both directions). If the drug's true effect is exactly what that trial saw, what is the chance that a second trial of the same size also wins? Then run the repeats. The lab's answer, about half, is the statistical reason a regulator may prefer replication; its second panel shows the other side, two independent wins at one-sided 0.025 by luck about once in 1,600 times.

3.1.5. What the standard implies for the numbers

Two adequate and well-controlled trials, each tested at a two-sided 0.05 level (one-sided 0.025), independent and each required to succeed in the prespecified beneficial direction, imply that a useless drug passes both only with probability $0.025 \times 0.025 = 1/1600$. A single trial at 0.025 passes a useless drug 1 time in 40. A single trial at a stricter level (the book's companion online lab, module 11, https://vsokolov.org/bayesian_trials_lab/11-substantial-evidence.html, uses 0.001) passes a useless drug 1 time in 1,000. In this sense, one very persuasive trial and two ordinary trials can offer comparable protection against a false positive, which is why FDA's willingness to accept "a highly persuasive trial" is consistent with the statute. For trials with an overall frequentist statistical analysis plan, the Bayesian methodology draft says that the familywise type I error rate is almost always required to be no greater than 0.025 one-sided (U.S. Food and Drug Administration 2026g). The draft also describes direct interpretation of posterior probabilities and decision-theoretic or benefit-risk criteria, so calibration is one of its routes (the book's default for confirmatory adaptive designs without borrowing), not a universal requirement. FDA's REBYOTA review used two posterior thresholds chosen as analogues of one-sided type I error rates of 0.00125 and 0.025 without borrowing (Section 3.11). The notation of this book follows suit: one-sided $\alpha = 0.025$ is the default, and we say so whenever we use two-sided.

3.2. What a traditional submission looks like

It is worth describing the traditional package accurately, because it is what a regulatory reviewer expects to see and what the Bayesian design is measured against.

The development path. A sponsor begins with nonclinical work, then files an Investigational New Drug application (IND). FDA has 30 days to object; if it does not, the first human trials can begin. Phase 1 examines tolerability and pharmacokinetics, phase 2 looks for efficacy and chooses a dose, and phase 3 is the pivotal comparison, usually randomized, controlled and blinded. The sponsor then files a New Drug Application (NDA) or, for biologics, a Biologics License Application (BLA), which FDA reviews, checking completeness

for 60 days first. At the major junctures FDA offers meetings: a pre-IND meeting, an end-of-phase 2 meeting to agree the phase 3 plan, and a pre-NDA or pre-BLA meeting to confirm the file is complete. Module 13 of the companion lab draws this road.

The phases themselves, what each is designed to learn and how Bayesian methods enter each, are the subject of Chapter 2, the chapter before this one. Bioequivalence studies, which establish that a generic or reformulated product delivers the same exposure as a reference product and so replace a new efficacy trial, follow a different pathway from the phase 1 to 3 efficacy sequence when they support a generic (ANDA) approval. They also occur inside an innovator program, for example to bridge formulations, and FDA's final bioequivalence guidance covers INDs and NDAs as well as ANDAs. Chapter 2 treats the Bayesian version of the bioequivalence question.

The pivotal trial protocol. The protocol fixes, before any patient is randomized, the primary endpoint and its definition, the estimand (the precise treatment effect the trial targets, with a stated strategy for intercurrent events such as rescue medication or treatment discontinuation, as set out in ICH E9(R1) (International Council for Harmonisation 2019)), the randomization scheme, blinding, the sample size computed from a power formula, the analysis populations, and the primary analysis. ICH E9 says the statistical methods must be described in the protocol and that "all interim analyses should be carefully planned in advance and described in the protocol" (International Council for Harmonisation 1998). The sample size is usually the number n that gives power $1 - \beta$ (often 80 or 90 percent) against a clinically relevant effect at one-sided $\alpha = 0.025$. Chapter 9 derives that formula.

The statistical analysis plan (SAP). Written before database lock and typically before unblinding, the SAP states the primary analysis, the handling of missing data, the multiplicity strategy (a single primary endpoint, a gatekeeping hierarchy, a Bonferroni or graphical procedure; see the FDA multiple endpoints guidance of October 2022 (U.S. Food and Drug Administration 2022b) and Chapter 10), the sensitivity analyses, and the timing and boundaries of any interim analyses, usually a group sequential design with alpha spending (Chapter 11). An independent Data Monitoring Committee (DMC) sees the unblinded interim results under a charter, and the sponsor's trial team does not.

The evidence. For the standard route to approval, the package contains two adequate and well-controlled trials, each with a statistically significant result on its pre-specified primary endpoint, or one such trial plus confirmatory evidence in the circumstances discussed above. The primary result is a point estimate, a confidence interval and a p-value. FDA's statistical reviewers re-run the analysis from the datasets the sponsor submits, in standardized formats, and check that what was done matches what the protocol and SAP said would be done.

What the regulator is really asking. The questions a statistical reviewer asks of a traditional trial are worth listing, because the same questions reappear in new form for Bayesian and adaptive designs. Was the primary analysis specified before the data were seen? Was the type I error rate controlled for the family of hypotheses that supports the claim? Is the effect estimate credible, free of selection bias, and clinically meaningful? Was the trial conducted with integrity: blinded, with limited access to comparative data, with deviations documented? The answers are almost entirely about pre-specification and error control, which the traditional package makes simple: the type I error rate is α by construction and the power is a formula.

Where the traditional design is the right choice. For a single-look trial with a well-understood endpoint and no usable prior information, the fixed-sample design is simple, transparent, and cheap to review. A sponsor with no meaningful prior information and no

appetite for the added simulation work should not adopt a more elaborate design merely because it is available. Chapter 21 shows a case (REWIND) where the scheduled design wins.

3.3. What changes with a Bayesian adaptive design

The short answer is that the legal standard does not change, the evidentiary bar does not change, and the demands for pre-specification and error control do not change. What changes is how the operating characteristics are established, who has to be convinced, and what the sponsor must hand over. It is helpful to separate three kinds of change.

The design becomes a rule set, and the rule set is the thing that is pre-specified. In a traditional trial the protocol pre-specifies a test and a sample size. In an adaptive trial it also pre-specifies decision rules that depend on accumulating data: when to stop for efficacy or futility, when to drop an arm, how to change allocation, and how to re-estimate the sample size. FDA's 2019 guidance defines an adaptive design as "a clinical trial design that allows for prospectively planned modifications to one or more aspects of the design based on accumulating data" (U.S. Food and Drug Administration 2019a), and the draft ICH E20 sharpens the point: "prospectively planned means that the potential trial adaptations are pre-specified in the clinical trial protocol prior to initiation of the trial" (International Council for Harmonisation 2025a).

Operating characteristics are established by simulation, not by formula. A group sequential design has analytic boundaries. A Bayesian adaptive design with predictive probability stopping, response-adaptive allocation and borrowing generally does not. The 2019 guidance says "a major use of simulations in adaptive trial design is to estimate trial operating characteristics" (U.S. Food and Drug Administration 2019a), and the Bayesian draft recommends "comprehensive" simulation covering the plausible range of assumptions, with the simulation code available to FDA for verification (U.S. Food and Drug Administration 2026g). Chapter 14 is devoted to this. This is the currency in which the book itemizes costs: each benefit of a Bayesian adaptive design is bought with a simulation burden, and the simulation report is how the sponsor pays.

The success criterion is stated on the posterior and calibrated to error. The Bayesian draft gives the standard form: success is declared when the posterior probability that the true treatment effect exceeds a threshold is high enough, $\Pr(\delta > a \mid \text{data}) > c$, where a is the minimum benefit and c the minimum probability (U.S. Food and Drug Administration 2026g). Section IV.A describes three routes: calibration of the thresholds to a type I error rate ("typically 0.025 one-sided" for the overall familywise error rate), direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA), and decision-theoretic or benefit-risk criteria. Calibration is this book's default for confirmatory adaptive designs without borrowing, and the draft says it may be appropriate for complex adaptive designs; it is one route, not a requirement. Against a single unadjusted analysis the calibrated Bayesian bar is not lower without an informative prior. The other routes require agreement with FDA, and their justification is heavier, not lighter (Chapter 8 and Chapter 19).

Table 3.1 lines the two submissions up.

Table 3.1.: What changes between a traditional and a Bayesian adaptive submission.

Element	Traditional fixed-sample or group sequential	Bayesian adaptive	Where developed
Legal standard	Substantial evidence (505(d), 314.126)	Identical	This chapter
Primary inference	p-value and confidence interval	Posterior probability and credible interval, usually also with a calibrated frequentist error rate	Chapter 6, Chapter 7
Type I error	α by construction; alpha spending for looks	Estimated by simulation; calibrate a and c to FWER 0.025 one-sided by default	Chapter 14
Prior information	None in the primary analysis	Pre-specified analysis prior; justified; sensitivity analyses	Chapter 8, Chapter 19
Sample size	Fixed from power formula (or event count)	Maximum plus expected size under scenarios; predictive stopping	Chapter 9, Chapter 15
Interim looks	Pre-planned group sequential, DMC	Pre-planned decision rules, DMC, firewalls, access limits	Chapter 11, Chapter 23
Allocation	Fixed randomization ratio	Fixed or response-adaptive (the latter needs strong justification)	Chapter 16
Documents beyond the protocol	SAP	SAP plus simulation report with code, prior documentation	Chapter 23
Meeting with FDA	Pre-IND, end-of-phase 2, pre-NDA	Same, plus CID paired meeting when design is novel	This chapter

The third row of the table is a regulatory choice and not a mathematical necessity: what makes the calibrated route the default is that it leaves FDA's reviewers where they were: with a number they know how to interpret.

3.4. ICH: the harmonized principles

The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) brings together regulators and industry to agree guidelines that

member regulators then adopt. A guideline reaches Step 4 when the ICH Assembly adopts the final text, and regional regulators then implement it (FDA publishes it as guidance, EMA as a scientific guideline). Step 2 is a draft out for consultation. Six ICH documents shape this book.

E8(R1), General Considerations for Clinical Studies (Step 4, 6 October 2021) (International Council for Harmonisation 2021). It replaces E8 of 17 July 1997. E8(R1) is the document where "quality by design" enters clinical research: it says quality should be "driven proactively by designing quality into the study protocol and processes" (Section 3.1). It states that "the major purpose of a control group is to separate the effect of the treatment(s) from the effects of other factors" (Section 5.3). Its critical-to-quality factors, for a Bayesian adaptive trial, include the prior, the calibration of the success criterion and the integrity of the interim data, though E8(R1) does not say so.

E9, Statistical Principles for Clinical Trials (Step 4, 5 February 1998) (International Council for Harmonisation 1998). After more than 25 years E9 remains the foundation. It describes randomization, blinding, the analysis populations, the primary variable, interim analysis, and multiplicity. On randomization it says that the deliberate element of chance "tends to produce treatment groups in which the distributions of prognostic factors, known and unknown, are similar" (Section 2.3.2), and on analysis populations that "Preservation of the initial randomisation in analysis is important in preventing bias" (Section 5.2.1). On multiplicity it says that "the usual frequentist approach to the analysis of clinical trial data may necessitate an adjustment to the type I error" (Section 5.6). On interim analyses it asks for pre-planning: "all interim analyses should be carefully planned in advance and described in the protocol" (Section 4.5), and group sequential designs are "the usual way" to look repeatedly (Section 3.4). E9 mentions Bayesian approaches only briefly, in Section 1.2 and the Glossary, and does not elaborate them. Pre-specification applies to them as to any analysis; the point for this book is that E9 neither prohibits nor privileges them.

E9(R1), Addendum on Estimands and Sensitivity Analysis (Step 4, 20 November 2019) (International Council for Harmonisation 2019). FDA labels its own publication of this document May 2021. E9(R1) defines an estimand as "a precise description of the treatment effect reflecting the clinical question posed by a given clinical trial objective", requires that "the targets of estimation are to be defined in advance of a clinical trial", and introduces intercurrent events, events after treatment initiation that affect the interpretation or the existence of the measurements. It names five strategies for handling them: treatment policy, hypothetical, composite variable, while on treatment and principal stratum. A posterior probability is only as interpretable as the estimand it is posterior about. Estimands are developed in Chapter 5.

E20, Adaptive Designs for Clinical Trials (draft, Step 2, endorsed 25 June 2025) (International Council for Harmonisation 2025a). This is the first international guideline on adaptive designs. It remains a draft: the EMA consultation ran from 30 June to 30 November 2025, the EMA overview of comments was published on 16 February 2026, and the ICH work plan of July 2026 targets Step 3 and Step 4 sign-off in February 2027. Until final it binds no region, but it shows where regulators are heading. Its content: the definition quoted above; the expectation that a "proposed adaptive design requires a clear and compelling justification"; the expectation that in confirmatory trials "the number and complexity of adaptations should generally be limited"; the statement that "controlling the chances of false positive efficacy conclusions" is "an essential element of regulatory decision-making"; the warning that naive analyses can raise the type I error probability to "more than double" its nominal value; and a section on simulation (5.2). E20 also has a Bayesian section (5.3),

which says that Bayesian adaptive designs "can achieve targeted operating characteristics" while "maintaining Type I error probability control" and defines Bayesian methods as those that "combine a prior probability distribution with current trial data to obtain a posterior probability distribution". Where E20 and the FDA Bayesian draft meet, the expectation is calibration to type I error for confirmatory trials. Chapter 14 uses E20 section 5 for the contents of a simulation report. The adaptive combination methods behind confirmatory adaptive analyses go back to Bauer (1989) and Bauer and Köhne (1994), as the review of Bauer and colleagues recounts (Peter Bauer et al. 2016); the interim-look machinery is in Chapter 11.

E11A, Pediatric Extrapolation (Step 4, August 2024) (International Council for Harmonisation 2024). E11A sets out when and how evidence from an adult or other source population can be extrapolated to children, and it promotes quantitative approaches including Bayesian borrowing and modeling and simulation to do so. It matters for this book because pediatric extrapolation is the clearest case where a regulator has asked for, rather than tolerated, a quantitative statement of how much prior information counts. It is taken up in Chapter 19, and belimumab in children is a worked case in Section 3.11. The Bayesian draft itself uses "adult data for children" as an example of borrowing (U.S. Food and Drug Administration 2026g).

M15, General Principles for Model-Informed Drug Development (Step 4, January 2026) (International Council for Harmonisation 2026a). FDA published this document in June 2026. M15 defines model-informed drug development (MIDD) as "the use of computational modeling and simulation (M&S) methods that can include and integrate nonclinical data, clinical data, prior information" to inform decisions. Its structure is a credibility framework built on the question of interest, the context of use, model influence, the consequence of a wrong decision and model risk, where model risk is "derived by combining model influence and consequence of wrong decision" and "is essential for determining the requirements for model evaluation". The more a decision rests on a model, and the worse a wrong decision would be, the more evidence the model needs. A Bayesian adaptive design is in part a model on which stopping decisions rest, and the simulation report is the evaluation, so the M15 vocabulary is a useful way to think about how much documentation FDA will want.

Three older ICH documents round out the picture and are referenced elsewhere: E3 (Step 4, November 1995; the structure of the clinical study report) (International Council for Harmonisation 1995), M4 (M4(R4), Step 4, 2016; the common technical document format of the submission) (International Council for Harmonisation 2016), and E6(R3) on good clinical practice (Step 4, January 2025) (International Council for Harmonisation 2025b). A sponsor writes the SAP and the simulation report to fit into the first two structures. E6(R3) governs conduct and applies to adaptive trials, including the integrity of interim data.

3.5. FDA design guidances

FDA guidance documents "contain nonbinding recommendations". The agency's own convention is that the word "should" in a guidance means "suggested or recommended, but not required". A sponsor may depart from a guidance, but must be prepared to show that the alternative approach meets the statute and regulations. In practice a departure without prior discussion with the agency is a risk to the program, and that is why the meetings of the next section exist. A guidance is first issued as a draft, comments are taken (60 days after

the Federal Register notice is usual), and a final follows, or the draft is revised or withdrawn. A draft is labeled as not for implementation, but sponsors often follow it because it tells them what reviewers will look for.

3.5.1. Adaptive Designs for Clinical Trials of Drugs and Biologics (final, November 2019)

This is the founding FDA document for the book (U.S. Food and Drug Administration 2019a). It is final. Its four principles for adaptive designs are the ones to remember: "the chance of erroneous conclusions should be adequately controlled, estimation of treatment effects should be sufficiently reliable, details of the design should be completely prespecified, and trial integrity should be appropriately maintained" (Section III). It says plainly that "adaptive features can inflate the Type I error probability of a trial", with the simplest example a repeated look at accumulating data (Section III.A), and it works through the example of a single interim look at 50 percent of the information. It names simulation for operating characteristics (Section VI.A), limited access to comparative interim data (Sections III and VII), and an example of effectiveness rules based on a Bayesian posterior probability (Section V). It recommends that the four principles be addressed in the protocol and submission, and that complex designs be discussed with FDA beforehand. For frequentist and Bayesian adaptive designs alike, the 2019 document is the baseline, and the 2026 Bayesian draft builds on it.

3.5.2. Bayesian methods at the regulators before 2026

The January 2026 draft is not the first regulatory text on Bayesian trials. ICH E9 (1998) already allowed Bayesian approaches when the reasons for their use are clear and the conclusions are sufficiently robust (Section 3.4), and the device side went first. CDRH issued a draft Bayesian guidance on 23 May 2006 and the final version on 5 February 2010 (U.S. Food and Drug Administration 2010). The final text treats type I error as one operating characteristic among others and says that when prior information is used it may be appropriate to control it at a less stringent level, case by case (Section 4.8). The CDRH statisticians have written up the experience: Pennello and Thompson on reviewing Bayesian device trials (Pennello and Thompson 2008a), Campbell on the Bayesian innovation the FDA sparked (Campbell 2011) and on the similarities between Bayesian and adaptive designs (Campbell 2013), and Campbell, Irony, Pennello and Thompson on progress since 2010 (Campbell et al. 2023).

On the drug side, the complex innovative trial design program gave Bayesian designs a forum for early dialogue (Section 3.7), and Ruberg and colleagues argued in 2023 that wider use of Bayesian approaches in drug development could start a virtuous cycle of experience and acceptance (Ruberg et al. 2023). The practical lesson is the same on both sides of the Agency: agree the prior and the operating characteristics with the reviewers before the trial starts.

3.5.3. Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft, January 2026)

This is the document that most changes the conversation for a sponsor considering a Bayesian design (U.S. Food and Drug Administration 2026g). It is a draft guidance for industry, marked "Contains Nonbinding Recommendations", issued by CDER and CBER under docket FDA-2025-D-3217 in January 2026. It does not apply to devices, which have their own Bayesian guidance (Section 3.5.2); two device approvals under it are in Section 3.11. Its scope is INDs, NDAs, BLAs and supplements, with a "primary focus" on "the use of Bayesian methods to support primary inference". Its main elements, in the order a sponsor will meet them:

- **Success criteria.** The criterion is "most often based on the posterior probability that the true treatment effect size exceeds some threshold", $\Pr(\delta > a \mid \text{data}) > c$. For trials with an overall frequentist analysis plan, the thresholds a and c are almost always calibrated so that the familywise type I error rate across all primary and secondary estimands is no greater than 0.025 one-sided; the draft also describes posterior-probability and benefit-risk routes.
- **Priors.** "It is always critical for Bayesian analyses to have a prespecified prior." The sponsor should "pre-specify and justify the full details of the proposed prior distribution in the protocol". Skeptical priors "express skepticism about the presence of very large treatment effects", and design priors that explore pessimistic assumptions should be considered. Where the data do not dominate the prior, "the accuracy of conclusions depends strongly on the choice of prior distribution". Chapter 8 takes up these points.
- **Borrowing.** Static discounting fixes how much external information counts; dynamic methods "borrow less when the data are less similar and more when the data are more similar". Borrowing needs "strong justification that considers feasibility … and the relevance of the available information", information should typically come from patient-level data, and a prior that is too informative "risks overwhelming the data collected in the target population". The draft says that type I error inflation is "not recommended" as the measure of borrowing, and prefers effective sample size and similar influence metrics. The draft's named examples are summarized in Section 3.11. Chapter 19 develops borrowing.
- **Operating characteristics and simulation.** "Simulations should comprehensively cover the plausible range of assumptions", and the plan should compare complex designs with simpler alternatives and request an FDA meeting before the trial starts.
- **Documentation.** The sponsor should submit protocol, SAP and simulation report "as early as possible" and "during the design stage". The report on the trial should state the estimates and credible intervals, the success criteria, the sensitivity analyses, the computational details (software versions, code, the MCMC seed numbers). On software the draft says that "FDA does not require use of any specific software for statistical analyses".

It requires nothing by itself, since it is a draft; it permits both the calibrated and the non-calibrated pathway, the latter with agreement. Chapter 23 turns the draft into a protocol and SAP checklist.

3.5.4. The response to the draft

Three commentaries on the draft appeared together in JAMA on 19 May 2026 (J. J. Lee et al. 2026; S. R. Evans et al. 2026; Gelman, Zwet, and Więcek 2026). The routes the draft leaves open are described in Chapter 3; this book takes calibration as the default for confirmatory adaptive designs without borrowing, and direct posterior criteria with FDA agreement when prior information is borrowed, as the draft allows. The older cautions about prior dependence, type I error and operational bias are collected in Section 1.9.

3.5.5. Master Protocols for Drug and Biological Product Development (revised draft, June 2026)

A first draft appeared in December 2023; a revised draft that "revises and replaces" it was announced in the Federal Register on 24 June 2026 (91 FR 37991), with comments closing on 24 August 2026 (U.S. Food and Drug Administration 2026f). FDA's description of the changes is "additional recommendations on basket trials and minor changes for clarity on topics such as randomization, choice of control, and informed consent". The document is issued under the Food and Drug Omnibus Reform Act of 2022 (U.S. Congress 2022). It defines a master protocol as "a protocol designed with multiple substudies, which may have different objectives and involve coordinated efforts to evaluate one or more drugs"; an umbrella trial "evaluate[s] multiple drugs concurrently for a single disease or condition", a basket trial "evaluate[s] a drug for multiple diseases, conditions, or disease subtypes", and a platform trial evaluates multiple drugs "in an ongoing manner, with drugs entering or leaving the platform".

Its recommendations are specific. A shared control arm improves efficiency, and the draft discusses allocating more participants to the control arm than to each drug arm (the EMA platform-trials draft gives Dunnett's square-root-of-k rule as one example). The primary comparison should use "only concurrently eligible control participants"; nonconcurrent controls are allowed only in rare circumstances agreed with FDA in advance, with concurrent-only sensitivity analyses. On multiplicity, FDA "generally does not recommend the use of multiplicity adjustments to strongly control the probability of making at least one type I error across the multiple comparisons of different drugs", because the expected number of type I errors is the same whether the comparisons are made in independent trials or in one master protocol. Adjustment is still expected across doses or formulations of the same drug. For a new master protocol the draft points sponsors to early discussion with FDA, and a master protocol pre-IND meeting is requested as Type B (see the meetings section below). It complements the March 2022 oncology master protocols guidance (U.S. Food and Drug Administration 2022a). Chapter 20 develops all of this.

3.5.6. Multiple Endpoints in Clinical Trials (final, October 2022)

This final guidance (U.S. Food and Drug Administration 2022b) describes the multiplicity problem and the strategies for handling it when a trial has several endpoints, doses or comparisons, and the statistical procedures for strong control of the familywise error rate: Bonferroni, Holm, fixed-sequence and gatekeeping procedures, and graphical methods. It is the regulatory partner of ICH E9 section 5.6 and the basis of Chapter 10. For the Bayesian reader the guidance matters in two ways. It sets the expectation that a claim supported by several endpoints must come with a stated multiplicity strategy, which the

Bayesian draft echoes by recommending that the FWER be controlled "across all primary and secondary estimands". And it leaves room for hierarchical modeling as a complement, not as a replacement, for the confirmatory strategy.

3.5.7. AI and new approach methodologies

Two further FDA drafts share this book's logic. *Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products* (draft, January 2025) (U.S. Food and Drug Administration 2025c) sets a seven-step, risk-based credibility framework (question of interest, context of use, model risk, credibility plan, execution, documentation of results and deviations, adequacy determination), in which model risk combines "model influence" and "decision consequence", the structure of M15. It covers AI that produces information supporting decisions on safety, effectiveness or quality, not AI used only for operational efficiency, so a machine-learned model that selects patients or imputes outcomes in a trial falls under it. *General Considerations for the Use of New Approach Methodologies in Drug Development* (draft, March 2026) (U.S. Food and Drug Administration 2026e) concerns nonclinical methods that replace or reduce animal testing. It says a NAM "does not necessarily need to be validated" and is judged in the weight of evidence.

3.5.8. Other FDA documents that recur

Several other FDA documents recur and are listed in Table 3.4: the final guidance *Optimizing the Dosage of Human Prescription Drugs and Biological Products for the Treatment of Oncologic Diseases* (August 2024, Project Optimus) (U.S. Food and Drug Administration 2024d) in Chapter 17, the February 2023 draft guidance on externally controlled trials (U.S. Food and Drug Administration 2023a) in Chapter 19, the expedited programs guidance (final, May 2014) (U.S. Food and Drug Administration 2014), and the guidance on data monitoring committees (final, March 2006; revised draft, February 2024) (U.S. Food and Drug Administration 2006), the template for the DMC firewalls of Chapter 23. Expedited programs change the speed and intensity of FDA's interaction, not the substantial-evidence standard: a faster review is not a lower bar.

3.6. Requires, recommends, permits

Because the force of a document varies, it helps to sort the documents above, and a book on design should say precisely what it means by "FDA requires". Table 3.2 does the sorting.

Table 3.2.: What each kind of regulatory source requires, recommends or permits.

Force	Source	Examples in this chapter	What it means for a sponsor
Requires (statute)	Act of Congress	FD&C Act 505(d) substantial evidence; FDAMA 1997 single-trial provision; FDORA 2022	Binding. The standard to be met
Requires (regulation)	Code of Federal Regulations	21 CFR 314.126 characteristics of adequate and well-controlled studies	Binding. Defines what the evidence must look like
Recommends (final guidance)	FDA guidance, final	Adaptive Designs 2019; Multiple Endpoints 2022; ICH E8(R1), E9, E9(R1) as adopted by FDA; Project Optimus 2024	Nonbinding, but FDA's current thinking. Depart only with a reason and, ideally, with agreement
Recommends (draft guidance)	FDA guidance, draft	Bayesian Methodology (January 2026); Master Protocols (June 2026); Substantial Evidence (June 2026); AI (January 2025); NAMs (March 2026)	Nonbinding and subject to change. Shows where review is heading
Recommends (ICH Step 2 draft)	ICH draft, out for comment	E20 (June 2025)	Not yet adopted in any region
Recommends (ICH Step 4)	ICH final, adopted regionally	E11A (2024); M15 (January 2026)	Adopted by FDA as guidance; see the issuer's page for the regional status
Permits	Statute or guidance giving discretion	One trial plus confirmatory evidence; non-calibrated Bayesian success criteria with FDA agreement; nonconcurrent controls in rare circumstances	Available only if the agency concurs, so the route goes through a meeting

Process	Procedural guidance, PDUFA commitments	Formal Meetings (August 2026); CID and MIDD paired meeting programs	Defines how to ask. Not a source of evidentiary standards

Two corollaries follow. A sponsor does not need a guidance to be final before using it; but a sponsor should never describe a draft as "FDA requirements" in a protocol or a meeting package. And a *permission* is not an entitlement. The permissions that matter most to Bayesian adaptive designs are all of the form "if FDA agrees", and the practical meaning of "regulation as the spine" is that those agreements are sought in writing, early, and on the record.

3.7. Interacting with FDA

This section explains how a sponsor finds out whether FDA agrees. FDA's meeting types and programs differ in what they can promise, and the differences matter.

3.7.1. Formal meetings under PDUFA

The procedural guidance *Formal Meetings Between the FDA and Sponsors or Applicants of PDUFA Products* (August 2026) is the current text; it states that it withdraws the December 2017 formal meetings guidance (U.S. Food and Drug Administration 2026c). PDUFA, the Prescription Drug User Fee Act, is the law, periodically reauthorized, under which sponsors pay fees and FDA commits to review goals and meeting timelines; the current commitments are in the PDUFA VII goals letter for fiscal years 2023 through 2027 (U.S. Food and Drug Administration 2021c). The guidance defines six meeting types (Section III): "Type A, Type B, Type B (end of phase (EOP)), Type C, Type D, and Initial Targeted Engagement for Regulatory Advice on CDER and CBER ProducTs (INTERACT)". In plain terms:

- **Type A** is "necessary for an otherwise stalled drug or biological product development program to proceed or to address an important safety issue". It is for a program that is stuck, for instance after a disagreement with the review division.
- **Type B** covers the milestone meetings: pre-IND, and pre-NDA or pre-BLA meetings. Type B (end of phase) covers certain end-of-phase 1 meetings (for products to be considered under 21 CFR part 312 subpart E or part 314 subpart H, or similar products) and end-of-phase 2 meetings, where the sponsor shows the phase 3 plan (Section III.C). A master protocol pre-IND meeting is requested as Type B.
- **Type C** is the residual category: any meeting other than A, B, B (EOP), D or INTERACT (Section III.D). It is the usual route for design questions between milestones, such as a novel endpoint or a Bayesian analysis plan.
- **Type D** is small and focused: "often one, but typically not more than two issues". It is meant for a narrow question that does not need a multidisciplinary meeting.
- **INTERACT** is for very early engagement: it is "intended for novel drug and biological products and development programs that present unique challenges in early development". It comes before the first IND.

The timelines in the guidance (Tables 1 and 2) are how a trialist plans a calendar. FDA responds to a meeting request within 14 calendar days of receipt for Type A, Type B (EOP) and Type D, and within 21 for Type B and Type C. The meeting or written response is scheduled within 30 days of receipt for Type A, 60 for Type B, 70 for Type B (EOP), 75 for Type C and 50 for Type D; the INTERACT clocks should be read from the guidance tables. Formats include face to face, video, teleconference and "written response only", in which written answers replace a meeting (Section IV). Module 13 of the companion lab turns these tables into a clock, and the sensible planning conclusion is that a Type C meeting on a novel design takes roughly three months from request to answer, so it must be requested well before the protocol is final.

Meeting minutes matter. FDA's written responses and the sponsor's record of agreements are the main evidence at the time of review that a design was discussed. They are not a binding contract: a review division can change its position if the science changes or if the data raise new concerns. But an agreement documented in minutes is persuasive, and a sponsor that departs from it should expect a question.

3.7.2. The Complex Innovative Trial Design (CID) paired meeting program

The 21st Century Cures Act of 2016 (United States Congress 2016) directed FDA to promote novel trial designs, including complex adaptive and Bayesian designs, through public meetings and guidance. FDA's response, under its PDUFA commitments, was the CID paired meeting program (U.S. Food and Drug Administration, n.d.d), and the guidance *Interacting with the FDA on Complex Innovative Trial Designs for Drugs and Biological Products* (final, December 2020) (U.S. Food and Drug Administration 2020a) describes how it works. The program is the main route for a sponsor proposing a Bayesian or otherwise complex adaptive design. In outline: a sponsor applies with a description of the proposed design, and accepted sponsors receive a pair of meetings with extra FDA attention, the second after the sponsor has developed the simulation work discussed in the first. Admission is selective, and the program page is the place to check current eligibility and application windows. Two features matter for the reader. The meetings do not replace the ordinary milestones, and the program asks the sponsor to bring the thing that matters most to a Bayesian design, the simulations: a CID package normally includes the proposed design, the scenarios, the operating characteristics, and the comparison with a simpler alternative.

3.7.3. The MIDD paired meeting program

The model-informed drug development paired meeting program (U.S. Food and Drug Administration, n.d.g) offers paired meetings with FDA on the use of modeling and simulation approaches in a specific development program, for instance dose selection, trial design simulation, or use of an exposure-response model to support a labeling claim. It is the procedural counterpart of ICH M15 (International Council for Harmonisation 2026a): M15 sets the principles for assessing model credibility, and the program is where a sponsor tests a particular modeling plan against FDA's reading. A Bayesian adaptive design with a model-based dose-response component (see Chapter 17) can fall naturally under both the CID and the MIDD programs, and a sponsor should ask which fits the question: the CID program for the trial's decision rules, the MIDD program for the models that feed them.

3.7.4. Critical Path Innovation Meetings (CPIM)

CPIM is a different kind of meeting (U.S. Food and Drug Administration, n.d.e). It is a nonbinding, informal discussion of methods or technologies that are not tied to a single product or an active application. FDA's stated aim is to respond to a request for a CPIM within 14 days. For a group developing a general methodology, such as a class of Bayesian borrowing models, a CPIM is a low-cost way to get an informal view early.

3.7.5. Drug Development Tool (DDT) qualification

DDT qualification (U.S. Food and Drug Administration, n.d.f) is a formal process by which FDA qualifies a tool, such as a biomarker, a clinical outcome assessment or an animal model, for a stated context of use. Once qualified, the tool can be used by any sponsor within that context of use without FDA re-reviewing its suitability in each application. The context of use is the key idea: a tool is qualified for a purpose, not in the abstract. The 21st Century Cures Act (United States Congress 2016) put the process on a statutory footing. For trial design the relevance is endpoints: a qualified tool is the kind of "well-defined and reliable method of assessing response" that 21 CFR 314.126 asks for, and using it in its context removes one reviewer question. DDT qualification is slow and is typically pursued by consortia.

3.7.6. How to choose the meeting

Table 3.3 sorts the options by the question being asked. The "stuck, early, or small" rule of thumb in the takeaways of the companion lab (module 13) is a useful first filter.

Table 3.3.: Which meeting fits which question.

Question	Meeting or program	Binding?	Chapter
Is this first-in-human plan acceptable for a novel product?	INTERACT, then pre-IND (Type B)	No (advice)	This chapter
Is the phase 3 plan acceptable?	Type B (end of phase 2)	No, but minutes matter	Chapter 23
One narrow question, one or two issues	Type D	No	This chapter
Our program is stalled, or there is a safety issue	Type A	No	This chapter
Our design is Bayesian, complex adaptive or simulation-driven	CID paired meeting program; Type C otherwise	No	Chapter 14, Chapter 23
Our dose or design depends on a model	MIDD paired meeting program	No	Chapter 17
A general method, not tied to a product	CPIM	No	This chapter

Question	Meeting or program	Binding?	Chapter
A biomarker or outcome assessment for many sponsors	DDT qualification	Qualified for stated context of use	This chapter
Is one trial plus confirmatory evidence acceptable?	Type B (EOP) or Type C, before the trial starts	No	Chapter 23
Our master protocol	Type B pre-IND, labeled as a master protocol	No	Chapter 20

Try it in the lab

Open Module 13, *The regulator's map* (https://vsokolov.org/bayesian_trials_lab/13-regulators-map.html). Before you click, write down which meeting you would request for each of the five scenarios in the lab (a new pill at phase 2, a gene therapy before the first IND, a stalled program, a new kind of trial design, and a dose picked by a computer model). Then pick each scenario and compare. Two scenarios (the new design and the model-picked dose) point to the paired-meeting programs, not to a standard meeting type. Then read the meeting clocks, which are as printed in Tables 1 and 2 of the August 2026 guidance.

3.8. EMA and MHRA counterparts

The European Medicines Agency (EMA) and the UK Medicines and Healthcare products Regulatory Agency (MHRA) apply the same statistical principles, since both adopt ICH guidelines, but their documents on adaptive and complex designs differ in emphasis and date. A sponsor seeking approval in several regions needs to read them together.

EMA adaptive-design reflection paper (2007) (European Medicines Agency, Committee for Medicinal Products for Human Use 2007). The reference paper on methodological issues in confirmatory trials planned with an adaptive design (CHMP/EWP/2459/02) was adopted in October 2007 and is still current until E20 reaches its final step in the EU. It is more conservative than the 2019 FDA guidance: adaptations need justification and seamless phase 2/3 designs are treated with caution.

EC/EMA/HMA Questions and Answers on complex clinical trials (2022) (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). This final document (EMA/298712/2022, version of 23 May 2022, published in June 2022) is the first European text to deal with complex clinical trials as a class: master protocols, platform, basket and umbrella designs, and trials using Bayesian methods. It is practical rather than statistical, covering the clinical trial application under the EU Clinical Trials Regulation, sub-protocols, transparency, and use in marketing authorization applications. On Bayesian methods it asks sponsors to justify and document the priors, to use prior-predictive simulation, and to engage in early dialogue.

EMA concept paper on Bayesian methods (2026) (European Medicines Agency, Methodology Working Party 2026). The Methodology Working Party agreed a concept paper

for a future reflection paper on Bayesian methods in clinical development in November 2025 (EMA/CHMP/1813/2026); the CHMP released it for consultation on 19 January 2026 and the consultation ran from 30 January to 30 April 2026. A concept paper is not guidance: it proposes that a reflection paper be written and lists the questions it should answer, among them what defines technical success for a Bayesian trial, how to assess error control for primary and secondary endpoints without frequentist inference, how much weight the prior carries relative to the data, and which sensitivity analyses are required. The tentative timetable puts a draft reflection paper out for consultation in September 2027 and the final text in June 2028; an EMA workshop on the topic was held in June 2025. Until then a sponsor should read the 2022 Q&A and ICH E9 together.

EMA reflection paper on methodological aspects in confirmatory platform trials (draft, October 2026) (European Medicines Agency 2026a). This is a draft (EMA/217509/2026) that follows the 2022 concept paper. The CHMP adopted it for consultation on 5 October 2026, and the consultation runs from 19 October 2026 to 31 January 2027. It concerns confirmatory platform trials only and deliberately avoids the broader term "master protocol". It says that objectives, estimands and the overall testing strategy must be documented at the start "as far as possible", warning that "lack of pre-specification could render a confirmatory platform trial as exploratory only". It prefers concurrent controls, saying that "non-concurrent controls are usually not acceptable for the primary analysis of the primary endpoint", and it holds the arm-to-control ratio for each comparison fixed, with the consequence stated plainly: "response adaptive randomisation (RAR) is not supported". It does not discuss Bayesian methods. FDA's June 2026 draft permits non-concurrent controls in rare cases by agreement, while the EMA draft is stricter on both controls and allocation, so a global platform should treat the stricter position as the design constraint until the paper is final. Chapter 20 and Chapter 16 discuss the consequences.

MHRA draft guideline on external control arms (2025) (Medicines and Healthcare products Regulatory Agency 2025). The MHRA consulted in 2025 on a draft guideline on the use of external control arms based on real-world data to support regulatory decisions. It is relevant here because external controls are a form of borrowing. It parallels FDA's February 2023 draft guidance on externally controlled trials (U.S. Food and Drug Administration 2023a) and is taken up in Chapter 19.

The three regulators share the principles: pre-specification, control of false positive conclusions, simulation, integrity. They differ in how much flexibility they will accept in the primary analysis of a confirmatory trial, and the EMA drafts are the more cautious. A global program should plan European scientific advice in parallel with the US meetings.

3.9. The document table

Table 3.4 collects the documents discussed in this chapter. "Status" is as of November 2026 and the dates are those printed on the documents or in the issuer's notice. The last column gives the chapters that rely on the document; Appendix Appendix B lists the links.

Table 3.4.: Regulatory documents in this book: issuer, date, status and use.

Document	Issuer	Date and status	Used in
FD&C Act 505(d), substantial evidence	US Congress	Law since 1962; amended 1997 (FDAMA)	Chapter 1, Chapter 23
21 CFR 314.126, adequate and well-controlled studies	FDA	Regulation	Chapter 5, Chapter 23
21st Century Cures Act	US Congress	Law, December 2016	This chapter
FDORA	US Congress	Law, December 2022	Chapter 20
ICH E8(R1), General Considerations for Clinical Studies	ICH	Step 4, 6 October 2021	Chapter 5, Chapter 23
ICH E9, Statistical Principles	ICH	Step 4, February 1998	Chapter 4, Chapter 5, Chapter 11
ICH E9(R1), Estimands addendum	ICH	Step 4, 20 November 2019	Chapter 5, Chapter 23
ICH E20, Adaptive Designs	ICH	Draft, Step 2, endorsed 25 June 2025; sign-off targeted February 2027	Chapter 14, Chapter 23
ICH E11A, Pediatric Extrapolation	ICH	Step 4, August 2024	Chapter 19
ICH E6(R3), Good Clinical Practice	ICH	Step 4, January 2025	Chapter 23
ICH M15, Model-Informed Drug Development	ICH (FDA publication)	Step 4, January 2026; FDA June 2026	Chapter 17, Chapter 14
Adaptive Designs for Clinical Trials of Drugs and Biologics	FDA	Final, November 2019	Chapter 11, Chapter 14, Chapter 16
Use of Bayesian Methodology in Clinical Trials	FDA	Draft, January 2026	Chapter 7, Chapter 8, Chapter 14, Chapter 19
Master Protocols for Drug and Biological Product Development	FDA	Revised draft, June 2026 (first draft December 2023)	Chapter 20
Master Protocols: Oncology Drugs and Biologics	FDA	Final, March 2022	Chapter 20

Document	Issuer	Date and status	Used in
Demonstrating Substantial Evidence of Effectiveness	FDA	Revised draft, June 2026 (first draft December 2019); on finalization replaces the May 1998 guidance	Chapter 23, Chapter 19
ICH E3, Structure and Content of Clinical Study Reports	ICH	Step 4, November 1995	Chapter 23
ICH M4(R4), Common Technical Document	ICH	Step 4, 2016	Chapter 23
Bayesian Statistics in Medical Device Clinical Trials	FDA (CDRH, CBER)	Final, February 2010	Chapter 19, Section 3.11
PDUFA VII commitment letter	FDA and industry	Goals for FY2023 to FY2027	This chapter
Multiple Endpoints in Clinical Trials	FDA	Final, October 2022	Chapter 10
Optimizing the Dosage (Project Optimus)	FDA	Final, August 2024	Chapter 17
Considerations for Externally Controlled Trials	FDA	Draft, February 2023	Chapter 19
Expedited Programs for Serious Conditions	FDA	Final, May 2014	This chapter
Establishment and Operation of Clinical Trial Data Monitoring Committees	FDA	Final, March 2006; revised draft, February 2024	Chapter 23
Considerations for the Use of AI to Support Regulatory Decision-Making	FDA	Draft, January 2025	Chapter 23
General Considerations for the Use of NAMs	FDA	Draft, March 2026	This chapter
Formal Meetings Between FDA and Sponsors (PDUFA)	FDA	Final procedural guidance, August 2026	Chapter 23
Interacting with FDA on Complex Innovative Trial Designs	FDA	Final, December 2020	Chapter 14, Chapter 23

Document	Issuer	Date and status	Used in
CID, MIDD paired meeting programs; CPIM; DDT qualification	FDA	Active programs; see program pages	Chapter 23, Chapter 17
Reflection paper on adaptive designs (CHMP/EWP/2459/02)	EMA	Adopted October 2007	Chapter 11
Complex clinical trials Q&A (EMA/298712/2022)	EC, EMA, HMA	Final, 2022	Chapter 20, Chapter 23
Concept paper, reflection paper on Bayesian methods (EMA/CHMP/1813/2026)	EMA	Concept paper, consultation January to April 2026	Section 3.8
Reflection paper on confirmatory platform trials (EMA/217509/2026)	EMA	Draft, October 2026; consultation to 31 January 2027	Chapter 20, Chapter 16
External control arms based on real-world data	MHRA	Draft, 2025	Chapter 19

3.10. Reading a regulatory document

Two habits prevent most misuse. Check the date and status on the document itself, and again before submission: FDA media URLs can serve a later revision, and the address of the 2019 substantial-evidence draft now serves the June 2026 revision. And read the context-of-use language of M15 and the AI draft as a rule of proportion: a design whose stopping rule decides approval is highly influential, so its simulation report should be proportionately thorough. None of this replaces the conversation: the review division that examines an application has the last word on what is adequate for that product, and the way to learn its view is to ask, with a concrete design and simulation report in hand.

3.11. In practice

The standard of this chapter has already been met with Bayesian arithmetic, in a vaccine, a biologic approved on one trial plus borrowing, a pediatric extension and two device approvals, and the January 2026 draft names further examples.

3.11.1. REBYOTA: one trial standing in for two

Ferring's RBX2660 (REBYOTA) for recurrent *C. difficile* infection was approved on 30 November 2022 (US Food and Drug Administration, Center for Biologics Evaluation and Research 2022) on one phase 3 trial (PUNCH CD3) plus a Bayesian hierarchical model that

borrowed from a phase 2b study, an approach FDA called "acceptable" (U.S. Food and Drug Administration, Center for Biologics Evaluation and Research 2022). Success was a posterior probability of superiority with two thresholds chosen as analogues of one-sided type I error without borrowing: 0.9993275 (analogue 0.00125, a stricter bar for one trial standing in for two) and 0.9750338 (analogue 0.025, one adequate and well-controlled trial). The observed posterior probability of superiority was 0.991, which clears the second threshold and misses the first, so the evidence counts as one adequate and well-controlled trial, not two. Approval rested on FDA's assessment of the totality of evidence, and the thresholds were case-specific evidentiary criteria agreed with FDA, not numerical standards set by the 1997 amendment (Section 3.1). The 2026 draft cites it as its example of borrowing from an earlier trial of the same drug (U.S. Food and Drug Administration 2026g). The planning history, the borrowing model and the data-alignment lesson are in Section 19.17.1.

3.11.2. BNT162b2: a Bayesian criterion in an EUA and a BLA

The pivotal efficacy trial of the Pfizer and BioNTech vaccine (C4591001) had a Bayesian primary criterion written in its statistical analysis plan (Pfizer Inc. 2022): success was a posterior $\Pr(\text{VE} \geq 30\% \mid \text{data}) > 0.995$ at an interim analysis or > 0.986 at the final analysis, thresholds calibrated so that the overall one-sided type I error is 0.025. The trial observed 8 cases on vaccine against 162 on placebo, vaccine efficacy (VE) 95.0%, 95% credible interval 90.3 to 97.6 (Polack et al. 2020). FDA issued the emergency use authorization on 11 December 2020 for ages 16 and older (U.S. Food and Drug Administration 2021a) and approved the BLA (Comirnaty) on 23 August 2021. The beta-binomial model and the prior are worked in Section 6.12. The lesson is that a pivotal trial can use a posterior criterion calibrated to 0.025 (calibration is one of the three routes in the January 2026 draft, and this book's default), and that both an EUA and a BLA accepted it.

3.11.3. Belimumab in children: a tipping point on the borrowing weight

PLUTO (C1109), a small trial of belimumab in pediatric lupus, was not powered to stand alone (Brunner et al. 2020). At FDA's request the sponsor borrowed from the adult trials through a robust mixture prior and reported a tipping point: the posterior probability of efficacy was at least 97.5% whenever the borrowing weight was 0.55 or more, a weight the review division judged reasonable, and FDA approved the pediatric indication (U.S. Food and Drug Administration, Center for Drug Evaluation and Research 2026; Pottackal et al. 2025). The analysis was post hoc, and FDA's case study recommends prespecifying such borrowing with the agency. The full case is in Section 19.17.2.

3.11.4. Devices: Superion and OPTIMIZER Smart

CDRH has accepted Bayesian primary analyses for years under its 2010 guidance (U.S. Food and Drug Administration 2010). The Superion interspinous spacer (PMA P140004, approved 20 May 2015) was a randomized 1:1 non-inferiority trial against X-STOP with a Bayesian adaptive sample size, a 10% margin, independent Beta(1, 1) priors, and success declared if the posterior probability of non-inferiority was at least 0.958. The final analysis gave 0.9927 (U.S. Food and Drug Administration, CDRH 2015). The OPTIMIZER Smart system (P180036, approved 21 March 2019) analyzed FIX-HF-5C ($n = 160$) with a Bayesian model and 30% fixed borrowing from the earlier FIX-HF-5 study, with success if $\Pr(\Delta > 0) > 0.975$

for the change in peak VO2. With borrowing the posterior probability was 0.977 to 0.989 across imputation models; the new trial alone gave 0.911 to 0.960 (the lower value when last-observation-carried-forward imputation is included), below the bar (U.S. Food and Drug Administration, Center for Devices and Radiological Health 2019). The lesson is that the borrowing weight can decide the outcome, which is why the draft asks for justification and sensitivity analysis.

3.11.5. The other precedents in the draft

The draft names the GBM AGILE (NCT03970447) and Precision Promise (NCT04229004) platforms, which use Bayesian models for temporal drift when augmenting concurrent controls (Benjamin R. Saville et al. 2022); a pediatric multiple sclerosis non-inferiority study through the CID program, with a prespecified Bayesian analysis borrowing historical active-comparator data and modeling relapse rate by age (U.S. Food and Drug Administration, CDER, n.d.b); supportive Bayesian analyses for empagliflozin and linagliptin in pediatric type 2 diabetes, accepted because the pediatric pathophysiology resembles the adult; an epilepsy with myoclonic-atonic seizures (EMAS) placebo-controlled study proposed through the CID program, with a hierarchical model borrowing from other epilepsy types (U.S. Food and Drug Administration, CDER, n.d.a); and hierarchical subgroup borrowing in the LEADER trial of liraglutide and the Rinvoq Drug Trial Snapshot (U.S. Food and Drug Administration 2026g). For the last three we rely on the draft's description.

Key points

- The legal standard is substantial evidence of effectiveness: adequate and well-controlled investigations (FD&C Act 505(d)), characterized by 21 CFR 314.126. The statute is a standard of evidence, not of method; it names no test, level or prior.
- Since FDAMA 1997, one adequate and well-controlled trial plus confirmatory evidence can suffice at FDA's discretion. The June 2026 revised draft says this calls for a highly persuasive trial or strong confirmatory evidence, and that the plan should be discussed with FDA before the trial.
- A traditional submission rests on pre-specification and error control: a protocol with a fixed sample size, a SAP, a one-sided 0.025 level (two-sided 0.05), a stated multiplicity strategy, and a DMC for interim looks. Bayesian adaptive designs keep all of that and add a simulation report and a justified prior. For confirmatory adaptive designs without borrowing, this book uses calibration of the success criterion to one-sided 0.025; FDA's draft also describes direct posterior interpretation and decision-theoretic or benefit-risk criteria (Section IV.A).
- The Bayesian draft (January 2026) states success as $\Pr(\delta > a \mid \text{data}) > c$ and describes three routes, by agreement with FDA: calibration to a familywise error rate (typically 0.025 one-sided), direct posterior interpretation, and decision-theoretic or benefit-risk criteria. Calibration is this book's default, not the draft's. Against a single unadjusted analysis the calibrated bar is not lower without an informative prior.
- Documents differ in force: statutes and regulations bind; guidances recommend; drafts are subject to change; permissions depend on FDA's agreement. Never describe a draft as a requirement.

- The statuses that matter as of November 2026: Adaptive Designs final (2019); Bayesian, Master Protocols and Substantial Evidence drafts (2026); ICH E20 at Step 2; ICH M15 at Step 4; EMA platform draft in consultation to 31 January 2027.
- Real approvals already use the arithmetic: REBYOTA's two posterior thresholds (0.9993275 for one trial standing in for two, 0.9750338 for one adequate and well-controlled trial; observed 0.991), BNT162b2's calibrated posterior criterion, belimumab's tipping-point analysis, and CDRH's Superion and OPTIMIZER Smart approvals (Section 3.11).
- FDA offers Type A, B, B (EOP), C, D and INTERACT meetings, the CID and MIDD paired meeting programs, CPIM, and DDT qualification. A novel design should go to FDA early, with simulations in hand.
- EMA's drafts are more cautious than FDA's on allocation and nonconcurrent controls; a global platform should design to the stricter position.

Discussion questions

1. The statute asks whether experts "could fairly and responsibly" conclude that the drug works, and does not mention a significance level. Why has 0.025 one-sided become so entrenched, and what would have to be true for a regulator to accept a different working standard in a given program?
2. The June 2026 substantial-evidence draft allows one trial with confirmatory evidence. In what ways is an informative Bayesian prior similar to confirmatory evidence, and in what ways is it different? Who decides the weight, and when?
3. Suppose your sponsor is considering a Bayesian adaptive phase 3 design with response-adaptive randomization across three arms, to be submitted in both the US and the EU. Using the FDA master protocol draft and the EMA platform draft, which design elements would you keep, and which would you change?
4. A colleague says "FDA requires a simulation report for any Bayesian design". Is that accurate? Classify the statement using the requires, recommends, permits distinction, and name the document you would cite.
5. Pick one of the six formal meeting types and one of the programs (CID, MIDD, CPIM, DDT). For a trial of your own, say which you would request first, what you would bring, and what answer you would need to proceed.

Part II.

Basics: Comparing Groups, Randomizing, and Updating Beliefs

4. Are They Different? Testing From the Ground Up

A trial ends with two rows of numbers, one for each arm, and a question that sounds simple: are they different? Everything that follows in this book, the Bayesian machinery included, is an answer to a sharper version of that question: how much should the gap between the rows change what we do? This chapter rebuilds the traditional answer from nothing. We assume you have forgotten testing, or learned it as a set of recipes without the picture behind them. We use one trial, simulated many ways, so that every formula arrives after the thing it summarizes has been seen.

The plan is short. First we look at how patients differ from one another, because that variation is the whole reason a trial needs statistics. Then we rerun the trial thousands of times on the computer and look at how the observed difference wobbles. That wobble is the sampling distribution, and every test and interval in the chapter is a way of reading it. The first test we build, the randomization test, needs nothing beyond shuffling labels. The z test is the same idea with a bell curve in place of the shuffling. Confidence intervals, error rates, power and the choice between one-sided and two-sided tests then follow, and we finish with what regulators write down about each.

! Regulatory anchor

- **ICH E9, Statistical Principles for Clinical Trials** (Step 4, February 1998) (International Council for Harmonisation 1998), section 5.5, Estimation, Confidence Intervals and Hypothesis Testing: the protocol must specify the hypotheses and the statistical model, estimates of treatment effects should be accompanied by confidence intervals, and one-sided tests must be justified prospectively. The guideline prefers a one-sided type I error at half the conventional two-sided level.
- **21 CFR 314.126(b)(7)** (U.S. Food and Drug Administration 2024b): an adequate and well-controlled study uses methods of analysis adequate to assess the effects of the drug, and the protocol and report describe them. The test and the interval of this chapter are those methods in their simplest form.
- **FDA draft guidance on Bayesian methodology** (January 2026, draft, not binding) (U.S. Food and Drug Administration 2026g), section IV.A: for trials with an overall frequentist statistical analysis plan, success criteria are almost always chosen so that the familywise type I error rate is no greater than 0.025, one-sided. The draft recommends this calibration as one route among three (it also describes direct interpretation of posterior probabilities, for example with borrowing and by agreement with FDA, and decision-theoretic criteria). Calibration is the default of this book, and the bar that its confirmatory Bayesian designs without borrowing are measured against. The regulatory map is in Chapter 3.

4.1. One trial, two endpoints

Throughout the chapter we use two versions of the same trial: an investigational drug against control, with 100 patients per arm, allocated by a fair coin. For these calculations, we condition on 100 patients in each arm. The first version has a binary endpoint, whether the patient responded. Suppose the control response rate is 30% and the drug's true response rate is 45%, a difference of 15 percentage points. The second version has a continuous endpoint, the fall in systolic blood pressure in mmHg after twelve weeks. Suppose the true mean fall is 4 mmHg larger on the drug, and that patients vary around their arm's mean with a standard deviation of 12 mmHg. These numbers are chosen to be plausible and round. They are the truth of the simulated world, which no real trial ever gets to see, and that is the point of simulating: we can compare what the analysis says with what we know is true.

The notation follows the book. The response probability in arm j is p_j, with $j = 0$ for control and $j = 1$ for the drug. The treatment effect is θ, here $\theta = p_1 - p_0$ for the binary endpoint and $\theta = \mu_1 - \mu_0$ for the continuous one. The number of patients in each arm is $n_0 = n_1 = n$, so the total is $2n$; this chapter writes n for the common arm size.

The first thing to see is variation between patients. Figure 4.1 shows one simulated trial of each kind. In the binary version each square is a patient, filled if they responded. Nobody's response is a fixed property of the drug: among 100 patients on control, about 30 respond, and which 30 is not something we can predict. In the continuous version each dot is a patient's blood pressure fall. The two clouds overlap heavily. Many patients on the drug do worse than the average patient on control, and many on control do better than the average patient on the drug.

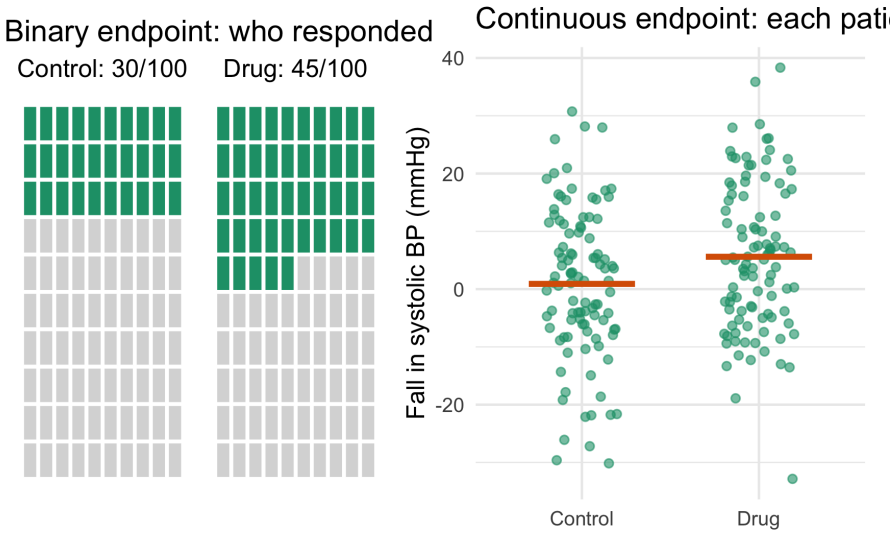


Figure 4.1.: One simulated trial with 100 patients per arm. Left: binary endpoint (filled square = responded; 30 of 100 on control, 45 of 100 on drug). Right: continuous endpoint (each dot is one patient's fall in systolic blood pressure, mmHg; horizontal bar = arm mean). The arms differ on average, but the individual patients overlap heavily.

In this particular trial the binary data show 30 responders on control and 45 on the drug, so the observed difference is exactly the true 15 points. That is a coincidence we arranged on purpose, so that the arithmetic stays clean. The continuous data show means of 0.90 mmHg on control and 5.60 mmHg on the drug, an observed difference of 4.69 against a truth of 4. That is the more typical situation: the data are close to the truth but not equal to it.

This overlap is what D. R. Cox had in mind when he wrote about the comparison of treatments in the presence of what he called large uncontrolled variations between experimental units (Cox 1958). If every patient responded identically to a given treatment, one patient per arm would settle the matter. Because they do not, a single difference in averages comes with a question attached: would we have seen a gap like this anyway, from the luck of which patients landed where? A test is a procedure for answering that question honestly.

A word on what the data cannot tell us. Each patient has, in principle, two potential outcomes, one under each treatment, and we see only one of them. The laboratory module that accompanies this chapter draws the unseen outcomes as faded cells for this reason. Randomization is what lets the observed gap stand in for the unobservable average difference between the two potential outcomes, and Chapter 5 takes that argument up properly. For now it is enough that patients were assigned by a coin, so the two arms are alike apart from chance and the drug.

4.2. The same trial, rerun ten thousand times

Here is the experiment that makes the rest of the chapter easy. Take the binary trial and rerun it. Draw 100 new control patients with a 30% chance of responding, 100 new drug patients with a 45% chance, and compute the observed difference in response rates. Do it again. Do it ten thousand times. We cannot run a real trial ten thousand times, but a computer can, and the resulting histogram is the sampling distribution of the observed difference: the pattern of results the trial design produces, given a truth.

Three things stand out. First, the histogram on the left is centered on the truth, 0.15, so the observed difference is an unbiased estimate of the effect: right on average, wrong in any single trial. Second, it is spread out. The standard deviation is about 0.067, so a trial that estimates 0.15 could easily have estimated 0.08 or 0.22 and nothing about the drug would have been different. In about 1.7% of reruns, a drug that truly works shows no benefit at all or a harm. Third, the histogram on the right, for a drug that does nothing, is not a spike at zero. It is a hill of about the same width centered on zero, and its upper tail is not empty: in roughly 1.0% of those reruns the observed difference is 0.15 or larger, which is the gap we saw. A drug that does nothing produces a result as good as ours about 1 time in 98. The p-values of the next sections (0.014 to 0.021, depending on the method) are the same idea computed from the data of one trial, and they differ from this 1.0% because a real trial does not know that both arms are at 30%.

That last sentence is, in essence, a p-value, built from the picture and not from a formula. The same logic applied to RECOVERY's 28-day mortality comparison is in Section 4.11. To turn it into a method we must find a way to learn the right-hand histogram from the data of one trial, because in a real trial we do not know that both arms are 30%. The next two sections give two ways to do this.

The width of the histogram has a formula worth knowing. The standard deviation of the observed difference, called the standard error, is

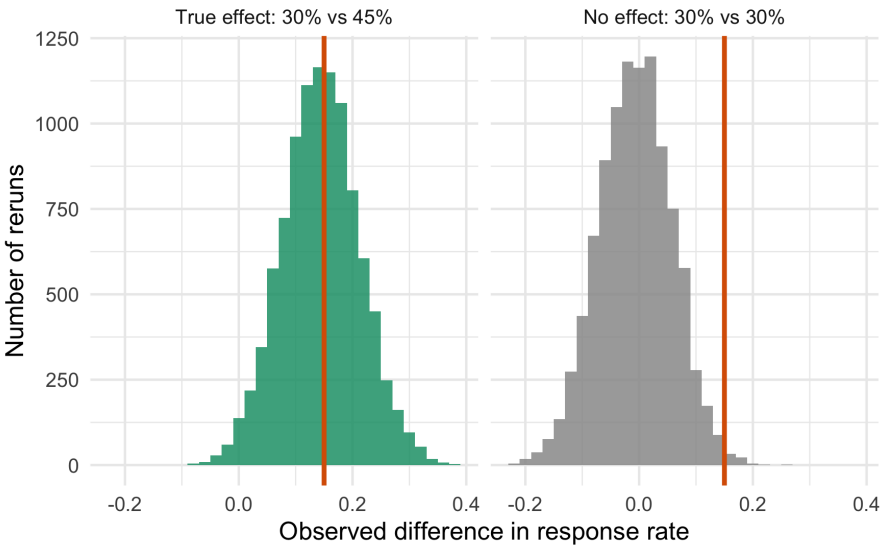


Figure 4.2.: The observed difference in response rates in 10,000 reruns of the same trial (100 per arm). Left: the world where the drug raises the response rate from 30% to 45%. Right: the world where the drug does nothing (both arms 30%). The vertical line is the difference we observed, 0.15.

$$\mathrm{SE}(\hat{\theta}) = \sqrt{\frac{p_0(1-p_0)}{n} + \frac{p_1(1-p_1)}{n}}, \tag{4.1}$$

which for our numbers is $\sqrt{0.3 \times 0.7/100 + 0.45 \times 0.55/100} = 0.0676$, matching the simulation. For the continuous endpoint, with common standard deviation σ, it is $\sigma\sqrt{2/n} = 12\sqrt{2/100} = 1.70$ mmHg. Two features matter for planning. The standard error shrinks like $1/\sqrt{n}$, so halving it requires quadrupling the trial. And it grows with the variability of the outcome: noisier patients mean wider histograms, which is why trials work so hard on reducing measurement noise.

4.3. The randomization test: shuffle the labels

The simplest honest test uses nothing but the randomization the trial already performed. Suppose the drug does nothing at all, a claim that Fisher called the null hypothesis (Fisher 1935). Then each patient's outcome would have been what it was whatever arm they were assigned to (Fisher's sharp null, which is stronger than a zero average effect, because opposing individual effects can average to zero; Chapter 5 gives the full definition), and the label "drug" or "control" on a patient is just the result of the coin. Under that hypothesis, a different toss of the coins would have produced the same 200 outcomes, dealt into the two arms differently. So we can imitate the coin: pool the 200 outcomes, shuffle the labels, deal 100 into a "drug" arm and 100 into a "control" arm, and compute the difference. Repeat many times. The histogram of those differences shows what chance alone does to this particular set of 200 patients.

Figure 4.3 first shows the idea on a handful of shuffles, then the histogram from 20,000 of them, with the observed difference marked. The share of shuffles at least as large as the observed difference is the randomization p-value.

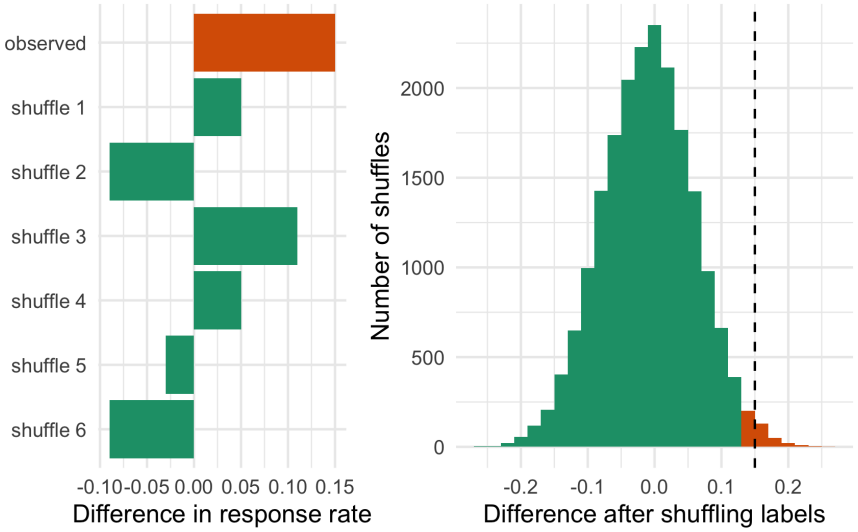


Figure 4.3.: The randomization test for the binary trial. Left: the observed gap (0.15) and six individual re-deals of the same 200 outcomes into two arms of 100. Right: the differences from 20,000 re-deals; the orange area is the share at least as large as the observed 0.15, which is the one-sided randomization p-value.

The one-sided randomization p-value is about 0.021 and the two-sided one about 0.041. In words: if the drug had no effect on any patient, then only about 2.1% of the possible ways the coins could have fallen would give the drug arm 15 or more additional responders. The two-sided version counts gaps of 15 points in either direction, so it roughly doubles.

Notice what this test assumed. It did not assume a bell curve, or a model for how responses arise, or a large sample. It assumed that the labels were assigned by chance and that, if the drug is useless, the outcomes do not depend on the label. The analysis is justified by the randomization in the design, which is exactly the position Fisher took, and which ICH E9 endorses when it calls randomization "a deliberate element of chance" in the assignment (section 2.3.2). The same shuffle works for the continuous endpoint. Pool the 200 blood pressure changes, deal them into two arms of 100, and compute the difference in means. For the continuous trial of Figure 4.1 the observed difference is 4.69 mmHg, and the one-sided randomization p-value is about 0.005 (two-sided about 0.011).

Table 4.1.: Randomization test for the continuous endpoint (20,000 shuffles).

Quantity	Value
Observed difference (mmHg)	4.693
One-sided p-value	0.005
Two-sided p-value	0.011
SD of shuffled differences (mmHg)	1.850
Planning formula, sigma*sqrt(2/n) (mmHg)	1.697

Table 4.1.: Randomization test for the continuous endpoint (20,000 shuffles).

Quantity	Value

Table 4.1 gives the result. The shuffled differences have a standard deviation of 1.85 mmHg. That is a little above the 1.70 of the formula because this sample's standard deviations (13.0 and 12.9) exceed the planning value of 12; it is close to the data-based standard error of 1.83. This anticipates the next section: the shuffling distribution is a bell curve with a width we can write down.

Two practical remarks. A shuffle test is a clean way to check any other test, because it makes so few assumptions. And in the binary case the shuffling can be done exactly, by counting, instead of by simulation. The exact count is Fisher's exact test, whose two-sided p-value here is 0.0405, which matches our simulation. In real trials, with stratified randomization or a covariate-adjusted analysis, the shuffle must respect the randomization scheme, so only labels within a stratum are exchanged. That connection between the design and the analysis is the subject of Chapter 5.

4.4. The normal approximation and the z test

Shuffling is correct but heavy, and it was laborious before computers (exact enumeration and manual randomization tests existed long before them, but they were practical only for very small trials). The histograms we have drawn all look like bell curves. This is not an accident: the observed difference is an average of many small independent contributions, one from each patient, and the central limit theorem says that such sums settle into a normal distribution as n grows. This lets us replace the shuffling with a formula.

Under the null hypothesis, where the drug does nothing, the difference in response rates is approximately normal with mean zero and standard error computed from the pooled response rate $\bar{p} = (x_0 + x_1)/(2n)$, where x_j is the number of responders in arm j:

$$Z = \frac{\hat{p}_1 - \hat{p}_0}{\sqrt{2\bar{p}(1-\bar{p})/n}}. \tag{4.2}$$

The pooling uses the null hypothesis that both arms have the same p. For the continuous endpoint the statistic is the difference in means divided by $\sqrt{s_1^2/n + s_0^2/n}$, with s_j the sample standard deviation in arm j. We then ask how far out in the tail of a standard normal curve Z falls. A value of Z means "this many standard errors from zero", and the book's notation z_α is the upper α quantile of the standard normal, so $z_{0.025} = 1.96$.

For the binary trial, $\bar{p} = 0.375$, the standard error is 0.0685, and $Z = 0.15/0.0685 = 2.19$. The upper normal tail beyond 2.19 is 0.0142, so the one-sided p-value is 0.014 and the two-sided one is 0.029. For the continuous trial, $Z = 4.69/1.83 = 2.56$, with one-sided p-value 0.0052.

Table 4.2 puts the three methods side by side. The randomization test and the z test agree closely for the continuous endpoint. For the binary endpoint they disagree more than a reader might expect: one-sided 0.021 by shuffling against 0.014 by the normal curve. The reason is visible in the left panel of Figure 4.4. The shuffled differences can only take values

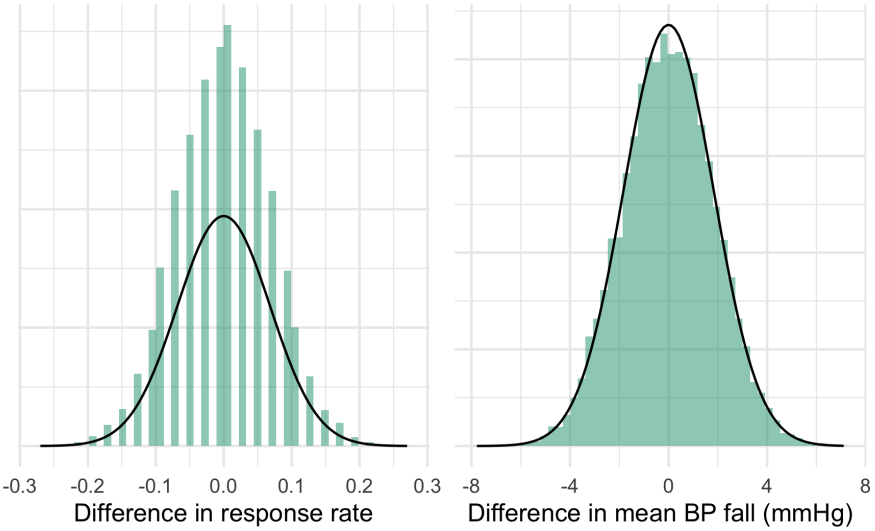


Figure 4.4.: The shuffling distributions of the two endpoints (histograms) against the normal curve the z test uses (line). The curve is almost on top of the histogram for the continuous endpoint. For the binary endpoint the histogram moves in steps, because the response counts are whole numbers.

in steps of 2/100 on a difference scale, and the "at least 0.15" tail includes the whole step at 0.15, while the normal curve effectively counts only half of that step. The standard remedy is a continuity correction, which moves the cutoff half a step, from 0.15 to 0.14 (which gives $Z = 2.04$ and a one-sided p-value near 0.021). The lesson for a protocol is that the normal approximation is an approximation, and with a binary endpoint, moderate n and a p-value close to the threshold, the choice of test statistic can change the verdict. This is why statistical analysis plans name the test exactly (Pearson chi-square, Fisher exact, with or without continuity correction) before any data exist.

Table 4.2.: The same trial analyzed by several methods. Shuffle values carry Monte Carlo error of about 0.001.

Endpoint	Method	One-sided p	Two-sided p
Binary (30 vs 45 of 100)	Randomization (shuffle)	0.021	0.041
Binary	Fisher exact	0.020	0.041
Binary	z test, pooled (no correction)	0.014	0.028
Continuous (mean difference 4.69)	Randomization (shuffle)	0.005	0.011
Continuous	z test	0.005	0.010
Continuous	t test	0.006	0.011

> **Technical note: why Z is approximately standard normal**
>
> Write the difference in means as $\bar{Y}_1 - \bar{Y}_0$. Each mean is an average of n independent terms, so by the central limit theorem it is approximately normal with mean μ_j and variance σ_j^2/n. The difference of independent normals is normal with variance $\sigma_1^2/n + \sigma_0^2/n$. Dividing by the standard error gives a statistic that is approximately $N(0,1)$ under the null. For the binary endpoint the variance of a single response is $p(1-p)$, which is what Equation 4.1 uses. When the variance is estimated from the data and n is small, the reference distribution is a Student t with slightly heavier tails, but with 100 patients per arm the difference is in the third decimal (the 97.5th percentile is 1.972 on 198 degrees of freedom against 1.960).
> For the clinician: the approximation fails when responses are rare, when n is small, or when the outcome has extreme values that dominate the mean. In those situations use the exact or shuffle versions, and make the choice in the protocol.

4.5. The p-value, defined carefully

Both tests produced a number called the p-value. It is worth defining it once with care, because everything about misuse starts with sloppy definitions. The ingredients are three: a null hypothesis, a test statistic, and the data.

The **null hypothesis** H_0 is the claim that the treatment effect is zero, $\theta = 0$ (or, for a one-sided test, $\theta \le 0$), a statement about the average effect and weaker than the sharp null of the randomization test. The **test statistic** is a summary of the data that is large when the data favor the drug: here, the observed difference or the Z derived from it. The **p-value** is then

$$p \;=\; \mathbb{P}(\text{test statistic at least as favorable to the drug as the one observed} \;\big|\; H_0 \text{ true}), \tag{4.3}$$

where the probability is over the repetition of the trial, in the sampling sense of the previous sections. That is the share of the right-hand histogram of Figure 4.2 beyond the observed value. It is the answer to the question "if nothing is going on, how often would luck look this good?"

The definition has several consequences that are easy to forget.

- The probability is about the data, given the hypothesis. It is not the probability of the hypothesis given the data. A p-value of 0.014 does not mean a 1.4% chance that the drug does not work, and its complement is not the probability that the drug works. Getting from data to the probability of a hypothesis needs a prior, which is the subject of Chapter 6.
- It depends on the plan, not only on the data. The same sequence of 9 heads and 3 tails has different p-values if the experimenter planned 12 flips or planned to stop at the third tail, because "as extreme or more" is evaluated over the trials one could have run. In trial language, a p-value depends on the stopping rule and on the number of analyses. This is the reason interim looks must be pre-specified, and it is the starting point of Chapter 11.

- It depends on how many questions were asked. A p-value of 0.03 for the one endpoint the protocol named in advance and a p-value of 0.03 for the best of twenty endpoints are not the same evidence. ICH E9 section 5.6 states that multiplicity "may necessitate an adjustment to the type I error", and Chapter 10 treats the methods.
- It mixes the size of the effect with the size of the trial. A tiny and clinically irrelevant difference gives a tiny p-value in a huge trial, and a large and important difference can give an unimpressive p-value in a small one. The p-value says nothing about whether the effect is worth having. That is the job of the estimate and its interval.
- A large p-value is not evidence of no effect. It says the data were not surprising under the null. They may be even less surprising under a real and useful effect, and in a trial of 50 per arm the 30% versus 45% drug has a power near one in three (the next sections explain what that means).

The American Statistical Association's statement on p-values collects these points (Wasserstein and Lazar 2016), and Goodman's "dirty dozen" lists twelve misreadings that survive in the clinical literature (S. Goodman 2008). We return to the evidential meaning of a p-value, and to what a Bayes factor or a posterior probability would say about the same data, in Chapter 7.

To turn a p-value into a decision, we need a threshold. The protocol fixes a level α in advance, and rejects H_0 if $p \le \alpha$. That is a decision rule with a known long-run error rate, and the next sections look at what the rule does when the drug works and when it does not.

4.6. Confidence intervals

A p-value says whether the data are compatible with no effect. It does not say what effect is likely, and a decision about a drug needs both. A **confidence interval** reports the range of effect sizes that the data do not rule out. For a difference with an approximately normal estimator the 95% interval is the estimate plus and minus 1.96 standard errors:

$$\hat{\theta} \pm 1.96 \times \mathrm{SE}(\hat{\theta}). \tag{4.4}$$

For the binary trial the standard error with unpooled rates (Equation 4.1, estimated) is 0.0676, so the 95% interval for the risk difference is 0.15 ± 0.132, that is, 0.017 to 0.283. The data are compatible with a gain as small as 2 percentage points and as large as 28. For the continuous trial the interval is $4.69 \pm 1.96 \times 1.83$, that is, 1.10 to 8.28 mmHg. Neither interval contains zero, which agrees with the two-sided tests that gave p-values below 0.05.

The interval is wide, and this is not a flaw of the method. It tells the truth about a trial of 100 per arm: a response-rate gain of 2 points and one of 28 points are both consistent with what we saw, and they would lead to different decisions about whether the drug is worth developing. ICH E9 section 5.5 asks that estimates of treatment effects "should be accompanied by confidence intervals, whenever possible". The reason is that a bare p-value hides this width.

4.6.1. What the 95% refers to

The correct reading of "95%" is about the procedure, not about one interval. If the same trial were rerun many times and an interval computed each time, about 95% of the intervals would contain the true effect. Figure 4.5 shows 50 reruns of the binary trial, with the true difference of 0.15 marked.

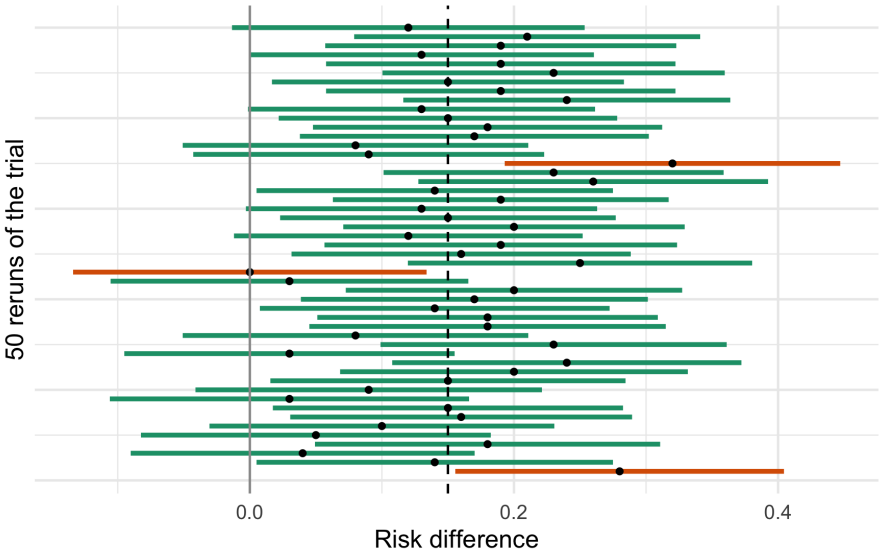


Figure 4.5.: Fifty reruns of the binary trial, each with its own 95% Wald interval for the risk difference. The vertical line is the truth (0.15). Orange intervals miss it. In the long run about 5% of intervals miss; in 50 trials the count is itself random.

Over 100,000 reruns the Wald interval covers the truth in 94.8% of reruns, close to the nominal 95%. That shortfall is a property of the normal approximation again, as with the p-values; better-behaved intervals for proportions exist, for example the score-based interval that Newcombe compared with ten other methods (Newcombe 1998). The practical rule is the one for tests: name the interval method in the analysis plan.

Notice also what the interval is not. It is not a statement that there is a 95% chance that the true difference lies in 0.017 to 0.283. Once computed, the interval either contains the truth or does not, and the 95% describes the method's long-run record. A statement of the form "given these data, the probability that the effect lies in this range is 95%" is a different object, a credible interval, and requires a prior. For a large trial with little prior information the two usually agree numerically, which is why the common misreading rarely does damage, but the distinction matters when the sample is small, the prior is informative, or when we want to evaluate a decision rule. Chapter 6 returns to it.

4.6.2. Tests and intervals are the same thing

A confidence interval and a test are two views of one calculation. The 95% two-sided interval contains exactly the values θ_0 that a two-sided test at the 5% level would not reject: if you ask "can we reject that the true difference is θ_0?" for every θ_0, the ones you cannot reject

form the interval. This is the duality. It has a protocol consequence. A test of no effect asks only whether 0 is in the interval, but the same machinery also lets you test other hypotheses, such as "the benefit is at least 5 points", by checking whether the relevant confidence bound (here the lower limit of a one-sided interval) is above 5, not merely whether 5 lies inside a two-sided interval. Non-inferiority testing rests on this: the null hypothesis is that the new treatment is worse than control by at least the margin, and the test is whether the lower end of a one-sided interval clears the margin.

It also explains why ICH E9 links the two. The same section that calls for confidence intervals, section 5.5, says the one-sided type I error should be set at half of the two-sided one, and gives the reason: "This promotes consistency with the two-sided confidence intervals that are generally appropriate for estimating the possible size of the difference between two treatments." A one-sided test at 0.025 rejects essentially when the lower limit of the two-sided 95% interval is above zero, so the result in the CSR (a p-value) and the figure in the abstract (an interval) cannot disagree in any material way. (The equivalence is exact only when the test and the interval use the same standard error. Our binary trial uses 0.0685 in the pooled test and 0.0676 in the interval, so a result very close to the line could fall on different sides.)

For the binary trial, the lower limit of the interval is 0.017, above zero, and the one-sided p-value from the normal test (0.014) is below 0.025. For the continuous trial the lower limit is 1.10 and the one-sided p-value is 0.005. Both trials "succeed" on the regulatory convention, though the binary one only just, and the width of its interval is a more honest summary than the verdict.

4.7. Two kinds of error, and power

So far we have asked what the data say about one trial. A regulator, a sponsor and a patient also want to know how often the decision rule, "reject H_0 if $p \leq \alpha$", gives the wrong answer. There are two ways to be wrong, and they are not symmetric. The language of type I and type II errors and of power is Neyman and Pearson's (Neyman and Pearson 1933). Fisher's p-value as a measure of evidence against a null hypothesis, and the 0.05 convention, come from his work (Fisher 1935, 1946).

A **type I error** is rejecting H_0 when the drug has no effect: approving something useless. A **type II error** is failing to reject H_0 when the drug does work: abandoning something useful. The probability of a type I error is the level of the test, α. The probability of a type II error is β, and the probability of avoiding it, $1 - \beta$, is the **power**. Notice that α is a probability computed under the null world and that β is computed under a specific alternative. Power is not a single number for a trial but a function of the true effect, and when someone quotes "90% power" the first question is: for what effect?

We can read both off the simulation of Figure 4.2. Run the same trial many times, apply the decision rule each time, and count rejections. In the world where the drug does nothing the rejection rate is the type I error rate, and it is close to the nominal 2.5% when we reject for $Z > 1.96$. In the world with the true 15-point effect the rejection rate is the power.

The type I error is held at about 2.5%, because that is what the threshold 1.96 was chosen to do. The power at 100 patients per arm is 60%: if the drug truly lifts the response rate from 30% to 45%, this trial would detect it about six times in ten. The other four times, a drug that works would be missed. Our observed trial, with $Z = 2.19$, happens to be one of the six.

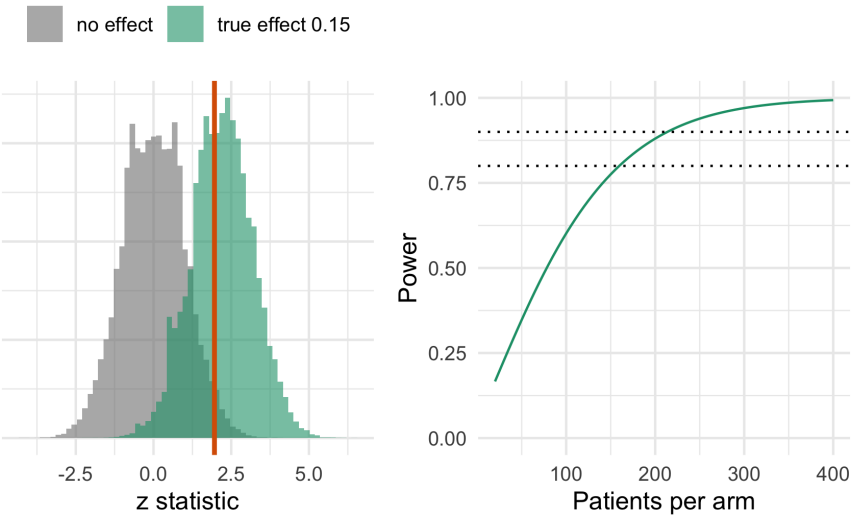


Figure 4.6.: Left: the z statistic in 100,000 reruns with no effect (grey) and with the true 15-point effect (teal), 100 patients per arm. Reruns to the right of 1.96 reject the null. The grey area beyond 1.96 is the type I error (2.5%); the teal area beyond 1.96 is the power (60%); the teal area to its left is the type II error. Right: power of the one-sided 0.025 test against the number of patients per arm.

It would be wrong to read that as evidence that the trial was well designed. The right-hand panel shows that power rises with the number of patients: roughly 35% at 50 per arm, 60% at 100, 78% at 150, 88% at 200, 94% at 250 and 97% at 300. The customary targets of 80% or 90% need roughly 160 to 215 patients per arm for this effect. The continuous trial, with a 4 mmHg effect and standard deviation of 12 mmHg, has power of about 65% at 100 per arm.

The two errors trade against each other. Lower α to reduce false positives and the rejection region shrinks, so power falls. Raise α and power rises, along with the chance of approving something useless. The only way to improve both at once is to make the standard error smaller, with more patients, a less noisy endpoint, or better use of information, and this is where the later chapters of the book make their case. The size of a trial is dictated by the question “how many patients make both errors acceptably small?” and the answer is a formula in α, β, the effect and the variance that Chapter 9 develops. A guide for the reader who wants the shape now: for a one-sided $\alpha = 0.025$ and 90% power, the number of patients needed is proportional to $(z_{0.025} + z_{0.10})^2 = 10.51$ divided by the squared standardized effect.

It is worth stating who bears each error. A type I error falls mostly on future patients, who receive a drug that does not work, and on the regulator who approved it. A type II error falls on future patients too, who will not receive a drug that would have helped, and on the sponsor who abandoned it. Regulators fix α by convention and leave β to the sponsor’s judgment, which is why the traditional design is organized around the type I error and why sponsors, who pay for the type II error, ask for more efficient designs. Module 03 of the laboratory, errors and power (https://vsokolov.org/bayesian_trials_lab/03-errors-power.html), lets you move the slider and see the two areas trade off.

4.8. One-sided or two-sided, and why 0.025

We have used both kinds of p-value, so we must say which one counts. A **two-sided** test counts departures from the null in either direction as evidence: the drug might help or harm, and the test is alert to both. A **one-sided** test counts only departures in the direction of benefit. In the shuffle picture the one-sided p-value is the orange tail on the right, and the two-sided is that tail plus its mirror image on the left (twice as large, when the reference distribution is symmetric).

The argument for the two-sided test is that a trial should be able to detect harm, and that a one-sided test with a generous level appears to lower the bar. The argument for the one-sided test is that the regulatory question for a new drug is asymmetric: approval depends on benefit being demonstrated, and a harmful drug is, for the purpose of the efficacy decision, not approved either way. Harm shows up in the safety analyses and in the interim monitoring by the data monitoring committee.

The regulatory convention resolves the matter in a way that keeps both arguments. ICH E9 section 5.5 asks the sponsor to say in the protocol whether one- or two-sided tests will be used and to "justify prospectively the use of one-sided tests". It then states the rule that decides the practical question: "The approach of setting type I errors for one-sided tests at half the conventional type I error used in two-sided tests is preferable in regulatory settings." With the conventional two-sided level of 0.05, half is 0.025. That is the origin of the book's default: a one-sided level of $\alpha = 0.025$ is the same evidential bar as a two-sided level of 0.05, as the confidence interval duality of the previous section shows: both reject when the 95% interval excludes zero on the favorable side. The more recent FDA draft guidance on Bayesian methodology uses exactly this language, describing the usual standard as performing a statistical hypothesis test at a one-sided significance level of 0.025 (section IV.A, draft, January 2026, non-binding) (U.S. Food and Drug Administration 2026g).

Two practical consequences follow. First, when a paper reports "p = 0.04, two-sided" for a drug that helped, the one-sided equivalent is 0.02, and the trial meets the 0.025 bar. When a paper reports "one-sided p = 0.04", it does not, and the usual regulatory reading would be that the trial missed. Always check which p-value is quoted. Our binary trial is a live example of this edge: the normal z test gives a one-sided p-value of 0.014, the Fisher exact test 0.020 and the randomization test 0.021. All three clear 0.025, but the last two with little room. The claim "the drug works" is the same in each, but the margin is slim. Second, the sample size formula contains $z_{0.025} = 1.96$ for one-sided designs, which is the same number as the two-sided 5% critical value, so the trial size is the same under either label provided the conventions are matched.

It is also possible to be tempted by the one-sided test in the wrong way. If the sponsor looks at the data, sees the drug is ahead, and then declares a one-sided test at 0.05, the type I error rate has doubled to 5% for a drug that does nothing, and the test is no longer prospectively justified. This is why E9 says what it says, and why the statistical analysis plan fixes the side, the level and the test before unblinding. The trap in the other direction is that a one-sided 0.025 test gives no formal weight to results in the harmful direction. A trial can end with a strongly negative difference and a one-sided p-value near 1, which is true but uninformative; the safety review and monitoring plan, and not the efficacy test, are where harm is handled.

4.9. Noninferiority and equivalence: testing against a margin

Every test so far asked whether the drug beats a comparator. Many trials ask a different question. When an effective treatment already exists, withholding it in favor of placebo is often unethical, so the new drug is compared with the active control, and the claim is that it is not unacceptably worse, perhaps because it is safer, cheaper, easier to take, or because the trial is an outcomes trial in which the new drug is expected to help the heart as well as lower glucose. This is a **noninferiority** trial, and its logic is the logic of this chapter with the null hypothesis moved.

Let $\gamma = \log \mathrm{HR}$ be the raw log hazard ratio of the new drug against the active control, a separate symbol from the effect θ of the book's notation because here values above zero mean the new drug is worse. The superiority hypotheses are $H_0 : \gamma \geq 0$ against $H_1 : \gamma < 0$. For noninferiority with margin $M > 1$ on the hazard ratio scale, the hypotheses are

$$H_0 : \ \gamma \geq \log M \qquad \text{against} \qquad H_1 : \ \gamma < \log M. \tag{4.5}$$

Rejecting H_0 means the new drug is shown to be no worse than the margin allows. The test is one-sided at 0.025, exactly as in Section 4.8, and it is equivalent to checking whether the upper limit of the two-sided 95% interval lies below $\log M$ (Section 4.6). Nothing about the machinery is new; what is new is that the answer depends entirely on M, a number the data do not supply. Putting the burden of proof on the claim of similarity, by reversing the roles of null and alternative, goes back to Dunnett and Gent for equivalence (Dunnett and Gent 1977) and to Blackwelder for noninferiority (Blackwelder 1982).

4.9.1. Where the margin comes from

The FDA guidance on noninferiority trials (final, November 2016) (U.S. Food and Drug Administration 2016) and ICH E10 (International Council for Harmonisation 2000) give the reasoning. A noninferiority trial with no placebo arm cannot show by itself that the new drug works: if the new drug is "not much worse" than a control, and the control was never shown to work in this trial, the comparison is informative only if the control's effect is known from earlier placebo-controlled trials and is assumed to hold now. The guidance builds the margin in two steps. The first, usually written M_1, is the whole effect of the active control over placebo that the sponsor can reasonably assume in the present trial, estimated from the historical trials (conventionally the conservative limit of a meta-analytic confidence interval, not the point estimate). The second, M_2, is the margin actually used, chosen to preserve a clinically meaningful fraction of that effect, so that $M_2 \leq M_1$. How much to preserve is a clinical judgment, and the guidance says the margin should be justified for the setting, not read from a convention. Two ways of using the historical estimate are described (Rothmann and colleagues set out both for noninferiority mortality trials in oncology (Rothmann et al. 2003)): the fixed-margin approach, which sets the margin from the historical confidence limit and then runs an ordinary test, and the synthesis approach, which combines the historical and current estimates directly and carries the uncertainty of both into one test (the Bayesian version is in Section 19.11).

The scheme rests on the **constancy assumption**: the effect of the active control against placebo, estimated years ago, is the effect it would have had in this trial's population, with this background care and this endpoint definition. The assumption cannot be checked within the trial, which is why the guidance and ICH E10 attach so much weight to **assay sensitivity**, the ability of the trial to distinguish an effective treatment from an ineffective

one. A noninferiority trial that is sloppy (poor adherence, crossover, loosely defined events) pulls the two arms together and makes any new drug look noninferior, so the usual protections against bias in a superiority trial work in the opposite direction here. Sloppiness helps the sponsor, which is the reason the analysis population also matters. The guidance stresses trial quality and conduct; it does not prescribe a rule that intention-to-treat and per-protocol analyses must agree, though reporting both and examining any discrepancy is common practice.

The margin must be fixed in the protocol and justified before unblinding. A margin that is changed after the data are seen is not a margin.

4.9.2. Equivalence

An **equivalence** trial asks for the difference to lie inside a two-sided margin, $-\log M < \gamma < \log M$ on the log scale (the hazard ratio between $1/M$ and M). The standard analysis is two one-sided tests (TOST), each at the one-sided level, one against each end of the margin, and equivalence is declared only if both reject; this is the same as the confidence interval lying entirely inside $(-\log M, \log M)$. Bioequivalence studies and some biosimilar comparisons use this form. Most clinical efficacy trials that say "equivalence" mean noninferiority, and the one-sided test is the more common. The upper bound of a two-sided 95% interval is the standard one-sided 0.025 noninferiority test, and noninferiority imposes no corresponding lower-bound requirement. A sponsor who wants noninferiority should say so, and should not frame the trial as equivalence, which adds a lower-bound requirement that the question does not need.

4.9.3. SURPASS-CVOT: a noninferiority margin in practice

SURPASS-CVOT randomized adults with type 2 diabetes and established atherosclerotic cardiovascular disease to tirzepatide or dulaglutide 1.5 mg, with a primary composite of cardiovascular death, myocardial infarction and stroke (Nicholls et al. 2025). Dulaglutide was chosen as the control because REWIND had shown that it reduces cardiovascular events against placebo (Gerstein et al. 2019), which is what makes it a legitimate active comparator. The pre-specified noninferiority margin was 1.05 for the upper limit of the 95.3% confidence interval of the hazard ratio, and superiority would be claimed only if the upper limit were below 1.00. The trial randomized 13,299 patients; the modified intention-to-treat population was 13,165 (6586 on tirzepatide, 6579 on dulaglutide). The primary outcome occurred in 801 patients (12.2%) on tirzepatide and 862 (13.1%) on dulaglutide, with a hazard ratio of 0.92 (95.3% CI 0.83 to 1.01).

Table 4.3.: SURPASS-CVOT primary outcome, tested against the noninferiority margin 1.05 and against superiority, from the published hazard ratio and its 95.3% interval (normal approximation on the log scale).

Hypothesis	z statistic	One-sided p	Upper limit below the bound?
Noninferiority: H0 is HR at least 1.05	-2.68	0.0037	Yes (1.01 < 1.05)
Superiority: H0 is HR at least 1.00	-1.69	0.0460	No (1.01 > 1.00)

The standard error implied by the published interval is 0.049 on the log scale. Against the margin the one-sided p-value is 0.0037, in line with the reported 0.003 (the reported value comes from the trial's own analysis, which can differ a little from a calculation on the rounded interval), so noninferiority is established. Against zero the one-sided p-value is 0.05, in line with the reported two-sided 0.09 for superiority, so superiority is not. Both statements come from the same interval. The two tests have two different nulls, and the trial can win the first and lose the second without contradiction. Notice also that the margin did the work: the upper limit 1.01 is above 1.00 and below 1.05, and a margin of 1.00 would have been a superiority trial.

The traditional report stops here: noninferiority met, superiority not met. What it does not give is the probability that the hazard ratio lies below 1.05, below 1.00, or below 0.90, nor the effect against placebo that the dulaglutide comparison implies. The probability that the hazard ratio lies below a given value is a posterior probability, which is where the Bayesian distinction lies. The effect against placebo has frequentist treatments as well, such as the synthesis approach above. Both are taken up in Section 19.11 and Section 19.12.

4.10. Where the traditional trial stands, and what changes next

Put the pieces together and you have the traditional single-look trial. The protocol states the endpoint, the effect to be detected, the one-sided level of 0.025 and the power. It fixes n from these. The trial runs to completion. The analysis computes a test statistic and a p-value from the pre-specified method, and an interval for the effect. The decision is "success" if the p-value is at most 0.025, one-sided, and the CSR reports the interval alongside. The method is transparent, its error rates are known by construction (type I by the choice of the level, power by calculation or simulation), and it has a regulatory history of decades. It is the benchmark against which every design later in the book is judged. When there is one look, a single primary endpoint and no useful prior information, it is also hard to improve on, and the book says so where it matters (Chapter 21).

What this design does not do is answer the questions a decision maker actually has. The p-value does not give the probability that the drug works, the interval's 95% is a statement about the procedure, and the error rates describe a procedure that is run once at a fixed sample size, so they change when the trial is monitored, when several endpoints or arms are examined, or when earlier information is brought in. Each of these is a place where the traditional answer needs repair, and where a Bayesian formulation offers another route at a price that the book will itemize. The table summarizes what this chapter established and where the book goes next.

Question	Traditional answer (this chapter)	Where it goes next
Is there a difference?	p-value against H_0: the share of reruns or shuffles at least this extreme	Evidence beyond p-values and Bayes factors (Chapter 7)
How big is it?	Estimate and 95% confidence interval	Posterior and credible interval (Chapter 6)

Question	Traditional answer (this chapter)	Where it goes next
What error rates does the rule have?	Type I fixed at 0.025 one-sided; power by formula	Power and sample size (Chapter 9); operating characteristics by simulation (Chapter 14)
What if we look more than once?	The p-value is no longer calibrated	Group sequential designs (Chapter 11)
Is the new drug not much worse than an active control?	One-sided test against a margin (Section 4.9)	Bayesian noninferiority and margin synthesis (Section 19.11)
What justifies the test?	Randomization (shuffling labels)	Randomization and the design (Chapter 5)
What does the regulator expect?	Pre-specified test, side, level, interval (E9 section 5.5)	The regulatory map (Chapter 3)

Where do the two approaches agree numerically? In a simple setting with a flat prior, the Bayesian posterior probability that the drug is better than control is close to one minus the one-sided p-value. For our continuous trial this is a statement about 0.995, and for the binary trial about 0.986. The agreement is not a coincidence, and it tells us that the Bayesian bar and the traditional bar coincide when there is a single look and no prior information. They diverge once the design is made more flexible, and the point of the book is to show how and by how much.

In the protocol

A statistical section that follows this chapter would state: the primary endpoint (response at week 12, or change from baseline in systolic blood pressure); the null and alternative hypotheses ($H_0 : \theta \leq 0$ against $H_1 : \theta > 0$); the test and its exact form (for the binary endpoint, the pooled z test or Fisher exact test and whether a continuity correction is used; for the continuous endpoint, the t test or an analysis of covariance); the one-sided significance level of 0.025; the effect against which power is calculated and the power; the two-sided 95% confidence interval and its method; the rule on how the randomization scheme enters the analysis; and the statement that none of these will change after unblinding.

Try it in the lab

Module 00 at https://vsokolov.org/bayesian_trials_lab/00-are-they-different.html lets you decide "same or different" for rows of 0s and 1s before it reveals the truth, then re-randomize the same data 1,000 times to see how often chance alone makes a gap that large. Predict the p-value from the picture first, then compare with the box; the "Why the coin?" section at https://vsokolov.org/bayesian_trials_lab/00-are-they-different.html#why-coin shows what the shuffling is protecting against. In module 02 at https://vsokolov.org/bayesian_trials_lab/02-p-values.html set 20 successes in 30 tries with a luck-only chance of 50%, press "Simulate 1,000 experiments" and compare the orange share with the exact binomial tail; then try the same data under the two stopping plans and the "peeking" panel, and notice which of the two changes

the p-value and which does not.

4.11. In practice

Two real trials show the machinery of this chapter at work. One demonstrated a mortality benefit; the other remained inconclusive despite an estimate compatible with substantial benefit.

4.11.1. RECOVERY: a two-group comparison, a rate ratio and an interval

The RECOVERY platform trial randomized hospitalized patients with Covid-19 to usual care or to one of several treatments. Its dexamethasone comparison is, statistically, our trial: two groups, a binary outcome (death within 28 days) and a decision about whether the gap is more than chance (RECOVERY Collaborative Group et al. 2021). Of 2104 patients assigned to dexamethasone, 482 (22.9%) died, against 1110 of 4321 (25.7%) on usual care, a difference of 2.8 percentage points. The headline estimate was the age-adjusted rate ratio 0.83, with 95% confidence interval 0.75 to 0.93 and $P < 0.001$. The analysis was frequentist: no prior and no posterior probability, only a p-value and an interval computed from the repetition-of-the-trial logic of this chapter.

Two features connect it to what we built. The interval is as informative as the p-value: the data are compatible with a relative reduction in mortality of 7% to 25%, which is the range a clinician needs. And the size of the trial, thousands of patients rather than our 100 per arm, is what makes a 2.8 point gap decisive; the same gap in a trial of 100 per arm would be buried in a standard error of about 0.06. The lesson for this chapter: a clean randomized two-group comparison with a pre-specified analysis needs no more than the tools above, and when it is large enough the traditional answer is a good one. The design side of the same trial (concealed central randomization, open label, an objective endpoint) is in Section 5.12, and RECOVERY returns as a platform in Chapter 20.

4.11.2. ANDROMEDA-SHOCK: what "not significant" does and does not mean

ANDROMEDA-SHOCK randomized 424 patients with septic shock in 28 intensive care units to resuscitation targeting capillary refill time (212 patients) or serum lactate (212) (Hernández et al. 2019). The primary outcome was 28-day mortality: 74 deaths (34.9%) against 92 (43.4%), a risk difference of -8.5 points (95% CI -18.2 to 1.2) and a hazard ratio of 0.75 (95% CI 0.55 to 1.02), $P = .06$. On the convention of this chapter, a two-sided P above 0.05 is a miss: the interval contains no effect.

That verdict is correct, and it is not the same as "capillary refill targeting does nothing". The same interval contains a 45% relative reduction in the hazard of death and a 2% increase. A p-value of .06 says the data are not very surprising if there is no effect, and, as the definition above warns, they may be just as unsurprising under a substantial benefit. The trial was inconclusive between two clinically different possibilities, and the p-value alone cannot say which. A post hoc Bayesian reanalysis of the same trial is in Chapter 22. It shows why later chapters ask for a probability statement about the effect, which a p-value cannot give. The

lesson: report the estimate and the interval, read a large p-value as "inconclusive" unless the interval excludes effects that matter, and decide in the protocol what will count as success. Chapter 7 takes this example further.

Key points

- A trial compares groups of patients who vary, so a difference in averages is not yet evidence. The sampling distribution (the observed difference over many imagined reruns) shows how much a difference can move by chance alone. Its width is the standard error, which shrinks like $1/\sqrt{n}$.
- The randomization test needs only the coin that allocated the patients: pool the outcomes, shuffle the labels and count how often chance matches the observed gap. The z test is the same idea with a bell curve in place of shuffling, and it is an approximation that is least reliable for binary outcomes at moderate n close to the threshold.
- The p-value is the probability, under the null and over repetitions of the design, of a result at least as extreme as the one observed. It is not the probability that the null is true, it depends on the planned analyses and stopping rule, and it says nothing about the size or importance of the effect.
- A confidence interval reports the effects that the data do not rule out. The 95% refers to the long-run record of the method. The interval and the test are the same calculation: the interval is the set of null values a test would not reject. ICH E9 section 5.5 asks for intervals alongside tests.
- The type I error rate α is a property of the rule under the null world, and power $1 - \beta$ is a property under a stated alternative. At fixed information, lowering α generally increases β and reduces power. More information can reduce both error probabilities. In our example, 100 patients per arm give 60% power for 30% against 45%.
- Regulators set the confirmatory bar at a one-sided 0.025 (half of a conventional two-sided 0.05), prefer that the one-sided choice be justified before the trial (ICH E9 section 5.5), and the 2026 FDA Bayesian draft guidance uses the same bar for success criteria. For confirmatory adaptive designs without borrowing, this book uses calibration to one-sided 0.025. FDA's draft also describes direct posterior interpretation and decision-theoretic or benefit-risk criteria (section IV.A).

i Discussion questions

1. In Figure 4.2, the no-effect histogram has about 1.0% of reruns at or above the observed difference, yet the observed trial is called significant at one-sided 0.025. Explain, in terms of reruns, why 1% of a null world and "the probability that the drug works" are different things. What additional information would you need to convert one into the other?
2. The binary trial has a normal-approximation one-sided p-value of 0.014 and a randomization p-value of 0.021. Both clear 0.025, but suppose they had been 0.024 and 0.031. Which should the statistical analysis plan have named, when, and who decides? What does that imply for how close to the threshold a "pass" can be trusted?
3. A confidence interval of 0.017 to 0.283 and a p-value of 0.014 come from the same trial. A sponsor says the trial proved the drug works. A clinician says it may help only 2 patients in 100. Which statement does the interval support, and

which does the p-value support? How should the CSR present both?

4. A colleague proposes switching from a two-sided 0.05 to a one-sided 0.05 because "the drug can only help". Using E9 section 5.5 and the duality between tests and intervals, explain why this is not a harmless change.
5. Power of 60% was enough for our observed trial to succeed. Why is it still a poor design? Who bears the cost of the 40%?

5. Randomization From the Ground Up

Every trial in this book rests on one act that is so familiar it is easy to overlook: a patient is assigned to a treatment by chance. The rest of the apparatus, the p-value of Chapter 4, the posterior probabilities of Chapter 6, the interim looks of Chapter 11, all assume that the two groups being compared differ only because of the treatment, and by luck of the draw. This chapter explains why that assumption is credible when the assignment is random and not credible otherwise, how randomization is carried out in practice, what it means for a sequence to be random, and what a trialist must do after randomization to keep what it bought. We start with a doctor who chooses patients, because the failure of that arrangement is the reason the rest of the chapter exists.

The standard practice comes first. Everything here, simple and blocked randomization, stratification, minimization, concealment, blinding, intention to treat, is what a protocol written today for a regulator already contains. Response-adaptive randomization, where the allocation probabilities change as data accumulate, is a departure from this standard and is taken up in Chapter 16. A reader who knows the standard well can go straight to the sections on randomness and on why a Bayesian still randomizes.

> ! Regulatory anchor
>
> **ICH E9, Statistical Principles for Clinical Trials** (Step 4, February 1998). Section 2.3.2 on randomization: "Randomisation introduces a deliberate element of chance into the assignment of treatments to subjects in a clinical trial," and it "also tends to produce treatment groups in which the distributions of prognostic factors, known and unknown, are similar." Section 2.3.1 on blinding defines the double-blind trial as one in which neither the subject nor any of the investigator or sponsor staff involved in treatment or evaluation are aware of the treatment received. Section 5.2.1 says that the intention-to-treat principle implies that the primary analysis should include all randomized subjects and says that preserving the initial randomization in analysis is important in preventing bias.
> **ICH E9(R1), Addendum on Estimands and Sensitivity Analysis** (Step 4, November 2019). Requires the treatment effect of interest to be defined, before the trial, as an estimand, with an explicit strategy for each intercurrent event (treatment discontinuation, rescue medication, death).
> **ICH E8(R1), General Considerations for Clinical Studies** (Step 4, October 2021). Lists randomization, blinding or masking, and control of confounding among the approaches to minimize bias, and says that randomized studies play a central role in drug development because they can better control biases.
> **21 CFR 314.126(b)(4) and (b)(5)**. An adequate and well-controlled investigation must use a method of assigning patients to treatment that minimizes bias and ensures comparability of the groups on pertinent variables such as age, sex and severity of disease, and must take adequate measures to minimize bias on the part of subjects, observers and analysts. See Chapter 3 for the map of these documents.

5.1. Why allocate by chance

5.1.1. The counterfactual problem

Suppose a new drug is given to some patients and not to others. For any one patient we see one outcome. We never see what would have happened to the same patient under the other treatment. Write $Y(1)$ for the outcome a patient would have if treated and $Y(0)$ for the outcome if not. The treatment effect for that patient is $Y(1) - Y(0)$, and we can never observe both terms. This is the fundamental problem of causal inference in the potential outcomes language of Rubin (Rubin 1974), and every trial design is a way of getting around it.

What we can do is compare groups. Let $D = 1$ mean "was assigned to the drug" and $D = 0$ mean "was not", and assume complete adherence in this opening example (the intention-to-treat section below separates assignment from receipt). The difference in average observed outcomes between the groups can be written as

$$\underbrace{\mathbb{E}[Y \mid D=1] - \mathbb{E}[Y \mid D=0]}_{\text{what the trial shows}} = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid D=1]}_{\text{effect on those who were treated}} + \underbrace{\mathbb{E}[Y(0) \mid D=1] - \mathbb{E}[Y(0) \mid D=0]}_{\text{selection bias}}. \tag{5.1}$$

The last term asks how different the two groups were before the drug did anything: what would have happened to the treated patients had they not been treated, compared with what did happen to the untreated. If the groups were chosen in a way that depends on prognosis, the term is not zero, and the observed difference mixes the effect of the drug with the effect of the choice. If the assignment is made by a coin that ignores everything about the patient, then the treated and untreated groups are two random samples from the same population, the selection term is zero on average, and the difference in observed means estimates the average effect of the drug. That is the whole argument, and it needs no assumption about which patient characteristics matter. R. A. Fisher made it central to experimental design (Fisher 1935), and it is the reason the regulation asks for a method of assignment that minimizes bias.

5.1.2. Confounding by indication: a doctor picks the patients

The commonest way for the selection term to be nonzero in medicine is *confounding by indication*: the reasons a doctor chooses a treatment are themselves related to the outcome. A physician who believes a new drug is promising may offer it to patients who seem strong enough to benefit and keep the old treatment for the frail. No bad faith is required. The physician is doing what a physician is supposed to do, using judgment on behalf of each patient, and that is exactly what ruins the comparison.

We can see the damage in a simulation. Each patient has a hidden health score h that the doctor senses but that is not recorded. Healthier patients are more likely to recover whatever they receive. The new drug truly adds 10 percentage points to the probability of recovery. In the first scheme the doctor offers the drug with probability that rises with h. In the second, a coin decides. Each trial has 400 patients and we run 1,000 trials of each kind.

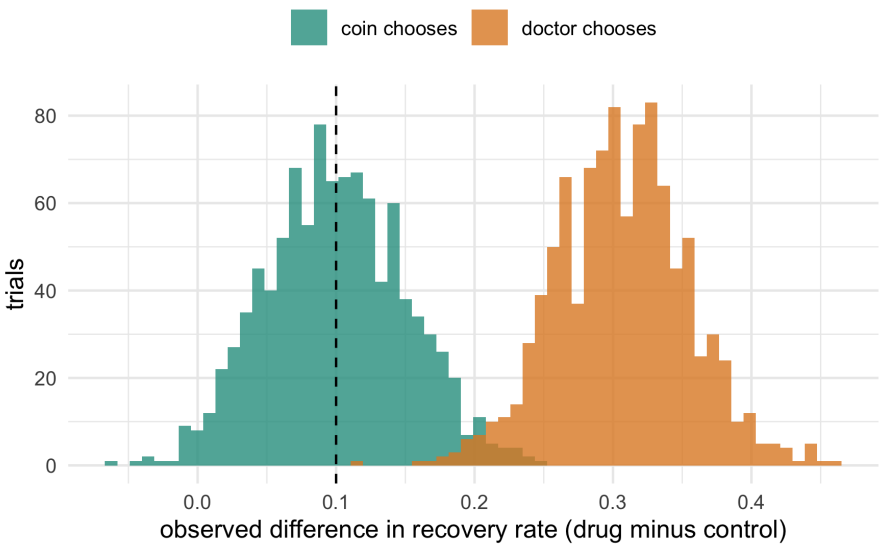


Figure 5.1.: Estimated effect of a drug that truly adds 10 percentage points to recovery, in 1,000 simulated trials of 400 patients. When a doctor chooses who gets the drug (orange) the healthier patients end up on it and the estimates are centred far above the truth. When a coin chooses (teal) they are centred on the truth. The dashed line marks the true effect.

The doctor's trials do not scatter around the truth. They scatter around roughly three times the truth (the mean estimate is 0.31), and in 100 percent of the 1,000 trials the drug looks better than it is. The reason is in the health gap: the patients given the drug were 1.06 standard deviations healthier than the patients who were not, and nothing in the recorded outcome tells you so. The standard error of the estimate is about the same in the two schemes (0.048 for the doctor, 0.05 for the coin), so a larger trial would not help. It would only make the wrong answer more convincing.

A natural reaction is that the analyst should adjust for health. That works when health has been measured well. Suppose we do record h and compare treated and untreated patients within five bands of h. Then the estimate of the doctor-chosen trials, in the chunk below, falls from about 0.31 to about 0.12, much closer to the truth, but still not equal to it, because patients within a band still differ and the doctor's choice depends on h itself.

```
set.seed(8)
r <- t(replicate(300, sim_adjusted_vs_naive(2000)))
knitr::kable(data.frame(Estimate = c("Truth", "Naive", "Stratified on
 ↪  health"), Value = c(0.100, mean(r[, "naive"]), mean(r[, "adjusted"]))),
 ↪  digits = 3, align = "lr")
```

Estimate	Value
Truth	0.100
Naive	0.306
Stratified on health	0.117

Estimate	Value

And in real data the doctor's impression is rarely a clean recorded number. The coin needs no such measurement. It balances health as recorded, health as sensed and health that nobody has thought of, which is the content of the sentence in ICH E9 about prognostic factors "known and unknown".

5.2. What goes wrong without it

The simulation is a cartoon. The history of medicine supplies the real thing, in four recurring forms.

Historical controls. The new treatment is given to patients now and compared with patients treated in the past. Patients change over time: referral patterns, supportive care, diagnostic criteria and even the way outcomes are recorded move. Sacks, Chalmers and Smith reviewed studies of six therapies, each of which had been studied with both randomized and historical controls, and found that the historically controlled studies were much more likely to report the new therapy as better than the randomized ones (Sacks, Chalmers, and Smith 1982). The pattern, that the nonrandomized comparison finds benefit more often than the randomized one, is what Equation 5.1 predicts when the investigator's access to patients and the trend over time both favor the new treatment. FDA's regulation recognizes the problem: 21 CFR 314.126(b)(2)(v) lists the historical (external) control as a type of control that is considered, and the Substantial Evidence draft guidance (June 2026) notes that differences between groups are a particular concern "because of the lack of randomized treatment assignment". We return to when external controls are nonetheless useful, and how to borrow from them, in Chapter 19.

Alternation and other predictable schemes. Giving every other patient the new drug looks like a fair procedure. It is not, because the next assignment is known to the person who decides whether the next patient is eligible. A clinician who knows that the next patient will receive the experimental drug can steer a frail patient away (or toward it) by deciding that the patient is not quite eligible, or by delaying enrollment by a day. The scheme is not random, and it is not concealed. The Medical Research Council's 1948 streptomycin trial solved this problem with concealed allocation (see "In practice" below).

Unconcealed allocation. Even a perfectly random list is useless if the enrolling clinician can read it. Schulz and colleagues examined 250 controlled trials from 33 meta-analyses in pregnancy and childbirth and found that trials with inadequately concealed allocation reported odds ratios that were exaggerated, on average, by about 41 percent, and trials whose concealment was unclearly reported by about 30 percent, compared with trials with adequate concealment (Schulz et al. 1995). These are averages over one body of trials, an association and not a randomized comparison of concealment methods. A separate paper from the same group describes the many ways clinicians have been seen to subvert allocation schedules: holding envelopes up to bright lights and opening several envelopes at once, among others (Schulz 1995). These are results about the process by which patients are assigned, not about the mathematics of randomness. We return to concealment in a later section.

Self-selection by patients. If patients decide which arm to join, the decision is again related to prognosis. A trial in which sicker patients seek the experimental treatment, or

healthier, more motivated patients adhere to the offered regimen, compares people who differ in more than the treatment. The Women's Health Initiative is the instructive example, because its randomized result reversed an observational finding about hormone therapy (see "In practice" below).

The first lesson of this section is that these failures are failures of *assignment*, not of analysis. No statistical adjustment applied afterward can repair an assignment mechanism that depended on unrecorded information. The second is that randomization is only as good as its concealment: a random list that the investigator can see is a way of choosing patients with extra steps. The two together are the content of the phrase "adequate and well-controlled".

5.3. Methods of randomization

All the methods below assign each patient to treatment or control with some chance. They differ in how they use constraints to avoid unlucky draws. In this section we assume two arms with 1:1 allocation. Rosenberger and Lachin give a full treatment of the theory (Rosenberger and Lachin 2016); ICH E9 (section 2.3.2) advises that the block size be kept unknown to the investigators and that the randomization schedule be filed securely by the sponsor; the specification should be written so that the schedule is reproducible.

5.3.1. Simple randomization

The simplest method assigns each patient by a fair coin, independently of every other patient. Its strength is that it is unpredictable: knowing all previous assignments tells you nothing about the next one. Its weakness is that, in a small trial, chance may give unequal arm sizes and unequal covariate distributions. With $n = 40$ patients, the number in the treatment arm has standard deviation $\sqrt{40}/2 \approx 3.2$, so a split of 15 and 25 is not rare. The loss is not catastrophic (it costs a little efficiency), but the covariate imbalance is more serious, as we will see.

5.3.2. Permuted block randomization

Permuted block randomization fixes the arm sizes at intervals. Patients are grouped in consecutive blocks of, say, four. In each block, two get the drug and two get control, in a random order (there are six possible orders, each equally likely). After every block the arms are exactly balanced in size, and at any moment the imbalance cannot exceed half a block. This also protects the trial against a time trend: if the standard of care slowly improves, both arms have patients from each period. (Zelen (Zelen 1974) reviews stratified permuted blocks; Efron's biased coin (Efron 1971) and Wei's adaptive version (Wei 1978) are early alternatives.)

The price is predictability. If the block length is fixed at four and the investigator knows it, then after the first three patients of a block have been assigned, the fourth assignment is always determined. Suppose the first three patients were assigned drug, drug, control. The fourth must be control. A clinician who can see past assignments (open-label trials, or trials with unmasked packaging) can use this to select patients. The standard remedy is to vary

the block length at random (for example, blocks of 4 and 6, drawn at random) and to keep the block sizes out of the protocol text and in the unblinded randomization plan.

5.3.3. Stratified randomization

If a prognostic factor is known to matter, such as disease stage, prior treatment or study center, we can balance it by design. We run a separate permuted-block list inside each *stratum*, the combination of levels of the factors. Every stratum then has near-equal arm sizes. ICH E9 (section 5.7) asks that the factors used to stratify the randomization be accounted for in the analysis. Stratification has a ceiling: with k binary factors there are 2^k strata, and each stratum has its own incomplete block at the end of enrollment. With 20 patients percenter and four factors, most blocks are unfinished and the stratification has bought little. A good rule is to stratify on few factors (two or three) that are strongly prognostic, and to analyze the primary endpoint adjusting for those same factors, a point we come back to below. Kernan and colleagues review the method and its practical limits (Kernan et al. 1999).

5.3.4. Minimization

Minimization is the method for small trials with many prognostic factors. Taves (Taves 1974) and, independently, Pocock and Simon (Pocock and Simon 1975) proposed assigning each new patient in a way that depends on what has happened so far: look at the patient's levels of each factor, count how many earlier patients with those same levels are in each arm, and assign the new patient to the arm that makes the total imbalance smaller. In its deterministic form it is not random at all, and the next assignment can be predicted by anyone who knows the current counts. Pocock and Simon therefore proposed to assign to the preferred arm with a probability p larger than a half (0.8 is a common choice), and to a fair coin when the arms tie. The package function `assign_minimization` implements exactly this. The Pocock-Simon method balances the *margins* of each factor (the number of women in each arm, the number of patients over 65 in each arm) but not necessarily the combinations. Reviews of the method (N. W. Scott et al. 2002) stress that the analysis should include the minimization factors as covariates, because in a minimized trial the assignment is not independent of those factors.

5.3.5. Seeing the difference

Each method has a different effect on how balanced the arms are. We simulate trials of 40 and 200 patients with three binary traits: two known and used in stratification and minimization (for example, sex and age above 65, with the second being less common) and one unmeasured (a hidden trait with prevalence 0.3 that nobody recorded). For each method we compute the absolute difference between arms in the proportion of patients who have the second known trait, and in the proportion who have the hidden trait.

Three things stand out. First, at $n = 40$ the average gap in the known trait under simple randomization is about 0.12, and in one trial in twenty it is larger than about 0.3. That is imbalance big enough to matter if the trait is prognostic. At $n = 200$ it falls by a factor of $\sqrt{5}$, to about 0.05. Second, stratified blocks and minimization cut the imbalance in the known trait to a fraction of that (at $n = 40$, 0.04 for stratified blocks and 0.04 for minimization; at

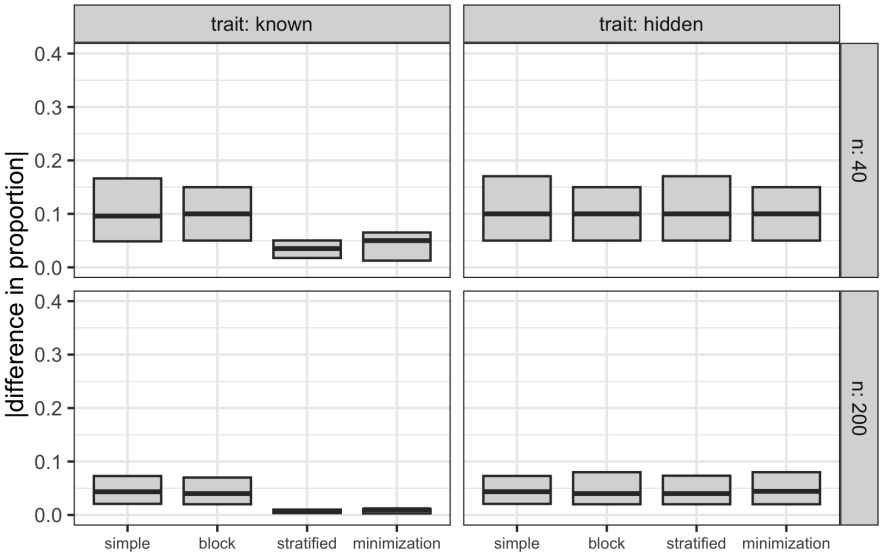


Figure 5.2.: Imbalance between arms in the proportion with a binary trait, 1,000 simulated trials per method. Left: the trait is known and used by the method (stratified blocks of 4, or Pocock-Simon minimization). Right: the trait is hidden and nobody balances it. Stratification and minimization remove the imbalance in the known trait and do nothing for the hidden one, which is as imbalanced as under simple randomization.

$n = 200$, 0.01 and 0.01). Third, nothing helps the hidden trait: its imbalance is the same under all four methods, as large as under simple randomization. Randomization does not guarantee balance in any single trial. What it guarantees is that the imbalance is due to chance, with a known distribution, and that this distribution is what the confidence interval and the p-value account for. This is why a randomized trial can be analyzed honestly without balance, and a confounded one cannot.

i Technical note: why blocks do nothing for covariate imbalance

A permuted block makes the *number* of patients in each arm equal. It says nothing about which patients. For a covariate $x_i \in \{0, 1\}$ unrelated to the order of enrollment, the assignment within a block is a random permutation, so each patient's arm is still a fair coin independent of x_i and the covariate imbalance has the same distribution as under simple randomization apart from a small finite-population correction. The simulation shows the effect: the "block" boxplots match the "simple" ones. Blocks help with arm size and with time trends. For covariates, stratification inside blocks, or minimization, is needed.

5.3.6. How the schedule is actually made

In a trial that uses an interactive response technology (IRT, formerly IVRS and IWRS), the investigator does not see a list. The site enters the patient's stratification factors into a web

or phone system, and the system returns a kit or bottle number. Behind it is a randomization schedule produced by an unblinded statistician (or the vendor) before the trial starts, from a pseudo-random number generator and a stored seed. The schedule is a table: stratum, sequence number within the stratum, treatment, kit number. It is generated once, checked for balance and validated, and then locked. The protocol should state the allocation method, the strata and the access controls, so that site staff, monitors and the blinded statistical team cannot see the schedule. A separate restricted randomization specification should record the block sizes, generator settings and seed, so that the schedule can be reproduced by an auditor (ICH E9 advises keeping details that would let assignments be predicted, block length in particular, out of the protocol).

```
strata <- rep(c("stage II", "stage III", "stage IV"), each = 30)
arm <- assign_stratified(strata, sizes = c(4, 6), seed = 20261108)     #
 ↪  seed recorded in the randomization specification
sched <- data.frame(stratum = strata, seq = ave(arm, strata, FUN =
 ↪  seq_along),
                    arm = ifelse(arm == 1, "drug", "control"))
set.seed(1); sched$kit <- sample(1001:(1000 + nrow(sched)))           # kit
 ↪  number reveals nothing
knitr::kable(head(sched, 8), col.names = c("Stratum", "Sequence in
 ↪  stratum", "Arm", "Kit number"), align = "lrlr")
```

Stratum	Sequence in stratum	Arm	Kit number
stage II	1	control	1068
stage II	2	control	1039
stage II	3	drug	1001
stage II	4	drug	1034
stage II	5	control	1043
stage II	6	drug	1014
stage II	7	control	1082
stage II	8	drug	1059

Two properties matter. The schedule is *reproducible* from the seed, which lets an auditor confirm that nobody altered it. And the site staff see only the kit number the system returns after the patient is irreversibly enrolled, which is what allocation concealment means in practice and which we turn to next.

5.4. Allocation concealment

Randomization has two parts that are easy to confuse. *Sequence generation* produces the unpredictable list. *Allocation concealment* keeps that list hidden from the people who decide whether, and when, a patient enters the trial, until the moment the patient is irreversibly enrolled. Concealment is about the time before the assignment. Blinding, which we cover next, is about the time after it. A trial can be concealed and open-label (the patient is randomized by a central system and then sees that they are getting an injection), and in this case the bias of the investigator in selecting patients is prevented but the bias in treatment

and assessment is not. A trial cannot be properly blinded without concealment, because an investigator who knows the next assignment already knows the arm.

The practical means are central randomization by IRT, sequentially numbered opaque sealed envelopes (opened only after the patient's name is written on them and the patient is enrolled), and pharmacy-controlled coded packs. Of these, central telephone or web randomization is the strongest, because it cannot be opened in advance, and is the usual choice in pivotal trials. The evidence summarized earlier (Schulz et al. 1995) shows why: concealment is a property of the process and is something that can be assessed. CONSORT reporting (and the SPIRIT 2025 protocol statement (Chan et al. 2025)) asks authors to say how it was done. Altman and Schulz summarize why and how to conceal allocation (Altman and Schulz 2001).

In the protocol

A concealment paragraph a reviewer can check: "Patients will be randomized centrally through the IRT system after eligibility has been confirmed and informed consent signed. The system returns a kit number only after the patient has been entered. The randomization schedule, block sizes and seed are held by the unblinded statistician at the vendor and are not available to the sponsor's blinded team, to site staff or to monitors. Block sizes vary at random. Any request to unblind goes through the emergency unblinding procedure of the IRT and is logged."

5.5. Randomization inference: the test the design justifies

Chapter 4 introduced the permutation test: take the observed outcomes, shuffle the group labels many times, and ask how often a shuffled difference is as large as the observed one. At that point the test was presented as a device. Here we can say what it is for. The randomization test is *justified by the design*. When treatment was assigned by a known random mechanism, then under the hypothesis that the drug has no effect on any patient (the sharp null: $Y_i(1) = Y_i(0)$ for every i), each patient's outcome would have been the same whatever arm they were assigned to, and the only thing that varied from one possible trial to another was the assignment. The permutation distribution is therefore the exact distribution of the statistic over the possible assignments, computed under the same randomization procedure that was used. It requires no model for the outcome (no normality, no equal variance), and the type I error is controlled by construction. Fisher put this forward in 1935 as the logical basis of significance testing in experiments (Fisher 1935), and the normal-theory z and t tests of Chapter 4 are approximations to it.

The phrase "under the same randomization procedure" is important. Shuffling labels freely is correct if the original assignment was by simple randomization. If it was by permuted blocks, the shuffle should respect the blocks: the possible assignments are only those with two drug and two control in each block. A test that ignores this treats as possible many assignments that the design excluded. We can see what that costs under a time trend. We simulate a null trial of 40 patients, where the drug does nothing but the average outcome drifts upward during enrollment by two standard deviations (a plausible effect if the standard of care changes or the mix of referring centers evolves). We analyze each trial by the one-sided permutation test with a nominal level of 0.025, once unrestricted and once restricted to the blocks.

Table 5.3.: Empirical type I error (one-sided, nominal 0.025) of the permutation test under a time trend and no treatment effect, 2,000 simulated trials of 40 patients per row, 500 permutations each. Monte Carlo standard error is about 0.003.

assignment used	unrestricted shuffle	shuffle within blocks
simple	0.025	NA
block	0.011	0.022

With simple randomization the unrestricted shuffle is the right test, and its error rate (0.025) is close to nominal. With blocks, the unrestricted shuffle treats as possible many assignments the design excluded: with this drift its error rate is 0.011, well below the nominal 0.025 (the Monte Carlo standard error is about 0.003), so the test is conservative and gives away power. The shuffle within blocks gives 0.022, close to nominal; it is the exact test for the design, up to the discreteness of a 40-patient permutation distribution and Monte Carlo error. The lesson for the statistical analysis plan is that the analysis should mirror the randomization, and that factors used in stratification should be in the model for the primary analysis. ICH E9 makes the same point for covariates: the analysis should reflect the stratification used in the design.

The randomization test also answers a question that the normal-theory tests answer only by assumption: *what population does the p-value refer to?* The patients in a trial are not a random sample from anything. They are the people who consented and were eligible at the participating sites. Randomization inference does not need them to be a sample. It asks only about the patients actually enrolled, and says whether the difference between the arms is larger than the randomization alone would produce. Generalizing to future patients is a separate step that rests on clinical judgment, and it is a step that the p-value does not take.

5.6. What randomness is

So far "random" has meant "by a coin". For a protocol that is enough. But a trialist who asks what makes a schedule random, and whether a computer generates it properly, is asking a real question. There are two answers, one from information theory and one from geometry, and a practical consequence.

5.6.1. Randomness as incompressibility

Look at two strings of 0s and 1s of the same length. One is 0101010101… repeated. The other was obtained by tossing a coin. We feel the first is not random, because it can be described in a few words: "repeat 01". The second cannot be described more briefly than by writing it out. Kolmogorov turned this feeling into a definition (Kolmogorov 1965): a string is random if it cannot be described by anything much shorter than the string itself (its *complexity* is close to its length).

Two facts follow that matter for trials. First, randomness in this sense is a property of a *sequence*, not of a procedure, and no particular finite sequence can be certified random: Kolmogorov complexity cannot be computed. The best a trialist can do is to apply tests.

Second, a sequence produced by a short program (a pseudo-random number generator started from a seed) has low complexity by definition, because the program and the seed describe it. That is not a flaw for a clinical trial, because what matters is that nobody can predict the sequence without the seed. It is why the seed belongs in the sealed randomization specification and is a reason to use a generator of good statistical quality.

Technical note: complexity, and a compression experiment

Formally, the complexity of a string is the length of the shortest computer program that outputs it. This definition concerns finite strings and program length without restriction. Martin-Löf defined randomness for infinite sequences through effective statistical tests (Martin-Löf 1966), and the equivalence with incompressibility holds for infinite sequences whose prefixes all have high prefix-free complexity, not for the finite-string definition above; Li and Vitányi give the precise theorem (Li and Vitányi 2019).

A crude illustration uses an everyday compressor in place of the shortest program. A compressor finds only some patterns, so its output length is an upper bound on the complexity.

```
set.seed(7)
coin <- rbinom(2000, 1, 0.5)                                    # fair
 ↪  coin
alternating <- rep(0:1, 1000)                                   #
 ↪  alternation: drug, control, drug, ...
formula_bits <- rand_bits_lcg(2000, seed = 1, a = 75, m = 65537)   # a
 ↪  one-line formula, no obvious pattern
anu <- as.integer(strsplit(trimws(readLines("data/anuqrng.txt", warn =
 ↪  FALSE)[1]), "")[[1]])[1:2000]  # ANU quantum generator
cr <- sapply(list(coin = coin, alternation = alternating, formula =
 ↪  formula_bits, ANU_quantum = anu), compress_ratio)
knitr::kable(data.frame(Sequence = names(cr), `Compressed size /
 ↪  original size` = unname(cr), check.names = FALSE), digits = 3,
 ↪  align = "lr")
```

Sequence	Compressed size / original size
coin	0.176
alternation	0.012
formula	0.182
ANU_quantum	0.179

The coin sequence compresses to about 0.18 of its size, which is as far as it will go (the floor is 0.125, since each stored character carries one bit and the rest is overhead). The alternation compresses to about 1 percent. The bits of a one-line multiplicative generator ($x_{k+1} = 75\,x_k \bmod 65537$) are also not compressed, although a very short program generates them: the compressor does not see the structure, and this is a small demonstration that *passing a compression test is not a proof of randomness*. The last entry is the first 2,000 bits of a string from the ANU quantum random number generator, which is physically random and compresses as the coin does; a compressor cannot tell it from the one-line formula. A sequence that looks random to a particular test may have a short generator. For clinical use the requirement is therefore practical: a generator with documented statistical quality, a recorded seed, and a schedule that the people who enroll patients cannot see.

5.6.2. Arnold's measure of randomness

V. I. Arnold proposed a different, geometric, way to judge how random a set of points on a circle looks (Arnol'd 2006). His measure asks whether the points are spread the way independent draws would be spread: evenly spaced points are too regular to be random, and clumped points are too attracted to one another. Applied to a sequence of integers modulo a number, it flags the overly regular ones. For allocation the consequence is practical. A schedule generated by a linear congruential generator with a short period or poor lattice structure fails such tests, and this is a reason to use a modern generator such as the Mersenne Twister (the R default) or PCG64 with a long period, and not to write one's own.

i Technical note: Arnold's stochasticity parameter

Arnold's measure takes T points on a circle of length L (Arnol'd 2006). Sort them and let $x_1, \dots, x_T$ be the gaps between neighbours around the circle, so that $\sum x_i = L$. His *stochasticity parameter* is the normalized sum of squared gaps,

$$s \;=\; T \sum_{i=1}^{T} \left(\frac{x_i}{L} \right)^2, \qquad 1 \le s \le T. \tag{5.2}$$

If the points are evenly spaced ("barracks-like"), $s = 1$, which Arnold calls strong repulsion. If all of them are clustered in one place, $s = T$, strong attraction. For T points placed independently and uniformly at random, s has expected value $s_1 = 2T/(T+1)$, close to 2 for large T, which he calls the "freedom-loving" value: neither repulsion nor attraction. A sequence of T integers modulo L is thus judged by where its s falls relative to s_1. Values close to 1 mean the points are too evenly spread to be random, values much above s_1 mean they clump.

Here is the example in R: 15 residues modulo 100 from the formula 3^i, from the formula $37i$, and from the random number generator, and then the distribution of s over many random sequences.

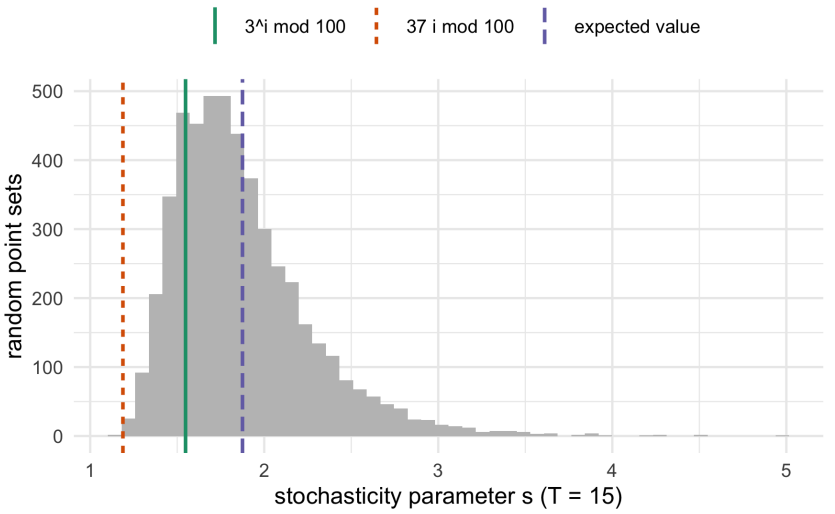


Figure 5.3.: Arnold's stochasticity parameter s for 15 points on a circle of length 10^6. The histogram shows s for 5,000 independent uniformly random point sets. The dashed line is the expected value $2T/(T+1) = 1.875$, the value 1 is a perfectly regular arrangement. The two deterministic sequences of the text are marked on a circle of length 100.

The arithmetic progression $37i$ mod 100 sits at about 1.19, near the regular-polygon floor of 1, which is unmistakably too regular. The sequence 3^i mod 100, a power sequence that looks more complicated, is at about 1.55, which is on the regular side of the random range but inside the central 95% interval of random sets, and so Arnold's measure on 15 points does not reject it. The value for a draw from R's generator (1.72) is in the middle of the distribution. The measure is a diagnostic and not a theorem. With $T = 15$ the random distribution is wide, and a single sequence falling in the interval proves nothing. Its attraction is that it tests the one thing that matters for allocation: whether the points are spread the way independent draws would spread. It is a cousin of the spacing tests (and Arnold's own motivation was the orbits of a^t mod n, used in some congruential generators).

Randomness is a standard for the *process*, and the protocol's job is to document it: a well-tested generator, a seed that is recorded and then sealed, and a schedule checked for balance before it is locked.

5.7. Blinding

Randomization protects the comparison against bias in *who* receives each treatment. *Blinding*, also called masking, protects it against bias in *what happens afterward*: how patients report symptoms, how investigators care for them and how outcomes are judged. ICH E9 (section 2.3.1) defines a double-blind trial as one in which neither the subject nor any of the investigator or sponsor staff involved in the treatment or clinical evaluation of the subjects are aware of the treatment received. The common levels are these.

Table 5.5.: Levels of blinding.

Level	Who is kept unaware	What it protects against	Limitation
Open label	No one	Nothing beyond randomization	Performance and assessment bias possible
Single blind	Participants	Participant expectation effects	Investigators may treat or assess differently
Double blind	Participants and investigators	Expectation and assessment bias	Side effects may reveal the arm
Triple blind	Also the data analysts	Analytic bias	Maximum protection, harder to run
Blinded outcome assessment	Outcome assessors only	Assessment bias when other blinding is impossible	Covers specific outcomes only

The first row deserves a fair word. An open-label trial can be informative when the outcome is objective. RECOVERY, which established dexamethasone in hospitalized COVID-19 patients, compared it with usual care without blinding; the primary outcome was death within 28 days, an objective endpoint that limits assessment bias, and randomization was concealed and central (RECOVERY Collaborative Group et al. 2021) (see "In practice" below). Blinding matters most when outcomes are subjective (pain, depression, symptom scores, many adjudicated events) and when patients or investigators can change what they do. It is not always possible: surgery, devices and behavioral interventions often cannot be masked, and then the protocol should use blinded outcome assessment, objective endpoints and standard rules for co-interventions.

Maintaining the blind takes work. Placebos must match in taste, smell and appearance. Side effects can unblind (a drug that causes nausea, a rash or a lab change that is visible to the clinician). The usual precautions are a double-dummy design when two active comparators differ in form, independent unblinded staff who handle the dosing adjustments, and a pre-specified procedure for emergency unblinding. A trial with a data monitoring committee (Chapter 11) has an additional layer: the DMC and its supporting statistician see unblinded data, while the sponsor and the investigators do not, and the charter should state who sees what and when. Chapter 23 returns to these firewalls, which matter even more for adaptive designs, where the decision to change something depends on data.

5.8. Intention to treat, and what the trial estimates

Randomization is performed at one moment. After it, things happen: patients stop the drug, take rescue medication, move away, die of something else. If the analysis then drops these patients or moves them to another arm, the arms are no longer the groups that the coin created.

5.8.1. The ITT principle

The *intention-to-treat* (ITT) principle says to analyze every patient in the arm they were randomized to, whatever happened afterward. ICH E9 (section 5.2.1) puts it this way: the primary analysis should include all randomized subjects, and "preservation of the initial randomisation in analysis is important in preventing bias". A patient assigned to the drug who never took a pill is still analyzed as a drug patient. That seems odd the first time you meet it, and the following simulation shows why it is right. The drug adds 10 percentage points to recovery for anyone who takes it. In the drug arm, about 15 percent of patients stop the drug early, and the patients who stop are more likely to be the ones who feel worse (low hidden health). We compare three analyses on the same 4,000 patients in each of 300 trials: ITT (everyone as randomized), per-protocol (drop the patients who stopped), and as-treated (put stoppers with the controls).

Table 5.6.: Average estimate over 300 trials of 4,000 patients, in percentage points of recovery. Truth for a patient who takes the drug: +10. Stopping is more likely in sicker patients, and about 15% of the drug arm stops on average.

	analysis	estimate_points
itt	ITT (all as randomized)	8.4
pp	per-protocol (drop stoppers)	12.6
at	as-treated (stoppers to control)	14.7

ITT is diluted: it estimates the effect of *assigning* the drug, which is smaller than the effect on those who take it because about one in seven did not. The per-protocol and as-treated analyses go to the other side. They remove or reshuffle exactly the patients who differ in prognosis, so the groups are no longer comparable, and the estimates exceed what the drug does. Neither bias can be seen in the data of a single trial. It is the randomization that is being thrown away.

5.8.2. Estimands and intercurrent events

ITT answers one question well: what happens if we offer this treatment to patients like these, as people actually use it? Sometimes that is the right question, and sometimes it is not. For a regulator, a clinician and a patient, "the effect of the drug in patients who take it" is a different quantity from "the effect of prescribing it". The ICH E9(R1) addendum (Step 4, November 2019) made the choice explicit. It asks the sponsor to define an *estimand*, "a precise description of the treatment effect reflecting the clinical question posed by a given clinical trial objective", before the trial, with five attributes: the population, the treatment conditions being compared, the variable (endpoint), the population-level summary, and the way *intercurrent events* are handled. Intercurrent events are events after treatment initiation that affect either the interpretation or the existence of the measurements: stopping the drug, taking rescue medication, switching treatment, death before the endpoint.

The addendum names five strategies for an intercurrent event. We illustrate with the drug-discontinuation event of our simulation.

Table 5.7.: The five ICH E9(R1) strategies for intercurrent events.

Strategy	Question answered	How the event is handled	Randomization intact?
Treatment policy	Effect of being assigned the drug, with whatever follows	Use the outcome regardless of the event (the ITT spirit)	Yes
Hypothetical	Effect if the event had not happened	Model what would have been observed	Relies on a model
Composite variable	Effect on a combined outcome	Include the event in the endpoint (stopping counts as failure)	Yes
While on treatment	Effect up to the event	Use the outcome until the event	Usually partly
Principal stratum	Effect in patients who would not have the event	Condition on a latent subgroup	Relies on assumptions

The treatment-policy and composite strategies are the ones that fit the randomization directly, because they compare the groups the coin made. The hypothetical and principal-stratum strategies ask a more targeted question and must pay for it by modeling or assuming something about patients who stopped. The per-protocol analysis above is not one of the five strategies at all: it drops patients on the basis of a post-randomization event, and without a model of why they stopped, it estimates nothing well defined. One consequence is that a trial with substantial discontinuation can give a perfectly valid answer to the treatment-policy question and a doubtful one to the hypothetical question. A protocol that names the estimand and says why each intercurrent event strategy was chosen also tells the reviewer what the analysis can and cannot show, and the choice of the strategy has to match the label the sponsor hopes to obtain.

i Technical note: how a Bayesian analysis respects the estimand

Estimands are about what is estimated, not about how. This note uses the terms prior and posterior, which Chapter 6 defines. A frequentist and a Bayesian analysis answer the same estimand differently only in the machinery. For the treatment-policy estimand, the Bayesian model is fit to all randomized patients by arm, and the posterior of the difference θ in arm means is the posterior of the estimand. Nothing about the intercurrent event enters, other than through the outcome as it was observed. This requires the outcome to remain defined after the event; terminal events such as death may require another strategy. For the hypothetical strategy, the model must include a submodel for what the outcome would have been without the event, with priors on the unidentifiable part. That prior is part of the analysis, and it deserves a sensitivity analysis, because the data cannot correct it. The ICH E9(R1) addendum requires such sensitivity analyses of the frequentist analysis anyway, and the Bayesian formulation makes them explicit: change the prior on the missing-data mechanism and report how the posterior probability of benefit moves. The decision rule (for instance

$\Pr(\delta > a \mid \text{data}) > c$ from the FDA draft guidance on Bayesian methodology, January 2026), which requires the posterior probability that the effect exceeds a minimum relevant value a to be above a threshold c, applies to the posterior of the *stated* estimand. A posterior probability computed for one estimand and quoted for another is the same error as quoting an ITT p-value for a per-protocol claim. Chapter 7 covers the decision rule itself.

5.9. Why a Bayesian still randomizes

This section uses the words prior, likelihood and posterior, which Chapter 6 defines (the prior is what is believed about the effect before the trial, the likelihood is how probable the observed data are for each possible value of the effect, and the posterior is the belief after seeing the data). A reader meeting them for the first time can return here after that chapter.

A natural question for the rest of the book is whether randomization is a frequentist habit. The likelihood principle says that inference should depend on the data only through the likelihood, and the likelihood for a given dataset does not change with the method that allocated the patients. If so, why not let the doctor choose, which is cheaper and perhaps kinder to patients? The question was debated seriously in the 1970s and 1980s, and the answer is: because the Bayesian wants the assignment to be *ignorable*, and randomization is the only way to make it so without assumptions.

5.9.1. Ignorability of the assignment mechanism

Let d_i be the arm of patient i, x_i the covariates recorded, and y_i the outcome. The model is a joint distribution for the outcomes, governed by parameters θ of interest, and for the assignment, governed by some mechanism ϕ. Rubin (Rubin 1978) showed that the posterior for θ can be computed from the outcome model alone, ignoring how patients were assigned, if the assignment depends only on recorded data (it is "unconfounded", and strictly it needs also that θ and ϕ are independent a priori). Then the likelihood factorizes as

$$p(d, y \mid x, \theta, \phi) \;=\; \underbrace{p(d \mid x, \phi)}_{\text{assignment}} \times p(y \mid d, x, \theta), \tag{5.3}$$

and the first factor drops out of the posterior of θ. If the assignment depends on something that was not recorded, such as the doctor's impression of the patient's strength, then ϕ is tied to the outcome through the unrecorded quantity, the factorization fails, and the posterior for θ can be badly biased whatever the prior. Randomization makes the first factor a known constant, because the coin ignores everything. It makes the assignment ignorable for free.

The condition "depends only on recorded data" is the Bayesian counterpart of the frequentist selection term of Equation 5.1. It is also why a Bayesian trial can use covariate-dependent assignments such as minimization without changing the likelihood. Response-adaptive randomization needs one more step, because the assignment of patient i depends on the outcomes of earlier patients. Let H_i be the recorded history (earlier arms, covariates and outcomes). The sequential likelihood is $\prod_i q_i(d_i \mid H_i)\, p_\theta(y_i \mid d_i, x_i, H_i)$, and when the assignment probabilities q_i are known and do not depend on θ except through the recorded

history, they can be omitted from the posterior for θ. The posterior is valid in this sense, and the operating-characteristic issues of that method are separate ones, treated in Chapter 16 and Chapter 14.

5.9.2. The Bayesian arguments for randomizing

Three arguments in the literature are worth knowing, because they show that the Bayesian case for randomization does not rest on the fragile likelihood-principle position.

The first is Rubin's, just given: randomization is a way to make ignorability true by construction, so the analysis needs no model of the selection process and no assumptions that the unrecorded drivers of selection have been captured (Rubin 1978).

The second is the *robustness* argument of Kadane and Seidenfeld (Kadane and Seidenfeld 1990). For a Bayesian with a single, fully specified model, randomization is not needed; the optimal design is whichever one best informs about θ. But no trialist has a model that all other readers, in particular a regulator and a skeptical clinician, accept. Randomization yields an analysis whose validity does not depend on the particular prior beliefs of any one person about the unmeasured determinants of the outcome: any two Bayesians who agree that the assignment was by a coin and the data are as recorded agree that the assignment is ignorable. In their analysis the case for randomization is that it protects a decision-maker against a model that is wrong in a way that she cannot anticipate, and that it makes the result persuasive to others.

The third is the practical position of Berry, whose review of Bayesian clinical trials (D. A. Berry 2006) treats randomization as compatible with, and in adaptive designs a component of, the Bayesian approach: the allocation probabilities may depend on the accrued data, but chance remains in the assignment, so the selection effects that would need modeling are absent. Randomizing with probabilities that depend on accrued data is the subject of Chapter 16. Within all of these, the statement is the same: the Bayesian randomizes for the sake of credibility and robustness, and the reasons are not different from the reasons of a frequentist.

> **i** Technical note: what changes and what does not
>
> A Bayesian conditions on the data actually observed, so the permutation distribution (a statement about assignments that did not happen) does not appear in the posterior. What randomization buys a Bayesian is the factorization in Equation 5.3 with a known assignment factor, not a sampling distribution. It also does the job without covariates: the posterior is computed from the outcome model and the covariates one chooses to include, and unmeasured prognostic factors are balanced in distribution across arms, so omitting them does not bias the marginal treatment effect under an appropriate outcome model, and it generally costs precision (see Section 5.10, where noncollapsibility shows that adjusted and unadjusted coefficients can target different quantities). Under confounding, omitting them biases the posterior, and more data concentrate the posterior around the wrong value. The simulation of Figure 5.1 is a Bayesian failure as well as a frequentist one: the posterior for the doctor-chosen trial concentrates near 0.31, the naive estimate of the doctor-chosen simulation.

5.10. Covariate adjustment

Randomization makes the unadjusted comparison of the arms unbiased: the arms differ, on average, only by the treatment. It does not make that comparison efficient. Two arms of 100 patients will differ by chance in age, disease severity and the baseline value of the outcome, and that chance imbalance is part of the noise in the estimate. A baseline covariate that predicts the outcome explains some of that noise, and an analysis that uses it gives a narrower interval and more power from the same patients. This is the cheapest source of precision a trial has. It needs no prior, no interim look and no extra patients, and regulators on both sides of the Atlantic have written guidance on it. The rest of the section follows the order of this book: the traditional analysis, its regulatory basis, the pitfalls in nonlinear models, robust estimators, and then the Bayesian version.

5.10.1. Continuous outcomes: ANCOVA

For a continuous outcome with a prognostic baseline covariate, the standard adjusted analysis is analysis of covariance (ANCOVA): regress the outcome on treatment and the covariate, and read the treatment coefficient. If the covariate explains a fraction R^2 of the outcome variance within arms, the variance of the estimated treatment effect is multiplied by about $1 - R^2$, and so is the sample size needed for a given power. The gain is the same as a reduction in the outcome standard deviation by the factor $\sqrt{1 - R^2}$.

Take a trial that wants 90 percent power at a two-sided 5 percent level for a difference of 0.4 standard deviations. The unadjusted analysis needs 132 patients per arm. If the baseline value of the outcome explains 30 percent of the variance (a correlation of about 0.55, typical of symptom scores measured a few weeks apart), the adjusted analysis needs about 93 per arm, a saving of 30 percent, or 78 patients in total. The chunk below is the whole calculation.

```
n0 <- ceiling(2 * (qnorm(0.975) + qnorm(0.90))^2 / 0.4^2)  # per arm,
 ↪  unadjusted
n1 <- covadj_ancova_gain(r2 = 0.30, n_unadjusted = n0)$n_adjusted  # per
 ↪  arm, ANCOVA
```

The gain has a limit that the sample-size formula hides. It applies to the covariate-adjusted analysis that is planned: the sponsor has to commit to the covariate in the protocol and the sample size has to be computed under it. FDA's guidance says that power and sample size calculations can be based on adjusted or unadjusted methods, and that the latter will often be more conservative (U.S. Food and Drug Administration 2023d). A conservative sample size with an adjusted analysis is a legitimate and common choice: the trial is then overpowered if the covariate behaves as expected, and protected if it does not.

5.10.2. Change from baseline or ANCOVA

When the outcome is measured at baseline and again at follow-up, there are three common analyses of the same data: the final value alone, the change from baseline, and the final value adjusted for the baseline (ANCOVA). With equal variances and a baseline-to-follow-up

correlation ρ, the variance of the estimated treatment effect, relative to the final-value analysis, is 1, $2(1-\rho)$ and $1-\rho^2$ respectively.

Table 5.8.: Variance of the estimated treatment effect relative to the final-value analysis, by baseline-to-follow-up correlation, for the three analyses of a baseline and follow-up outcome. ANCOVA is never worse than either alternative. The change score beats the final value only when the correlation exceeds 0.5.

Correlation	Final value	Change score	ANCOVA
0.3	1	1.4	0.91
0.5	1	1.0	0.75
0.7	1	0.6	0.51
0.9	1	0.2	0.19

The change score is better than the raw final value only when $\rho > 0.5$, and ANCOVA is better than both at every ρ (Senn 2006; Vickers and Altman 2001). Vickers and Altman put the practical conclusion in the title of their note: analyze the follow-up value with the baseline as a covariate. In a randomized trial the change-score analysis is unbiased, since baseline is balanced in expectation, but it spends precision. Presenting the result as a mean change from baseline is a separate question from how it is estimated: an ANCOVA estimate can be reported on the change scale and, for a linear model, the treatment effect is identical whether the outcome is the raw follow-up value or the change (European Medicines Agency, Committee for Medicinal Products for Human Use 2015).

5.10.3. Analyze as you randomized

If randomization was stratified, the analysis should include the stratification factors. ICH E9 section 5.7 asks for it, and the reason is statistical: stratification removes variation between strata from the treatment comparison, and an analysis that ignores the strata does not give that variation back to the standard error. The result is a standard error that is too large, confidence intervals that are too wide, a type I error below its nominal level and a loss of power (Kahan and Morris 2012). The FDA guidance says that an analysis ignoring stratified randomization is likely to overestimate standard errors and can be unduly conservative, and that the Agency recommends that the standard error computation account for it (U.S. Food and Drug Administration 2023d). This is the same argument as for the permutation test earlier in the chapter, which shuffles within blocks. Minimization has the same logic, with the factors in the minimization algorithm playing the role of strata.

5.10.4. Do not test baseline balance

A common sponsor habit is a table of baseline p-values, with the conclusion that the arms are balanced if none is below 0.05. Senn pointed out that this tests a null hypothesis that is true by construction: any baseline difference in a properly randomized trial is due to chance, so a significant difference only tells us that chance happened (Senn 1994). Randomization does not guarantee balance on any covariate. Randomization provides the probability distribution over allocations on which a design-appropriate analysis can base valid error rates, which is a different and more useful property (Senn 2013). The remedy for a worrying imbalance is the

one of the preceding sections: prespecify the prognostic covariates and adjust for them, as ICH E9 asks, and do not select covariates by looking at the baseline table.

One alternative design deserves a mention for its estimand. Zelen's original single-consent proposal randomized patients before seeking consent for experimental treatment, and only those assigned to the new treatment were asked to accept it (Zelen 1979). Contemporary use requires a justified consent arrangement, IRB approval, and any applicable regulatory waiver or exception. The analysis by allocation is an intention-to-treat comparison, and its estimand is the effect of being offered the new treatment, diluted by those who decline, not the effect of receiving it. The loss in power from the dilution is the price of the design, and its estimand has to be said plainly in the protocol.

5.10.5. Binary and time-to-event outcomes: what is being estimated

For a continuous outcome and a linear model, adding a covariate improves precision without changing what the treatment coefficient means. For a binary or time-to-event outcome analyzed with a logistic or Cox model this is no longer true, and a sponsor that adds a prognostic covariate to the logistic model can find that the odds ratio changed in the absence of any confounding. The reason is that the odds ratio and the hazard ratio are non-collapsible: the odds ratio within each level of a prognostic covariate is not the odds ratio of the population, even when the covariate is perfectly balanced between the arms and the effect is the same in every subgroup (Greenland, Pearl, and Robins 1999; Hernán 2010). Risk differences and risk ratios are collapsible. The FDA guidance illustrates this with a hypothetical trial in which the odds ratio is 8.0 in each of two equal-sized biomarker subgroups (success rates 80.0 percent against 33.3 percent, and 25.0 percent against 4.0 percent) and 4.8 in the combined population (52.5 percent against 18.7 percent) (U.S. Food and Drug Administration 2023d).

This is not a defect of the adjusted analysis. It is a reminder that the two analyses answer different questions, and that the question has to be fixed first. The conditional effect is the odds ratio for patients with the same covariate values. The marginal (unconditional, population-averaged) effect is the effect of moving the whole target population from control to treatment, and it is the quantity the unadjusted analysis estimates. In the language of ICH E9(R1) (International Council for Harmonisation 2019) this is a choice of population-level summary, one of the attributes of the estimand, and the protocol should name it. FDA asks sponsors to specify whether the treatment effect of interest is conditional or unconditional, and EMA says that in nonlinear models the adjusted and unadjusted parameters have different interpretations and that the applicant must explain the meaning of the estimated effect clearly (U.S. Food and Drug Administration 2023d; European Medicines Agency, Committee for Medicinal Products for Human Use 2015).

The marginal effect can be estimated with the precision gain of adjustment by standardization (also called g-computation or the plug-in estimator). FDA lays it out in six steps: fit a logistic model of the outcome on treatment and the prespecified covariates; for every patient, whatever their arm, predict the risk under treatment and under control; average each set of predictions over all the patients in the trial; and compute the risk difference, risk ratio or odds ratio from the two averages (U.S. Food and Drug Administration 2023d). The standard error comes from a bootstrap or from a delta-method formula. The package computes the latter, with the influence function of the plug-in estimator, so that the variation of the covariates over the sample is accounted for.

A simulation shows all of this at once. Each of 500 trials has 300 patients, 1:1 complete randomization, a standard normal prognostic covariate and a conditional log odds ratio of 0.7 (a conditional odds ratio of 2.01). Because the covariate is prognostic, the marginal log odds ratio is smaller: 0.495, a marginal odds ratio of 1.64.

Table 5.9.: Estimates of the log odds ratio in 500 simulated randomized trials of 300 patients. The unadjusted and standardized estimates target the marginal log odds ratio; the adjusted logistic coefficient targets the larger conditional one. The standardized estimator has the same target as the unadjusted analysis and a smaller standard error. Monte Carlo error of the means is about 0.01.

Analysis	Target	Mean	Empirical SD	Mean SE
Unadjusted logistic	0.495	0.495	0.251	0.234
Adjusted logistic (conditional)	0.700	0.699	0.297	0.283
Standardized (marginal)	0.495	0.490	0.208	0.197

The adjusted logistic coefficient is centered on 0.7, not on the marginal value, and nobody is confounded: the two analyses simply estimate different things. The standardized estimate is centered on the same value as the unadjusted one, 0.495, and its standard deviation is 17 percent smaller, which is a gain equivalent to enrolling about 46 percent more patients. The standard errors track the standard deviations, so the interval from the delta-method formula is about the right width. Applied to one trial the call is two lines.

```
d <- covadj_sim_trial(300, seed = 21)
fit <- glm(y ~ trt + x, binomial, data = d)
knitr::kable(covadj_standardize(fit, d, trt = "trt"), digits = 3)
```

Table 5.10.: Standardized marginal risk difference, risk ratio and odds ratio with delta-method standard errors (log scale for the ratios) and 95 percent intervals for one simulated trial of 300 patients.

measure	estimate	se	lower	upper
rd	0.130	0.049	0.034	0.226
rr	1.310	0.104	1.068	1.608
or	1.688	0.200	1.140	2.500

The output is a table of three rows (risk difference, risk ratio and odds ratio) with estimates, standard errors and 95 percent intervals. For time-to-event outcomes the same ideas apply, with the marginal effect expressed as a difference in survival probabilities or restricted mean survival time at a prespecified time; a Cox model with prognostic covariates and strata is the usual conditional analysis, and the survival chapter shows how it is used (Section 18.1).

5.10.6. Estimators that do not depend on the model being right

A concern with regression adjustment is that the model is never exactly correct. For linear models the concern is small: the treatment coefficient from ordinary least squares remains

a valid estimate of the average treatment effect even when the covariate relationships are misspecified, provided the standard error is robust (the Huber-White sandwich estimator) when the model has no treatment by covariate interactions, and nominal standard errors are acceptable in two-arm trials with 1:1 allocation (U.S. Food and Drug Administration 2023d). For nonlinear models, a literature that began in the 2000s showed how to construct estimators that are consistent and asymptotically normal under any outcome model, because the randomization probabilities are known and are not estimated from the data. Tsiatis, Davidian, Zhang and Lu derived a family of semiparametric estimators that augment the unadjusted estimator with a function of the covariates, and showed that the most efficient one has asymptotic variance no larger than the unadjusted one (Tsiatis et al. 2008). Moore and van der Laan developed targeted maximum likelihood estimation for randomized trials with binary outcomes, an estimator whose asymptotic variance is no larger than that of the unadjusted analysis (K. L. Moore and Laan 2009), and Rosenblum and van der Laan extended simple standardization-type estimators to generalized linear models (Rosenblum and Laan 2010). Benkeser, Díaz, Luedtke and colleagues applied this approach to binary, ordinal and time-to-event outcomes in COVID-19 treatment trials, with software, and evaluated the precision gains by simulation (Benkeser et al. 2021). Ye, Shao, Yi and Zhao treated the case that matters for this chapter, covariate-adaptive randomization (stratified blocks and minimization), and gave a robust analysis that is never less efficient than the unadjusted one and has valid standard errors when the randomization used the covariates (Ye et al. 2023).

What a trialist should take from this is practical. The point estimate of a standardized or augmented estimator is valid whether or not the working outcome model is right, so a misspecified working model costs precision but does not bias the estimate, and a better model buys more precision. (FDA states that its guidance does not cover machine-learning adjustment, so a flexible learner is a topic to discuss with the review division.) The price is that the inference rests on large-sample theory, which is best understood when the number of covariates and strata is small relative to the sample size, a caveat that both regulators repeat.

5.10.7. What the regulators say

Both documents below were read in full from the primary PDF. The FDA document is the final guidance for industry of May 2023, and it states that it contains nonbinding recommendations and uses “should” to mean suggested or recommended, not required. The EMA document is a CHMP guideline adopted on 26 February 2015, in effect from 1 September 2015, replacing the earlier Points to Consider (CPMP/EWP/2863/99). ICH E9 section 5.7 is the common root of both: it encourages identifying covariates expected to have an important influence on the primary variable, specifying how they will be handled in the analysis, and cautions against covariates measured after randomization because they could be affected by the treatment.

Table 5.11.: What the FDA guidance (May 2023, final, nonbinding) and the EMA guideline (EMA/CHMP/295050/2013, adopted 2015) say about covariate adjustment. Quotations and paraphrases of the primary documents; the verb is the document's own.

Topic	FDA	EMA
Unadjusted analysis	Acceptable for the primary analysis of efficacy; sponsors can adjust	Adjustment generally improves efficiency; a prespecified primary analysis carries most credibility
Prespecification	Sponsors should specify procedures before any unblinding of comparative data	Covariates in the primary analysis must be prespecified in the protocol
Choice of covariates	Recommends adjusting for covariates anticipated to be most strongly associated with the outcome; may use earlier trials to form prognostic indices	Covariates with a known or expected strong or moderate association with the outcome; the baseline value of a continuous outcome should usually be included
Covariates after baseline	Cites ICH E9 caution against adjusting for covariates measured after randomization	Should not be included in the primary analysis
Selection from the data	Discuss data-adaptive covariate selection with the review division	Methods that select covariates by their association with the outcome should be avoided in confirmatory trials; post hoc baseline imbalance is not a reason to include a covariate
Number of covariates	Discuss with the division if large relative to the sample size	No more than a few covariates in the primary analysis
Stratification factors	Model should generally include the strata variables; standard errors should account for stratified randomization	Should normally be included as covariates or stratification variables, unless used only for administrative reasons
Nonlinear models	Specify whether the estimand is conditional or unconditional; standardization is one reliable method for the unconditional effect	Adjusted and unadjusted parameters have different interpretations and the meaning must be explained
Robust standard errors and model checking	Sponsors should consider a robust method, such as the sandwich estimator, when there are no interactions	Model assumptions must be checked; alternative prespecified analyses if they fail
Treatment by covariate interactions	May be included; the primary analysis can still be the average treatment effect	The primary model should not include them, unless substantial interactions are expected, in which case design for subgroups

Three points in the table deserve a sentence each. First, the two agencies differ on interactions: FDA allows a model with treatment by covariate interactions as long as the primary estimate is the average treatment effect, and EMA tells sponsors to leave them out of the primary model; the interaction model is where the estimators of the previous section gain the most, so a trial intended for both agencies should discuss it. Second, neither document requires adjustment. FDA says an unadjusted primary analysis is acceptable, and EMA says adjustment is not always necessary and is advisable when a covariate is strongly or moderately associated with the outcome; what both ask is that whatever is done be prespecified. Third, the FDA guidance states that it does not address the use of Bayesian methods or of machine-learning methods for covariate adjustment. The Bayesian analysis below is therefore a prespecified model whose operating characteristics should be shown by simulation, in the way the rest of the book argues.

5.10.8. The Bayesian version

A Bayesian adjusts for covariates by putting them in the outcome model. For the logistic model with treatment indicator t_i and covariates x_i, the model is logit $\Pr(Y_i = 1) = \alpha + \gamma t_i + \beta^\top x_i$, with weakly informative priors on the coefficients, for example normal with mean zero and standard deviation 2.5 on the log-odds scale after the covariates are standardized. These priors are not borrowing: they carry almost no information, they only regularize, and they do not change the target. The coefficient γ is, again, a conditional effect. The marginal effect is obtained as in the frequentist case, by averaging, and the Bayesian version of the average is a posterior. For each draw of the coefficients, predict every trial patient's risk under treatment and under control, average over the patients in the trial, and compute the risk difference or log odds ratio. The collection of these draws is the posterior of the marginal effect (Bayesian g-computation, or posterior predictive standardization).

Table 5.12.: Posterior of the marginal risk difference and marginal log odds ratio for one simulated trial of 300 patients, with and without the prognostic covariate in the outcome model (normal priors with standard deviation 2.5, Laplace approximation, 4,000 draws). Adjustment shrinks the posterior standard deviation and moves the mean very little.

	RD mean	RD SD	RD 2.5%	RD 97.5%	log OR SD
Treatment only	0.124	0.058	0.012	0.239	0.237
Treatment + covariate	0.127	0.050	0.028	0.229	0.207

The posterior SD of the risk difference falls from 0.058 to 0.05 when the covariate is added, with the same data and the same prior on the treatment coefficient. The gain is the same one that the frequentist standardized estimator obtains, and the two analyses agree closely on a trial of this size. When the outcome model is more flexible than a linear predictor, Bayesian additive regression trees (BART) are a natural choice for the outcome model, with the same averaging step (J. L. Hill 2011). A flexible model has more room to be wrong in a way that the regulators have not seen, so the prespecification (covariates, model, priors, summary) and the simulation evidence matter more, not less.

Adjustment is an efficiency source that needs no informative prior. This has a consequence for comparisons of designs elsewhere in the book. When a Bayesian adaptive design is said to be more efficient than a traditional design, the traditional design in the comparison should

include the covariate-adjusted analysis that FDA and EMA recommend. A reduction in sample size that adjustment alone delivers (30 percent in the example above) should not be credited to the Bayesian machinery. The sample-size arithmetic is that of Chapter 9 with the variance multiplied by $1 - R^2$.

5.10.9. Where else adjustment appears

The prognostic-score adjustment of Section 19.15 is a special case: the covariate is a single score built from external data, entered in the outcome model as one more baseline covariate, so that the arguments of this section about prespecification, estimand and precision apply to it without change. The stratified Cox model for time-to-event outcomes is covered in Section 18.1. The statistical analysis plan fixes everything that matters here (the covariates, the model, the estimand, the handling of strata and of missing covariate values) before unblinding, which is the subject of Chapter 23.

5.11. What to put in the protocol and what the regulator will ask

Table 5.13.: Randomization in the protocol and the statistical analysis plan.

Item	Standard practice (this chapter)	What a reviewer asks
Sequence generation	Validated generator, stratified permuted blocks with random block sizes, seed sealed	Is the schedule reproducible and checked for balance?
Stratification factors	Two or three prognostic factors, same factors in the analysis	Does the primary analysis include them?
Allocation concealment	Central IRT, kit number after enrollment	Could any site staff see or infer the next assignment?
Blinding	Double blind where possible, blinded outcome assessors otherwise, unblinding procedure	Who can unblind, and is it logged? Could side effects reveal the arm?
Analysis population	All randomized patients for the primary analysis (ITT or treatment policy)	Are exclusions defined before the data are seen?
Estimand	Five attributes written down, one strategy per intercurrent event	Does the estimand match the claim in the label?
Randomization test or model	Analysis mirrors the design	Is the permutation or model-based analysis consistent with the blocks and strata?

Item	Standard practice (this chapter)	What a reviewer asks
Bayesian analysis	Posterior for the stated estimand, assignment ignorable by design	What is the success criterion, and is it on the estimand?

What does not change under a Bayesian design is nearly everything in this table. A Bayesian analysis does not make allocation concealment or blinding less important, because they protect against bias, and bias is not a matter of which inferential paradigm one holds. What changes is the analysis in the last row, and, in adaptive designs, the possibility of changing allocation probabilities as data accrue, which raises the questions about time trends and unblinding that Chapter 16 will take up.

Try it in the lab

Open module 00, Why the coin?. First predict what happens to the gap between the groups when a doctor chooses who gets the drug and when a coin chooses, and how it changes with a larger trial. Then run the simulation and check. Next open module 06, Fair comparisons: placebo, blinding and who counts, and use the "Quit the drug? Still counted" panel to see how the intention-to-treat estimate dilutes as more patients stop, before reading the estimand discussion above.

5.12. In practice

5.12.1. The MRC streptomycin trial (1948): concealment before the word existed

The Medical Research Council trial of streptomycin in acute pulmonary tuberculosis (Medical Research Council Streptomycin in Tuberculosis Trials Committee 1948) is the chapter's first example of the sequence-generation and concealment distinction. Hill, who designed the trial, used random allocation, as Fisher had in agricultural experiments (Fisher 1935), chiefly to prevent selection bias in the choice of patients for each arm, and paired it with concealment of the allocation (A. B. Hill 1952); Yoshioka documents how the random numbers and envelopes were actually used (Yoshioka 1998). The allocation was determined from a series of random numbers and kept in sealed envelopes, so the admitting physician decided whether a patient was eligible before knowing which treatment the patient would receive. The comparison was streptomycin with bed rest against bed rest alone. The lesson for the chapter is that the contribution of the trial was not the coin but the concealment: a list that the admitting doctor could see would have recreated the doctor-chosen arm of Figure 5.1. The regulatory descendant is the requirement of 21 CFR 314.126(b)(4) for a method of assignment that minimizes bias.

5.12.2. RECOVERY (2020): randomization at scale, open label, objective endpoint

RECOVERY, a platform trial in UK hospitals, randomized patients with COVID-19 to usual care or to usual care plus dexamethasone, among other arms, without blinding (RECOVERY Collaborative Group et al. 2021). Of 11,303 patients randomized from 19 March to 8 June 2020, 2,104 were assigned to dexamethasone and 4,321 to usual care in the comparison reported. The mortality comparison itself, with its rate ratio and interval, is read as a two-group test in Section 4.11. The preliminary results were announced on 16 June 2020, nearly 100 days after the protocol was first drafted, and were adopted into UK practice the same day. The analysis was frequentist, so this is not a Bayesian trial. It is the standard of this chapter at scale: central concealed randomization, an objective endpoint that limits assessment bias (open-label care can still introduce differences in co-interventions), and an intention-to-treat analysis of all randomized patients. The lesson is the one of the first row of Table 5.5: when the endpoint is objective, open-label randomized evidence can be decisive, and the speed came from simple randomization with a small number of data items, not from any change in how assignment works.

5.12.3. The Women's Health Initiative (2002): randomization against observational evidence

The Women's Health Initiative randomized healthy postmenopausal women to estrogen plus progestin or placebo and reported its principal results in 2002 (Rossouw et al. 2002). Observational studies had suggested that hormone therapy protected against coronary heart disease, and the randomized trial did not confirm it: it found an increase in coronary events and in strokes, and the trial arm was stopped early because the risks outweighed the benefits. The paper reports the hazard ratios; what matters here is the direction of the result. The regulatory lesson is that the observational association was an instance of Equation 5.1: women who chose hormone therapy differed from those who did not before any pill was taken, and the discrepancy illustrated the risks of confounding and of comparing different treatment histories. Subsequent analyses substantially reconciled the estimates after accounting for time since treatment initiation and confounding (Prentice et al. 2005). The trial's answer was credible because assignment was by chance and because the analysis kept all randomized women in their assigned arm.

Key points

- Randomization makes the selection term of Equation 5.1 zero on average, for factors measured or not. Bias from a doctor's choice of patients does not shrink with sample size, and adjustment repairs it only for what was recorded.
- The standard methods are simple, permuted block (with random block sizes), stratified, and minimization (Pocock-Simon, with a probability of 0.8 or so of following the minimizing arm). They balance arm sizes and known factors, and they do nothing for hidden ones; chance does that, with a distribution that the inference accounts for.
- Concealment (the next assignment is hidden until the patient is enrolled) is a different protection from blinding (nobody knows the arm afterward). Inadequate concealment is empirically associated with exaggerated treatment effects.

- The permutation test is justified by the design and has the type I error of the randomization procedure. The analysis should mirror the design: shuffle within blocks, and adjust for stratification factors.
- Real trials show the same chain at work: concealed sealed-envelope allocation (MRC 1948), central randomization with an objective endpoint at scale (RECOVERY), and randomization overturning observational evidence (WHI).
- Randomization validity depends on the assignment process. Kolmogorov complexity concerns individual sequences and cannot be computed, and statistical diagnostics such as Arnold's stochasticity parameter detect pattern but do not certify an allocation mechanism. Use a validated generator, record and seal the seed, and check the schedule.
- Analyze all randomized patients. ICH E9(R1) asks for an explicit estimand and a strategy per intercurrent event. A per-protocol analysis is not among the strategies and can be biased badly.
- A Bayesian randomizes so that the assignment mechanism is ignorable by construction and the result is persuasive to those who do not share the analyst's model (Rubin 1978; Kadane and Seidenfeld 1990).
- Covariate adjustment for prespecified prognostic baseline covariates is the cheapest precision a trial can buy: ANCOVA multiplies the variance by about $1 - R^2$, and the stratification factors belong in the analysis. For binary and time-to-event outcomes name the estimand: the conditional odds ratio is not the marginal one even without confounding, and standardization gives the marginal effect with the adjusted precision. FDA (May 2023, final) and EMA (2015) recommend prespecified adjustment, and neither requires it.
- A Bayesian analysis adjusts by putting covariates in the outcome model and averaging predictions over the trial sample; because adjustment needs no informative prior, a fair benchmark for a Bayesian adaptive design is an adjusted traditional analysis (Section 5.10).

i Discussion questions

1. A sponsor proposes to compare a new therapy in 200 patients with 500 matched historical patients and argues that matching on ten measured covariates removes the need for randomization. Using Equation 5.1, say what the matching can and cannot remove, and what you would ask the sponsor to show.
2. A trial uses blocks of four with no variation in block size and is open label. A site investigator has enrolled 7 patients. Describe how the investigator could predict the eighth assignment, and propose two changes to the protocol.
3. In the ITT simulation, per-protocol overshoots the truth. Would it always? Construct a scenario in which patients who stop the drug are healthier than those who remain, and say what happens to the per-protocol estimate.
4. A DMC statistician sees unblinded data and the trial team does not. List three ways the firewall could leak in a platform trial, and which stage of the chain (sequence generation, concealment, blinding) each affects.
5. A colleague says, "I am a Bayesian, so randomization is irrelevant to me." Use Equation 5.3 to explain in two sentences what is wrong, and say in what situation the colleague would have a point.

6. Prior, Likelihood, Posterior

Chapter 4 asked one question of a two-group trial: could the difference we saw be produced by chance alone? The answer was a p-value, a number computed on the assumption that nothing is going on. A trialist usually wants the answer to a different question: given what we saw, how likely is it that the new treatment is better, and by how much? Bayesian inference answers that question directly. It does so by adding one ingredient the traditional analysis leaves out, a probability distribution describing what was known about the effect before the trial, and by combining it with the data through a single rule.

This chapter introduces that rule and the vocabulary around it: prior, likelihood, posterior, credible interval, predictive distribution. We stay with the example of Chapter 4, a control response rate of 30% and a treatment response rate of 45%, and we redo its analysis the Bayesian way. Each Bayesian quantity is introduced next to the traditional quantity it parallels, and we will see that with a flat prior the two give nearly the same numbers. The difference is in what the numbers are allowed to mean, and in what they let us do later: monitor, borrow, predict and decide.

! Regulatory anchor

ICH E9, Statistical Principles for Clinical Trials (Step 4, February 1998) (International Council for Harmonisation 1998). Section 1.2 observes that clinical-trial methodology has been based on frequentist methods, then states that this "should not be taken to imply that other approaches are not appropriate: the use of Bayesian (see Glossary) and other approaches may be considered when the reasons for their use are clear and when the resulting conclusions are sufficiently robust." The Glossary defines Bayesian approaches as those that provide "a posterior probability distribution for some parameter (e.g. treatment effect), derived from the observed data and a prior probability distribution for the parameter." Section 5.5 asks for estimates of treatment effect with confidence intervals in addition to tests. E9 permits Bayesian analysis and asks for reasons and robustness; it does not forbid it.
ICH E20, Adaptive Designs for Clinical Trials (Step 2 draft, June 2025) (International Council for Harmonisation 2025a). Section 5.3 describes Bayesian methods as those that "combine a prior probability distribution with current trial data to obtain a posterior probability distribution," and expects Bayesian adaptive designs to show targeted operating characteristics, including control of the type I error probability where that is the objective. A draft, not binding.
FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft guidance, January 2026) (U.S. Food and Drug Administration 2026g). Section II defines the prior as "the pre-study probability distribution for model parameters." Section IV.B says "It is always critical for Bayesian analyses to have a prespecified prior." A draft, not binding; Chapter 3 describes all three documents and their status.

6.1. Bayes' rule with a diagnostic test

The rule itself is old arithmetic, and a clinician has used it for years under another name. The rule is Bayes's (1763) and Laplace's, and it became a theory of decision with de Finetti and Savage (Savage 1954). Suppose a screening test for a disease has sensitivity 90% (it is positive in 90% of people who have the disease) and specificity 95% (it is negative in 95% of people who do not), and the disease has prevalence 2% in the population being screened. A patient tests positive. What is the probability the patient has the disease?

The sensitivity is a probability about the test given the disease. The question asks for the reverse, a probability about the disease given the test. To reverse the conditioning, write the joint probability of "disease and positive" in two ways and equate them:

$$\mathbb{P}(D \mid +)\,\mathbb{P}(+) = \mathbb{P}(+ \mid D)\,\mathbb{P}(D), \qquad \text{so} \qquad \mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\,\mathbb{P}(D)}{\mathbb{P}(+ \mid D)\,\mathbb{P}(D) + \mathbb{P}(+ \mid \bar{D})\,\mathbb{P}(\bar{D})}.$$

This is Bayes' rule. The three pieces have names that carry through the rest of the book. $\mathbb{P}(D)$ is the **prior**, what was believed before the test result. $\mathbb{P}(+ \mid D)$, viewed as a function of the unknown state, is the **likelihood** of the observed result under each state. $\mathbb{P}(D \mid +)$ is the **posterior**, what is believed after.

```
sens <- 0.90; spec <- 0.95; prev <- 0.02
p_pos <- sens * prev + (1 - spec) * (1 - prev)
ppv <- sens * prev / p_pos
knitr::kable(data.frame(Quantity = c("P(positive)", "P(disease |
 ↪  positive)", "P(disease | negative)"),
                        Probability = c(p_pos, ppv, (1 - sens) * prev / (1
                         ↪  - p_pos))), digits = 4, align = "lr")
```

Table 6.1.: Screening test with sensitivity 90%, specificity 95% and prevalence 2%: probability of a positive result, and probability of disease after a positive and after a negative result.

Quantity	Probability
P(positive)	0.0670
P(disease | positive)	0.2687
P(disease | negative)	0.0021

Table 6.1 gives the numbers. A patient with a positive test has about a 27% chance of having the disease, not 90%. The test is good, but the disease is rare, so most positives come from the large healthy group. The only thing that changes in a trial is that the unknown is no longer "disease or not" but a continuous quantity, the response rate or the log hazard ratio, and the prior is a distribution over its possible values rather than a single prevalence.

In symbols, for an unknown parameter θ and data y,

$$\pi(\theta \mid y) = \frac{p(y \mid \theta)\,\pi(\theta)}{\int p(y \mid \theta')\,\pi(\theta')\,d\theta'} \;\propto\; p(y \mid \theta)\,\pi(\theta), \tag{6.1}$$

where $\pi(\theta)$ is the prior, $p(y \mid \theta)$ the likelihood and $\pi(\theta \mid y)$ the posterior. The denominator is the probability of the data averaged over the prior. It does not depend on θ, so it only rescales the product to integrate to one; the shape of the posterior is the prior times the likelihood, "posterior is proportional to prior times likelihood."

The traditional analysis uses the same likelihood. A frequentist estimate, the maximum likelihood estimate, is the value of θ at which $p(y \mid \theta)$ peaks, and a z test is a calculation on its sampling distribution. What the Bayesian adds is the prior and the multiplication. What the Bayesian gives up is nothing mathematical, but something practical: the answer now depends on a choice, the prior, that has to be made, defended and written into the protocol. That is the subject of Chapter 8 and the reason the FDA draft is emphatic that the prior be prespecified.

6.2. Exchangeability and de Finetti, informally

A trial analysis nearly always begins with the sentence "patients' outcomes are independent and identically distributed with response probability p." The sentence is a perfectly good model when θ is treated as a fixed unknown constant: given θ, outcomes are independent. The Bayesian view adds a distribution for θ, and that addition changes what the model says about outcomes we have not yet seen, which is the cleanest route into it.

Consider a thumbtack tossed repeatedly, with $y_i = 1$ if it lands point up. If we knew the probability θ of landing point up, the tosses would be independent: the first result would tell us nothing about the second. But we do not know θ. After 500 tosses with 499 point up, nobody believes the next toss is as uncertain as the first one was. So, once θ is given a distribution and integrated out, the predictive distribution of the tosses is not independent; they are independent only given θ, and a given θ is exactly what a trial does not have. The same applies to patients: ten consecutive responders in an arm tell us something about the next patient, because they tell us about the arm's response rate.

What is plausible in place of independence is a weaker symmetry, **exchangeability**: the probability we assign to a set of outcomes does not depend on the order in which patients are listed. Eight tosses with four point up are equally probable in the order 1,0,1,0,1,0,1,0 and in the order 0,1,1,0,1,0,0,1. Independent and identically distributed outcomes are always exchangeable, but not conversely, and exchangeability makes no reference to a parameter at all. It is a statement about our beliefs concerning data not yet seen.

de Finetti's representation theorem (de Finetti 1937) says what an exchangeable sequence of binary outcomes looks like. If any number of further outcomes could be added and the sequence stays exchangeable, then there exists a probability distribution Π on $[0, 1]$ (the mixing distribution, which need not have a density) such that

$$\mathbb{P}(y_1, \dots, y_n) = \int_0^1 \prod_{i=1}^{n} \theta^{y_i} (1-\theta)^{1-y_i} \, d\Pi(\theta). \tag{6.2}$$

Read it left to right: the joint probability of the data is a mixture, over the unknown response rate, of the independent-coin probabilities. The theorem does not assume a parameter and then a prior. It produces both from the single judgment of exchangeability. The parameter θ is the limiting fraction of responders, and Π is our uncertainty about it (when it has a

density we write $\pi(\theta)\,d\theta$, as in the Beta examples below). For the trialist the content is practical:

- The model "responses are Bernoulli(p) given p" and the prior on p come as a package. A protocol that specifies a likelihood without a prior has specified half of a Bayesian model; the traditional protocol specifies the likelihood half and handles the other half through the sampling distribution.
- Exchangeability is a judgment about the patients, and it can fail. If patients enrolled in the first year are systematically different from those enrolled later, or the sites differ, the order does matter. Time trends, site effects and covariates are ways of relaxing exchangeability to "exchangeable within strata," and the same judgment returns when we borrow information in Chapter 19.
- Randomization (Chapter 5) makes treatment assignment independent of prognosis and potential outcomes. The response model separately assumes exchangeability within each arm, or within specified covariate strata.

A full treatment, including the proof and the connection to subjective probability, is in Gelman et al. (2013) and Dennis V. Lindley (1985).

6.3. The coin and the Beta family

The simplest trial has one arm and a binary outcome, and the simplest model for it is the coin. The prior-and-posterior module of the book's companion online lab (https://vsokolov.org/bayesian_trials_lab/01-prior-posterior.html, described in the Preface) uses a coin of unknown bias. After 10 tosses, 7 are heads. The traditional estimate of the heads probability is $7/10 = 0.70$, with a standard error of about $\sqrt{0.7 \times 0.3/10} = 0.145$. The Bayesian analysis starts with a prior distribution over the bias θ, which Figure 6.1 shows at work on a coarse grid of values. (In this and the next two sections θ is the success probability of a single arm, which the two-group sections call p_j.)

Take the grid value $\theta = 0.7$. The prior bar is the Beta$(2, 2)$ density there, $6 \times 0.7 \times 0.3 = 1.26$, rescaled so that the eleven bars sum to one: 0.127. The likelihood bar is the probability of 7 heads in 10 tosses at $\theta = 0.7$, 0.267, rescaled the same way: 0.293. Multiply the two bars at every grid value, then divide each product by the sum of the eleven products so that the posterior sums to one. At $\theta = 0.7$ the result is 0.300. That is all Bayes' rule does. In the top row the prior still counts: it pulls the posterior toward one half, and the bar at 0.6 is nearly as tall as the bar at 0.7. In the bottom row the likelihood is so narrow that any smooth prior is nearly constant across it, and the posterior is the likelihood. Make the grid finer and the bars become thin slivers along a smooth curve. With a Beta prior and a binomial likelihood those curves are Beta densities, which the rest of this section uses.

The convenient family of priors for a probability is the **Beta distribution**, $\theta \sim \text{Beta}(a, b)$, with density proportional to $\theta^{a-1}(1-\theta)^{b-1}$ on $[0, 1]$. Its mean is $a/(a+b)$. Beta$(1, 1)$ is the flat distribution on $[0, 1]$: every value of θ is equally believable. Beta$(2, 2)$ is a gentle bell centered at one half. Larger a and b give narrower bells.

The convenience is that the likelihood of h heads and t tails, $\theta^h(1-\theta)^t$, has the same algebraic form as the Beta density. Multiplying them as in Equation 6.1 gives

$$\theta \sim \text{Beta}(a, b), \quad h \text{ heads}, \ t \text{ tails} \quad \Longrightarrow \quad \theta \mid y \sim \text{Beta}(a + h,\ b + t). \tag{6.3}$$

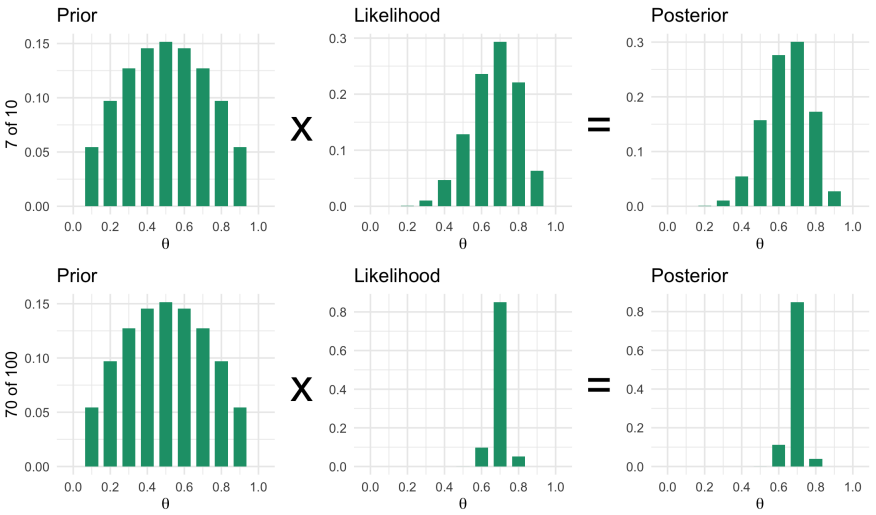


Figure 6.1.: Bayes' rule on a grid of ten-percent steps for the coin, with the Beta(2, 2) prior. Top: 7 heads in 10 tosses. Bottom: 70 heads in 100 tosses. In each row the prior times the likelihood, each rescaled to sum to one, gives the posterior. The likelihood sharpens with more data and the prior's influence fades.

A prior with this property, that the posterior belongs to the same family, is called **conjugate** to the likelihood. For the Beta prior and a binomial likelihood the update is addition: add the responders to a and the non-responders to b. One reads Beta(a, b) as "as if $a-1$ responders and $b-1$ non-responders had already been seen, starting from a flat prior" (with a convention that makes the rule exact, shown in the Technical note below).

```
post <- beta_post(7, 10, 2, 2)                          # prior Beta(2, 2), 7
 ↪  heads in 10 tosses
flat <- beta_post(7, 10)
ecmo <- beta_post(11, 11)                               # the lab's second
 ↪  scenario: 11 of 11 newborns survive on ECMO
summ <- function(label, f) data.frame(Scenario = label, Posterior =
 ↪  sprintf("Beta(%g, %g)", f$a, f$b), Mean = f$a / (f$a + f$b),
  Interval = sprintf("(%.3f, %.3f)", qbeta(0.025, f$a, f$b), qbeta(0.975,
   ↪  f$a, f$b)), Prob = pbeta(0.8, f$a, f$b, lower.tail = FALSE))
knitr::kable(rbind(summ("Coin, Beta(2, 2) prior", post), summ("Coin, flat
 ↪  prior", flat), summ("ECMO 11/11, flat prior", ecmo)),
             digits = 3, col.names = c("Scenario", "Posterior", "Posterior
              ↪  mean", "95% credible interval", "Pr(theta > 0.8)"), align
              ↪  = "llrrr")
```

Table 6.2.: Beta posteriors for the coin (7 heads in 10 tosses) under the Beta(2, 2) prior of the companion lab's coin module and a flat prior, and for the ECMO trial (11 of 11 survivors) under a flat prior.

Scenario	Posterior	Posterior mean	95% credible interval	Pr(theta > 0.8)
Coin, Beta(2, 2) prior	Beta(9, 5)	0.643	(0.386, 0.861)	0.099
Coin, flat prior	Beta(8, 4)	0.667	(0.390, 0.891)	0.161
ECMO 11/11, flat prior	Beta(12, 1)	0.923	(0.735, 0.998)	0.931

As Table 6.2 shows, with the companion lab's Beta(2,2) prior and 7 heads and 3 tails, the posterior is Beta(9,5) with mean $9/14 = 0.643$. With the flat prior it is Beta(8,4) with mean 0.667. The traditional estimate 0.70 is the value at the peak of the likelihood; the Bayesian posterior mean sits between the data and the prior's center, and by how much depends on the prior's strength, which we quantify shortly.

i Technical note: why the update is addition

Multiply the prior density by the binomial likelihood:

$$\underbrace{\theta^{a-1}(1-\theta)^{b-1}}_{\text{prior}} \times \underbrace{\theta^{h}(1-\theta)^{t}}_{\text{likelihood}} = \theta^{a+h-1}(1-\theta)^{b+t-1},$$

which is the kernel of $\text{Beta}(a+h, b+t)$. The normalizing constant in Equation 6.1 is whatever makes the result integrate to one, namely $B(a+h, b+t)$, so nothing needs to be integrated by hand. The statement "$a-1$ prior responders" holds because the flat $\text{Beta}(1,1)$ corresponds to none; some authors instead describe $\text{Beta}(a,b)$ as $a+b$ pseudo-observations, which is the count used for effective sample size below. The two descriptions differ by the two pseudo-observations of the flat prior and are both in use.

The lab's second scenario is a real one: Bartlett and colleagues' 1985 trial of extracorporeal membrane oxygenation (ECMO) in newborns with respiratory failure (R. H. Bartlett et al. 1985). In the play-the-winner design, all 11 newborns assigned to ECMO survived (the single control patient did not). With a flat prior the posterior is $\text{Beta}(12, 1)$, which puts almost all its mass near survival rates close to one; Table 6.2 gives $\Pr(\theta > 0.8) = 0.931$. This example comes back in Chapter 16, where the design rather than the analysis is the issue.

6.4. Posterior mean as a weighted average, and the prior's effective sample size

The posterior mean of a Beta-binomial model has a form every trialist can read. Let $n = h+t$ be the number of patients and $\hat{p} = h/n$ the observed proportion. Then

$$\mathbb{E}[\theta \mid y] = \frac{a+h}{a+b+n} = \frac{a+b}{a+b+n} \cdot \frac{a}{a+b} + \frac{n}{a+b+n} \cdot \hat{p}. \tag{6.4}$$

It is a weighted average of the prior mean $a/(a+b)$ and the sample proportion $\hat{p}$, with weights proportional to $a+b$ and n. The prior acts as if it were a sample of size $a+b$. This is the **effective sample size** (ESS) of a Beta(a,b) prior: a flat Beta$(1,1)$ is worth two patients, a Beta$(2,2)$ four, a Beta$(30,70)$ one hundred. More refined definitions exist, which match the prior to the amount of information in the data under different criteria (Morita, Thall, and Müller 2008), and Chapter 8 treats them; for the Beta-binomial, $a+b$ is the answer all of them roughly agree on. The prior of the BNT162b2 vaccine trial in Section 6.12, Beta$(0.700102,1)$, has an ESS of 1.7 cases (the trial's patients enter through the cases they produce): a prior of real shape and almost no weight.

A prior with an ESS of 10 in a trial of 200 contributes 5% of the weight; the same prior in a trial of 10 contributes half. As n grows, the data dominate and every reasonable prior gives the same posterior. This is why Bayesian analysis of a large trial with a weak prior looks like the traditional analysis, and why the debate over priors matters most in small trials, rare diseases, pediatric extrapolation and anywhere a prior of substantial ESS is deliberately used to borrow information. The FDA draft's warning (Section IV.B.2) that conclusions can depend strongly on the prior is a statement about exactly this weight (U.S. Food and Drug Administration 2026g).

```
rows <- list()
for (pr in list(c(1, 1), c(3, 7), c(30, 70))) for (d in list(c(20, 9),
 ↪  c(100, 45), c(1000, 450))) {
  ess <- ess_beta(pr[1], pr[2]); n <- d[1]; h <- d[2]
  rows[[length(rows) + 1]] <- data.frame(Prior = sprintf("Beta(%g, %g)",
 ↪  pr[1], pr[2]), ESS = ess, n = n, w = ess / (ess + n),
                                         pm = (pr[1] + h) / (ess + n), sp
                                          ↪  = h / n)
}
knitr::kable(do.call(rbind, rows), digits = c(0, 0, 0, 3, 3, 3), align =
 ↪  "lrrrrr",
             col.names = c("Prior", "Prior ESS", "Patients n", "Weight on
              ↪  prior", "Posterior mean", "Sample proportion"))
```

Table 6.3.: Weight of a Beta prior in the posterior mean for three priors and three trial sizes (45% observed response). Weight on prior is ESS/(ESS + n).

Prior	Prior ESS	Patients n	Weight on prior	Posterior mean	Sample proportion
Beta(1, 1)	2	20	0.091	0.455	0.45
Beta(1, 1)	2	100	0.020	0.451	0.45
Beta(1, 1)	2	1000	0.002	0.450	0.45
Beta(3, 7)	10	20	0.333	0.400	0.45
Beta(3, 7)	10	100	0.091	0.436	0.45
Beta(3, 7)	10	1000	0.010	0.449	0.45
Beta(30, 70)	100	20	0.833	0.325	0.45
Beta(30, 70)	100	100	0.500	0.375	0.45
Beta(30, 70)	100	1000	0.091	0.436	0.45

Table 6.3 tabulates the weights. For a prior centered at 0.30 with ESS 100 and a sample of 100 patients with a 45% response, the posterior mean is 0.375, halfway between, because the prior and the data are worth the same number of patients. With 1,000 patients the same prior moves the answer by less than 0.015.

6.5. The two-group comparison, redone

Return to the trial of Chapter 4: a control arm with a true response probability $p_0 = 0.30$, a treatment arm with $p_1 = 0.45$, and suppose 100 patients per arm are randomized 1:1. For the worked numbers, let the trial come out at its design values: 30 of 100 control patients respond, and 45 of 100 treatment patients respond.

6.5.1. The traditional analysis

The traditional analysis is the one Chapter 4 derived. The observed difference is $\hat{p}_1 - \hat{p}_0 = 0.15$. The pooled response is 0.375, the standard error of the difference under the null is $\sqrt{0.375 \times 0.625 \times (1/100 + 1/100)} = 0.0685$, so $z = 2.19$ and the one-sided p-value is 0.014. Against the usual one-sided level of 0.025, this is a (modest) success. The 95% confidence interval for the difference, using the unpooled standard error 0.0676, runs from 0.017 to 0.283. Fisher's exact test gives a one-sided p of 0.020.

6.5.2. The Bayesian analysis

Put an independent flat Beta(1, 1) prior on each arm's response rate. By Equation 6.3 the posteriors are $p_0 \mid y \sim \text{Beta}(31, 71)$ and $p_1 \mid y \sim \text{Beta}(46, 56)$, independent. The quantity of interest is a function of both parameters, the difference $p_1 - p_0$ or the event $p_1 > p_0$. There is no neat formula for its distribution, but there does not need to be: draw many samples from each posterior and compute the quantity of interest on the pairs. This is **Monte Carlo** and it is how almost every Bayesian trial quantity in this book is computed.

```
x1 <- 45; x0 <- 30; n <- 100
z <- z_test_binary(x0, x1, n); p_one <- pnorm(z, lower.tail = FALSE)
 ↪  # traditional
ci <- wald_ci(x0, x1, n)

p1 <- beta_post(x1, n); p0 <- beta_post(x0, n)
 ↪  # Bayesian, flat Beta(1,1) priors
set.seed(2026)
diff <- rbeta(2e5, p1$a, p1$b) - rbeta(2e5, p0$a, p0$b)
sup <- function(m = 0) prob_superior(p1$a, p1$b, p0$a, p0$b, margin = m)
knitr::kable(data.frame(Quantity = c("z statistic", "One-sided p-value", "1
 ↪  - p", "Difference in response rates", "95% confidence interval",
 ↪  "Posterior mean difference",
    "Pr(p1 > p0 | data), exact", "Pr(p1 > p0 | data), Monte Carlo",
     ↪  "Pr(difference > 0.05 | data)", "Pr(difference > 0.10 | data)",
     ↪  "95% credible interval"),
  Value = c(sprintf("%.2f", z), sprintf("%.4f", p_one), sprintf("%.4f", 1
   ↪  - p_one), sprintf("%.2f", (x1 - x0) / n), sprintf("(%.3f, %.3f)",
   ↪  ci$lower, ci$upper),
```

```
    sprintf("%.3f", mean(diff)), sprintf("%.4f", sup()), sprintf("%.4f",
     ↪  mean(diff > 0)), sprintf("%.4f", sup(0.05)), sprintf("%.4f",
     ↪  sup(0.10)),
    sprintf("(%.3f, %.3f)", quantile(diff, 0.025), quantile(diff,
     ↪  0.975)))), align = "lr")
```

Table 6.4.: The running trial (30/100 control and 45/100 treatment responders): traditional z test and confidence interval, and the Bayesian analysis with flat Beta(1,1) priors on each arm.

Quantity	Value
z statistic	2.19
One-sided p-value	0.0142
1 - p	0.9858
Difference in response rates	0.15
95% confidence interval	(0.017, 0.283)
Posterior mean difference	0.147
Pr(p1 > p0 | data), exact	0.9855
Pr(p1 > p0 | data), Monte Carlo	0.9853
Pr(difference > 0.05 | data)	0.9259
Pr(difference > 0.10 | data)	0.7600
95% credible interval	(0.015, 0.277)

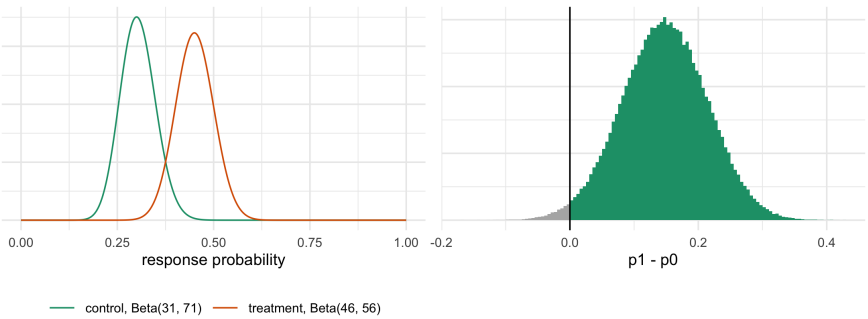


Figure 6.2.: Posterior distributions after 30/100 and 45/100 responders with flat priors (left), and the posterior of the difference p1 - p0 (right). The shaded area to the right of zero is Pr(p1 > p0 | data).

The posterior probability that the treatment's response rate exceeds the control's is $\Pr(p_1 > p_0 \mid y) = 0.9855$ (Table 6.4, Figure 6.2; numerical integration; the Monte Carlo estimate with 200,000 draws is within 0.001 of it). The one-sided p-value is 0.0142, and one minus it is 0.9858. The two numbers are close because in this problem they are nearly the same calculation, as the next section shows.

The Bayesian analysis also answers questions a p-value cannot. The posterior probability that the effect exceeds 5 percentage points is 0.926; that it exceeds 10 points is 0.760. These are the quantities that enter a success criterion of the form $\Pr(\delta > a \mid \text{data}) > c$ used

in the FDA draft (Section IV.A), where a is the smallest benefit worth having and c is a pre-specified probability. Chapter 7 treats such criteria, and Chapter 14 shows how c is chosen so the design has the type I error the regulator expects. Two real trials in Section 6.12 used exactly this form: the BNT162b2 vaccine trial (success if $\Pr(\text{VE} > 30\%) > 0.986$ at the final analysis) and REMAP-CAP (efficacy if the posterior probability of benefit exceeds 99%).

6.5.3. A side-by-side

Table 6.5 sets the two analyses next to each other.

Table 6.5.: Same trial (30/100 control and 45/100 treatment responders), analyzed the traditional way and the Bayesian way with flat priors.

	Traditional	Bayesian (flat prior)
Unknown	$p_1 - p_0$, a fixed number	$p_1 - p_0$, described by a distribution
Input	likelihood	likelihood and prior
Main output	estimate, 95% CI, one-sided p	posterior distribution, credible interval, $\Pr(p_1 > p_0 \mid y)$
Probability statement is about	the procedure, over repeated trials	the parameter, given these data
Result here	$\hat{d} = 0.15$, CI (0.017, 0.283), $p = 0.014$	mean 0.147, 95% CrI (0.015, 0.277), Pr = 0.9855
Decision rule	reject if $p < 0.025$	declare success if $\Pr(\delta > a \mid y) > c$
What the regulator asks for	prespecified test, α control	prespecified prior and (a, c), calibrated operating characteristics

6.6. Why the two nearly agree

With a flat prior, the posterior of $p_1 - p_0$ is centered, approximately, at the observed difference $\hat{d}$ with standard deviation approximately equal to its standard error s, and approximately normal (this is the Bernstein-von Mises phenomenon: with enough data the posterior looks like the normalized likelihood). So

$$\Pr(p_1 - p_0 > 0 \mid y) \approx \Phi\left(\frac{\hat{d}}{s}\right) = 1 - \Phi\left(-\frac{\hat{d}}{s}\right).$$

The one-sided p-value for testing $H_0 : p_1 \le p_0$ is $1 - \Phi(\hat{d}/s_0)$, with s_0 the standard error under the null. If $s \approx s_0$ the p-value is $1 - \Phi(\hat{d}/s)$, and so $1 - p \approx \Phi(\hat{d}/s)$. The two expressions are the same up to which standard error (pooled or unpooled) is used and the slight skewness of the Beta. The match is not a theorem of the model, it is a property of a flat prior and a nearly normal likelihood, and it fails when the prior is informative, when the

sample is small or when the response rates are near 0 or 1. We check it numerically on many simulated trials.

Across 500 simulated trials with 100 patients per arm and true rates 0.45 and 0.30, the largest gap between $1 - p$ and the flat-prior posterior probability (computed exactly) is 0.002, and the average gap is 0.0005.

Two cautions come with this near-equality. First, it is a numerical coincidence that has no logical force: the p-value is a statement about the data under the null, and $\Pr(p_1 > p_0 \mid y)$ is a statement about the parameter given the data. A reader who says "$1 - p$ is the probability that the drug works" is correct in this one setting only because the prior is flat, which is itself a strong choice (it gives a prior probability of one half to the claim that treatment is better, and puts substantial mass on differences nobody believes). Second, the near-equality is what makes the Bayesian bar not lower. A trial that declares success when $\Pr(p_1 > p_0 \mid y) > 0.975$ with a flat prior is, to a good approximation, the trial that declares success when the one-sided p-value is below 0.025. The Bayesian design's savings, to be shown in later chapters, come from monitoring, multiple arms and prior information, not from a lower threshold at a single final analysis. Change the prior and the two diverge (Table 6.6):

```
priors <- list("flat Beta(1,1)" = c(1, 1), "Jeffreys Beta(.5,.5)" = c(0.5,
 ↪  0.5), "skeptical Beta(3,7), ESS 10" = c(3, 7))
pp <- sapply(priors, function(ab) prob_superior(ab[1] + 45, ab[2] + 55,
 ↪  ab[1] + 30, ab[2] + 70))
knitr::kable(data.frame(Prior = names(priors), P = unname(pp)), digits = 3,
 ↪  col.names = c("Prior on each arm", "Pr(p1 > p0 | data)"), align = "lr")
```

Table 6.6.: Posterior probability that the treatment is better than control for the running trial (45/100 against 30/100), under three priors applied to both arms.

Prior on each arm	Pr(p1 > p0 | data)
flat Beta(1,1)	0.985
Jeffreys Beta(.5,.5)	0.986
skeptical Beta(3,7), ESS 10	0.983

Applied to both arms alike, even the skeptical Beta(3,7) prior (ESS 10, mean 0.30) moves the answer only from 0.9855 to about 0.983, because 200 patients swamp 10 pseudo-patients and the prior is the same on both arms. A prior that differs between arms, or one that centers the treatment effect at zero, is a different matter and is treated in Chapter 8.

6.7. Credible intervals and confidence intervals

The 95% confidence interval for the difference is (0.017, 0.283). The 95% credible interval, the central 95% of the posterior of $p_1 - p_0$, is (0.015, 0.277). They are numerically close, and they mean different things.

A **confidence interval** is a procedure. If the trial were repeated many times and the interval computed each time by the same recipe, 95% of the intervals would contain the true $p_1 - p_0$. The probability statement is about the procedure; for the one interval in the CSR, the true

value is either inside or it is not, and the "95%" does not say how likely it is that this interval covers the truth. A **credible interval** is a statement about the parameter: given the data and the prior, the probability that $p_1 - p_0$ lies in the interval is 95%. There is also a choice of which 95% region, and two are common: the equal-tailed interval, leaving 2.5% in each tail, and the highest posterior density (HPD) interval, the shortest region with 95% probability. For a skewed posterior they differ; for the symmetrical one above they nearly coincide.

Many trialists read confidence intervals as if they were credible intervals, and in most large trials with weak priors they get away with it. The distinction matters in two places. One is when the prior is informative: the credible interval shifts and narrows, which is the intention, and the confidence interval cannot. The other is near boundaries, where a confidence interval built from the normal approximation may stray outside $[0, 1]$ or cover poorly, while a credible interval built from the Beta posterior stays inside by construction.

```
set.seed(11); N <- 20000; n <- 100
cover <- function(x, p) {                              # coverage of Wald and
 ↪  flat-prior credible intervals
  ph <- x / n; s <- sqrt(ph * (1 - ph) / n)
  lo <- qbeta(0.025, 1 + x, 1 + n - x); hi <- qbeta(0.975, 1 + x, 1 + n -
 ↪  x)
  c(wald = mean(abs(ph - p) < 1.96 * s), credible = mean(lo < p & p < hi))
}
rows <- NULL
for (p in c(0.05, 0.30, 0.50)) {
  cv <- cover(rbinom(N, n, p), p)
  rows <- rbind(rows, data.frame(Truth = sprintf("fixed p = %.2f", p), wald
 ↪  = cv[["wald"]], cred = cv[["credible"]]))
}
p <- rbeta(N, 1, 1)                                    # Bayesian calibration:
 ↪  truth drawn from the prior
rows <- rbind(rows, data.frame(Truth = "p drawn from Beta(1,1)", wald = NA,
 ↪  cred = cover(rbinom(N, n, p), p)[["credible"]]))
options(knitr.kable.NA = "")
knitr::kable(rows, digits = 3, col.names = c("True response rate", "Wald
 ↪  interval", "Flat-prior credible interval"), align = "lrr")
```

Table 6.7.: Coverage of the Wald 95% confidence interval and the flat-prior 95% credible interval with n = 100: at fixed true response rates, and averaged over true rates drawn from the Beta(1,1) prior (20,000 simulated trials per row).

True response rate	Wald interval	Flat-prior credible interval
fixed p = 0.05	0.880	0.964
fixed p = 0.30	0.953	0.952
fixed p = 0.50	0.944	0.944
p drawn from Beta(1,1)		0.949

Table 6.7 shows the two notions of calibration. At a fixed true rate of 0.30 or 0.50, both intervals cover 94% to 95% of the time. At a rate of 0.05 the Wald interval covers only 88%, while the flat-prior credible interval covers 96.4%. Averaged over draws of p from the prior,

the credible interval covers 95% by construction (94.9% in the run): if the prior is a faithful description of where true rates come from, the 95% is exact, and that is the Bayesian sense of calibration. If the prior is wrong, the coverage at a specific p can be off, which is why the FDA draft asks for operating characteristics under a range of scenarios.

6.8. The posterior predictive distribution

A trial is not only about the parameter. Decisions depend on what will happen next: how many of the next 50 patients will respond, how likely it is that the trial, if continued to its planned size, will end in success. The Bayesian answer is the **posterior predictive distribution**, the distribution of future data after averaging over the uncertainty about the parameter:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta)\, \pi(\theta \mid y)\, d\theta. \tag{6.5}$$

For a Beta posterior $\mathrm{Beta}(a', b')$ and m future patients, the number of future responders follows a **beta-binomial** distribution, which has a closed form, and its mean is $m a'/(a'+b')$ (the posterior mean times m). The traditional analogue is a prediction interval, or more often a plug-in calculation: assume $p = \hat{p}$ and compute a binomial. The plug-in ignores the uncertainty in $\hat{p}$ and is therefore too narrow.

```
a <- 46; b <- 56; m <- 50; k <- 0:m                  # treatment-arm
 ↪  posterior after 45/100, next 50 patients
pred <- dbetabinom(k, m, a, b)                        # beta-binomial pmf
plug <- dbinom(k, m, 0.45)
mom <- function(pm) c(mean = sum(k * pm), sd = sqrt(sum(k^2 * pm) - sum(k
 ↪  * pm)^2))
int <- function(pm) k[c(which(cumsum(pm) >= 0.025)[1], which(cumsum(pm) >=
 ↪  0.975)[1])]
draws <- rbinom(2e5, m, rbeta(2e5, a, b))            # Monte Carlo check:
 ↪  parameter first, then data
mp <- mom(pred); mq <- mom(plug); ip <- int(pred); iq <- int(plug)
knitr::kable(data.frame(Method = c("Posterior predictive (beta-binomial)",
 ↪  "Plug-in binomial, p = 0.45", "Monte Carlo check"),
  Mean = c(mp[[1]], mq[[1]], mean(draws)), SD = c(mp[[2]], mq[[2]],
   ↪  sd(draws)),
  Lower = c(ip[1], iq[1], unname(quantile(draws, 0.025))), Upper = c(ip[2],
   ↪  iq[2], unname(quantile(draws, 0.975)))),
  digits = c(0, 2, 2, 0, 0), col.names = c("Method", "Mean", "SD", "95%
   ↪  interval, lower", "95% interval, upper"), align = "lrrrr")
```

Table 6.8.: Number of responders among the next 50 patients, given 45/100 so far and a flat prior: posterior predictive distribution against the plug-in binomial with p = 0.45.

Method	Mean	SD	95% interval, lower	95% interval, upper
Posterior predictive (beta-binomial)	22.55	4.27	14	31
Plug-in binomial, p = 0.45	22.50	3.52	16	29
Monte Carlo check	22.56	4.28	14	31

In Table 6.8 the predictive mean is 22.5 responders out of 50 and its standard deviation is 4.27, against 3.52 for the plug-in binomial. The extra spread is the contribution of not knowing p: a prediction made as if $\hat{p} = 0.45$ were the truth would be overconfident by about 20 percent. The 95% predictive interval for responders among the next 50 is 14 to 31, which the Monte Carlo agrees with.

The same two-stage logic, "draw the parameter from its posterior, then draw the future data from the model," gives the predictive probability that a trial in progress will succeed at its final analysis. Chapter 15 builds on exactly Equation 6.5 to define predictive probability, futility rules and Goldilocks-type sample-size choices.

6.9. Normal-normal on the log hazard ratio

Many pivotal trials, in oncology, cardiovascular disease and neurology, do not have a binary endpoint but a time to event, and the treatment effect is a hazard ratio (HR). The Beta-binomial model does not apply. The normal-normal model, which does, rests on a standard approximation.

For 1:1 randomization with d total events, the log hazard ratio estimate has standard error $\approx 2/\sqrt{d}$. (Each arm contributes about $d/2$ events and the variance of a log hazard ratio is about $1/d_1 + 1/d_0 = 4/d$.) With $\theta = -\log \mathrm{HR}$, so that a positive θ means benefit, the data are summarized by

$$\hat{\theta} \mid \theta \;\sim\; N\!\left(\theta,\; \frac{4}{d}\right). \tag{6.6}$$

This is the likelihood, an approximate one: the Cox partial likelihood in large samples is close to a normal in θ, and the approximation is the standard route taken by Bayesian design in survival studies. In the traditional trial the logrank test is, to a good approximation, the z test $z = \hat{\theta}\sqrt{d}/2$, and the one-sided p-value is $1 - \Phi(z)$. The number of events, not patients, is the sample size, in both frameworks; this chapter uses the same events-as-information language that Chapter 9 and Chapter 11 use.

Take a trial that observes $d = 200$ events and $\widehat{\mathrm{HR}} = 0.74$, so $\hat{\theta} = -\log 0.74 = 0.30$ with standard error $2/\sqrt{200} = 0.141$. Then $z = 2.12$ and the one-sided p-value is 0.017, a success at the 0.025 level.

For the prior take a normal distribution, $\theta \sim N(m_0, \tau^2)$. A normal prior and a normal likelihood give a normal posterior (normal-normal conjugacy), with precision (the reciprocal of the variance) equal to the sum of the precisions:

$$\theta \mid \hat{\theta} \sim N(m_1, s_1^2), \quad \frac{1}{s_1^2} = \frac{1}{\tau^2} + \frac{d}{4}, \quad m_1 = \frac{m_0/\tau^2 + \hat{\theta}\, d/4}{1/\tau^2 + d/4}. \tag{6.7}$$

This is the same weighted average of Equation 6.4, now with weights given by precision. The data have precision $d/4$, so each event carries a quarter of a unit of precision, and a prior of variance τ^2 carries $1/\tau^2$. The effective sample size of the prior, in events, is therefore $4/\tau^2$: a prior $N(0, 0.2^2)$ is worth 100 events, and $N(0, 0.35^2)$ is worth about 33.

```
d <- 200; theta_hat <- 0.30; se <- 2 / sqrt(d)
margin <- log(1 / 0.9)                                  # effect margin a: HR
 ↪  below 0.9
z <- theta_hat / se
priors <- c("flat N(0, 10^2)" = 10, "skeptical N(0, 0.35^2)" = 0.35,
 ↪  "strong skeptical N(0, 0.20^2)" = 0.20)
tab <- do.call(rbind, lapply(names(priors), function(nm) {
  s0 <- priors[[nm]]; u <- normal_update(0, s0, theta_hat, se)
  ci <- exp(-(u$mean + c(1, -1) * 1.96 * u$sd))
  data.frame(nm, 4 / s0^2, u$mean, exp(-u$mean), pnorm(0, u$mean, u$sd,
   ↪  lower.tail = FALSE),
             pnorm(margin, u$mean, u$sd, lower.tail = FALSE), sprintf("%.2f
              ↪  to %.2f", ci[1], ci[2]))
}))
knitr::kable(tab, digits = c(0, 0, 3, 2, 3, 3, 0), align = "lrrrrrr",
  col.names = c("Prior", "Prior ESS (events)", "Posterior mean", "Posterior
   ↪  HR", "Pr(theta > 0)", "Pr(HR < 0.9)", "95% credible interval for
   ↪  HR"))
```

Table 6.9.: Posterior for theta = -log HR with 200 events and an observed hazard ratio of 0.74, under a flat, a skeptical and a strong skeptical normal prior. Pr(HR < 0.9) is the posterior probability that the hazard ratio is below the margin 0.9.

Prior	Prior ESS (events)	Posterior mean	Posterior HR	Pr(theta > 0)	Pr(HR < 0.9)	95% credible interval for HR
flat N(0, 10^2)	0	0.300	0.74	0.983	0.916	0.56 to 0.98
skeptical N(0, 0.35^2)	33	0.258	0.77	0.975	0.878	0.60 to 1.00
strong skeptical N(0, 0.20^2)	100	0.200	0.82	0.958	0.794	0.65 to 1.03

Table 6.9 gives the three posteriors. With a flat prior (Figure 6.3 shows the skeptical case), the posterior is $N(0.300, 0.141^2)$ and $\Pr(\theta > 0 \mid \hat{\theta}) = 0.983$. That is $1 - p$ again, to within the third decimal place ($1 - 0.017 = 0.983$), and now exactly: in the normal-normal model with a flat prior, $\Pr(\theta > 0 \mid \hat{\theta}) = \Phi(\hat{\theta}/\mathrm{se})$ is identically one minus the one-sided p-value, so

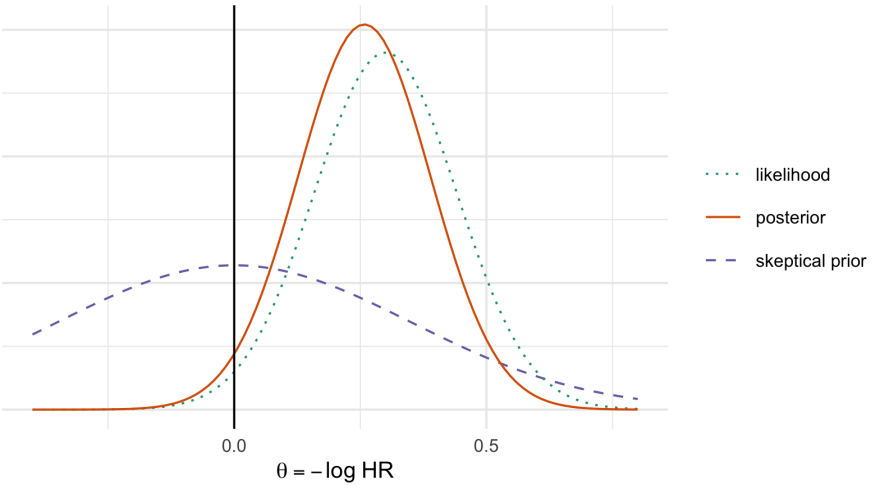


Figure 6.3.: Prior, likelihood and posterior for theta = -log HR with 200 events and an observed hazard ratio of 0.74. The skeptical prior N(0, 0.35^2) pulls the posterior toward no effect.

the agreement of the previous section is an identity here. With a skeptical prior of ESS 33 events, the posterior mean is 0.258 (HR 0.77), $\Pr(\theta > 0) = 0.975$ and the 95% credible interval for the hazard ratio is about 0.60 to 1.00. With the stronger skeptical prior (ESS 100) the posterior mean drops to 0.200 (HR 0.82) and the credibility of any benefit falls to 0.958. The same data under three priors give 0.983, 0.975 and 0.958: at a threshold of 0.975 the first passes, the second passes only in the fourth decimal, and the third fails. This is why the prior must be specified before the data, and why Chapter 8 is about how to choose one that a regulator will accept.

> **Technical note: normal-normal in one line**
>
> Complete the square. The product of $\exp\{-(\theta - m_0)^2/2\tau^2\}$ and $\exp\{-(\hat{\theta} - \theta)^2 d/8\}$ is proportional to $\exp\{-\frac{1}{2}(1/\tau^2 + d/4)\,\theta^2 + (m_0/\tau^2 + \hat{\theta}\,d/4)\,\theta\}$, which is the kernel of a normal with precision $1/\tau^2 + d/4$ and mean the ratio of the linear to the quadratic coefficient. The approximation in Equation 6.6 is a modeling choice, not an exact result: it treats $\hat{\theta}$ as sufficient for the log hazard ratio, which holds only approximately for a Cox model, and it ignores the discreteness of event times. For design purposes it is the usual choice; the final analysis in a Bayesian trial that wants the exact partial or full likelihood can use a piecewise exponential model.

6.10. Sequential updating: the posterior today is the prior tomorrow

A trial accrues over months or years. Suppose data arrive in batches, y_1 and then y_2. Bayes' rule can be applied twice: first to the prior and y_1, giving $\pi(\theta \mid y_1)$, then to $\pi(\theta \mid y_1)$ taken as the prior and y_2:

$$\pi(\theta \mid y_1, y_2) \;\propto\; p(y_2 \mid \theta)\,\pi(\theta \mid y_1) \;\propto\; p(y_2 \mid \theta)\,p(y_1 \mid \theta)\,\pi(\theta), \tag{6.8}$$

provided y_1 and y_2 are conditionally independent given θ (which is what exchangeability delivers). The last expression is the posterior obtained from all data at once with the original prior. Updating in batches, one patient at a time or all at once, gives the same posterior: the posterior today is the prior tomorrow, and the path by which the data arrived does not matter.

```
r <- c(11, 9, 14, 11); m <- rep(25, 4)              # responders and
 ↪  patients per batch, total 45/100
a <- 1; b <- 1; grid <- seq(0, 1, length.out = 500)
curves <- data.frame(x = grid, y = dbeta(grid, a, b), stage = "prior")
for (k in 1:4) {
  a <- a + r[k]; b <- b + m[k] - r[k]
  curves <- rbind(curves, data.frame(x = grid, y = dbeta(grid, a, b), stage
 ↪  = paste("after", 25 * k)))
}
stopifnot(a == 1 + sum(r), b == 1 + 100 - sum(r))   # same as the one-step
 ↪  update with all 100
curves$stage <- factor(curves$stage, unique(curves$stage))
ggplot(curves, aes(x, y, colour = stage)) + geom_line() +
 ↪  scale_colour_brewer(palette = "Dark2") +
  labs(x = "treatment response probability", y = NULL, colour = NULL) +
   ↪  theme_minimal(base_size = 9) +
  theme(axis.text.y = element_blank())
```

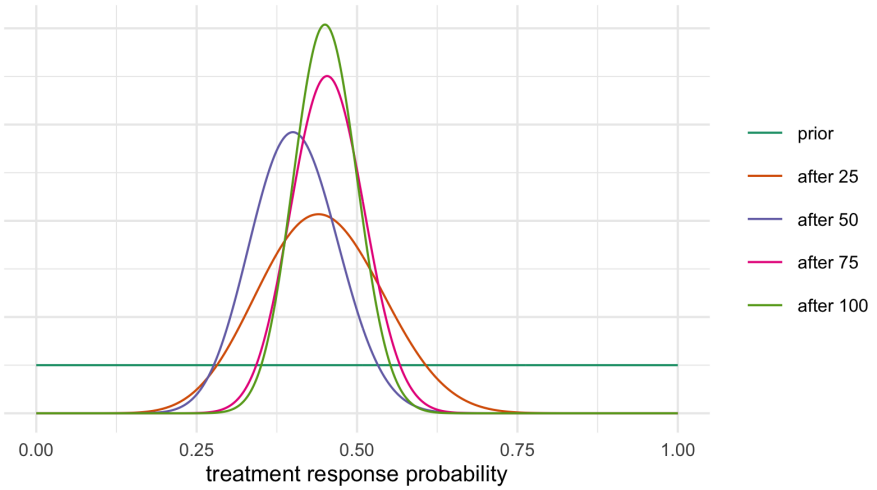


Figure 6.4.: Posterior of the treatment response rate as 100 patients accrue in four batches of 25, starting from a flat prior. The last curve is also the posterior from all 100 patients at once.

The assertion at the end of the chunk (and the last curve of Figure 6.4) is the whole point: the four-step posterior is $\mathrm{Beta}(46, 56)$, the same as the one-step posterior of Section 6.5.

For conjugate models the batch-wise update is just a running total of responders and non-responders, which is why Bayesian interim analyses are cheap computationally: the interim posterior is a handful of arithmetic operations, and a Data Monitoring Committee can be given it at any time.

6.10.1. Looking does not change the posterior

The traditional trialist's reaction to the last paragraph is that something has been skipped. In a group sequential design (the subject of Chapter 11) every look at the data costs something: the nominal level at each look is reduced, because testing five times at 0.025 does not preserve a type I error of 0.025. Yet the Bayesian posterior at any interim analysis is computed with exactly the same formula as the posterior at the end, and it does not know how many looks preceded it. How can both statements be true?

They are answers to different questions. The **likelihood principle** says that, given the data, the evidence about θ in an experiment depends on the data only through the likelihood function, and not on the stopping rule or on what other results might have been observed but were not (J. O. Berger and Wolpert 1988). The posterior depends on the data through the likelihood only, so it is unchanged by the decision to look, or by the decision to stop because the posterior crossed a threshold. A p-value, by contrast, is a statement about all the data sets the procedure could have produced, and which data sets those are depends on the stopping rule: "stop at the first of five looks where $z > 1.96$" is a different procedure from "analyze once at the end," and has a different sampling distribution. The type I error rate is a property of the procedure over repeated trials under the null, so it is changed by looking. The posterior probability, a statement about this trial's parameter given this trial's data, is not.

We can see both statements in one simulation of the trial of this chapter. Five looks at 50 patients per arm (cumulative n = 50, 100, 150, 200, 250 per arm).

```
res <- sim_sequential_looks(n_trials = 4000, n_look = 50, n_looks = 5, seed
 ↪  = 11)
knitr::kable(data.frame(Setting = c(rep("Null true (p1 = p0 = 0.30),
 ↪  one-sided z > 1.96", 2), rep("Truth drawn from flat priors, stop at
 ↪  first look with Pr(p1 > p0 | y) > 0.975", 3)),
  Quantity = c("Rejection rate, one look at the end", "Rejection rate, any
   ↪  of 5 looks", "Trials that stopped (of 4000)",
                "Mean posterior probability at stopping", "Fraction of
                 ↪  stopped trials where p1 > p0 truly"),
  Value = c(sprintf("%.3f", res$final_only), sprintf("%.3f", res$any_look),
   ↪  sprintf("%d", res$n_stopped), sprintf("%.4f", res$mean_post_at_stop),
   ↪  sprintf("%.4f", res$frac_truly_better))),
  align = "llr")
```

Table 6.10.: Simulation of five looks at 50 patients per arm (4000 trials): frequentist rejection rates under the null, and the behavior of a Bayesian stopping rule when the truth is drawn from the prior used in the analysis.

Setting	Quantity	Value
Null true (p1 = p0 = 0.30), one-sided z > 1.96	Rejection rate, one look at the end	0.024
Null true (p1 = p0 = 0.30), one-sided z > 1.96	Rejection rate, any of 5 looks	0.069
Truth drawn from flat priors, stop at first look with Pr(p1 > p0 | y) > 0.975	Trials that stopped (of 4000)	1776
Truth drawn from flat priors, stop at first look with Pr(p1 > p0 | y) > 0.975	Mean posterior probability at stopping	0.9969
Truth drawn from flat priors, stop at first look with Pr(p1 > p0 | y) > 0.975	Fraction of stopped trials where p1 > p0 truly	0.9955

Table 6.10 has two blocks. The first is the frequentist finding: with the null true and five looks at the nominal one-sided 0.025 level, the chance of declaring success at some look is 0.069, not 0.025 (the final-look-only rate in this simulation is 0.024). That is the reason group sequential designs adjust the thresholds, and the reason regulators insist that the monitoring plan be prespecified (ICH E9 Section 4.5 on interim analysis). The second is the Bayesian finding. When the true response rates are drawn from the prior that the analysis assumed, the trials that stop at a looked-at threshold of 0.975 have, on average, a posterior probability of 0.9969 at the moment they stop, and the fraction of those trials in which the treatment is truly better is 0.9955. The posterior is right on average among the trials that stopped, despite the five opportunities to stop.

Two cautions keep this from being an overstatement.

- Stopping-rule invariance follows from the likelihood factorization: for an ignorable stopping mechanism and a fixed analysis prior, the stopping factor cancels. Calibration across trials additionally depends on the assumed prior and sampling model. The calibration in the second block was produced by drawing truth from the same prior the analysis used. If the true effects are not drawn from that prior, then the posterior at the moment of stopping is not calibrated to anything in the real world, and the frequentist inflation reappears as an operating characteristic: the fraction of null drug trials that stop for success is what the regulator cares about, and it is a property of the design and the threshold, not of the posterior at a single look. For this reason one of the routes the FDA draft describes, and this book's default for confirmatory adaptive designs without borrowing, is to calibrate Bayesian success criteria by simulation to the type I error probability, and E20 Section 5.3 describes the same operating characteristics (U.S. Food and Drug Administration 2026g; International Council for Harmonisation 2025a). Chapter 14 shows how.
- Having a valid posterior at every look does not make every look free. Looks cost in the currency of operating characteristics: a threshold chosen to be right at a single look must be raised, or the number of looks limited, if the design is to hold the type I error at 0.025 across the monitoring plan. Chapter 11 quantifies the price. What the likelihood principle removes is the conceptual objection that Bayesian inference "does not know about" the looks; what it does not remove is the regulator's requirement, in practice, that it be shown that it does not matter.

There is a debate, spanning decades, about what the likelihood principle implies for the way trials are run and judged (J. O. Berger 1985; Edwards, Lindman, and Savage 1963).

6.11. What a protocol has to say

In the protocol

A Bayesian primary analysis needs, at minimum, the following statements in the protocol or the statistical analysis plan. They follow from the definitions of this chapter and from the FDA draft's emphasis on prespecification.

1. The model: the sampling distribution of the endpoint given the parameter (for example, binomial with response probability p_j in arm j; or $\widehat{\theta} \sim N(\theta, 4/d)$ for $\theta = -\log \mathrm{HR}$).
2. The prior for every parameter, with its form and values (for example, $p_j \sim \mathrm{Beta}(1,1)$ for both arms), its effective sample size, and the reasoning behind it. The prior is specified before the data are unblinded.
3. The quantity on which the decision is made, with its threshold (for example, $\Pr(p_1 - p_0 > 0 \mid \text{data}) > 0.975$, or the Pfizer-BioNTech rule $\Pr(\mathrm{VE} > 30\%) > 0.986$ at the final analysis and > 0.995 at interims), so that the success criterion is explicit.
4. The computation: whether the posterior is analytic or by Monte Carlo (and with how many draws, and with what seed and software), so that the result can be reproduced.
5. The operating characteristics: type I error and power by simulation under the stated scenarios, referred to in the simulation report (Chapter 14).
6. A statement of how the traditional analysis (estimate, confidence interval, p-value) will be reported alongside, if the sponsor wants the Bayesian analysis to be read against it.

6.12. In practice: two trials that stated their success rule as a posterior probability

6.12.1. Pfizer-BioNTech BNT162b2, phase 3 (C4591001)

The phase 2/3 trial of the BNT162b2 mRNA vaccine is the cleanest large example of this chapter's machinery. Its statistical analysis plan [Pfizer Inc. (2022), Section 7.1 and Appendix 2; the document at the cited ClinicalTrials.gov address is version 10, dated 7 October 2022, and is not an archived 2020 version] used a beta-binomial model, conditional on the total number of first COVID-19 cases s, of whom s_1 occur in the vaccine arm. With equal allocation and equal follow-up, a case falls in the vaccine arm with probability $p = (1 - \mathrm{VE})/(2 - \mathrm{VE})$, where VE is the vaccine efficacy, one minus the ratio of incidence rates. Then $s_1 \mid s \sim \mathrm{Binomial}(s, p)$, and the prior is $p \sim \mathrm{Beta}(0.700102, 1)$. Its mean is 0.4118, the value of p at $\mathrm{VE} = 30\%$, so the prior is centered on the efficacy margin, and the plan calls it minimally informative and says it can be considered pessimistic. Its 95% interval

for p is $(0.005, 0.964)$, which is $(-26.2, 0.995)$ for VE written as a fraction, about $-2{,}620\%$ to 99.5%: the lower limit corresponds to an incidence rate ratio of 27.2, a vaccine arm with 27 times the cases of the placebo arm, and the interval is nearly the whole admissible range, as a weak prior should be. The ESS of Section 6.4 is $a + b = 1.7$ cases.

The success criterion was a posterior probability, as in the form $\Pr(\delta > a \mid \text{data}) > c$ above: efficacy is declared if $\Pr(\text{VE} > 30\% \mid \text{data}) > 0.986$ at the final analysis, and if it exceeds 0.995 at an interim analysis. The plan states that the two thresholds were calibrated so that the overall one-sided type I error is 2.5%, which is the regulator's language from Section 6.11 applied to a Bayesian rule. The final analysis was planned at 164 cases, with interim looks at 62, 92 and 120 cases (a look at 32 cases was in the plan and not performed, and the thresholds were unchanged). The trial reported 8 cases with onset at least 7 days after the second dose in the vaccine group against 162 in the placebo group (Polack et al. 2020), so $s = 170$ and $s_1 = 8$. The posterior is $\text{Beta}(0.700102 + 8,\ 1 + 162)$, and everything follows from base R.

```
a0 <- 0.700102; b0 <- 1                                  # prior on p = (1 -
 ↪  VE) / (2 - VE)
ve <- function(p) (1 - 2 * p) / (1 - p)   # VE as a function of p
post <- beta_post(8, 170, a0, b0)                        # 8 vaccine cases out
 ↪  of 170 (8 vs 162)
p30 <- 0.7 / 1.7                                      # p at VE = 30%
pr_fail <- pbeta(p30, post$a, post$b, lower.tail = FALSE)   # Pr(VE < 30% |
 ↪  data)
q <- 100 * ve(qbeta(c(0.975, 0.5, 0.025), post$a, post$b))      # VE is
 ↪  decreasing in p
knitr::kable(data.frame(Quantity = c("Prior mean of p", "Prior effective
 ↪  sample size", "Prior 95% interval for VE (as a fraction)", "Posterior
 ↪  of p",
    "Pr(VE < 30% | data)", "Posterior median VE", "95% credible interval
     ↪  for VE", "Plug-in VE, 1 - IRR"),
  Value = c(sprintf("%.4f", a0 / (a0 + b0)), sprintf("%.1f", ess_beta(a0,
   ↪  b0)), sprintf("(%.1f, %.3f)", ve(qbeta(0.975, a0, b0)),
   ↪  ve(qbeta(0.025, a0, b0))),
    sprintf("Beta(%.6f, %g)", post$a, post$b), sprintf("%.1e", pr_fail),
     ↪  sprintf("%.1f%%", q[2]), sprintf("(%.1f%%, %.1f%%)", q[1], q[3]),
     ↪  sprintf("%.1f%%", 100 * ve(8 / 170)))),
  align = "lr")
```

Table 6.11.: Reproduction of the BNT162b2 analysis: prior and posterior for p = (1 - VE)/(2 - VE) with 8 vaccine cases among 170, and the implied vaccine efficacy (VE).

Quantity	Value
Prior mean of p	0.4118
Prior effective sample size	1.7
Prior 95% interval for VE (as a fraction)	(-26.2, 0.995)
Posterior of p	Beta(8.700102, 163)
Pr(VE < 30% | data)	2.0e-28
Posterior median VE	94.9%

Table 6.11.: Reproduction of the BNT162b2 analysis: prior and posterior for p = (1 - VE)/(2 - VE) with 8 vaccine cases among 170, and the implied vaccine efficacy (VE).

Quantity	Value
95% credible interval for VE	(90.4%, 97.6%)
Plug-in VE, 1 - IRR	95.1%

Table 6.11 gives the prior and the posterior. The posterior probability that VE exceeds 30% is $1 - 2 \times 10^{-28}$, which clears 0.986. The 95% credible interval for VE, the equal-tailed interval of the posterior of p transformed through the monotone map $p \mapsto \text{VE}$, is about $(90.4\%, 97.6\%)$, and the point estimate is 95.1% (plug-in) or 94.9% (posterior median). The paper reports 95.0% with 95% credible interval 90.3 to 97.6 (Polack et al. 2020), so the chapter's calculation reproduces it to within a tenth of a point on the lower limit; we did not model the small difference in surveillance time between the arms, which is the likely source. Two lessons for the chapter's method. First, a Bayesian rule can be stated entirely as a threshold on a posterior probability and still be calibrated to a type I error, so the regulator reads it as it would read a boundary on a test statistic. Second, with ESS = 1.7 and 170 cases the prior is irrelevant to the conclusion, and a different prior, such as a flat $\text{Beta}(1, 1)$, would shift the interval by well under a point. The prior was chosen to be defensible, not to carry the result. The FDA authorized the vaccine for individuals 16 years and older on 11 December 2020 (U.S. Food and Drug Administration 2021a), and approved the biologics license on 23 August 2021. The public record used here is the statistical analysis plan, the trial report and the authorization notice; the agency's own assessment of the Bayesian criterion is not summarized. Chapter 8 returns to pessimistic priors on the margin, and Chapter 15 to the futility rule in the same plan.

6.12.2. REMAP-CAP: posterior probabilities of benefit as the reported result

REMAP-CAP is a platform trial (Chapter 20) whose primary analysis is a Bayesian cumulative logistic model for organ support-free days to day 21. The report on interleukin-6 receptor antagonists (REMAP-CAP Investigators, Gordon, et al. 2021) states the rules as posterior probabilities, fixed in advance: efficacy if the posterior probability that the odds ratio exceeds 1 is above 99%, futility if the posterior probability that the odds ratio exceeds 1.2 is below 5%, and equivalence if the probability that the odds ratio between two non-control interventions lies between 0.83 and 1.2 exceeds 90%. Priors on the individual treatment effects were neutral. At an interim analysis in 2020, tocilizumab crossed the efficacy threshold with posterior probability 99.75%. The chronology and final estimates are in Chapter 20. The lesson is the one of Section 6.5 at scale: the result is a posterior probability checked against a prespecified threshold, the stopping time is determined by when it crosses and not by a fixed sample size, and the reader is told, in the same units, how far above the threshold the evidence was. The posterior being valid at each look (Section 6.10) is what lets the platform report while it continues.

6.13. Summary: what changed, what stayed

Starting from the traditional two-group comparison, the Bayesian version changed three things and left a fourth alone.

- **The unknown becomes a distribution.** The treatment effect is described by a prior, updated by data into a posterior, and statements are probabilities about the effect.
- **The inputs gain a prior.** The prior has to be chosen, prespecified and defended; it is worth a stated number of patients (its effective sample size).
- **The outputs change meaning.** The credible interval is a probability statement about the parameter, $\Pr(p_1 > p_0 \mid y)$ is a probability that the treatment is better, and posterior predictive distributions say what future patients will do.
- **The data's role does not change.** The likelihood is the same one the traditional trial uses, and with a flat prior and a large trial the numbers agree: $\Pr(p_1 > p_0 \mid y) = 0.9855$ against $1 - p = 0.9858$ here, and exactly $1 - p$ in the normal-normal model.

The regulator's position, stated in ICH E9 and reaffirmed in the FDA draft and E20, is that the Bayesian approach is acceptable when the reasons for its use are clear and the conclusions are robust. Robust here is a technical word: the conclusion should survive a reasonable change of prior, and the design should have operating characteristics, demonstrated by simulation, that a regulator can read in the language of error rates. Chapter 7 takes up the weighing of evidence, Chapter 8 the choice of priors, and Chapter 11 and Chapter 15 the use of the posterior and the predictive distribution in monitoring.

> Try it in the lab
>
> Open module 01 of the companion lab, Prior and posterior. Before pressing "Toss 10," predict where the posterior will peak after 7 heads and 3 tails starting from a flat prior and from the Beta(2,2) prior (the modes are $7/10 = 0.70$ and $8/12 = 0.667$; the means are $8/12 = 0.667$ and $9/14 = 0.643$, as in Section 6.3). Then load the "Newborn trial (real)" example and note how a single arm of 11 successes concentrates the posterior near 1 even from a flat start.

Key points

- Bayes' rule reverses a conditional probability: posterior is proportional to prior times likelihood. The diagnostic-test example, with a 27% positive predictive value from a 90% sensitive, 95% specific test at 2% prevalence, is the same logic as a trial posterior.
- Exchangeability, the judgment that the order of patients does not matter, replaces independence; de Finetti's theorem turns it into a likelihood and a prior. It can fail. Randomization makes treatment assignment independent of prognosis; the response model separately assumes exchangeability within each arm or stratum.
- The Beta prior is conjugate to the binomial: the update adds responders and non-responders. The posterior mean is a weighted average of prior mean and sample proportion, and the prior's effective sample size is $a + b$.
- With flat priors, $\Pr(p_1 > p_0 \mid y) = 0.9855$ matches $1 - p = 0.9858$ for the trial of 45/100 versus 30/100, and across simulated trials the two differ by at most 0.002. In

the normal-normal model on $\theta = -\log \mathrm{HR}$ the match is exact. A Bayesian threshold of 0.975 with a flat prior is not a lower bar than a one-sided 0.025.

- A confidence interval describes a procedure and a credible interval describes a parameter. Predictive distributions, which add the uncertainty about the parameter to the sampling variability, are wider than plug-in predictions and are the basis of predictive probability.
- The posterior at an interim look is the same function of the data whatever the stopping rule, but the type I error of a stopping rule is not; the first is a statement about this trial, the second about repeated trials. Regulators judge a Bayesian design by the second, which is why the FDA draft recommends prior prespecification and evaluated operating characteristics, with calibration to type I error as one of its three routes and this book's default.

Discussion questions

1. A sponsor states that "the p-value of 0.014 means there is a 98.6% probability that the drug works." Under what assumptions is the number right, what is the claim's problem, and how would you rephrase it for the clinical study report?
2. A Beta(30, 70) prior is proposed for the control arm from a historical trial of 100 patients. What is its effective sample size, how much of the posterior mean would it contribute in a current trial with 50 control patients, and what would you ask the sponsor to show about the exchangeability of the historical and current patients?
3. A Data Monitoring Committee is shown, after 150 events, a posterior probability of benefit of 0.96 under a skeptical prior and 0.99 under a flat prior. What would you need to know to decide which number to rely on, and what does the protocol have to have said beforehand?
4. A colleague says Bayesian monitoring is free because the posterior is unaffected by looking. Using the simulation of the previous section, explain what is true in that sentence and what is not.
5. ICH E9 permits Bayesian approaches when "the reasons for their use are clear and when the resulting conclusions are sufficiently robust." Draft two sentences for a protocol that meet that expectation for the normal-normal analysis of Section 6.9.

7. Weighing Evidence: p-values, Bayes Factors, and Posterior Probabilities

A trial ends with a number, and the number has to be turned into a sentence: the drug works, the drug does not work, we cannot tell. Three numbers compete to play that role. The p-value is on the first page of nearly every clinical study report. The Bayes factor is the number a statistician reaches for when asked how strongly the data favor one hypothesis over another. The posterior probability of benefit is the number a physician wishes the p-value were, and it is the quantity the January 2026 FDA draft guidance on Bayesian methods puts in its success criterion. They answer three different questions, they can disagree sharply on the same data, and a protocol that names one of them has made a choice about what counts as evidence.

This chapter lines them up on one example and one set of numbers. We start with the traditional quantity, the p-value, and say precisely what it is and what it is not. We then ask the question every sponsor and every regulator actually cares about, how often a "significant" result is a false alarm, and see that the answer depends on something the p-value does not contain. The Bayes factor and its built-in penalty for vague alternatives follow, with Lindley's paradox as the cleanest demonstration that a fixed p-value does not mean a fixed amount of evidence. A first look at the arithmetic of "how much evidence is enough" (the first two of the three deviation scales of Polson and Sokolov) comes next. The chapter closes on the criterion that regulators now write down for Bayesian trials, the posterior probability that the effect exceeds a threshold, and on how its two dials map onto the familiar one-sided 0.025.

! Regulatory anchor

- **ICH E9, Statistical Principles for Clinical Trials** (Step 4, February 1998). Section 1.2 and the Glossary acknowledge that the guideline is written mostly in frequentist language and do not exclude Bayesian approaches when their use is justified and the conclusions are robust. Section 5.5 sets out estimation, confidence intervals and hypothesis testing. Section 5.6 states that, when multiplicity is present, the usual frequentist approach "may necessitate an adjustment to the type I error." Section 2.2.2 requires the primary variable to be defined in the protocol before the data are seen.
- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products** (draft guidance, January 2026), section IV.A. For trials with an overall frequentist statistical analysis plan, success criteria are "almost always" chosen so that the familywise type I error is no greater than 0.025, one-sided. The draft describes three routes for Bayesian success criteria: calibration to a type I error rate, direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA), and decision-theoretic or benefit-risk criteria. A Bayesian criterion takes the form Pr(delta > a) > c, the posterior

probability that the true effect exceeds a threshold, and "a and c can be chosen such that the overall FWER is controlled at a given level, typically 0.025 one-sided" (section IV.A.1). A draft recommends and is not binding.

- **ICH E20, Adaptive Designs for Clinical Trials** (Step 2 draft, endorsed 25 June 2025), section 5.3. A Bayesian adaptive design "can achieve targeted operating characteristics … while maintaining Type I error probability control." Section 3.3 calls control of false positive conclusions "an essential element of regulatory decision-making."
- **ASA Statement on p-values** (Wasserstein and Lazar, 2016) is not a regulation, but it is the consensus statement of the profession on what a p-value can bear, and regulators and journals cite it.

The map of all these documents and their status is in Chapter 3.

7.1. What a p-value is

Chapter 4 built the p-value from a two-group comparison. Here is the definition again, stated with the care that the rest of the chapter needs. A trial has a null hypothesis H_0 (the new treatment has no effect, $\theta = 0$) and a test statistic T that is large when the data look unlike H_0 in the direction of benefit. If the observed value is t_{obs}, then

$$p = \Pr(T \geq t_{\text{obs}} \mid H_0),$$

computed over the repeated trials one could imagine running if H_0 were true and the same design were repeated. For a one-sided z test of a treatment effect, $T = Z_n = \hat{\theta}_n \sqrt{I_n}$, where I_n is the Fisher information after n observations (one over the squared standard error of $\hat{\theta}_n$), and $p = 1 - \Phi(z_{\text{obs}})$. The regulatory convention is a one-sided $\alpha = 0.025$, which corresponds to $z_\alpha = 1.96$ and to the familiar two-sided 0.05 (this book uses one-sided 0.025 throughout unless it says otherwise).

Section 4.5 discusses what the definition does and does not say. Three points from it matter here. The p-value is a statement about the data under H_0, not about H_0 under the data. It depends on the experiment that would have been run, not only on the data in hand (the 9 heads and 3 tails of that section have different p-values under different stopping plans; module 2 of the companion lab makes this concrete). And it measures surprise at the null without saying how well any alternative explains the data.

None of this makes the p-value a bad tool. It is the basis of the type I error control that regulators require, and its reading as "how often would a trial of this design, run on an ineffective drug, look this good" is exactly the property a regulator wants to bound. The trouble starts when it is asked to answer a question it was not built for.

7.1.1. The ASA statement and the dirty dozen

The ASA statement (Wasserstein and Lazar 2016) and Goodman's "dirty dozen" (S. Goodman 2008) are summarized in Section 4.5. Three of the twelve misreadings recur in every trial team and each matters for what follows.

- **"p = 0.03 means there is a 3% chance the null hypothesis is true."** False. The p-value is a probability of data given the null. The probability of the null given the data requires a prior probability for the null, and the p-value has none. The next section computes what the answer actually is.
- **"A non-significant result means there is no effect."** False. EOLIA reported $p = 0.09$ for 60-day mortality and was labeled "not significant," yet a later Bayesian reanalysis put the probability that ECMO lowers mortality between 88% and 99%, depending on the prior (full account in Section 7.10). A p-value above 0.025 or 0.05 says the trial did not clear the bar. It does not say the treatment does nothing.
- **"p = 0.05 in one trial and p = 0.001 in another means the second effect is bigger."** False. The p-value mixes the size of the effect with the size of the trial. A modest effect in a huge trial can have a tiny p-value, and a large effect in a small trial can have a p-value above 0.05.

The fix the ASA statement suggests, and that the regulatory documents already implement, is not to abolish the p-value but to surround it: a pre-specified primary analysis, an effect estimate with an interval, and a clear account of everything that was looked at.

7.2. The false positive risk

Suppose a sponsor tells you its phase 3 trial produced $p = 0.02$ in a one-sided test. What is the probability that the drug does not work? The p-value cannot answer, but one more piece of information can: how often do drugs of this kind work to begin with?

The calculation is the screening-test calculation that every clinician has done for a diagnostic test. A test with sensitivity 90% and false alarm rate 2.5% applied to a population where 1% have the disease will produce many more false positives than true positives, because the 99% without disease generate 0.025 times 99 false alarms for every 0.90 times 1 true detections. The trial is the test, the drug is the patient, and:

- **sensitivity** is the power $1 - \beta$ of the trial, the chance of a significant result when the drug works;
- **false alarm rate** is the type I error α;
- **prevalence** is the **prior probability π_a that the drug works**, which is the fraction of drugs entering a trial of this kind that are in fact effective.

Among all trials that end "significant," the fraction in which the drug does not work is

$$\mathrm{FPR} \;=\; \Pr(\text{no effect} \mid \text{significant}) \;=\; \frac{\alpha\,(1-\pi_a)}{\alpha\,(1-\pi_a) + (1-\beta)\,\pi_a}. \tag{7.1}$$

This is Bayes' theorem applied to the event "the trial was significant," and it is the calculation behind Ioannidis's argument that most published research findings can be false (Ioannidis 2005) and behind Colquhoun's estimates of the false discovery rate in the medical literature (Colquhoun 2014). The same Bayes-theorem arithmetic appeared earlier in Soric's bound on the proportion of false discoveries (Sorić 1989) and, in Sterne and Davey Smith (Sterne and Davey Smith 2001) and, as the false positive report probability, in Wacholder and colleagues (Wacholder et al. 2004); Colquhoun later called it the false positive risk (Colquhoun 2017). Note that Equation 7.1 conditions on the trial being significant at level α, not on the exact

p-value; this is the version that bounds the long-run share of false claims among approved drugs, which is the version that matters for a regulator.

Table 7.1.: False positive risk after a significant trial, by the prior probability that the drug works. One-sided alpha = 0.025.

prior P(works)	FPR, power 90%	FPR, power 80%	FPR, power 50%
0.50	0.027	0.030	0.048
0.30	0.061	0.068	0.104
0.20	0.100	0.111	0.167
0.10	0.200	0.220	0.310
0.05	0.345	0.373	0.487
0.01	0.733	0.756	0.832

If half of all drugs entering a confirmatory trial work, a significant result at one-sided 0.025 with 90% power is wrong about 2.7% of the time, close to the nominal 0.025. If one in ten works, the false positive risk after a significant result is 20%, eight times the nominal level. If one in a hundred works, which is not an absurd figure for an early-stage mechanism with a poor track record, a significant result is more likely to be false than true even with a well-powered trial: 73% (Figure 7.1 plots the whole curve).

Two practical lessons follow. First, power is not only about not missing a real effect; it also protects against false positives, because the denominator of Equation 7.1 contains $(1-\beta)\pi_a$. An underpowered trial that happens to cross the line is more likely to have done so by chance. Second, the same type I error level means different things in different development programs. A phase 3 trial confirming a dose chosen after two positive phase 2 studies sits at the top of Table 7.1. A trial of a novel mechanism in a disease with a long record of failures sits near the bottom. The empirical record bears this out: in a review of phase III oncology trials that repeated a regimen already positive in phase II, only 28% were themselves positive (Zia et al. 2005) (Section 7.10). The regulator's decision to demand two trials, or one trial and confirmatory evidence (the FDA substantial-evidence draft, see Chapter 3), can be read as a way of pushing the prior probability of a true effect up before approval, and thus pushing the false positive risk down.

i What a sponsor should take from this

The false positive risk is not an alternative to the type I error rate. The type I error rate is Pr(significant | no effect) and is what a design controls. The false positive risk is Pr(no effect | significant) and is what a decision-maker wants. They are connected by the prior probability of an effect, and the Bayesian framework is the one that lets that prior enter the analysis openly instead of silently. In a regulatory submission, the prior still has to be defended (Chapter 8) and, for a confirmatory design without borrowing, this book calibrates it to a one-sided type I error of 0.025 (Chapter 14). Calibration is one of the routes the FDA draft describes, alongside direct interpretation of posterior probabilities and benefit-risk criteria, not a universal requirement.

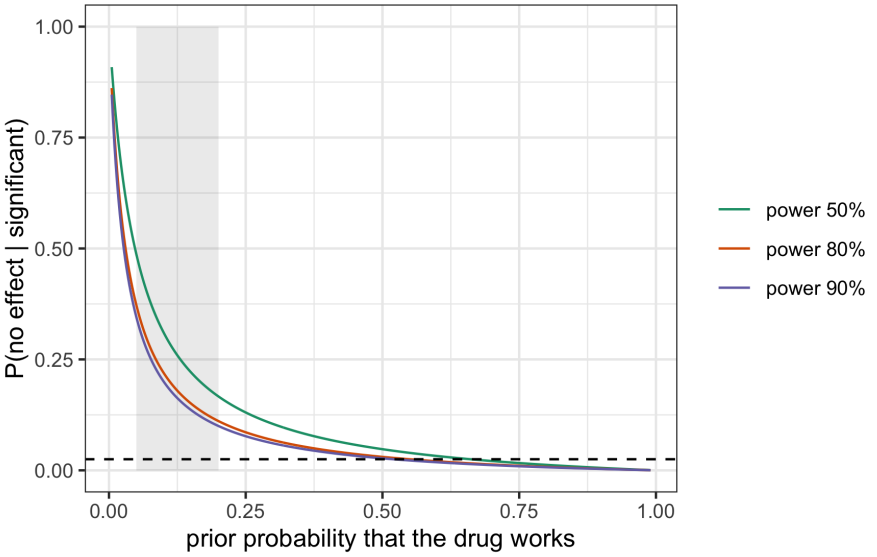


Figure 7.1.: False positive risk as a function of the prior probability that the drug works, for three levels of power. The dashed line is the nominal alpha = 0.025. The shaded band marks priors between 5% and 20%.

7.3. The exact p-value and the Bayes factor

The false positive risk above conditions on "significant at α." One can do better by conditioning on the actual data, and this requires comparing the null with a specific alternative.

The Bayesian measure of the evidence in the data for one hypothesis against another is the **Bayes factor**. For a null $H_0 : \theta = 0$ and an alternative H_1 under which θ has a prior distribution $\pi(\theta)$,

$$\mathrm{BF}_{10} \;=\; \frac{p(y \mid H_1)}{p(y \mid H_0)} \;=\; \frac{\int p(y \mid \theta)\, \pi(\theta)\, d\theta}{p(y \mid \theta = 0)},$$

the ratio of the probabilities the two hypotheses assigned to the data that actually occurred. It updates prior odds into posterior odds:

$$\underbrace{\frac{\Pr(H_1 \mid y)}{\Pr(H_0 \mid y)}}_{\text{posterior odds}} \;=\; \mathrm{BF}_{10} \times \underbrace{\frac{\pi_a}{\pi_0}}_{\text{prior odds}}, \tag{7.2}$$

where π_0 and π_a are the prior probabilities of the null and the alternative. The notation is the book's (see Appendix A); the sources in the Bayesian literature sometimes write the ratio the other way up, as BF_{01}, and the two are reciprocals.

Two things are worth noticing at once. The Bayes factor weighs the data against *both* hypotheses, which is what the p-value does not do. And it averages the likelihood under the alternative over the prior $\pi(\theta)$, rather than evaluating it at the best-fitting $\hat{\theta}$. This

averaging is where the Bayes factor differs most from the likelihood ratio statistic familiar from frequentist testing, and it is where its penalty for vague alternatives comes from.

7.3.1. A Gaussian example with closed form

To keep the algebra short, work on the scale where the data reduce to the estimate $\hat{\theta}_n$ with $\hat{\theta}_n \mid \theta \sim N(\theta, 1/I_n)$, so that $Z_n = \hat{\theta}_n \sqrt{I_n}$ is the usual standardized statistic and I_n is the Fisher information after n observations (for a time-to-event endpoint with d events and $\theta = -\log \mathrm{HR}$, $I_n \approx d/4$). Take the alternative to be $\theta \sim N(0, \tau^2)$, a prior centered at no effect with scale τ that expresses how large an effect is plausible. Then the integral can be done exactly. With $k = I_n \tau^2$ (the information in the data relative to the information in the prior),

$$\mathrm{BF}_{10} \;=\; (1+k)^{-1/2} \exp\!\left\{ \tfrac{1}{2} Z_n^2 \, \frac{k}{1+k} \right\}, \qquad 2 \log \mathrm{BF}_{10} \;=\; Z_n^2 \, \frac{k}{1+k} \;-\; \log(1+k). \quad (7.3)$$

This is the form given by Polson and Sokolov (Polson and Sokolov 2026b), and it is the finite-sample version of the Schwarz criterion (Schwarz 1978). For large k the factor $k/(1+k)$ tends to one and

$$2 \log \mathrm{BF}_{10} \;\approx\; \underbrace{Z_n^2}_{\text{fit}} \;-\; \underbrace{\log(1 + I_n \tau^2)}_{L_n,\ \text{Ockham penalty}} \; . \quad (7.4)$$

Read Equation 7.4 as a ledger. The first term is how well the best-fitting alternative explains the data, in standardized squared units, and it is the same quantity that the z test examines. The second term is a charge for flexibility. An alternative that spreads its prior over a wide range of effects θ gives most of that range to effects that the data rule out, so on average it predicts the data worse than a sharp alternative would. The more data we collect (the larger I_n), the more sharply the data pin down θ, and the larger the share of the prior that is wasted: the penalty grows like $\log I_n$. This is the **Ockham penalty**, the automatic, parameter-free version of "do not prefer a more complex hypothesis unless the data pay for it."

The practical content for a trial: with $\tau = 0.25$ on the log hazard ratio scale and $I_n = 400$ (about 1,600 events), the penalty is $L_n = \log(26) = 3.26$. The Bayes factor reaches 1 only when $Z_n^2 \approx 3.26$, that is $|Z_n| \approx 1.8$, and requires more than that for any substantial evidence. The p-value at $|Z_n| = 1.8$ is 0.036 one-sided, which is not "significant" at 0.025 and yet is, on this measure, only just break-even evidence for the alternative.

7.3.2. Lindley's paradox

Lindley (D. V. Lindley 1957), following Jeffreys (Jeffreys 1939) and Bartlett's reply (M. S. Bartlett 1957), noticed that the penalty can turn a rejection of the null into evidence for the null (the Jeffreys-Lindley paradox). Consider a sequence of trials of increasing size, every one of which ends with the same two-sided p-value of exactly 0.01, so that $|Z_n| = 2.576$ in every case. A frequentist reading says the evidence against the null is the same in each trial:

significant at 1%. The Bayes factor says something else, because the Ockham penalty keeps growing while Z_n^2 is stuck at 6.63.

Table 7.2.: Lindley's paradox. Every row has the same two-sided p-value, 0.01 (z = 2.576). Prior N(0, 0.25^2) on the effect under the alternative, unit standard deviation per observation, prior probability of the null 0.5.

n	BF10	P(H0 | data)
1e+01	2.81	0.262
1e+02	6.48	0.134
1e+03	3.29	0.233
1e+04	1.10	0.477
1e+05	0.35	0.741
1e+06	0.11	0.901

Table 7.2 and Figure 7.2 show the effect. At $n = 100$ the Bayes factor is 6.5 in favor of the alternative, and the posterior probability of the null is 13%: not as small as the p-value of 0.01 suggests, but pointing the same way. By $n = 10{,}000$ the Bayes factor is 1.1, the evidence is a wash, and the posterior probability of the null is 48%. At $n = 10^6$ the same p-value of 0.01 corresponds to a posterior probability of the null of 90%. The data that "reject the null at the 1% level" have become strong evidence *for* it.

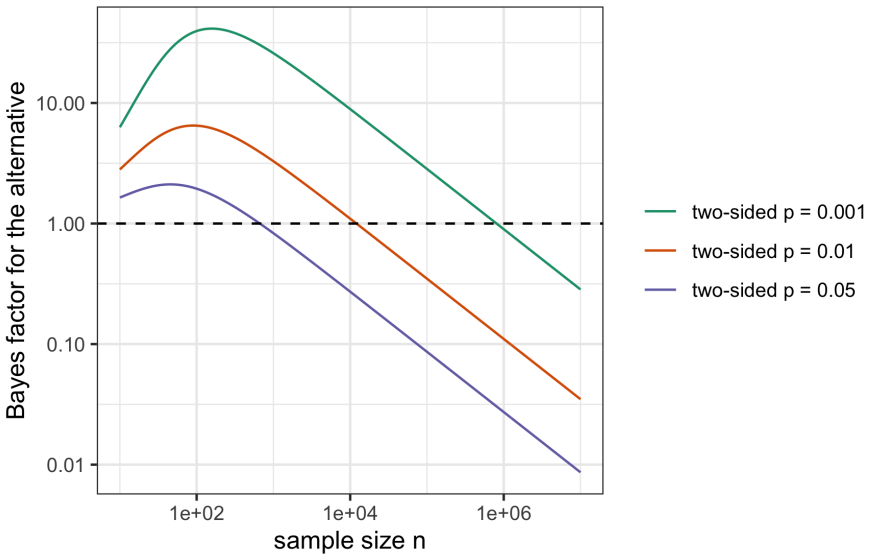


Figure 7.2.: Lindley's paradox. Holding the two-sided p-value at 0.01 (and also at 0.05 and 0.001), the Bayes factor for the alternative first rises and then falls as the sample size grows, crossing below 1 for large n. The dashed line is BF = 1.

How should a trialist react? There are two honest reactions and one dishonest one.

The honest reactions are these. First, the paradox lives at sample sizes far larger than those of most confirmatory trials of a large effect, and with a vague prior (τ large) on a point null. It is a warning that a fixed p-value threshold cannot mean a fixed amount of evidence as

the trial gets larger, which is why very large trials (cardiovascular outcome trials with tens of thousands of patients, platform trials with shared controls) sometimes find "significant" effects that are clinically trivial. The Bayesian remedy is a better question: instead of asking whether θ is exactly zero, ask whether θ exceeds a clinically relevant a. That is precisely the form of the FDA's criterion, as the last section of this chapter shows. Second, the paradox is a reminder that the answer depends on the prior τ under the alternative. A Bayes factor with τ ten times larger needs a much larger Z_n to reach the same conclusion. This sensitivity is real, it is not hidden by the frequentist approach but simply left out of it, and it is the reason Chapter 8 treats prior choice and sensitivity analysis at length.

The dishonest reaction is to pick τ after seeing the data to give the result you want. A Bayes factor is as open to that abuse as a p-value is to choosing the test statistic after the fact, and the remedy is the same: pre-specify it in the protocol.

Edwards, Lindman and Savage (Edwards, Lindman, and Savage 1963) gave lower bounds over all priors on the alternative, showing that a result significant at 5% is weak evidence against a point null at any sample size, and Berger and Sellke (J. O. Berger and Sellke 1987) sharpened the point with lower bounds on the posterior probability of a point null over classes of priors, the Sellke-Bayarri-Berger bound used in the next section. Jeffreys, who developed the posterior-odds test of a point null in the 1930s (Jeffreys 1935) and set it out in his book (Jeffreys 1998) (the Oxford reprint of the third edition), wrote that the use of the p-value implies that "a hypothesis that may be true may be rejected because it has not predicted observable results that have not occurred".

7.4. How small can the Bayes factor be? Calibrating the p-value

The Bayes factor depends on the prior under the alternative. Can we say anything that does not? Yes, at the cost of being deliberately generous to the alternative. For any prior on the alternative, the Bayes factor against the null cannot be larger than what the most favorable choice produces. Sellke, Bayarri and Berger (Sellke, Bayarri, and Berger 2001) showed that for a p-value $p < 1/e$ and for a class of alternatives (the p-value is uniformly distributed under the null, and under the alternative the distribution of $-\log p$ has a non-increasing hazard for $Y = -\log p$ (their Section 3.1.3), of which the Beta$(\xi, 1)$ family of p-value densities is the leading example), the Bayes factor in favor of the null is at least

$$\mathrm{BF}_{01} \;\geq\; -e\, p \log p. \tag{7.5}$$

This is a **minimum Bayes factor**, a bound on how strongly the data can favor the alternative within that class. The class matters: monotonicity of the p-value density alone is not enough, and the bound is not a statement about every prior. For $Z \sim N(\theta, 1)$ with $Z = 1.96$ and prior mass one half at each of $\theta = \pm 1.96$, the Bayes factor in favor of the null is about 0.29, below the bound of 0.41 at $p = 0.05$. The bound calibrates a p-value against the stated class, and it should be quoted with that qualification. Combined with prior odds on the null of one to one ($\pi_0 = \pi_a = 0.5$), it gives a lower bound on the posterior probability of the null:

$$\Pr(H_0 \mid y) \;\geq\; \left(1 + \frac{1}{-e\, p \log p}\right)^{-1}.$$

Table 7.3.: Calibration of two-sided p-values by the Sellke-Bayarri-Berger bound. 'Max BF10' is the largest Bayes factor for the alternative that any member of the class can produce. 'Min P(H0 | data)' assumes prior probability 0.5 on the null.

two-sided p	min BF01	max BF10	min P(H0|data)
0.100	0.626	1.6	0.385
0.050	0.407	2.5	0.289
0.025	0.251	4.0	0.200
0.010	0.125	8.0	0.111
0.005	0.072	13.9	0.067
0.001	0.019	53.3	0.018

A two-sided p-value of 0.05 corresponds at best to a Bayes factor of about 2.5 for the alternative, and to a posterior probability of the null of at least 29% if the null and alternative were equally likely at the outset. So the "5% significance" that is so often read as "95% sure" is, from this angle, at best "71% sure." A p-value of 0.01 corresponds to a Bayes factor of at most 8 and a posterior probability of the null of at least 11%. Only at $p = 0.001$ does the bound get to the neighbourhood of "strong evidence" (a Bayes factor of 53). This is the calculation behind the teal bar in module 2 of the companion lab ("the smallest chance that luck alone could still explain the result, given that p-value, assuming a 50:50 chance before the trial"). It is also an argument that has been made for tightening the default threshold for claims of new discoveries (Benjamin et al. 2018), a proposal that is controversial and does not apply to regulatory trials, where the one-sided 0.025 is set by decades of agency practice and by the substantial-evidence standard, not by a statistician's preference.

What the bound does not do is replace a design-specific calculation. It is generous to the alternative within its class (it asks which alternative would make the data look best for H_1), so it understates what a reasonable analysis would find. In a trial, the prior under the alternative comes from the phase 2 data, the mechanism, and the effect size the trial was powered for, and Chapter 8 and Chapter 14 show how to build and defend it.

The bound is a best case over a class of alternatives. A different route to the same kind of calibration fixes a Bayes factor threshold and asks which test reaches it most often. Johnson's uniformly most powerful Bayesian tests (Johnson 2013b) choose, for a one-parameter exponential family such as a normal mean, the alternative that maximizes the probability that the Bayes factor exceeds a stated threshold, and their link to classical uniformly most powerful tests gives an approximate calibration between p-values and Bayes factors. Johnson used it to argue for tighter standards of evidence (Johnson 2013a), a proposal in the direction of Benjamin and colleagues' 0.005 (Benjamin et al. 2018). Cornfield (Cornfield 1966a) put a prior probability on the null in a Bayesian sequential test for clinical trials, and Kass and Raftery (Kass and Raftery 1995) give the scale on which such Bayes factors are usually read. Goodman's two papers (S. N. Goodman 1999a, 1999b) explain to clinicians why a p-value is not the probability that the null is true and how a Bayes factor answers a different question. What the two bars of Section 7.5.1 add is that the threshold is not fixed: for a normal mixture alternative the Bayes factor bar grows with the information, so no single p-value cutoff corresponds to one Bayes factor as a trial grows.

7.5. One toll, three scales: how much evidence is enough?

The Ockham penalty in Equation 7.4 says that a Bayesian who uses a point null and a diffuse alternative has to see more than a one-sided z test's $Z_n^2 = 1.96^2 = 3.84$ before acting once the information is large (for a one-sided claim, with $Z_n > 0$). The following is the first part of the framework in Polson and Sokolov's working paper (Polson and Sokolov 2026b); the rest of it appears in later chapters, as flagged below.

A reader may object that tuning a Bayesian rule until its type I error is 0.025 gives up the Bayesian position. Frequentist evaluation of Bayesian procedures has a long applied tradition. Rubin argued that the applied statistician should use Bayesian methods to build procedures and frequency calculations to judge them (Rubin 1984). Lewis and Berry applied exactly that program to clinical trials, judging Bayesian decision-theoretic designs by their classical operating characteristics (Lewis and Berry 1994a), and Grieve's "idle thoughts of a well-calibrated Bayesian" bring the argument to drug development and discuss how closely Bayesian procedures should match conventional frequentist targets (Grieve 2016). Dawid's forecast-calibration result, that a coherent forecaster should expect to be well calibrated against the frequencies of the events forecast, concerns a different property than fixed-null type I error control (Dawid 1982). Evaluating a Bayesian procedure by its frequentist behavior is therefore Bayesian practice and not a compromise with the other school, and it is what the regulator asks for when the design is calibrated.

7.5.1. The two bars

In plain terms: a fixed-level one-sided test acts when $Z_n \geq z_\alpha$, equivalently (for $Z_n > 0$) when Z_n^2 passes a constant bar, 3.84 at one-sided $\alpha = 0.025$, whatever the sample size. A Bayes rule that acts when the alternative is at least as probable as the null passes a bar that grows with the logarithm of the information, plus a constant that is positive when the null is favored at the outset. At small samples the Bayes bar is the lower one, at large samples it is the higher one, and whether a given trial is past the point where they cross depends on the prior scale. The comparison is for the specified uncalibrated point-null rule with a symmetric normal alternative. Figure 7.3 shows the two bars, and the note below derives them.

> **i** Technical note: the two bars and where they cross
>
> A decision rule that says "act when the evidence passes a bar" can be written as a condition on Z_n^2, together with the positive-direction condition $Z_n > 0$ when the claim is one-sided (the symmetric Bayes factor measures departure in either direction). Two rules, and what each charges:
>
> - **The fixed-level test** acts when $Z_n \geq z_\alpha$, that is, when $Z_n > 0$ and $Z_n^2 \geq z_\alpha^2$. The bar on Z_n^2 is a constant: 3.84 at one-sided $\alpha = 0.025$, whatever the sample size. This is the traditional rule, calibrated to a type I error rate that does not depend on n.
> - **The Bayes rule** acts when the posterior odds in favor of the alternative reach one, $\pi_a \mathrm{BF}_{10} \geq \pi_0$ (act when the alternative is at least as probable as the null, which is optimal if wrongly acting and wrongly not acting cost the same). By Equation 7.2 this is $\mathrm{BF}_{10} \geq \pi_0/\pi_a$, and by Equation 7.4 it is, in the large-information approximation and with $Z_n > 0$ for a one-sided claim,

$$Z_n^2 \;\geq\; L_n + C, \qquad L_n \sim \log n, \qquad C = 2\log\frac{\pi_0}{\pi_a}. \tag{7.6}$$

The exact bar for the normal mixture of Equation 7.3 carries the factor $(1 + 1/k)$: $Z_n^2 \geq (1 + 1/k)\,(L_n + C)$ with $k = I_n\tau^2$. The bar of the Bayes rule is not constant. It grows with the logarithm of the sample size (more precisely, of the information $I_n\tau^2$), plus a constant C that is zero if the null and the alternative are equally probable a priori, positive if the null is favored (the usual case for a drug that has to prove itself), and negative if the alternative is favored.

Each bar is a price. The fixed-level test charges a flat z_α^2. The Bayes rule charges $\log n + C$, which at small n is less than the flat price and at large n is more. Where do they cross? Set $\log n + C = z_\alpha^2$ (asymptotically). With $C = 0$ and $\alpha = 0.025$,

$$n^* \;=\; e^{z_\alpha^2} \;=\; e^{3.84} \;\approx\; 46. \tag{7.7}$$

Here n is measured in units of the prior's information, so that $L_n = \log(1+I_n\tau^2) \approx \log n$. The exact Gaussian expression of Equation 7.3 gives the crossing at $I_n\tau^2 = 41.6$, close to the asymptotic 46. In plain terms: once the data carry about forty to fifty times the information in the prior, the uncalibrated point-null Bayes rule asks for more than the fixed-level test does. Whether a given trial is past that point depends on the prior scale. For a prior of $\tau = 0.25$ on the log hazard ratio, the asymptotic crossing of 46 corresponds to $I_n > 730$ and about 2,900 events (the exact 41.6 gives 666 and about 2,660), the size of a large cardiovascular outcome trial and not of every confirmatory trial; the example of this section, with $I_n\tau^2 = 25$, is below the crossing. For a prior with a larger τ the crossing arrives sooner, and for a smaller τ later.

Two consequences for the reader who runs trials.

The Bayesian bar is not lower. A recurring claim is that Bayesian trials "need less evidence." Against a single unadjusted look at a fixed-level test, the opposite holds for the uncalibrated point-null rule once information times prior variance exceeds about 46 (exactly, about 41.6; a few dozen observations for a unit-information prior, far more at the prior scales used for hazard ratios): the Bayes rule with a point null charges more. A Bayesian bar calibrated to the same type I error is never lower at a single look without an informative prior. The gains of Bayesian adaptive designs come from elsewhere, namely monitoring, multiple arms and prior information, and each is costed later in the book (Chapter 11, Chapter 10, Chapter 19). This is the thesis of the book and it is first visible in Equation 7.7.

A third toll exists, and it is the one that matters for monitoring. Both bars above are for a decision made once, at a pre-specified sample size. If the trial is looked at repeatedly and one wants the type I error to hold at *every* stopping time, the bar has to rise again. For the symmetric normal-mixture e-process of this chapter, comparing the threshold $b = 1/\alpha$ with the baseline $b = 1$, the increase in the squared-Z bar is $2\log(1/\alpha)(1 + 1/k)$, which tends to the constant $2\log(1/\alpha) = 7.38$ at $\alpha = 0.025$ as the information grows. This is the anytime-valid surcharge for this construction, and it comes from Ville's inequality for test martingales. It is a large-information approximation for this normal mixture, not a universal minimum cost of sequential validity: other e-processes, such as those behind iterated-logarithm confidence sequences, behave differently. It is developed in Chapter 12, and the sequential story, where the three tolls z_α^2, $\log n$ and $\log n + 2\log(1/\alpha)$ become the three deviation scales and meet the group sequential boundaries (Pocock, O'Brien-Fleming, Lan-DeMets), is told in Chapter 11.

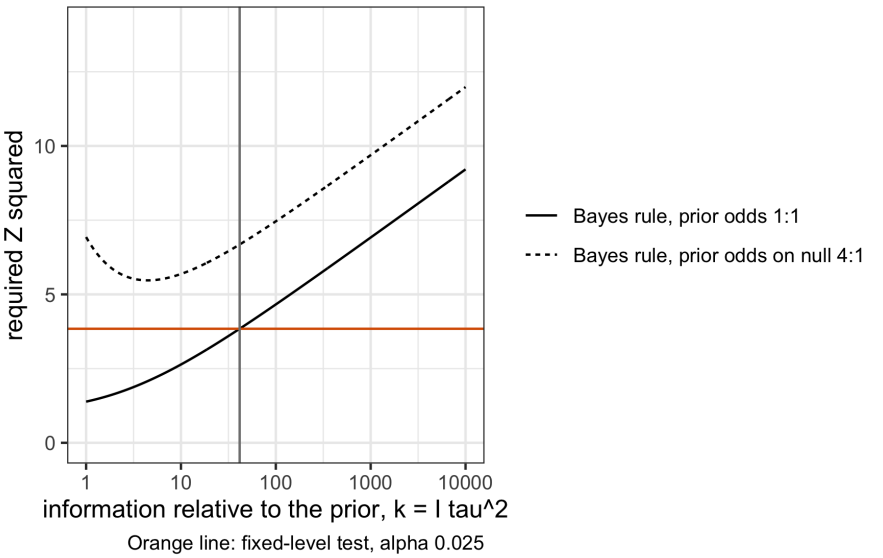


Figure 7.3.: The first two tolls. The fixed-level test acts when Z squared exceeds the constant 3.84 (one-sided alpha 0.025). The Bayes rule with prior odds 1 to 1 acts when Z squared exceeds the Ockham bar, which grows like the logarithm of the information. The two cross where the information is about 42 times the prior information. The dotted curve shows the bar when the prior odds on the null are 4 to 1 ($C = 2\log 4$).

Technical note: why log n, and where the constant C comes from

Under the Gaussian model, the evidence that θ is nonzero accumulates in the statistic $S_n = \hat{\theta}_n I_n$, which behaves like Brownian motion in information time with drift θ. The Bayes factor of Equation 7.3 integrates the exponential likelihood ratio over the prior. A Laplace approximation of that integral gives the maximized log likelihood ratio, $Z_n^2/2$, minus the log of the ratio of the prior's width to the posterior's width, $\frac{1}{2}\log(1 + I_n\tau^2)$. The posterior width shrinks like $I_n^{-1/2}$, so the ratio grows like $I_n^{1/2}$ and its logarithm like $\frac{1}{2}\log I_n$. Doubling gives the L_n of Equation 7.4.

For the constant, the Bayes rule for a two-point decision with equal costs acts when the posterior probability of the alternative exceeds one half, that is, when $\mathrm{BF}_{10}\,\pi_a/\pi_0 \geq 1$. Taking logarithms, $2\log \mathrm{BF}_{10} \geq 2\log(\pi_0/\pi_a)$, and substituting Equation 7.4 gives Equation 7.6. If the costs of the two errors differ, C absorbs their ratio as well. A regulator's preference for protecting the public from false approvals can be written as a larger C.

Setting b in "act when $\mathrm{BF}_{10} \geq b$" to $1/\alpha$ turns the Bayes factor into an e-value test and gives the anytime-valid bar, exactly $(1 + 1/k)(L_n + 2\log(1/\alpha))$ and approximately $L_n + 2\log(1/\alpha)$ for large information, discussed in Chapter 12. The Bayes rule at $b = 1$ and the anytime-valid rule at $b = 1/\alpha$ are built on the same statistic and differ only in where the bar is placed.

7.5.2. What this means for the protocol

A protocol written in the traditional style states a fixed-level test and a sample size computed from it. A protocol with a Bayes factor decision (rare in confirmatory trials, more common in early phase and in decision-theoretic designs) states the prior under the alternative, the prior odds or threshold, and the resulting bar. The table below compares the two on the same trial; the regulator's question for either is the same one, namely what the operating characteristics are.

	Fixed-level z test	Bayes factor rule
Quantity compared with a bar	Z_n against z_α (Z_n^2 against z_α^2 when $Z_n > 0$)	BF_{10} against π_0/π_a, or Z_n^2 against $L_n + C$
Bar depends on n?	No	Yes, like $\log n$
Needs a prior?	No	Yes: τ and π_0/π_a
Type I error	α, by construction	Falls with n; must be computed by simulation for the actual design
Where it agrees	One-sided p-value < 0.025	At the crossing n^*, where the two bars are equal
What the regulator asks	Is α spent correctly?	What are the operating characteristics, and why this prior?

7.6. A test is a Bayes rule for an implied loss

The previous section priced evidence with a bar. This section asks what a bar is worth, and the answer ties the chapter back to the type I error and power of Chapter 4. The frequentist ledger holds two numbers, the type I error α and the power $1 - \beta_\theta$ at each effect θ, each computed with the parameter held fixed. The Bayesian ledger holds one, the Bayes risk, which weighs the same two errors by how likely each state of the world is and by how much each mistake hurts. The two ledgers are not rivals: one is built from the other.

7.6.1. The Bayes risk of a test

Let $D(y) = 1$ declare the treatment effective. Let $\pi_0 = \Pr(H_0)$, let the effect have density $g(\theta)$ under H_1, and let the loss be ℓ_1 for approving a useless drug and ℓ_0 for withholding a useful one (zero otherwise). The Bayes risk of D is its expected loss over both the parameter and the data,

$$r(D) \;=\; \pi_0\,\ell_1\,\alpha(D) \;+\; (1-\pi_0)\,\ell_0 \int \beta_\theta(D)\,g(\theta)\,d\theta. \tag{7.8}$$

A frequentist who reports α and β has reported the ingredients of the risk, and the weights are what is missing: the prior probability of each state times the loss of the corresponding mistake. The rule that minimizes $r(D)$ is found one data set at a time. For each y the contribution to the risk is linear in $D(y) \in \{0, 1\}$, so the optimal rule picks the action with the smaller coefficient, which gives

$$D^B(y) = \mathbf{1}\{\mathrm{BF}_{10}(y) > b\}, \qquad b = \frac{\pi_0}{1-\pi_0} \cdot \frac{\ell_1}{\ell_0}, \tag{7.9}$$

equivalently, reject when the posterior probability of H_0 falls below $\ell_0/(\ell_0 + \ell_1)$. This is the rule for choosing between a point null and a symmetric alternative, with its threshold read as a loss ratio, and it uses only the data that occurred. It is two-sided model selection: the alternative $N(0, \tau^2)$ puts mass on harmful effects, so $\Pr(H_1 \mid y)$ is not the benefit probability $\Pr(\theta > a \mid y)$ of the FDA criterion in Equation 7.11. A rule for approving on benefit needs hypotheses and losses that separate beneficial from harmful effects, and restricting this rule to $Z_n > 0$ does not preserve its optimality. The averaging over data that did not occur defines the risk it minimizes, but it is not needed to find the rule.

The Neyman-Pearson lemma (Neyman and Pearson 1933) is a statement about the same rules. Minimizing the type II error at a simple alternative θ_1 subject to $\alpha(D) \le \alpha$ is the same as minimizing $\beta_{\theta_1}(D) + \lambda\,\alpha(D)$ for some multiplier $\lambda \ge 0$, which is the Bayes risk with g a point mass, $(1-\pi_0)\ell_0 = 1$ and $\pi_0\ell_1 = \lambda$. So every most powerful test is a Bayes rule for some prior odds and loss ratio, and the likelihood-ratio cutoff k_α is the product in Equation 7.9. A trialist who fixes α has chosen this product without saying so. The useful question is how the implied choice behaves as the trial grows.

7.6.2. What a fixed level implies as information grows

Take the normal model of Equation 7.3, $\theta \sim N(0, \tau^2)$ under H_1, with $q = I_n\tau^2$. The fixed-level test that acts when $|Z_n| > z_{\mathrm{crit}}$ is exactly the Bayes rule $\mathrm{BF}_{10} > b_n$ with

$$2\log b_n \;=\; z_{\mathrm{crit}}^2\,\frac{q}{1+q} - \log(1+q), \qquad \text{so} \qquad b_n \approx \frac{e^{z_{\mathrm{crit}}^2/2}}{\sqrt{q}} \to 0. \tag{7.10}$$

The implied threshold is simply the Bayes factor at the boundary, so the first column of Table 7.5 is both the Lindley paradox at $Z = 1.96$ and the implied Bayes-factor threshold of the 5% test (a two-sided point-null quantity, not a one-sided benefit bar). A regulator that holds α fixed behaves, as trials grow, as if the prior odds on the null or the harm of a false approval were shrinking toward zero. Here the prior is the unit-information prior, $\tau = 1$ on the scale of one observation, so $q = n$; the table of Section 7.3.2 used $\tau = 0.25$ and a p-value of 0.01, and the pattern is the same.

Table 7.5.: A result on the conventional boundary, Z = 1.96, as the trial grows. Prior N(0, 1) on the effect (unit information, q = n), equal prior odds. The Bayes factor at the boundary is the Bayes-factor threshold implied by the 5% test (Equation 7.10); P(H0) is the posterior probability of the null.

n	Implied threshold = BF10 at Z = 1.96	P(H0 | Z = 1.96)
1e+01	1.728	0.37
1e+02	0.666	0.60
1e+03	0.215	0.82
1e+04	0.068	0.94
1e+05	0.022	0.98

At $n = 10$ a result on the boundary of significance mildly favors an effect (Bayes factor 1.73). At $n = 100$ it already favors the null (0.67), and at $n =$ 1e+05 it corresponds to a posterior probability of the null of 0.98. The 5% test is a Bayes rule at every n, but for an implied loss that drifts with n.

7.6.3. The type I error of the Bayes rule

The converse question is what type I error the Bayes rule has. With equal odds and loss ($b = 1$), the exact bar on $|Z_n|$ is $\sqrt{(1 + 1/q)\log(1 + q)}$, the bar of Equation 7.6 with the factor of the technical note above, so the type I error is $2\{1 - \Phi(\text{bar})\}$. It exceeds 5% for small trials, passes 5% at the crossing $q = 41.6$ of Section 7.5.1, and falls thereafter, roughly like $(q \log q)^{-1/2}$ in the large-information approximation. Table 7.6 gives the exact values, with the Bayes risk of the two rules at $\pi_0 = 1/2$ and equal losses ($\ell_0 = \ell_1 = 1$).

Table 7.6.: The Bayes rule (BF10 > 1, equal prior odds, unit-information prior N(0, 1), q = n) against the fixed 5% test. The exact bar is sqrt((1 + 1/q) log(1 + q)). Risk is the Bayes risk of Equation 7.8 with pi0 = 0.5 and equal losses.

n	Bayes bar	Type I error, Bayes rule	Risk, 5% test	Risk, Bayes rule
10	1.62	0.1044	0.2477	0.2400
100	2.16	0.0309	0.1023	0.1005
1,000	2.63	0.0085	0.0497	0.0374
10,000	3.04	0.0024	0.0328	0.0133

The two rules have the same risk near the crossing and then separate. The fixed-level test has Bayes risk at least $\pi_0 \ell_1 \alpha$ for every n, because its type I error is α at every n. For the Bayes rule both error rates tend to zero, the type I error by the display above and the type II error at each $\theta \neq 0$ because the statistic grows like $\sqrt{n}\,\theta$ while the bar grows like $\sqrt{\log n}$, so its risk tends to zero. A fixed bar spends a constant error forever, and by Lindley's paradox the constant is spent on exactly the cases where the evidence favors the null. This is the point-null, two-sided calculation. The one-sided confirmatory rule of the protocol is a different decision problem, and deriving its Bayes rule needs hypotheses and losses that distinguish beneficial from harmful effects, which this section does not do.

7.6.4. Which error matters more

The threshold-as-loss-ratio derivation and its table are in Section 1.6; here only the prior odds enter the new part. By Equation 7.9, with equal prior odds, acting when $\Pr(H_1 \mid y) > p^\star$ is optimal when a false approval is $p^\star/(1 - p^\star)$ times as costly as a false rejection, whatever the prior odds: with $p^\star = \ell_1/(\ell_0 + \ell_1)$ the prior odds are already inside the posterior, and they multiply only the Bayes-factor threshold b.

Table 7.7.: Posterior probability thresholds and the loss ratio they imply (false approval relative to false rejection).

Threshold p*	Implied ratio l1/l0
0.975	39
0.990	99
0.995	199

A threshold of 0.975 (which equals a one-sided 0.025 only under the flat-prior, normal, single-look conditions of Section 1.6, and which is a threshold on the benefit probability of a continuous model, not on $\Pr(H_1 \mid y)$ in the point-null model, so the two 0.975 thresholds are not interchangeable) therefore says that approving a useless drug is 39 times worse than withholding a useful one, 0.99 says 99, and 0.995 says 199. Writing the threshold as a loss ratio forces the question that a fixed α hides: is a false approval in this indication really 100 or 200 times worse than a false rejection?

The ratio is not a statistical quantity. In a refractory cancer with no effective therapy, delay costs months of life and a false approval of a drug that does little harm costs relatively little, so the ratio is small and the threshold can be lower. For a preventive drug taken for years by millions of healthy people, a false approval exposes a large population to side effects for no benefit while delay costs each person little, so the ratio is large and the threshold should be higher. In a pandemic vaccine the cost of delay is counted in deaths per week, and the ratio is small even though the population is large. Isakov, Lo and Montazerhodjat (Isakov, Lo, and Montazerhodjat 2019) carry out this exercise across diseases with burden-of-disease data and find that the best threshold differs by disease, so that a single conventional α cannot be the Bayes rule for all of them. A protocol that states its threshold with a loss ratio, and a reason, is more informative than one that states α alone (the loss-ratio reading of a posterior threshold returns in Chapter 15).

7.6.5. The Jeffreys table, in terms of q

The growth of the bar is old. Jeffreys (Jeffreys 1939) derived it for a normal mean with the Cauchy prior he recommended on invariance grounds. To leading order in n the Bayes factor in favor of the null is $K = \sqrt{\pi n/2}\,\exp(-t^2/2)$, which falls through one at $t_{\text{crit}} = \sqrt{\log(\pi n/2)}$. This is a large-information approximation, not an identity: evaluated exactly, the Cauchy Bayes factor at t_{crit} is 1.30 at $n = 10$ and 1.008 at $n = 10^3$. Table 7.8 restates the criterion in terms of q and sets it beside the exact bar of the normal-mixture rule of Section 7.5.1, which is what the rest of the chapter uses.

Table 7.8.: Jeffreys' criterion in standardized units, with unit prior scale so that q=n. t_crit is the large-information approximation to where the Cauchy-prior Bayes factor turns against the null (the exact crossing is somewhat higher at small n); p is its two-sided tail probability; t (K^-1=10) is the value for Jeffreys' grade of strong evidence; the last two columns are the exact bar of the normal-mixture Bayes rule at BF10 = 1 and at BF10 = 10, and the conventional 1.96.

q = n	t_crit	p	t (K^-1 = 10)	Normal bar, BF = 1	Normal bar, BF = 10	Conventional
1e+01	1.66	0.0970	2.71	1.62	2.78	1.96
1e+02	2.25	0.0245	3.11	2.16	3.05	1.96
1e+03	2.71	0.0067	3.46	2.63	3.39	1.96
1e+04	3.11	0.0019	3.78	3.04	3.72	1.96
1e+05	3.46	0.0005	4.07	3.39	4.01	1.96
1e+06	3.78	0.0002	4.34	3.72	4.29	1.96

Two cautions keep the table from being misread. First, the bar is a function of the prior scale through q, not of n alone. The Cauchy scale and the normal τ are different priors (at $q = 10$ the two bars in the table are approximately 1.66 and 1.62, the first being the asymptotic Cauchy threshold, whose exact crossing at $n = 10$ is 1.83), and a prior concentrated on the small effects that cardiovascular trials target moves the same bar to much larger n: with $\tau = 0.25$ the rows read at $q = n/16$, and the crossing with 1.96 arrives at the thousands of events found in Section 7.5.1, not at $n \approx 30$. The statement that the fixed level is too lenient for every trial beyond a phase I cohort holds only for a unit-information prior. Second, the calculation prices a point null, and an exact zero effect is a convenience. Against an interval null of clinically negligible effects the dimensional penalties of the two continuous models can cancel, and the resulting bar depends on the interval and the priors chosen; no general growth rate is claimed here. Both cautions bear on how the bar should behave, which is why the prior scale and the odds belong in the protocol and not in a convention.

7.6.6. Lindley information and the Ockham penalty

Lindley (Dennis V. Lindley 1956) defined the information in an experiment as the expected reduction in entropy from prior to posterior, the mutual information between θ and the data. For a normal mean with prior variance τ^2 and data of precision n/σ^2 it is $\frac{1}{2}\log(1+q)$, half of the Ockham penalty $L_n = \log(1 + I_n\tau^2)$ that sets the Bayes bar. The coincidence is not an accident: the Bayes factor charges the alternative the amount of information that an experiment of this size could deliver about θ, and the bar rises like $\log n$ because the information does. With $\tau = 0.25$ and $\sigma = 1$ a trial delivers 1.0, 1.9, 2.8, 3.8 bits at $n = 50, 200, 800, 3200$: each quadrupling of the trial adds about one bit. Information returns are logarithmic, so the evidence a larger trial must clear grows only slowly, and so does what it buys.

7.7. From p-values to posterior probabilities of benefit

The Bayes factor compares two hypotheses. For most clinical purposes there is a more useful number: the posterior probability that the treatment effect is large enough to matter.

Chapter 6 built the posterior distribution of θ; here the question is how to turn it into a go/no-go criterion that a regulator can accept.

7.7.1. The FDA's success criterion

The January 2026 FDA draft guidance on the use of Bayesian methodology (U.S. Food and Drug Administration 2026g), section IV.A, describes the standard way a Bayesian trial defines success. It says that Bayesian success criteria are typically "based on the posterior probability that the true treatment effect size exceeds some threshold." In the notation of the guidance and of this book,

$$\Pr(\delta > a \mid \text{data}) \;>\; c, \tag{7.11}$$

where δ is the treatment effect, a is the threshold (a minimum clinically relevant benefit, or zero if any benefit counts) and c is the minimum posterior probability. The notation δ is used when quoting the criterion; elsewhere the book writes θ for the same quantity. Two dials, then. The threshold a says how big an effect counts. The probability c says how sure one must be that it is at least that big. For trials with an overall frequentist statistical analysis plan, the same guidance says the success criteria are "almost always" chosen so that the familywise type I error is no greater than 0.025, one-sided, and that "a and c can be chosen such that the overall FWER is controlled at a given level, typically 0.025 one-sided" (section IV.A.1). It also describes two other routes: direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA) and decision-theoretic or benefit-risk criteria (section IV.A.3). Section IV.A is thus clear that a Bayesian criterion is a *rule* whose operating characteristics are to be reported. Calibration to the one-sided 0.025 is one of FDA's routes and this book's default for confirmatory adaptive designs without borrowing; it is not presented as FDA's universal requirement.

7.7.2. The flat-prior case: where Bayes and the p-value agree

The simplest instance connects the criterion to what the reader already knows. Let the estimate be $\hat{\theta} \sim N(\theta, s^2)$, with standard error $s = 1/\sqrt{I_n}$, and let the prior on θ be flat. The posterior is $N(\hat{\theta}, s^2)$, so

$$\Pr(\theta > a \mid \hat{\theta}) \;=\; \Phi\left(\frac{\hat{\theta} - a}{s}\right).$$

For the threshold $a = 0$, this is $\Phi(Z_n)$, which is exceeded by $c = 0.975$ exactly when $Z_n > 1.96$, that is, exactly when the one-sided p-value is below 0.025. A flat-prior posterior probability of benefit is one minus the one-sided p-value. The two numbers differ in how they are read (probability about the effect, given the data; probability of the data, given no effect), but the posterior probability is one minus the one-sided p-value, so the two thresholds agree and a trial that succeeds by one succeeds by the other. This is a useful cross-check to keep in the SAP: whatever a Bayesian analysis with a very weak prior reports, it should be close to the frequentist analysis of the same trial. It is also why the passage of Chapter 6 on credible intervals matches the confidence interval of the CSR to the second decimal for large trials.

7.7.3. The mapping from (a, c) to type I error

With a flat prior, the criterion $\Pr(\theta > a \mid \hat{\theta}) > c$ is equivalent to $\hat{\theta} > a + z_c\, s$, where $z_c = \Phi^{-1}(c)$. At $\theta = 0$ (no effect) the probability that a trial meets the criterion is

$$\alpha(a, c) \;=\; 1 - \Phi\Big(\frac{a}{s} + z_c\Big)\,. \tag{7.12}$$

It follows that:

- With $a = 0$, $\alpha = 1 - c$. A criterion of $c = 0.975$ gives $\alpha = 0.025$. A criterion of $c = 0.95$ gives 0.05, one-sided.
- With $a > 0$, the type I error is *smaller* than $1 - c$, because clearing a positive threshold is harder. To keep $\alpha = 0.025$ at $\theta = 0$ one can lower c: the calibrating value is $c = \Phi(z_\alpha - a/s)$. Lowering c in this way controls the error under the null $\theta \le 0$, not under $\theta \le a$: with $a = s = 0.1$ the calibrating $c \approx 0.83$ gives about 17% rejection at $\theta = a$. The larger the threshold a is relative to the standard error, the lower c can be while the type I error stays at 0.025. When a is large compared with s one can end up at a c of 0.8 or so, a figure that looks lenient. This calibration reproduces the ordinary zero-null test: the rejection condition reduces to $\hat{\theta} > z_\alpha s$ whatever a is, so it establishes superiority over zero, not benefit exceeding a. To support a claim that the effect exceeds a, calibrate at $\theta = a$; in this flat-prior normal model that gives $c = 0.975$. FDA draft section IV.A.1 uses $a = 0$ for calibrated superiority designs.
- The calibration depends on s, that is, on the sample size. A given pair (a, c) gives a different α in a trial with 200 patients than in one with 2,000.

Table 7.9.: Mapping the posterior criterion Pr(theta > a | data) > c to the one-sided type I error at theta = 0 and to the calibrating c, for a flat prior. The standard error is s = 0.10 (so a/s is the threshold in standard errors). Calibration target: one-sided 0.025.

a	a/s	alpha at c=0.975	alpha at c=0.90	c giving alpha=0.025
0.00	0.0	0.0250	0.1000	0.975
0.05	0.5	0.0069	0.0374	0.928
0.10	1.0	0.0015	0.0113	0.831
0.15	1.5	0.0003	0.0027	0.677
0.20	2.0	0.0000	0.0005	0.484

In Table 7.9, with $a = 0$, $c = 0.975$ gives exactly 0.025; $c = 0.90$ gives 0.10. When the threshold rises to one standard error ($a = 0.10$), the $c = 0.975$ rule has a type I error of 0.0015, so it is much stricter than needed, and the calibrating c falls to 0.83. At two standard errors it falls to 0.48, close to a coin flip: a threshold of two standard errors above zero is itself about as demanding as the frequentist boundary, so little extra probability is needed on top of it. The numbers illustrate why guidance asks for calibration by simulation rather than by a rule of thumb about c: the pair (a, c) means nothing until it is attached to a sample size and a model.

7.7.4. An informative prior changes the mapping

A flat prior is the exception. The point of a Bayesian design is often to use prior information, or at least a skeptical prior that shrinks the estimate toward zero. With a prior $\theta \sim N(0, \tau^2)$ the posterior is normal with mean $w\hat{\theta}$ and variance ws^2, where $w = \tau^2/(\tau^2 + s^2)$ is the weight on the data. The criterion becomes $\hat{\theta} > (a + z_c\sqrt{w}\, s)/w$, and a skeptical prior therefore raises the bar on $\hat{\theta}$ at a given c: the trial must show a larger observed effect to convince a skeptic. The type I error at $\theta = 0$ falls, and so does the power.

Table 7.10.: The same criterion Pr(theta > 0 | data) > 0.975 under three priors, for s = 0.10: flat, a skeptical N(0, 0.2^2) prior, and a more skeptical N(0, 0.1^2) prior. Type I error at theta = 0 and power at theta = 0.3, the effect the trial was designed to detect (3 standard errors).

prior	type I error	power at 0.3	c for alpha=0.025
flat (tau = 100)	0.0250	0.851	0.975
skeptical tau = 0.2	0.0142	0.791	0.960
skeptical tau = 0.1	0.0028	0.590	0.917

In Table 7.10, the first row is the traditional trial in Bayesian clothing: $\alpha = 0.025$ and the power of a one-sided z test at three standard errors, 0.85. With the skeptical priors the same $c = 0.975$ is a stricter criterion: the type I error falls to 0.014 under $\tau = 0.2$ and to 0.003 under $\tau = 0.1$, and power falls with it (0.79 and 0.59). Calibrating by lowering c (to 0.96 and 0.92) returns the type I error to 0.025. Once c is lowered to restore the type I error of 0.025, power returns to the flat-prior value of 0.85, and in this particular case the rule is *exactly* the flat-prior test, because a normal prior with a known variance and a one-sided criterion is equivalent to a z test with a different cut-off. The equivalence also holds for a normal prior with a shifted mean: once c is calibrated, the rejection region is again the same Z region. More generally, in this scalar normal model with known variance, any fixed prior that makes the posterior benefit probability strictly increasing in the estimate (any nondegenerate prior with mass on both sides of zero, mixtures included) yields, after threshold calibration, exactly the same rejection region as the Z test. More complex sampling models, nuisance parameters, and multi-arm or sequential decision rules require separate evaluation, by simulation, which is the subject of Chapter 14.

The consequence is important enough to state plainly. If the Bayesian criterion is calibrated to $\alpha = 0.025$ and the prior is centered at no effect, then at a fixed sample size the trial has the same type I error and, to a first approximation, the same power as the traditional trial. The Bayesian version buys nothing at the single analysis. What it buys appears when the same posterior is used repeatedly, with multiple arms, or with an informative prior from earlier data, at which point the calibration is a simulation and the prior becomes a design element (Chapter 8, Chapter 19). If, on the other hand, an informative prior is used without calibration, the trial's type I error can be higher or lower than 0.025 depending on the prior and the criterion (a prior $N(0.01, 0.1^2)$ with $s = 0.1$ and $c = 0.975$ gives about 0.004), and a regulator will ask for the justification the FDA draft recommends in sections V.A and V.F (a pre-specified prior, and a demonstration that conclusions do not depend too strongly on it).

7.7.5. A worked comparison

Consider a two-arm trial of a new therapy with a continuous outcome in which the estimated effect is 0.28 (in units of the outcome's standard deviation) with standard error $s = 0.10$, so $Z = 2.8$ and the one-sided p-value is 0.0026. Table 7.11 gives the summaries.

Table 7.11.: One trial, six summaries. Estimate 0.28, standard error 0.10. The Bayes factor uses the N(0, 0.25^2) alternative of the Lindley example scaled to this problem (k = tau$^{2/s}$2 = 6.25) and prior probability 0.5 on the null; the SBB column is the bound of the previous section applied to the two-sided p-value.

summary	value
one-sided p-value	0.002555
flat-prior Pr(theta > 0 | data)	0.997400
flat-prior Pr(theta > 0.1 | data)	0.964100
Bayes factor BF10 (tau = 0.25)	10.900000
P(H0 | data) from that BF	0.084030
SBB max BF10 over priors	13.640000

The trial clears every bar a traditional reviewer would set: the one-sided p-value of 0.0026 is below 0.025, and the lower confidence limit is above zero. The flat-prior posterior probability of benefit, 0.9974, is one minus that p-value. If the criterion has a minimum clinically relevant benefit of $a = 0.10$, the posterior probability that the effect exceeds it is 0.964, which clears $c = 0.95$ but not 0.975, so the *same data* meet or miss the success criterion depending on how the sponsor set (a, c) in advance. That is why the FDA draft recommends that the criterion be pre-specified, and why this book calibrates it. Sensitivity to the prior, as opposed to (a, c), appears in the ANDROMEDA-SHOCK reanalysis of Section 7.10. The Bayes factor of about 11 in favor of the alternative (under the stated prior) corresponds to a posterior probability of the null near 8% from even prior odds, far larger than the p-value of 0.0026 suggests and within the calibration bound of the previous section, which allows at most 13.6 for BF_{10}.

7.8. Side by side: what changes in the protocol

The table compares the traditional and Bayesian handling of the same decision.

	Traditional (fixed sample, one-sided 0.025)	Bayesian posterior criterion
Success statement	$p < 0.025$ for $H_0 : \theta \leq 0$	$\Pr(\delta > a \mid \text{data}) > c$
What is fixed in the protocol	α, test statistic, sample size from power	a, c, prior, model, sample size by simulation
What the number means	Chance of data this extreme if no effect	Probability, given data and prior, that the effect exceeds a
Needs a prior	No	Yes, pre-specified (FDA draft V.A)
Flat prior, $a = 0$, $c = 1 - \alpha$	The same decision	The same decision

	Traditional (fixed sample, one-sided 0.025)	Bayesian posterior criterion
Type I error	By construction	By calibration, reported with simulation (Chapter 14)
Threshold for a clinically relevant effect	Separate: the effect estimate and its interval	Built into the criterion through a
Behavior as $n \to \infty$	Can declare trivial effects significant	With $a > 0$, declares only effects exceeding a
What the regulator asks	Was α preserved? Multiplicity?	Operating characteristics, prior justification, sensitivity analysis
Where it is the better choice	Single look, no usable prior, sponsor not committing to simulation	Informative prior, adaptive monitoring, multiple arms

To see the numbers on one trial, take the running trial of Section 6.5 (45 of 100 against 30 of 100, $z = 2.19$) and line up every quantity of this chapter in Table 7.13.

Table 7.13.: The running trial of Chapter 4 read through each quantity of this chapter. The minimum Bayes factor is the Sellke-Bayarri-Berger bound at the two-sided p-value.

Quantity	Value
One-sided p-value	0.0142
Two-sided p-value	0.0285
Flat-prior Pr(p1 > p0 | data)	0.9855
Minimum BF01 (null over alternative)	0.275
Maximum BF10 over priors	3.6
Minimum P(H0 | data), even prior odds	0.22

The flat-prior posterior probability is close to one minus the one-sided p-value (the two agree exactly in the normal model, and approximately in this binary example), the same information read two ways. The minimum Bayes factor calibrates the two-sided p-value: no prior in the class can give more than about 3.6 to 1 for the alternative, and from even prior odds the null keeps at least 22%. A result that clears 0.025 in the posterior-probability reading carries only modest evidence in the Bayes factor reading.

The traditional design is not the weaker partner. For a single-look, fixed-sample confirmatory trial with no relevant prior information, the flat-prior Bayesian analysis is the p-value in a different vocabulary, and the traditional design is the simpler thing to defend. The Bayesian criterion earns its place in the cases the later chapters take up, and its price, the extra specification and simulation burden, is paid in the currency regulators accept.

> **In the protocol**
>
> A statistical analysis plan that uses a posterior-probability success criterion should state, before data are seen:

- the effect scale and the definition of δ (for example, the log hazard ratio with benefit positive, or the risk difference);
- the threshold a and the probability c, with the clinical justification for a;
- the prior distribution, with its source;
- the calibration: the simulation that shows the familywise type I error at $\delta = 0$ (and at the least favorable point of the null) is no greater than 0.025 one-sided, with the number of simulated trials and the Monte Carlo error;
- the frequentist analysis that will be reported alongside (p-value and confidence interval), so that a reviewer can compare;
- a sensitivity analysis to the prior, including a weakly informative or flat prior, in line with FDA draft section V.F.

7.9. Reading the four quantities together

The four quantities of this chapter answer four different questions, and a clinician or reviewer who keeps the questions straight will rarely be misled.

- The **p-value** asks: if the treatment did nothing, how often would a trial like this look this good? It is the tool for controlling the rate of false approvals across a program, and it is what the type I error requirement is about.
- The **false positive risk** asks: among trials that look this good, how many are for drugs that do not work? It needs a prior probability that the drug works, and it explains why a fixed alpha is a different level of protection in different development settings.
- The **Bayes factor** asks: how much do the data favor a specific alternative over the null? It needs a prior under the alternative, it carries an Ockham penalty that grows with the information, and it is the quantity behind Lindley's paradox.
- The **posterior probability of benefit** asks: how probable is it, given everything, that the effect is at least a? It is the quantity a physician or a patient wants, it can be made to match the p-value exactly under a flat prior, and with an informative prior this book calibrates it by simulation, which is one of the draft's three routes and the one a regulator will most readily recognize.

A good SAP lets a reader recover all four from the same analysis. The p-value and the interval are what ICH E9 section 5.5 asks for. The posterior probability and its calibration are what the FDA draft and ICH E20 section 5.3 recommend reporting when the design is Bayesian. Where they disagree, the disagreement is information: a trial where the p-value is just under 0.025, the Bayes factor is barely above one, and the posterior probability of exceeding the clinically relevant threshold is 60% is a trial in which the regulator, the sponsor and the clinician should talk about what the evidence amounts to, not only about whether it met the criterion.

Try it in the lab

Open module 2, "p-values" at https://vsokolov.org/bayesian_trials_lab/02-p-values.html. Before pressing the button, predict the share of luck-only experiments that land in the orange tail, then check; then find the teal bar for $p = 0.05$ and compare it with the 29% in Table 7.3. Module 0, https://vsokolov.org/bayes

ian_trials_lab/00-are-they-different.html (the "Why the coin?" section is at https://vsokolov.org/bayesian_trials_lab/00-are-they-different.html#why-coin), shows the shuffle test that the p-value of this chapter is built on; predict how often a 12-versus-12 comparison with no true difference gives a "significant" result, then run it.

7.10. In practice

Bayesian reanalysis of reported trials has a method literature. Harrell and Shih (Harrell and Shih 2001) argued that decision makers need probabilities of actual interest, such as the probability of any benefit and of a clinically meaningful benefit; Zampieri and colleagues (Zampieri et al. 2021) wrote a critical care tutorial, with a reanalysis of the ART trial; and Wijeysundera and colleagues (Wijeysundera et al. 2009) applied Bayesian inference to contemporary randomized trials. Such reanalyses report the posterior probability of benefit, the posterior probability of a clinically meaningful benefit and the sensitivity to the prior, and they do not replace the prespecified analysis.

Three published results show the chapter's numbers at work. None of the Bayesian analyses below was a regulatory submission: two are post hoc reanalyses of trials already reported with a p-value, and the third is a survey of how often promising results are confirmed.

7.10.1. EOLIA: a p-value of 0.09 and its posterior probabilities

EOLIA randomized 249 patients with very severe acute respiratory distress syndrome to early extracorporeal membrane oxygenation (ECMO) or conventional management and was stopped early for futility (Combes et al. 2018). Sixty-day mortality was 35% (44 of 124) with ECMO and 46% (57 of 125) with control, a relative risk of 0.76 (95% CI 0.55 to 1.04), $p = 0.09$. Thirty-five control patients (28%) crossed over to ECMO. By the convention of this chapter's Section 7.1.1 the trial was "not significant."

Goligher and colleagues reanalyzed it post hoc, to show the effect of the prior (Goligher et al. 2018). The posterior probability that ECMO lowers 60-day mortality (relative risk below 1) was 96% under a minimally informative prior (RR 0.78, 95% credible interval 0.56 to 1.04), 99% under a moderately enthusiastic prior worth 264 patients at RR 0.78, 88% under a strongly skeptical prior worth 264 patients at RR 1.0 (RR 0.88, 0.71 to 1.09), and 99% under a prior from a downweighted meta-analysis of earlier studies. The skeptical prior is the weakest case and it still puts 88% on some benefit. The trial had been designed to detect a 20-point absolute reduction in mortality, and for that target the picture is different: the posterior probability of an absolute reduction of at least 20 points was about 2% under the minimally informative prior.

The lesson for this chapter is twofold. A flat-prior posterior probability is one minus a one-sided p-value (here about 0.96 for the observed RR of 0.76, up to the scale of the analysis), so a posterior of 88% to 99% is not a contradiction of $p = 0.09$ but the same data read as a probability about the effect. And the choice of a matters: "any benefit" and "the benefit the trial was powered for" give very different answers from the same posterior, which is the point of the threshold a in Equation 7.11. No regulatory action followed from the reanalysis.

7.10.2. ANDROMEDA-SHOCK: a borderline trial under four priors

ANDROMEDA-SHOCK (Hernández et al. 2019), a trial of 424 patients with septic shock, missed its primary end point narrowly (two-sided $p = 0.06$), and a post hoc reanalysis under four priors (Zampieri et al. 2020) found the probability of benefit high under all of them; the trial and the reanalysis are told in full in Chapter 22.

The lesson is the one of Table 7.10 in a clinical setting: a result that sits just beyond the frequentist line ($p = 0.06$ two-sided, against 0.05, or equivalently 0.03 one-sided against 0.025) can be robust to the prior at a moderate c, and the reanalysis also shows where that robustness stops: a pessimistic prior, the analogue of the skeptical priors of that table, is what moves the conclusion (Chapter 22). A sponsor who wants such a statement to count must specify the priors and (a, c) before the data, as the FDA draft recommends (sections IV.A and V.A).

7.10.3. Zia and colleagues: false positive risk in practice

Zia and colleagues found that only 28% of phase III studies that repeated a regimen already tested in phase II were positive (Zia et al. 2005); the full account is in Section 9.6.3.

A positive phase II result is the evidence that makes a phase III worthwhile, and yet fewer than a third of these confirmatory trials were positive. This is an empirical warning about predicting phase III results from phase II. It does not identify the fraction of ineffective treatments or an appropriate borrowing weight: a positive rate of 28% is produced, for example, by a development program in which half of the treatments are truly effective, average power is 53.5% and the type I error is 2.5%. The demand for adequate and well-controlled confirmation (Chapter 3) is a response to this kind of attrition, and a Bayesian design that borrows a phase II result as a prior (Chapter 19) has to justify its borrowing weight by other means.

Key points

- A p-value is the probability of data at least this extreme if the null were true. It is not the probability that the null is true, it depends on the design that produced the data, and it carries no information about the alternative (ASA 2016; Goodman 2008).
- The false positive risk, $\alpha(1-\pi_a)/[\alpha(1-\pi_a)+(1-\beta)\pi_a]$, depends on the prior probability π_a that the drug works. At $\pi_a = 0.1$ and 90% power, a result significant at one-sided 0.025 is wrong about one time in five.
- The Bayes factor weighs the data against both hypotheses and averages over the prior on the alternative, which creates an Ockham penalty of about $\log n$ in $2 \log \mathrm{BF}$. Lindley's paradox is the consequence: a fixed p-value of 0.01 gives a Bayes factor that rises and then falls below one as n grows.
- Within the Sellke-Bayarri-Berger class of alternatives (a non-increasing hazard for $Y = -\log p$, not every prior), a two-sided p-value of 0.05 corresponds to a Bayes factor of at most about 2.5, and a posterior probability of the null of at least 29% from even prior odds. The bound is $-e\, p \log p$ (Sellke, Bayarri and Berger 2001).

- The fixed-level test charges a flat $z_\alpha^2 = 3.84$. The Bayes rule charges $\log n + C$ with $C = 2\log(\pi_0/\pi_a)$. They cross at about $n = e^{3.84} \approx 46$ prior-information units (that is, $I_n\tau^2 \approx 46$, exactly about 41.6; for a unit-information prior $I_n\tau^2 = n$), after which the uncalibrated point-null Bayesian bar against a single unadjusted look is the higher one. For this normal-mixture construction the anytime-valid surcharge is $2\log(1/\alpha)$ for large information and $2\log(1/\alpha)(1 + 1/k)$ exactly; it is not a universal cost of sequential validity. It is the subject of Chapter 12, and the sequential story of Chapter 11.
- A Neyman-Pearson most powerful test between simple hypotheses has a Bayes-risk interpretation. In the symmetric normal example, the fixed-level two-sided test is a Bayes rule with an information-dependent Bayes-factor threshold that falls like $1/\sqrt{q}$ as information grows. The exact Bayes rule has type I error that falls with n, and a posterior threshold of 0.975, 0.99 or 0.995 is a loss ratio of 39, 99 or 199 that the protocol should defend.
- The FDA draft's success criterion is $\Pr(\delta > a \mid \text{data}) > c$. With a flat prior and $a = 0$, $c = 0.975$ is the one-sided p-value test at 0.025. With $a > 0$ or an informative prior the type I error must be calibrated by simulation when calibration is the chosen route, typically at 0.025 one-sided for trials with a frequentist analysis plan.
- A Bayesian criterion calibrated to the same type I error is not a lower bar at a single look. Its benefits come from monitoring, multiple arms and prior information, and each carries a price that later chapters itemize.

i Discussion questions

1. A phase 3 trial for a drug with a novel mechanism reports $p = 0.02$, one-sided, with 90% power. The sponsor believes about one in ten drugs with this mechanism in this disease works. What is the false positive risk? How would you describe it to a clinical colleague, and what would change if the trial had been preceded by two positive phase 2 studies?
2. A cardiovascular outcome trial of 20,000 patients finds a hazard ratio of 0.95 with a two-sided p-value of 0.01. Use Lindley's paradox and the Ockham penalty to discuss whether this is strong evidence of an effect, and what a criterion of the form $\Pr(\delta > a \mid \text{data}) > c$ with a clinically relevant a would say.
3. A sponsor proposes the success criterion $\Pr(\delta > 0 \mid \text{data}) > 0.90$ with a flat prior and a single analysis. What is the one-sided type I error? What would you ask the sponsor to do before this criterion is acceptable in a confirmatory protocol, in light of FDA draft section IV.A.1?
4. The crossing point $I_n\tau^2 \approx 46$ ($n^* \approx 46$ for a unit-information prior) says that an uncalibrated point-null Bayes rule asks for more than a fixed-level test once the data carry enough information. Is it reasonable, in your therapeutic area, to take the Bayesian bar as the more conservative one? What does "conservative" mean for a regulator, and what does it mean for a patient waiting for the drug?
5. The minimum Bayes factor bound is generous to the alternative. Why might a regulatory reviewer nonetheless find it useful when looking at a borderline p-value, and why might a sponsor object to its use?

8. Choosing and Defending Priors

Every statistical analysis of a trial rests on assumptions that are not in the data, and the Bayesian analysis is distinguished mainly by the fact that it writes one of them down. That is also the objection to it. A sponsor who proposes a prior is proposing a number that a reviewer can attack, and a reviewer who has never been asked to attack the implicit assumptions of a power calculation may find the explicit ones easier to see. This chapter is about making the prior something a sponsor can state, justify, quantify and defend, in the form a regulator will ask for it.

We begin with the traditional design, which has a prior of its own that rarely gets that name. We then ask where a prior comes from in a sponsor's program, and why the apparently safe choice of a flat prior is often the least defensible one. We describe the three classical reference priors (skeptical, enthusiastic and noninformative), the conditional means prior that lets a clinician specify a prior on the scale she can think on, and the effective sample size that says in patients how much a prior weighs. We close with prior-data conflict, sensitivity analysis and the tipping-point prior, and with what the FDA's January 2026 draft says about all of it. Chapter 6 supplied the machinery (prior, likelihood, posterior); Chapter 19 treats the case in which the prior is built from external trials; here we stay with the prior as a design choice.

! Regulatory anchor

- **FDA draft guidance, *Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products* (draft, January 2026)**, Section IV.B and the section on prior choice. It states that "It is always critical for Bayesian analyses to have a prespecified prior" and that without calibration "the accuracy of conclusions depends strongly on the choice of prior distribution." It asks sponsors to justify the prior in the protocol, to describe skeptical priors as an option, to quantify the influence of the prior through an effective sample size, and to plan sensitivity analyses over alternative reasonable priors. U.S. Food and Drug Administration (2026g) is a draft and is not binding; it describes current FDA thinking.
- **ICH E20, Adaptive Designs for Clinical Trials (Step 2 draft, June 2025), Section 5.3.** Defines the Bayesian approach as one that will "combine a prior probability distribution with current trial data to obtain a posterior probability distribution" and sets expectations for pre-specification and for reporting the operating characteristics of the design. International Council for Harmonisation (2025a) has not reached Step 4.
- **FDA, *Guidance for the Use of Bayesian Statistics in Medical Device Clinical Trials* (final, February 2010).** The precedent: a final guidance that has allowed priors built from earlier studies in device submissions for more than fifteen years, on the condition that the prior is agreed with the Agency before the trial and its influence is examined. U.S. Food and Drug Administration (2010)
- **ICH E9 (Step 4, February 1998)** requires that the analysis plan and the

assumptions it relies on be fixed before unblinding; a prior is one of those assumptions. See Chapter 3 for the full map.

8.1. The traditional design has no stated prior

Take the ordinary fixed-sample design of Chapter 4. The protocol states a one-sided type I error of 0.025, a power of 90 percent, and a design alternative: a hazard ratio of 0.75, say. From these the sample size follows, and the protocol says nothing further about what the sponsor believes. In the protocol table of the working paper behind this book, the traditional column for the prior reads: "Not stated. The design alternative enters only through the power calculation." (Polson and Sokolov 2026b)

That sentence is accurate, and it hides something. The design alternative assigns no prior probability to any effect: it is a planning assumption. The power calculation evaluates the probability of success at one value of the treatment effect and treats that value as if it were known. For time-to-event data with 1:1 allocation, the standard deviation of the estimated log hazard ratio is approximately $\sqrt{4/d}$ for d events, and with $\theta = -\log \mathrm{HR}$ the design needs

$$d = \frac{4(z_\alpha + z_\beta)^2}{\theta_a^2} \tag{8.1}$$

events (Chapter 9 derives this and asks what it predicts). A hazard ratio of 0.75 gives $\theta_a = 0.288$ and 508 events. The power calculation is conditional performance: the chance of success if the effect is exactly 0.288. It assigns no prior probability to that value, but it is silent about the rest of the sponsor's uncertainty. Nobody believes the effect is exactly 0.288. The sponsor's actual belief is a distribution, wide enough that the real hazard ratio might be 0.90 or 0.65.

The program's real uncertainty can be put back in. If the sponsor describes its uncertainty about θ as a normal distribution centered at the design alternative with standard deviation s_0, the probability that the final z test rejects is no longer the nominal 90 percent. Table 8.1 computes it, using the package function `prob_success`. This prior-averaged performance is called assurance or Bayesian power (O'Hagan, Stevens, and Campbell 2005), and Chapter 9 develops it.

Table 8.1.: Probability that the planned z test rejects, when the sponsor's uncertainty about the log hazard ratio is N(0.288, s_0^2). The first row is the conventional power of 90 percent.

Prior sd s0 on theta	Probability of success
0.00	0.900
0.05	0.868
0.10	0.803
0.20	0.698

Design: HR 0.75, 508 events, $\theta_a = 0.288$, standard error of the estimate = 0.089.

Two consequences follow, and they organize the rest of the chapter. First, a Bayesian protocol is not adding uncertainty to a design that had none; it asks the sponsor to state the uncertainty about the effect that the conditional power calculation sets aside, and assurance averages power over it. Second, the regulator's question about the Bayesian prior ("why this one, and how much does it matter?") is matched by a fair question to the traditional design (what effect was assumed, and how sensitive is the conclusion to it?), and the Bayesian answer, a stated prior with a sensitivity analysis, is more transparent than the traditional one.

The traditional analysis is not a bad choice for this reason. When the primary analysis is a single test of a pre-specified hypothesis in a trial large enough that any reasonable prior is overwhelmed, the traditional design is simple, well understood by every reviewer, and sufficient. The Bayesian prior earns its place when the trial is small, when external information is genuinely available and relevant, when interim decisions must be made on little data, or when several subgroups, arms or doses must be estimated at once.

8.2. Where priors come from in a sponsor's program

A prior is not a mood. In a drug development program it has an identifiable provenance, and the guidance asks the sponsor to document it. Four sources recur.

Phase 2 data from the same program. The most natural source: a phase 2 trial of the same drug, in a similar population, on an endpoint that is the same or close. The estimate and its standard error give a normal prior on θ or a beta prior on a response rate. This is also the source most likely to mislead, for a reason that is hardly specific to Bayesian analysis: programs advance when the phase 2 result looks good. We return to the selection problem below.

The published literature and earlier programs. Trials of drugs of the same class, in the same disease, with the same control. A meta-analysis of such trials yields a distribution for the control response rate, which is how historical control information enters a modern design, and a distribution for effects in the class. Two cautions apply. A literature search is a sampling mechanism, and the prior should be built from a pre-specified search, not from the studies the sponsor happens to like. The FDA draft is explicit that the construction should guard against "preferential selection of favorable studies" (Section V.D.3). How much weight a historical-trial prior should carry, and how the answer moves the posterior, is worked through for the GUSTO trials in Chapter 22. Second, the populations, background therapy and endpoint definitions drift between decades, a problem we take up under borrowing in Chapter 19.

Expert elicitation. When data are thin, as in a rare disease or a new mechanism, the sponsor may elicit the beliefs of clinicians. Elicitation is a technique with a literature of its own, and its main finding is that experts are overconfident (Garthwaite, Kadane, and O'Hagan 2005; O'Hagan et al. 2006): intervals stated as 90 percent contain the truth far less often. Structured protocols exist to blunt this. The Sheffield Elicitation Framework (SHELF) (Oakley and O'Hagan 2019; Gosling 2018) has experts give individual judgments on quantiles or probabilities of observable quantities, share and discuss the evidence, and then agree on a single distribution that a "rational impartial observer" would find reasonable; the facilitator fits a parametric family to the agreed judgments and shows the fit back to the group. The full treatment of the field is O'Hagan et al. (2006). Three rules protect the clinician and the sponsor alike: elicit observables (a response rate in a defined group of patients) and

not parameters (a log odds ratio); elicit tails as well as the middle, because the tails drive the posterior when data disagree; and document the dissenters, since a single "consensus" can hide a bimodal opinion that a mixture prior would represent honestly. Chaloner and colleagues elicited priors graphically from clinicians for a real trial (Chaloner et al. 1993), and Dallow, Best and Montague argue for formal elicitation as routine in drug development (Dallow, Best, and Montague 2018).

Mechanistic and pharmacometric models. An exposure-response model or a disease-progression simulator can produce a distribution of the response at a given dose. The Statistics in Medicine manuscript on priors and predictive stopping (Polson and Sokolov 2026a) describes converting such output into the beta parameters of a conditional means prior by matching moments: if the simulator gives a mean $\hat{\mu}$ and variance $\hat{v}$ for the response rate, the matched beta has $a+b = \hat{\mu}(1-\hat{\mu})/\hat{v} - 1$. A tight simulator is worth many patients; a diffuse one, few. The ICH M15 guideline on model-informed drug development (Step 4, January 2026) governs the credibility of the model that feeds such a prior.

8.2.1. The phase 2 selection problem

A phase 2 estimate that has been selected for being large is biased upward as an estimate of the true effect, because the program carried on only if the data looked good. A prior built naively from that estimate imports the bias into the confirmatory analysis. A small simulation makes the size of the problem concrete. The numbers below are illustrative and not drawn from any program: suppose the true log hazard ratio across the sponsor's candidate drugs varies with mean 0.10 and standard deviation 0.10 (a typical true hazard ratio near 0.90), and that each drug gets a phase 2 trial with 100 events, so the estimate has standard error 0.20. Programs advance to phase 3 when the phase 2 estimate exceeds 0.20 (a hazard ratio below 0.82).

About a third of drugs (0.33) advance. Among them the average phase 2 estimate corresponds to a hazard ratio of 0.71 while the average true effect corresponds to 0.86. A prior centered on the phase 2 estimate would be centered on an effect more than twice the size of the truth, on the log scale. This is the Bayesian form of the winner's curse and of regression to the mean, and it is why the FDA draft warns that analyzing the external data further "risks overfitting to observed random highs" (Section VI). Three remedies are standard and all of them belong in the protocol. Discount the phase 2 information, by a factor or by inflating its variance, so that it counts for fewer patients than it enrolled. Center a prior on the distribution of true effects in the class, an estimate that the sponsor can often obtain from the pooled history of similar programs and not from its own favorable result. Or use the phase 2 data only through a skeptical prior of the kind described next, so that it cannot carry the confirmatory result on its own.

8.3. Skeptical, enthusiastic and noninformative priors

Spiegelhalter, Freedman and Parmar (1994) proposed that a trial should be reported against a small set of reference priors, each standing for a recognizable audience, so that a reader can see which readers the data convince. The 1994 paper demonstrated the idea on actual trials and remains the clearest statement of it. The device grew out of Freedman and Spiegelhalter's earlier work on eliciting clinical opinion for stopping rules (L. S. Freedman and Spiegelhalter 1983), was used in the CHART trials (Mahesh K. B. Parmar et al. 1994) and in the Lancet

proposal for monitoring large trials (Mahesh K. B. Parmar et al. 2001); the later book Spiegelhalter, Abrams, and Myles (2004) is the standard reference. Freedman, Spiegelhalter and Parmar (1994) applied the same device to monitoring, the subject of Chapter 11, where Freedman and Spiegelhalter compared Bayesian and group sequential monitoring (Laurence S. Freedman and Spiegelhalter 1989). For a treatment effect $\theta = -\log \mathrm{HR}$ (positive is benefit), with design alternative θ_a, the three reference priors are the following.

- A **noninformative** (or reference) prior, such as a normal with a very large variance. It adds essentially no information, and for a normal likelihood the posterior is the likelihood: the posterior probability that $\theta > 0$ equals one minus the one-sided p-value.
- A **skeptical** prior, centered at zero with a spread chosen so that the prior probability that the effect reaches the design alternative is small: $\theta \sim N(0, s^2)$ with $s = \theta_a/1.645$, so that $\Pr(\theta > \theta_a) = 0.05$. It represents a colleague who thinks the drug does nothing, and who will concede only that it might work as well as the sponsor hoped, with probability one in twenty.
- An **enthusiastic** prior, centered at the design alternative with the same spread, so that the prior probability that the effect is zero or harmful is 0.05: $\theta \sim N(\theta_a, s^2)$. It represents the investigator who designed the trial.

The enthusiastic prior is a calibration device and not an analysis prior. If a trial is positive even under the skeptical prior, the skeptic is convinced. If it is negative even under the enthusiastic prior, then even the investigator must accept that the drug is not delivering what it promised, which is the logic of a futility rule. The FDA draft states the position about skeptical priors in these terms: “Skeptical priors express skepticism about the presence of very large treatment effects.” The draft notes that they “have not been standard practice in drug development but could be considered”, in particular after failed related programs or where only an incremental gain is plausible, and that in adaptive designs they can serve to temper early stopping for efficacy in place of a more severe boundary (Section V.C). Under the draft’s other axis, a skeptical prior is a way to ask for more evidence without changing the threshold: a skeptic requires a larger observed effect to reach the same posterior probability, which is how a Bayesian design can reproduce the early conservatism of an O’Brien-Fleming boundary (Chapter 11).

The set is completed by a range of equivalence, a band of effects around zero that clinicians regard as no different from no effect. Freedman, Lowe and Macaskill (L. S. Freedman, Lowe, and Macaskill 1984) stopped trials by reference to the opinion of clinicians about such a range, and the 1994 paper reports the posterior probability that the effect lies below the range, inside it and above it. This gives three decision regions in place of one threshold: the drug is favored when the posterior mass above the range is large, the control when the mass below is large, and practical equivalence when the mass inside is large. The middle region is the link to the rest of the book. A noninferiority test also uses an acceptable-loss boundary, but its margin requires a separate justification based on the active control’s established effect and the clinically acceptable loss (Section 4.9). Some futility rules use a low posterior probability of exceeding a prespecified clinically relevant effect. Others use conditional power or predictive probability of final success (Section 15.2). The threshold need not equal the design alternative. The same idea is now called a region of practical equivalence. Tutorials with worked examples of reference priors and of monitoring against them are Abrams, Ashby and Errington (Abrams, Ashby, and Errington 1994), Fayers, Ashby and Parmar (Fayers, Ashby, and Parmar 1997) and the health technology assessment review of Spiegelhalter and colleagues (Spiegelhalter et al. 2000).

The skeptical and enthusiastic reference priors express informative positions about the

treatment effect; the noninformative prior provides a baseline comparison. Where a prior is only meant to regularize a nuisance parameter, a weakly informative default is common: a half-normal or half-t prior on a between-trial standard deviation (Gelman 2006), and a heavy-tailed Student-t (Cauchy) prior on logistic regression coefficients (Gelman et al. 2008). Penalized complexity priors (Simpson et al. 2017) give a principled way to set the scale of such a prior, formal rules for noninformative priors are reviewed by Kass and Wasserman (Kass and Wasserman 1996), and Berger argues the case for objective Bayesian analysis (J. Berger 2006). A reviewer's practical question about any of these defaults is the one the book asks of every prior: how many patients is it worth (Section 8.6), and does the conclusion survive a change of scale? A default whose prior sample size is a small fraction of the trial and whose conclusions do not move under a doubling of its scale is easy to defend. One that carries the result is an informative prior and is to be justified as one.

The next chunk reanalyses the design of the previous section. We suppose the trial ran to its 508 events and observed a hazard ratio of 0.80, so $\widehat{\theta} = 0.223$ with standard error 0.089 and a z statistic of 2.51. The traditional result is a one-sided p-value of 0.006, comfortably below 0.025. Table 8.2 gives the posterior under each reference prior, along with the prior's weight in events. The weight is computed with the formula of Section 8.6.

Figure 8.1 shows the mechanism on a grid of the treatment effect, before the table gives the numbers. The likelihood is the same in both rows (508 events, $\widehat{\theta} = 0.223$ with standard error 0.089), and only the prior changes.

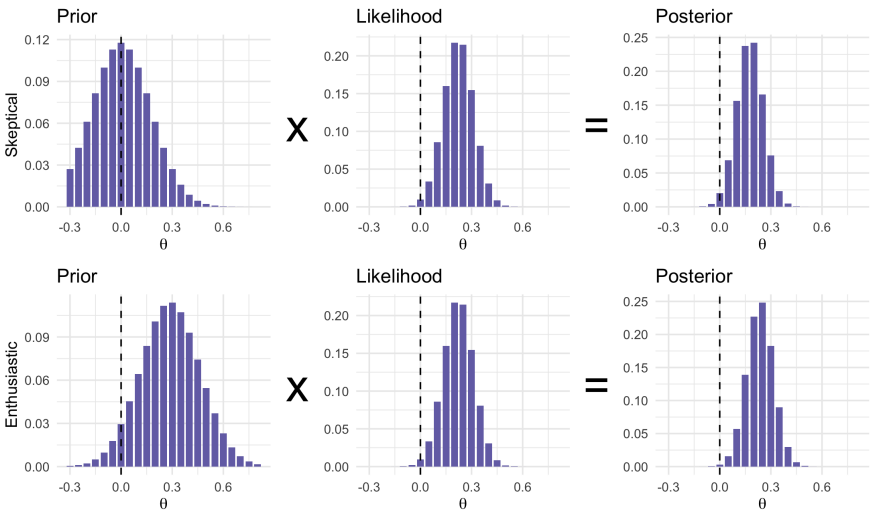


Figure 8.1.: Prior times likelihood equals posterior on a grid of the log hazard ratio effect theta = -log HR (step 0.05), for 508 events and observed HR 0.80. Top: skeptical prior N(0, 0.175^2). Bottom: enthusiastic prior N(0.288, 0.175^2). The dashed line marks no effect.

Reading the figure from left to right, the likelihood is identical in the two rows and sits near 0.22, with a spread much narrower than either prior. The skeptical posterior is pulled toward zero and the enthusiastic posterior toward the design effect, but both end up close to the likelihood, because the data carry about four times the weight of either prior. The posterior probability of benefit, $\Pr(\theta > 0 \mid y)$, is 0.988 under the skeptical prior and 0.999

under the enthusiastic one (exact normal calculation, not the grid sum). The prior moves the posterior, and in this trial it moves the conclusion very little.

Table 8.2.: Posterior for a trial with 508 events and observed HR 0.80, under three reference priors. Pr(theta > 0) is the posterior probability of any benefit. ESS is the prior's weight in events.

Prior	ESS (events)	Posterior median HR	95% interval for HR	Pr(theta > 0)
Noninformative N(0, 100^2)	0	0.800	0.67 to 0.95	0.9940
Skeptical N(0, 0.175^2)	131	0.837	0.72 to 0.98	0.9875
Enthusiastic N(0.288, 0.175^2)	131	0.789	0.68 to 0.92	0.9986

The posterior probability of benefit under the noninformative prior is 0.994, which is one minus the one-sided p-value of 0.006 (the agreement between the traditional and the flat-prior Bayesian answer that Chapter 7 emphasized). The skeptic, who started with only a 5 percent chance that the drug works as designed, ends with a posterior probability of 0.988 that it does something, and a posterior hazard ratio of 0.84. The skeptical prior shrinks the estimate by about a fifth of the way back toward the null, and the evidence is still above 0.975. In this trial the conclusion does not depend on who is asked.

Two warnings apply to this table. The skeptical prior here is not weak: it is worth 131 events, a quarter of the trial. The mean is zero by construction, and the spread is tied to the sponsor's own design alternative, so a sponsor with an optimistic alternative has a correspondingly wide skeptic. A regulator who thinks the realistic effect is smaller than the design alternative would ask for a different skeptic, centered at the same value with a smaller spread, and the fair response is the sensitivity analysis of the next sections and not an argument about whose skeptic is right. The EOLIA trial is the standard real example of how far a skeptic and an enthusiast can sit from each other on one dataset (88 percent against 99 percent posterior probability of benefit; Section 8.13).

8.4. Why default flat priors fail on the scale that matters

The noninformative prior in the table was harmless because the likelihood for θ was normal and the prior was flat on the same scale. It is not harmless in general, and the cases in which it fails are the ones in which a sponsor most needs the prior to behave: small trials, binary endpoints, regression models, and dose-response. The difficulty is that "noninformative" is a statement about one parameterization. A prior that is vague on a parameter is generally informative about every nonlinear function of it, and the treatment effect a trial is designed to estimate is nearly always a function of the parameters. The fullest statement is in the Statistics in Medicine manuscript on priors and predictive stopping (Polson and Sokolov 2026a), whose treatment of this topic this section follows.

8.4.1. Independent beta priors imply an extreme prior on the risk difference

Consider a two-arm trial with a binary endpoint and response rates p_1 (treatment) and p_0 (control). Give each arm a Jeffreys prior (Jeffreys 1946; see Kass and Wasserman 1996 for formal-rule priors), Beta$(1/2, 1/2)$, or a uniform prior, Beta$(1, 1)$. Each is a respectable noninformative choice for a single proportion. What do they say about the risk difference $\theta = p_1 - p_0$? The next chunk (Figure 8.2) answers by simulation and adds a symmetric informative Beta$(20, 20)$ for comparison.

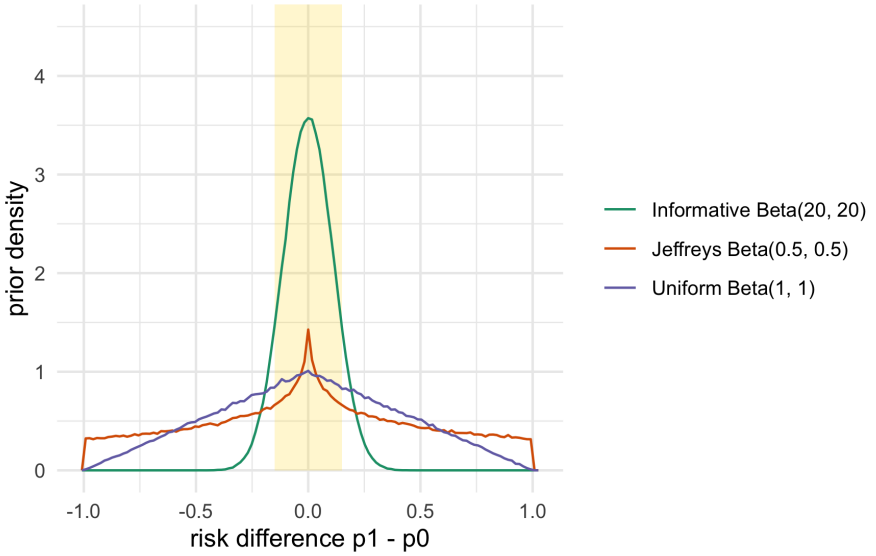


Figure 8.2.: Implied prior density of the risk difference p1 - p0 under independent priors on the two arms. The shaded band is |delta| <= 0.15, a generic range for illustration and not an indication-specific margin.

Under independent Jeffreys priors, the prior standard deviation of the risk difference is 0.50, and there is a 56 percent prior probability that the two arms differ by more than 30 percentage points. Under independent uniform priors the standard deviation is 0.41 and the same probability is 49 percent. A prior that looks like "no opinion" about each response rate is, about the contrast, the opinion that the drug is very likely either dramatically effective or dramatically harmful. Treatment effects that reach a confirmatory trial are mostly modest (D. A. Berry 2012; A. Sokolova, Sokolov, and Polson 2026a), so this is a prior that disagrees with the sponsor's own experience. These priors permit clinically implausible differences, so their posterior influence and the stopping operating characteristics should be evaluated under the proposed design: with equal arm sizes the posterior mean difference is shrunk toward zero relative to the observed difference, and whether early stops for efficacy are in excess depends on the truth and the monitoring rule.

Beta priors were used in trial analysis by Novick and Grizzle (1965), whose convention reads Beta$(z + 1, m - z + 1)$ as m prior patients with z responders, the practical way to express a prior in patients, related to the conjugate prior-sample reading of Raiffa and Schlaifer (Raiffa and Schlaifer 1961); Morita, Thall, and Müller (2008) give the general effective sample size.

The opposite deliberate choice, a prior that is weak on purpose and a success threshold calibrated by simulation, is the one the BNT162b2 vaccine trial used (Section 8.13).

8.4.2. Flat priors on logistic coefficients

The same issue arises with more force in regression. A logistic dose-response model, $\operatorname{logit} p(x) = \beta_0 + \beta_1(x - 30)$, has coefficients on the log-odds scale, a scale on which most clinicians have no intuition. The common default is a vague normal prior on each coefficient, for example $N(0, 10^2)$. The vagueness is an illusion on the probability scale, because the inverse logit saturates: a coefficient on the log-odds scale with standard deviation 10 pushes almost all of the prior mass to the boundaries. Simulating the prior makes the point.

For a patient at the center of the dose range, about 77 percent of the prior mass on the response probability lies below 0.05 or above 0.95. The prior says that the drug at that dose either almost always works or almost never does, which is not the "no information" the investigator meant. Sensible diffuse priors on the coefficient scale exist (the familiar recommendation of Gelman and colleagues (Gelman et al. 2008) is a Cauchy prior with scale 2.5 on coefficients of covariates standardized as they describe), but they are a decision to be defended with the same rigor as any other, and the correct way to defend it is to show what it implies on a scale the clinician understands. That is the idea of the next section.

8.5. The conditional means prior

Bedrick, Christensen and Johnson (1996) proposed to move the elicitation from the coefficient scale to the scale where beliefs already exist. The investigator specifies prior distributions for the response probability at several covariate values (dose levels, say), and the prior on the regression coefficients is whatever distribution those statements induce through the link function. A clinician cannot say what the log-odds slope of the dose-response curve is. She can say what she expects at 10 mg and at 50 mg.

For a model with p coefficients, take p covariate vectors $\tilde{x}_1, \ldots, \tilde{x}_p$ that are linearly independent, so that the map from coefficients to the success probabilities at those points is invertible. Specify independent beta priors, $\tilde{m}_i \sim \text{Beta}(a_{1i}, a_{2i})$, for the success probabilities $\tilde{m}_i = F(\tilde{x}_i'\beta)$, where F is the inverse link. The induced prior on the coefficients is

$$\pi(\beta) \propto \prod_{i=1}^{p} F(\tilde{x}_i'\beta)^{a_{1i}-1} \left[1 - F(\tilde{x}_i'\beta)\right]^{a_{2i}-1} f(\tilde{x}_i'\beta), \tag{8.2}$$

where $f = F'$ is the density associated with the link. (Bedrick, Christensen, and Johnson 1996; Polson and Sokolov 2026a)

> **i** Technical note: why the conditional means prior is a data-augmentation prior
>
> Each factor in Equation 8.2 has the form of a binomial likelihood contribution for the success probability at the design point $\tilde{x}_i$, with $a_{1i} - 1$ successes and $a_{2i} - 1$ failures. The conditional means prior is therefore equivalent to adding $a_{1i} + a_{2i} - 2$ pseudo-observations at each of the p design points to the data, with the factor $f(\tilde{x}_i'\beta)$ playing the part of the Jacobian of the transformation from probabilities to coefficients.

For the logistic link $f = F(1-F)$, so the factors combine to $F^{a_{1i}}(1-F)^{a_{2i}}$, which is exactly the likelihood of a_{1i} successes and a_{2i} failures. A conditional means prior on a logistic model is thus literally $a_{1i} + a_{2i}$ pseudo-patients, and its weight in the analysis is countable. This is the property that makes the effective sample size of the next section exact for this prior.

An elicitation that fits in a protocol: the clinician gives for each design point a mean m_i and a prior sample size ν_i, and the hyperparameters are $a_{1i} = \nu_i m_i$ and $a_{2i} = \nu_i(1-m_i)$. Suppose the clinician says, for a new agent in a dose range from 10 to 50 mg, that she expects about 30 percent response at 10 mg and 70 percent at 50 mg, with the confidence of 10 patients each. That gives Beta$(3,7)$ at 10 mg and Beta$(7,3)$ at 50 mg. (This example is a numerical illustration, not an indication-specific elicitation.) The chunk below (Figure 8.3) samples from the induced prior and compares it with the default $N(0, 10^2)$.

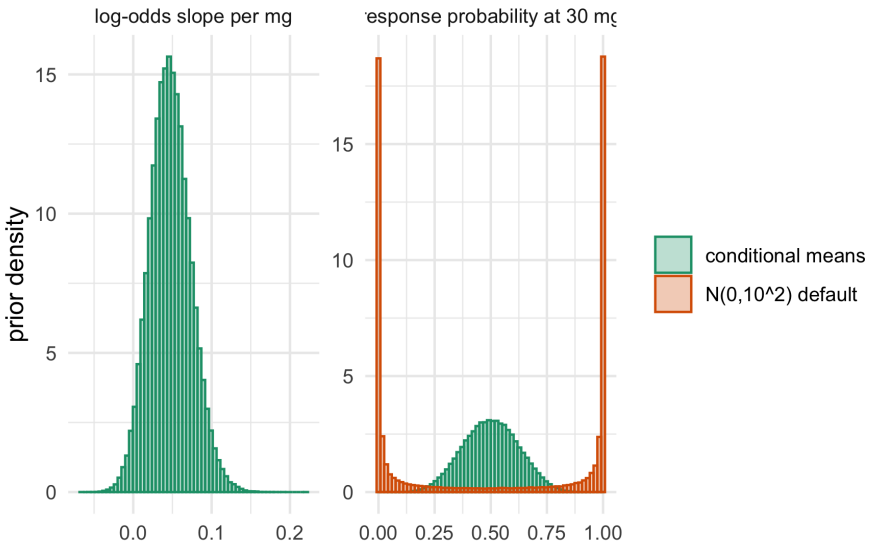


Figure 8.3.: Conditional means prior for a logistic dose-response model with dose centered at 30 mg. Left: induced prior on the response probability at 30 mg, under the elicited Beta(3,7) at 10 mg and Beta(7,3) at 50 mg and under the default N(0, 10^2) coefficients. Right: induced prior on the log-odds slope per mg.

The elicited prior concentrates the response probability at the middle dose near one half and gives the slope a positive mean of 0.047 per mg, with a prior probability of only 3 percent that the slope is negative. The default prior puts nearly all of its mass at the extremes. Two features of this construction help in a protocol. First, the investigator can vary the means to test sensitivity to location, and the sample sizes ν_i to test sensitivity to the amount of information, each in units he understands. Second, the induced prior differs between logit and probit links, which means that the *same scientific belief* gives different coefficient priors for different links without the investigator reasoning about the model (Bedrick, Christensen, and Johnson 1996). One limit is worth recording: independent beta statements at two doses do not enforce a monotone dose-response; if monotonicity is scientifically required, the elicitation must constrain the ordered probabilities or the induced slope. The slope probability reported above (3 percent negative) is the prior's residual tolerance for a non-monotone response, and

it is worth showing to the clinician.

The method is not confined to dose-response models. With a covariate such as baseline severity, the design points are the typical patients at which a clinician has expectations (a mild and a severe patient, in a model with intercept and severity), and the same construction applies. Hanson, Branscum and Johnson (2014) extended the idea through informative g-priors for logistic regression.

8.6. How much does a prior weigh: effective sample size

A skeptical prior in the previous table was said to be worth 131 events. The idea that a prior has a weight in patients is old (it is the content of the Beta$(z+1, m-z+1)$ parameterization), and Morita, Thall and Müller (2008) made it precise. They define the effective sample size (ESS) of a parametric prior as the number of observations whose information about the parameter is equivalent to the prior's, using a minimally informative reference prior as the baseline. The practical content for the common models is simple.

For a response probability with a Beta(a, b) prior the answer is the one Section 6.4 reached from the weighted-average form of the posterior mean, and the other definitions roughly agree with it:

$$\text{ESS} = a + b. \tag{8.3}$$

(The older convention of Novick and Grizzle, under which Beta$(z+1, m-z+1)$ stands for $m = a+b-2$ patients and a uniform prior is worth none, is discussed in Section 6.3; it differs by a constant 2, irrelevant for a prior worth 50 patients and not for one worth 3.) For a normal prior $\theta \sim N(\mu_0, s_0^2)$ on the mean of observations with variance σ^2 per observation,

$$\text{ESS} = \frac{\sigma^2}{s_0^2}. \tag{8.4}$$

For the log hazard ratio with 1:1 allocation, each event carries variance 4, so ESS $= 4/s_0^2$ events, which is the package function `ess_events` used above. A skeptical prior with $s_0 = 0.175$ is worth $4/0.0306 = 131$ events, and the enthusiastic prior with the same spread is worth the same. If the trial plans 508 events, these priors carry about 26 percent as much information as the data.

The working paper comparing Bayesian and standard design puts the guidance in a sentence: "A prior worth five patients in a trial of five hundred is a modelling choice; a prior worth two hundred is an assumption that needs defending." (Polson and Sokolov 2026b) The point of reporting ESS is to turn the argument about the prior into an argument about patients. A reviewer may or may not accept that a phase 2 trial of 100 events deserves to count for 50 events in the confirmatory analysis, but both sides can see what they are disagreeing about.

The conditional means prior of the previous section gives a prior ESS of ν_i at each design point, here 10 at each of two doses. A power prior on an external dataset of n_0 patients, discounted by a factor a_0 between 0 and 1, has an ESS of about $a_0 n_0$ (plus the baseline prior's own weight). Table 8.3 shows the pattern for a phase 2 trial of 100 patients with 30 responders, discounted at four levels; Chapter 19 returns to how a_0 is chosen.

Table 8.3.: A phase 2 result (30 responders of 100) discounted by a power-prior factor a0, as a Beta prior for the control or treatment response rate. Beta(1,1) is the baseline.

a0	Prior	Prior mean	ESS = a + b	Share of a 400-patient trial
1.00	Beta(31.0, 71.0)	0.304	102	26%
0.50	Beta(16.0, 36.0)	0.308	52	13%
0.25	Beta(8.5, 18.5)	0.315	27	7%
0.10	Beta(4.0, 8.0)	0.333	12	3%

Three cautions. First, the formulas give the ESS of a prior in the model's own information currency, and they are exact for the conjugate beta and normal families. For a mixture prior, such as the robust prior of the next section, the ESS depends on what the data do, because the mixture weights move. The FDA draft recognizes this by asking sponsors to report the ESS "for the entire plausible range of outcomes", together with its maximum and its mean (Section V.E), and to compare it with the ESS of a noninformative prior. Second, ESS is a summary of the prior's *variance*, not its location, so two priors with the same ESS and different means have very different effects on a trial whose truth sits away from the prior mean. Third, a prior can be small in ESS and still decisive: if the trial is at a decision boundary, a prior worth five patients can change the decision. This is why ESS is reported alongside a sensitivity analysis and not instead of one.

The evaluation should also report how the posterior and the operating characteristics change with the prior's weight (Morita, Thall, and Müller 2010). That is what the next two sections do.

8.7. Prior-data conflict and robust priors

Suppose the prior is informative and well justified, and the data turn out to disagree with it. A fully Bayesian analysis with a single normal prior will compromise between the two, weighted by their precisions, and it will do so *however far apart they are*. A skeptical prior with a standard deviation of 0.175 will drag a very large observed effect back toward the null by a fixed fraction, and an enthusiastic prior will drag an observed null result toward benefit. For a prior based on external information, such as a phase 2 trial or an earlier study in adults, the same arithmetic says that the posterior is pulled toward information that the data are telling us is irrelevant. The FDA draft recommends that sponsors examine prior-data conflict (Section V.D) and not assume it away.

The standard defense, which this chapter only previews (Chapter 19 develops it, with its connection to power priors and hierarchical models), is a robust mixture prior. The informative component is combined with a weakly informative component, with a prior weight on each. For example, the prior is $0.8\,N(0.29, 0.10^2) + 0.2\,N(0, 1)$: an 80 percent chance that the effect looks like the phase 2 estimate, and a 20 percent chance that nothing is known. After the data, the weights update: the component that predicted the data better gains weight. If the data agree with the informative component, the posterior stays with it, and if they disagree, the weight shifts to the vague component and the posterior follows the data. Schmidli and colleagues (2014) developed this construction for historical controls, and it is the form the FDA draft calls a mixture prior. The FDA applied it in a real submission, the pediatric belimumab trial in lupus (Section 8.13).

Table 8.4.: Robust mixture prior 0.8 N(0.29, 0.10^2) + 0.2 N(0, 1^2) against the pure informative prior N(0.29, 0.10^2), for four observed log hazard ratio estimates with standard error 0.12. Weight is the posterior weight of the informative component.

Observed estimate	HR	Weight on informative	Mixture posterior mean	Informative only posterior mean	Pr(theta > 0), mixture	Pr(theta > 0), informative only
+0.29	0.75	0.96	+0.290	+0.290	1.000	1.000
+0.15	0.86	0.95	+0.228	+0.233	0.993	0.999
+0.00	1.00	0.82	+0.141	+0.171	0.900	0.987
-0.20	1.22	0.16	-0.151	+0.089	0.182	0.877

In Table 8.4, when the observed effect equals the informative mean, the mixture and the single prior agree and the informative component keeps 96 percent of its weight. When the observed effect is $\theta = -0.20$ (the negative log hazard ratio), a harmful hazard ratio of 1.22 that lies about three prior-predictive standard deviations from the informative prior's center, the pure informative prior still reports a positive posterior mean of 0.089 (the data pull it only part of the way, and it gives a posterior probability of benefit of 0.88), while the mixture has moved 84 percent of its weight to the vague component and reports a posterior mean of -0.151, much closer to the data, with a posterior probability of benefit of 0.18. The mixture is, in effect, a prior that has an opinion about what would count as disagreement, which a single normal cannot have. That is what the regulator has in mind when it asks whether the method will "borrow less when the data are less similar" (the draft's description of dynamic discounting methods, Section V.D).

The robust prior has a price. The weight 0.2 on the vague component is itself a prior judgment, and the operating characteristics (type I error when the informative prior is wrong, and the extra patients needed to detect it) have to be simulated. The simulation is the business of Chapter 14, and the justification of the informative component, which is the part a regulator will contest, is the business of Chapter 19.

8.8. Sensitivity analysis over the prior scale

The FDA draft asks that "sensitivity analyses should be planned that utilize a range of alternative reasonable choices for the prior distributions" (Section V.F), for instance by varying the strength of borrowing. The working paper comparing Bayesian and standard design says that it reports a sensitivity analysis for every Bayesian number (Polson and Sokolov 2026b), and the protocol table calls for "a prespecified sensitivity analysis over the prior scale". For a normal prior the natural one-parameter family is the prior standard deviation s_0 at a fixed mean, because s_0 sets the ESS through Equation 8.4 and spans the range from a dogmatic prior ($s_0 \to 0$) to a noninformative one ($s_0 \to \infty$).

The curve in Figure 8.4 is flat to the right and falls only when the prior becomes strong. For prior standard deviations above about 0.15 (ESS up to about 180 events) the posterior probability of benefit stays above 0.98; it falls below the 0.975 threshold only when the skeptical prior is as tight as $s_0 = 0.11$, which is worth 328 events, about two thirds of the information in the trial. This shape is the typical answer when the data are decisive:

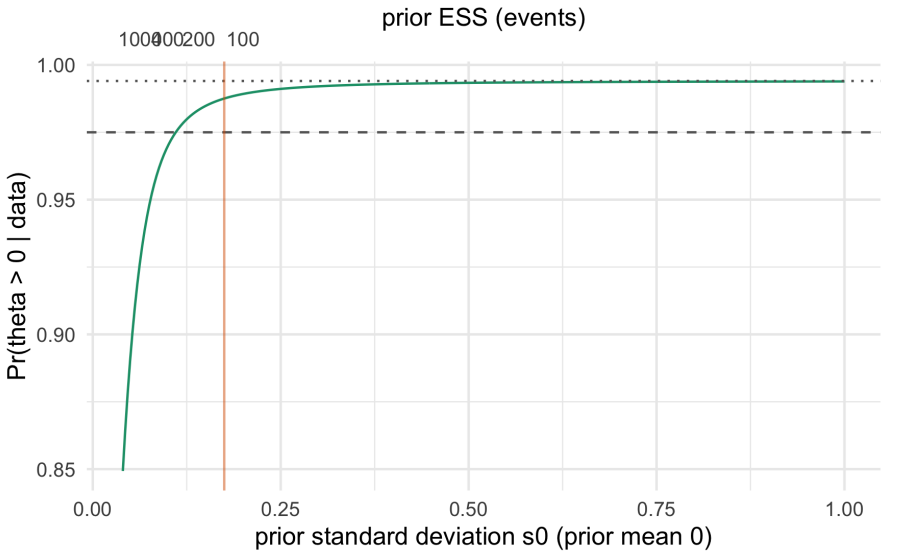


Figure 8.4.: Posterior probability of benefit Pr(theta > 0 | data) for the 508-event trial with observed HR 0.80, as the standard deviation of a prior centered at zero (the skeptical family) varies. The upper axis gives the corresponding prior ESS in events. The dashed line is the 0.975 threshold.

the conclusion is robust to a wide family of priors and breaks only for priors that are themselves almost as informative as the trial. The multiplatform heparin trials pre-specified an enthusiastic prior as exactly this kind of check (Section 8.13).

8.8.1. The tipping-point prior

The sensitivity curve suggests a sharper way of reporting, borrowed from the handling of missing data: instead of asking what the conclusion is under a given prior, ask which prior would reverse it. A **tipping-point prior** is the least extreme prior in a stated family for which the success criterion is no longer met. In the family above, the tipping point is a standard deviation of 0.11, or an ESS of 328 events. A second family varies the prior mean at a fixed spread, which asks how pessimistic the skeptic must be to overturn the result. Holding the skeptical spread of 0.175, the posterior probability of benefit drops below 0.975 only if the prior mean is -0.109, a prior expectation of a hazard ratio of 1.12, which is to say a skeptic who believed that the drug was harmful on average before seeing the data.

The tipping-point report is useful to the regulator because it replaces a debate about the right prior with a statement about which priors the regulator would have to hold to disagree with the sponsor, and a regulator can say plainly whether she holds one. A tipping point that lies in an incredible region (a skeptic who thought the drug harmful, a prior worth two thirds of the trial) supports the claim that the conclusion is driven by the data. A tipping point close to the sponsor's own prior is a warning that the prior drives the answer, which is the case that point 6 of the next section addresses.

The tipping-point analysis belongs in the statistical analysis plan, with the family and the success criterion written down before unblinding; a family chosen after seeing the data would

be a way of looking for a prior that works.

8.9. What the FDA draft says

The FDA's January 2026 draft guidance is the first FDA statement on the Bayesian method in drug trials in general, and its position on priors can be stated in six points. The text of the draft governs; the points below are a summary of Section IV.B and the section on priors.

1. **Pre-specification.** "It is always critical for Bayesian analyses to have a prespecified prior" (Section IV.B), and the sponsor should pre-specify and justify the full details of the prior in the protocol. The prior used for analysis (the analysis prior) is distinguished from design (sampling) priors used only in planning simulations; the second may be skeptical or optimistic about the truth to explore performance, but only the first enters the inference.

2. **Calibration or direct use of the posterior.** The draft describes three routes for the success criterion $\Pr(\delta > a \mid \text{data}) > c$: calibration to type I error, direct interpretation of posterior probabilities, and benefit-risk or decision-theoretic criteria. In a *calibrated* design, a and c are chosen by simulation so that the familywise type I error is at most 0.025 one-sided, and the prior is used as a modeling device whose choice affects power and not the error rate. In a *non-calibrated* design, in which the prior is meant to summarize real external information, the posterior probability is read directly (a posterior probability of 0.98 means a 2 percent probability of ineffectiveness given the prior), and this is acceptable where the prior is specified so that it "accurately and comprehensively reflects the external data" (Section IV.A), by agreement with the Agency. Here the prior matters in the second sense: without calibration, "the accuracy of conclusions depends strongly on the choice of prior distribution" (Section IV.B.2). The third, decision-theoretic route chooses the threshold from stated losses (Section 1.6). The calibration route is the subject of Chapter 14.

3. **Noninformative and minimally informative priors.** No single approach is universally preferred. They can still influence the posterior when data are sparse, and they are not invariant to transformation, so a prior that is flat on one scale must be checked on the scale of the estimand: the point of the beta and logistic examples above. They also serve as the reference against which ESS is reported.

4. **Skeptical priors.** "Skeptical priors express skepticism about the presence of very large treatment effects." They may be appropriate after failed related programs or where only an incremental gain is plausible, and they can stand in for adjusting an early-stopping boundary. The same section also discusses enthusiastic priors.

5. **Informative priors need more.** "Informative priors will generally need a greater amount of justification" (Section V.A), with strong justification for borrowing on the grounds of relevance and feasibility, most commonly in pediatrics and rare diseases. Details of prior construction (a pre-specified search, criteria against preferential selection of favorable studies, relevance of the estimand and population) are the substance of Chapter 19.

6. **Quantify the influence.** Report the prior mean and the ESS, over the plausible range of outcomes, and plan sensitivity analyses over alternative reasonable priors. The "when the prior drives the answer" scenario is the one in which a sponsor must

> show, by the sensitivity curve and the tipping point, that the conclusion survives or must justify the prior at the level of evidence that a regulator accepts for any other analytic assumption.

The draft leaves several things open. It does not say how large an ESS is acceptable, how wide a family of sensitivity priors is enough, or how to resolve a disagreement between a sponsor's prior and the Agency's. These are the points on which a sponsor should seek agreement in a formal meeting before the trial (the programs are in Chapter 3 and the practicalities in Chapter 23). The draft is also not binding, and a sponsor who relies on it should say in the briefing document that it does.

ICH E20 in its Step 2 draft treats the prior in the same spirit, in Section 5.3: the Bayesian analysis combines a prior with the trial data, within the draft's general expectation that adaptations and analysis methods are pre-specified and that the design's performance is evaluated by simulation (see Chapter 14). As a Step 2 text, E20 was open for public consultation until 30 November 2025 and may change before final adoption.

8.9.1. The device precedent

None of this is new at the Agency. The Center for Devices and Radiological Health issued its Bayesian guidance for medical device trials as a draft in 2006 and in final form in February 2010 (U.S. Food and Drug Administration 2010), and several device approvals have rested on priors built from earlier studies, often a previous generation of the same device. The device guidance's recipe is the one this chapter describes in different words: agree on the prior with the Agency before the trial, justify the exchangeability of the earlier data with the current trial, quantify how much the earlier data count, and show by simulation how the design performs when the prior is wrong. On the type I error it says (Section 4.8, "Assessing the operating characteristics of a Bayesian design"): "When using prior information, it may be appropriate to control type I error at a less stringent level than when no prior information is used," and the degree of relaxation "is a case-by-case decision that depends on many factors, primarily the confidence we have in the prior information." This parallels the distinction the 2026 drug draft draws between calibrated and non-calibrated designs. The device precedent shows that the regulatory machinery for an informative prior exists and works, and it shows the limits: devices are often iterations of a platform with a plausible exchangeability argument, an argument that is much harder for a new molecular entity.

8.10. Side by side: what changes in the protocol

Item	Traditional design	Bayesian design
Prior information	Implicit in the design alternative and the assumed variance or event rate	Stated as a distribution, with source and justification
Where the sponsor's uncertainty appears	Nowhere in the analysis; sometimes in an assurance calculation	In the prior and in the sensitivity analysis

Item	Traditional design	Bayesian design
How the effect of the assumption is reported	Power at several alternatives	Prior ESS; posterior under a skeptical, enthusiastic and noninformative prior; tipping point
Agreement with the regulator	Design alternative and sample size	Prior, success criterion, calibration, and sensitivity family
When the traditional and Bayesian answers agree	Flat prior and a normal likelihood: Pr(effect > 0) equals one minus the one-sided p-value	
Main regulatory question	"Is the assumed effect realistic?"	"Why this prior, how much does it weigh, and does it drive the answer?"

What stays the same is the standard of evidence: the trial is still judged by whether the conclusion would survive an informed skeptic, and a Bayesian design for a confirmatory trial is calibrated to the same one-sided 0.025 when the prior is not meant to carry information. What changes is that the assumptions are written down and can be argued with.

8.11. Writing the prior into the protocol

In the protocol

A prior section of a protocol or statistical analysis plan should contain, at minimum:

1. **The analysis prior**, by name and parameters, on the scale of the estimand, together with its implied distribution for the estimand (for example, the implied prior on the risk difference or the hazard ratio) so that a reviewer does not have to compute it.
2. **Its provenance**: the data, search or elicitation exercise from which it was built, with the dates and the people involved, and the criteria for including or excluding sources, written before the data from the current trial are seen.
3. **Its effective sample size**, as defined in Morita, Thall and Müller (2008), by formula for conjugate priors and by simulation for mixtures, over the plausible range of outcomes, and as a share of the planned sample size.
4. **The comparison priors**: a noninformative prior (reference), a skeptical prior, and where useful an enthusiastic prior, with the posterior summaries under each.
5. **The sensitivity analysis**: the family of priors (for example, standard deviation from the ESS-zero reference to twice the proposed ESS; mean over a stated range), the success criterion, and the tipping-point report.
6. **The robustness provision**: if the prior is informative, the mixture or discounting rule applied under prior-data conflict, with its parameters, and the simulation that evaluated it.
7. **The success criterion and its calibration**: whether a and c are calibrated to a familywise type I error of 0.025 one-sided by simulation, or the posterior

> probability is to be read directly (with the agreement with the Agency that supports it), or a benefit-risk or decision-theoretic criterion is used, with its losses.

8.12. What can go wrong

Four failures recur, and a sponsor can guard against each.

The prior chosen after the data are seen. A prior that is fixed after unblinding is a free parameter, and an analysis plan that allows it is a plan that allows the conclusion to be selected. Pre-specification is not a formality (the first quotation in the regulatory anchor says "always critical"), and the sensitivity family and the tipping-point report are fixed in the same document.

A single published estimate treated as a prior. One phase 2 result, selected for a good look, is the case simulated above. The cure is to discount, to center on a population of programs and not on the sponsor's own estimate, or to cap the prior's ESS at a level the sponsor is willing to defend.

A flat prior mistaken for no prior. A flat prior is a choice and one that carries implicit commitments on every other scale. Compute what it implies about the estimand before claiming that it is neutral. In a large trial with a normal likelihood the issue disappears, and for small binary trials, dose-response and subgroup models it does not.

A prior that matters presented as a prior that does not. A skeptical prior of ESS 131 in a trial of 508 events changes the posterior hazard ratio from 0.80 to 0.84. That is a prior that matters to the estimate, even if the conclusion survives. The report should say so, and let the reader judge.

8.13. In practice

Four trials show the prior at work: one where the regulator asked for a tipping-point analysis, one family of platform trials that fixed neutral priors in advance and tested an enthusiastic one, one reanalysis that shows the prior moving the conclusion, and one pivotal trial that chose a deliberately weak prior.

8.13.1. Pediatric belimumab in lupus: a tipping point over the borrowing weight

The pediatric trial of intravenous belimumab in lupus (C1109, PLUTO) was not powered to stand on its own, so FDA requested a post hoc analysis that borrowed from the adult trials through a robust mixture prior with borrowing weight w (Brunner et al. 2020; U.S. Food and Drug Administration, Center for Drug Evaluation and Research 2026). The sponsor was not asked to defend a single w: the posterior probability of efficacy was at least 97.5 percent whenever w was 0.55 or more, and the review division judged such a weight reasonable (Pottackal et al. 2025). The data, the mixture and the approval are told in Section 19.17.2.

The lesson is the tipping-point logic of Section 8.8.1 applied to the borrowing weight instead of the prior spread: the report names the weakest reliance on adult data under which the conclusion holds, and the argument moves from "what is the right weight" to "is a weight of 0.55 or more credible". The FDA case study also stresses that the analysis was post hoc, and recommends that a future program agree and pre-specify the Bayesian approach early (pre-IND), which is the point of the regulatory anchor of this chapter. Chapter 19 returns to the mixture itself.

8.13.2. REMAP-CAP and the multiplatform heparin trials: neutral priors, enthusiastic sensitivity

The REMAP-CAP platform fitted a Bayesian cumulative logistic model for organ support-free days, with neutral prior distributions for the individual treatment effects, and fixed its efficacy threshold (a posterior probability of an odds ratio above 1 exceeding 99 percent) before the data (REMAP-CAP Investigators, Gordon, et al. 2021). The tocilizumab results are in Section 6.12. The point here is the choice of prior: it favored neither arm, so that the data had to do the work.

The therapeutic heparin trials (ATTACC, ACTIV-4a and REMAP-CAP, run as a multiplatform trial) used the same style: weakly informative priors for the baseline probabilities, a hierarchical prior with the severe and moderate cohorts nested around an overall effect with a neutral mean, superiority at a posterior probability above 99 percent and futility at above 95 percent for an odds ratio below 1.2. The pre-specified sensitivity analysis used an enthusiastic prior with a mean odds ratio of 1.75 (95 percent credible interval 0.74 to 4.15, prior probability of superiority 90 percent), the enthusiastic reference prior of this chapter (REMAP-CAP Investigators, ACTIV-4a Investigators, et al. 2021; ATTACC Investigators et al. 2021). Enrollment in critically ill patients stopped for futility (adjusted odds ratio 0.83, 95 percent credible interval 0.67 to 1.03), and in non-critically ill patients the neutral-prior analysis gave a 98.6 percent posterior probability that therapeutic dosing increased organ support-free days (odds ratio 1.27, 1.03 to 1.58) (ATTACC Investigators et al. 2021).

The lesson is the structure and not a single number. Neutral analysis priors, a calibration of the thresholds by simulation, and an enthusiastic prior as a stress test: a conclusion that holds with a skeptical-to-neutral analysis prior and is not overturned by the optimist is hard to attack. Chapter 20 treats the platform designs themselves.

8.13.3. EOLIA: how far the prior moves the conclusion

EOLIA, whose design and p-value of 0.09 are described in Section 7.10 and whose full reanalysis is in Section 22.3, is the standard real example of how far the prior can move a conclusion. In the post hoc reanalysis (Goligher et al. 2018) the posterior probability of any mortality benefit ranged from 88 percent under a strongly skeptical prior to 99 percent under an enthusiastic or a published-study prior.

The same reanalysis shows what the prior does not change. The probability that the absolute risk reduction reached the 20 points the trial was designed to detect stayed small under every prior (see Section 22.3). The lesson is the one of the sensitivity curve of this chapter: report the posterior probability of the clinically relevant effect (and not only of any benefit) under a skeptical, a neutral and an enthusiastic prior, and read the range, not a single value. The range 88 to 99 percent says that the answer about "some benefit" depends on who is asked,

and the small probability of the large benefit says that the trial was not built for the effect size it assumed.

8.13.4. BNT162b2: a deliberately weak prior with a calibrated threshold

The pivotal Pfizer-BioNTech vaccine trial (analysis and numbers in Section 6.12) made the opposite choice: a deliberately weak prior with a threshold calibrated by simulation. Its Beta$(0.700102, 1)$ prior on $p = (1 - \text{VE})/(2 - \text{VE})$, the probability that a case falls in the vaccine arm, has mean 0.4118 (the value at VE = 30 percent) and an ESS by Equation 8.3 of $a + b = 1.7$ cases, against a final analysis planned at 164 cases. Success required a posterior probability of VE > 30 percent above 0.995 at an interim analysis and above 0.986 at the final one, with the thresholds set by simulation so that the overall one-sided type I error was 0.025 (Pfizer Inc. 2022).

The lesson is the calibrated route of the FDA draft (point 2 of Section 8.9): the prior is a modeling device, small enough that no reviewer could say it carried the result, and the evidence standard sits in the threshold, whose frequentist behavior was checked in advance. The posterior calculation for this prior is in Section 6.12.

Key points

- The traditional design conditions on a design alternative in the power calculation and states no uncertainty about it. A Bayesian protocol asks the sponsor to state that uncertainty as a prior, and assurance averages power over it.
- Priors come from phase 2 data, earlier programs and literature, structured expert elicitation (SHELF) and mechanistic models. Phase 2 estimates are inflated by selection: in the illustration, programs that advanced had a phase 2 hazard ratio of 0.71 and a true one of 0.86.
- Skeptical, enthusiastic and noninformative priors are reference points that show which audiences the data convince. A skeptical prior with a 5 percent chance of the design effect is worth 131 events here, a quarter of a 508-event trial.
- "Flat" is a statement about one scale. Independent uniform or Jeffreys priors on two arms put 49 to 56 percent prior probability on a risk difference larger than 30 points in either direction, and N(0, 10^2) priors on logistic coefficients put 77 percent of mass on probabilities below 0.05 or above 0.95. The conditional means prior moves the elicitation to the probability scale, and it is a data-augmentation prior with countable pseudo-patients.
- Effective sample size reports the weight of a prior in patients: $a + b$ for a beta, σ^2/s_0^2 for a normal, $4/s_0^2$ events for a log hazard ratio. A prior worth 5 patients in a trial of 500 is a modeling choice and one worth 200 is an assumption.
- A robust mixture prior defends against prior-data conflict by moving weight to a vague component when data disagree. It has its own price and is treated in Chapter 19.
- Report a sensitivity analysis over the prior scale and a tipping point: the weakest prior in a stated family that reverses the conclusion.
- In practice: the belimumab pediatric analysis reported a tipping point over the borrowing weight (efficacy at least 97.5 percent once $w \geq 0.55$), the multiplatform heparin trials stress-tested neutral priors with an enthusiastic one, EOLIA's posterior

probability of benefit ranged from 88 to 99 percent with the prior, and BNT162b2 used a Beta(0.700102, 1) prior worth 1.7 cases with calibrated thresholds.

- The FDA's January 2026 draft recommends a pre-specified prior, distinguishes calibrated from non-calibrated success criteria, and asks for the ESS and sensitivity analyses. It is a draft; the 2010 device guidance is the final precedent.

Discussion questions

1. Your phase 3 protocol uses a hazard ratio of 0.75 for the sample size. The design alternative is a planning assumption, not a prior. What prior would you actually defend for the hazard ratio, which uncertainty would it carry, and where would that uncertainty come from?
2. A sponsor proposes to use a phase 2 estimate with a hazard ratio of 0.65 as the center of an informative prior in the confirmatory trial. List three questions you would ask about how the phase 2 result came to be in the program. How would you change the prior, in the light of the selection simulation of this chapter?
3. A reviewer says that the skeptical prior in your analysis is too weak, and proposes one centered at a hazard ratio of 1.05. How should the tipping-point report be designed so that this conversation can be settled with a number?
4. For a rare-disease trial of 60 patients, a prior of ESS 20 is proposed from a registry. The posterior probability of benefit is 0.96 under the prior and 0.88 under a noninformative prior. Is the prior driving the answer? What does the FDA draft say about the non-calibrated route, and what would you want to see before agreeing?
5. Why does the conditional means prior make it easier to check an elicited prior against the clinician's beliefs than a prior on regression coefficients? Which statements would you ask the clinician to confirm before the protocol is locked?

Part III.

The Standard Apparatus Reconsidered

9. Error Rates, Power, and Sample Size

Every protocol contains a sentence of the form "a sample size of N patients provides 90% power to detect a hazard ratio of 0.80 at a one-sided significance level of 0.025." It is the most-read sentence in the statistical section, the one a finance officer uses to budget the trial, a regulator uses to judge whether the study is adequate, and a data monitoring committee uses to understand what the trial can and cannot show. It is also, as this chapter will argue, a sentence that hides a great deal. The number N is the output of a formula whose inputs are two error rates and a guess about the treatment effect, and the formula is silent about how the guess was made, how much the answer moves when the guess is wrong, and what the trial would look like if the sponsor believed something different about the drug before it started.

This chapter does three things. First it presents the traditional calculation correctly and fairly: the formula, its binary, continuous and time-to-event versions, worked numbers, its sensitivity to the design alternative, and the power curve that shows what a fixed trial can detect. Second it presents the Bayesian reading of the same calculation. A conventional power calculation can be represented as assurance under a point-mass design prior, and averaging the power curve over an explicit prior gives a quantity called assurance (or probability of success) that answers the question the sponsor actually has. Third it presents a sample size equation derived by Polson and Sokolov (Polson and Sokolov 2026b) from the Bayes decision rule, which replaces the constant z_α in the textbook formula by a boundary that grows with the sample size. That equation contains a result that runs against the thesis of this book, stated here up front: a fixed-sample design built on the Bayes decision boundary (the moderate deviation scale) is larger than the textbook design, not smaller. The Bayesian saving in patients and time lies elsewhere, in the monitoring of the trial, and 11 and 21 show where.

! Regulatory anchor

- **ICH E9, Statistical Principles for Clinical Trials** (Step 4, 1998) (International Council for Harmonisation 1998), section 3.5, Sample Size. Requires that the number of subjects be determined by the primary objective, that the method of calculation and the assumptions behind it (variability, the two error rates, the size of the treatment difference) be given in the protocol, and encourages investigating how sensitive the estimate is to departures from those assumptions. The difference to be detected is to be based on clinical relevance or on the anticipated effect.
- **21 CFR 314.126** (U.S. Food and Drug Administration 2024b). The regulation that defines an adequate and well-controlled investigation, and so what counts as substantial evidence under section 505(d) of the FD&C Act. It does not prescribe a sample size formula; it requires that the methods of analysis be adequate to assess the effect of the drug. A sample size justification is part of that adequacy.
- **FDA draft guidance, Use of Bayesian Methodology in Clinical Trials**

of Drug and Biological Products (draft, January 2026) (U.S. Food and Drug Administration 2026g). The draft describes three routes for a Bayesian success criterion: calibration to a type I error rate (one-sided 0.025 is the typical target, section IV.A), direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA), and decision-theoretic or benefit-risk criteria (section IV.A.3). Calibration is this book's default for confirmatory designs without borrowing and one of the draft's routes, not a universal requirement, and a draft recommends and does not require. Sample size for such a design is obtained by simulating its operating characteristics under pre-specified scenarios (Chapter 14).

- **ICH E9(R1)** (Step 4, November 2019) (International Council for Harmonisation 2019). The sample size is for an estimand, not for a test. The design alternative must be an effect on the estimand that the trial reports, with its handling of intercurrent events.

The map of all of these documents, with status and dates, is in Chapter 3.

9.1. The textbook calculation

9.1.1. Two errors and one guess

A trial that ends in a one-sided test can be wrong in two ways. It can declare a useless drug effective, a type I error, whose probability under the null hypothesis the protocol fixes at α, conventionally 0.025 one-sided (Chapter 4 explained why 0.025 and not 0.05). Or it can fail to declare an effective drug effective, a type II error, with probability β at a stated effect. The complement $1-\beta$ is the power. The two errors are not symmetric in regulatory practice. Type I error control is a central regulatory design criterion, and calibration to one-sided 0.025 is this book's default for confirmatory adaptive designs without borrowing: the sponsor must demonstrate adequate control. Type II error is a business and ethical quantity: the sponsor chooses it, usually at 0.10 or 0.20, and bears the cost of a failed trial. A regulator will ask about it because an underpowered trial exposes patients to an experimental treatment without a realistic prospect of an answer, but power is not a condition of approval in the way that error control is.

For a treatment effect θ estimated by $\hat{\theta}$ with standard error $1/\sqrt{nI}$, where n is the sample size and I the Fisher information per observation, the standardized statistic is $Z = \hat{\theta}\sqrt{nI}$. Under $\theta = 0$ it is standard normal, and the test rejects when $Z > z_\alpha$. Under a true effect θ, Z is normal with mean $\theta\sqrt{nI}$ and variance one, so

$$\text{Power}(\theta) = \Phi\left(\theta\sqrt{nI} - z_\alpha\right). \tag{9.1}$$

Setting this equal to $1-\beta$ gives $\theta\sqrt{nI} - z_\alpha = z_\beta$, and solving for the sample size,

$$n = \frac{(z_\alpha + z_\beta)^2}{I\,\theta^2}. \tag{9.2}$$

This is the formula the book writes as $n = (z_\alpha + z_\beta)^2/\theta^2$ when $I = 1$. At $\alpha = 0.025$ and power 90%, $z_\alpha = 1.960$ and $z_\beta = 1.282$, so $(z_\alpha + z_\beta)^2 = 10.51$. The numerator is the whole of the error specification. The denominator is the whole of the science: the effect θ the trial is built to detect, on the scale of information. At 80% power the numerator is 7.85. A trial designed for 90% rather than 80% power therefore needs a third more patients (10.51 over 7.85 is 1.34), whatever the effect.

Two features of Equation 9.2 deserve attention before any numbers. The sample size is inversely proportional to θ^2, so halving the effect quadruples the sample size. And the formula takes the effect as given. The guess about θ is the one input the data of the trial have not yet informed, and it is the input to which the answer is most sensitive.

9.1.2. The same formula for three kinds of endpoint

What changes between endpoints is only what $I\theta^2$ means.

Continuous endpoint. For a difference in means Δ between two arms with common standard deviation σ and n patients per arm, the standard error of the difference is $\sigma\sqrt{2/n}$ and

$$n_{\text{per arm}} = \frac{2\,\sigma^2 (z_\alpha + z_\beta)^2}{\Delta^2}. \tag{9.3}$$

The standardized effect is Δ/σ. This is the calculation behind the worked design of the book's companion online lab (module 03, https://vsokolov.org/bayesian_trials_lab/03-errors-power.html) (an improvement of 3 points on a score with patient-to-patient standard deviation of 10 gives a standardized effect of 0.3): 10.51 times 4 over 0.09 is 467 patients in total at 90% power.

Binary endpoint. For response probabilities p_0 and p_1 the variance of the difference in observed proportions is $p_0(1-p_0)/n + p_1(1-p_1)/n$ and

$$n_{\text{per arm}} = \frac{(z_\alpha + z_\beta)^2 \left[p_0(1-p_0) + p_1(1-p_1)\right]}{(p_1 - p_0)^2}. \tag{9.4}$$

The variance now depends on the unknown rates, so a second assumption, the control rate, enters the calculation alongside the effect. A trial that assumed a 30% control response rate and finds 40% has a different information per patient than the plan assumed, before any question of treatment effect arises. Equation 9.4 is an approximation. It uses the alternative variances in both the critical value and the power term, whereas software that uses a pooled variance under the null for the critical value gives a different answer, and the difference need not be small when the rates are far apart. The two versions use the variance at different roles and neither is the exact one.

Time-to-event endpoint. Here the information is carried by events, not patients. Schoenfeld (D. Schoenfeld 1981) showed that for a logrank test with 1:1 allocation the statistic is approximately normal with mean $\log(\text{HR})\sqrt{d}/2$, where d is the total number of events, so $I = 1/4$ per event and

$$d = \frac{4\,(z_\alpha + z_\beta)^2}{\log(\text{HR})^2}. \tag{9.5}$$

With the book's sign convention $\theta = -\log \mathrm{HR}$ so that benefit is positive, this is Equation 9.2 with $I = 1/4$. Freedman's (Laurence S. Freedman 1982) formula, $d = (z_\alpha + z_\beta)^2(1 + \mathrm{HR})^2/(1 - \mathrm{HR})^2$, is the other version that software commonly prints, and it differs from Equation 9.5 in the second-order terms: at $\mathrm{HR} = 0.80$, 90% power and one-sided $\alpha = 0.025$ it gives 851 events against 844 for Schoenfeld's, a difference of about one percent. The choice between them is not what drives a time-to-event design. The number of patients is then obtained from d by a second, separate calculation: how many patients must be enrolled, and followed for how long, so that d events have occurred by the planned end of the trial. That second calculation depends on the control event rate, the accrual pattern, dropout and the follow-up time, and a trial that accrues slower than planned or has a lower event rate than assumed does not lose power through its enrollment but through its calendar. A time-to-event trial is complete when it has its events.

```
ss3 <- data.frame(
  endpoint = c("Continuous, delta 3, sd 10 (total)", "Binary, 30% vs 45%
   ↪  (total)", paste("Events, HR", c(0.5, 0.7, 0.8, 0.9))),
  power80 = c(2 * n_continuous(3, 10, power = .8), 2 * n_binary(.30, .45,
   ↪  power = .8), events_schoenfeld(c(0.5, 0.7, 0.8, 0.9), power = .8)),
  power90 = c(2 * n_continuous(3, 10, power = .9), 2 * n_binary(.30, .45,
   ↪  power = .9), events_schoenfeld(c(0.5, 0.7, 0.8, 0.9), power = .9)))
   ↪  |>
  transform(power80 = round(power80), power90 = round(power90))
knitr::kable(ss3, col.names = c("Endpoint", "80% power", "90% power"),
 ↪  align = "lrr")
```

Table 9.1.: Sample sizes from the textbook formula, one-sided 0.025, for three kinds of endpoint.

Endpoint	80% power	90% power
Continuous, delta 3, sd 10 (total)	349	467
Binary, 30% vs 45% (total)	319	427
Events, HR 0.5	65	87
Events, HR 0.7	247	330
Events, HR 0.8	631	844
Events, HR 0.9	2828	3786

The binary example is the companion lab's scenario (module 03) (30% on the old treatment, 45% on the new): 214 patients per arm at 90% power. The event counts show the other face of the square law. Moving from a hazard ratio of 0.50 to 0.80 multiplies the required events by 9.6, from 87 to 844. For a trial whose effect is a modest 10% reduction in hazard (HR 0.90), the answer at 90% power is nearly 3,800 events. Chapter 21 takes the SELECT cardiovascular outcomes trial (Lincoff et al. 2023) as its main example (see also Section 9.6.2). SELECT was designed for a hazard ratio of 0.83, and the formula gives 1211 events at 90% power for that design alternative, close to the 1,225 events planned. At the hazard ratio of 0.80 that the trial later observed the formula gives 844 events, a number the later chapters use for the retrospective reconstruction.

9.1.3. Sensitivity to the design alternative

ICH E9 asks sponsors to consider how sensitive the sample size is to the assumptions. The square law makes that sensitivity easy to quantify. If the trial is sized for an effect θ_d but the truth is $\theta = \rho\,\theta_d$, power is $\Phi(\rho(z_\alpha + z_\beta) - z_\alpha)$. This depends only on the ratio ρ and not on θ_d or n.

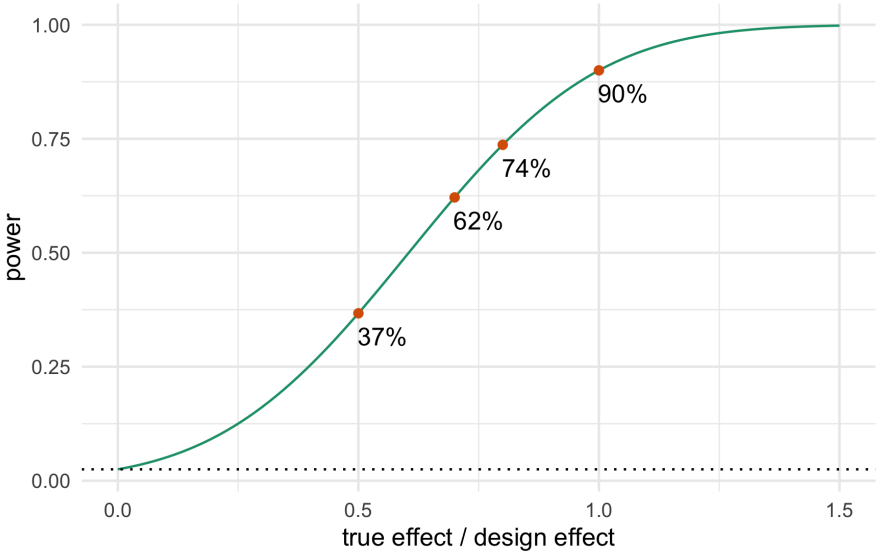


Figure 9.1.: Power of a trial sized for 90% power (one-sided 0.025) when the true effect is a fraction of the design alternative. A true effect of 70% of the design value leaves a power of about 62%.

A trial sized for 90% power has 83% power if the true effect is 90% of the design alternative, 62% power at 70% of it, and 37% at half. The converse is equally striking and less discussed: if the drug is better than assumed, the trial is overpowered and ends later than it needed to. Both errors are expensive, and the traditional design has no means of correcting either once the first patient is enrolled, other than the sample size re-estimation and interim analyses of Chapter 11.

The next table varies the design alternative over a range typical of a standardized effect and gives the information requirement, which the moderate deviation section below uses again.

Table 9.2.: Textbook information requirement $n = 10.51/\theta^2$ at one-sided 0.025 and 90% power, for a range of design alternatives.

Design effect	Information required
0.05	4203
0.10	1051
0.15	467
0.20	263
0.30	117
0.50	42

A halving of θ from 0.20 to 0.10 takes the requirement from 263 to 1,051. A statistician who is asked "what effect should I design for" has an unusually consequential question on her hands, and E9 section 3.5 names two ways to answer it: the smallest effect that would be clinically relevant, or the effect the sponsor anticipates. Those are different numbers. A design for the minimal clinically important difference is conservative: it gives the trial the chosen power to detect superiority when the true effect equals that minimum. It does not establish that the effect exceeds the minimum, because the null being tested is still an effect of zero, and demonstrating benefit beyond a minimum requires a different success criterion. A design for the anticipated effect is optimistic and, if the anticipation is wrong, produces a failed trial. In practice the sponsor picks something between, informed by phase 2, and that act of picking is a prior elicitation that the protocol never calls by that name. The price of getting it wrong is concrete: pivotal trials are expensive, and cost rises steeply with patients (Section 9.6.1).

9.1.4. The power curve

A single power number hides the shape. The power curve plots Power(θ) for fixed n and α over the whole range of effects. It equals α at $\theta = 0$, rises through 50% at $\theta = z_\alpha/\sqrt{nI}$ and passes the design value θ_d at $1 - \beta$.

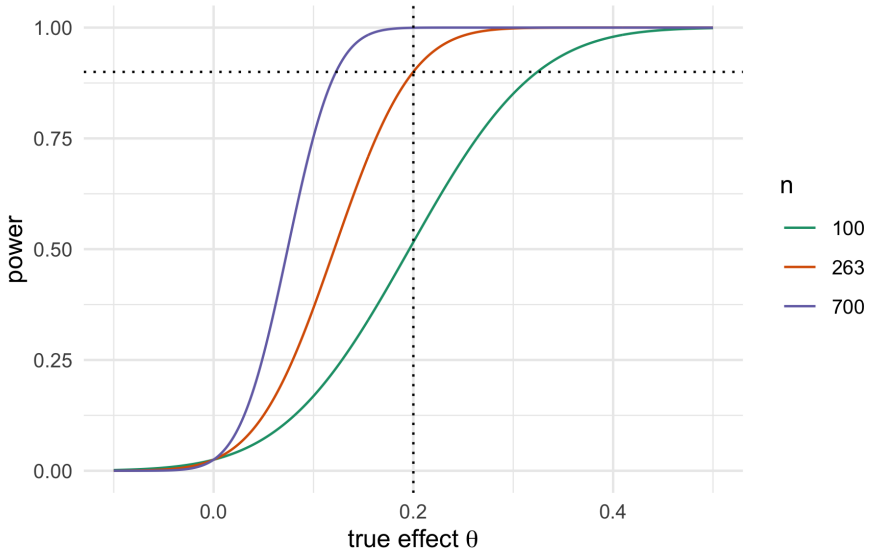


Figure 9.2.: Power curves for fixed sample sizes at one-sided 0.025, with information nI of 100, 263 and 700. The middle curve is the design for $\theta = 0.20$ at 90% power.

Two things the curve shows that a single number does not. The trial has a threshold effect, the value at which power is 50%, and effects below it are more likely to be missed than found. That value is $z_\alpha\sqrt{1/(nI)}$, which for a design with 90% power at θ_d is $z_\alpha/(z_\alpha + z_\beta) = 60.5\%$ of the design effect: a trial with 90% power at a hazard ratio of 0.80 is, for an effect smaller than about a 13% hazard reduction, more likely than not to fail. And the curve is flat near the design point. Moving from 90% to 95% power at the design effect needs 24% more information, so the last points of power are bought at a steep price.

One more observation concerns the left end of the curve. The probability that the trial reports a positive result for a drug that does nothing is α. It does not depend on n. The protocol writer who increases the sample size buys power and nothing else; the type I error stays at 0.025 however large the trial, even if a trial of 20,000 patients would, for the same observed data, support a much more skeptical conclusion than a trial of 200. The moderate deviation analysis below challenges exactly this.

9.1.5. What the regulator sees, and what the sponsor can do wrong

An E9 section 3.5 justification, as a regulator reads it, has five parts: the primary endpoint and estimand; the effect assumed and its source; the variability or event rate and its source; the error rates; and a sensitivity analysis. The common failures are well known to reviewers. The assumed effect is the observed effect from a small phase 2 trial, which is biased upward by selection (the drug was taken forward because it looked good). The variability is taken from the same small trial, with the uncertainty ignored. The assumed control event rate is taken from a historical cohort that no longer resembles the enrolled population. And the sponsor presents no sensitivity at all. The traditional calculation is correct as arithmetic. Its weakness is that it treats the inputs as known.

One legitimate source of a smaller required sample is a prespecified covariate-adjusted analysis: a baseline covariate that explains a fraction R^2 of the outcome variance lowers the required n by about the factor $1 - R^2$, as worked in Section 5.10, and the sample size paragraph should say whether the calculation assumes the adjusted or the unadjusted analysis.

> In the protocol
>
> A sample size paragraph that survives review names: the estimand and primary analysis; the design alternative and its provenance (minimal clinically important effect or anticipated effect, with the source study); the nuisance parameters (standard deviation, control rate, event rate) with their sources; α one-sided and $1 - \beta$; the resulting n or number of events and the accrual and follow-up assumptions that deliver it; and a table of power for effects at 70%, 80%, 100% and 120% of the design alternative, together with the rule for what the trial does if the nuisance parameters turn out wrong (blinded re-estimation, in an adaptive protocol under E20).

9.2. A conventional power calculation is assurance under a point-mass prior

9.2.1. Power is conditional

Power is a conditional probability: it is the probability of a significant result given that the true effect equals θ_d. The sponsor, the investors and the DMC are not conditioning on the truth. They want to know the probability that the trial succeeds, taking into account that they do not know the effect. A statement such as “90% power” is regularly heard in the boardroom as “90% chance of success”, which is a different claim. It would be the same claim only if the sponsor were certain that the effect equals the design value.

The traditional literature responds to this by sensitivity analysis, as above. The Bayesian response is to make the uncertainty explicit and average. Let $\pi(\theta)$ be a prior distribution over the effect, representing everything known before the trial, such as phase 2 results, mechanism and class experience. The probability that the trial will be successful is the average of the power curve under that prior,

$$\text{Assurance} = \int \text{Power}(\theta)\, \pi(\theta)\, d\theta. \tag{9.6}$$

O'Hagan, Stevens and Campbell (O'Hagan, Stevens, and Campbell 2005) named this quantity assurance, and earlier versions go back to Spiegelhalter and Freedman (Spiegelhalter and Freedman 1986) and Spiegelhalter, Freedman and Blackburn (Spiegelhalter, Freedman, and Blackburn 1986) under the name predictive power, and to Brown and colleagues (Brown et al. 1987) under the name expected power. Kunzmann and colleagues (Kunzmann et al. 2021) review the variants, including probability of success defined with a Bayesian rather than a frequentist final analysis. Chapter 15 uses the same logic at interim analyses, where the prior is updated by the data so far.

9.2.2. A closed form

For a normal prior $\theta \sim N(m, s^2)$ and the z test above, with standard error $\text{se} = 1/\sqrt{nI}$ of $\hat{\theta}$, the trial's estimate is marginally normal with mean m and variance $s^2 + \text{se}^2$. The test succeeds when $\hat{\theta} > z_\alpha \,\text{se}$, so

$$\text{Assurance} = \Phi\!\left(\frac{m - z_\alpha \,\text{se}}{\sqrt{s^2 + \text{se}^2}} \right). \tag{9.7}$$

Take a trial sized for $\theta_d = 0.20$ at 90% power ($n = 263$, $\text{se} = 0.0617$). Three sponsors with the same design have three beliefs. The first centers its prior at 0.20 with a standard deviation of 0.10, which is a sponsor that believes the design alternative but admits that it is uncertain by plus or minus half of itself. The second believes the drug is a bit worse, with prior mean 0.14 (70% of the design value) at the same standard deviation. The third is skeptical, with prior mean 0.10.

```
n <- n_fixed(0.20); se <- 1 / sqrt(n); sds <- c(0.02, 0.05, 0.10, 0.20)
tab <- sapply(sds, function(s) assurance(c(1, .7, .5) * 0.20, s, se))
knitr::kable(data.frame(c(1, .7, .5) * 0.20, tab), digits = 3, col.names =
 ↪  c("Prior mean", paste("SD", sds)), align = "r")
```

Table 9.3.: Assurance of a trial sized for theta = 0.20 at 90% power, for three priors and several prior standard deviations.

Prior mean	SD 0.02	SD 0.05	SD 0.1	SD 0.2
0.20	0.889	0.840	0.750	0.647
0.14	0.616	0.595	0.564	0.536
0.10	0.373	0.396	0.429	0.460

Several things stand out in this table. With a point-mass prior (standard deviation 0.02 is close to a point) at the design value the assurance is about 0.89, near the 90% power the trial was sized for. As the sponsor becomes uncertain, assurance falls: at standard deviation 0.10 it is 0.75, and at 0.20 it is 0.65. This is not a flaw but a property of averaging. The power curve flattens as it approaches one, so the effects below θ_d lose more power than the effects above gain. Uncertainty about the effect, even when it is symmetric around the design value, reduces the probability of success. And a skeptical sponsor, whose prior mean is 0.10 or 0.14, sees a probability of success of 43% to 56% for the very same design, although the protocol says 90%. This is not a thought experiment: when chemotherapy regimens were taken unchanged from phase II into phase III, only 28% of the phase III trials were positive (Zia et al. 2005) (see Section 9.6.3).

The "90% power" in the protocol is a statement conditional on one value of θ, and the assurance is the unconditional version. For a phase 3 trial with a program-level decision pending, assurance is the better planning quantity, and it can be computed in a spreadsheet. Spiegelhalter and Freedman's argument, which is recognizable from the earliest days of Bayesian trial design, is that the traditional power calculation is already Bayesian in its inputs: the sponsor's choice of θ_d summarizes belief, but it summarizes it as a point mass and discards the uncertainty.

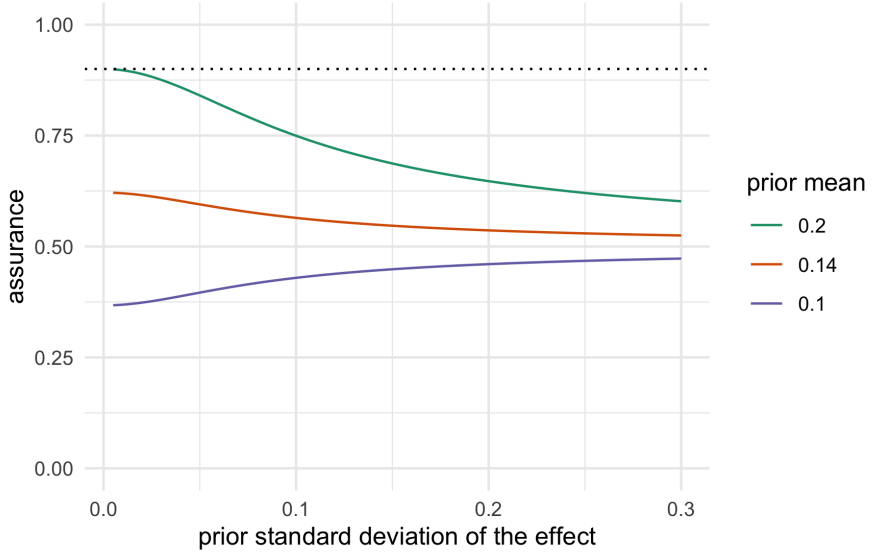


Figure 9.3.: Assurance of a trial sized for a design effect of 0.20 at 90% power as a function of the prior standard deviation, for three prior means. Assurance approaches the power at the design value only when the prior is concentrated on it.

9.2.3. What assurance is not

Assurance is not a measure of evidence. It is a forecast, computed before the data, of the frequency with which a trial of this size will cross its pre-specified threshold. The threshold is still the frequentist one in the classical version above: $z > z_\alpha$ with α fixed at 0.025. A fully Bayesian version computes the probability that the posterior probability of benefit exceeds the threshold c in the FDA draft's criterion $\Pr(\delta > a \mid \text{data}) > c$, with the expectation taken

over the design prior. This is sometimes called Bayesian assurance, and it separates the roles of two priors. The design prior (also called the sampling prior) describes what the sponsor believes about the effect and is used to forecast. The analysis prior is the one used in the final analysis, and it is the one that has to be pre-specified and defended to the regulator (Chapter 8). A sponsor can be optimistic in the design prior and skeptical in the analysis prior; there is no inconsistency, and indeed that combination is a sound one. What the regulator will not accept is calibration of the Type I error under the design prior. Type I error is computed under the null, whatever the sponsor believes.

The FDA draft (U.S. Food and Drug Administration 2026g) describes three routes (calibration, direct posterior, decision-theoretic), and this book's default is calibration. Under the calibration route the operating characteristics reported are the type I error under the null scenarios (calibrated, typically to 0.025 one-sided), and power or probability of success under a range of alternative scenarios, shown with Monte Carlo error. Assurance is the average of the latter over the sponsor's own prior, and it is a legitimate additional output, but it does not replace the scenario table. The other two routes justify the design differently (a direct posterior criterion, or a decision-theoretic one), but the draft still expects operating characteristics evaluated by simulation, and the draft recommends rather than requires.

9.2.4. A comparison

Table 9.4.: Power and assurance side by side.

	Traditional power	Assurance
Treats the effect as	a single design value θ_d	uncertain, with prior $\pi(\theta)$
Answers	If the effect is θ_d, how often does the test succeed?	How often will the trial succeed, given what we believe?
Inputs	α, β, θ_d, nuisance parameters	the same, plus a prior for θ and its source
Typical value	the target, 80% or 90%	lower, often 50% to 70% for a phase 3 sized on a phase 2 estimate
Regulatory status	required justification (E9 3.5)	optional; a planning aid. The scenario table of operating characteristics is what is reported

This is how assurance is used in sponsor practice, which is where a reader will meet it. Rufibach, Burger and Abt (Rufibach, Burger, and Abt 2016) study the choice of prior for Bayesian predictive power used as a probability of success in drug development and make recommendations for its use, and Crisp and colleagues (Crisp et al. 2018) report practical experience of adopting assurance as a framework for decisions in drug development. The usual pattern is a go or no-go decision on a phase 3 program: the prior is built from the phase 2 estimate and its uncertainty, often with a discount for the optimism of a selected result, elicited from the team where the data are thin, and the assurance is reported beside the design power. In the example above, a phase 2 estimate of 0.14 with standard deviation 0.10 (a 30% discount on the design effect of 0.20) turns a 90% power trial into one with

assurance 56%. A phase 3 program with that number may still be worth running, but the decision is now explicit. Two cautions. Assurance is a planning quantity: it is not an error rate and it does not replace the scenario table of operating characteristics. And it is only as good as the prior, so the elicitation record and a sensitivity analysis to the discount belong with it.

9.2.5. Bayesian sample size criteria

Assurance is one of several ways to turn a prior into a sample size. Inoue, Berry and Parmigiani (Inoue, Berry, and Parmigiani 2005) separate the Bayesian rules into three families and show how each relates to the power calculation. The first two are named by what they average. Most of these criteria involve two priors, the design prior that predicts the data and the analysis prior that turns the data into a posterior (F. Wang and Gelfand (2002) call them the sampling prior and the fitting prior), and keeping the two apart (the distinction of Chapter 8) is the point of the distinction; the exception is the known-variance length criterion below, which needs no design prior. The taxonomy and the two-prior framing used here are this book's presentation of that literature, not a development in Inoue and colleagues. Take a two-arm trial with outcome standard deviation $\sigma = 1$, an effect θ with design prior $N(0.3, 0.1^2)$ and a flat analysis prior.

The next three criteria were introduced for binomial proportions by Joseph, Wolfson, and Berger (1995) and set out for means and proportions, with mixed Bayesian and likelihood versions, by Joseph, Berger, and Bélisle (1997) and, for normal means and their differences, by Joseph and Bélisle (1997); the CRAN packages SampleSizeProportions and SampleSizeMeans implement them, and Joseph and Bélisle (2019) extend them to a community of priors held by different stakeholders, sizing the trial so that skeptics and enthusiasts alike end with a narrow enough interval.

Average length criterion (ALC). Choose n so that the posterior 95% interval has expected length at most ℓ. With a flat analysis prior this is the familiar $2 \times 1.96\sqrt{2\sigma^2/n} \le \ell$, so $n \ge 2(3.92\sigma/\ell)^2$, which is 342 per arm for $\ell = 0.3$. It needs no design prior at all when the variance is known, and it sizes a trial for estimation, not for a decision.

Average coverage criterion (ACC) and worst outcome criterion (WOC). Fix the length ℓ of the interval, centered at the analysis posterior mean, and choose n so that its coverage under the design prior (the coverage actually delivered if the design prior were true) is at least a target, either on average over the data (ACC) or in the worst case over the central region of the data that contains 90% of the prior predictive probability (a modified WOC). This is the design-prior version of the criteria; in the formulation of Joseph and colleagues the coverage is the analysis-posterior mass, which with a flat analysis prior and known variance does not depend on the data, so that ACC and WOC coincide there. At $\ell = 0.3$ the ACC is 0.87, 0.93 and 0.97 at 200, 300 and 400 per arm, and the WOC is lower, 0.68, 0.85 and 0.92, because it protects against the unlucky data and not the typical ones. When design and analysis priors coincide and the variance is known the two criteria agree.

Posterior probability criterion. Choose n so that the preposterior probability of a convincing posterior, $\Pr\{\Pr(\theta > 0 \mid X) > 0.975\}$, reaches a target such as 0.8. With the design prior above and a flat analysis prior this is the assurance of the previous section, and it is 0.77, 0.86 and 0.91 at 200, 300 and 400 per arm. It first reaches 0.8 at 226 per arm. If the same informative prior is also the analysis prior, the probability is above 0.99 already at 50 per arm, because the prior alone gives $\Pr(\theta > 0) = \Phi(3) = 0.9987$. The trial is then unnecessary

for an analyst who holds that prior, which is the right conclusion for her and a warning against letting a sponsor's design prior into a regulator's analysis.

The result that ties this to the first half of the chapter is that the power calculation is the posterior probability criterion with a point-mass design prior at the alternative and a flat analysis prior (Inoue, Berry, and Parmigiani 2005). A power calculation is a Bayesian sample size calculation with an unusually confident design prior. At $\theta = 0.3$ a two-arm trial needs 234 per arm for 90% power (the criterion evaluates to 0.90 at that size), while the assurance at 234 per arm, which acknowledges the uncertainty about θ, is 0.81. A decision-theoretic criterion, which chooses n to maximize expected utility net of the cost of sampling, is the third family; it is the one that asks how much information is worth buying, and it is the subject of Section 9.5.

The Bayesian sample size literature is wider than these families. Adcock (Adcock 1988, 1997) gave early average-criterion treatments and a review of the field, Lindley (D. V. Lindley 1997) set out the decision-theoretic choice of sample size, and Chaloner and Verdinelli (Chaloner and Verdinelli 1995) review Bayesian experimental design more generally. Wang and Gelfand (F. Wang and Gelfand 2002) made the simulation-based version practical. De Santis (De Santis 2006) asks for sample sizes that remain adequate across a class of priors, and Brutti and De Santis (Brutti and De Santis 2008) extend this to avoiding a range of equivalence. Gittins and Pezeshk (J. Gittins and Pezeshk 2000) and Sahu and Smith (Sahu and Smith 2006) mix Bayesian planning with a frequentist analysis, and Pezeshk (Pezeshk 2003) reviews the field. Read by the question each answers, the coverage and length criteria size an estimate, the posterior probability criterion and assurance size a decision, the expected-utility criteria size a decision together with its cost, and the robust criteria protect against the choice of prior. For a confirmatory trial the regulator still asks for the operating characteristics of the final analysis, so these criteria serve planning and the calibrated error rates of Chapter 14 serve the claim.

A rare-disease example: lupus

Moore and Joseph (A. D. Moore and Joseph 1999) make the case for estimation-based sample sizes in systemic lupus erythematosus, where trials are small because patients are few and follow-up is long. Their dichotomous example is progression to dialysis over five years in lupus nephritis, 50% on standard therapy and a hoped-for 25% on a new regimen. The textbook calculation with a two-sided 5% test and 90% power needs 77 patients per arm (the paper reports 77, from the normal approximation with rounded quantiles; `power.prop.test` gives the same after rounding up). That is a fair answer to the question it asks. The difficulty is that a 25-point reduction is a large effect, and a drug that delivers a more modest one, say 35% against 50%, needs 227 per arm, 454 in all, which may exceed the number of eligible patients in a registry of a rare disease.

The alternative the paper recommends is to size the trial to estimate the difference to a stated accuracy. A total interval width of 0.3 (plus or minus 0.15) for a 95% interval gives 75 per arm by the frequentist formula, which is the paper's 75. Its Bayesian counterparts take a Beta prior on each proportion, here worth four prior patients per arm and centered on the design values, and ask for the same width on average (ALC), with average coverage of 95% (ACC), or with coverage of at least 95% for every possible dataset (WOC; restricting the guarantee to the central 90% of the prior predictive distribution, as in the previous section, gives the modified WOC). They were computed with Joseph and Bélisle's package SampleSizeProportions (Joseph, Bélisle, and du Berger 2023), and the ACC is a Monte

Carlo estimate with a fixed seed. The paper only mentions the Bayesian route and gives no Bayesian sample sizes, so these numbers have no published counterpart. They are smaller than the frequentist width calculation for the ALC and ACC, 55 and 57 per arm, for two reasons: the prior contributes a few patients' worth of information, and the design prior averages $p(1-p)$ over its support, which is below the value 0.25 that the plug-in formula uses at $p = 0.5$. The WOC, 81, is larger than the plug-in answer because it protects against unlucky data. The width is also the lever that matters: widening it to 0.4 (plus or minus 0.2) cuts the Bayesian sizes to 33, 33 and 43 per arm for the modest scenario (Table 9.5).

Table 9.5.: Patients per arm for a two-arm trial with a binary outcome (50% on control). The large-effect column is the lupus nephritis example of Moore and Joseph (25% on treatment, interval width 0.3); the modest-effect column is 35% on treatment with interval width 0.4. Bayesian criteria use Beta priors worth four patients per arm centered on the design proportions, and a 95% interval.

Method	Large effect	Modest effect
Power, 90% at 5% two-sided	77	227
Interval width, frequentist	75	46
Bayesian ALC	55	33
Bayesian ACC	57	33
Bayesian WOC	81	43

What the smaller trial buys and costs is the point. It buys feasibility: a trial of 66 patients in all can be run where one of 454 cannot. It costs the guarantee of a decision. At that size the trial estimates the difference to within plus or minus 0.2, so if the true effect is 15 points the interval will often include zero, and the trial is underpowered for a test of no difference. It is not underpowered for the question it was sized to answer, how large the benefit is, and an interval of that width, read with the prior in hand, is the evidence a small trial can honestly offer. The criteria also take their inputs from the design prior, so the sensitivity analysis recommended for the power calculation applies to them in the same way.

9.3. The moderate deviation sample size

9.3.1. The fixed-n test has no memory of n

In the textbook calculation the critical value z_α is fixed. A trial of 200 patients and a trial of 20,000 patients both declare success at $Z > 1.96$. The consequences are known to every reader who has met a very large trial that reports a result with $p = 0.04$ and a clinically trivial effect. A statistically significant result in a huge trial is weak evidence against the null when measured the Bayesian way: the Bayes factor in favor of the alternative at $Z = 1.96$ shrinks as the sample size grows, which is Lindley's paradox (Chapter 7). A Bayesian decision maker who places a prior mass on the null and balances the costs of the two errors will use a critical value that increases with n.

Polson and Sokolov (Polson and Sokolov 2026b), building on a result for the Bayes risk of a regular testing problem (Polson and Sokolov 2026e), work out how fast. The Bayes-optimal boundary on the standardized scale is

$$t_{\text{crit}} = \sqrt{\log n + C}\,(1 + o(1)), \qquad C = 2\log \frac{\pi_0/\pi_a}{\pi(\theta_0)\sqrt{2\pi/I(\theta_0)}}, \tag{9.8}$$

where π_0 and π_a are the prior probabilities of the null and the alternative and $\pi(\theta_0)$ is the prior density of the effect at the null. For a unit-information prior the denominator is one and $C = 2\log(\pi_0/\pi_a)$; with equal prior odds $C = 0$ and the boundary is $\sqrt{\log n}$. This is called the *moderate deviation* scale because the boundary grows with n: it lies above the fixed value z_α of the central limit regime and below the large-information approximation to the anytime-valid (large deviation) bar of the normal-mixture construction, $\sqrt{\log n + C + 2\log(1/\alpha)}$ (an approximation for that construction, not a universal cost of anytime validity), and Chapter 11 sets the three scales side by side.

i Technical note: where the boundary comes from

The Bayes rule rejects the null when the posterior probability of the alternative exceeds one half, after weighting by the prior probabilities. Compare the two marginal likelihoods. Under the null, the standardized statistic has the standard normal density $\phi(t)$. Under the alternative, with a prior π smooth at θ_0, the marginal density of the data is, by Laplace's method, proportional to $\pi(\theta_0)/\sqrt{nI(\theta_0)}$, because the prior mass that is compatible with the data has width of order $1/\sqrt{nI}$. The balance condition is

$$\pi_0\,\phi(t_n) = \pi_a\,\frac{\pi(\theta_0)}{\sqrt{n\,I(\theta_0)}},$$

which equates a Gaussian density to an $n^{-1/2}$ term. Taking logarithms, $t_n^2/2 = \frac{1}{2}\log n + \text{constant}$, hence $t_n^2 = \log n + C$. The constant collects the prior odds and the prior density at the null. For a standard Cauchy prior on a unit-variance Gaussian mean the asymptotic approximation is $t_{\text{crit}} = \sqrt{\log(\pi n/2)}$, which is $C = \log(\pi/2) = 0.452$. The expression holds asymptotically; it is an approximation in n and the $o(1)$ matters at small sample sizes.

Two quick numbers fix ideas. The boundary with $C = 0$ exceeds the textbook 1.96 for every n above 46 (since $\sqrt{\log 46} = 1.96$). At $n = 1000$ it is 2.63 and at $n = 10{,}000$ it is 3.03. The induced type I error, which is no longer fixed, is exactly $1 - \Phi(t)$ one-sided and twice that two-sided; the Mills-ratio approximation $2\phi(t)/t$ for the two-sided value is accurate only for large t, and the error falls like $(n \log n)^{-1/2}$. This chapter reports the exact one-sided value, to match the one-sided 0.025 of the textbook design.

9.3.2. The power function and the sample size equation

The power of the moderate deviation rule against an effect θ follows by the same normal approximation as before, with the fixed z_α replaced by the boundary:

$$\text{Power}_n(\theta) = \Phi\Big(\sqrt{nI}\,\theta - \sqrt{\log n + C}\Big) + o(1), \qquad \theta > 0 \text{ for the one-sided benefit rule.} \tag{9.9}$$

Setting this equal to $1 - \beta$ gives the sample size equation

$$n\,I\,\theta^2 = \left(\sqrt{\log n + C} + z_\beta\right)^2 . \tag{9.10}$$

Set it beside Equation 9.2. The structure is identical, but z_α has been replaced by $\sqrt{\log n + C}$, a quantity that depends on n itself. The equation is implicit, but only through a logarithm, and a fixed-point iteration converges in a handful of steps: compute the textbook n, put it into $\sqrt{\log n + C}$, recompute, and repeat. In the code the equation is solved with a root finder.

```
zb <- qnorm(0.90); n <- (qnorm(0.975) + zb)^2 / 0.2^2; it <- n
for (k in 1:4) { n <- (sqrt(log(n)) + zb)^2 / 0.2^2; it <- c(it, n) }
knitr::kable(data.frame(0:4, it), digits = 1, col.names = c("Iteration",
 ↪  "Information"), align = "rr")
```

Table 9.6.: Fixed-point iteration for the moderate deviation sample size at theta = 0.20, C = 0, 90% power, starting from the textbook answer with z = 1.96 (two-sided 5%, the convention of POL's tables).

Iteration	Information
0	262.7
1	331.6
2	340.5
3	341.5
4	341.7

The iterates are 262.7, 331.6, 340.5, 341.5 and 341.7, and the exact root is 341.7, which agrees with the values reported by Polson and Sokolov.

A note on units, since the tables can otherwise confuse. In Equation 9.10 the quantity n is the sample size on the scale where one unit carries information I, so that nI is the total information. For a one-sample or paired standardized mean difference with $I = 1$, n is the number of patients. For a two-arm trial with 1:1 allocation and a standardized effect, the information is $N/4$, so the number of patients is $N = 4n$. For a logrank test there are two equivalent conventions: count in events with $n = d$ and $I = 1/4$, or count in information units with $n = d/4$ and $I = 1$. The boundary $\log n + C$ must describe the same rule in both, so the constant moves with the unit: $\log(d/4) + C_{\text{info}} = \log d + C_{\text{events}}$, that is $C_{\text{info}} = C_{\text{events}} + \log 4$. Keeping $C = 0$ while switching units changes the implied prior. The package function `events_md()` uses the information convention ($n = d/4$, $I = 1$), so its $C = 0$ is the unit-information prior at equal prior odds on that scale, and $C = 2\log(\pi_0/\pi_a)$ holds in that convention. The same numerical $C = 0$ read in the events convention means $C_{\text{info}} = \log 4 = 1.39$, the constant of two-to-one odds in the information convention, and at a hazard ratio of 0.80 it calls for 1255 events rather than 1067. Any stated value of C must therefore name its unit. All tables below use the information convention, and the comparison with the textbook formula is made in the same units.

9.3.3. The surprising result

Table 9.7 compares the textbook information requirement with the moderate deviation requirement. The textbook column uses the two-sided 5% convention, $z = 1.96$, because that is the convention of the comparison in POL; at one-sided 0.025 the critical value is the same.

```
th <- c(0.05, 0.10, 0.15, 0.20, 0.30, 0.50); nt <- n_fixed(th); nm <-
 ↪  n_md(th); tb <- sqrt(log(nm))
knitr::kable(data.frame(th, round(nt), round(nm), round(nm / nt, 2), tb,
 ↪  pnorm(-tb)), digits = c(2, 0, 0, 2, 3, 5), col.names = c("Design
 ↪  effect", "Textbook", "Moderate deviation", "Ratio", "Boundary", "Type I
 ↪  error (one-sided)"), align = "r")
```

Table 9.7.: Information required for 90% power, textbook versus moderate deviation (C = 0, information convention). The last two columns are the boundary at the moderate deviation sample size and the exact one-sided type I error it induces, 1 - Phi(boundary).

Design effect	Text-book	Moderate deviation	Ratio	Bound-ary	Type I error (one-sided)
0.05	4203	7271	1.73	2.982	0.00143
0.10	1051	1598	1.52	2.716	0.00330
0.15	467	651	1.39	2.545	0.00546
0.20	263	342	1.30	2.415	0.00786
0.30	117	136	1.16	2.216	0.01333
0.50	42	41	0.98	1.928	0.02690

The moderate deviation design needs more, not fewer, patients than the nominal textbook calculation for every effect smaller than about 0.5. At $\theta = 0.20$ the ratio is 1.30, so the moderate deviation fixed design is about 30% larger; at $\theta = 0.10$ it is 1.52 and at $\theta = 0.05$ it is 1.73. Only for large effects, at which the requirement is below 46 units and the boundary is below 1.96, does the ordering reverse.

The reason is that the Bayes boundary is more demanding than 1.96 for any trial of moderate or large size: a test that demands $Z > 2.42$ rather than $Z > 1.96$ needs more information to reach 90% power.

What the extra patients buy is in the last column. The one-sided type I error is no longer held at 2.5% at all sample sizes but falls as n grows: 0.79% at $\theta = 0.20$ (the two-sided equivalent is twice that), 0.33% at $\theta = 0.10$ and 0.14% at $\theta = 0.05$. At a thousand units of information the induced one-sided error is 0.43% and at ten thousand 0.12%. POL's reading is that a fixed error is the right error for no sample size in particular. Whether one accepts that reading or not, the arithmetic is clear: a design that maintains a lower error as information accumulates cannot be cheaper than one that does not.

9.3.4. The prior odds in the constant

The constant C makes the dependence on prior belief explicit. With the unit-information prior, $C = 2\log(\pi_0/\pi_a)$, and moving from even odds to two-to-one and four-to-one against the alternative raises C from 0 to 1.39 and 2.77.

```
o <- c(1, 2, 4); C <- 2 * log(o); nm <- n_md(0.20, C); tb <- sqrt(log(nm)
 ↪  + C)
knitr::kable(data.frame(o, C, round(nm), tb, pnorm(-tb)), digits = c(0, 2,
 ↪  0, 3, 5), col.names = c("Odds against", "C", "Information", "Boundary",
 ↪  "Type I error (one-sided)"), align = "r")
```

Table 9.8.: Effect of prior odds (null to alternative) on the moderate deviation requirement at theta = 0.20, 90% power, information convention. Type I error is the exact one-sided value.

Odds against	C	Information	Boundary	Type I error (one-sided)
1	0.00	342	2.415	0.00786
2	1.39	400	2.716	0.00330
4	2.77	454	2.982	0.00143

The requirement rises from 342 to 400 and 454 information units, and the induced one-sided error falls from 0.79% to 0.33% and 0.14%. Skepticism is paid for in patients, at a rate the formula makes explicit. This is the property a protocol reviewer wants: the sponsor who claims the drug is a long shot is told how many more patients it will take, and the sponsor who claims equal prior odds is told what that claim is worth.

It also shows what this approach does not do. It is not a method of getting out of a regulatory error-rate expectation. Where the sponsor takes the FDA draft's calibration route, with a one-sided type I error of 0.025 verified by simulation, that target stays; the moderate deviation analysis is a design-stage theory of why a trial of a given size should be held to a stricter standard than the nominal one, and of the cost of holding it there.

9.3.5. Many parameters and many endpoints

Two extensions carry over into design (Polson and Sokolov 2026e). For a d-dimensional parameter, for instance a trial with $d+1$ arms, the boundary is $\sqrt{d\log n}$, so at $n = 1000$ the radius is 2.63 for one parameter, 3.72 for two and 4.55 for three. This is the moderate deviation form of the Schwarz penalty $d\log n$, a Bayesian counterpart of a multiplicity correction arrived at without reference to a family of tests (Chapter 10). Both extensions are model-based approximations (a regular parametric model with d free parameters for the first, sparsity of the true signals for the second), and neither displayed expression by itself controls the familywise error rate across clinical claims. For p simultaneous endpoints or subgroups under sparsity the boundary is the universal threshold $\sqrt{2\log p}$, which is 3.03 at a hundred endpoints and 4.45 at twenty thousand, the latter being the scale of a genome-wide scan.

Table 9.9.: Boundaries for d parameters at n = 1000, and for p endpoints under sparsity.

Case	Boundary
d = 1 , n = 1000	2.63
d = 2 , n = 1000	3.72
d = 3 , n = 1000	4.55
p = 100	3.03
p = 2000	3.90
p = 20000	4.45

9.3.6. Reading SELECT against the formulas

SELECT is the natural trial against which to read these formulas, with a caution stated first: it does not test the moderate deviation theory. SELECT was designed for a hazard ratio of 0.83 (a 17% reduction), with 90% power, one-sided 0.025, 1,225 planned events and one superiority interim analysis at two thirds of the planned events, followed by the final analysis (D. H. Ryan et al. 2020). The observed hazard ratio was 0.80 on 1,270 events (Lincoff et al. 2023) (Chapter 21 gives the trial in detail). The design hazard ratio, not the observed one, is the design alternative, and the table sets both beside each calculation.

```
options(knitr.kable.NA = "")
knitr::kable(data.frame(calculation = c("textbook, HR 0.83 (design)",
 ↪  paste0("moderate deviation, HR 0.83, ", c(1, 2, 4), ":1 odds"),
                                         "textbook, HR 0.80 (observed)",
                                          ↪  paste0("moderate deviation, HR
                                          ↪  0.80, ", c(1, 2, 4), ":1
                                          ↪  odds"),
                                         "SELECT as planned", "SELECT as
                                          ↪  observed"),
           events = round(c(ev83, md83["events", ], ev80, md80["events", ],
            ↪  1225, 1270)),
           t_crit = round(c(qnorm(0.975), md83["t", ], qnorm(0.975),
            ↪  md80["t", ], NA, NA), 2)),
  col.names = c("Calculation", "Events", "Critical value"), align = "lrr")
```

Table 9.10.: Events for 90% power: textbook against moderate deviation (information convention), at the design hazard ratio 0.83 and at the retrospective hazard ratio 0.80, with the events planned and observed in SELECT.

Calculation	Events	Critical value
textbook, HR 0.83 (design)	1211	1.96
moderate deviation, HR 0.83, 1:1 odds	1603	2.45
moderate deviation, HR 0.83, 2:1 odds	1867	2.74
moderate deviation, HR 0.83, 4:1 odds	2119	3.01
textbook, HR 0.80 (observed)	844	1.96
moderate deviation, HR 0.80, 1:1 odds	1067	2.36
moderate deviation, HR 0.80, 2:1 odds	1255	2.67

Table 9.10.: Events for 90% power: textbook against moderate deviation (information convention), at the design hazard ratio 0.83 and at the retrospective hazard ratio 0.80, with the events planned and observed in SELECT.

Calculation	Events	Critical value
moderate deviation, HR 0.80, 4:1 odds	1433	2.94
SELECT as planned	1225	
SELECT as observed	1270	

At the design hazard ratio of 0.83 the textbook formula gives 1211 events against the 1,225 planned, a difference of 14 events, about 1% of the plan. There is nothing substantial left over for a moderate deviation standard to explain. The moderate deviation formula at the same hazard ratio and equal prior odds gives 1603 events, far above the plan, so it does not predict SELECT's size. The textbook 844 events at the observed 0.80 is a retrospective number and was never SELECT's design target.

The moderate deviation sample-size theory stands on its own terms and does not need SELECT. It says what a fixed design would cost if the type I error were allowed to fall with information, and what the prior odds in C are worth in patients. Whether sponsors and regulators in practice demand that kind of stricter standard from very large trials is an empirical question for many trials, and one trial whose size is explained by its own stated design assumptions does not answer it. A protocol's event target also reflects considerations that no formula captures: safety database size, subgroup precision, competing events and secondary endpoints.

9.3.7. Where the saving is, and where it is not

A fixed-sample Bayesian design using the uncalibrated moderate deviation boundary is larger than the nominal textbook design, by about 30% at $\theta = 0.20$, and it buys a type I error that falls with n. The Bayesian saving in patients and time does not come from the fixed design at all. Against a single unadjusted analysis at a fixed sample size, the calibrated Bayesian bar is never lower without an informative prior, and the uncalibrated Bayes boundary is higher once the information $q = n\tau^2$ exceeds 41.6 (that is, $n > 41.6$ in the unit-information case); that is the thesis of this book and this chapter is its arithmetical base. The saving comes from monitoring. Once the trial is allowed to stop when the posterior is decisive, the same boundary that was expensive to reach at the design stage is reached early along most trajectories. In the SELECT reconstruction of Chapter 12 (a counterfactual with true hazard ratio 0.80 and at most 1,270 events), among simulated trials that declare efficacy the median stopping point is 464 events for a calibrated Bayesian monitoring rule (power 0.95), 612 for an e-value rule (power 0.83) and 952 for a four-look O'Brien-Fleming design (power 0.98), against 817 for SELECT's planned two-look design (power 0.97); Table 12.2 gives the medians over all trials, the realized type I errors and the conditioning. The same arithmetic, applied to whole programs rather than single trials, is the platform-trial claim of Saville and Berry (Benjamin R. Saville and Berry 2016) (see Section 9.6.4 and Chapter 20). A sponsor who goes to the regulator with a Bayesian fixed-sample design and a claim that it is cheaper than the traditional one has made a claim that this chapter's arithmetic contradicts.

Equally, the traditional design deserves its fair credit. If the trial will be run once, analyzed once, at a fixed time, with no interim looks, the textbook design is simpler, its operating

characteristics are known in closed form, and its regulatory path is established. POL itself concedes the case of REWIND, in which a scheduled design does better (Chapter 21). A sponsor that cannot commit to a monitoring plan has no reason to move.

9.4. Side by side

Table 9.11.: What changes in the protocol, and what stays. Where the two agree numerically: with a normal likelihood and a flat prior and the same one-sided threshold, $\Pr(\theta > 0 \mid y) = 1 - p_{\text{one-sided}}$, so the corresponding decision thresholds agree (Chapter 7).

Aspect	Traditional fixed design	Bayesian, same fixed sample	Bayesian, monitored
Sample size rule	$n = (z_\alpha + z_\beta)^2/(I\theta^2)$	$nI\theta^2 = (\sqrt{\log n + C} + z_\beta)^2$, for the pure Bayes rule; the textbook n if calibrated to 0.025	maximum sample size and boundaries chosen jointly to achieve target power
Type I error	0.025 at every n	calibrated to 0.025 by choice of c, or falling with n under the pure Bayes rule	calibrated to 0.025 by simulation (Chapter 14)
Required size, same power	baseline	larger (about 30% at $\theta = 0.20$ for the pure Bayes rule)	smaller in expectation than a fixed design after matching power and error targets, because of early stopping
Design alternative	point value θ_d, implicit prior	explicit design prior, assurance	explicit design prior, predictive probability at interims
What the regulator asks	E9 3.5 justification and sensitivity	calibration and operating characteristics report (FDA draft IV.A)	the same, plus the monitoring plan and simulation report (E20, FDA adaptive 2019)

When the Bayesian criterion is calibrated so that the familywise type I error is 0.025, which is this book's default for confirmatory designs without borrowing and one of the routes the FDA draft describes (U.S. Food and Drug Administration 2026g), the fixed-sample Bayesian and traditional designs reach the same threshold and need the same sample size for the same power, because calibration reproduces the traditional critical value in the one-look case. The difference in sample size between the "pure Bayes" boundary and the textbook arises when the error rate is allowed to fall with n. That distinction is worth keeping in mind when reading claims about Bayesian sample sizes: the claim must say which of the two it refers to.

9.5. Sizing for the population

Every calculation so far takes the two error rates as given. A decision-theoretic planner has no error rates to take. Consider the problem of Anscombe (Anscombe 1963a) and Colton (Colton 1963a). A population of N patients will be treated over the relevant horizon. Before treating them the planner randomizes $2n$, n to each of two treatments, with a normal outcome of known standard deviation σ, so the difference in means is $N(\theta, 2\sigma^2/n)$. The effect has prior $\theta \sim N(0, \tau^2)$ (a simplifying symmetry assumption that treats the two arms as exchangeable before the trial; a planner who currently prefers one arm can still value the trial, and a nonzero prior mean would shift the optimum), and after the trial the remaining $N - 2n$ patients receive the arm with the higher posterior mean. Welfare is linear in θ, and the cost of the trial is the welfare of its participants, half of whom receive the inferior arm. The expected loss against full information is

$$L(n) = n\,\tau\sqrt{2/\pi} + (N - 2n)\,\frac{\tau(1-\rho)}{\sqrt{2\pi}}, \qquad \rho = \frac{\tau}{\sqrt{\tau^2 + 2\sigma^2/n}}, \tag{9.11}$$

where the first term is the price of experimentation and the second is the price of residual ignorance. For large N the minimizer has a closed form,

$$n^\star = \frac{\sigma}{\tau}\sqrt{\frac{N}{2}}, \qquad L(n^\star) \approx 2\sigma\sqrt{\frac{N}{\pi}}, \tag{9.12}$$

the square root rule: the optimal trial enrolls $\sqrt{N}$-order patients, not a multiple of N. The leading-order approximation balances the experimentation and residual-ignorance terms, the usual signature of a loss of the form $an + b/n$. The same problem was posed for clinical trials by Cornfield, Halperin and Greenhouse (Cornfield, Halperin, and Greenhouse 1969), who proposed an adaptive procedure for sequential clinical trials. It is the clinical-trial counterpart of the Anscombe and Colton problem, in which the horizon is the population that will be treated, and Chapter 16 returns to that view as the bandit problem.

Table 9.12.: The square root rule against direct minimization of the exact loss, tau = 1.3, sigma = 2 (outcome units). Rel. error is the excess of the approximate minimum loss over the exact one.

N	n* (rule)	n (exact)	Loss (exact)	Loss (rule)	Rel. error
100,000	344.0	340	707.5	713.6	0.86%
1,000,000	1087.9	1084	2250.6	2256.8	0.27%

The rule is accurate where it is used: at $N = 10^6$ the closed form is within 0.27% of the exact loss and at $N = 10^5$ within 0.86%, the larger error being the $O(1)$ correction to an $O(\sqrt{N})$ leading term. Three features matter for the contrast with practice. First, the optimal trial is small relative to the population it serves: at $N = 10^6$ and $\sigma/\tau = 2$ it is 1414 per arm. Second, it contains no significance level, no power and no design alternative, so one-sided 0.025 and 80% or 90% power appear as conventions and not as decision-theoretic primitives. Third, prior dispersion shrinks the optimal trial: a larger τ makes the sign of the effect easier to learn, and a small τ (effects known in advance to be small) makes $n^\star$ large. Trials are

large because effects are small, which is the same message as the square law of Equation 9.2 arrived at from the other end.

The planner's rule and the regulator's rule answer different questions. The planner sizes the trial to the population that will use the result and trades the participants' welfare against the value of the information; the regulator asks whether this trial, by its own data, controls the probability of a false claim. They are not reconciled by a sampling cost. Adding a monetary cost per patient to Equation 9.11 changes the optimal sample size (the first-order condition becomes $n^\star = \sigma\sqrt{N/(2\tau^2 + 2\sqrt{2\pi}\,c\,\tau)}$ for a cost c per patient in the units of θ, as `size_sqrt_rule()` implements), but not the terminal decision: choosing the arm with the higher posterior mean selects the experimental arm with probability one half under the null whatever n is, so the rule controls no error rate. Population welfare and regulatory success are separate criteria. A planner who also wants error control would impose a calibrated success rule as a constraint and re-optimize n under it, and the sample sizes the textbook formula produces may sit in the same flat region of the loss when the design alternative is sensible. Neither the cost nor the loss is a regulatory requirement; the first-order condition is the model's, and Chapter 15 and Chapter 12 take the same decision-theoretic logic into monitoring.

9.6. In practice

9.6.1. What a sample size costs

The square law has a price list. Moore and colleagues estimated the cost of the 138 pivotal trials behind the 59 novel agents the FDA approved in 2015 and 2016 (T. J. Moore et al. 2018): a median of US$19.0 million per trial (interquartile range 12.2 to 33.1 million), a mean of US$157.2 million for the five cardiovascular trials (the costliest single trial, a noninferiority trial with clinical-benefit endpoints, was US$346.8 million), and by design a mean of US$13.5 million for trials without a control group, 28.8 million with a placebo and 48.9 million with an active comparator. A second study of 225 pivotal trials for 101 drugs approved in 2015 to 2017 (T. J. Moore et al. 2020) put the median at US$41,413 per patient (interquartile range 29,894 to 75,047), with patients per trial ranging from 4 to 8,442, and found that cost rose exponentially with patients and clinic visits ($R^2 = 0.696$). These are estimates assembled from public sources, not sponsor accounts, and they are medians over very different trials. The lesson for this chapter is the multiplier. A design alternative chosen 30% too optimistic does not cost 30% more when the shortfall is discovered: by the square law of Equation 9.2, a true effect of 70% of the design value needs about twice the events (1/0.49), and at the per-patient prices above a trial that must double is a nine-figure decision. The sample size sentence in the protocol is a budget line.

9.6.2. SELECT: sizing a cardiovascular outcomes trial

SELECT (Lincoff et al. 2023; D. H. Ryan et al. 2020) is worked through above, where the design hazard ratio, the observed one and each formula are set side by side (Table 9.10), and the trial is described in full in Chapter 21. In the language of this chapter: the sponsor designed for a more modest effect than the one it observed, and the planned size is what the textbook formula gives for that design alternative, 1211 events. The regulator did not ask

for a Bayesian design, and the moderate deviation sizes are reported for comparison and do not explain the plan.

9.6.3. Phase II to phase III: why the design alternative is too large

The usual source of the design alternative is a phase II trial, and Zia and colleagues examined how that fares (Zia et al. 2005). Of 181 phase III studies in advanced solid tumors published from July 1998 to June 2003, 43 used regimens identical to those in 49 preceding phase II studies. Twelve of the 43 phase III studies (28%) were positive, 81% had lower response rates than the phase II study that preceded them, and the mean difference was 12.9 percentage points. The mechanism is the selection effect of the "What the regulator sees" subsection above: a drug is taken into phase III because phase II looked good. A design alternative equal to the phase II estimate is therefore biased upward, a trial sized on it is underpowered at the true effect (Figure 9.1), and the probability that it succeeds is the assurance of Table 9.3, not the 90% in the protocol. This is the case for assurance. A prior that shrinks the phase II estimate, as in the sponsors with prior mean 0.14 or 0.10 above, reports the probability of success the program will actually face, and if the answer is uncomfortable the time to learn it is before the trial is funded. The caveat is scope: the study concerns chemotherapy in solid tumors, and other areas differ.

The sponsor's question is then not only how likely the trial is to succeed but what size maximizes the expected value of the program. Suppose, as a synthetic example, that the phase II posterior is flat, $\theta \sim N(0.30, 0.141^2)$ on a standardized scale, and the pivotal trial randomizes n per arm and succeeds if the one-sided z statistic exceeds 1.96. Power at θ is $\Phi(\theta\sqrt{n/2} - 1.96)$, and the assurance averaged over a posterior $N(m, v)$ is Equation 9.7 with $\mathrm{se}^2 = 2/n$. Take a fixed cost of US\$20 million plus US\$0.1 million per patient (so $0.2n$ for $2n$ patients), approval after success with probability 0.9, and a value of US\$1 billion for an approved drug. The expected net present value, in millions, is $\mathrm{ENPV}(n) = 0.9 \times 1000 \times A(n) - (20 + 0.2n)$. A skeptical sponsor shrinks the same estimate to $N(0.20, 0.115^2)$.

Table 9.13.: Sizing a pivotal trial (synthetic example). Power is evaluated at the posterior mean; money in millions of US dollars. Flat posterior N(0.30, 0.141^2), skeptical posterior N(0.20, 0.115^2).

n per arm	Cost	Power (flat)	Assurance (flat)	ENPV (flat)	Power (skeptical)	Assurance (skeptical)	ENPV (skeptical)
200	60	0.85	0.73	594	0.52	0.51	399
300	80	0.96	0.80	644	0.69	0.61	470
400	100	0.99	0.85	662	0.81	0.68	508
500	120	>0.99	0.87	665	0.89	0.72	527
600	140	>0.99	0.89	661	0.93	0.75	535
800	180	>0.99	0.91	640	0.98	0.79	533
1000	220	>0.99	0.92	612	0.99	0.82	517

Two readings. First, power at the estimate overstates the chance of success: at 300 per arm the flat posterior gives power 0.96 but assurance 0.80, because the posterior puts weight on smaller effects, and under the skeptical posterior the same trial has assurance 0.61. Second, the optimum is flat and the prior moves it: the ENPV peaks near 482 per arm for the flat posterior and near 671 for the skeptical one, and both curves are within a few percent of

their maximum from 400 to 800 per arm. The skeptical sponsor runs the larger trial because it needs more information to detect a smaller effect. The figures are an illustration of the arithmetic with invented costs and value, not a forecast, and the go decision is easy here because ENPV is several hundred million dollars in both columns; with a value of US$300 million the skeptical ENPV at 600 per arm falls to 63 million, still positive but thin, and the decision then depends on alternative uses of the budget.

9.6.4. Platform arithmetic

Saville and Berry simulated 5,000 drug-development programs under stated assumptions (200 patients enrolled per year, three months between trials) (Benjamin R. Saville and Berry 2016). A traditional sequence of two-arm trials used 1,966 patients, had 1,357 patient failures, took 12.0 years and found an effective treatment with probability 0.78. An open adaptive platform with response-adaptive randomization and ten active arms used 640 patients, had 431 failures, took 3.2 years and had probability 0.87. That platform compares each arm with all control patients, including those enrolled before the arm entered; their concurrent-control strategy 3d used 912 patients and took 4.6 years, with success probability 0.83. It also differed in arm count, sample-size caps and efficacy threshold, so this contrast does not isolate control borrowing. These are simulations, not empirical results, but they show where the saving this chapter locates in monitoring becomes large: it compounds over a program through shared controls and early stopping. The designs and their operating characteristics are the subject of Chapter 20.

Key points

Key points

- The textbook formula is $n = (z_\alpha + z_\beta)^2/(I\theta^2)$, with $(z_\alpha + z_\beta)^2 = 10.51$ at one-sided 0.025 and 90% power and 7.85 at 80%. Binary, continuous and time-to-event endpoints differ only in what $I\theta^2$ is: for events, $d = 4(z_\alpha + z_\beta)^2/\log(\text{HR})^2$, which gives 844 events at HR 0.80 and 90% power.
- Sample size scales as $1/\theta^2$, so the design alternative is the most consequential and least certain input. A trial sized for 90% power has about 62% power if the true effect is 70% of the design value.
- A conventional power calculation can be represented as assurance under a point-mass design prior. The conventional calculation itself does not assert certainty about the effect or specify an analysis prior. Averaging the power curve over an explicit prior gives assurance, which is typically well below the nominal power (0.75 for a prior standard deviation of half the design effect in the worked example). Assurance is a planning forecast, not a replacement for the scenario table of operating characteristics.
- The moderate deviation sample size equation $nI\theta^2 = (\sqrt{\log n + C} + z_\beta)^2$, with $C = 2\log(\pi_0/\pi_a)$, replaces the fixed z_α by a boundary growing like $\sqrt{\log n}$. At $\theta = 0.20$ it asks for 342 information units against 263, about 30% more, and it drives the type I error down with n.
- The Bayesian saving is not in the fixed design. A calibrated fixed-sample Bayesian

design is at best equal to the traditional one and, on the pure moderate deviation scale, larger. The saving is in monitoring, the subject of 11 and 21.

- Bayesian sample size criteria (average length, average coverage, worst outcome, posterior probability) generally depend on a design prior and an analysis prior; the known-variance average-length criterion needs no design prior. The power calculation is the posterior probability criterion with a point-mass design prior and a flat analysis prior. For a design prior $N(0.3, 0.1^2)$, 90% power needs 234 per arm and assurance first reaches 0.8 at 226.
- A decision-theoretic planner needs no error rates: the square root rule $n^\star = (\sigma/\tau)\sqrt{N/2}$ sizes the trial to the population it serves, and it is within 0.86% of the exact minimum loss at $N = 10^5$. In a program, the size that maximizes expected net present value is flat and moves with the prior.
- SELECT was designed for a hazard ratio of 0.83 with 1,225 events, and the textbook formula at that design gives about 1,211. The moderate deviation formula does not predict SELECT's size; the observed hazard ratio of 0.80 is a retrospective scenario.
- Regulators ask for the sample size justification (E9 3.5), for sensitivity to its assumptions, and, for a Bayesian design, for simulated operating characteristics, with the success criterion either calibrated to a type I error or justified by another route the January 2026 FDA draft describes.

Try it in the lab

Open module 03, Errors and power, go to the worked design (effect 3 points, standard deviation 10) and write down the number of patients you expect for half the effect before moving the slider; then check it, and repeat for the hazard ratio of 0.8 in the events panel.

Discussion questions

1. A sponsor sizes a phase 3 trial for 90% power at the effect observed in a 60-patient phase 2 trial. Using what you know about selection and the square law, which direction is the error in the design alternative likely to be, and how would you quantify the effect on the probability of success?
2. The protocol states 90% power and the investor reads 90% chance of success. Write the two sentences you would give the sponsor so that the board knows what the number means, and say what extra input you would need to compute the assurance.
3. The moderate deviation calculation (with $C = 0$ in the information convention) calls for 30% more information at $\theta = 0.20$ than the textbook formula. A regulator accepts a fixed-sample Bayesian design whose success criterion is calibrated to a type I error of 0.025. Does that design need the extra 30%? Why or why not?
4. SELECT was designed for a hazard ratio of 0.83 and 1,225 events, and the textbook formula gives about 1,211 at that design. A colleague says the moderate deviation formula explains the plan. What is wrong with that argument, and what data on many trials would you need before claiming that sponsors already design on the moderate deviation scale?
5. When is a fixed-sample traditional design the right choice for a confirmatory

trial, even for a sponsor who prefers Bayesian methods?

10. Multiplicity and Shrinkage

A trial that asks one question once is rare. Most trials have a primary endpoint and several secondary ones, two or three doses, a handful of prespecified subgroups, and an interim analysis or two. Every additional question is another chance for luck to produce a result that looks like a discovery. A sponsor who reports the best of twenty results as though it were the only one asked has not told the reader how the result was found, and a regulator who approves on it has accepted a risk that nobody measured. Multiplicity is the name for this problem and the family of procedures that controls it is one of the oldest and most settled parts of trial statistics.

This chapter does two things. First, it sets out the traditional apparatus properly, because it is what regulators expect and because it works: familywise error, Bonferroni, Holm, Hochberg, closed testing, fixed-sequence and graphical procedures, and the way guidance asks for the plan to be written. Second, it shows a different way to pay for looking at many things at once. Instead of raising the bar that each result must clear, a hierarchical model lowers the estimate of each result toward what the other results suggest. Both approaches make the sponsor pay for the extra questions. They pay in different currencies, and the difference matters most when the questions are related, which in a clinical program they nearly always are.

The centerpiece is an illustration motivated by a trial that Novick and Grizzle used in 1965 to argue for Bayesian analysis of trial data: four operations for duodenal ulcer. We work through it with the traditional tools and then with a hierarchical model, on the same illustrative recurrence counts (set at the scale of that trial, not transcribed from the 1965 paper).

> **!** Regulatory anchor
>
> **ICH E9, Statistical Principles for Clinical Trials (1998) (International Council for Harmonisation 1998), section 5.6 (multiplicity).** Multiplicity arises from "multiple primary variables, multiple comparisons of treatments, repeated evaluation over time and/or interim analyses". When it is present, "the usual frequentist approach to the analysis of clinical trial data may necessitate an adjustment to the type I error". The guideline asks that the strategy be identified in the protocol and that adjustment always be considered. Section 2.2 asks for the precise definition of the primary variable to be specified in the protocol, to avoid multiplicity concerns from post hoc definitions.
> **FDA, Multiple Endpoints in Clinical Trials, guidance for industry (final, October 2022) (U.S. Food and Drug Administration 2022b).** The agency's working document on endpoints, with the standard procedures (Bonferroni, Holm, Hochberg, fixed-sequence or hierarchical testing, gatekeeping and graphical approaches) and the expectation that the multiplicity strategy is prespecified and that strong control of the familywise error rate is demonstrated for the endpoints that support a claim.

FDA, Master Protocols for Drug and Biological Product Development (revised draft, June 2026), section on type I error. Draft position: FDA "generally does not recommend the use of multiplicity adjustments to strongly control the probability of making at least one type I error across the multiple comparisons of different drugs" in a master protocol, because the expected number of type I errors is the same as in separate trials. The draft's scope is wider than unrelated drugs: it also generally does not recommend adjustment across comparisons of one drug in different diseases, conditions or disease subtypes when these are distinct clinical objectives (section III.H). Drafts recommend and are not binding.
FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft, January 2026), and ICH E20 (Step 2 draft, June 2025). Both place the operating characteristics of a Bayesian design, simulated under a prespecified plan, at the center of the submission, which includes the false positive rate of the decision rule over the family of claims the trial can make. Shrinkage across arms or subgroups is a modeling choice that must be justified and examined in sensitivity analyses (see Chapter 3 for the full map and Chapter 14 for how to run the simulations).

10.1. The problem: many chances for luck

Suppose a drug does nothing and we test it on k independent endpoints, each at one-sided level $\alpha = 0.025$. The chance that at least one test crosses the bar is one minus the chance that none does:

$$\Pr(\text{at least one false positive}) = 1 - (1 - \alpha)^k. \tag{10.1}$$

This probability is the **familywise error rate** (FWER) when all k null hypotheses are true. For $k = 1$ it is 0.025 by design. The next chunk shows how it grows.

```
k <- c(1, 2, 3, 5, 6, 10, 20, 50)
knitr::kable(data.frame(k, 1 - (1 - alpha)^k, k * alpha), digits = 3,
 ↪  col.names = c("Number of tests", "Familywise error", "Bonferroni bound
 ↪  k alpha"), align = "r")
```

Table 10.1.: Probability of at least one false positive among k independent tests of true nulls, each at one-sided 0.025, with the Bonferroni upper bound k alpha.

Number of tests	Familywise error	Bonferroni bound k alpha
1	0.025	0.025
2	0.049	0.050
3	0.073	0.075
5	0.119	0.125
6	0.141	0.150
10	0.224	0.250
20	0.397	0.500
50	0.718	1.250

With twenty independent tests of a useless drug the chance of at least one "significant" result is about 0.40. The book's companion online lab (module 04, https://vsokolov.org/bayesian_trials_lab/04-multiplicity.html) makes the same point with twenty drugs that are exactly as good as placebo. The right-hand column, $k\alpha$, is the upper bound used by Bonferroni and it is a good approximation while $k\alpha$ is small.

There are two further points before the procedures. First, independence is neither the best nor the worst case: dependence can make the probability of at least one false positive either larger or smaller than Equation 10.1, depending on its structure. If the tests share a control group or the endpoints are positively correlated, it is typically smaller, but the number of false positives, when there is one, tends to be larger: the errors come in clumps. Second, the formula is about a drug that does nothing. What a regulator wants to prevent is a false claim, and a claim is made on one result or on a combination of results. The family is therefore defined by the claims the sponsor may make, not by the number of tests the statistician happens to run. A statistician who runs forty tests to understand the data but whose protocol says the claim rests on one prespecified comparison has a family of one.

10.1.1. What counts as the family

ICH E9 lists three sources (multiple primary variables, multiple treatment comparisons, repeated evaluation over time and interim analyses) and the practical definition follows from them. The family is the set of hypotheses such that a rejection of any one of them would be presented as evidence of an effect that supports approval or labeling. That makes three things part of the family: alternative primary endpoints (success on any one is enough), the key secondary endpoints for which a labeling claim is sought, and the dose or population comparisons that could each be the basis of approval. Interim looks on the same endpoint are handled by a group sequential boundary, which Chapter 11 covers, and are combined with the cross-endpoint procedures of this chapter as in the protocol paragraph below.

Co-primary endpoints are a different case and need a different explanation. When the claim requires success on all of them, the trial succeeds only if every endpoint is individually significant, and testing each at the full level α controls the error for that conjunctive claim without any Bonferroni division (an intersection-union test, as in section III.C.1 of the FDA multiple-endpoints guidance). Dividing α among co-primary endpoints is therefore unnecessary and costs power; the cost of co-primary endpoints lies instead in the power to win on all of them at once, which is lower than the power on any one. Alternative primary endpoints, where success on any one supports the claim, are the case that needs the adjustments of this chapter.

Exploratory endpoints, safety endpoints and descriptive subgroup displays are outside the family because no claim is made on them, and the protocol should say so. Their results can generate hypotheses, and the honest way to report them is with estimates and intervals and no label of significance.

10.1.2. What preserving the option to look costs in patients

The inflation of the bar can be converted into the currency a sponsor understands. Suppose a trial must keep K analyses open (endpoints, subgroups, per-protocol against intention-to-treat, adjusted against unadjusted) and guarantees at most $\alpha = 0.025$ for the best of them, whichever one is reported. For independent statistics the exact threshold is the Sidak value

$z_K = \Phi^{-1}((1-\alpha)^{1/K})$ (Bonferroni's $\Phi^{-1}(1-\alpha/K)$ is slightly larger). Because the sample size for a given effect and power is proportional to $(z_{\text{crit}} + z_\beta)^2$, the trial must grow by the factor $n_K/n_1 = ((z_K + z_\beta)/(z_1 + z_\beta))^2$.

```
sc <- decide_subset_cost()
knitr::kable(data.frame(K = sc$K, `z_K` = round(sc$z_K, 2), `n_K / n_1` =
 ↪  round(sc$ratio, 2),
                         `Extra per 1000` = round(sc$extra_per_1000),
                          ↪  check.names = FALSE))
```

Table 10.2.: Cost of keeping K analyses open: Sidak threshold at one-sided 0.025, sample size ratio at 80 percent power, and the extra patients required per 1000 enrolled under a single prespecified analysis.

K	z_K	n_K / n_1	Extra per 1000
1	1.96	1.00	0
5	2.57	1.48	485
10	2.80	1.69	693
50	3.29	2.17	1172
100	3.48	2.38	1377
1000	4.05	3.05	2052

Keeping five analyses open costs about 485 extra patients per 1000, and keeping a hundred open costs a trial about 2.4 times the size. With a standardized effect of 0.1 and 80 percent power, a single prespecified analysis needs about 1570 patients per arm, so the same effect with $K = 100$ needs roughly 3,730. For Gaussian statistics with nonnegative correlations the Sidak threshold is conservative, so correlated analyses have an effective K below the nominal count and the table is an upper bound; under negative correlations the Sidak error can exceed its nominal level, and Bonferroni, which needs no dependence assumption, is the safe choice. The table is the argument for prespecifying few analyses, and it is also the reason to price the many analyses that cannot be avoided by modeling (Section 10.4).

10.2. The traditional toolbox

10.2.1. Bonferroni

The simplest procedure: with k hypotheses and a familywise level α, test each at α/k (the inequality is Bonferroni's, Dunn (Dunn 1961) made it a multiple-comparison method, and Dunnett (Dunnett 1955) gave an exact procedure based on the multivariate t distribution for several treatments against one control, the multi-arm trial case, which is less conservative than Bonferroni because it uses the correlation induced by the shared control). Equivalently, multiply each p-value by k and compare to α. The union bound guarantees that the probability of any false rejection is at most α, whatever the dependence between the tests. That guarantee is the reason Bonferroni is in every guidance and every statistical analysis plan: it needs no assumption and its logic can be explained to a clinician in one sentence.

Its cost is power. For $k = 6$ the per-test level is $0.025/6 = 0.00417$, and the critical value moves from $z = 1.96$ to $z = 2.64$. A trial sized for 90% power at the unadjusted level has noticeably less than that after adjustment, and the lost power is the price of protecting against the case where all six nulls are true. When one or two of the hypotheses are in fact true effects, Bonferroni is paying to guard against errors that cannot occur.

10.2.2. Holm

Holm (1979) is a uniform improvement. Order the p-values $p_{(1)} \leq ... \leq p_{(k)}$. Compare $p_{(1)}$ to α/k; if it is rejected, compare $p_{(2)}$ to $\alpha/(k-1)$; continue until a p-value fails, and then stop and retain that hypothesis and all remaining ones. Holm controls the familywise error rate in the strong sense, with no assumption on dependence, and it rejects everything Bonferroni rejects and sometimes more. There is no good reason to use plain Bonferroni when Holm is available, other than that Bonferroni's simplicity helps in a protocol that must be read by many people.

10.2.3. Hochberg

Hochberg (1988) runs the order in the other direction: start from the largest p-value and compare it with α, the next largest with $\alpha/2$, and so on; at the first $p_{(j)} \leq \alpha/(k-j+1)$ reject that hypothesis and all with smaller p-values. It is a step-up version of Holm's step-down test and rejects at least as many hypotheses. The price is an assumption: the guarantee holds when the test statistics are independent or positively dependent in a specific sense (positive dependence such as nonnegative correlation among one-sided normal test statistics, which covers comparisons with a shared control). Its step-up logic rests on the Simes inequality (Simes 1986), and Hommel (Hommel 1988) gave a more powerful procedure under the same conditions. FDA's guidance on multiple endpoints states these conditions and does not treat Hochberg as assumption-free.

Here are the three procedures on one set of p-values, with the package functions `mult_bonferroni()`, `mult_holm()` and `mult_hochberg()`.

```
p <- c(E1 = 0.004, E2 = 0.011, E3 = 0.019, E4 = 0.022, E5 = 0.041, E6 =
 ↪  0.30)   # six one-sided p-values
yn <- function(r) ifelse(r, "reject", "")
knitr::kable(data.frame(names(p), p, yn(p <= alpha),
 ↪  yn(mult_bonferroni(p)), yn(mult_holm(p)), yn(mult_hochberg(p))),
             col.names = c("Hypothesis", "p-value", "Unadjusted",
              ↪  "Bonferroni", "Holm", "Hochberg"), align = "lrcccc")
```

Table 10.3.: Hypotheses rejected at one-sided 0.025 by the unadjusted, Bonferroni, Holm and Hochberg procedures for six p-values.

	Hypothesis	p-value	Unadjusted	Bonferroni	Holm	Hochberg
E1	E1	0.004	reject	reject	reject	reject
E2	E2	0.011	reject			
E3	E3	0.019	reject			
E4	E4	0.022	reject			

Table 10.3.: Hypotheses rejected at one-sided 0.025 by the unadjusted, Bonferroni, Holm and Hochberg procedures for six p-values.

	Hypothesis	p-value	Unadjusted	Bonferroni	Holm	Hochberg
E5	E5	0.041				
E6	E6	0.300				

At the unadjusted level four of the six pass. All three adjusted procedures reject only E1, whose p-value 0.004 is just under $0.025/6 = 0.0042$; after that Holm's next threshold is $0.025/5 = 0.005$, which 0.011 misses, so it stops. The three procedures give the same answer here and separate when several p-values are small, for example $0.004, 0.005, 0.006$ and the rest large.

10.2.4. Closed testing

The procedures above are special cases of one principle due to Marcus, Peritz and Gabriel (1976). A **closed test** starts from the family of hypotheses $H_1, \dots, H_k$ and forms all intersection hypotheses (for example, "H_1 and H_3 are both true"). Each intersection is tested with any valid level-α test. An individual hypothesis H_i is rejected if every intersection that contains H_i is rejected at level α. This controls the familywise error rate strongly, because a true intersection of nulls is only wrongly rejected with probability at most α, and any false rejection of an individual hypothesis requires rejecting the intersection of all true nulls. Holm's procedure is the closed test that uses Bonferroni for each intersection. Choosing a more powerful intersection test, such as one that uses the correlation between endpoints, gives a more powerful closed procedure, and that is where the modern literature on multiplicity lives (Dmitrienko and D'Agostino 2013 gives a full survey).

For a clinician the lesson is that "the multiplicity adjustment" is not a single formula. It is a rule for how a result about one hypothesis is allowed to depend on results about the others, and it is chosen at the protocol stage. The same logic yields the gatekeeping and fixed-sequence procedures that dominate practice.

10.2.5. Fixed-sequence (hierarchical) testing

Order the hypotheses in advance by clinical importance: $H_1, H_2, \dots, H_k$. Test H_1 at the full α. If it is rejected, test H_2 at the full α, and so on. Stop at the first failure and do not test the rest. The familywise error rate is controlled at α without any division of the level, because at most one hypothesis can be the first true null, and it is tested at α.

This is the most common strategy for a primary endpoint followed by key secondaries, and it is what "primary/secondary hierarchy" means in a protocol. The price is that the order is final. A secondary endpoint late in the sequence can be tested at full level only if everything before it succeeded, so a failure on the primary, or on an early secondary, makes all later results exploratory regardless of how impressive they look. This is the right behavior for a regulator and it often surprises a clinical team, which has seen a large effect on endpoint 4 and expects to say something about it.

Two consequences follow for how the hierarchy is written. First, the sponsor should put first the endpoint on which it is most confident of success and which carries the claim that matters

most, since a failure ends the chain. Second, the sequence can be a tree and not a line: "if the primary succeeds, test secondary A and secondary B each at $\alpha/2$, and if A succeeds, test A's follow-up at $\alpha/2$ plus whatever passes from B." That is a graphical procedure.

10.2.6. Graphical procedures

Bretz, Maurer, Brannath and Posch (2009) gave a framework that unifies Bonferroni, Holm, fixed sequence, fallback and gatekeeping procedures in one picture. Each hypothesis is a node with a weight w_i (its share of α, with $\sum w_i \le 1$). Directed edges carry weights g_{ij} giving the fraction of H_i's level that passes to H_j if H_i is rejected. The algorithm is: reject any H_i with $p_i \le w_i \alpha$; remove it, pass its weight along the edges, update the edge weights, and repeat. The result controls the familywise error rate strongly, because each step is a closed-test shortcut. Burman, Sonesson and Guilbaud (Burman, Sonesson, and Guilbaud 2009) developed a parallel recycling framework, Bretz and colleagues (Bretz et al. 2011) extended graphs to Simes and parametric tests, and Maurer and Bretz (Maurer and Bretz 2013) showed how to combine a graph with group sequential looks, which is where this chapter meets Chapter 11.

The method is popular because the graph is the protocol: a statistician can draw it for the clinical team, the team can see which results unlock which, and the same picture goes into the statistical analysis plan. Here is a small example with two primary endpoints and two secondaries. Each primary gets half the level; if a primary is rejected, its level passes to the secondary that clinically follows it; once a secondary is rejected, its level passes to the other primary.

```
w0 <- c(0.5, 0.5, 0, 0)                          # nodes P1, P2, S1, S2:
 ↪  alpha/2 on each primary
G  <- rbind(c(0, 0, 1, 0), c(0, 0, 0, 1),        # P1 -> S1, P2 -> S2
            c(0, 1, 0, 0), c(1, 0, 0, 0))        # S1 -> P2, S2 -> P1
scen <- list("P1 strong, P2 weak, S1 moderate" = c(0.004, 0.020, 0.010,
 ↪  0.300),
             "P1 weak, others strong" = c(0.020, 0.003, 0.004, 0.002),
             "all marginal" = rep(0.015, 4))
knitr::kable(data.frame(Scenario = names(scen), Rejected = sapply(scen,
 ↪  function(p)
  paste(c("P1", "P2", "S1", "S2")[mult_graph(p, w0, G, alpha)], collapse =
   ↪  ", "))), row.names = FALSE)
```

Table 10.4.: Hypotheses rejected by the graphical procedure in three scenarios of p-values for two primary and two secondary endpoints.

Scenario	Rejected
P1 strong, P2 weak, S1 moderate	P1, P2, S1
P1 weak, others strong	P1, P2, S1, S2
all marginal	

In the first scenario $P1$ is rejected at 0.0125, which passes its weight to $S1$; $S1$ ($p = 0.010$) is rejected at 0.0125 and passes its weight to $P2$, which now has the full level and rejects at

$p = 0.020$. $S2$ ($p = 0.30$) stays. In the second, $P1$ at 0.020 fails alone, but $P2$ is rejected and passes to $S2$, which is rejected and returns its level to $P1$, which then clears 0.025, and $S1$ follows. The third is the reminder that a single common marginal p-value of 0.015 is not enough when each primary has only half of the level, 0.0125, and nothing unlocks. The operating characteristics of the whole graph can be simulated as in Chapter 14.

Gatekeeping procedures, in which a family of primary hypotheses must be passed before a secondary family is tested (Dmitrienko, Offen and Westfall (Dmitrienko, Offen, and Westfall 2003) is the standard reference for the case where not every primary effect must be significant), are special cases of such graphs. The harder practical question is how a graph meets the interim analyses of Chapter 11, since an outcomes trial has both a key secondary hierarchy and a group sequential design. Take a trial with H1 (the primary composite) and H2 (all-cause death, tested only after H1). At the start the whole level 0.025 sits on H1 and H2 has weight zero. If H1 is rejected at an interim look, at the level its spending function allows there, its weight moves to H2, which is then tested from that look onward, with its own spending function applied to the information H2 has accumulated and not to the information H1 had at the time of the transfer. This is the setting of Maurer and Bretz (Maurer and Bretz 2013), who showed how to combine a graph with group sequential looks, and of the recycling framework of Burman, Sonesson and Guilbaud (Burman, Sonesson, and Guilbaud 2009); the details of the spending rule are theirs to fix and the protocol has to state the version chosen. For the protocol this means three statements that the graphical description above must be supplemented by: the spending function for each hypothesis, the rule that a transferred weight is spent from the look at which the transfer happens, and the graph with its weights printed. A design that says only "alpha is spent by O'Brien-Fleming" has left the key secondary undefined.

10.2.7. What the guidance asks for

The EMA's guidance is older and shorter. The Points to Consider on multiplicity issues (CPMP/EWP/908/99) was adopted in 2002 and took legal effect on 1 April 2003 (European Medicines Agency, Committee for Proprietary Medicinal Products 2002). A draft replacement guideline (EMA/CHMP/44762/2017) went to public consultation in 2017 and, as far as the EMA's page shows, remains a draft (European Medicines Agency, Committee for Medicinal Products for Human Use 2017); its summary says it adds or updates the treatment of multiplicity in safety, drug-response studies, secondary endpoints, subgroup analysis and estimation. A draft recommends and does not bind, so a European submission is still read against the 2002 document and ICH E9. ICH E9 section 5.6 is short and gives the principle: identify multiplicity in the protocol and consider adjustment. It also asks that the key primary variable and critical treatment contrast be identified in advance, which is a form of multiplicity control without any adjustment at all: a family of one. The FDA guidance on multiple endpoints (final, October 2022) is the detailed companion. It describes the situations (multiple primary endpoints and the claims that follow from them, co-primary endpoints that must all succeed (each tested at the full level), primary and secondary endpoints, composite endpoints with components) and the procedures above, and it expects strong control of the familywise error rate for the endpoints that will support a claim.

In the protocol: the multiplicity paragraph

State, before unblinding: (1) the family of hypotheses, by name, with the claim each supports; (2) the testing procedure (fixed sequence, Holm, a graph; include the graph); (3) the familywise level, usually one-sided 0.025; (4) how interim analyses on each endpoint consume that level (reference the alpha-spending function from Chapter 11); (5) which endpoints are exploratory and will be reported without significance claims; (6) the rule for what happens to later hypotheses when an earlier one fails; (7) if subgroups are prespecified for claims, where they sit in the graph. Strong control of the familywise error rate needs a proof for an analytically controlled procedure, or calibration by simulation over the relevant mixed null configurations (some hypotheses true, others false), effect magnitudes and nuisance parameters, with a stated justification for the least favorable scenarios; the global null alone is not enough. That simulation, and the power for each hypothesis under the planned scenarios, belong in the SAP appendix.

10.3. Subgroups and the forking paths

Of all the sources of multiplicity, the subgroup analysis is the one that most often misleads. A trial that fails overall will be examined by age, sex, region, baseline severity, biomarker status and prior therapy. With twelve ways to slice the data, luck has twelve chances. The classic warning is the ISIS-2 trial (ISIS-2 (Second International Study of Infarct Survival) Collaborative Group 1988), which enrolled 17,187 patients with suspected myocardial infarction. To illustrate the problem, its authors showed that when patients were split by astrological birth sign, aspirin appeared to have no benefit for patients born under Gemini or Libra, although in the trial as a whole aspirin reduced vascular mortality very strongly. The point is not that Gemini is special. It is that twelve signs gave twelve chances and two came up as an apparent exception. Module 04 of the companion lab repeats the exercise with a made-up useless drug split twelve ways.

A subgroup finding has a structure that makes it worse than a simple count suggests. The analyst who sees a promising subgroup usually chose how to cut: age above or below 65, or 60, or 70; ejection fraction by tertiles or by a threshold; region by country or by continent. Gelman and Loken (2014) called this the garden of forking paths: even if the analyst ran only one test, the choice of which test depended on the data, and the effective number of comparisons is the number of tests that would have been run had the data looked different. The multiplicity is hidden in decisions, not in the number of p-values. This is the practical reason ICH E9 insists on defining the primary variable and analysis in the protocol, and why post hoc subgroup claims are treated as hypothesis-generating.

A reader who needs a standard for reporting subgroup analyses in a trial paper will find one in Wang and colleagues (2007): say how many subgroup analyses were done, state which were prespecified, and test for interaction between the subgroup variable and treatment rather than comparing p-values inside subgroups (a significant effect in one subgroup and a non-significant effect in another is not evidence that the effects differ). The interaction test has little power at the usual sample size, so the honest result of most subgroup analyses is "consistent with a common effect".

The traditional remedies are to prespecify a small number of subgroups with a rationale, to include the subgroup variable in the multiplicity graph when a claim is intended, to

test interaction first, and to treat everything else as exploratory. Those remedies work and should be used. They have a limitation that the Bayesian alternative addresses: they are all-or-nothing. A subgroup is either in the family or out of it, and the estimate in a subgroup that is outside the family is reported at face value, with all the noise that implies.

10.3.1. Why a Bayesian must also ask how the result was chosen

A Bayesian analysis conditions on the data it is shown, and that is where the forking paths matter. The regulator does not see the sponsor's data X. It sees a message $M = \varsigma(X)$, where the reporting rule ς (which subgroup, which endpoint, which analysis) is chosen by the sponsor.

Proposition (selected likelihood)

Let X have density $p(x \mid \theta)$ and let the sponsor report $M = \varsigma(X)$, with ς possibly randomized. The regulator's posterior is $p(\theta \mid M = m) \propto p(\theta)\, \mathcal{L}_\varsigma(\theta; m)$, where $\mathcal{L}_\varsigma(\theta; m)$ is the density at m of the distribution of $\varsigma(X)$ under θ (the pushforward of $p(x \mid \theta)$ through ς, with respect to a dominating measure for the reports, and averaging over any randomization), so it depends on ς. If ς reports the largest of K independent components $X_1, \dots, X_K$, the likelihood of the reported value m is

$$\mathcal{L}_\varsigma(\theta; m) = \sum_{k=1}^{K} p(m \mid \theta_k) \prod_{j \neq k} F(m \mid \theta_j), \tag{10.2}$$

with F the distribution function of a component. In the upper tail $F(m \mid \theta_j) \to 1$, so for exchangeable components the ratio of $\mathcal{L}_\varsigma$ to the naive likelihood $p(m \mid \theta_k)$ approaches K.

The proof is the density of a maximum of independent variables. The consequence is that the naive likelihood $p(m \mid \theta_k)$ is the wrong likelihood for the reported value. A factor that does not depend on θ cancels from posterior odds, so the ratio approaching K is not itself a K-fold distortion of the evidence. The distortion is in how $\mathcal{L}_\varsigma$ depends on θ. Take the simplest case, a common effect $\theta \sim N(0, 1)$ and K independent $N(\theta, 1)$ components, of which only the maximum m is reported. The correct likelihood is $K\phi(m - \theta)\Phi(m - \theta)^{K-1}$ and the naive one is $\phi(m - \theta)$, so the correct one carries the extra factor $\Phi(m - \theta)^{K-1}$, which falls as θ rises and pulls the posterior down. For $m = 2$ and $K = 5$ the posterior mean of θ is 1 under the naive likelihood and 0.59 under the correct one, and the posterior probability that $\theta > 0$ falls from 0.92 to 0.85. The regulator cannot compute the correct posterior without knowing ς. This, and not any claim that posteriors depend on intentions, is the Bayesian reason for a protocol: the protocol is the specification of ς.

It also shows what is right and what is beside the point in the familiar Bayesian retort that optional stopping does not affect the posterior (see Chapter 12 for the calibration side of that argument). The retort is correct when the full sample path is reported, because then $M = X$ and ς is the identity. It is beside the point when the report is a summary chosen after the fact, because then ς is not the identity and the sponsor knows something the regulator does not. Shrinkage, below, is the Bayesian way to put the selection into the model by giving the whole family of subgroups one prior, instead of reporting the winner alone.

10.4. The Bayesian alternative: shrinkage instead of threshold inflation

10.4.1. The idea

A Bonferroni correction makes the evidence required for each result higher, and it does so independently of what the other results say. That is a strange property when the results are related. If six subgroup effects look similar, that is information that each of them is probably close to their average. If one is large and five are near zero, that is information that the large one is probably a mix of effect and luck. A hierarchical (multilevel) model uses this information. It treats the effects $\theta_1, \dots, \theta_k$ of the k arms, doses or subgroups as draws from a common distribution,

$$\hat{\theta}_j \mid \theta_j \sim N(\theta_j, s_j^2), \qquad \theta_j \mid \mu, \tau_g \sim N(\mu, \tau_g^2), \tag{10.3}$$

where $\hat{\theta}_j$ is the observed effect (for example a log odds ratio), s_j its standard error, μ the common mean and τ_g the between-group standard deviation. For fixed μ and τ_g, the posterior mean for each group is a weighted average of its own estimate and the common mean,

$$\mathbb{E}[\theta_j \mid y] = (1 - B_j)\,\hat{\theta}_j + B_j\,\mu, \qquad B_j = \frac{s_j^2}{s_j^2 + \tau_g^2}. \tag{10.4}$$

This is **partial pooling**. A group with a small standard error (much information) stays near its own estimate. A group with a large standard error (little information) is pulled toward the mean. When τ_g is large, the groups are nearly unrelated and little pooling occurs; when τ_g is small, the groups behave as one, and the model reproduces the pooled analysis. The data inform τ_g, but the amount of pooling also depends on the exchangeability assumption, the distributional form of the group effects and, in a full Bayesian analysis, the hyperprior; it is not free of assumptions. In a full Bayesian analysis the posterior mean integrates over the posterior of μ and τ_g, so Equation 10.4 is its conditional form. (In a full Bayesian analysis, μ and τ_g have priors. With few groups, the prior for τ_g should be weakly informative and examined in sensitivity analysis. Chapter 8 discusses how. Chapter 19 returns to the same hierarchical structure for borrowing across trials and baskets.)

The consequence for multiplicity is that the correction for looking at many things is built into the estimate. The largest of k observed effects is on average too large, because it was selected as the largest, and the hierarchical estimate pulls it back toward the mean by about the right amount. A posterior statement about the selected group is then made with honest uncertainty, without any change to the threshold. Gelman, Hill and Yajima (2012; see also Gelman and Tuerlinckx 2000 for the Type S error analysis that preceded it) make this argument in the language of applied statistics: when estimates are partially pooled, multiple comparisons stop being a problem that needs a separate fix, because the problem was that unpooled estimates are noisier than they should be. Berry and Berry (2004) applied the same idea to adverse events in a drug safety trial, where hundreds of event types are examined and the hierarchical model borrows strength across them.

10.4.2. Stein, Efron and Morris, and the batting averages

There is a classical theorem behind shrinkage, and its scope should be stated exactly. Stein (1956) showed that for three or more means in a multivariate normal model with known variances, the usual estimate, each group's own average, is inadmissible under total squared-error loss: there is an estimator that has smaller total squared error whatever the true means are, and James and Stein (W. James and Stein 1961) gave one explicitly. The theorem concerns that model and that loss. It does not establish that the empirical-Bayes fit used for the ulcer analysis below improves every arm, the selected contrast, posterior coverage or a clinical decision; that analysis is a separate model-based procedure whose performance has to be assessed, as the simulation below does for one case. Efron and Morris (1975, 1977) made this concrete with a data set of baseball players' early-season batting averages, where the estimates shrunk toward the league mean predicted the rest of the season better than the raw averages did for most players, and the improvement came largely from the extreme players. The same logic applies to the player who hits 1 for 19 against one pitcher: against a bar of "a probability of one in a hundred if his true average is 0.300", it looks like evidence he cannot hit that pitcher, but there are many pitchers, and a hierarchical analysis of the player's record across all of them gives an estimate for that pitcher that is lower than average but much nearer to it than 1/19. The batting average problem is a multiplicity problem in a form everyone has intuition for: look at enough batter and pitcher pairs and some pair will look like a curse.

10.4.3. A numerical illustration: the winner's curse

The reason shrinkage answers selection bias can be seen in a short simulation. Six arms have true effects drawn from $N(0, 0.2^2)$ on the log odds scale, each estimated with standard error 0.28 (about what the ulcer trial's arms give, below). The trial reports the arm with the largest observed effect. How wrong is that arm's raw estimate, and how wrong is the shrunken one?

```
set.seed(3); k <- 6; reps <- 20000; tau <- 0.2; se <- 0.28
th <- matrix(rnorm(reps * k, 0, tau), reps); y <- th + matrix(rnorm(reps *
 ↪  k, 0, se), reps)
i <- cbind(1:reps, max.col(y)); B <- se^2 / (tau^2 + se^2)    # shrinkage
 ↪  factor, mu = 0 known
err <- data.frame(raw = y[i] - th[i], shrunk = (1 - B) * y[i] - th[i])
wc <- sapply(err, function(e) c(mean_error = mean(e), rmse =
 ↪  sqrt(mean(e^2))))
ggplot(stack(err), aes(values, fill = ind)) + geom_histogram(aes(y =
 ↪  after_stat(density)), bins = 60, alpha = 0.6, position = "identity") +
  geom_vline(xintercept = 0) + scale_fill_brewer(palette = "Dark2", name =
   ↪  NULL) +
  labs(x = "estimate minus true effect (log odds)", y = "density") +
   ↪  theme_minimal()
```

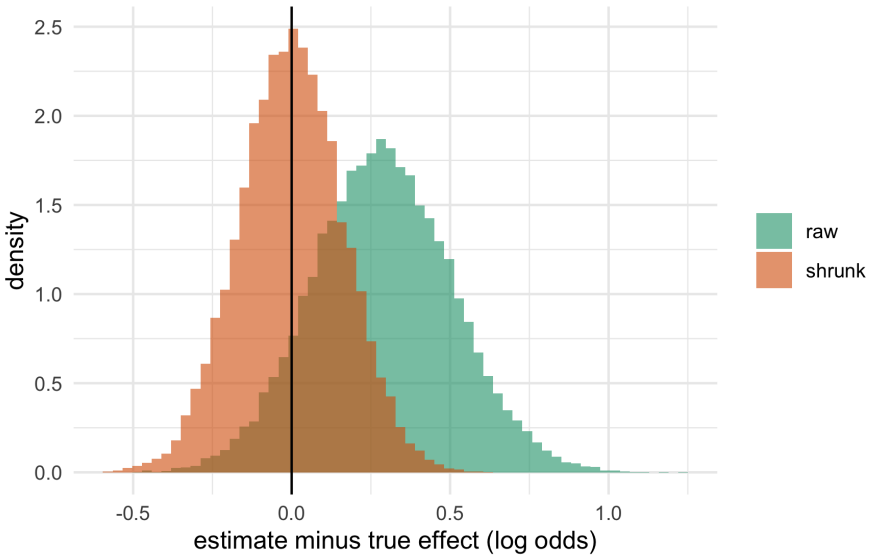


Figure 10.1.: Error of the estimate for the apparently best of six arms. The raw estimate is biased upward because the arm was selected for looking best. The shrunken estimate is centered on the truth with a much smaller spread.

The raw estimate of the selected arm is too high by about 0.29 on the log odds scale on average and its root mean squared error is about 0.36. The shrunken estimate has no average bias and a root mean squared error of about 0.16. Here the true between-arm spread τ_g was known; in a real trial it is estimated, which adds uncertainty, and the fitted model is only as good as the exchangeability assumption behind it. The qualitative point stands. The correction for selection is an estimate that is smaller and more honest, and not a harder test applied to an estimate that is still biased.

10.4.4. What shrinkage does not do

It does not control the familywise error rate, and a regulator who requires strong control of that rate for a claim will not be satisfied by a posterior probability alone. Chapter 14 shows how to calibrate the Bayesian decision rule, with its shrinkage in place, so that the familywise error rate over the family of claims is controlled at the required level, and reports the result as simulated operating characteristics. When this is done, the hierarchical rule usually has more power than Bonferroni at the same simulated familywise error rate if the effects are related, because the threshold does not have to be raised by the full factor of k. When the effects are unrelated (a single true effect among many nulls), shrinkage toward the common mean works against the one real effect, and a good hierarchical design uses a heavier-tailed or mixture distribution for the group effects so that one large outlier is not pulled all the way in. That robustness is also a choice to be justified in the protocol and tested by simulation. Two platform trials that handle multiplicity by modeling and not by adjustment, I-SPY 2 and REMAP-CAP, are described in the “In practice” section below.

The comparison, then, is not “Bayes avoids multiplicity” but a different division of labor. The traditional procedure sets a threshold that guarantees the error rate, whatever the effects are, and pays in power. The Bayesian procedure uses the structure of the effects to estimate

more accurately, and then, if the regulator requires it, is calibrated to the same error rate by simulation, with the power gain realized to the extent that the structure is real.

10.4.5. Where familywise control is not the target: false discovery rate and prior probabilities

The two halves of this chapter, thresholds and shrinkage, meet when the aim is a screen and not a claim. For hundreds of biomarkers, signatures or adverse event terms the aim is to find most of the real signals and to tolerate a known share of false leads. Benjamini and Hochberg (Benjamini and Hochberg 1995) defined the false discovery rate, the expected proportion of false rejections among the rejections, and gave a step-up procedure: order the p-values, find the largest j with $p_{(j)} \le jq/m$, and reject the hypotheses with the j smallest p-values. It controls the rate at q for independent p-values and under positive dependence of the kind that supports Hochberg's procedure (Benjamini and Yekutieli 2001). Table 10.5 applies the procedures of this chapter and the Benjamini-Hochberg procedure to simulated screens of 1000 hypotheses, 100 of them with a true effect (a z statistic centered at 3), all at the level 0.025, averaged over 1000 replications.

Table 10.5.: Average results over 1000 simulated screens of 1000 hypotheses, 100 with a true effect (z statistic centered at 3), at the level 0.025 for each procedure. Rejected, False and True are the mean numbers of hypotheses rejected, of rejections that are true nulls, and of real effects found. Realized FDR is the mean over screens of the proportion of false rejections among the rejections (zero when nothing is rejected).

	Rejected	False	True	Realized FDR
Unadjusted	107.6	22.7	85.0	0.209
Bonferroni	14.3	0.0	14.3	0.002
Holm	14.4	0.0	14.4	0.002
Benjamini-Hochberg	50.1	1.2	48.9	0.022

On average Bonferroni and Holm reject 14.3 and 14.4 hypotheses with 0.02 and 0.02 false rejections, Benjamini-Hochberg rejects 50.1 with 1.2 false, a realized false discovery rate of 0.022 against the nominal 0.025, and the unadjusted test rejects 107.6 with 22.7 false, a realized rate of 0.21.

The Bayesian reading of the same screen is Efron's two-groups model (Efron 2008): each statistic comes from the null with probability π_0 and from an alternative otherwise, and the local false discovery rate is the posterior probability of the null given the statistic. The Benjamini-Hochberg procedure thresholds an estimate of the tail-area false discovery rate, with the null proportion set to one. Berry and Hochberg (D. A. Berry and Hochberg 1999) set out the Bayesian view of multiple comparisons, in which a hierarchical prior supplies the correction without a change of threshold. Scott and Berger (J. G. Scott and Berger 2006, 2010) show where the correction comes from: if the prior probability that a hypothesis is a true effect is itself unknown and is learned from all the hypotheses, the posterior imposes a penalty that grows with the number of hypotheses when few signals are present among many hypotheses, and the penalty relaxes as more signals appear.

In practice false discovery rate control is used for screening, exploratory endpoints and signal detection, where no confirmatory claim is made, for example the double false discovery rate flagging of adverse events of Mehrotra and Adewale (Mehrotra and Adewale 2012), and it is not used for a confirmatory claim, which stays under familywise control as in the guidance above.

10.4.6. A large trial with many subgroups: RECOVERY

The same machinery applies when the groups are prespecified subgroups of one large trial. Chapter 22 reanalyzes the RECOVERY dexamethasone result with a hierarchical model across its prespecified subgroups, with the prior on the between-subgroup standard deviation τ_g stated explicitly, the posterior median of τ_g reported (the posterior of τ_g depends on its prior), and age-adjusted rate ratios. The subgroup estimates move toward the overall effect in proportion to how little information each carries, which is the winner's-curse correction of the previous section applied to a trial that actually did report subgroups. The reader should take from it the sensitivity of the shrinkage to the prior on τ_g as much as the shrunken estimates themselves.

10.5. An illustration motivated by the four-operation ulcer trial

10.5.1. The original

Novick and Grizzle (1965) made one of the first cases for Bayesian analysis of clinical trial data using an experiment then in progress that compared four operative treatments for duodenal ulcer. The structure is the one on which the traditional apparatus does worst: four arms, hence six pairwise contrasts; a trial in progress, so repeated analysis; and a modest effect on a binary endpoint. They argued that a Bayesian analysis conditions on the data in hand and therefore needs neither a multiplicity correction nor a spending schedule, and that prior information about operative outcomes can be stated rather than suppressed. The feature of the trial we use is structural: the four operations differ in how much stomach is removed, so they are ordered.

The counts below are illustrative, set at the scale of that trial and not transcribed from the 1965 paper, with recurrence decreasing in the extent of resection. They are the counts used in Polson and Sokolov's working paper (Polson and Sokolov 2026b) and ship with the package as the dataset `ulcer`.

```
l <- shrink_logodds(ulcer); y <- l$y; v <- l$v
knitr::kable(transform(ulcer, rate = round(100 * recurrences / n, 1),
 ↪  log_odds = round(y, 3), se = round(sqrt(v), 3)), col.names =
 ↪  c("Operation", "Patients", "Recurrences", "Recurrence rate (%)", "Log
 ↪  odds", "Standard error"), align = "r")
```

Table 10.6.: Illustrative duodenal ulcer data: recurrences by operation, with the recurrence rate, log odds and its standard error.

Operation	Patients	Recurrences	Recurrence rate (%)	Log odds	Standard error
Drainage and vagotomy	96	20	20.8	-1.335	0.251
25 per cent resection and vagotomy	104	16	15.4	-1.705	0.272
50 per cent resection and vagotomy	110	12	10.9	-2.100	0.306
75 per cent resection	107	9	8.4	-2.388	0.348

i A note on the companion lab

Module 05 of the companion lab uses a different table for the same trial: four treatments (named A to D) with 100 patients each and a three-level outcome (excellent, fair, death), described there as illustrative counts at the scale of Novick and Grizzle (1965), not transcribed from Polson and Sokolov (2026). The working paper itself says its counts are illustrative and recurrence-based, so the companion lab's table and this chapter's table are not the same data. Both illustrate the same lesson, and the numbers in this chapter come from the code shown.

10.5.2. What the traditional reading does with it

```
d <- y[1] - y[4]; se <- sqrt(v[1] + v[4]); z <- d / se; p2 <- 2 * (1 -
 ↪  pnorm(z))
ulcer_std <- c(log_OR = d, se = se, Z = z, p_two_sided = p2, bonferroni_6 =
 ↪  min(1, 6 * p2), bar_z = qnorm(1 - 0.025 / 6))
```

The best against worst contrast, drainage and vagotomy against 75 percent resection, is a difference of 12.4 percentage points in the recurrence rate, a log odds ratio of 1.053 with standard error 0.43. The test statistic is $Z = 2.45$ and the unadjusted two-sided p-value is 0.0142. That is the answer if this single contrast was the only one asked. It was not: with four arms there are six pairwise contrasts, and the best against worst is the one that was chosen because it is the largest.

The Bonferroni-adjusted p-value is 0.0855, and the finding is not reportable. A group sequential design with four interim analyses at the corrected level would put the first boundary at 5.34 (an O'Brien-Fleming boundary at the Bonferroni level, from Polson and Sokolov's script), which $Z = 2.45$ does not approach. An e-value procedure at the same corrected level, with a normal-mixture alternative of prior scale 0.5 on the log odds ratio, would need $Z \geq 4.53$ (the construction is in Chapter 12). Every one of these procedures returns nothing from a twelve point difference in recurrence. It is worth being fair to the traditional tools here: they are doing what they promise. If the question really were "does any of these six contrasts show a difference, with a guarantee that I will make a false claim at most one time in forty", the answer from this trial is "not established". The difficulty is

that this is the wrong question for the clinicians who ran the trial, and the correction is not informative about the question they care about.

Holm does not rescue it. With six contrasts and the best one's p-value adjusted by 6, Holm and Bonferroni agree on the smallest p-value; Holm's gain would come from the other contrasts, none of which is stronger. The pairwise Bonferroni and Holm procedures do not use the ordering visible in the table (the recurrence rate falls steadily as more stomach is removed); a prespecified trend test does.

10.5.3. What the hierarchical reading does with it

```
f <- shrink_eb_fit(y, v); ds <- f$shrunk[1] - f$shrunk[4]; ss <-
 ↪  sqrt(f$sd[1]^2 + f$sd[4]^2)
ulcer_hier <- c(mu = f$mu, tau = f$tau, shrunk_contrast = ds, sd = ss,
 ↪  pr_drainage_worse = pnorm(ds / ss))
op <- c("drainage", "25%", "50%", "75%")
df <- data.frame(op = factor(rep(op, 2), op), kind = rep(c("raw",
 ↪  "hierarchical"), each = 4),
                 est = c(y, f$shrunk), half = 1.96 * c(sqrt(v), f$sd))
ggplot(df, aes(op, est, colour = kind, shape = kind)) +
 ↪  geom_pointrange(aes(ymin = est - half, ymax = est + half), position =
 ↪  position_dodge(0.4)) +
  geom_hline(yintercept = f$mu, linetype = 2, colour = "grey50") +
   ↪  scale_shape_manual(values = c(16, 1), name = NULL) +
  scale_colour_brewer(palette = "Dark2", name = NULL) + labs(x = NULL, y =
   ↪  "log odds of recurrence") + theme_minimal()
```

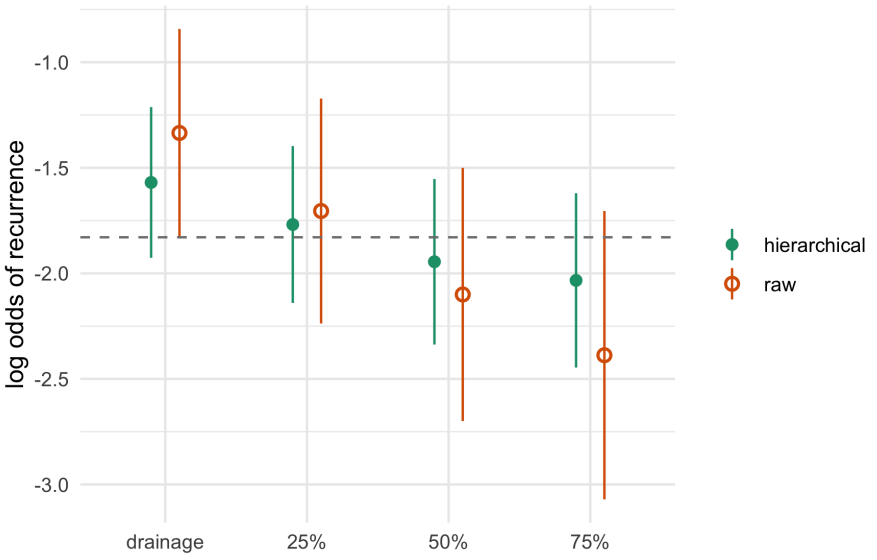


Figure 10.2.: Four operations for ulcer: raw log odds of recurrence (open circles) and hierarchical posterior means (filled), with 95% intervals. The arms with the least information are pulled toward the common mean.

The Bayesian analysis also pays for having four arms, but it pays in the estimate and not in the threshold. A hierarchical model across the four operations, as in Equation 10.3, with τ_g estimated by marginal likelihood, gives a between-arm standard deviation of 0.264 on the log odds scale, which shrinks each arm roughly half way to the common mean. The fit is `shrink_eb_fit()` in the package.

The best against worst contrast falls from 1.053 to 0.464 with posterior standard deviation 0.278, and the posterior probability that drainage and vagotomy is worse than 75 percent resection is 0.952. This is the number a clinician can use: the chance that the less extensive operation really has the higher recurrence rate. It was computed from the same data as the Bonferroni-adjusted p-value of 0.0855 and on a scale where it can be compared with the success criterion $\Pr(\delta > a \mid \text{data}) > c$ of the FDA Bayesian draft (Chapter 7).

Notice that the Bayesian answer is not permissive. The Bayesian point estimate of the difference is less than half the raw one. What differs is that the analysis reports where the evidence stands instead of discarding it, and that it reports the quantity a decision maker needs. A posterior probability of 0.952 is not a claim of statistical significance under strong familywise error control, and it should not be presented as one. Whether it would meet a prespecified threshold c is a question for the design, answered by simulation (Chapter 14).

The estimate of τ_g here rests on four arms, which is thin, and it is the sensitive part of the analysis. With four groups the data say little about τ_g and a plug-in estimate treats it as known. A fully Bayesian analysis would integrate uncertainty in the hierarchy's parameters; its interval and its sensitivity to the prior scale should be reported, and the sensitivity of the 0.952 to the prior on τ_g is the first thing a reviewer will ask about. This is part of the cost of the approach and is itemized in the side-by-side table below.

10.5.4. Structure that the correction cannot see

The four operations are ordered by extent of resection, so the scientific question is a monotone trend. A single prespecified contrast carrying a linear trend carries no multiplicity burden at all.

```
w <- c(-1.5, -0.5, 0.5, 1.5)                    # equally spaced scores for
 ↪  extent of resection
tr <- sum(w * y); se_tr <- sqrt(sum(w^2 * v)); z_tr <- tr / se_tr
pp <- shrink_post_prob(-tr, se_tr, 0.5)$prob    # prior sd 0.5 on the trend
lbf <- -0.5 * log(1 + 0.5^2 / se_tr^2) + 0.5 * (tr / se_tr^2)^2 / (1 /
 ↪  se_tr^2 + 1 / 0.5^2)
ulcer_trend <- c(trend = tr, se = se_tr, Z = z_tr, p_one_sided =
 ↪  pnorm(z_tr), post_prob = pp, log_BF = lbf)
```

The linear trend is -1.777 with standard error 0.676, so $Z = -2.63$ and the one-sided p-value is 0.0043. It is immune to the Bonferroni argument, since it is a single comparison chosen from the design and not from the data. Under a prior of scale 0.5 on the weighted sum (a prior standard deviation of 0.1 per step, which is a skeptical choice) the posterior probability that the prespecified linear trend contrast is negative is 0.941, with $\log \text{BF} = 1$. This is a statement about one weighted sum of log odds. It can hold even when adjacent recurrence rates reverse direction, so it is not the posterior probability that recurrence is monotone in extent of resection; that would need a joint model and the joint ordering event.

This result is available to a traditional analysis too, and it is the point of the section: a single prespecified trend test (or any contrast that encodes the ordering, such as the Williams or Bartholomew procedures used for ordered dose-response) is a traditional remedy and a good one. What the Bayesian framework adds is that the same structure can be built into the model, with a monotone dose-response or an ordered hierarchical prior on the arm effects, so that the final analysis uses the ordering and still reports the pairwise comparisons that a clinician wants. Dose finding (Chapter 17) takes this idea much further.

10.5.5. The two ways to pay for looking

Both paradigms must pay for examining four arms, and for examining them repeatedly. The standard design pays by raising the bar. Bonferroni divides the level, spending functions raise the early boundaries, and the normal-mixture e-value construction adds an approximately constant surcharge. The Bayesian design pays by moving the estimate: hierarchical shrinkage pulls each arm toward the common mean in proportion to how little information it carries. Multiplicity adjustment limits confirmatory claims; the estimates and intervals remain available. Shrinking the estimate uses the information in the comparison, and uses the other arms as well.

	Traditional (Holm, graph, fixed sequence)	Hierarchical Bayesian
What the protocol fixes	The family, the procedure and graph, the level α	The model, the prior on τ_g, the decision rule on a posterior probability
How looking at many things is paid for	A raised bar (smaller per-test level) or an ordered gate	A shrunken estimate, in proportion to the evidence in each arm
Error guarantee	Strong control of FWER by construction	By simulation under the prespecified design (Chapter 14)
Uses the relatedness of arms	Only through the correlation in some closed tests	Directly, through τ_g
Uses ordering of doses	Trend test or a gate; or not at all	Monotone or ordered prior
Main risk	Low power when many hypotheses are tested and effects are related	Over-shrinkage when one arm is truly different; sensitivity to the prior on τ_g
What the regulator asks	Is the graph prespecified and is the FWER controlled?	Is the exchangeability assumption plausible, what is the simulated FWER and power, how sensitive are results to τ_g?
Where they agree	With a flat prior, a normal likelihood and one comparison, the posterior probability of benefit is one minus the one-sided p-value, so the thresholds agree	

Where the traditional design is the better choice: a confirmatory trial with two or three prespecified endpoints and a sponsor who needs a clean familywise error statement for each claim. A fixed-sequence or graphical procedure delivers exactly that, is understood by every review division, and needs no modeling decision that the agency can dispute. The Bayesian hierarchical approach earns its place when there are many related groups (doses, subgroups, baskets, arms of a platform), when an estimate for each group is wanted and not only a decision on a family, and when borrowing across groups is in the sponsor's interest and the exchangeability assumption can be defended.

10.6. Across arms and across drugs: platforms

Multiple arms in one trial raise a question that a single-drug trial does not: what is the family? FDA's June 2026 revised draft guidance on master protocols takes a position that surprises people who learned multiplicity from the textbook. For comparisons of different drugs against a common control, the agency "generally does not recommend the use of multiplicity adjustments to strongly control the probability of making at least one type I error across the multiple comparisons of different drugs". The reason given is that the expected number of type I errors is the same whether those comparisons are made in independent trials or in one master protocol. A hypothetical sponsor of two unrelated drugs that run two separate trials does not adjust either one for the other, so there is no reason to adjust when they happen to share a protocol. The argument has a long history outside master protocols. Rothman (Rothman 1990) argued that no adjustments are needed for multiple comparisons at all, and Proschan and Waclawiw (Proschan and Waclawiw 2000) gave practical guidelines for when adjustment is warranted. The book's rule, here and in the rest of the chapter, is that the claim defines the family: adjust when one claim rests on the best of several comparisons. The different-drug comparisons described here can constitute separate families. Distinct endpoint claims for one drug do not, by themselves, justify dispensing with multiplicity control. The position is for unrelated drugs, often from different sponsors, and, in the draft's wording, generally also for comparisons of one drug in different diseases, conditions or disease subtypes when these represent distinct clinical objectives, as in separate basket substudies. It is not for multiple doses or regimens of one drug, which are alternative ways to support one claim and sit within the usual family, and it is not a licence to select subgroups or baskets and then combine them to support a single claim, which returns to the usual family. It is a draft. Chapter 20 develops the shared control, the non-concurrent control question and the way hierarchical models are used across arms and across subgroups in platform trials.

The consequences are two. First, in a platform with several drugs the multiplicity question becomes a question about which comparisons form a family, and that is decided by claims, as in the rest of this chapter. Second, shrinkage across drugs, where the model treats different drugs as exchangeable draws, is a different question from multiplicity adjustment, and it needs its own justification: unrelated drugs are not exchangeable in any scientific sense, and shrinking one drug's effect toward the average of others borrows from the wrong place. Hierarchical modeling is natural for doses of the same drug, for subgroups of the same population, and for baskets of tumor types with a common mechanism. It is not natural for a list of unrelated products. The hierarchical designs actually used for subgroups and baskets, the Thall and colleagues model for diseases with several subtypes, the Berry and colleagues subpopulation models and the exchangeability-nonexchangeability model, are described with their priors on the between-group variance and their simulated error rates in Section 19.13.

10.7. Seeing it in the lab

> Try it in the lab: module 04, many tests, false alarms
>
> Go to https://vsokolov.org/bayesian_trials_lab/04-multiplicity.html. Before you press anything, predict how many false alarms you will see in 1,000 trials of a useless drug with 5 tests at the unadjusted cutoff, then run it. Repeat with Bonferroni and with the fixed order, and compare the climb curve with Table 10.1. The package function `mult_simulate()` runs the same experiment in R.

> Try it in the lab: module 05, the ulcer trial
>
> Go to https://vsokolov.org/bayesian_trials_lab/05-ulcer-trial.html. Pick the best and worst treatments in the table, predict whether the chi-square test and the pairwise comparison agree, and then check. Note that the lab uses its own three-outcome table for the trial, so the numbers differ from Table 10.6; the lesson, that picking the extreme pair after the fact fools you far more often than the nominal level suggests, is the same.

10.8. In practice: multiplicity handled by modeling

Large platform trials ask many questions at once, and the two examples below are among the best documented cases in which the multiplicity is addressed in the statistical model and a high decision threshold, and not by a familywise adjustment. Neither claims strong familywise control, and neither should be read as showing that the Bayesian machinery of this chapter is a regulatory substitute for it.

10.8.1. I-SPY 2: ten signatures per drug

I-SPY 2, a neoadjuvant breast cancer platform, evaluates each experimental drug in ten prospectively defined biomarker signatures (all patients, HR+, HR-, HER2+, HER2-, MammaPrint high-2, and the four HER2 by HR combinations). A drug is therefore tested in ten overlapping subgroups, which is exactly the subgroup problem of Section 10.3. The design does not adjust the threshold ten ways. Estimates come from a covariate-adjusted logistic model for pathologic complete response (covariates HER2, HR and MammaPrint), and a drug graduates in a signature when its Bayesian predictive probability of success in a hypothetical 300-patient equally randomized phase 3 trial reaches 85% (J. W. Park et al. 2016). The threshold is set high on purpose, and the authors state that a drop between the interim and final predictive probability "was anticipated in the I-SPY 2 design and led to our setting a high threshold of 85%". A drug is dropped for futility only if its predictive probability is below 10% in all ten signatures.

Neratinib shows what this looks like. With 115 neratinib patients and 78 concurrent controls it graduated in one signature only, HER2+/HR-: estimated pCR 56% (95% probability interval 37 to 73%) against 33% (11 to 54%) for the trastuzumab control, probability of superiority 95%, final predictive probability of phase 3 success 79%. It did not graduate in HER2+/HR+ (30% against 17%, predictive probability 65%) or in all HER2+ patients (39%

against 23%, predictive probability 73%). The lesson for this chapter: the answer a clinician needs, in which of the ten signatures does the drug work, is reported as an estimate with an interval for each, and the selection of the best signature is guarded by a demanding decision rule and the shared model, with the operating characteristics established by simulation before the trial (Chapter 14). The graduation is a screening decision that routes a drug to a confirmatory trial, and not a claim of efficacy for labeling. The exact graduation rule is revisited in Chapter 20.

10.8.2. REMAP-CAP: hierarchical modeling across domains and arms

REMAP-CAP, the adaptive platform for severe pneumonia, evaluates interventions in several domains at once with a single Bayesian cumulative logistic model for organ support-free days. In its interleukin-6 receptor antagonist comparison, the model adjusts for site, age, sex and time epoch, uses neutral priors on the treatment effects, and nests tocilizumab and sarilumab under a shared hierarchical prior on an overall IL-6 antagonist effect, so each drug borrows from the other in proportion to how alike their data are (REMAP-CAP Investigators, Gordon, et al. 2021). The decision triggers are posterior probabilities: efficacy if the probability that the odds ratio exceeds 1.0 against control is above 99%, futility if the probability that it exceeds 1.2 is below 5%. The chronology and the effect estimates are in Chapter 20.

The two-drug hierarchy is the shrinkage of Section 10.4 in a small case: sarilumab, with few patients, is held nearer the common IL-6 effect than its own data would put it, and the 99% efficacy bar is the high threshold that stands in for a per-comparison adjustment. The lesson is the one in Section 10.6: the two drugs share a mechanism, so borrowing between them is defensible, whereas the domains evaluating unrelated treatments are not pooled in the same way. FDA's draft master protocol guidance does not generally recommend adjusting across different drugs, so the justification for the shrinkage must come from the science of the shared mechanism.

10.8.3. The FDA master protocol position

As set out in Section 10.6, the June 2026 revised draft on master protocols does not recommend strong familywise control across comparisons of different drugs. It does not remove the need to define the family for doses, regimens and subgroups that support a single claim (separate basket substudies with distinct objectives are different from selecting subgroups to support one claim), and it does not make a posterior probability a substitute for a calibrated error rate. I-SPY 2 and REMAP-CAP both fit within that position because their per-drug decisions are not claims of a familywise kind, but any labeling claim based on a signature or a pooled effect would return to the questions of the family and of simulated operating characteristics raised earlier in this chapter.

10.9. Reporting and review: what to expect

A reviewer who sees a multiplicity section will ask a short list of questions, and the answers should be written into the protocol and SAP before unblinding.

What is the family, and what claim does each hypothesis support? Is the testing procedure prespecified in enough detail that two statisticians would reach the same set of rejections from the same p-values? If a graph is used, does the SAP show the initial weights and transition weights, and has the graph been simulated for the planned scenarios? How do interim analyses for each endpoint consume the level? What is the status of the secondary endpoints if the primary fails? Which subgroups are prespecified, and is there an interaction test?

For a design that uses hierarchical shrinkage, add: what is the exchangeability assumption and why is it plausible; what prior is used for the between-group standard deviation τ_g and what happens to the decisions if that prior is changed; what is the simulated probability of a false claim over the family, across the relevant mixed null configurations (some groups null, others active), effect magnitudes and nuisance parameters, with a stated justification for the least favorable scenarios (hierarchical borrowing can make a mixed configuration worse than the global null); and what is the power for each group when effects are related and when only one group is active. These are the simulated operating characteristics that the FDA Bayesian draft and ICH E20 ask for, and they are the currency in which the saving from shrinkage is demonstrated. A hierarchical design that has not been calibrated in this way is an analysis plan for a regulator to doubt, however sensible the model.

Estimates matter as much as tests. A forest plot of subgroup results should show the shrunken estimates next to the raw ones when a hierarchical model is used, and should show intervals, not asterisks. The extreme subgroups are the ones that move most, and a reader can see at once which findings survive the correction for looking.

Key points

- The chance of at least one false positive among k independent tests at level α is $1-(1-\alpha)^k$; for $k=20$ at 0.025 it is about 0.40. The family is defined by the claims the trial may make, not by the number of tests run.
- Holm uniformly beats Bonferroni with the same dependence-free guarantee. Hochberg is more powerful but needs positive dependence. Closed testing is the principle behind both, and fixed-sequence and graphical procedures are closed tests with a clinically ordered structure that the protocol can draw.
- ICH E9 section 5.6 and the FDA multiple endpoints guidance (final, October 2022) ask for the strategy to be prespecified, for one key primary variable where possible, and for strong control of the familywise error rate on the claims. Hypotheses outside the family are exploratory.
- Subgroups and the forking paths hide multiplicity in analyst choices. Prespecify few, test for interaction, report intervals, and treat the rest as hypothesis generating.
- Hierarchical shrinkage pays for looking at many groups in the estimate, using the other groups as information; Bonferroni pays in the threshold and ignores them. Stein's result says the raw estimates are inadmissible under total squared-error loss for three or more normal means, which is a statement about that model and loss, not a guarantee for any fitted procedure.
- In the four-operation ulcer example the Bonferroni-adjusted p-value for best against worst is 0.0855, while the hierarchical posterior probability that drainage is worse than 75 percent resection is 0.952 on a contrast shrunk from 1.053 to 0.464; a prespecified monotone trend has $p=0.0043$ and needs no correction.

- Shrinkage does not give a familywise error guarantee. When a regulator requires one, calibrate the Bayesian rule by simulation and report the operating characteristics (Chapter 14).
- FDA's master protocols draft generally does not recommend multiplicity adjustment across different drugs in a platform, or across one drug's distinct clinical objectives; shrinkage across unrelated drugs is not justified by that position and is a separate modeling choice (Chapter 20).
- Prefer a single prespecified contrast that encodes the structure (ordered doses, a monotone trend) to a bag of pairwise tests.
- I-SPY 2 (ten signatures per drug, 85% predictive-probability graduation, neratinib graduating only in HER2+/HR-) and REMAP-CAP (hierarchical IL-6 antagonist effect, 99% efficacy trigger) handle multiplicity by modeling and a demanding threshold, not a familywise adjustment.

i Discussion questions

1. A trial has a primary endpoint and four key secondary endpoints. The team proposes testing the four secondaries at $\alpha/4$ each, regardless of the primary. Why might a fixed-sequence or graphical procedure be better for the claims the sponsor cares about, and what would the team lose by moving to it?
2. In the ulcer example the Bonferroni-adjusted p-value is 0.0855 and the hierarchical posterior probability is 0.952. A regulator asks which number the sponsor would put in the label. What would you say, and what additional work would you do to make the Bayesian number acceptable?
3. A subgroup analysis shows a large benefit in patients under 65 and none above. List the questions you would ask before believing the interaction, including the ones about how the cut point was chosen.
4. When is the exchangeability assumption behind partial pooling plausible across doses, subgroups, baskets and drugs, and what would you examine to check it? What happens to the shrunken estimate of the one truly active group when the others are inert?
5. FDA's master protocols draft does not generally recommend adjustment across different drugs on a platform. Why does that argument not extend to several doses of one drug tested against the same control, if the sponsor wants to claim any of them?

11. Interim Looks and the Three Deviation Scales

A trial that runs for years should not be forced to run its full course when the answer is already clear, or when it is clear that no answer is coming. Patients on the inferior arm deserve a way out, patients outside the trial deserve the result sooner, and sponsors deserve the chance to stop spending on a drug that will not work. Everyone who runs trials agrees on this. The difficulty is arithmetic: every look at accumulating data is another chance for chance to produce a convincing-looking result, and a trial that looks often enough will eventually see one.

This chapter has two halves. The first is the traditional apparatus that data monitoring committees (DMCs) use today: why repeated testing at a fixed level fails, the exact calculation of boundary crossing probabilities, the O'Brien-Fleming and Pocock boundaries, Lan-DeMets spending, futility rules, and how a committee actually works. We treat it fully, because it is the baseline against which any Bayesian monitoring plan is judged, and because it is good. The second half asks what changes when the same monitoring problem is posed as a decision. The posterior does not care how the trial was stopped, but the error rates a regulator reads do. We then put the whole subject on one scale, the "three deviation scales", and obtain three prices of evidence: a constant z_α^2 for a single look, a slowly growing $\log n + C$ for the Bayes rule, and, for the normal-mixture construction, approximately $\log n + C + 2\log(1/\alpha)$ for a rule that is valid at every stopping time (these are large-information approximations, with exact finite-information forms given below). The gap between the last two, measured against the uncalibrated Bayes-factor bar, is the seventy percent result that frames the asymptotic cost of that construction, and the SELECT reconstruction of Chapter 21 shows how much of it calibration removes.

! Regulatory anchor

ICH E9, Statistical Principles for Clinical Trials (1998, final), sections 3.4 and 4.5. Interim analyses are defined as any analysis intended to compare arms on efficacy or safety before formal completion. They "should be carefully planned in advance and described in the protocol" (section 4.5), and the statistical methods "should be fully specified in advance of the availability of information on treatment outcomes" (section 3.4). Group sequential designs are the usual vehicle, because assessing grouped outcomes at periodic intervals is more practicable than continuous review.

FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics (November 2019, final), section III. Looking more than once can inflate the type I error probability, "the most obvious examples" being cases in which multiple hypothesis tests are performed. The guidance gives a worked example of an interim look at 50 percent of the patients, and names four principles: error control, reliable estimation, full prespecification, and trial integrity. Group sequential designs are a standard adaptive

design under it.
ICH E20, Adaptive Designs for Clinical Trials (Step 2 draft, June 2025). Adaptations must be pre-specified in the protocol, and the type I error control must hold for the design as a whole. A draft, not binding.
FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (January 2026, draft), section IV.A. The draft describes three routes for Bayesian success criteria: calibration to a type I error rate (one-sided 0.025 typically), direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA), and decision-theoretic or benefit-risk criteria. Its sentence that criteria are "almost always" set to a familywise type I error of at most 0.025 concerns trials with a frequentist analysis plan. Calibration is this book's default for confirmatory adaptive designs without borrowing, and a Bayesian stopping rule has its operating characteristics shown by simulation (U.S. Food and Drug Administration 2026g). A draft recommends and does not bind.
FDA, Establishment and Operation of Clinical Trial Data Monitoring Committees (final guidance, March 2006). Describes when a DMC is appropriate, how it is constituted, and why access to unblinded interim comparisons must be restricted (section 4.2 on confidentiality of interim data). FDA issued a draft revision, *Use of Data Monitoring Committees in Clinical Trials*, in February 2024; a draft does not replace the 2006 text.
The map of these documents is in Chapter 3.

11.1. The traditional starting point: one look, one test

Take the familiar fixed-sample trial of Chapter 4 and Chapter 9. Patients are randomized, the primary endpoint is measured, and at the planned end a test statistic Z is compared with $z_\alpha = 1.96$ at one-sided $\alpha = 0.025$. The sample size was fixed by the power formula so that, if the true effect is θ, the probability of $Z > 1.96$ is $1 - \beta$. Under the null, Z is approximately standard normal and the chance of a false positive is 0.025 because the experiment has exactly one opportunity to produce one.

The notation of the book is that Z_n is the standardized statistic after n observations, I is Fisher information per patient (or per event for a time-to-event endpoint), and nI is total information. In a group sequential trial it is convenient to measure progress by the information fraction $t = nI/(N_{\max}I)$, the share of the planned total information that has accrued. For a trial with a time-to-event endpoint, t is the number of events observed divided by the number planned, and for a binary endpoint with equal arms it is the fraction of the planned patients with outcome data. The final analysis is at $t = 1$. Writing the interim analyses at $t_1 < t_2 < \cdots < t_K = 1$ is the whole of the design language needed in what follows.

11.2. Why repeated testing at a fixed level fails

Suppose a committee reviews the data K times and each time declares success if $Z > 1.96$. Under the null the K statistics are positively correlated but not identical: the statistic at the second look contains the first half of the data that produced the first, but also a new half. The chance that at least one of them exceeds 1.96 is larger than 0.025, and it increases

with K. This is the problem that P. Armitage, McPherson, and Rowe (1969) quantified, and the numbers that appear in textbooks are theirs (their tables are for two-sided 5% tests and are about twice the one-sided figures below).

We can see the inflation by simulation before deriving anything. Under the null the score S_n after n patients is a random walk with mean zero, and $Z_n = S_n/\sqrt{n}$. We simulate K equally spaced looks and record how often at least one exceeds 1.96.

Table 11.1.: Probability that a trial with no effect is declared positive at least once, when each of K equally spaced looks is tested at the one-sided nominal level 0.025. Simulation (200,000 trials, with Monte Carlo standard error below 0.001) next to the exact recursion of the next section (gs_power).

K	simulated	exact
1	0.0247	0.0250
2	0.0411	0.0416
3	0.0537	0.0536
4	0.0641	0.0631
5	0.0716	0.0709
10	0.0962	0.0968
50	0.1616	0.1619

Two equally spaced looks already take the false positive rate from 2.5% to about 4.2%, which is the kind of number the FDA guidance has in mind when it uses an interim look at 50 percent of patients to illustrate inflation. Five looks give about 7%, ten about 10%, and fifty about 16%. The increase slows but it does not stop. A committee that reviewed the data continuously and applied the fixed level would, with probability one, eventually see Z exceed 1.96, because a random walk standardized by $\sqrt{n}$ recurrently exceeds any constant (the law of the iterated logarithm). Applied repeatedly, the fixed cutoff is not a test at all. Every sequential method in use is a response to this table, and the methods differ only in what they charge for the right to look.

At fifty looks the exact figure is already 16.2 percent, and at 2000 equally spaced looks, a close stand-in for continuous monitoring, it is 30.5 percent. That last number is still not one: for a finite grid the maximum of the standardized statistic is bounded in probability, and it reaches certainty only as the grid is refined without limit, at a rate governed by the iterated logarithm (the iterated-logarithm discussion later in the chapter returns to that rate).

The inflation does not come from a flaw in the data. It comes from the fact that “was Z large at any of the looks” is a different event from “was Z large at the end”, and the null distribution that was computed for the second event is the wrong one for the first. This is also why the guidance is firm that the number and timing of looks must be written down before the data are seen. The rule that turns the data into a decision is part of the experiment, and the error rate belongs to the rule.

11.3. The Armitage recursion

What is needed is the distribution of the maximum of the standardized statistic at a finite number of times, and this has an exact numerical solution. Armitage, McPherson and

Rowe (P. Armitage, McPherson, and Rowe 1969) observed that the accumulated score is a Markov process: given the score at look k the next score is a normal increment, so the density of the score at look $k+1$ among trials that have not yet stopped is an integral of the density at look k against a normal kernel, truncated at the boundary. The probability of first crossing at look $k+1$ is a one-dimensional integral too. Everything is a sequence of numerical convolutions.

Let $S(t)$ be Brownian motion with drift ξ per unit of information fraction and variance t, so that $Z_k = S(t_k)/\sqrt{t_k}$ and $\xi = \theta\sqrt{N_{\max} I}$ is the expected value of the final statistic ($\xi = 0$ under the null). Let b_k be the boundary on the Z scale at look k and let $f_k(s)$ be the density of $S(t_k)$ on the event that the trial has not stopped at looks $1, \dots, k$. Then

$$
\begin{aligned}
f_1(s) &= \phi_{t_1}(s - \xi t_1)\, \mathbf{1}\{s < b_1\sqrt{t_1}\}, \\
f_{k+1}(s) &= \int_{-\infty}^{b_k\sqrt{t_k}} f_k(u)\, \phi_{\Delta_{k+1}}(s - u - \xi\Delta_{k+1})\, du, \qquad s < b_{k+1}\sqrt{t_{k+1}},
\end{aligned}
\tag{11.1}
$$

where $\Delta_{k+1} = t_{k+1} - t_k$ and ϕ_v is the $N(0, v)$ density. The probability of stopping for efficacy at look k is

$$
\begin{aligned}
p_1 &= 1 - \Phi(b_1 - \xi\sqrt{t_1}), \\
p_k &= \int_{-\infty}^{b_{k-1}\sqrt{t_{k-1}}} f_{k-1}(u)\, \Pr\!\left(u + \sqrt{\Delta_k}\,\varepsilon + \xi\Delta_k \geq b_k\sqrt{t_k}\right) du \quad (k \geq 2).
\end{aligned}
\tag{11.2}
$$

Here $\varepsilon \sim N(0,1)$. The type I error of the design is $\sum_k p_k$ computed at $\xi = 0$, and its power is $\sum_k p_k$ computed at the ξ that corresponds to the alternative. A lower (futility) boundary enters by truncating the integral in the other direction and recording the mass that leaves through it.

The whole method fits in a few lines of numerical integration, and it is what `gs_crossing_prob` in the `bayestrials` package does (with Gauss-Legendre nodes, so that it is accurate to many digits). The call below gives the probability of crossing at each of two looks and the total, and the exact naive inflation for four looks.

```
t <- c(0.5, 1); bounds <- c(1.96, 1.96)
r <- gs_crossing_prob(bounds, t)          # crossing probability at each
 ↪ look, drift 0
knitr::kable(data.frame(quantity = c("Crossing at look 1", "Crossing at
 ↪ look 2", "Total, two looks", "Total, four equally spaced looks at
 ↪ 1.96"),
  probability = round(c(r$cross[1], r$cross[2], r$total,
   ↪ gs_naive_inflation(K = 4)), 4)))
```

quantity	probability
Crossing at look 1	0.0250
Crossing at look 2	0.0166
Total, two looks	0.0416
Total, four equally spaced looks at 1.96	0.0631

The total for $K = 2$ agrees with the exact column of Table 11.1, and four unadjusted looks at 1.96 give a type I error of 0.0631, more than two and a half times the nominal 0.025. The

same package solves for boundaries, spending-function designs and power, and does the work in the rest of the chapter.

> **i** Technical note: why the recursion is exact
>
> The recursion uses only two facts. The increments of Brownian motion over disjoint intervals are independent normal variables, and the standardized statistic at look k is a deterministic function of the score at look k. Hence the pair (look index, score) is a Markov chain, and the joint distribution of the first-passage time and score is obtained by the Chapman-Kolmogorov equation with absorption. There is no approximation in the recursion itself; the only error is numerical integration. Real trials depart from the Brownian model, because the statistic after n patients is only approximately $S_n/\sqrt{n}$ and because event-driven endpoints have information that is estimated rather than known. Jennison and Turnbull (2000) show that the canonical joint distribution holds asymptotically for a very wide class of endpoints (normal means, proportions, logrank and Cox statistics, repeated measures), and this is why one table of boundaries serves for all of them. The simulation report of Chapter 14 is where the departures are checked for a particular endpoint.

11.4. O'Brien-Fleming, Pocock, and the family between them

With the recursion in hand the design question becomes: which boundaries $b_1, \dots, b_K$ should be used so that the type I error is 0.025 in total? Infinitely many sequences work, and the design is the choice among them. Two shapes dominate practice, and both belong to the family of Wang and Tsiatis (1987), $b_k = c\, t_k^{\Delta - 1/2}$.

- Pocock (Pocock 1977) uses the same critical value at every look ($\Delta = 1/2$). Early stopping is easy, but the final critical value is well above 1.96.
- O'Brien-Fleming (O'Brien and Fleming 1979) uses a critical value proportional to $1/\sqrt{t_k}$ ($\Delta = 0$). Early looks are nearly impossible to cross, and the final critical value is close to the single-test value.

A third, informal rule uses a very small fixed nominal level, for example a two-sided $|Z| > 3$ at every interim look and the unadjusted level at the end (Haybittle 1971; R. Peto et al. 1976).

For the Pocock and O'Brien-Fleming shapes the constant c is the one number to be found, and it is found by solving $\sum_k p_k(c) = 0.025$ at $\xi = 0$ with the recursion. The cell below does that for four equally spaced looks.

Table 11.3.: Four-look group sequential boundaries on the Z scale at one-sided alpha = 0.025, and what they cost. 'Inflation' is the maximum sample size relative to a fixed design with 90% power; E(t) is the expected fraction of the planned information used, under the alternative for which power is 90% and under the null.

design	look1	look2	look3	look4	inflation	E_H1	E_H0
O'Brien-Fleming	4.049	2.863	2.337	2.024	1.022	0.751	0.997

Table 11.3.: Four-look group sequential boundaries on the Z scale at one-sided alpha = 0.025, and what they cost. 'Inflation' is the maximum sample size relative to a fixed design with 90% power; E(t) is the expected fraction of the planned information used, under the alternative for which power is 90% and under the null.

design	look1	look2	look3	look4	inflation	E_H1	E_H0
Pocock	2.361	2.361	2.361	2.361	1.183	0.590	0.989
Lan-DeMets OBF-type	4.333	2.963	2.359	2.014	1.018	0.763	0.997
Lan-DeMets Pocock-type	2.368	2.368	2.358	2.350	1.178	0.592	0.989

The O'Brien-Fleming row reproduces the boundaries usually quoted for four looks, $4.05, 2.86, 2.34, 2.02$. (Polson and Sokolov (2026b) print them as $4.05, 2.87, 2.34, 2.03$, from the rounded constant 2.026; the exact solution of the recursion has $c = 2.024$, and the third decimal is not something a protocol should hinge on.) The first look requires $Z > 4.05$, a one-sided p of about 2.6×10^{-5}, and the final look requires $Z > 2.02$, a p of about 0.021. Alpha is "distributed" across the looks in the sense that the earlier looks hold almost none of it, and the unadjusted 0.025 at the end becomes 0.021.

The Pocock row has the same value, 2.36, at every look. Early stopping is much easier (a one-sided p of 0.009 at the first of four looks), and the final look must clear 2.36 rather than 1.96. The price appears in the inflation column: to have 90 percent power when the alternative is true, the Pocock design needs about 18 percent more patients than the fixed design if the trial runs to the end, against about 2 percent for O'Brien-Fleming. The reward appears in $E(t)$: when the treatment works, Pocock stops on average at about 59 percent of the planned information and O'Brien-Fleming at about 75 percent. Under the null the two designs run almost to the end, because the null is hardly ever rejected and there is no futility rule yet.

This is the trade-off a clinical team faces in choosing between the two. The choice is not a statistical question alone. A cardiovascular outcomes trial with an event-driven analysis and a durable commercial case wants a final result that carries the weight of a fixed trial, so O'Brien-Fleming is almost universal. A trial in which an early answer is worth much more than a marginally larger sample, such as one in a serious acute illness, may choose a boundary closer to Pocock.

11.4.1. Other boundary families

The Wang-Tsiatis family is not the only one. Armitage's sequential plans (Peter Armitage 1954) were among the first proposed for medicine: two boundaries on a plot of the accumulated treatment difference against the accumulated information. The closed (restricted) plans, which end at a maximum sample size, are in Armitage's later work (Peter Armitage 1960). Whitehead's triangular test (Whitehead and Stratton 1983) is a closed alternative to Wald's test of Section 11.8.4: the boundaries converge to a point at the maximum information, so the trial always ends with a decision, and the expected sample size under the design alternative is well below that of a fixed design. EOLIA (Section 11.12) is the book's real trial that used it.

Pampallona and Tsiatis (Pampallona and Tsiatis 1994) added futility boundaries to the Wang-Tsiatis family, Emerson and Fleming (Emerson and Fleming 1989) gave symmetric designs

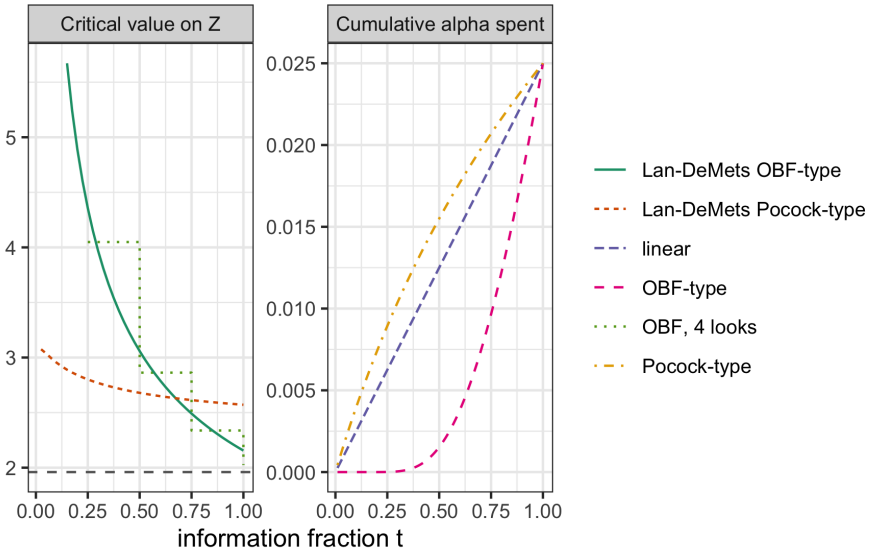


Figure 11.1.: Left: critical values on the Z scale against the information fraction for four designs with one-sided alpha 0.025 and a fixed 1.96 for reference (the dashed line is the unadjusted cutoff that the repeated-testing table shows to be invalid). Right: cumulative alpha spent as a function of the information fraction by the two Lan-DeMets spending functions. Lan-DeMets boundaries were computed at 40 looks to approximate continuous monitoring.

between the O'Brien-Fleming and Pocock shapes, and Kittelson and Emerson (Kittelson and Emerson 1999) unified these in one family with efficacy and futility boundaries. Every member is evaluated by the recursion above and compared by the same properties: the maximum sample size, the expected sample size under the null and the alternative, and the stopping probabilities at each look. A design is chosen by weighing those properties against the protocol's needs, not by ranking the families. Emerson, Kittelson and Gillen (Emerson, Kittelson, and Gillen 2007) set out this frequentist evaluation of group sequential designs (power, sample size and stopping probabilities across the family), and it is the same one the `bayestrials` package applies to the book's Bayesian rules, so that a Bayesian design is compared with these on the same terms.

11.5. Spending functions and the information fraction

The designs above assume that the looks are at exact multiples of one quarter of the information. Real trials are not like that. Enrollment is irregular, events arrive when they arrive, and a DMC meeting is scheduled for the first Tuesday of a quarter and finds 47 percent of the planned events, not 50. A boundary computed for a t that did not occur is wrong.

Lan and DeMets (1983; see also K. Kim and DeMets 1987 for the power family $\alpha^*(t) = \alpha t^\rho$) resolved this by changing what is fixed in advance. Instead of fixing the boundaries, fix a spending function $\alpha^*(t)$, a nondecreasing function with $\alpha^*(0) = 0$ and $\alpha^*(1) = \alpha$, which says how much of the total type I error may have been used up by the time the information

fraction reaches t. At the look with information fraction t_k the boundary b_k is the one for which the cumulative probability of crossing so far equals $\alpha^*(t_k)$. Two spending functions mimic the classical designs (the right panel of Figure 11.1):

$$\alpha^*_{\mathrm{OBF}}(t) = 2\{1 - \Phi(z_{\alpha/2}/\sqrt{t})\}, \qquad \alpha^*_{\mathrm{Pocock}}(t) = \alpha \log\{1 + (e-1)\,t\}. \tag{11.3}$$

The first spends almost nothing until t is large (at $t = 0.5$ it has spent about 0.0015 of the 0.025), and the second spends steadily. The rule is that the function is in the protocol, not the number or times of the looks, and the boundary is computed at each look from the realized information fraction, using the previous boundaries as they were actually applied. Because the recursion needs only the past boundaries and the current t, the calculation does not require knowing how many looks will occur. The only restriction is that the timing of looks must not depend on the observed treatment effect.

The calculation shows the point. Four looks planned at $t = 0.4, 0.6, 0.8, 1$ (the design used in module 07 of the book's companion online lab, https://vsokolov.org/bayesian_trials_lab/07-interim-looks.html) have an O'Brien-Fleming-type boundary at the second look of 2.68. If enrollment falls behind and the second look takes place at $t = 0.55$, the boundary at that look becomes 2.82, and the later boundaries move slightly. The total error is 0.025 and 0.025 in the two cases.

Two cautions belong with this flexibility. First, the information fraction at the final analysis is 1 only if the trial reaches its planned size; a trial that stops short, for example because event accrual is slower than planned, must decide in the protocol whether the remaining alpha is spent at the last look actually held. Second, the final analysis of a Lan-DeMets design is valid even if the interim looks turn out not to occur at the planned fractions, but the *sample size* was chosen for a particular plan, and the power calculation assumed a particular boundary.

> **Alpha is distributed, not consumed**
>
> Interim looks are widely believed to "cost alpha". They do not in total: the trial still has one-sided 0.025 for the whole experiment, and the spending function says where in the trial it lives. A final critical value of 2.02 instead of 1.96 records where the rest of the alpha went. What requires adjustment is the *action* taken on the interim data. A stop for superiority spends alpha. A stop for futility does not, and, if binding, it gives some back. A DMC that reviews safety without an efficacy comparison does not touch alpha at all.

11.6. Futility boundaries: binding and non-binding

An efficacy boundary is the upper edge of the continuation region. A futility boundary is the lower edge: if the interim statistic is below it, the trial is stopped because the treatment is unlikely to show benefit. Futility stops reduce the chance of a false positive (a trial stopped for futility cannot later cross the upper boundary), and they shorten trials of useless drugs. They differ in what the protocol says about whether the DMC *must* obey them.

- A binding futility boundary is part of the design. If it is crossed the trial stops, and the type I error is computed on the assumption that the trial does. The reduction in type I error can then be used to lower the efficacy boundaries a little.

- A non-binding futility boundary is a guideline. The DMC may recommend stopping, or may decline to, and the type I error is computed as though the futility boundary did not exist. Because ignoring it can only increase the chance of crossing the efficacy boundary, the type I error is at most α whether the committee follows the guideline or not.

The conservative position of the traditional world, and what the regulatory documents generally encourage, is the non-binding rule, because a DMC's decision to continue a trial should never be what invalidates its type I error. The cost shows up in the power calculation, which should be done as if the futility rule were followed (otherwise the stated power is not what the trial would achieve if it were). The cell below uses a simple rule on a four-look O'Brien-Fleming design: stop for futility at looks 1, 2, 3 if the standardized statistic is below zero.

Table 11.4.: Effect of a futility boundary (stop if Z < 0 at looks 1 to 3) on an O'Brien-Fleming design with four equally spaced looks. 'Binding' lowers the efficacy boundaries to restore alpha = 0.025 assuming the rule is obeyed; 'non-binding' keeps the efficacy boundaries.

design	type1_if_obeyed	type1_if_ignored	efficacy_bounds	power	E_H1	E_H0
no futility rule	0.0250	0.0250	4.05, 2.86, 2.34, 2.02	0.900	0.751	0.997
non-binding	0.0229	0.0250	4.05, 2.86, 2.34, 2.02	0.869	0.713	0.544
binding	0.0250	0.0274	3.98, 2.81, 2.3, 1.99	0.874	0.705	0.543

Under the null, the futility rule stops the trial early about 69 percent of the time (50 percent at the first look alone), and the expected information used falls from 100 percent of the planned total to about 54 percent. Under the alternative the rule costs 3 percentage points of power in this example, the difference between 0.90 and 0.87, because some trials with a true benefit have a negative interim statistic by chance. A binding rule recovers about 0.5 of a percentage point of that loss by lowering the efficacy boundaries (to $3.98, 2.81, 2.30, 1.99$). It does so at the price that, if the committee overrides it, the type I error is no longer controlled, which is why many sponsors and most regulators prefer non-binding rules for confirmatory trials. A threshold of "$Z < 0$" is crude. In practice the futility criterion is usually a conditional power or a predictive probability, which is the subject of Chapter 15. Conditional power, introduced for trials as stochastic curtailment by Lan, Simon and Halperin (Lan, Simon, and Halperin 1982), is defined and compared with the predictive probability in Section 15.1. Two real trials that stopped for futility, EOLIA (under a triangular sequential design) and the critically ill cohort of the heparin multiplatform trial (under a Bayesian rule), are described in Section 11.12.

11.7. What a DMC does

A DMC (the older terms are DSMB, data safety monitoring board, and IDMC) is a small independent group, typically three to seven members, consisting of clinicians in the therapeutic area and at least one statistician, which reviews accumulating data during

the trial and recommends to the sponsor whether to continue, modify, or stop. Ellenberg, Fleming and DeMets (2019) is the standard book, and FDA's 2006 guidance (U.S. Food and Drug Administration 2006), and its February 2024 draft revision, not final (U.S. Food and Drug Administration 2024e). The practice that matters for the statistical design is as follows.

Separation of roles. The DMC sees unblinded comparative data. The sponsor and the investigators do not. An independent statistical group (not the trial's analysis team) prepares the closed reports for the DMC, and its work is walled off from the rest of the sponsor. The FDA guidance and the adaptive designs guidance both stress that access to interim comparative results by people who can influence the trial's conduct compromises its integrity, and in a confirmatory trial a compromise of that kind can make the result uninterpretable. The 2019 guidance names trial integrity as the fourth principle for adaptive designs.

The charter. Before the first patient is enrolled, the DMC charter fixes the committee's membership, meeting schedule, the content of the open and closed reports, the stopping guidelines, and the communication path to the sponsor. The stopping guidelines are the boundaries of this chapter: efficacy (for example, an O'Brien-Fleming-type Lan-DeMets spending function at the planned information fractions), futility, and safety. For safety the boundaries are usually not formal tests of an alpha-controlled hypothesis, and are typically reviewed with different weight.

Boundaries are guidelines. A DMC is not an automaton (the stop for harm of the Women's Health Initiative estrogen plus progestin trial, in Section 11.12, was a committee recommendation). A boundary crossed by a small margin at an early look, with consistency across secondary endpoints and no safety signal, will often be treated differently from the same crossing with an inconsistent picture. The reverse also holds: a DMC may recommend stopping for harm without a boundary having been crossed. Discretion does not by itself inflate the efficacy error. Declining to stop after an efficacy crossing, or stopping for safety, leaves the type I error at most α as long as a claim of success remains restricted to crossings of the prespecified boundary. What can damage the guarantee is a decision that uses comparative data outside the rule, and conduct that compromises integrity. The type I error is a statement about the rule as written, and the discretion the DMC retains is a reason that the boundaries are stringent early and permissive late. A stringent early boundary means that a DMC's judgment will be exercised in few cases, at times when the evidence is overwhelming.

Why early stops overestimate. A trial that stops at an early look because it crossed a boundary stopped because the observed effect was large, and a large observed effect is more likely to be a positive deviation from the truth. The point estimate at an early stop is biased away from the null, an effect documented by Pocock and Hughes (1989). The bias is bigger the smaller the trial and the earlier the stop. Adjusted estimators exist, but the practical message is to read an early-stopped estimate with the sample size and the confidence interval in view, and to say so in the report. Module 07 of the companion lab makes the same point in its takeaways. The empirical size of the effect is documented in Section 11.12: in a meta-epidemiological study of 91 trials stopped early for benefit, the ratio of relative risks against matched trials that were not stopped was 0.71 (Bassler et al. 2010).

Confidentiality and what the sponsor learns. The sponsor learns only the recommendation. A recommendation to continue unchanged discloses little. A recommendation to stop discloses much, and a recommendation to modify, for example to drop an arm or enrich a population, can disclose a great deal. This is why a design in which the sponsor can see the

interim effect (for example, to resize the sample) needs the extra controls that Chapter 23 discusses.

Stopping boundaries in the 2019 guidance. The group sequential design with a prespecified spending function is the simplest case of an adaptive design in the sense of the 2019 guidance, and one for which the type I error is controlled by a known calculation without simulation. For this reason regulators treat it as well understood, and it is the standard against which a Bayesian monitoring plan will be compared: sponsors who propose a Bayesian monitoring rule are asked to show its type I error under the null and its power, by simulation, and the group sequential design is the benchmark that sets the expectation of "what good looks like".

In the protocol: interim analysis section (traditional)

Interim analyses. One interim analysis for efficacy and futility will be performed when 50% of the planned primary events have been observed, and a second at 75%. Efficacy will be assessed using a Lan-DeMets spending function of the O'Brien-Fleming type at overall one-sided $\alpha = 0.025$, with boundaries recomputed from the observed number of events at each analysis. A non-binding futility guideline of conditional power below 10% under the design alternative applies at the first analysis. The Independent Data Monitoring Committee will review unblinded data prepared by an independent statistician and will recommend continuation, modification or termination; it is not bound by these guidelines. The sponsor will not have access to comparative data before database lock.

11.8. The Bayesian sequential problem

We now pose the monitoring problem the Bayesian way. The question is not "how do we divide up α", but "what should be done at each look, given what we have seen and what stopping costs". Three observations frame the answer: the posterior is unaffected by the stopping rule, the error rate is not, and the stopping rule itself is a decision problem with a solution.

11.8.1. The posterior does not depend on the stopping rule

Let the trial stop at a random time T determined by a rule that depends only on the data so far. The likelihood of the observed data under a parameter θ is, for the observations that were made, the product of the densities of those observations times the probability that the stopping rule would have continued until T and stopped there. The second factor depends only on the data, not on θ, so it cancels in the posterior. The posterior after stopping at T with data $y_{1:T}$ is exactly the posterior one would have computed had the sample size been fixed at T in advance. This is the likelihood principle in its sequential form, the stopping rule principle (Edwards, Lindman, and Savage 1963; Cornfield 1966b; D. A. Berry 1987; J. O. Berger and Wolpert 1988), and it is the foundation of the Bayesian claim that "you may look as often as you like". The heparin multiplatform trial of Section 11.12 monitored on exactly this kind of posterior rule.

The same is not true of a frequentist p-value or a confidence interval, because these are tail probabilities over the *other* data sets that might have been seen, and which other data sets are possible depends on the stopping rule. That is why the rest of this chapter has needed a boundary table, and why a Bayesian analyst and a frequentist analyst looking at the same interim data will report different things. Armitage (P. Armitage, McPherson, and Rowe 1969) was the long-standing voice of the other side, holding that the logic of repeated significance tests applies to the trial and that a conclusion must account for how often the data were examined.

A small example makes the difference concrete. A trial ends with 15 responders among 20 patients, and the null response probability is one half. If 20 was the planned sample size, the one-sided p-value is 0.021. If instead the plan was to keep enrolling until the fifth non-responder, the same sequence of 20 patients ends the trial at the same place, but the p-value is now 0.01, because the sample space of other trials that might have been run is different. The likelihood is the same up to a constant, so under a flat prior the posterior is Beta(16, 6) in both cases and $\Pr(\theta > 0.5 \mid \text{data}) = 0.987$ (for the response probability θ). That one result is the heart of the dispute in the sources cited above: the frequentist says the stopping plan is part of the evidence, the Bayesian says it is not, and the clinician needs only to know which of the two the protocol and the regulator are using. For the calibrated designs used here, simulate type I error and power under the planned stopping rule. Other Bayesian success criteria are discussed in Chapter 3.

The simulation below shows both facts (Table 11.5). The true effect θ is drawn from a prior $N(0, 0.5^2)$, data are $N(\theta, 1)$ per patient, and the committee monitors after every patient to $n = 400$, stopping at the first time the posterior probability that $\theta > 0$ reaches 0.975.

Table 11.5.: A Bayesian monitoring rule (stop at the first patient at which Pr(theta > 0 | data) >= 0.975, at most 400 patients). Calibration under the prior: among trials that stop for benefit, the mean claimed posterior probability matches the fraction in which theta really is positive. Frequentist error under a point null: the same rule declares benefit far more often than 2.5%.

Quantity	Value
P(stop for benefit), prior-averaged	0.453
Mean claimed posterior probability at stop	0.981
Fraction with theta > 0 among stopped	0.979
P(declare benefit), point null theta = 0	0.175

Among the trials that stop, the claim made at the stopping time (the mean posterior probability, about 0.98) matches the fraction of those trials in which the effect is truly positive. Bayesian calibration under the prior survives optional stopping untouched. But if the null is exactly true, the same rule declares benefit in about 17.5% of trials: an error rate seven times the usual 2.5% for a rule whose posterior probabilities are all correct. Both statements are right and they answer different questions. The first is a statement averaged over the prior, the second is conditional on one value of θ, which is the way a type I error is defined, and the requirement of the error-calibrated approach used in this book (Chapter 3) is on the second. This is the central tension of the Bayesian approach to monitoring. Its frequentist error rate must be computed by simulation under the design, which is the work of Chapter 14, and, if the rule is to be a basis for approval, the threshold c is tuned until that error is acceptable. Note that the point null in the table is an extreme case: under a

continuous prior, an exactly-zero effect has no prior mass, and the first three rows are what the sponsor is protected by when the prior is a fair description of the program.

11.8.2. A short history of Bayesian monitoring

The Bayesian monitoring rule of this chapter has a lineage. Freedman, Lowe and Macaskill (L. S. Freedman, Lowe, and Macaskill 1984) proposed stopping rules that incorporate clinical opinion as a prior, and Freedman and Spiegelhalter (Laurence S. Freedman and Spiegelhalter 1989) compared Bayesian monitoring with group sequential methods. Spiegelhalter, Freedman and Parmar (Spiegelhalter, Freedman, and Parmar 1994) proposed the skeptical and enthusiastic reference priors of Section 8.3 and a range of equivalence, and a monitoring rule that stops when the posterior probability of benefit, or of exceeding that range, passes a threshold. Grossman and colleagues (Grossman et al. 1994) gave a unified method for monitoring and analyzing controlled trials, and Fayers, Ashby and Parmar (Fayers, Ashby, and Parmar 1997) wrote the tutorial. Berry (D. A. Berry 1985, 1987) argued from the likelihood principle that interim analyses should not be penalized, and Rosner and Berry (Rosner and Berry 1995) gave a Bayesian group sequential design for a multiple-arm trial. The rule of this chapter keeps their posterior and adds the calibration of its operating characteristics by simulation (Chapter 14) that a regulator asks for. The wider history of Bayesian trial design is in Section 1.9.

11.8.3. The Bayes sequential decision problem

Writing the stopping rule as part of the decision gives the Bayes sequential problem. A design is a pair (T, d): a stopping time $T \le N_{\max}$ adapted to the data and a terminal decision d made at T. Let $\ell(\theta, d)$ be the loss from the terminal decision (the cost of approving an ineffective drug, or of abandoning an effective one) and let each enrolled patient carry a cost $\kappa(\theta)$, so the total loss is $\ell(\theta, d) + \kappa(\theta)\, T$. The Bayes rule minimizes the expected total loss under the prior.

The solution is dynamic programming on the posterior. Because the posterior depends on the data only through a sufficient statistic (the count of successes for binary data, the score S_n for a normal mean), the state at look n is the pair (n, S_n). Let $V(n, S)$ be the minimal expected future loss in that state. At the horizon only the terminal decision remains. Before the horizon the committee compares stopping now with taking one more observation and acting optimally thereafter,

$$V(n, S) = \min\Big\{ \min_{d} \mathbb{E}[\ell(\theta, d) \mid n, S],\;\; \mathbb{E}[\kappa(\theta) \mid n, S] + \mathbb{E}_{\text{pred}}\big[V(n+1, S + X_{n+1})\big] \Big\}, \quad (11.4)$$

with $V(N_{\max}, S) = \min_d \mathbb{E}[\ell(\theta, d) \mid N_{\max}, S]$. This is the construction given by Sokolova, Sokolov and Polson (2026a). A decision tree over $N_{\max}$ steps has exponentially many branches, but indexed by the sufficient statistic it becomes a table over states, filled by backward induction. For a single-arm binary endpoint the states (n, S) form a lattice with order $N_{\max}^2$ entries. For a two-arm binary trial the state is a pair of success counts at each n, and the lattice is cubic in $N_{\max}$ (Chapter 15). For normal data the statistic is continuous and the recursion is evaluated on a numerical grid. The set of states (n, S) in which stopping is optimal is the stopping region, and its boundary is the Bayes boundary. Two properties follow in the examples of this book. The continuation region narrows toward the horizon, because the option value of another observation shrinks as the number of remaining observations does;

its exact shape depends on the loss and the sampling cost, and it need not be triangular. And early in the trial the boundary is wide, since a decision that is premature is expensive and the patient cost is small relative to what another look can buy.

The reduction to a sufficient statistic is what makes this computable in a trial setting, and it is the subject of Chapter 15, where the loss is replaced by the sponsor's operational criterion (predictive probability of success) and the backward induction is implemented. For this chapter the point is the contrast. The traditional design chooses the boundary to control a type I error and then asks what it costs. The Bayes design chooses the boundary to minimize an expected loss and then asks what error rates it has. For the error-calibrated approach used in this book, the second must be turned into the first, by simulation, before it can be a primary analysis. The FDA's draft Bayesian guidance of January 2026 describes calibration to a type I error rate as one of its routes, beside direct interpretation of posterior probabilities and decision-theoretic criteria (see Chapter 14).

11.8.4. Wald's test and the large deviation ledger

The oldest sequential procedure, and the reference point for everything above, is Wald's sequential probability ratio test (SPRT) (Wald 1945), which Barnard (Barnard 1946) developed independently. For the simple hypotheses H_0 and H_1 with densities p_0 and p_1, accumulate the log likelihood ratio $\Lambda_n = \sum_{i \le n} \log\{p_1(X_i)/p_0(X_i)\}$ and stop at the first time it leaves the interval $(\log B, \log A)$ with $A = (1-\beta)/\alpha$ and $B = \beta/(1-\alpha)$, accepting H_1 at the upper edge and H_0 at the lower. This is exactly the posterior odds rule of a Bayesian with a two-point prior, and Wald and Wolfowitz (1948) proved that among all tests with the same error probabilities it has the smallest expected sample size under both hypotheses.

For the normal mean with unit variance, $H_0 : \theta = 0$ against $H_1 : \theta = 0.2$, with $\alpha = 0.025$ and $\beta = 0.10$, a fixed-sample test needs $(z_\alpha + z_\beta)^2/\theta^2 = 263$ patients. Wald's approximation to the expected sample size under H_1 is

$$\mathbb{E}_1[T] \approx \frac{(1-\beta)\log\{(1-\beta)/\alpha\} + \beta\log\{\beta/(1-\alpha)\}}{\mathrm{KL}(P_1 \| P_0)}, \tag{11.5}$$

where $\mathrm{KL} = \theta^2/2 = 0.02$ per patient is the Kullback-Leibler divergence. The cell below evaluates it and checks it by simulation.

Table 11.6.: Wald's SPRT for a normal mean, theta=0.2 against 0, alpha=0.025, beta=0.10. Wald's approximation against 20,000 simulated trials; the fixed-sample size for the same error rates is 263. Simulation overshoots the boundary, which is why its error rates are slightly better than nominal and its sample sizes slightly higher than the approximation.

	fixed_n	wald_approx	simulated_mean	p95	rejects_H0
under H1	263	150	159	360	0.913
under H0	263	107	113	297	0.023

In the simulation the SPRT stops on average at about 159 patients when the drug works (a 40 percent saving against 263; Wald's approximation says 150) and at about 113 when it does not (approximation 107). Three features of Equation 11.5 define what we call the large

deviation ledger. The numerator, $\log\{(1-\beta)/\alpha\}$ and its companion, is a fixed quantity of evidence, measured in nats and set by the error levels alone, not by the size of the trial. The denominator is a Kullback-Leibler divergence, the rate at which evidence accrues, linearly in n and at a speed fixed by the alternative. And the governing relation $n\,\mathrm{KL} \asymp \log(1/\alpha)$ is the Chernoff-Stein relation of large deviation theory. All of it is a statement at exponential speed: to rule out a null with probability error α requires evidence $\log(1/\alpha)$, and evidence comes at rate KL.

Two cautions apply to the SPRT in a trial. Its sample size has no upper bound: the 95th percentile under H_1 is above the fixed-sample size, and a few trials run for several times the fixed size. A trial must have a maximum, which is why group sequential designs, with a hard end, are the standard. And it tests two simple hypotheses, whereas a trial's alternative is composite. The remedy is to place a prior on the effect and use the mixture likelihood ratio, which brings us back to the Bayes factor and to the three tolls.

11.9. The three deviation scales

11.9.1. Set up

Let I_n be the Fisher information about θ after n observations, $\hat{\theta}_n$ the efficient estimator, $Z_n = \hat{\theta}_n\sqrt{I_n}$ the standardized score, and $S_n = \hat{\theta}_n I_n$ the accumulated score, which behaves like Brownian motion with drift θ in information time. For a time-to-event endpoint with d events and $\theta = -\log \mathrm{HR}$, the usual approximation is $I_n = d/4$. Put a prior $\theta \sim N(0, \tau^2)$ on the alternative. The mixture Bayes factor for the alternative against the point null is

$$\mathrm{BF}_n = (1+I_n\tau^2)^{-1/2}\exp\Big\{\frac{Z_n^2}{2}\cdot\frac{I_n\tau^2}{1+I_n\tau^2}\Big\}, \qquad 2\log\mathrm{BF}_n = Z_n^2 - L_n - \frac{Z_n^2}{1+q}, \qquad q = I_n\tau^2, \tag{11.6}$$

where

$$L_n = \log(1+I_n\tau^2) \sim \log I_n \tag{11.7}$$

is the Ockham penalty of Chapter 7, the price for integrating the parameter out. The remainder $Z_n^2/(1+q)$ is exact, not $o(1)$ in general (under a fixed nonzero effect it converges to $-\theta^2/\tau^2$ in $2\log\mathrm{BF}_n - Z_n^2 + L_n$); it is negligible near the boundaries below, where $Z_n^2 = O(\log q)$. This is the finite-sample form of the Schwarz penalty (Schwarz 1978). Under the null $\mathbb{E}_0[\mathrm{BF}_n] = 1$ and run over time BF_n is a nonnegative martingale (a test martingale, which is the subject of Chapter 12), so that Ville's inequality (Ville 1939) gives $\Pr_0(\sup_n \mathrm{BF}_n \geq 1/\alpha) \leq \alpha$. The same statistic is therefore the Bayes factor of a Bayesian and the e-process of an anytime-valid analyst. They differ in where they put the bar.

11.9.2. The three tolls

Result: evidence tolls (Polson and Sokolov)

With L_n as in Equation 11.7 and $q = I_n\tau^2$, the three calibrations act, for a one-sided claim (so $Z_n > 0$), when

$$\begin{aligned} &\text{fixed level } \alpha: && Z_n \geq z_\alpha, \\ &\text{Bayes, threshold } b: && Z_n^2 \geq \left(1+\tfrac{1}{q}\right)\{\log(1+q) + 2\log b\} \approx L_n + 2\log b, \\ &\text{anytime valid } \alpha: && Z_n^2 \geq \left(1+\tfrac{1}{q}\right)\{\log(1+q) + 2\log(1/\alpha)\} \approx L_n + 2\log(1/\alpha). \end{aligned} \tag{11.8}$$

The middle and last forms are exact for the symmetric normal mixture, and the approximations on the right are large-information forms ($q \to \infty$). For this normal-mixture construction the anytime-valid rule requires $2\log(1/\alpha)(1 + 1/q)$ more, in squared standardized units, than the Bayes rule at $b = 1$, which is approximately the constant $2\log(1/\alpha) = 7.38$ at large information. These are properties of this construction and not universal costs of anytime validity.

Technical note: proof of the tolls

The first is the definition of the one-sided fixed-level test. For the second, the rule acts when $\mathrm{BF}_n \geq b$, and Equation 11.6 gives $2\log \mathrm{BF}_n = \{q/(1+q)\}Z_n^2 - \log(1+q)$, so the event is $Z_n^2 \geq (1+1/q)\{\log(1+q)+2\log b\}$, which is $Z_n^2 \geq L_n + 2\log b + o(1)$ as $q \to \infty$ (the factor $1 + 1/q$ multiplies a quantity that grows only logarithmically). Setting $b = 1/\alpha$ gives the third. The ratio of the two squared bars is $\{L_n + 2\log(1/\alpha)\}/L_n = 1 + 2\log(1/\alpha)/\log(1 + I_n\tau^2)$, which tends to one, but only as $1/\log n$: the *additive* excess never shrinks.

In words. A single fixed-level test asks for a score of 1.96 however much information there is, so that its squared bar is the constant $z_\alpha^2 = 3.84$. A Bayesian who wants posterior odds of b for the alternative needs a squared score that grows like $\log n$, because a bigger sample makes a precisely-zero null a sharper target against a diffuse alternative. And, in this normal-mixture construction, a rule that must stay valid however and whenever it is stopped pays approximately $2\log(1/\alpha)$ on top of the Bayesian bar at large information, with $2\log 40 = 7.38$ at $\alpha = 0.025$ (exactly $7.38(1 + 1/q)$).

Table 11.7.: The Ockham penalty L_n = log(1 + I tau^2) and the bars on Z at prior scale tau = 0.25, with alpha = 0.025. The exact bars of the symmetric normal mixture carry the factor (1 + 1/q), q = I tau^2; the columns labeled approx. are the large-information forms sqrt(L_n + 2 log b) and sqrt(L_n + 7.378), shown for comparison. The unadjusted bar is 1.96 throughout. Ratio is the ratio of the squared anytime-valid and Bayes bars.

Information	Ockham L	Fixed	Bayes (exact)	Anytime (exact)	Bayes (approx.)	Anytime (approx.)	Ratio
25	0.94	1.96	1.24	3.69	0.97	2.88	8.84
50	1.42	1.96	1.37	3.41	1.19	2.97	6.21
100	1.98	1.96	1.52	3.29	1.41	3.06	4.72
200	2.60	1.96	1.68	3.28	1.61	3.16	3.83

Table 11.7.: The Ockham penalty L_n = log(1 + I tau^2) and the bars on Z at prior scale tau = 0.25, with alpha = 0.025. The exact bars of the symmetric normal mixture carry the factor (1 + 1/q), q = I tau^2; the columns labeled approx. are the large-information forms sqrt(L_n + 2 log b) and sqrt(L_n + 7.378), shown for comparison. The unadjusted bar is 1.96 throughout. Ratio is the ratio of the squared anytime-valid and Bayes bars.

Information	Ockham L	Fixed	Bayes (exact)	Anytime (exact)	Bayes (approx.)	Anytime (approx.)	Ratio
400	3.26	1.96	1.84	3.33	1.81	3.26	3.26
1000	4.15	1.96	2.05	3.42	2.04	3.40	2.78
4000	5.53	1.96	2.36	3.60	2.35	3.59	2.34

Two things stand out. The Bayes bar with $b = 1$ is *below* the unadjusted bar at small information, because the Ockham penalty L_n is small while the information is small relative to the prior precision $1/\tau^2$, but it overtakes 1.96 once L_n exceeds $z_\alpha^2 = 3.84$, that is asymptotically for $I_n\tau^2 > 46$ (for a unit-information prior, $I_n\tau^2 = n$ and the condition is $n > 46$, while at $\tau = 0.25$ it needs $I_n > 730$, about 2,900 events); the exact bar of the stated mixture crosses 1.96 at $q \approx 41.6$. From there on it keeps rising. Realistic confirmatory trials need not be past this point. Past that point the *uncalibrated* Bayes-factor bar is above the unadjusted bar, and the Bayesian saving is *not* a saving against a single unadjusted look. What matters in practice is the comparison of calibrated rules, which Chapter 21 and Chapter 12 make for the SELECT reconstruction: a single unadjusted look would cross at 309 events on the mean path, while among simulated trials that declare efficacy the calibrated Bayesian rule stops at a median of 464 events (power 0.95, realized type I error 2.5%). This is the book's thesis on its sharpest edge: the saving comes from monitoring and from using prior information, not from a lower bar. Second, the exact anytime-valid bar for this construction is higher again, at 3.3 to 3.7 across the range, by a surcharge of $7.38(1 + 1/q)$ that is approximately constant in n and that dominates L_n over the whole range of information that trials reach.

11.9.3. Where the surcharge comes from

For this normal-mixture construction the approximate boundary surcharge is not a convention: it comes from fixing the bar at $\log(1/\alpha)$ in log-evidence, and the mechanism is the exponential bound of the preceding section. It is a cost of this construction and of this comparison, not a lower bound for all anytime-valid procedures. Realizable boundaries with iterated-logarithm rates exist (Robbins 1970) and behave differently.

i Technical note: the surcharge is the Chernoff toll

Let M_n be a nonnegative martingale with $M_0 = 1$. Ville's inequality $\Pr(\sup_n M_n \geq 1/\alpha) \leq \alpha$ is Markov's inequality applied to M_n at a stopping time. For the exponential martingale $M_n = \exp(\lambda S_n - n\lambda^2/2)$ the first passage of M_n to $1/\alpha$ is the first time the cumulative log likelihood ratio reaches $\log(1/\alpha)$. Under an alternative with per-observation Kullback-Leibler divergence KL, Wald's identity gives $\mathbb{E}[T_\alpha] = \{\log(1/\alpha)/\mathrm{KL}\}(1 + o(1))$. The bar $\log(1/\alpha)$ is fixed and evidence accrues

linearly at rate KL, so the governing relation is $n\,\mathrm{KL} \asymp \log(1/\alpha)$: the Chernoff-Stein relation, the large deviation ledger of the previous section.

The Bayes bar, by contrast, is not fixed. It is $L_n \sim \log I_n$, and the governing relation is

$$n\theta^2 \asymp \log n, \tag{11.9}$$

the *moderate deviation* relation. Under it the type I error induced by a Bayes rule at threshold one tends to zero polynomially at a single analysis at n, as $(n \log n)^{-1/2}$, rather than being pinned at a constant, a result of Polson and Sokolov (2026e). The three scales line up as follows.

Table 11.8.: The three deviation scales.

scale	bar on Z_n^2	information required to decide
central limit, standard design	constant z_α^2	$n\theta^2 \asymp (z_\alpha + z_\beta)^2$
moderate deviation, Bayes	$\log n + C$, grows slowly	$n\theta^2 \asymp \log n + C$
large deviation, anytime valid	$\log n + C + 2\log(1/\alpha)$	$n\theta^2 \asymp \log n + C + 2\log(1/\alpha)$

The sample size formula of Chapter 9 is the first row; the second row is the sample size equation of that chapter (Equation 9.10) with the power margin z_β removed, as in Equation 11.12 below, and the third is the price of being valid at every stopping time.

11.9.4. The seventy percent result

i Result: required information

With per-observation standardized effect θ, the information needed to act satisfies $n_{\mathrm{MD}}\theta^2 = \log n_{\mathrm{MD}} + C$ for the Bayes rule and $n_{\mathrm{LD}}\theta^2 = \log n_{\mathrm{LD}} + C + 2\log(1/\alpha)$ for the anytime-valid rule, so the excess is the constant

$$n_{\mathrm{LD}} - n_{\mathrm{MD}} \approx \frac{2\log(1/\alpha)}{\theta^2}, \tag{11.10}$$

and relative to the fixed-sample requirement $n_{\mathrm{fix}} = (z_\alpha + z_\beta)^2/\theta^2$,

$$\frac{n_{\mathrm{LD}} - n_{\mathrm{MD}}}{n_{\mathrm{fix}}} \approx \frac{2\log(1/\alpha)}{(z_\alpha + z_\beta)^2} = \frac{7.38}{10.51} \approx 0.70 \tag{11.11}$$

at $\alpha = 0.025$ and 90 percent power, a large-information approximation of the boundary terms. The baseline is the uncalibrated Bayes-factor bar (posterior odds of one). The figure compares boundary terms asymptotically for this normal-mixture construction and is not an exact sample-size identity: subtracting the two sample-size equations adds a positive $\log(n_{\mathrm{LD}}/n_{\mathrm{MD}})$ term. A calibrated Bayesian rule has already paid part of the surcharge, so calibration narrows the realized gap (see below).

For this normal-mixture construction the approximate surcharge amounts to 70 percent of an entire conventional trial, measured against the uncalibrated Bayes-factor bar (posterior odds of one). That is the asymptotic ledger result, and calibration narrows it: a Bayesian rule whose threshold has been raised until its type I error is 2.5 percent pays part of the surcharge itself. In the SELECT reconstruction of Chapter 12 (true hazard ratio 0.80, at most 1,270 events), among simulated trials that declare efficacy, the calibrated Bayesian rule stops at a median of 464 events (power 0.95) and the e-value rule at 612 (power 0.83, realized type I error 0.6%), which is 32 percent more events and not seventy percent of a trial; SELECT's planned two-look design stops at 817. Over all trials, counting those that never cross, the medians are 486 and 714. The result is stated per unit $1/\theta^2$ and so does not depend on the effect size: it says the surcharge is a fixed multiple of the conventional sample size. It sits beside two consequences. First, because the ratio in the technical note tends to one, it is sometimes read as saying the anytime-valid penalty is asymptotically negligible. It is not, in practice, because the convergence is in $1/\log n$ and the additive excess never shrinks. Second, none of this makes continuous monitoring a bad idea. It says what unrestricted monitoring costs for this construction, in the same currency as the group sequential designs of the first half: a design with four planned looks has a final bar of 2.02 and pays an inflation of 2 percent, and a design valid at every stopping time pays a surcharge that is a large fraction of a trial. A sponsor chooses how much flexibility to buy, and Chapter 12 takes up when it is worth buying all of it.

11.9.5. Four boundaries compared

The tolls say what each calibration asks for at a given information level. The same comparison can be made as a function of n on the Z scale, and it contains one surprise.

Table 11.9.: Four boundaries on the standardized score. Fixed level z_alpha; moderate deviation sqrt(log n); iterated-logarithm shape sqrt(2 log log n); and the Robbins normal-mixture boundary, a realizable anytime-valid boundary at alpha = 0.025 with unit mixing scale, in the form used by Polson and Sokolov.

n	fixed	sqrt_log_n	sqrt_2loglog_n	robbins
5e+01	1.96	1.98	1.65	3.40
1e+02	1.96	2.15	1.75	3.48
5e+02	1.96	2.49	1.91	3.69
1e+03	1.96	2.63	1.97	3.78
1e+04	1.96	3.03	2.11	4.07
1e+05	1.96	3.39	2.21	4.35
1e+06	1.96	3.72	2.29	4.60

The ordering is not the obvious one. The iterated-logarithm rate, which is the shape any boundary valid at all times must have, grows *more slowly* than the moderate-deviation Bayes boundary: the ratio is 1.34 at a thousand observations and 1.44 at ten thousand. So the cheapest conceivable anytime-valid boundary, in shape, grows more slowly than the Bayes risk-optimal boundary for a fixed sample. Yet the realizable anytime-valid boundary, the Robbins mixture, sits well above both, at 3.78 where the Bayes boundary is 2.63 at $n = 1000$. The gap between the iterated-logarithm column and the Robbins column is not merely the constant surcharge: the iterated logarithm supplies the *shape* that uniform validity requires,

while the level at which a particular α is attained, and the finite-n structure of this particular mixture boundary, supply the rest (Robbins 1970). Boundaries with iterated-logarithm rates and tuned constants (Robbins 1970) can sit between the two columns, so the gap is a property of the Robbins construction and not an unavoidable cost.

Three logarithms are easily run together. The $\log\log n$ of the iterated logarithm is the envelope of a random walk, a fact about paths that has nothing to do with a loss. The $\log n$ of the Ockham penalty is the price of integrating a parameter out of a likelihood. The $\log(1/\alpha)$ of Ville and Wald is a quantity of evidence fixed in advance by an error level. The first is geometry, the second is integration, and the third is a budget. They coincide in no regime, and the four boundaries differ because each is built from a different subset of them.

The group sequential boundaries of the first half fit into this picture as a finite-horizon compromise. At four looks they are valid at four times only, and pay correspondingly less: O'Brien-Fleming at the final look asks for 2.02, close to the fixed-sample 1.96, which is a long way below the 3.78 that the unrestricted Robbins boundary asks at $n = 1000$.

11.9.6. The sequential power calculation

The design question in the sequential problem is not the sample size but the distribution of the information at which the decision is reached. For a rule with boundary b_n on the Z scale and true effect θ, the decision occurs at the first n at which $\sqrt{nI}\,\theta + W_n/\sqrt{n} \geq b_n$ (the drift is signed, so for a beneficial treatment $\theta > 0$), where W_n is the noise. Setting the noise to zero gives the mean-path crossing approximation for $\theta > 0$,

$$n\,I\,\theta^2 = b_n^2, \tag{11.12}$$

the fixed-sample requirement of Chapter 9 with the power margin z_β removed. It is a deterministic approximation and not a median stopping time: the first-passage time of a noisy path depends on where monitoring starts and on the noise. For the Bayes boundary $b_n^2 = \log n$ this gives $nI\theta^2 = \log n$, to be compared with the fixed-sample requirement $(\sqrt{\log n} + z_\beta)^2$. The margin z_β, which the fixed design must buy in advance and which is worth $z_\beta^2 + 2z_\beta\sqrt{\log n}$ in squared units, is not bought at all on the mean path. Actual stopping-time summaries come from simulation, as below.

At a per-observation effect of 0.20 the fixed design with the Bayes boundary needs 342 observations and the mean-path crossing is at 120. The ratio of about 2.9 is the mean-path sequential dividend. A simulation of 10^4 paths monitored continuously from $n = 20$ (the starting point matters, because the bar $\sqrt{\log n}$ is low at small n) gives a median first crossing of 57 observations among the 100 percent of paths that cross within 600 observations. These numbers are for an uncalibrated Bayes boundary used as a monitoring rule, whose type I error is far above 2.5 percent, so they say how the boundary behaves and not what a calibrated design achieves; the calibrated comparison is the SELECT reconstruction of Chapter 12. The statement is about typical trials and not every trial finishes early, and the length of the tail is the other half of the sequential design: the SPRT table above shows a 95th percentile well above the mean.

11.10. Traditional and Bayesian monitoring side by side

Table 11.10.: What changes and what stays the same.

	Group sequential (traditional)	Bayesian sequential
What is fixed in the protocol	number and timing (or spending function) of looks, boundaries, futility guideline	decision rule, e.g. stop if $\Pr(\delta > a \mid \text{data}) > c_k$ at each look, with c_k possibly varying
Type I error	exact by the Armitage recursion, in the protocol	by simulation under the null (Chapter 14), threshold c tuned
Stopping rule and the analysis	affects p-values and confidence intervals (adjusted estimates needed)	does not affect the posterior; reported as is
Flexibility in the number of looks	Lan-DeMets: free, provided timing is independent of the effect	no restriction for the posterior; error control must hold for the realized schedule
Where they agree numerically	with a flat prior and one look, $\Pr(\theta > 0 \mid \text{data}) = \Phi(Z)$, one minus the one-sided p-value	the same quantity, now compared with the calibrated threshold c_k
Bar at the final analysis	2.02 (O'Brien-Fleming, four looks)	$\Phi^{-1}(c_k)\sqrt{1 + 1/(I_n\tau^2)}$, with c_k calibrated by simulation (asymptotically, a point-null Bayes factor at odds one has bar $\sqrt{L_n}$, above 1.96 once $I_n\tau^2 > 46$)
What the regulator asks about	pre-specification, spending function, DMC charter	pre-specification, simulation report, calibration of c, prior
Where it is the better choice	single or few planned looks, committed schedule, no relevant prior information	many looks, rapidly accruing data, informative prior, decision-theoretic stopping

What stays the same is more important than what changes. In both worlds the rule is written before the data, the error rates are demonstrated before approval, the DMC is walled off from the sponsor, and the boundary is a guideline rather than an automaton. What changes is the language in which the rule is stated (a posterior probability rather than a critical value) and the method by which its operating characteristics are obtained (simulation rather than a closed-form recursion). Where only a few looks at fixed fractions are planned and no prior information will be used, the traditional design is simple, transparent, and hard to improve on, and the Bayesian version offers no gain beyond a change of vocabulary. Where the looks are numerous, the schedule irregular, or the decision depends on more than a single test (a utility that trades benefit against harm, or a predictive criterion), the Bayesian formulation is the natural one, and the three-tolls analysis shows what the price of the flexibility is.

In the protocol: Bayesian interim rule

Interim analyses. After 100, 200, 300 and 400 patients per arm have been followed to the primary endpoint, the DMC will be given $\Pr(\theta > 0 \mid \text{data})$ computed under the prespecified prior $N(0, 0.5^2)$. Early stopping for efficacy is recommended if this exceeds the threshold c_k in the SAP's schedule of thresholds, where the thresholds were calibrated by simulation so that the one-sided type I error under the null is at most 0.025 in the scenarios of the simulation report (the values of c_k come from that calibration and are not tabulated here; the wording is schematic). Early stopping for futility is recommended, non-bindingly, if the predictive probability of success at the planned end is below 10%. The type I error and power table, and the code used to produce it, are in Appendix X of the statistical analysis plan.

Try it in the lab

Open Module 7, Peeking at the data. Before you press Run, predict the false-alarm rate for a useless drug tested at the usual cutoff with four looks, then run 100 trials with the raw cutoff and again with an O'Brien-Fleming or Pocock rule and compare with your prediction. The lab uses two-sided 0.05 and the 40, 60, 80 and 100 percent design of this chapter.

11.11. One outcomes trial under three rulebooks

A DMC reasons informally about how likely the treatment is to work, how likely it is to harm, and what the rest of the trial will show, while the protocol gives it a group sequential boundary. To see what the rules say at each look, take one cardiovascular outcomes trial, invented for teaching (it is not SELECT or any real trial, though its numbers are typical of the field). It targets a hazard ratio of 0.8 with analyses at 300, 600, 900 and 1200 events, the last being final. The observed hazard ratios are 0.78, 0.83, 0.84 and 0.85, an early promise that attenuates slowly. Three rulebooks are applied at every look. The first is the O'Brien-Fleming boundary at one-sided $\alpha = 0.025$. The second is a posterior-probability rule: stop for efficacy if $\Pr(\text{HR} < 1 \mid \text{data})$ exceeds a threshold, under a flat prior and under a skeptical normal prior $N(0, 0.1^2)$ on the log hazard ratio (which puts about 14.6 percent of its mass on a hazard ratio below 0.9). The third is an anytime-valid rule built on the Bayes factor of $\theta \sim N(0, 0.2^2)$ against $\theta = 0$. To these we add a fourth column, a calibrated Bayes rule: the skeptical posterior-probability rule with its threshold chosen so that the exact null probability of ever crossing at these four looks is 0.025.

Table 11.11.: One invented outcomes trial, four looks, four rules. z is the one-sided statistic for the log hazard ratio. OBF is the O'Brien-Fleming boundary on z. Flat and Skeptical are posterior probabilities of benefit, Pr(HR $<$ 1), under a flat prior and under N(0, 0.1^2) on the log hazard ratio. BF is the Bayes factor of N(0, 0.2^2) against the null, for which the anytime-valid bar is 40. Calibrated bound is the z-scale boundary of the skeptical posterior rule at the calibrated threshold.

Events	HR	z	OBF	Flat	Skeptical	BF	Calibrated bound
300	0.78	2.15	4.05	0.984	0.921	2.8	2.87
600	0.83	2.28	2.86	0.989	0.961	3.5	2.42
900	0.84	2.62	2.34	0.996	0.985	6.9	2.25
1200	0.85	2.81	2.02	0.998	0.993	10.7	2.17

At 900 events $z = 2.62$ crosses the O'Brien-Fleming bar of 2.34, and the traditional rulebook permits a stop for efficacy. The skeptical posterior probability of benefit is 0.985, the Bayes factor is 6.9 against a bar of 40, and the calibrated Bayes rule, whose bar on z is 2.25, also stops. The rulebooks disagree because they answer different questions. The boundary asks how often a trial monitored this way would cross by now if the drug did nothing. The skeptical posterior asks how sure one should be, given a reasonable starting doubt. The Bayes factor asks how strongly the data favor an effect in a form whose error guarantee survives any monitoring scheme, and that guarantee is bought at a high bar: by Ville's inequality, stopping the first time the Bayes factor exceeds 40 keeps the null probability of ever stopping below $1/40 = 0.025$, and since a stop for benefit also requires the estimate to point the right way ($Z > 0$), the one-sided error is at most 0.025 (in fact about half of it, by symmetry, so the rule is conservative for a one-sided claim). Without the $Z > 0$ condition the bar controls the two-sided event, which is not the claim a one-sided efficacy rule makes.

The thresholds in a posterior-probability rule are not self-evidently right. The skeptical rule of 0.99 that a protocol would naturally write has, over these four looks, an exact null probability of crossing of 0.0057: it is conservative, well below 0.025, and that is invisible from the posterior alone. Calibration by the exact recursion (no simulation is needed for a normal statistic) gives a skeptical threshold of 0.97, with z-scale bars 2.87, 2.42, 2.25, 2.17. Under a flat prior the calibrated threshold is 0.991, a constant bar on z and so a Pocock-like rule. The skeptical prior makes early stops harder and late stops easier than the flat one, and the calibrated skeptical rule sits between the O'Brien-Fleming and Pocock shapes. Calibration, not the choice of 0.99 or 0.975, is what turns a posterior probability into a rule whose error a regulator can read; this is the same point as Table 11.5, now at four looks.

Early stopping also attenuates in the opposite direction from what the table suggests: the trial that stops at 900 events reports a hazard ratio of 0.84 where the full trial would report 0.85, and the skeptical posterior median hazard ratio is 0.886 at that look (the posterior mean is 0.888, since the log hazard ratio is normal with variance $1/(I + 1/0.1^2)$ and $I = 900/4$). In real trials the gap is often larger (Section 11.12, Bassler and colleagues).

Harm and futility

The same machinery monitors for harm. A DMC might ask for $\Pr(\text{HR} > 1.2 \mid \text{data})$ on a safety endpoint under a flat prior and be prepared to recommend stopping if it exceeds a modest threshold such as 0.5, because the burden of proof for harm is lower than for benefit. This is a guideline for discussion, not an alpha-controlled test, and its false-alarm rate needs the same simulation report as any Bayesian rule. With a flat prior on the log hazard ratio the posterior is centered at the observed log hazard ratio, so the probability exceeds one half exactly when the observed hazard ratio exceeds 1.2, whatever the number of events: 0.5 at an observed 1.2 and 0.62 at 1.3 with 60 events. What the number of events changes is how quickly the probability moves away from one half. For futility the relevant quantity is the predictive probability of final success: at the second look the flat-prior predictive probability of crossing the final boundary of 2.02 is 0.89, so there is no case for futility. A real example of a monitored stop for harm after an excess of deaths is torcetrapib: Pfizer stopped the ILLUMINATE outcomes trial in December 2006 on the DMC's recommendation, and the signal (a rise in blood pressure) had been visible in earlier phases (see Chapter 2).

A reporting template for the DMC

The comparison suggests a short monitoring section that a sponsor can write and a regulator can accept.

1. State a group sequential efficacy boundary (or a calibrated Bayesian rule with its simulated or exact null error) that controls the type I error for regulatory purposes.
2. Report to the committee at each look the posterior probabilities of benefit and of clinically meaningful benefit under a flat and a skeptical prior, the posterior probability of harm, and the predictive probability of final success, each next to the boundary it is read against.
3. Report the anytime-valid Bayes factor with its direction condition ($Z > 0$) so that unscheduled looks prompted by safety concerns do not compromise the error guarantee.
4. State that the committee may decline to stop at a crossed boundary, and that it will explain its reasoning in the final report.

None of this requires a change in regulation. It requires that the committee be shown the quantities it uses to reason, instead of reconstructing them informally from a z statistic.

11.12. In practice

Four real trials show the chapter's ideas at work: what stopping early does to the estimate, a posterior probability used as a stopping rule, a futility stop under a group sequential design, and a stop for harm decided by a DMC.

11.12.1. Truncated trials overestimate: Bassler and colleagues (2010)

Bassler and colleagues (Bassler et al. 2010) matched 91 randomized trials that had been stopped early for benefit (asking 63 distinct questions) to 424 nontruncated trials from systematic reviews of the same questions. The pooled ratio of relative risks, truncated over nontruncated, was 0.71 (95% confidence interval 0.65 to 0.77): the pooled relative-risk

estimate in the truncated trials was on average 29 percent lower than in the matched nontruncated trials. The difference was independent of whether a statistical stopping rule was present and of methodological quality (allocation concealment, blinding), and it was greatest in smaller studies.

The size of the problem is debated. Montori and colleagues (Montori et al. 2005) documented the pattern in a systematic review of trials stopped early for benefit, and Freidlin and Korn (Freidlin and Korn 2009) argue that the bias is smaller than such reviews suggest when the boundaries are conservative.

The lesson is the one in the DMC section. A boundary controls the type I error, but it does nothing for the point estimate at the moment of stopping, and a design with an O'Brien-Fleming boundary is not immune. The regulatory consequence is practical: a trial stopped early should report its estimate with the information fraction and the interval in view, and the protocol should say how the estimate will be read. A shrinkage prior is the Bayesian answer, and it is the subject of Chapter 8.

The frequentist answer is a bias-adjusted estimator. The maximum likelihood estimate after a sequential test is biased, which Whitehead (Whitehead 1986) studied, and adjusted or median-unbiased estimators and intervals that respect the stopping rule are the standard remedy in group sequential software. The size of the effect is easy to see in a two-look design (Table 11.12).

Table 11.12.: A two-look O'Brien-Fleming design (one-sided 0.025, looks at half and full information, bounds 2.797 and 1.977) with true drift 2.5 on the Z scale. Estimates are on the drift scale, Z divided by the square root of the information fraction.

Quantity	Value
P(stop at the interim)	0.15
Mean estimate, stopped at the interim	4.69
Mean estimate, all trials that reject	3.23
Mean estimate, all trials	2.66

Trials that stop at the interim report an estimate far above the truth of 2.5, and the trials that reject at either look report one that is still too high, although the average over all trials, including those that never cross, is much closer to the truth (it is still a little high, because the interim stop freezes the estimate at its peak). The bias belongs to the selection that stopping implies, and Bayesian shrinkage can reduce upward selection bias, but its bias and interval coverage depend on the prior and should be evaluated under the stopping rule. Either way, the protocol should say which estimate will be reported at an early stop.

11.12.2. A posterior probability as the stopping rule: therapeutic heparin in Covid-19 (2021)

The ATTACC, ACTIV-4a and REMAP-CAP trials ran as one multiplatform design, comparing therapeutic-dose heparin with usual-care thromboprophylaxis in two cohorts, with organ support-free days to day 21 (death scored -1) as the outcome. Randomization continued within each cohort until a posterior conclusion of superiority (more than 99 percent

probability that the proportional odds ratio exceeded 1) or futility (more than 95 percent probability that it was below 1.2) was reached. The model was Bayesian cumulative logistic, with the two cohorts nested under a hierarchical prior with a neutral overall effect.

In critically ill patients enrollment stopped on 19 December 2020 at the futility criterion (REMAP-CAP Investigators, ACTIV-4a Investigators, et al. 2021). Of 1207 randomized, primary outcome data were available for 1098. The adjusted odds ratio was 0.83 (95% credible interval 0.67 to 1.03), with a posterior probability of futility of 99.9 percent. Note what futility means here: the interval barely includes 1, and the criterion is defined against an odds ratio of 1.2, not against no effect. In non-critically ill patients the trial was stopped when its prespecified criteria for superiority were met (ATTACC Investigators et al. 2021). Of 2244 randomized, 2219 were analyzed, the probability that the therapeutic dose increased organ support-free days was 98.6 percent, and the adjusted odds ratio was 1.27 (95% credible interval 1.03 to 1.58).

The same rule gave opposite stops in two cohorts, and neither is a frequentist boundary. The comparison with this chapter is direct: the posterior is valid at any stopping time, but the operating characteristics of the rule (the chance of a false stop for benefit or futility) have to be shown by simulation, as in Chapter 14. The group sequential alternative would have fixed the looks and the spending in advance.

11.12.3. A futility stop and its reanalysis: EOLIA (2018)

EOLIA randomized 249 patients with very severe acute respiratory distress syndrome to early extracorporeal membrane oxygenation or conventional management (Combes et al. 2018). Sixty-day mortality was 35 percent (44 of 124) against 46 percent (57 of 125), a relative risk of 0.76 (95% confidence interval 0.55 to 1.04), $P = 0.09$. The trial was stopped for futility under a prespecified sequential design with a two-sided triangular test (Whitehead and Stratton 1983), the design of Section 11.4.1.

Goligher and colleagues (Goligher et al. 2018) reanalyzed the trial post hoc to show the effect of the prior. The posterior probability of any mortality benefit (relative risk below 1) was 96 percent under a minimally informative prior, 99 percent under a moderately enthusiastic prior, 88 percent under a strongly skeptical prior, and 99 percent under a prior built from published studies. But the probability of the 20-point absolute reduction the trial was powered for was only 2 percent under the minimally informative prior and 4 percent under the meta-analytic prior.

The lesson is that a frequentist futility stop and a Bayesian summary answer different questions. The first says the design target was unlikely to be reached. The second says benefit of some size is probable. Both can be true, and a decision maker needs both, which is the argument of Chapter 7 for reporting the posterior at several thresholds.

11.12.4. A stop for harm: the Women's Health Initiative (2002)

The estrogen plus progestin trial of the Women's Health Initiative, in healthy postmenopausal women, was stopped early for harm, on the recommendation of its data monitoring committee, and the principal results were published in JAMA in July 2002 (Rossouw et al. 2002). At the board's recommendation on 31 May 2002, after a mean of 5.2 years of follow-up, the test statistic for invasive breast cancer had exceeded the stopping boundary for that adverse effect and the global index statistic supported risks exceeding benefits. The estimated hazard

ratios (nominal 95% confidence intervals) were 1.26 (1.00 to 1.59) for invasive breast cancer (290 cases), 1.29 (1.02 to 1.63) for coronary heart disease (286 cases), 1.41 (1.07 to 1.85) for stroke, and 1.15 (1.03 to 1.28) for the global index, while total mortality was not affected (0.98, 0.82 to 1.18). The nominal interval for breast cancer reaches 1.00: the stop rested on the crossing of a sequential boundary, not on that interval.

Safety and harm monitoring in a trial of this kind is not a formal alpha-controlled test of a single endpoint. The committee weighs a pattern of outcomes, and a stop for harm is a decision under uncertainty made with discretion, which is what the DMC section of this chapter describes. The charter, the guidance on DMCs and the stopping guidelines exist so that such a decision can be made by people who see unblinded data and are independent of the sponsor.

Key points

- Testing repeatedly at a fixed level is not a test: two equally spaced looks raise one-sided 2.5% to about 4.2%, ten to about 9.7%, and, with monitoring continued indefinitely, the fixed cutoff is eventually crossed with probability one (finite-horizon crossing probabilities remain below one). The Armitage-McPherson-Rowe recursion computes the exact inflation and every boundary in the traditional toolbox.
- O'Brien-Fleming boundaries ($4.05, 2.86, 2.34, 2.02$ for four looks) cost about 2 percent in sample size and keep the final bar near 1.96. Pocock boundaries (about 2.36 throughout) stop earlier when the drug works and cost about 18 percent more patients if the trial runs to the end.
- Lan-DeMets spending fixes a function, not times: the boundaries follow the realized information fraction, provided the timing is not driven by the observed effect. Alpha is distributed, not consumed.
- Futility rules should be non-binding in confirmatory trials; power is computed as though they are obeyed. Binding rules buy back a little alpha at the price of validity if overridden.
- DMCs see unblinded data, recommend rather than decide, and treat boundaries as guidelines. Early-stopped estimates are biased upward.
- The posterior does not depend on the stopping rule, but the type I error does. A Bayesian monitoring rule can be perfectly calibrated under the prior and still reject a true null in about 17.5% of trials: its frequentist error is computed by simulation (Chapter 14).
- The Bayes sequential problem is solved by backward induction on a sufficient statistic. Wald's SPRT is its two-point special case, with the large deviation ledger $n\,\mathrm{KL} \asymp \log(1/\alpha)$.
- The three tolls: $Z_n \geq z_\alpha$ (fixed level), $Z_n > 0$ and $Z_n^2 \geq (1 + 1/q)\{\log(1+q) + 2\log b\}$ (Bayes, approximately $\log n + C$ at large information) and the same with $b = 1/\alpha$ and $Z_n > 0$ (anytime valid for the normal mixture, approximately $\log n + C + 2\log(1/\alpha)$). The Bayes-factor bar is above the unadjusted bar once $I_n\tau^2 > 46$ ($n > 46$ for a unit-information prior), so the Bayesian saving is not against a single look. For the normal-mixture construction, against the uncalibrated Bayes-factor bar, the approximate surcharge is $7.38/10.51 \approx 70\%$ of a conventional trial asymptotically (a comparison of boundary terms, not an exact sample-size identity); calibration narrows it (SELECT reconstruction, among trials that declare efficacy: a median of 464 events for the calibrated Bayesian rule against 612 for the e-value rule, with powers 0.95 and 0.83).

Discussion questions

1. A sponsor proposes ten Bayesian looks, each stopping for efficacy if $\Pr(\theta > 0 \mid \text{data}) > 0.975$, and argues that the posterior "is not affected by looking". What would you ask for before accepting the design, and what do the numbers in Table 11.1 and Table 11.5 suggest the answer will be?
2. A cardiovascular trial plans an O'Brien-Fleming-type design with looks at 50 and 75 percent of events, and the second look takes place at 68 percent because event accrual was slow. What does a Lan-DeMets design do, what would a fixed-boundary design have to do, and what would you check about why the look moved?
3. The DMC sees a crossing of the efficacy boundary at the first look, with a favorable trend on the primary endpoint and an unfavorable trend on all-cause mortality. Should the boundary be binding? How would the charter have to be written to give the committee the discretion it needs without compromising the type I error?
4. A futility rule is binding in the protocol and the DMC declines to follow it. What has happened to the type I error, and what can the sponsor say to the regulator?
5. The three tolls say the normal-mixture rule valid at every stopping time carries an approximate surcharge of seventy percent of a conventional trial against the uncalibrated Bayes-factor bar, and 32 percent more events at the median than a calibrated Bayesian rule in SELECT (among trials that declare efficacy). Under what circumstances is that worth paying, and under what circumstances is a four-look group sequential design the better purchase?

12. Anytime-Valid Inference and E-values

A group sequential design is a promise made in advance. The protocol states how many looks there will be, at what information fractions, and which boundary applies at each; the type I error is then controlled because the analyst did exactly what the protocol said. Most of the time that is how a trial is run, and Chapter 11 showed that the machinery behind it is sound. But anyone who has sat on a data monitoring committee knows the promise is sometimes hard to keep. A competing trial reports early and the committee wants an extra look. A safety signal prompts an unscheduled unblinded review. The sponsor asks whether enrollment can be extended by 200 patients because the interim estimate is promising but the boundary was not crossed. Each of these is a look the protocol did not plan. For a classical fixed boundary schedule an unplanned look is outside the guarantee, although a spending-function design can accommodate a look whose timing does not depend on the accumulating comparative results.

This chapter introduces a family of methods whose guarantee does not depend on the schedule. They are called anytime-valid, and the central object is the e-value: a nonnegative number that can be read as the outcome of a fair bet against the null hypothesis, and that can be updated as data arrive. The idea has a long history (Ville 1939 (Ville 1939), Wald's sequential probability ratio test (Wald 1945), Robbins's work on boundaries that hold for every sample size (Robbins 1970)) and a recent revival (Shafer and Vovk 2019; Shafer 2021; Vovk and Wang 2021). We follow the treatment of Sokolova and Sokolov (A. O. Sokolova and Sokolov 2026) and the working papers of Polson and Sokolov (Polson and Sokolov 2026e, 2026c).

The chapter has a second purpose. The guarantee has a price in the construction studied here, and the price can be stated in one line: for the normal-mixture e-value, at large information the anytime-valid rule needs approximately $2\log(1/\alpha)$ more units of squared evidence than a Bayesian rule at posterior odds of one, which at $\alpha = 0.025$ is 7.38. Measured against that uncalibrated Bayes-factor bar it amounts to about seventy percent of a conventional trial, as a comparison of boundary terms and not an exact sample-size identity; calibration narrows the realized gap (in the SELECT reconstruction below, among simulated trials that declare efficacy, a calibrated Bayesian rule stops at a median of 464 events and the e-value rule at 612). This is a cost of one construction, not a universal cost of anytime validity. Whether a sponsor should pay that price depends on whether the trial needs validity at stopping times that the protocol cannot anticipate. Most trials do not, and the chapter ends with a concrete way to decide.

> **!** Regulatory anchor
>
> No FDA or ICH guidance names e-values, test martingales or anytime-valid inference as of November 2026. What the guidances do require is control of the type I error probability, and e-values are a way to meet that requirement. A sponsor proposing one is proposing a method, not following a template, and should expect to justify it in

a Complex Innovative Trial Design (CID) paired meeting or a comparable interaction (Chapter 3).

- **FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics, final guidance, November 2019** (U.S. Food and Drug Administration 2019a). Lists four principles for adaptive designs: "the chance of erroneous conclusions should be adequately controlled, estimation of treatment effects should be sufficiently reliable, details of the design should be completely prespecified, and trial integrity should be appropriately maintained" (Section III, PDF p11). It also warns that "adaptive features can inflate the Type I error probability of a trial" (Section III.A, PDF p11).
- **ICH E20, Adaptive Designs for Clinical Trials, Step 2 draft, endorsed 25 June 2025** (International Council for Harmonisation 2025a). Calls "controlling the chances of false positive efficacy conclusions" "an essential element of regulatory decision-making" (Section 3.3, PDF p11) and defines a prospectively planned adaptation as one that is "pre-specified in the clinical trial protocol prior to initiation of the trial" (Section 1, PDF p6). A draft, not final.
- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products, draft guidance, January 2026** (U.S. Food and Drug Administration 2026g). The draft describes three routes for Bayesian success criteria: calibration to a type I error rate (one-sided 0.025 typically), direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA), and decision-theoretic or benefit-risk criteria (Section IV.A, PDF p10). Its phrase "almost always" describes trials with a frequentist analysis plan. A draft recommends and does not bind.
- **ICH E9 (1998), section 4.5**: the statistical principles under which interim analyses are to be planned and documented.

12.1. What a group sequential design does not cover

In Chapter 11 a one-sided test at $\alpha = 0.025$ was extended to K planned looks by spending the type I error over information fractions $t_1 < \cdots < t_K$. The O'Brien-Fleming and Pocock boundaries and the Lan-DeMets spending functions are different ways to do the spending. All of them share three properties. The number and approximate timing of looks are fixed before the data are seen (or the spending function is fixed and the realized looks are allowed to vary within it). The boundary at each look is computed from that plan. And the type I error guarantee is a statement about the plan: if the trial had been run to the plan, the probability of ever crossing a boundary under the null is at most α.

Four situations leave the plan behind.

1. **An unplanned look.** A Lan-DeMets spending function does accommodate looks at times that were not scheduled, provided the timing is not chosen on the basis of the observed effect. The question is whether the proviso holds: whether the request for the look used accumulating comparative results, and whether the spending or adaptation method remains valid for it. A look triggered by an interim signal in the trial's own data does not satisfy it.
2. **An external publication.** A different trial in the same disease reads out. The committee wants to know whether the accumulating data are now so convincing that

continuing is unethical, or so unpromising that it is pointless. Timing triggered entirely by independent external information need not depend on the trial's own observed treatment effect, and a Lan-DeMets spending function can accommodate such an additional look; the adaptive-design guidance also discusses acceptable changes based solely on external information. What has to be checked is whether the committee's request used accumulating comparative results, and whether the method keeps the error controlled if it did.

3. **Continuous review.** Some DMCs review unblinded safety and efficacy data continuously, or at a frequency far greater than the number of looks in the charter. A boundary designed for four looks says nothing about the 40th.
4. **Extending the trial.** The sample size was set from an assumed effect. At the interim the estimate is smaller, the trial is not futile, and the sponsor would like to continue to a larger sample. Doing so changes the sampling plan after the data were seen, which is precisely what the E20 and Adaptive Designs guidances treat as the source of inflated error, and which is why the standard solutions (combination tests, conditional error functions) are pre-specified in the protocol.

None of this is a criticism of group sequential designs. If the schedule is known and will be kept, a group sequential design is the more powerful choice, and we will see the numbers below. The point is that the type I error guarantee of a classical boundary is a property of the whole plan, whereas the guarantee we want in the four situations above is a property of the data path, whatever the monitoring did. A trial that ran its preplanned rule is the BNT162b2 vaccine trial, whose interim looks and thresholds were fixed in the statistical analysis plan, although one planned look was skipped and the realized case counts at the analyses differed from the listed ones (see the section on practice below).

The unplanned look is worth a picture before any theory. Suppose the drug does nothing and a naive analyst applies the ordinary one-sided z test at the 0.025 level to the accumulating data. How often is a false positive declared if the analyst is willing to look more than once?

With a single look the rate is 0.025 (Monte Carlo error aside), with five looks it is 0.071, with twenty it is 0.125, and with a look after every observation it is about 0.22. These are the Armitage, McPherson and Rowe numbers of Chapter 11 reproduced. The rate for the method of this chapter, monitored after every observation, is 0.006. How that is possible is the subject of the next section.

12.2. E-values as bets

12.2.1. The definition

An e-value for a null hypothesis H_0 is a nonnegative random variable E whose expectation under every distribution in H_0 is at most one,

$$\mathbb{E}_{H_0}[E] \leq 1.$$

An observed e-value of 20 means that the data have multiplied by 20 an amount of evidence that is fair under the null. The name suggests the natural reading: E is the payoff of a bet. A gambler stakes one dollar against the null (that is, on the drug working), the null being the claim that the bet is fair. If the null is true, the gambler's expected wealth afterwards is at

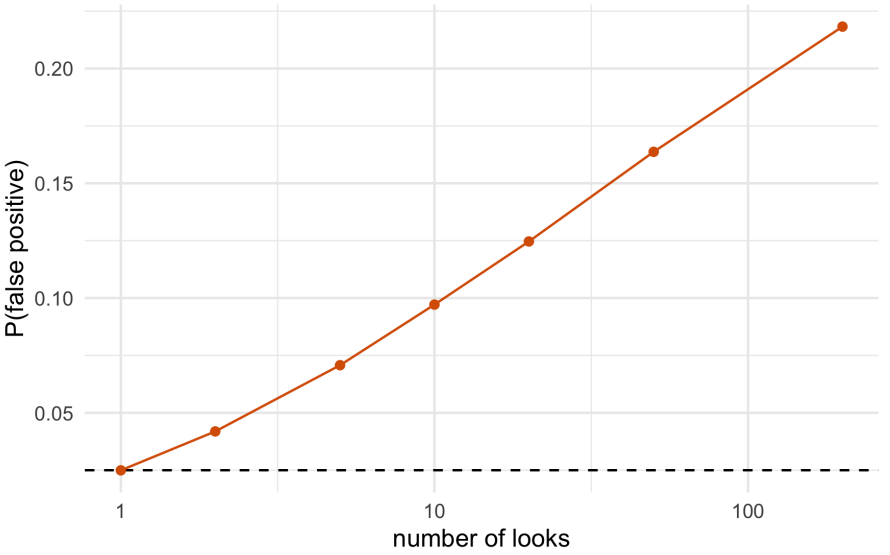


Figure 12.1.: Probability of at least one false positive when the drug does nothing, as a function of the number of equally spaced looks at 200 observations. The naive z test is applied at the 0.025 level at every look. The dashed line is 0.025.

most one dollar. If the gambler's wealth ends at $E = 20$, that outcome was an event which, under the null, has probability at most $1/20$. This follows from Markov's inequality,

$$\mathbb{P}_{H_0}(E \geq 1/\alpha) \leq \alpha\, \mathbb{E}_{H_0}[E] \leq \alpha.$$

So the rule "reject H_0 when $E \geq 1/\alpha$" is a level-α test. At $\alpha = 0.025$ the threshold is $E \geq 40$. A fixed-sample e-value is nothing more than a new way of writing a test. The new thing is what happens when the data keep coming.

12.2.2. Test martingales and Ville's inequality

Now let data arrive in a sequence, and let E_n be the gambler's wealth after n observations, starting from $E_0 = 1$. Suppose each bet is fair under the null: given the past, the expected wealth after the next observation equals the wealth now. A nonnegative process with this property and $E_0 = 1$ is called a test martingale (Shafer et al. 2011) (if the expected wealth can only fall, it is a test supermartingale, and everything below holds for it too). Ville's inequality, the mathematical heart of this chapter, says that such a process crosses a high level only rarely, and that this holds for the entire path,

$$\mathbb{P}_{H_0}\Big(\sup_n E_n \geq 1/\alpha\Big) \leq \alpha.$$

The word to notice is sup. For a fixed n, Markov's inequality gives the bound at that one time. Ville's inequality gives the same bound for the supremum over all times, which is the probability that the wealth is ever above $1/\alpha$. The gambler is allowed to look at the wealth after every patient, to pause and resume, to stop for any reason, including a reason the protocol never considered, and the probability of ever declaring success under the null is still at most α. This is what "anytime-valid" means. For likelihood ratios this bound is the

one Wald used for the error rates of his sequential probability ratio test (Wald 1945); Ville's version (Ville 1939) covers any nonnegative martingale.

i Technical note: why Ville's inequality holds

Let T be the first time $E_n \geq 1/\alpha$ (possibly infinite). The stopped process $E_{n\wedge T}$ is again a nonnegative martingale, so $\mathbb{E}[E_{n\wedge T}] = 1$ for every n. On the event $\{T \leq n\}$ the stopped wealth is at least $1/\alpha$, and it is nonnegative otherwise, so

$$1 = \mathbb{E}[E_{n\wedge T}] \geq \frac{1}{\alpha}\,\mathbb{P}(T \leq n).$$

Hence $\mathbb{P}(T \leq n) \leq \alpha$ for every n, and letting $n \to \infty$ gives $\mathbb{P}(T < \infty) \leq \alpha$. The argument is Markov's inequality applied at the stopping time, with the martingale property supplying the identity $\mathbb{E}[E_{n\wedge T}] = 1$. Note what is not used: the distribution of the data, the number of looks, and the reason for stopping. The same argument works for a test supermartingale, for which one has $\mathbb{E}[E_{n\wedge T}] \leq 1$.

12.2.3. A betting martingale for a binary endpoint

Take the two-arm binary trial of Sokolova and Sokolov (A. O. Sokolova and Sokolov 2026): patients are enrolled in matched pairs, one to treatment and one to control, and for pair i let $D_i = Y_{T,i} - Y_{C,i}$, which equals $+1$ if only the treated patient responded, -1 if only the control patient responded, and 0 otherwise. Under the null that the response probabilities are equal, $p_T = p_C = p$ for any common value p, we have $\mathbb{P}(D = +1) = \mathbb{P}(D = -1)$, so $\mathbb{E}[D] = 0$. Choose a betting fraction $\lambda \in (0, 1)$ and let

$$E_n = \prod_{i=1}^{n}(1 + \lambda D_i).$$

Then $\mathbb{E}[1 + \lambda D_i] = 1$ under the null whatever p is, so E_n is a test martingale for the composite null "the two arms have the same response probability". The nuisance parameter p never needs to be estimated. In the book's companion online lab (module 08, https://vsokolov.org/bayesian_trials_lab/08-e-values.html) the paths of this process under the null and under an alternative can be drawn directly. We reproduce the picture.

The left panel shows the feature that surprises people the first time. Under the null the *average* wealth stays at one dollar, yet the typical path drifts down: the average is held up by rare large excursions, and Ville's inequality says exactly how rare. The right panel shows what a working drug does: wealth grows, and the path that reaches 40 is declared a success.

With 20,000 simulated trials of 200 pairs the null rate of crossing 40 is 0.017, below the nominal 0.025 (we discuss why in a moment), and when the true rates are 45 and 30 percent about 74 percent of trials cross, with a median crossing at 104 pairs.

12.2.4. How big a bet? The growth rate

A bet that is fair under the null can still be a good or a bad bet under the alternative. If λ is small, wealth grows slowly; if it is large, the wealth wobbles and can collapse. For the

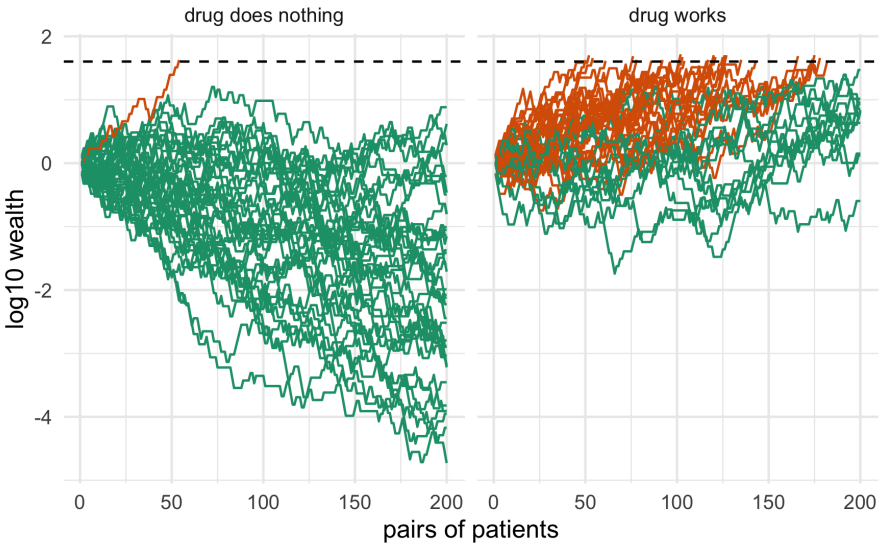


Figure 12.2.: Wealth (log scale) of the betting martingale with fraction 0.31 over 200 pairs, 40 paths per panel. Left: both arms respond at 30 percent, so the null is true and paths drift down. Right: 45 versus 30 percent. Paths are stopped when they reach the dashed line, $1/0.025 = 40$, and shown in color.

binary design above the expected log-growth per pair under the alternative is

$$g(\lambda) = \mathbb{P}(D = +1)\log(1+\lambda) + \mathbb{P}(D = -1)\log(1-\lambda),$$

and the expected number of pairs to reach $1/\alpha$ is roughly $\log(1/\alpha)/g(\lambda)$, which is Wald's identity from the sequential probability ratio test. In the 45 versus 30 percent design of Sokolova and Sokolov the best fraction is $\lambda^* = 0.312$ with $g = 0.0238$ nats per pair, giving an expected crossing near 155 pairs (A. O. Sokolova and Sokolov 2026). The trade-off is sharp: in their simulations a fraction of 0.10 gives power 0.119, 0.31 gives 0.723 and 0.50 gives 0.589 at 200 pairs. This is the e-value analogue of choosing a design effect size, and it has the same protocol implication: the bet is chosen in advance from the effect one hopes to detect, and an unrealistic choice costs power, not validity. The validity holds for every λ.

12.2.5. Cornfield and the betting odds of 1966

The idea is older than the vocabulary. Cornfield (1966a, 1969) proposed what he called relative betting odds as a sequential stopping criterion: the likelihood ratio of the alternative to the null, accumulated as the data arrive and averaged over a prior on the alternative. His companion argument was that the stopping rule should not change the strength of the evidence (Cornfield 1966b). For a simple null the betting odds are a test martingale in the sense of this chapter, so Ville's inequality bounds the probability that they ever cross a high level, a guarantee that does not need his argument. His proposal lost to frequentist group sequential boundaries (later alpha spending) for a reason that is not statistical. Betting odds require a declared prior on the alternative, and the reviewer has to accept it. The mixture e-value of this chapter makes the same trade, and the regulatory conversation about registering the prior (Chapter 3) is the modern version of that objection.

Shafer and colleagues (Shafer et al. 2011) connect test martingales, Bayes factors and p-values. Vovk and Wang (Vovk and Wang 2021) develop calibration and combination rules for e-values.

12.3. Likelihood-ratio, mixture and process e-values

12.3.1. Likelihood ratios

When the null and the alternative are each a single distribution, the likelihood ratio $\prod_i q(Y_i)/p_0(Y_i)$ is a test martingale: under p_0 the expected ratio of densities is $\int (q/p_0)p_0 = 1$. This is Wald's SPRT statistic, and for a simple alternative it is the best possible bet in the sense of growth. Its limitation is that it requires committing to one alternative. A bet on a hazard ratio of 0.70 pays badly if the truth is 0.90.

12.3.2. Mixtures

The standard remedy is to average over alternatives. If $L_n(\theta) = \prod_i p_\theta(Y_i)/p_0(Y_i)$ is a likelihood-ratio martingale for each θ in the alternative, then for any mixing distribution H

$$E_n = \int L_n(\theta)\, dH(\theta)$$

is again a test martingale, because an average of martingales with unit start is a martingale with unit start. For the Gaussian trial of Polson and Sokolov (Polson and Sokolov 2026e), let $X_i \sim N(\theta, 1)$ denote the standardized treatment-minus-control difference for pair i, $S_n = \sum_{i \le n} X_i$, and $H_0 : \theta \le 0$. Mixing $\exp(gS_n - ng^2/2)$ over a half-normal distribution for g with scale ρ gives, with $A_n = n + \rho^{-2}$, the closed form

$$E_n^{+} = \frac{2}{\rho\sqrt{A_n}} \exp\Big(\frac{S_n^2}{2A_n}\Big)\, \Phi\Big(\frac{S_n}{\sqrt{A_n}}\Big). \tag{12.1}$$

For $\theta < 0$ each integrand is a supermartingale (a martingale at $\theta = 0$), so Ville's inequality holds for the whole composite null. A mirror process, obtained by replacing S_n with $-S_n$, monitors the opposite one-sided null and can be used for futility.

The scale ρ plays the role of a prior standard deviation on the effect: it states how large an effect the bet is hedged against. It is a design parameter and not a nuisance. In the authors' sweep the integrated Bayes risk of the e-value rule changes by only a few percent across ρ from 0.10 to 0.32, so the choice matters less than the choice of threshold.

12.3.3. E-processes

Not every sequential statistic is a martingale. Often what one has is a process E_n that is dominated, under every null distribution, by some test martingale: an e-process. The definitions of this section are enough for the trials in this book, and the distinction matters mainly because it allows e-values to be built for composite nulls and with covariates in cases where no exact martingale exists. For the reader the working rule is the same: if $\sup_n E_n \ge 1/\alpha$ has probability at most α under every null, the process can be monitored continuously.

12.4. From e-values to p-values and confidence sequences

A regulator reads a p-value and a confidence interval. An e-process supplies both, in a form that survives continuous monitoring. The always-valid p-value after n observations is

$$p_n = \min\Big(1, \frac{1}{\max_{m \le n} E_m}\Big).$$

It is nonincreasing in n, it can only improve as evidence accrues, and $\mathbb{P}_{H_0}(\inf_n p_n \le \alpha) \le \alpha$: the probability that the always-valid p-value is *ever* below α is at most α. For a fixed-sample e-value the same relation, $p = \min(1, 1/E)$ (with $p = 1$ when $E = 0$), is a valid but conservative p-value by Markov's inequality, and the conservatism is part of what is paid. A running maximum of 15.3 gives $p = 0.065$, a running maximum of 45.8 gives 0.022. These are the numbers from the reanalysis of the Novick and Grizzle ulcer trial in Sokolova and Sokolov (A. O. Sokolova and Sokolov 2026), where the e-value of surgery A versus B is 15.3 and A versus C is 45.8 (these are the e-values of that paper's dataset, not computed from the ulcer data table of Chapter 10, and the patient order is simulated because the original order is not recorded; this is an illustration and not a valid test of that trial).

A confidence sequence is the interval analogue. It is a sequence of intervals C_n such that $\mathbb{P}(\theta \in C_n \text{ for all } n) \ge 1 - \alpha$, constructed by inverting the e-process over null values, and the idea goes back to the confidence sequences of Darling and Robbins (Darling and Robbins 1967; Robbins 1970), studied further by Lai (Tze Leung Lai 1976), and its group sequential counterpart in clinical trials is the repeated confidence interval of Jennison and Turnbull (Jennison and Turnbull 1984). A confidence sequence is wider than a fixed-sample confidence interval at the same n, and it can be read at any time, including at the time the DMC stopped the trial. The estimation principle in the Adaptive Designs guidance, that estimates be "sufficiently reliable", is what the confidence sequence is for.

12.5. The surcharge: what validity at every stopping time costs

The exact tolls and their derivation are in Chapter 11 (Equation 11.8, Table 11.7); this section only identifies the construction and the comparison baseline. The e-process rule at level α is the symmetric normal-mixture Bayes factor with $b = 1/\alpha$, with $q = I_n\tau^2$: the log of the symmetric normal-mixture e-value is $\frac{1}{2}\{qZ_n^2/(1+q) - \log(1+q)\}$, so for a one-sided claim ($Z_n > 0$) it acts when $Z_n^2 \ge (1 + 1/q)\{\log(1+q) + 2\log(1/\alpha)\}$, which exceeds the Bayes bar at $b = 1$ by $2\log(1/\alpha)(1 + 1/q)$, approximately $2\log 40 = 7.38$ at large information. The half-normal mixture of Equation 12.1 has the exact log e-value $\log 2 + \frac{1}{2}\{qZ_n^2/(1+q) - \log(1+q)\} + \log \Phi(Z_n\sqrt{q/(1+q)})$; at large information the correction reduces to $\log 2$, which lowers the bar by about $2\log 2 = 1.39$. The baseline is the uncalibrated Bayes-factor bar (posterior odds of one). The resulting figure of about seventy percent of a conventional trial (Equation 11.11) compares boundary terms asymptotically and is not an exact sample-size identity, and it is a cost of this construction, not of anytime validity in general (iterated-logarithm boundaries (Robbins 1970) behave differently). Calibration narrows it, as the SELECT comparison below shows. The figure is also not a statement that a sponsor who adopts an e-value design will run a trial 70 percent longer than a fixed design: stopping early at a large effect is built into the e-value design.

12.6. E-values and Bayes factors

The mixture e-value (Equation 12.1) looks like a Bayes factor, and it is one: it is the ratio of the marginal likelihood under a half-normal prior on the effect to the likelihood at the null value. The two ideas share their statistic and differ in how the statistic is read.

- A Bayes factor BF_{10} is read as a *measure of evidence*, on the scale of Jeffreys's table, or combined with prior odds to give a posterior probability. For a simple null, $\mathbb{E}_{H_0}[\mathrm{BF}_{10}] = 1$, so a Bayes factor *is* an e-value (Shafer et al. 2011; Shafer and Vovk 2019).
- An e-value is read as a *test statistic with a guaranteed error rate*: reject when it exceeds $1/\alpha$. The calibration is fixed by Markov's and Ville's inequalities and does not need a prior.

So the same number, say 40, is "strong evidence" on a Jeffreys scale and "a level 0.025 test" on the e-value scale. That is the sense in which the difference is one of calibration and not of statistic. A Bayes factor of 3 is a Bayes factor and an e-value of 3; it supports the alternative, but rejecting at 3 has a frequentist type I error as high as 1/3. The Bayesian reading is not wrong, it answers a different question, and Chapter 7 discussed which.

Two caveats. With a composite null, a Bayes factor with a prior on the null parameter is not an e-value in general, because its expectation is only an average under the null prior. And most e-values are not Bayes factors: the betting martingale of the binary trial multiplies $1 + \lambda D_i$, which is a likelihood ratio against no particular prior. The e-value framework is wider than Bayes factors, and the Bayes factor is the particular e-value that is calibrated for the prior used to mix it.

Polson and Sokolov (Polson and Sokolov 2026c) push the observation further. Their theory companion shows that the loss-optimal rejection boundary on the standardized scale grows like $\sqrt{\log n}$ (the moderate deviation scale), that the $1/\alpha$ rule is a fixed worst-case bound on a different (large deviation) scale, and that the Bayes factor is the e-variable calibrated at the first of these. The practical content is the next section: the statistic can be the same and the threshold different, and the threshold is where most of the cost lies.

12.7. Power and stopping time: e-value, Bayesian and group sequential designs

There are two comparisons, one on a binary endpoint (A. O. Sokolova and Sokolov 2026) and one on a time-to-event endpoint that reconstructs the SELECT trial of semaglutide (Polson and Sokolov 2026b). They produce different power orderings, and the difference is instructive. For SELECT there is one set of numbers in this book, those of the full reconstruction in Table 12.2.

12.7.1. Two-arm binary design

The design of Sokolova and Sokolov (A. O. Sokolova and Sokolov 2026) has $H_0 : p_T = p_C = 0.30$ against $H_1 : p_T = 0.45$, 1:1 randomization, at most 200 pairs, twenty looks every ten pairs, one-sided $\alpha = 0.025$, 50,000 replications. Table 12.1 reproduces their results. All figures are from their Table 2 and Table 5; we have not re-simulated them.

Table 12.1.: Binary two-arm trial, 20 looks at 10 pair intervals, from Sokolova and Sokolov (A. O. Sokolova and Sokolov 2026).

Method	Type I error	Power	Mean n under null	Mean n under alternative
E-value betting ($\lambda = 0.31$)	0.012	0.723	199.0	139.2
Group sequential, boundary $c/\sqrt{t}$, calibrated	0.025	0.861	198.9	139.8
Calibrated Bayesian posterior	0.020	0.688	197.1	133.9
Naive repeated p-values	0.148	0.933	178.4	74.2
Naive posterior threshold 0.975 (Jeffreys)	0.135	0.932	181.0	76.2

Three things stand out. The two naive rules are fast and powerful because their type I error is 14 percent: this is the repeated-look inflation of the first figure, and the table makes clear that a speed comparison means nothing unless the type I error is reported next to it. A naive posterior probability threshold of 0.975 under continuous monitoring is not safe either: the same inflation appears in the Bayesian column, a point we repeat in Chapter 14. Second, once each rule is calibrated, the group sequential design has the highest power (0.861), the e-value rule is next (0.723), and the calibrated Bayesian rule is last (0.688), stopping slightly earlier on average (133.9 against 139.2) but reaching its bar less often. Third, the e-value rule is conservative: its realized type I error is 0.012, about half the nominal level, because Ville's inequality is not tight: the wealth process overshoots the boundary when it crosses, so the bound is attained only in the limit of paths that stop exactly at the level, and a finite horizon leaves noncrossing some paths that would eventually have crossed.

Sokolova and Sokolov also vary the look schedule. With five planned looks the group sequential power is 0.870 against 0.686 for the e-value rule. With twenty looks it is 0.861 against 0.723. With five looks at random times it is 0.868 against 0.679. With continuous monitoring the e-value rule is at 0.750 and the group sequential design is not compared, because its boundary is not designed for 200 looks. The qualitative message is the one that matters: when the schedule is known and kept, the group sequential design is more powerful by 14 to 19 points of power in this design; the e-value rule keeps working when the schedule is not known.

12.7.2. Time-to-event design: the SELECT reconstruction

Polson and Sokolov (Polson and Sokolov 2026b) reconstruct SELECT, the cardiovascular outcomes trial of semaglutide (Lincoff et al. 2023), as a counterfactual: a Gaussian approximation in event time with a true hazard ratio held at the observed 0.80 (SELECT itself was designed for 0.83, at 90% power, with 1,225 planned events and one superiority interim at two thirds of them (D. H. Ryan et al. 2020)), a normal prior of scale $\tau = 0.25$ on the log hazard ratio, and a trial that ends at 1,270 events. Three rules are compared at a nominal 2.5 percent type I error under their own monitoring (the O'Brien-Fleming boundary and the Bayesian threshold are calibrated to it, and the e-value rule is bounded by it through Ville's inequality, with a lower realized rate): a four-look O'Brien-Fleming design at 25, 50, 75 and 100 percent of the events; the e-value mixture (the symmetric normal mixture, whose bar exceeds the Bayes bar by about the surcharge $2\log 40 = 7.38$), continuous, $E \geq 40$; and the continuous Bayesian posterior rule with a threshold raised until the type I error is 2.5 percent. For comparison the table includes SELECT's planned design: interim at 817 of 1,225 events, Lan-DeMets O'Brien-Fleming-type spending, then the final analysis, with nominal one-sided levels of about 0.00606 and 0.0231. Crossing a boundary is necessary but not sufficient for stopping: the DMC continued SELECT at its actual 831-event interim. We run the reconstruction with `select_simulate()` from the package, 40,000 simulated paths per hypothesis, and it is the only SELECT comparison used in this chapter; Chapter 21 returns to it and compares it with the authors' own script. The posterior bar that gives 2.5 percent is $p^* = 0.9952$, against a naive type I error of 11.8 percent if the posterior were simply required to exceed 0.975 under continuous monitoring.

Table 12.2.: SELECT reconstruction (true hazard ratio 0.80, a counterfactual; tau = 0.25; at most 1,270 events for the first three rules; 40,000 paths, seed 31415). Medians are given among trials that declare efficacy and over all trials, where a trial that never crosses is counted at its final read-out. Power is the probability of acting; realized type I is under the null at the rule's own monitoring.

Rule	Median events, declaring efficacy	Median events, all trials	Realized Power type I
O'Brien-Fleming, four scheduled looks	952	952	97.7% 2.55%
E-value mixture, continuous, $E \geq 40$	612	714	83.1% 0.58%
Bayes posterior, continuous, calibrated p^*	464	486	94.7% 2.50%
SELECT planned design (interim at 817 of 1,225 events, Lan-DeMets OBF-type spending, then final)	817	817	97.4% 2.54%

The ordering by stopping time is the clear result. Among simulated trials that declare efficacy, the median stopping point is 464 events for the Bayesian rule, 612 for the e-value rule and 952 for the four-look O'Brien-Fleming design, against 817 for SELECT's planned two-look design. Over all trials, counting those that never cross at the 1,270-event read-out, the medians are 486, 714 and 952. The powers sit beside them: 94.7 percent for the Bayesian rule, 83.1 for the e-value rule, 97.7 for the four-look design and 97.4 for SELECT's planned design. The Bayesian rule is calibrated to 2.5 percent, and the e-value rule is conservative, with a realized type I error of 0.58 percent against a nominal 2.5. The unadjusted single look at $z = 1.96$ would cross at about 309 events on the mean path (from $Z = |\log 0.80|\sqrt{d}/2$),

earlier than the calibrated Bayesian rule, which is the arithmetic that the thesis of this book requires us to acknowledge. Against the calibrated Bayesian rule the e-value rule needs 32 percent more events at the median among trials that declare efficacy. That is the realized price of validity at every stopping time once the Bayesian rule is calibrated, in this construction, and it is smaller than the asymptotic seventy percent measured against the uncalibrated Bayes-factor bar. The stopping times are events in a modeled counterfactual; any conversion to calendar time, patient-years or cost would inherit the same conditioning.

12.7.3. Why the orderings differ

The two comparisons give power orderings that appear to contradict each other. In the SELECT-like trial the calibrated Bayesian rule has more power than the e-value rule (94.7 against 83.1 percent in Table 12.2). In the binary trial it has less (68.8 against 72.3 percent). Both can hold, and the following are the plausible contributors, which neither Sokolova and Sokolov nor Polson and Sokolov isolates.

1. **Prior scale relative to the effect.** In SELECT the effect (a hazard ratio of 0.80) is of a plausible size relative to a prior with $\tau = 0.25$, so the calibrated Bayesian bar (the Z bar implied by p^*, which is above the uncalibrated $\sqrt{L_n}$) is low relative to the realized Z_n (about 4 at the end). In the binary design the effect is modest relative to the information, and the Bayes bar is not much below the e-value bar.
2. **Conservatism of the e-value.** The e-value rule's realized type I error is 0.58 percent in the SELECT design and 1.2 percent in the binary design, against a nominal 2.5. Some of the lost power is spent on that unused budget, and the amount depends on the design.
3. **The endpoint.** Time-to-event data accumulate on an information scale where L_n is modest (3.0 at 1,270 events with $\tau = 0.25$) and slowly growing, while the binary trial is small and its likelihood ratio has a different shape.

Polson and Sokolov (Polson and Sokolov 2026b) conclude that the robust finding across both studies concerns stopping time, not power, and that the direction of the power difference should be checked for the design at hand. We agree. In the language of Chapter 14, this is a statement that the operating characteristics have to be simulated for the specific design, and no theory replaces that report.

12.8. What a Bayes-risk comparison adds

The comparison so far is by type I error, power and stopping time, which are the currency of regulators. A different comparison asks what each rule costs in patients, and the working paper of Polson and Sokolov (Polson and Sokolov 2026e) makes it for a stylized mortality-type problem. Each pair is $N(\theta, 1)$ with a prior $\theta \sim N(0, 0.15^2)$, at most 1,000 pairs, a post-trial population of $M = 20{,}000$ patients, and a loss equal to $M|\theta|$ if the wrong terminal decision is made, plus $|\theta|$ per enrolled pair. The effect θ is a standardized effect, which is not an absolute mortality-risk difference, so the losses below are in illustrative loss units, not literal deaths; reading them as deaths would need a mortality scale and a conversion from the standardized effect. Relative reductions within the stipulated loss model are meaningful as they stand. The Bayes-optimal rule is computed by backward induction on the sufficient statistic (n, S_n) (Chapter 15). The e-value rule is $E_n^+ \geq 40$ with the mirror process for futility. We report their numbers, which we have not re-run.

Neither rule dominates. The risk functions cross. For marginally harmful treatments ($\theta \in [-0.05, -0.015]$) the e-value rule has the lower expected loss, because its bar of 40 almost never approves a harmful drug (approval probability at most 0.006, against 0.333 for the Bayes rule at $\theta = -0.015$). For beneficial treatments the Bayes rule is far better: at $\theta = 0.07$ the e-value rule approves 32.7 percent of the time against 98.2 percent, runs on average 849 pairs against 412, and has an expected loss of 1,001 against 55 loss units. The e-value rule is underpowered for effects of that size in a trial of 1,000 pairs.

Integrated Bayes risk. Averaged over the prior, the Bayes rule has an expected loss of 53.4 loss units and the e-value rule 237.6, a reduction of 77.5 percent (Monte Carlo standard errors 1.0 and 3.1). The reduction stays between 71 and 82 percent for horizons from 500 to 2,000 pairs, and widens as the stakes rise: 73.1 percent at $M = 10^4$ and 84.4 percent at $M = 10^5$. This is by construction, since the Bayes rule minimizes this criterion. It does not say that the Bayes rule is better by the criterion of a regulator.

Where the gap comes from. The gap decomposes into two parts. A rule that monitors the same e-process but stops at the loss-optimal threshold ($E^+ \geq 3.5$) instead of $E^+ \geq 40$ has Bayes risk 118.3, so *recalibrating the threshold* closes 119.3 of the 184.2 loss-unit gap (64.8 percent), and the remaining 35.2 percent is the value of the sampling-cost-aware continuation region. In the words of the authors, the e-process statistic is sound; it is the $1/\alpha$ calibration that carries most of the loss.

A caveat on the hybrid rule. The working paper recommends the loss-tuned rule as a default for a sponsor unwilling to give up validity. That recommendation needs care. By Ville's inequality a threshold of 3.5 bounds the type I error by $1/3.5 \approx 0.29$, so the tuned rule does *not* keep the 2.5 percent guarantee at its tuned threshold; what it keeps is the validity of the always-valid p-value computed from the same process, which can still be reported even if the trial is stopped on a different, loss-based criterion. A rule that keeps both the 2.5 percent guarantee *and* the loss-based stopping is the open problem the authors identify: a monitoring rule that is anytime-valid and calibrated to the moderate deviation scale at its realized stopping time.

The same paper makes a large-scale comparison, e-BH against empirical Bayes in 2,000 simultaneous tests with 5 percent signals: the e-BH procedure has a false discovery rate of 0.002 and power 0.174, while an empirical-Bayes rule that targets an FDR of 0.10 has FDR 0.098 and power 0.317. It is not a trial-level comparison and we note it only as a statement of the same pattern, robustness paid for in efficiency (see Chapter 10 for the multiplicity machinery the reader will meet in a trial).

12.9. When should a sponsor pay the surcharge?

The design question is not Bayesian against frequentist in the abstract, but whether the trial needs validity at stopping times the protocol cannot anticipate (Polson and Sokolov 2026b). If it does, the surcharge $2\log(1/\alpha)$ is the price, worth about seventy percent of a conventional trial by Equation 11.11 against the uncalibrated Bayes-factor bar (the realized gap to a calibrated Bayesian rule is smaller, 612 against 464 events among SELECT-reconstruction trials that declare efficacy), and it belongs in the sample size justification as an explicit line and not as a footnote about anytime validity. The advice of Polson and Sokolov is to itemize the toll before choosing the paradigm. A practical test has five questions.

1. **Is the monitoring schedule under the sponsor's control and will it be kept?** If the DMC charter fixes the looks and the committee will not deviate, a classical group sequential design is the more powerful choice (86 against 72 percent in the binary example), and the sponsor should not pay the surcharge.
2. **Is a continuously monitored or open-ended decision part of the design?** Platform trials with continuous or frequent interim reviews, adaptive enrichment with open-ended expansion, and trials whose end is defined by a data-driven event are the natural cases for an anytime-valid design. In Chapter 20 the reader will see that arm-level decisions in a platform repeat the same question many times.
3. **Can the sponsor foresee the plausible unplanned looks?** If an external readout, a safety review or an extension is likely, a classical design can, in part, accommodate it by pre-specification (a spending function, a pre-specified sample size re-estimation). If the sponsor cannot enumerate them, the e-value design is insurance.
4. **Is there a reason to expect off-schedule looks that the protocol cannot list?** An e-process protects against optional stopping, whatever the schedule, under its assumptions. It does not repair bias from unblinding, altered endpoint assessment, selective data collection or invalid model changes, so confidentiality and conduct safeguards are as necessary for an e-process design as for a group sequential one. A sponsor whose DMC has a history of ad hoc looks may still judge that a guarantee that does not depend on the look schedule is worth the price.
5. **Can the sponsor live with the power cost?** The cost is real. In the binary example the e-value rule lost 14 points of power to a group sequential design that knew its schedule, and in the Bayes-risk comparison the e-value rule is far behind for modest effects. Restoring it means a larger horizon or a better process while keeping the efficacy threshold at 40. A loss-based stopping decision, such as the threshold of 3.5 in the Bayes-risk comparison, does not itself establish efficacy at 0.025, and reporting the e-process p-value beside it does not restore that guarantee for the decision.

If the answers to the first and third questions are yes, do not pay. If the answers to the second and fourth are yes, pay and say so in the sample size section. If the answers are mixed, the middle route is a protocol that keeps a group sequential or Bayesian design as the primary analysis, with an e-process monitored in parallel as an always-valid ledger. Sokolova and Sokolov suggest exactly that: run the e-process alongside a Bayesian rule and report the e-value, its always-valid p-value and a confidence sequence as a frequentist evidential record of the trial (A. O. Sokolova and Sokolov 2026). The primary analysis is not displaced, and the ledger answers the inevitable question about the unplanned look.

There is also a Bayesian answer to optional stopping that does not concede the point: posterior and predictive probabilities condition on the data actually observed, so a Bayesian analysis is coherent regardless of when or why the data were examined, and the stopping rule does not enter the posterior (A. Sokolova, Sokolov, and Polson 2026a). That is true of the posterior. It is not true of a decision rule that has been *calibrated to type I error* under a stated monitoring scheme, which is the route this book takes as its default for confirmatory designs without borrowing (one of the routes the FDA draft describes), and the calibration is the part that does not survive a deviation. The Bayesian rule of the SELECT comparison holds 2.5 percent under the scheme it was calibrated to and some other number under another scheme. The two paradigms make different promises, and the comparison of the SELECT table is a comparison under the Bayesian's assumptions. The e-value rule trades some of the efficiency for a promise that does not need that assumption.

12.9.1. Many subgroups at once: a negative result for the mixture e-value

An e-value looks attractive when the analysis choices cannot be enumerated, as in the garden of forking paths of Section 10.3, because a mixture over a prespecified family of K candidate analyses is a valid e-value for the global null that all candidate subgroup effects are nonpositive. Rejecting it does not establish efficacy in a selected subgroup, and subgroup-specific claims need an appropriate multiplicity procedure. When K is known, declared and fixed, the comparison is with the Sidak-corrected maximum, and the answer is mixed. We simulate $K = 20$ standardized statistics with equicorrelation ϱ, and compare the naive rule (report the largest, test at 0.025), the Sidak maximum, and the equal-weight mixture e-value $\bar{e} = K^{-1} \sum_k 2(1+\lambda^2)^{-1/2} \exp\{\lambda^2 Z_k^2/(2(1+\lambda^2))\} \mathbb{1}\{Z_k > 0\}$ with $\lambda^2 = 9$ at the threshold $1/\alpha$. The indicator makes the mixture a positive-direction (one-sided) e-value, so that the maximum and the mixture test the same alternative, a positive effect in at least one subgroup; the squared symmetric form would also spend power on negative effects. Power is compared at thresholds recalibrated to realized size 0.025.

```
mk <- bayestrials::decide_max_of_k_sim(K = 20, nsim = 2e5, seed = 7,
 ↪  direction = "positive")
sz <- mk$size; pw <- mk$power
knitr::kable(data.frame(rho = sz$rho, Naive = round(sz$naive, 3), Sidak =
 ↪  round(sz$sidak, 3), `Mixture e-value` = round(sz$mixture, 4),
 ↪  check.names = FALSE))
knitr::kable(data.frame(Scenario = pw$scenario, rho = pw$rho, Sidak =
 ↪  round(pw$sidak, 3), `Mixture e-value` = round(pw$mixture, 3),
 ↪  check.names = FALSE))
```

Table 12.3.: K = 20 equicorrelated subgroup statistics, 200,000 replicates, positive-direction mixture with prior scale lambda^2 = 9. Upper block: realized type I error at each rule's nominal threshold under the global null. Lower block: power at thresholds recalibrated to realized size 0.025.

rho	Naive	Sidak	Mixture e-value
0.0	0.397	0.024	0.0007
0.5	0.211	0.018	0.0010
0.8	0.113	0.009	0.0012

Table 12.4.: K = 20 equicorrelated subgroup statistics, 200,000 replicates, positive-direction mixture with prior scale lambda^2 = 9. Upper block: realized type I error at each rule's nominal threshold under the global null. Lower block: power at thresholds recalibrated to realized size 0.025.

Scenario	rho	Sidak	Mixture e-value
one effect d=3.0	0.0	0.505	0.506
one effect d=3.5	0.0	0.694	0.694
three effects d=2.5	0.0	0.669	0.730
five effects d=2.2	0.0	0.693	0.795

Table 12.4.: K = 20 equicorrelated subgroup statistics, 200,000 replicates, positive-direction mixture with prior scale lambda^2 = 9. Upper block: realized type I error at each rule's nominal threshold under the global null. Lower block: power at thresholds recalibrated to realized size 0.025.

Scenario	rho	Sidak	Mixture e-value
one effect d=3.0	0.5	0.541	0.425
one effect d=3.5	0.5	0.725	0.612
three effects d=2.5	0.5	0.584	0.503
five effects d=2.2	0.5	0.554	0.500

The naive rule is as bad as advertised: it approves a useless drug 40 percent of the time at $\varrho = 0$. The mixture e-value at its nominal threshold is very conservative, with realized size at most 0.0012 against a nominal 0.025, because Markov's inequality is loose for a single look. Recalibrating to matched size removes that penalty. With no correlation and several true effects the positive-direction mixture is at least as powerful as Sidak (0.80 against 0.69 for five effects at $\varrho = 0$), because it pools evidence across the true effects, while for a single effect the two tie (0.51 against 0.51). Under correlation $\varrho = 0.5$ the maximum wins in every scenario, by 0.05 to 0.12. The reading is therefore narrow: the mixture is competitive when several subgroups carry effects and the statistics are nearly independent, and loses when they are correlated.

What this licenses is limited. It does not say mixture e-values dominate multiplicity corrections, nor that they are dominated, and the simulation covers one fixed look at known K and a few configurations. The mixture displayed here is a fixed-look e-value: it has no time index and no prespecified construction for further analyses, so its guarantee holds at the single look at which it is computed. Applying its constant threshold repeatedly to standardized cumulative data would cross under the null with probability one eventually. Validity under continued monitoring, or continuation into a second cohort, would need component e-processes with prespecified weights and a proof of their joint monitoring guarantee, which is not shown here. The Sidak guarantee also needs K declared and fixed. The guarantee is never free, and what a mixture buys beyond the maximum is a way to weight prespecified analyses, not an answer to unlisted ones.

12.10. What the regulator will ask

No guidance names e-values, so the sponsor is arguing from principle. The principle is available and it is the one the Adaptive Designs guidance and E20 state: the chance of erroneous conclusions should be adequately controlled, the design should be completely prespecified, and the trial's integrity should be maintained. An e-value design satisfies the first under its assumptions: the type I error is at most α for every stopping time, which is stronger than the guidance asks. That protection is against optional stopping; it does not repair bias from unblinding, altered endpoint assessment, selective data collection or invalid model changes. It satisfies the second in a particular way: what is prespecified is the e-process, the mixing distribution (or betting fraction), the threshold and the horizon, and not the look schedule. On the third, it removes the statistical consequence of an off-schedule look but not the need for the same confidentiality and conduct safeguards that a group sequential design requires.

What will not be familiar to the reviewer is the machinery, so the submission has to carry what a standard SAP would not: a statement of the null hypothesis as a composite null for which the process is a test supermartingale, the proof (or a reference) that the process is one, the choice of ρ or λ with its justification from the target effect, a simulation report showing type I error and power over the scenarios of interest (the same report as Chapter 14, with the difference that the type I error is expected to be below nominal and the report should say by how much), and the always-valid p-value and confidence sequence in the format of the CSR. The FDA's CID paired meeting program and the Type B and C meetings described in Chapter 3 are the forum. A sponsor should also expect the question "why not a spending function?", and the answer is the five-question test above, not a general preference.

In the protocol

Wording a sponsor could adapt for the statistical section of a protocol that uses an e-process, to be agreed with the agency in advance:
"The primary analysis is a mixture e-process E_n^+ for the null hypothesis $H_0 : \theta \leq 0$, defined in Appendix X with mixing distribution half-normal with scale $\rho = ...$ The null hypothesis is rejected, and the trial may stop for efficacy, at the first interim review at which $E_n^+ \geq 40$. By Ville's inequality the probability under H_0 that E_n^+ ever exceeds 40 is at most 0.025, irrespective of the number and timing of reviews. Always-valid p-values $p_n = \min(1, 1/\max_{m \leq n} E_m^+)$ and a 95 percent confidence sequence for θ will be reported in the clinical study report at the time of the final analysis. The operating characteristics under the design scenarios (Section Y) are those of the simulation report."

12.11. In practice

No trial has yet been run on an e-process as its primary analysis, and no regulator has said anything about the method. What real trials do offer is the situation the chapter is about: a monitoring schedule that was fixed in advance, and the question of what would have happened had it not been.

12.11.1. BNT162b2: several interim looks on a schedule fixed in the SAP

The phase 3 trial of the Pfizer and BioNTech COVID-19 vaccine (Protocol C4591001, NCT04368728) is a clear example of a Bayesian design with several interim looks, prespecified before the relevant unblinded analyses (Pfizer Inc. 2022). The linked document is a later amended version of the SAP (version 10, dated 7 October 2022), so it documents the rules but does not by itself establish their chronology. The model, prior, thresholds and the clinical result are in Section 21.5, and the worked posterior computation is in Chapter 6. What matters here is the schedule: the final analysis was planned at 164 primary cases, with interim looks planned at 32, 62, 92 and 120 cases and posterior thresholds of 0.995 at an interim analysis and 0.986 at the final one, calibrated to an overall one-sided type I error of 2.5 percent. According to the SAP the look at 32 cases was not performed; the remaining three looks and the thresholds were unchanged, and the SAP states that the overall type I error stays at 0.025.

The lesson for this chapter is about what the design needed and what it did not. The number of looks and the threshold at each were fixed before the first case, so the group sequential-type guarantee held and a method that is valid at every stopping time was not needed; a sponsor in this position should not pay the surcharge. The case supports prespecified decision rules. It does not show that an unlisted case count necessarily breaks validity: FDA's statistical review (section 6.1.9) reports that the first interim analysis actually performed was at 94 cases and the final analysis at 170, not at exactly the listed 92 and 164, which is the ordinary operational overrun between a planned trigger and the realized data cut. Distinguishing planned triggers from exact analysis counts matters: whether an analysis at a realized count differing from the plan stays within the type I error guarantee depends on whether the thresholds were defined on the planned counts or adjusted to the realized information, and that is a rule the SAP must state. The thresholds of 0.995 and 0.986 are a bar calibrated for the planned looks (four interim looks plus the final analysis), and the SAP's own handling of the dropped first look (the thresholds were kept) can only lower the type I error, since removing a look with unchanged thresholds shrinks the rejection region. An e-process monitored alongside would have supplied a validity statement that does not depend on the schedule.

Key points

- A classical group sequential design controls the type I error as a property of its plan. Unplanned looks that depend on the accumulating comparative results, data-dependent looks under a fixed boundary schedule, continuous DMC review and extensions of the trial are outside that plan, while independent external triggers such as a publication can be accommodated if the monitoring rule is valid at arbitrary times, as an e-value rule is. For each the question is whether the request used accumulating comparative results and whether the spending or adaptation method remains valid.
- An e-value is a nonnegative statistic with expectation at most one under the null. A test martingale is a nonnegative process starting at one whose next value has conditional expectation equal to its current value under every null distribution (replacing equality by an upper bound gives a test supermartingale), and Ville's inequality bounds the probability that it ever exceeds $1/\alpha$ by α, for every stopping time.
- Likelihood ratios and mixtures of them are test martingales. The always-valid p-value is $1/\max E_m$ and a confidence sequence is the matching interval.
- The same statistic is a Bayes factor (read as evidence) or an e-value (read with the threshold $1/\alpha$ as a level-α test). The difference is calibration, not statistic.
- For the normal-mixture e-value, the approximate price of validity at every stopping time is the surcharge $2\log(1/\alpha) = 7.38$ in squared standardized evidence (exactly $7.38(1 + 1/q)$), about 70 percent of a conventional trial against the uncalibrated Bayes-factor bar, as a comparison of boundary terms; it is not a universal cost of anytime validity. Calibration narrows it (SELECT reconstruction, among trials that declare efficacy: 464 events for the calibrated Bayesian rule, 612 for the e-value rule). The Bayes-factor bar is itself above a single unadjusted look once $I_n\tau^2 > 46$ asymptotically.
- The orderings are design-dependent: group sequential with a known schedule has the most power at full information; the calibrated Bayesian rule stops first; the e-value rule is conservative. The power ordering of Bayes and e-value reversed between the two studies' designs, and the stopping-time ordering did not.
- No FDA or ICH guidance names e-values as of November 2026. The argument is the type I error control principle of the Adaptive Designs guidance and E20, made in a

CID-type interaction.

- Pay the surcharge when the trial needs validity at looks the protocol cannot anticipate, and say so in the sample size justification.

Discussion questions

1. A DMC charter lists four looks. In the middle of the trial a competing trial reads out and the committee requests a fifth, unscheduled look. What does a Lan-DeMets spending function allow, what does it not allow, and what would an e-process add? What would a regulator ask to know about why the look was requested?
2. The always-valid p-value after a running maximum of $E = 15.3$ is 0.065. A reader says this is "not significant" and a colleague says the evidence is "fifteen-fold". Who is right, and what does each statement mean?
3. The surcharge in information is about 70 percent of a conventional trial (against the uncalibrated Bayes-factor bar). A sponsor says: "we will never look more than twice, so the surcharge is wasted." Is that argument correct? What would change it?
4. In the binary design the calibrated Bayesian rule has less power than the e-value rule, and in the SELECT-like design it has more. Name two properties of the designs that could explain the reversal, and say what simulation you would run to check them for your own trial.
5. The loss-tuned e-value rule with a threshold of 3.5 halves the Bayes risk of the default. What does it do to the type I error guarantee, and how could the sponsor report the trial honestly?

Try it in the lab

Open module 08 at https://vsokolov.org/bayesian_trials_lab/08-e-values.html. Before pressing anything, predict how many of 40 trials with a drug that does nothing will cross the \$40 line, then count them; then press "RECOVERY-like" and predict, and check, what fraction of trials with RECOVERY's survival rates (77.1 versus 74.3 percent, 2,000 per arm) reach \$40.

13. Safety Signals and Safety Monitoring

Efficacy questions arrive one at a time and are asked of a trial built to answer them. Safety questions arrive by the thousand, are asked of data collected for another purpose, and are answered too late if they wait for statistical significance. A trial is sized for its efficacy endpoint, so the rare adverse event that matters most is usually the one the trial is least able to detect. After approval the same drug is taken by millions of people, and the signal arrives as a trickle of spontaneous reports. This asymmetry makes safety the part of drug evaluation where continuous, cumulative reasoning is most natural, and where Bayesian shrinkage has been in routine regulatory use the longest.

The chapter has four parts. The first is the case that defined the field, the withdrawal of rofecoxib, read as a cumulative analysis with a traditional and a Bayesian account of what could have been said, and when. The second compares two ways of monitoring continuously, the posterior probability and the Bayes factor read as an e-value, and connects to Chapter 12. The third is the empirical Bayes shrinkage used to screen spontaneous reports. The fourth is the monitoring of toxicity inside a trial, with a Bayesian stopping boundary, its exact operating characteristics, and a Pocock-type repeated test as the traditional counterpart. The lifecycle setting is in Chapter 2, the sequential machinery in Chapter 11, and the regulatory background in Chapter 3.

! Regulatory anchor

The documents below govern what a sponsor must report and how safety data are monitored. Chapter 3 describes how to read the status of each.

- **21 CFR 312.32, IND safety reporting** (U.S. Food and Drug Administration 2024a). The sponsor must notify FDA and all participating investigators in an IND safety report of a suspected adverse reaction that is both serious and unexpected, as soon as possible and in no case later than 15 calendar days after the sponsor determines that the information qualifies for reporting, and of an unexpected fatal or life-threatening suspected adverse reaction as soon as possible and in no case later than 7 calendar days after the sponsor's initial receipt of the information. The regulation lists the evidence that suggests a causal relationship and so triggers a report, among it an aggregate analysis of specific events observed in a clinical trial that indicates they occur more frequently in the drug treatment group than in a concurrent or historical control group. It is a requirement, and it prescribes no statistical method for that aggregate analysis.
- **FDA, Sponsor Responsibilities: Safety Reporting Requirements and Safety Assessment for IND and BA/BE Studies (final guidance, December 2025; it finalizes the June 2021 draft of the same title)** (U.S. Food and Drug Administration 2025b). Recommends that the sponsor keep a safety surveillance plan, review accumulating events in aggregate and not one report at a time, and keep individual treatment assignments blinded where the trial is blinded, with unblinded aggregate review by a group independent of the

sponsor's study team. Guidance recommends; it does not require.

- **ICH E2A, Clinical Safety Data Management: Definitions and Standards for Expedited Reporting** (International Council for Harmonisation 1994). The definitions of serious, unexpected and related events that 21 CFR 312.32 and its counterparts in other regions build on.
- **FDA, Establishment and Operation of Clinical Trial Data Monitoring Committees (final guidance, March 2006; draft revision, February 2024)** (U.S. Food and Drug Administration 2006, 2024e). A DMC reviews unblinded safety data at intervals set in its charter. Interim data are confidential, and the sponsor sees only what the committee recommends. The 2024 text is a draft, not binding. Neither text prescribes a statistical stopping rule for safety.
- **EMA, Guideline on Good Pharmacovigilance Practices Module IX Addendum I: Methodological Aspects of Signal Detection from Spontaneous Reports (2017)** (European Medicines Agency 2017). Defines disproportionality statistics (the proportional reporting ratio, the reporting odds ratio and the Bayesian and empirical Bayes measures) as accepted screening tools, and says a statistical signal is only a hypothesis to be assessed clinically.
- **FDA, FAERS and its successor (public dashboard; AEMS announced March 2026)** (U.S. Food and Drug Administration 2024c, 2026b). The spontaneous reporting systems in which the FDA applies disproportionality screening, with empirical Bayes methods in use there since the late 1990s (Szarfman, Machado, and O'Neill 2002).

13.1. Rofecoxib: a cumulative signal

Rofecoxib, sold as Vioxx, was a selective inhibitor of cyclooxygenase 2 approved in 1999 for arthritis and pain. It was designed to relieve pain with less gastrointestinal bleeding than traditional nonsteroidal anti-inflammatory drugs (NSAIDs), and it was prescribed to tens of millions of patients. The VIGOR trial randomized 8076 patients with rheumatoid arthritis to rofecoxib 50 mg daily or naproxen 500 mg twice daily. It did show fewer confirmed upper gastrointestinal events on rofecoxib (2.1 against 4.5 per 100 patient-years), but myocardial infarction was reported in 0.4 percent of the rofecoxib patients and 0.1 percent of the naproxen patients, roughly a fourfold difference (Bombardier et al. 2000). The difference was attributed at the time to a protective effect of naproxen, not to harm from rofecoxib. A 2005 editorial expression of concern from the journal concerned cardiovascular events that had not been reported in the published paper (Curfman, Morrissey, and Drazen 2005). Merck withdrew the drug on 30 September 2004, after the placebo-controlled APPROVe trial in 2586 patients with a history of colorectal adenoma showed confirmed thrombotic events in 46 of 1287 rofecoxib patients (1.50 per 100 patient-years) against 26 of 1299 on placebo (0.78 per 100 patient-years), a rate ratio of 1.92 (95% CI 1.19 to 3.11), with the curves separating after about 18 months (Bresalier et al. 2005).

13.1.1. What the traditional analysis said

Later in 2004, Jüni and colleagues published a cumulative meta-analysis of the randomized trials, adding one trial at a time in the order of completion (Jüni et al. 2004). By the end of 2000 the trials had accumulated 52 myocardial infarctions among 20,742 patients, with a

pooled relative risk of 2.30 (95% CI 1.22 to 4.33, $p = 0.010$). A year later, with 64 events among 21,432 patients, it was 2.24 (1.24 to 4.02, $p = 0.007$). The relative risk was tested for heterogeneity by type of control group ($p = 0.41$) and by duration of treatment ($p = 0.82$). Neither test detected heterogeneity. Observational studies put the cardioprotective effect of naproxen at a relative risk of 0.86 (0.75 to 0.99), far too small to explain VIGOR. The authors concluded that the drug should have been withdrawn several years earlier.

The traditional reading is a significance test applied to a pooled estimate: by the end of 2000 the interval excluded 1. This is a fair account of the evidence, and it is the account a regulator can defend. Its weakness for decision making is in what it leaves out. A p-value of 0.010 does not say how large the harm probably was, how probable a harm of clinical importance was, or what to do while the evidence was still accumulating.

13.1.2. What a Bayesian reading adds

From the end-2000 pool, $\log \widehat{RR} = 0.833$ with standard error 0.323. Treat this as a normal likelihood for the log relative risk and take a normal prior centered at no effect. A flat prior gives the published estimate. A skeptical prior with standard deviation s_0 on the log scale represents a reviewer convinced of the drug's safety: with $s_0 = 0.3$ a relative risk above 1.5 has prior probability about 9 percent, and with $s_0 = 0.2$ about 2 percent. Table 13.1 gives the posterior.

Table 13.1.: Posterior median and posterior mean relative risk (RR) of myocardial infarction with rofecoxib from the cumulative trial evidence of Jüni et al. (2004), under a flat prior and two skeptical normal priors on log RR centered at 0 with standard deviation s0. The end-2000 inputs are log RR 0.833 and standard error 0.323; the end-2001 standard error, 0.300, is recovered from the published confidence interval.

Period	Prior	Posterior median RR	Posterior mean RR	Pr(RR > 1)	Pr(RR > 1.5)
End 2000	Flat	2.30	2.42	0.995	0.907
End 2000	Skeptical, s0 = 0.3	1.47	1.51	0.960	0.464
End 2000	Skeptical, s0 = 0.2	1.26	1.28	0.913	0.152
End 2001	Flat	2.24	2.34	0.996	0.909
End 2001	Skeptical, s0 = 0.3	1.50	1.53	0.971	0.496
End 2001	Skeptical, s0 = 0.2	1.28	1.30	0.932	0.172

The skeptical priors are deliberately strong. Even under the prior with $s_0 = 0.2$, the posterior probability that rofecoxib increased the risk of myocardial infarction was 0.91 at the end of 2000, and the posterior median and mean relative risks were 1.26 and 1.28. These are the book's computations from the published summary, not figures from Jüni and colleagues. They show something the p-value does not: the conclusion that a harm was probable did

not depend on the reviewer's prior. A reviewer entitled to disagree about its size could not reasonably disagree about its direction.

13.1.3. From inference to decision

The posterior probability does not by itself say what to do. The decision turns on the size of the risk and on what the drug offered in return. Let the baseline risk of myocardial infarction in the arthritis population be r_0 per year, let rofecoxib multiply it by RR, and let the gastrointestinal benefit relative to a traditional NSAID be a reduction of g serious bleeding events per patient-year, each with a disutility d_G relative to the disutility d_M of a myocardial infarction. Rofecoxib is preferred for a patient when

$$g\, d_G > r_0(\mathbb{E}[RR] - 1)\, d_M. \tag{13.1}$$

Whether the inequality holds depends on inputs this book does not supply from sources: the baseline risk r_0, the gastrointestinal benefit g against the relevant comparator (a traditional NSAID with or without a proton pump inhibitor changes g), and the disutilities. For a patient at high risk of bleeding and low cardiovascular risk it may hold; for others it may not, and the answer should be shown over a range of inputs (Exercise 2 gives one). What the cumulative evidence supplies is the term $\mathbb{E}[RR]$. Under every prior in Table 13.1 the posterior mean at the end of 2000 is at least 1.28, so the right-hand side was positive and not small, and a patient-level benefit-risk analysis of the kind above was warranted well before the withdrawal. This is a book-level illustration with stated inputs, not a regulatory benefit-risk analysis, which would use the full evidence on both sides and patient-level risk.

13.1.4. The naproxen hypothesis as a model comparison

The two explanations of VIGOR, harm from rofecoxib or protection from naproxen, are competing models, and Bayesian model comparison asks how well each predicts all the data. The naproxen model predicts an elevated relative risk only against naproxen; the harm model predicts it against every comparator, including placebo. Jüni and colleagues found no evidence that the relative risk differed by control group. We have not computed the Bayes factor between the two models, which would need the trial-level data, but its direction is clear: every placebo-controlled trial that showed a raised rate counted for harm and nothing for protection by naproxen. A frequentist analysis can compare comparator-specific effects with a meta-regression or an interaction test, as Jüni and colleagues did, and a significant interaction would favor the naproxen explanation. What posterior model probabilities add is a weight for each explanation that accounts for how well each predicts all the data at once, including the penalty for the flexibility of the rival, where a nonsignificant interaction test only fails to reject one of them. Explaining away an unwelcome signal with a plausible rival is how safety signals are most often lost, and writing the rival down as a model with its own predictions is the remedy.

13.2. Monitoring continuously: posterior probability and e-value

A safety monitor looks at the data whenever it changes. A test applied at each look inflates the false alarm rate with the number of looks (Section 12.1), and group sequential boundaries (Chapter 11) presume a schedule and a maximum sample size that pharmacovigilance does not have. Two statistics survive continuous looking, and they answer different questions.

The first is the posterior probability. It is coherent under any stopping rule, because the likelihood principle holds: the posterior after a hundred looks is the posterior after the last one, however the looks were chosen. Its frequentist false alarm rate is not controlled, and Chapter 12 says why that matters when the regulator's question is about error under the null.

The second is the Bayes factor read as an e-value (Section 12.6). The Bayes factor of an alternative against $RR = 1$, computed with a fixed prior under the alternative, is a nonnegative martingale under the null, so Ville's inequality (Ville 1939) gives

$$\Pr\left(\sup_t \mathrm{BF}_t \geq 1/\alpha \mid RR = 1\right) \leq \alpha. \tag{13.2}$$

Safety monitoring asks about harm, so the null is $RR \leq 1$ and the alternative is a harmful relative risk. Take the alternative $\log RR \sim N(0, 0.5^2)$ restricted to positive values, with density $2N(\theta; 0, 0.5^2)$ on $\theta > 0$. The likelihood ratio of any single $\theta > 0$ against $\theta = 0$ has expectation at most one under every $\theta \leq 0$, so the mixture is an e-value for the whole composite null and Equation 13.2 holds for it. (A two-sided Bayes factor with the unrestricted $N(0, 0.5^2)$ alternative also alarms when the risk is protective, so its bound is for the wrong null.) Acting when the harm-only Bayes factor first reaches twenty gives a false alarm probability below 5 percent however often one looks. For rofecoxib at the end of 2000, with posterior mean m and variance v of $\log RR$ under the $N(0, 0.5^2)$ prior,

$$\mathrm{BF} = 2\,\Phi\left(\frac{m}{\sqrt{v}}\right)\frac{N(0.833;\,0,\,0.323^2 + 0.5^2)}{N(0.833;\,0,\,0.323^2)} = 11.2,$$

where the second ratio is the two-sided Bayes factor 5.7. The value is substantial but short of the anytime-valid threshold of 20. The conclusion is different from the one a posterior probability gives, and not contradictory. The posterior probability answers how probable harm is given a prior. The e-value rule answers whether the evidence is strong enough to act with a guaranteed bound on false alarms, whatever the looking schedule. The factor calculated from the end-2000 summary is below the threshold. Whether a sequential monitor would have alarmed earlier requires the cumulative data path and a valid sequential likelihood for the successive meta-analyses, which are not supplied here. The summary would at least have marked the drug for urgent study, and the rational next step was a placebo-controlled trial designed to settle the cardiovascular question.

The false alarm behavior is easy to see by simulation. Table 13.2 gives the proportion of 60 monthly looks at a drug with no effect that ever cross each threshold, using the package's information-time simulation with 20,000 trials, in which each look adds the same information.

Table 13.2.: Probability of at least one alarm over 60 equally spaced looks at a drug with relative risk exactly 1 (simulation, 20,000 trials), for a rule that alarms when the posterior probability of harm first reaches 0.95 under a N(0, 0.5^2) prior, and a rule that alarms when the harm-only Bayes factor (half-normal alternative on log RR > 0, from the same prior) first reaches 20. The Ville bound on the second is 0.05.

Alarm rule	False alarm probability
Posterior probability of harm at least 0.95	0.232
Harm-only Bayes factor at least 20 (e-value rule)	0.022

The posterior probability rule raises a false alarm in about 23 percent of null drugs, and the e-value rule in about 2 percent, well under its bound (the two-sided Bayes factor, which also counts crossings in the protective direction, alarms in 2 percent). The price of that guarantee is a larger sample to reach the same alarm when a harm is present; Section 12.5 measures it and Section 12.9 discusses when it is worth paying. In safety it often is, because the looking schedule is not the sponsor's to fix. The two rules need not be used alone. A posterior probability can rank which signals to investigate first, while a threshold on the e-value, fixed in the charter, sets the level at which a DMC recommendation to stop or restrict is backed by a stated error guarantee.

13.3. Signals in spontaneous reports

Most rare adverse effects are first noticed in spontaneous reports, not in trials. The FDA Adverse Event Reporting System (FAERS) and the WHO's VigiBase together hold tens of millions of reports. The data are a large sparse table of counts n_{ij} of reports that mention drug i and event j, with an expected count E_{ij} computed from the margins under independence. Reports come from two streams, voluntary submissions by clinicians and patients and mandatory reports from manufacturers, but neither gives exposure denominators and ascertainment is incomplete, so the table cannot give incidence; it can only say which pairs are reported more than the margins predict (U.S. Food and Drug Administration 2026d). That is disproportionality.

13.3.1. The traditional measure

The simplest statistics are the relative reporting ratio n_{ij}/E_{ij} and its cousins, the proportional reporting ratio (S. J. W. Evans, Waller, and Davis 2001) and the reporting odds ratio (Puijenbroek et al. 2002; Rothman, Lanes, and Sacks 2004), with a confidence interval and a screening rule. Evans and colleagues (S. J. W. Evans, Waller, and Davis 2001) flagged a pair when the proportional reporting ratio is at least 2, the chi-square statistic at least 4 and there are at least three reports, and a common variant asks for a lower confidence bound above 1 with at least three reports. They are simple, transparent and still used, and the EMA guideline names them (European Medicines Agency 2017). Their weakness is at the sparse end. For a rare pair the ratio is dominated by noise: one report against an expected count of 0.05 gives a ratio of twenty, and a table with millions of cells will contain many such pairs by chance alone.

13.3.2. The gamma-Poisson shrinker

DuMouchel proposed the gamma-Poisson shrinker, the empirical Bayes method behind the FDA's data mining of its reports database (DuMouchel 1999; Szarfman, Machado, and O'Neill 2002). Each pair has a true reporting ratio λ_{ij}, with $n_{ij} \sim \text{Poisson}(\lambda_{ij} E_{ij})$, and the λ_{ij} are drawn from a mixture of two gamma distributions,

$$\lambda \sim P\,\text{Gamma}(\alpha_1, \beta_1) + (1 - P)\,\text{Gamma}(\alpha_2, \beta_2), \tag{13.3}$$

whose five parameters are estimated from the whole table by maximum marginal likelihood. The marginal distribution of n given E under each component is negative binomial, and the posterior of λ is again a two-component gamma mixture: component k becomes $\text{Gamma}(\alpha_k + n, \beta_k + E)$, with weights proportional to the prior weight times the negative binomial likelihood. Signals are ranked by the empirical Bayes geometric mean, $\text{EBGM} = \exp(\mathbb{E}[\log \lambda \mid n])$, and screened on its fifth percentile, EB05. The other main Bayesian disproportionality tool is the information component of the Bayesian confidence propagation neural network, which underlies the WHO's VigiBase screening (Bate et al. 1998), and Madigan and colleagues (Madigan et al. 2011) and Zorych and colleagues (Zorych et al. 2013) extend Bayesian pharmacovigilance to longitudinal databases.

Table 13.3 applies the shrinker with illustrative hyperparameters $P = 1/3$, $(\alpha_1, \beta_1) = (0.2, 0.1)$ and $(\alpha_2, \beta_2) = (2, 4)$. The first component has mean 2 and large variance and represents genuine associations; the second has mean 0.5 and represents background. These values illustrate the mechanics. In a real analysis they are fitted to the whole table (`safety_gps_fit()` maximizes the marginal likelihood), and they should be, since the shrinkage is only as good as the estimated mixture.

Table 13.3.: Gamma-Poisson shrinkage of disproportionality ratios with illustrative hyperparameters (P = 1/3, first component Gamma(0.2, 0.1), second Gamma(2, 4)). Weight is the posterior probability of the signal component; EBGM is the empirical Bayes geometric mean; EB05 is the posterior fifth percentile.

n	E	n / E	Weight on signal component	EBGM	EB05
1	0.05	20	0.561	2.00	0.26
3	0.50	6	0.804	3.37	0.72
12	3.00	4	0.969	3.70	2.12
40	10.00	4	0.997	3.93	3.00
2	2.00	1	0.167	0.62	0.23

The single report with a raw ratio of twenty is shrunk to an EBGM of 2.0, and its lower bound 0.26 is well below one: it is not a signal. Two pairs with the same raw ratio of four are treated very differently according to how much evidence supports them: the pair with 40 reports has an EB05 of 3.0 and would be flagged at once, the pair with 12 an EB05 of 2.1. A screening rule such as EB05 above 2 flags the pairs with $n = 12$ and $n = 40$ and nothing else. This is the same shrinkage that made the subgroup estimates in Chapter 22 trustworthy, applied to a table with millions of cells. Berry and Berry apply the hierarchical idea one level up, borrowing strength across adverse event categories within a trial, which is the within-trial version of the same remedy for multiplicity (Scott M. Berry and Berry 2004).

Two cautions belong with any use of the method. A high EB05 is a hypothesis for clinical and pharmacoepidemiologic assessment, not evidence of causation, since reporting is influenced by publicity, indication and the reporter (EMA says this in the guideline). And the Bayesian machinery protects against noise in rare cells, not against bias in the reporting mechanism. Neither the expected count nor the shrinkage can correct for a cause of reporting that is not in the margins.

13.4. Toxicity monitoring inside a trial

The rofecoxib analysis is retrospective and the shrinker is post-market. Inside a trial the question is prospective: given a toxicity that the protocol defines, when should enrollment pause? This is the case in which the Bayesian rule is simple, and in which it has a precise traditional counterpart.

13.4.1. The traditional counterpart

The traditional trial would apply a repeated exact binomial test of $H_0 : q \le \bar{q}$ after every five patients, up to 40, where $\bar{q} = 0.2$ is the acceptable toxicity rate. A Pocock-type version uses the same nominal level a at every look: pause at look n if $\Pr(X_n \ge x \mid \bar{q}) \le a$ (Pocock 1977 gives the normal-theory version for efficacy). The nominal level is then chosen so that the probability of ever stopping at $q = \bar{q}$ is a stated overall level. We use two calibrations. The first matches the overall stopping probability at $q = \bar{q}$ of the Bayesian rule introduced next, and the second uses a conventional overall level of 0.05.

13.4.2. The Bayesian rule

Bayesian monitoring of toxicity in single-arm trials goes back to Thall, Simon and Estey (P. F. Thall, Simon, and Estey 1995, 1996), and Ivanova, Qaqish and Schell (Ivanova, Qaqish, and Schell 2005) treat continuous toxicity monitoring in phase II trials; the rule below is a simple member of that family. Let the toxicity rate q in the single-arm study have a Beta(a_0, b_0) prior, with $a_0 = 0.5$ and $b_0 = 2.5$, a prior mean of 1/6 below the acceptable rate with the weight of three patients. After x toxicities in n patients the posterior is Beta($a_0 + x$, $b_0 + n - x$). The Bayesian rule, at the same looks: after every five patients, pause if

$$\Pr(q > \bar{q} \mid x, n) > \kappa, \qquad \kappa = 0.75. \tag{13.4}$$

Because the posterior probability is increasing in x at fixed n, the rule is a table of counts, one stopping count per look, which is how it goes into a protocol and a DMC charter. Table 13.4 gives it next to the traditional counterpart above.

Table 13.4.: Toxicity stopping counts at eight looks for a single-arm study of up to 40 patients with acceptable rate 0.2: the Bayesian rule (stop if $\Pr(q > 0.2 \mid \text{data}) > 0.75$ under a Beta(0.5, 2.5) prior), and the Pocock-type repeated exact binomial test with a common nominal level calibrated to an overall stopping probability of 0.05 at $q = 0.2$.

Patients treated	Bayesian rule: stop at toxicities of	Pocock-type (overall 0.05): stop at toxicities of
5	3	4
10	4	6
15	5	7
20	6	9
25	7	10
30	8	12
35	9	13
40	11	14

The Bayesian rule stops at 3 of the first 5 patients and 11 of 40. The Pocock-type rule at overall level 0.05 requires 4 of the first 5, and 14 of 40. There is one further fact that the table hides. The Bayes rule's stopping counts are identical to those of a Pocock-type test whose common nominal level is 0.26: the Bayesian rule is a constant-nominal-level repeated test at these looks, with overall level 0.384 at $q = 0.2$ (the same calibration, `tox_pocock_boundary()` at its default). This is not an accident of the example. The posterior probability of exceeding $\bar{q}$ is a monotone function of the tail probability under $\bar{q}$ at a given n, and the posterior threshold maps to a nominal level that varies slowly with n. The Bayesian rule does not escape the repeated-testing arithmetic; it chooses a point on it, with a threshold, 0.75, that is a statement about how much risk of a false pause the investigators will accept.

13.4.3. Operating characteristics

Both rules are fully described by their boundaries, so their operating characteristics are exact, from a dynamic program over the count of toxicities with no simulation. Table 13.5 gives the probability of stopping early and the expected number of patients enrolled for a range of true toxicity rates.

Table 13.5.: Exact operating characteristics of the two toxicity monitoring rules of the previous table: probability of pausing at some look before or at the 40th patient, and expected number of patients treated, by true toxicity rate.

True rate	Bayesian: P(stop)	Bayesian: E[n]	Pocock-type: P(stop)	Pocock-type: E[n]
0.10	0.034	39.1	0.001	40.0
0.15	0.154	36.5	0.008	39.8
0.20	0.384	31.5	0.046	39.1
0.30	0.843	19.7	0.373	33.8
0.40	0.986	12.2	0.827	23.9

At the acceptable rate 0.2 the Bayesian rule pauses the trial with probability 0.384, the Pocock-type rule with probability 0.046. If the true rate is 0.3, an unacceptably high rate by the protocol, they pause with probability 0.843 and 0.373, and the expected enrollment falls from 40 to 19.7 and 33.8. The choice between them is not one of Bayesian against frequentist, since they are members of one family. It is whether, for a safety rule, a pause at a false alarm probability of 0.38 under the boundary rate is acceptable. In many early-phase studies it is. A pause is a review, not an end, enrolled patients are exposed to the unknown, and a monitoring rule with a false pause probability of 0.05 has, in return, a probability of only 0.37 of pausing when the rate is 0.3.

The calibration is deliberate in either direction, and that is the point of a safety rule: it is the asymmetric loss between exposing patients and delaying a good drug that fixes the threshold, so the type I error of a safety boundary is a design choice stated with its consequences, not a number inherited from the efficacy analysis. The operating characteristics belong in the protocol and the DMC charter as the table above does (see Chapter 14 for the report every design carries). A DMC would not follow either table mechanically; the 2006 guidance expects the committee to use judgment, and the boundary is a flag that triggers a review.

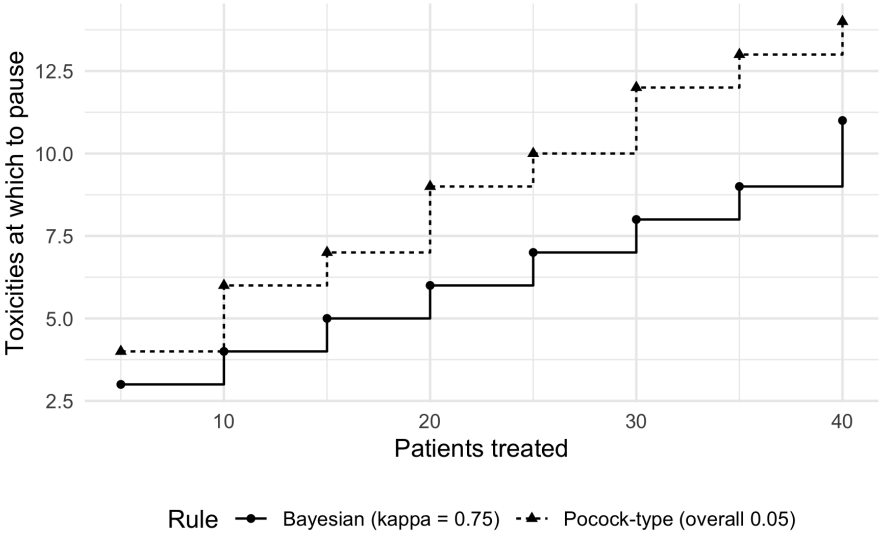


Figure 13.1.: Stopping counts of the Bayesian toxicity rule and the Pocock-type rule (overall 0.05 at q = 0.2) against patients treated. A trial pauses when the cumulative number of toxicities reaches the count.

13.5. Aggregate safety assessment

A sponsor's safety review is not one statistic. Four layers sit side by side, and each answers a different question. Blinded pooled review of the whole trial asks whether the overall event rate is higher than expected, without revealing the arms; given credible external information about adverse event rates in comparable untreated patients, Gould and Wang (Gould and Wang 2017) describe Bayesian monitoring for potential excess risk using pooled blinded data (their title calls it the canary in the coal mine). The unblinded DMC review compares arms,

with the boundaries and posterior rules of Section 13.2 and Section 13.4. A hierarchical model across adverse event terms, as in Xia, Ma and Carlin (Xia, Ma, and Carlin 2011), shrinks the many rare events toward a common distribution before flagging, in the same way that the gamma-Poisson shrinker does for spontaneous reports. And a false discovery rate flag, such as the double FDR of Mehrotra and Adewale (Mehrotra and Adewale 2012), then limits the expected false-discovery proportion among the flagged terms. Their procedure groups adverse events by body system, whereas Section 10.4.5 defines FDR and illustrates the ordinary Benjamini-Hochberg procedure.

Outside the trial, surveillance of marketed products and vaccines is sequential in the Wald sense. Kulldorff and colleagues (Kulldorff et al. 2011) proposed a maximized sequential probability ratio test for drug and vaccine safety surveillance. With a Poisson count c of events against an expected count μ_0 under no excess risk, the statistic is the log likelihood ratio maximized over relative risks above one, $c\log(c/\mu_0)-(c-\mu_0)$ when $c > \mu_0$, and a signal is declared when it crosses a critical value that has been calibrated for the planned surveillance. It is the likelihood-ratio machinery of Chapter 12 with a fixed alternative replaced by its maximum, so it needs a critical value from a calibration rather than the $1/\alpha$ of an e-value. Which of these layers is a trigger for action, and which only a hypothesis for review, is a decision for the safety plan to state, and the regulator will ask for it (Section 13.7).

13.6. Connecting the two views

The toxicity rule is a posterior probability monitor, so it inherits the properties in Section 13.2: its overall false alarm probability is a computed number for the stated schedule and is not guaranteed for looks off the schedule. An e-value version of the same boundary would instead use the likelihood ratio of a binomial alternative against $\bar{q}$. It would be valid at any look, including a fifth unscheduled one when a competing trial reads out or a serious event is reported. That is what Chapter 12 argued for DMCs in general, and safety is where its case is strongest, since the schedule of a DMC is almost never exactly the one in the charter. The price, again, is a larger sample at a given power. Which to use is a question of what the charter promises. A rule with a stated boundary at planned looks and a stated false pause probability fits a phase II toxicity review. An e-value threshold is the better statement for a long-running surveillance program or for a pooled monitoring of several trials of one drug, in which the number of looks cannot be fixed.

13.7. What the regulator will ask

- Which safety events are monitored, and who sees unblinded data. The IND safety reporting guidance and the DMC guidance ask for a plan, and the plan must say who reviews aggregate data and how.
- Which statistical rule is used, and whether it is a trigger for review or an automatic stop. Neither guidance prescribes a rule, and a Bayesian boundary is acceptable on the same terms as a frequentist one: prespecified, with operating characteristics that are reported.
- What the false alarm probability and the power are, under the stated schedule and, for a continuous monitor, under looks off the schedule.

- How signals from outside the trial (spontaneous reports, other trials of the same drug, the class) enter the review. A shrinkage screen is a hypothesis generator, and it is not a basis for a safety report by itself.
- For an event that triggers 21 CFR 312.32, the clock. For an unexpected fatal or life-threatening suspected adverse reaction that is interpretable as a single case, the 7-day limit runs from the sponsor's initial receipt of the information, whereas the 15-day limit for other potential serious risks runs from the date the sponsor determines that the information qualifies for reporting (21 CFR 312.32(c)). For events that need more than one occurrence or an aggregate analysis to establish a causal relationship, the sponsor makes the determination once it has evidence that the aggregate analysis meets the reporting criterion, and must then report as soon as possible. No monitoring rule shortens these limits or lengthens them (U.S. Food and Drug Administration 2025b, sec. VIII).

Key points

- Safety evidence accumulates, and the natural analysis is cumulative. The rofecoxib trials had pooled to a relative risk of about 2.3 for myocardial infarction by the end of 2000, and the pooled posterior probability of harm was above 0.91 even under a skeptical prior with standard deviation 0.2 on the log scale.
- A p-value measures how unusual the observed or more extreme data would be under the specified null model. A posterior says how large it probably is and how probable a harm of a given size is, which is what a benefit-risk decision uses. The decision itself needs the benefit side, as in the inequality of this chapter.
- Continuous monitoring has two statistics. The posterior probability is coherent under any stopping rule but its false alarm rate is not controlled (about 0.23 over 60 monthly looks at a threshold of 0.95 in the simulation). The Bayes factor read as an e-value is controlled by Ville's inequality at any look (about 0.02 against a bound of 0.05 for the harm-only factor).
- Empirical Bayes shrinkage, the gamma-Poisson shrinker, ranks spontaneous-report signals by evidence as well as by size. A raw ratio of 20 on one report is not a signal; a ratio of 4 on 40 reports is.
- A Bayesian toxicity stopping rule is a table of counts. Its operating characteristics are exact (at a true rate of 0.2 it pauses with probability 0.384, and at 0.3 expected enrollment is 19.7 of 40), and it is a constant-nominal-level repeated test, the same family as the Pocock-type rule it is compared with.
- No guidance prescribes a statistical stopping rule for safety. 21 CFR 312.32 is a requirement about what to report and when; the DMC and IND safety guidances recommend how to monitor.

Discussion questions

1. A sponsor says that the VIGOR excess was explained by naproxen. What data would distinguish that account from harm by rofecoxib, and what Bayes factor would you want to see before accepting it?
2. A DMC charter fixes a toxicity boundary at 5-patient looks. A serious event occurs between looks. What does the boundary table allow, and what would an e-value boundary have allowed?

3. A drug-event pair has 3 reports against an expected 0.5, a raw ratio of 6, and an EB05 of 0.72. A clinician says it is a "known" effect. What does the screen say and what does it not say?
4. The Bayesian toxicity rule pauses a null-rate trial with probability 0.38. Argue for and against a Pocock-type rule at 0.05 for a first-in-human oncology study and for a trial of a vaccine in healthy volunteers.

Exercises

1. Verify the end-2000 rows of Table 13.1. Show that with a $N(0, s_0^2)$ prior and a normal likelihood with standard error 0.323 the posterior is normal with precision $1/0.323^2 + 1/s_0^2$.
2. In Equation 13.1 take $r_0 = 0.01$ per year, $g = 0.01$ per year and $d_G/d_M = 0.3$. For which values of $\mathbb{E}[RR]$ is rofecoxib preferred? (Answer: below 1.3.)
3. Show that the marginal distribution of n under a Gamma(α, β) prior on λ and $n \sim \text{Poisson}(\lambda E)$ is negative binomial with size α and success probability $\beta/(\beta + E)$.
4. Derive the posterior weight on each gamma component in the shrinker and reproduce the third row of Table 13.3.
5. Use `safety_bf_monitor()` with `looks = 60` and different `pp_threshold` values to find the threshold at which the posterior probability rule has a false alarm probability of 0.05. The function simulates null paths only, so to compare the sample size needed to alarm when the true relative risk is 1.5, extend the simulator by adding drift log(1.5) to the information-time paths and recording the first crossing of the threshold.
6. Using `tox_bayes_boundary()` and `tox_oc()`, find the smallest threshold κ for which the Bayesian rule's probability of pausing at $q = 0.2$ is at most 0.05 (exactly 0.05 is not attainable with discrete boundaries, so report the adjacent attainable values), and compare the boundary with the third column of Table 13.4.

Part IV.

Bayesian Adaptive Designs

14. Operating Characteristics by Simulation

A regulator who is asked to accept a trial design wants to know how often it will go wrong and how often it will go right. For a fixed-sample trial, a formula answers both questions in a line. For a group sequential trial, a short numerical integration answers them. For the designs that make Bayesian methods worth the trouble (response-adaptive allocation, arm dropping, borrowing, predictive stopping, delayed outcomes), a closed formula is usually unavailable, and the answer usually has to be computed by running the design many times on a computer against assumed truths (some posterior and predictive stopping rules permit exact recursive enumeration, as in Chapter 15). That computation, written up and submitted, is the simulation report, and it is the deliverable of a Bayesian adaptive design in the way the sample size calculation is the deliverable of a traditional one.

This chapter opens Part IV. It starts where the reader already is, with the analytic type I error and power of a fixed and a group sequential design and what a traditional statistical analysis plan (SAP) says about them. It then takes the Bayesian success rule $\Pr(\delta > a \mid \text{data}) > c$ of Chapter 7 and shows why it has no operating characteristics until someone calibrates it, how the calibration is done, and what the regulator wants to see when it is finished. The worked example is a two-arm binary trial with four looks, and every number in it comes from the simulator in the `bayestrials` package (`oc_simulate` and its companions), which Chapter 23 reuses. The chapter closes with four real trials, from a vaccine to a spinal device, whose submissions state calibrated posterior thresholds (Section 14.11).

! Regulatory anchor

- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft guidance, January 2026), Section IV.A and IV.A.1.** For trials with an overall frequentist statistical analysis plan, success criteria are "almost always" chosen so that the familywise type I error rate is no greater than 0.025, one-sided. The draft describes three routes for Bayesian success criteria: calibration to a type I error rate, direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA), and decision-theoretic or benefit-risk criteria (Section IV.A.3). Calibration is the approach used in this chapter and this book's default for confirmatory adaptive designs without borrowing. For a posterior criterion $\Pr(\delta > a) > c$, "a and c can be chosen such that the overall FWER is controlled at a given level, typically 0.025 one-sided." Operating characteristics come from clinical trial simulations, and the draft asks for extensive documentation of the simulations, code and software. A draft: it is not binding, but it is the clearest statement of what a reviewer will ask for.
- **FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics (final guidance, November 2019), Sections III and VI.** Section III states the four

principles: the chance of erroneous conclusions should be adequately controlled, estimation of treatment effects should be sufficiently reliable, the details of the design should be completely prespecified, and trial integrity should be maintained. Section VI.A: "A major use of simulations in adaptive trial design is to estimate trial operating characteristics."

- **ICH E20, Adaptive Designs for Clinical Trials (Step 2 draft, June 2025), Sections 3.3, 5.2 and 5.3.** Section 3.3 calls control of the chance of false positive efficacy conclusions "an essential element of regulatory decision-making." Section 5.2 defines a simulation as "the repeated execution of a large number of hypothetical clinical trials" and says what a simulation report should contain. Section 5.3 says that Bayesian adaptive designs must still show their operating characteristics and maintain type I error probability control. A draft, not yet adopted.
- The status of each document, and how the documents relate, is in Chapter 3.

14.1. What a traditional design computes, and what goes in the SAP

14.1.1. The fixed-sample design

The traditional two-arm trial with a binary endpoint fixes the sample size n per arm in advance, and its operating characteristics follow from the sampling distribution of the test statistic (Chapter 4, Chapter 9). Suppose the control response rate is $p_0 = 0.30$ and the trial is powered to detect a treatment rate of $p_1 = 0.45$, a difference of 15 percentage points. With one-sided $\alpha = 0.025$ and power 80 percent, the standard formula is

$$n = \frac{(z_\alpha + z_\beta)^2 \left[p_0(1-p_0) + p_1(1-p_1) \right]}{(p_1 - p_0)^2},$$

where $z_\alpha = 1.960$ and $z_\beta = 0.842$. The calculation below gives $n = 159.6$ per arm, which the protocol rounds up to 160. The formula is a normal approximation, and a binary endpoint with 160 patients per arm is discrete, so the SAP of a careful sponsor also reports the exact calculation: enumerate every pair of success counts (x_0, x_1), weight it by its binomial probability, and add up the probability of the pairs for which the pooled z test rejects. This is what the second half of the chunk does, and it shows that the exact type I error of the nominal 2.5 percent test is 0.0252, and the exact power at $p_1 = 0.45$ is 0.794.

```
p0 <- 0.30; p1 <- 0.45
n_formula <- n_binary(p0, p1, alpha = 0.025, power = 0.80)
x <- 0:160                                              # exact enumeration
 ↪  at n = 160 per arm
z <- outer(x, x, z_test_binary, n = 160)                # pooled z; rows x0,
 ↪  columns x1
exact <- function(p0, p1) sum(outer(dbinom(x, 160, p0), dbinom(x, 160, p1))
 ↪  * (z > qnorm(0.975)))
knitr::kable(data.frame(Quantity = c("Formula n per arm", "Exact type I
 ↪  error at n = 160", "Exact power at n = 160"),
```

```
  Value = round(c(n_formula, exact(0.3, 0.3), exact(0.3, 0.45)), 4)))
```

Quantity	Value
Formula n per arm	159.5939
Exact type I error at n = 160	0.0252
Exact power at n = 160	0.7943

Two things about this calculation matter for what follows. First, every number is a property of the test statistic's distribution, not of the data: the type I error is computed before a single patient is enrolled, and it is the same for any control rate close to the one assumed (the normal approximation degrades only near 0 and 1). Second, the computation is the proof. The regulator can verify it in any package, and the SAP says "the sample size of 160 per arm gives 80 percent power at $p_0 = 0.30$ and $p_1 = 0.45$ with one-sided type I error 0.025," which is a complete statement of the design's operating characteristics. The 80 percent and 0.025 are the planning targets: the exact enumeration above gives power 0.794 and type I error 0.0252, and a careful SAP reports those beside the targets.

14.1.2. The group sequential design

Add three interim analyses at 40, 80 and 120 patients per arm, so that four looks are equally spaced in information, and the analysis changes. The test statistic at look k is Z_k, and the sequence $Z_1, \dots, Z_4$ is approximately multivariate normal with correlation $\sqrt{t_j/t_k}$ between looks $j < k$, where $t_k = k/4$ is the information fraction (Chapter 11). A rule that rejects at the first look where Z_k exceeds a boundary b_k has type I error $1 - \Pr(Z_1 < b_1, \dots, Z_4 < b_4)$, which is a four-dimensional normal probability, and the O'Brien-Fleming boundary $b_k = C/\sqrt{t_k}$ has its constant C chosen so that this equals 0.025 (O'Brien and Fleming 1979; Jennison and Turnbull 2000). The package does this with `gs_boundary` and `gs_crossing_prob`. The same integral with a nonzero mean gives the power, and the differences between successive "not yet crossed" probabilities give the probability of stopping at each look, hence the expected sample size. All of this is analytic: no random numbers are drawn.

```
tt <- (1:4) / 4; looks <- 40 * (1:4)                    # four equally
 ↪  spaced looks
bound <- gs_boundary(4, alpha = 0.025, type = "obf")
drift <- (p1 - p0) / sqrt(2 * 0.375 * 0.625 / 160)      # E[Z_K] at n = 160
 ↪  per arm
h1 <- gs_crossing_prob(bound, tt, drift); h0 <- gs_crossing_prob(bound, tt)
EN <- function(h) sum(h$cross * 2 * looks) + (1 - h$total) * 2 * 160
gsm <- round(rbind(Boundary = bound, `Nominal p` = pnorm(bound, lower.tail
 ↪  = FALSE), `Stop at look, H1` = h1$cross), 4)
colnames(gsm) <- paste("Look", 1:4)
knitr::kable(gsm)
```

	Look 1	Look 2	Look 3	Look 4
Boundary	4.0486	2.8628	2.3375	2.0243
Nominal p	0.0000	0.0021	0.0097	0.0215
Stop at look, H1	0.0039	0.1795	0.3469	0.2515

```
knitr::kable(data.frame(Quantity = c("Power", "E[N] under H1", "E[N] under
 ↪  H0"), Value = round(c(h1$total, EN(h1), EN(h0)), 3)))
```

Quantity	Value
Power	0.782
E[N] under H1	262.591
E[N] under H0	318.993

The O'Brien-Fleming design needs a final critical value of $z = 2.024$, which is a nominal one-sided p of 0.0215 rather than 0.025, and in exchange it stops at the first look (40 per arm) only for $z > 4.05$, so it has almost no chance of doing so. At the planned alternative the power is 0.782, a little below the fixed design's 0.794 because the final look has been slightly tightened, and the design uses an expected 263 patients against a fixed 320. That small difference in power is why traditional group sequential designs are usually inflated by a few percent in maximum sample size (Chapter 11). These are the classical O'Brien-Fleming values (4.049, 2.863, 2.337, 2.024), the ones this book quotes. The online simulator of the companion lab (https://vsokolov.org/bayesian_trials_lab/14-simulator.html) uses Lan-DeMets O'Brien-Fleming-type spending by default, whose boundaries are 4.333, 2.963, 2.359 and 2.014, and offers the classical boundaries as a toggle; the two differ slightly at the early looks and agree on the message. The full open simulator is at https://vsokolov.org/bayesian_trials_lab/14-studio.html.

14.1.3. What goes in the traditional SAP

For either design, the SAP and protocol statistical section state: the primary endpoint and estimand (ICH E9(R1)); the analysis population; the one-sided α and the sidedness convention; the sample size with its assumptions and software; the number and timing of looks, the spending function or boundary, and who sees the interim data; the multiplicity strategy (Chapter 10); and the primary test, with the estimator of the treatment effect and its confidence interval, adjusted for the interim looks if any. The type I error is guaranteed by the theorem behind the boundary. The power is computed under one stated alternative, with a sensitivity table if the sponsor is careful. Nothing else is needed, and the regulator can reproduce all of it.

That is the standard against which a Bayesian design is judged, and it also sets the limit of the traditional method. The canonical normal joint distribution holds when the looks are prespecified in information time, when the allocation does not change, when every patient's outcome is available at the look, and when only one hypothesis is tested. Drop any of these, add a response-adaptive allocation rule, an arm-dropping rule, a posterior or predictive stopping rule, a hierarchical model that borrows information across subgroups, or a delay between enrollment and outcome, and the analysis often leaves the standard normal setting.

Some adaptive designs permit analytic or exact recursive evaluation (Chapter 15 gives one); complex combinations generally require simulation. Both conventional and Bayesian designs require justification of the complete procedure: the analysis assumptions, the multiplicity handling, the estimation and the operations. FDA's 2019 adaptive guidance says what to do then: when analytic methods are not available, evaluate the operating characteristics by simulation (Section VI.A) (U.S. Food and Drug Administration 2019a). A Bayesian design is generally in this position from the start, which is the reason the simulation report is central to it and not a supplement.

14.2. The Bayesian success rule and why it has no error rate until calibrated

The Bayesian draft guidance describes a success criterion of the form $\Pr(\delta > a \mid \text{data}) > c$: the posterior probability that the treatment effect δ exceeds a threshold a must be larger than a level c (U.S. Food and Drug Administration 2026g). In the notation of this book, with response probabilities p_0 and p_1 and $\delta = p_1 - p_0$, the criterion is

$$\Pr(p_1 - p_0 > a \mid x_0, x_1) > c.$$

With independent Beta$(1, 1)$ priors on the two response probabilities (Chapter 6), the posterior is Beta$(1 + x_j, 1 + n_j - x_j)$ for each arm, and the probability is a one-dimensional integral over p_0 of the control posterior density times the treatment posterior tail,

$$\Pr(p_1 - p_0 > a \mid x_0, x_1) = \int_0^1 \pi(p_0 \mid x_0)\,[\,1 - F_1(p_0 + a \mid x_1)\,]\,dp_0,$$

where F_1 is the treatment-arm posterior distribution function. The function `oc_post_table` evaluates this integral once for every possible pair of success counts at each look, and stores the result in a table. A simulated trial then reads off its posterior probability from the table, so the cost of a trial is two binomial draws and a lookup, and a hundred thousand four-look trials take a fraction of a second.

The rule as written has no type I error until a, c, the prior and the schedule of looks are all fixed, because the error rate is a property of the whole procedure and not of the posterior. A single look at the end of the trial with a flat prior is the simplest case: the posterior probability that $p_1 > p_0$ is, to a good approximation, the one-sided p-value's complement, so $c = 0.975$ is the familiar one-sided 2.5 percent test, with a small discrepancy from the discreteness of the data and the prior's pull. This is the agreement noted in Chapter 7. Add a second look and the same c is no longer a 2.5 percent test, and, for a design whose success criterion is calibrated to error control, the draft guidance describes choosing a and c so that it is. The pivotal BNT162b2 vaccine trial did exactly this: its analysis plan used 0.995 at the interim analyses and 0.986 at the final one, chosen to keep the overall type I error at 2.5 percent (Section 14.11).

i Technical note: the table lookup

For n patients per arm the table $T_n[x_0, x_1]$ has $(n+1)^2$ entries. Each entry is the posterior probability of superiority for one pair of success counts, evaluated by adaptive integration on the rank scale (the exact integral, to a relative tolerance of 10^{-10}, so that exchangeable states give exactly 0.5 and concentrated posteriors are not truncated); with $a = 0$ only half of the table is integrated. Looks at which all trials have the same n share a table, and tables are cached by (n, a, prior), so every simulated trial is a lookup. The posterior probabilities are numerically evaluated integrals, whereas the predictive probabilities of the Goldilocks section below are exact beta-binomial sums over the unobserved counts, with the same posterior-superiority integrals at the end states.

14.3. Why the uncalibrated rule inflates the error

The inflation is the same phenomenon as in the group sequential case (Chapter 11): each look is a new chance for noise to cross the bar. The chunk below holds the maximum sample size fixed at 160 per arm, applies the rule "declare success if $\Pr(p_1 > p_0 \mid \text{data}) > 0.975$ at any look" under the null $p_0 = p_1 = 0.30$, and varies only the number of equally spaced looks.

Table 14.4.: Type I error of the uncalibrated rule Pr(p1 > p0 | data) > 0.975, flat priors, p0 = p1 = 0.30, maximum 160 per arm, K equally spaced looks. 40,000 simulated trials per row (Monte Carlo SE at most 0.0011). The last column is the threshold that would give exactly 2.5 percent.

Looks K	Type I error (%) at c = 0.975	Calibrated c
1	2.5	0.9749
2	4.2	0.9853
4	6.3	0.9909
8	8.7	0.9942
16	11.0	0.9956
32	12.8	0.9965

With one look the rule is the one-sided test, and its error is 2.5 percent. With four looks it is 6.3 percent, with 32 looks it is 12.8 percent. The error approaches a ceiling as looks approach continuous monitoring: the working paper behind this book reports 12.0 percent for a posterior threshold of 0.975 monitored continuously in a time-to-event setting built on the SELECT cardiovascular outcomes trial, "rather than 2.5" (Polson and Sokolov 2026b). The same paper notes that an earlier two-arm binary-endpoint study found about 13.5 percent for a naive posterior threshold. The settings differ (priors, endpoints, horizon), so the figures are not interchangeable with the 12.8 percent above, but they agree in order of magnitude, and any claim that a Bayesian rule "decides sooner at the conventional threshold" is an artifact unless the error is reported next to it.

The last column of Table 14.4 is the remedy. A posterior threshold of $c = 0.975$ is a 2.5 percent test at one look, and $c = 0.991$ is a 2.5 percent test at four looks. The fix is a higher

bar for the evidence, and calibration chooses a and c so that the error is controlled. In this book's terms: against a single unadjusted analysis the Bayesian bar is never lower without an informative prior, and the saving comes from monitoring and multiplicity, not from lowering the bar (Chapter 1).

14.4. Calibrating c and a by simulation

14.4.1. The algorithm

Calibration is a one-dimensional search that can be done without any search at all. Let M be, for one simulated null trial, the largest posterior probability $\Pr(p_1 - p_0 > a \mid \text{data})$ that the trial reaches at any look it attends. The rule "success if the probability exceeds c at any look" rejects exactly when $M > c$, so its type I error is $\Pr(M > c)$, and the smallest c with type I error at most α is the $(1-\alpha)$ quantile of M over the null simulations. The function `oc_calibrate` does this. A single simulation serves every candidate c (common random numbers), which is why the table above could print the calibrated threshold for each schedule at no extra cost.

```
looks <- c(40, 80, 120, 160)
null  <- oc_simulate(0.3, 0.3, looks, nsim = 1e5, seed = 11)    # 1.
 ↪  simulate under the null
c0    <- oc_calibrate(null, alpha = 0.025)                      # 2.
 ↪  calibrate c to 2.5 percent
alt   <- oc_simulate(0.3, 0.45, looks, nsim = 4e4, seed = 12)   # 3.
 ↪  simulate the alternative
w4 <- unlist(oc_summarize(alt, c0))[c("reject", "EN", "early_succ")]  # 4.
 ↪  report the operating characteristics
knitr::kable(data.frame(Quantity = c("Power", "Expected total N", "P(early
 ↪  success)"), Value = round(unname(w4), 3)))
```

Quantity	Value
Power	0.712
Expected total N	231.384
P(early success)	0.570

In the package the sponsor's workflow is four calls, run here for the example of this chapter (four looks, flat priors, $a = 0$), and the calibrated threshold is $c = 0.9907$.

The same logic extends to every efficacy-stopping rule. If the design also stops for futility at an interim look, futility removes some trials from further looks, and M is the maximum over the looks the trial actually reaches. The futility rule is simulated as part of the null run, so the calibration accounts for it. A design in which the futility rule is non-binding, so that the success rule can still be applied after a futility signal, would calibrate on the maximum over all looks, which is the more conservative choice and the one the FDA draft's emphasis on controlling error whatever the sponsor does at interims points to.

14.4.2. Choosing a and c

The criterion has two knobs. The threshold a is a clinical quantity: the minimum effect that is worth having, so that a posterior probability near one that the effect is above $a = 0.05$ says something stronger than "the drug is better than control." The level c is a statistical quantity, set by the calibration. For a given a, the type I error constraint determines c, and the power then follows. The examples with $a > 0$ (a margin) depart from the draft's $a = 0$ calibrated superiority formulation; they show the mechanics of the criterion and are not a form the draft prescribes. The chunk below calibrates c at $a = 0, 0.05, 0.10$ and 0.15 for the four-look design, then reports the power at $p_1 = 0.45$.

Table 14.6.: Calibrated pairs (a, c) for a one-sided type I error of 2.5 percent at p0 = p1 = 0.30, four looks (40, 80, 120, 160 per arm), flat priors. Calibration on 100,000 null trials; power on 40,000 trials at p1 = 0.45 (SE at most 0.0025).

a	Calibrated c	Power at p1 = 0.45
0.00	0.9907	0.712
0.05	0.9511	0.658
0.10	0.8492	0.595
0.15	0.6707	0.520

Raising a lowers the calibrated c (from 0.9907 to 0.671) because a posterior probability of exceeding a larger effect is harder to reach by chance, and it lowers the power at $p_1 = 0.45$ from 0.712 to 0.52, since the same data now have to clear a higher effect threshold. The pair $(a, c) = (0.15, 0.671)$ requires only a two-in-three posterior probability that the effect exceeds 15 points, and is no more lenient than $(0, 0.9907)$: both are 2.5 percent tests under the null. They differ in what they reward. The $a = 0$ rule succeeds whenever the data make any benefit nearly certain, and the $a = 0.15$ rule succeeds only when the observed effect is large. The choice of a is a clinical judgment that belongs in the protocol, with its rationale; the choice of c then follows from the calibration, and the draft guidance asks that both be prespecified. Real submissions land on such pairs: the Superion spinal device trial fixed its non-inferiority success threshold at a posterior probability of 0.958, a value that makes sense only with its margin, priors and adaptive sample size, and REBYOTA fixed two success thresholds in advance and then adjusted them at study end for the information fraction actually reached (Section 14.11). The rest of the chapter uses $a = 0$ and the four-look schedule.

14.5. The worked example: a two-arm binary trial with four looks

The design is: 1:1 randomization, maximum 160 patients per arm, looks after 40, 80, 120 and 160 patients per arm, flat Beta$(1, 1)$ priors, success if $\Pr(p_1 > p_0 \mid \text{data}) > c$ at any look, with c calibrated to 2.5 percent at the design null $p_0 = p_1 = 0.30$. Outcomes are available immediately (a restriction that Chapter 15 relaxes). In terms of sample size and alternative this is the fixed design and the O'Brien-Fleming design of the previous section, so the three can be compared on the same footing.

14.5.1. Type I error across the null space

The calibration was done at one control rate, and a regulator's first question is whether the error is still 2.5 percent at the others. The traditional design, being a test on a standardized difference, is nearly insensitive to the control rate. A Bayesian posterior probability is not a function of the standardized difference alone: for a binary endpoint, the posterior probability for a given observed difference depends on the success counts, through the variance, so the error rate can drift. The chunk simulates 100,000 trials at each control rate with $p_1 = p_0$ (the null grid) and applies the c calibrated at $p_0 = 0.30$. Its last column calibrates each control rate on its own, and the largest of them is the worst-case threshold.

Table 14.7.: Type I error of the four-look Bayesian rule across control rates p0 (p1 = p0), at the threshold c calibrated at p0 = 0.30, and the threshold that would give exactly 2.5 percent at each p0. 100,000 simulated trials per row.

Control rate p0	Type I error (%) at the design c	Calibrated c at this p0
0.1	2.27	0.9890
0.2	2.44	0.9904
0.3	2.43	0.9904
0.4	2.50	0.9907
0.5	2.49	0.9906
0.6	2.59	0.9910
0.7	2.52	0.9910
0.9	2.33	0.9895

Table 14.7 shows the error: 2.27 to 2.59 percent across the grid. It is lowest near the edges of the range, where the data are most nearly deterministic, and highest at $p_0 = 0.6$ to 0.7; the design null $p_0 = 0.30$ itself reads 2.43 percent on this fresh sample. The Monte Carlo standard error at 100,000 trials is 0.05 percentage points, so the excess of 0.09 points at the worst grid point is under two standard errors: probably noise, but a careful reviewer would want it removed. The last column shows what each control rate would need on its own, and the largest of them, $c = 0.991$, is the worst-case calibration: with that threshold every grid point is at or below 2.5 percent in the calibration sample. The design null is chosen by the sponsor, but the error constraint is over the whole null space, and the calibrated threshold is the largest of the thresholds that each point in the null space requires.

The difference between the calibration at 0.30 and the worst case is 0.0003 in c, which looks negligible, and for the power it is. The point is the procedure. A sponsor who reports only the type I error at the design null has reported the easy half; the guidance's phrase is a familywise error rate controlled "at a given level," and the reviewer can reasonably read it as "everywhere the null is true."

14.5.2. The null boundary

The null hypothesis in a superiority trial is composite: $p_1 \leq p_0$, not only $p_1 = p_0$. The simulation below examines treatment rates below control (Table 14.8).

Table 14.8.: Probability of declaring success when the treatment rate is below the control rate of 0.30, four-look Bayesian rule at the calibrated c. 100,000 simulated trials per row.

True effect p1 - p0 (p0 = 0.30)	Probability of declaring success (%)
0.00	2.455
-0.05	0.308
-0.10	0.033
-0.20	0.000

The probability falls from 2.5 percent at the boundary to 0.31 percent at -0.05, 0.03 percent at -0.10 and zero at -0.20. The supremum over the null hypothesis is attained on the boundary $p_1 = p_0$, and the null boundary search reduces to a one-dimensional search over the control rate. This is the usual result for a monotone rule, and it is why the null grid of the previous table is a grid over p_0 only. A design with a nonmonotone rule, such as one that selects among several arms or borrows from an external control with a conflicting rate, has no such guarantee and needs a grid in every nuisance dimension: the borrowed control rate, the time trend, the number of arms that are truly null (Chapter 19, Chapter 20).

14.5.3. Power, expected sample size, early stopping and bias

For each alternative from $p_1 - p_0 = 0$ to 0.25 the next chunk simulates 100,000 trials. It reports three designs: the Bayesian rule with calibrated $c = 0.9907$ and no futility stopping; the same rule with a futility stop at the first three looks if $\Pr(p_1 > p_0 \mid \text{data}) < 0.10$, recalibrated; and, from the analytic section, the fixed 160-per-arm design.

Table 14.9.: Operating characteristics of the four-look Bayesian design (flat priors, a = 0, maximum 160 per arm, 100,000 trials per row). Power is the probability of declaring success; E[N] is the expected total sample size (both arms); Early is the probability of stopping for success at looks 1 to 3; Fut is the probability of stopping for futility; bias is the average of (observed difference minus true difference) among all trials, and among trials that stopped early for success. The columns marked fut. are for the same design with the futility rule added.

Effect	Fixed power	Power	E[N]	Early	Bias, all	Bias, early	Power, fut.	E[N], fut.	Fut
0.00	0.025	0.025	317	0.020	0.002	0.217	0.025	282	0.183
0.05	0.158	0.127	305	0.094	0.009	0.163	0.127	293	0.060
0.10	0.470	0.383	277	0.281	0.019	0.118	0.383	274	0.016
0.15	0.794	0.713	232	0.569	0.028	0.080	0.713	231	0.004
0.20	0.956	0.927	181	0.823	0.031	0.052	0.927	181	0.001
0.25	0.995	0.990	140	0.955	0.029	0.035	0.990	140	0.000

Read the table row by row.

At $p_1 - p_0 = 0$ it is the type I error row: 2.5 percent in this particular run, within Monte Carlo error of 2.5, and a fixed 160-per-arm design has the same 2.5 percent. The expected

sample size under the null is 317 of a maximum 320, because the only way to stop early is by a false success.

At 0.15 (the alternative that the fixed design was powered for), the Bayesian four-look design has power 0.713, against 0.794 for the fixed design and 0.782 for the O'Brien-Fleming design. The cost of four looks at a flat threshold is about 8 percentage points of power at a fixed maximum sample size, more than the O'Brien-Fleming design pays. In return, the design stops early for success 57 percent of the time and uses an expected 232 patients instead of 320, about 28 percent fewer. The posterior threshold is constant across looks (0.9907 at every look), so it is closer to a Pocock boundary than to an O'Brien-Fleming one: it spends more of its error early (Chapter 11), which buys earlier stopping and costs power at the end. Changing the threshold by look, or setting the sample size so that the Bayesian design is powered to 80 percent, are both design decisions to be made by simulation, and they would move this table.

At large effects the picture is more favorable. At 0.20 the power is 0.927 with an expected 181 patients, and at 0.25 it is 0.99 with 140.

The futility rule does what a futility rule should do and nothing else. Under the null it stops 18 percent of trials, cutting the expected sample size from 317 to 282. At 0.10 it stops 1.6 percent of trials and costs no measurable power. At 0.15 or above it stops under 0.5 percent. The calibrated c does not change when this futility rule is added (the value $c_f = 0.9907$ against $c = 0.9907$ for efficacy only), because a trial that would stop for futility at an early look almost never would have crossed the efficacy bar later. A different futility rule, one stopping more aggressively, would change both the power and the calibrated c, and that is the situation in which the order "calibrate after the futility rule is fixed" matters.

The two bias columns are the ones the 2019 guidance's second principle asks for ("estimation of treatment effects should be sufficiently reliable"). Averaged over all trials, the observed difference is nearly unbiased under the null (bias +0.002) and overstates the true effect by about 0.03 at the large alternatives: because the trial stops when the data look good, the final estimate is taken at a favorable moment. Among trials that stopped early for success the overstatement is larger, 0.217 under the null (where an early success is a fluke), 0.08 at $p_1 - p_0 = 0.15$ and 0.035 at 0.25. A trial that stops early with only 40 or 80 patients per arm needs a large observed difference to cross a bar of 0.9907, so its estimate is high by construction. A report that gives only the posterior probability and the estimate at stopping, without this table, is not showing what the estimate means. Remedies are bias-adjusted estimators, shrinkage priors, and confirmatory follow-up; the guidance wants the bias quantified before any of them is chosen.

14.5.4. Where the calibration noise comes from

The calibrated c is itself an estimate: a quantile of a simulated distribution. How many simulations are needed depends on what is being estimated. The Monte Carlo standard error of a simulated rejection probability is $\sqrt{p(1-p)/B}$ for B trials. At $p = 0.025$, $B = 10{,}000$ gives 0.0016, which is 6 percent of the target (a reported 2.7 percent is within noise of 2.5); $B = 100{,}000$ gives 0.0005, and $B = 1{,}000{,}000$ gives 0.00016. To show a type I error of 0.0250 with a 95 percent interval of ± 0.0005 requires about 380,000 trials. The calibrated threshold inherits this noise, and a design calibrated on a small simulation can be miscalibrated on the fresh data that will actually decide the trial. The chunk below calibrates c on B trials,

repeats the exercise with 30 different seeds, and then evaluates the realized type I error of each calibrated c on one common independent sample of 400,000 null trials.

Table 14.10.: Monte Carlo error. Binomial standard error of an estimate of 0.025, and the effect of calibrating c on B null trials: mean and SD (over 30 independent calibrations) of c, and the mean and maximum realized type I error (percent) of the calibrated c on 400,000 fresh null trials (SE 0.00025).

B	Binomial SE at 0.025	Mean c	SD of c	Mean realized error (%)	Max realized error (%)
2,000	0.0035	0.9907	0.0013	2.46	3.24
10,000	0.0016	0.9906	0.0005	2.46	2.82
100,000	0.0005	0.9907	0.0002	2.45	2.58

With 2,000 calibration trials the realized error is spread across a wide range, and the worst of 30 calibrations is far above the target. At 100,000 the worst of the 30 calibrations realizes 2.58 percent. The practical conclusions are three. First, calibrate with at least 10^5 trials, and more near a boundary one needs to defend. Second, validate the final design on an independent simulation with a different seed: the validation estimate is unbiased, the calibration estimate is not (it was chosen to hit the target). Third, report the Monte Carlo standard error beside every rate in the simulation report, as the checklist at the end of this chapter requires. A reported "0.0252 against a target of 0.025" with no standard error invites the question of whether it is a violation.

14.5.5. The three panels: null grid, power and expected sample size

The three panels of Figure 14.1 show the null grid, the power curves and the expected sample size.

The worst-case threshold moves c by only 0.0003, so its effect on power is negligible, and it pulls the null curve down toward the line. In the middle panel the Bayesian design has power 0.713 at an effect of 0.15, below the fixed design's 0.794 and the O'Brien-Fleming design's 0.782; the right panel shows the expected sample size, which falls from 317 at the null to 140 at an effect of 0.25. In this example the Bayesian design is a group sequential design with a particular boundary: nothing about the posterior probability makes it more efficient than a well-chosen frequentist boundary, and the benefits that come from Bayesian adaptive designs in later chapters (borrowing, predictive stopping, dose selection, response-adaptive allocation) come from structure that a group sequential design does not have. The agreement of the single-look case with the exact enumeration is itself a check on the code (Section 14.9).

14.6. Scenario grids, the worst case, and what the grid must span

A simulation study is a set of scenarios, and the scenarios decide whether the report convinces. The FDA's 2019 guidance asks for a range of plausible parameter values, including the null space; the Bayesian draft asks that the operating characteristics be shown across the effect sizes of interest, and, for designs that are not calibrated to type I error, that the probability

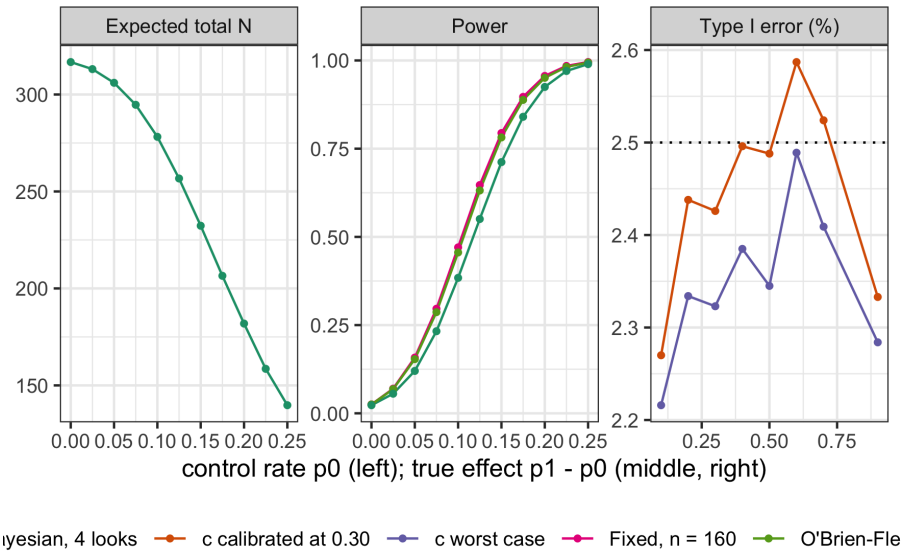


Figure 14.1.: Left: type I error across control rates, with the threshold calibrated at p0 = 0.30 and with the worst-case threshold; dotted line at 2.5 percent. Middle: power of the fixed design, the O'Brien-Fleming design (analytic) and the four-look Bayesian design (simulated) against the true effect. Right: expected total sample size of the Bayesian design.

of a positive conclusion be reported over the range (U.S. Food and Drug Administration 2019a, 2026g). The following structure covers what reviewers usually need.

Efficacy scenarios. The global null; the design alternative; an intermediate alternative (half of the design effect is the convention); a small-effect alternative close to the minimum clinically relevant effect a; and a large effect. For multi-arm designs add: one effective arm among several, a monotone dose response, a peaked dose response, and several near-maximal arms. A table of power and the probability of incorrect selection (choosing a wrong arm) by scenario is the standard report.

Nuisance scenarios. For a binary endpoint, the control rate (as in the null grid above); for a continuous endpoint, the standard deviation; for time-to-event, the control hazard and accrual pattern. These are the parameters that the type I error must be robust to, and where the "worst case null" is found: the supremum of the error over the nuisance space under the null. A grid is a pragmatic search, and one reports the grid spacing and where the maximum fell. If the maximum falls at the edge of the grid, the grid was too small.

Operational scenarios. Accrual rate (slow, expected, fast), dropout, outcome delay, site activation, and, for platform trials, arm arrival. A phase 2 platform screening design with fixed-size arms and a limit on concurrent arms can fail in this way: a single effective arm that arrives late may never be started, a failure of operations and not of statistics, visible only in a simulation that models arrivals. Operational assumptions can dominate the efficacy assumptions.

Misspecification scenarios. Time trends in the control rate (when non-concurrent controls are used), prior-data conflict (when a prior from external data is used), non-proportional

hazards, and heterogeneity. The point of these is to show how the operating characteristics degrade when the model is wrong, and what the monitoring plan does about it. The draft guidance's emphasis on a prespecified prior and on the effect of the prior on the operating characteristics is, in simulation terms, a request for a prior-conflict scenario grid (Chapter 8).

Sampled-effect scenarios. When the sponsor has a prior over effects, as in a portfolio or a phase 2 program, the true effect may be sampled from it, and the report gives probabilities by the true class of the effect (null, mediocre, good) in addition to the overall rates. This gives the assurance-style number (the probability of success averaged over the prior, Chapter 9), but the report must also give the rates conditional on the true state, because the type I error is a statement about the null only.

For each of these, the design is fixed first, then run across all scenarios without modification. Tuning a design to make the scenario table look good and then reporting the table is legitimate, if the final design is then frozen and revalidated on fresh seeds; the report should say that the tuning happened and which scenarios drove it. The "worst case" of the type I error is not itself a probability under a model: it is the maximum over a set of models the sponsor decides to consider, and the report should state what set that is.

14.7. The Goldilocks design

The industrial practice of Bayesian adaptive design (associated with Berry Consultants and the simulator FACTS) has one organizing principle: the design is chosen by simulation rather than derived in closed form. It describes its own method as flight-simulating a trial: a structure is specified, many assumed truths are simulated, and the design is judged by how it behaves across all of them (Polson and Sokolov 2026b; Scott M. Berry et al. 2010). The operating characteristics reported are power, type I error, expected sample size, expected duration, assurance and the probability of correct selection. The published comparison with conventional packages is explicit about the division of labor: those packages are built for analytic type I error control and closed-form sample size, while the simulator is built for structures that have no closed form, with conventional group sequential and promising-zone designs available as special cases.

The idea is older than the industry, and its history explains why a Bayesian design is judged by frequentist operating characteristics. Lewis and Berry's classical evaluation of Bayesian decision-theoretic designs (Lewis and Berry 1994b) illustrates the operating-characteristic perspective used here; Section 1.9 and Section 15.6 give the historical development and backward-induction construction. P. F. Thall and Simon (1994) chose Bayesian phase 2 stopping rules by their frequentist operating characteristics. P. Müller et al. (2007) formalized simulation-based sequential Bayesian design, the approach of the adaptive dose-ranging designs of D. A. Berry et al. (2002). The principle that a Bayesian procedure should be calibrated to long-run frequency behavior is stated by Rubin (1984), Spiegelhalter, Freedman, and Parmar (1994) and Little (2006), and the calibration of this chapter is that principle applied to a posterior threshold.

The signature construction is the Goldilocks sample size design of Broglio, Connor and Berry (K. R. Broglio, Connor, and Berry 2014): a trial that does not fix its sample size but lets accrual run until the data say the sample is "not too big, not too small." At each look it computes two predictive probabilities, $\mathrm{PP}_{\mathrm{stop}}$ (success if enrollment stops now and the pending outcomes are completed) and $\mathrm{PP}_{\mathrm{max}}$ (success at the maximum sample size). It

stops enrolling for expected success when the first is high and stops for futility when the second is low; the construction and a worked example are in Section 15.5. The full design of Broglio and colleagues also has an immediate-success rule, a higher threshold on $\mathrm{PP}_{\mathrm{stop}}$ at which success is declared without following the pending patients; it is not used here, where outcomes are immediate. The sample size is an output, bounded by $N_{\max}$, not a design input, and the thresholds are calibrated by simulating type I error, not derived. Where this construction sits among the calibration devices, relative to the spending boundary and the moderate-deviation boundary of the working paper, is an open question there: the working paper conjectures a moderate-deviation rule with a data-dependent constant, and does not claim a proof. Predictive probabilities, the choice of thresholds, and the delayed-outcome model are developed in Chapter 15.

For a trial with immediate outcomes, as in the example, $\mathrm{PP}_{\mathrm{stop}}$ collapses to an indicator of whether the posterior rule is already met, and what is left of Goldilocks is the futility rule based on $\mathrm{PP}_{\max}$. The chunk below uses the exact beta-binomial predictive probability of the posterior rule being met at 160 per arm, stops for futility if it falls below $F = 0.05$ at any of the first three looks, and recalibrates c by iteration, because c appears both in the success rule and inside the predictive probability.

Table 14.11.: Goldilocks-style futility (stop if the predictive probability of success at 160 per arm is below 0.05) against efficacy-only stopping, four looks, flat priors, c recalibrated to 2.5 percent. 40,000 trials per row for the null and 40,000 per alternative (SE at most 0.0025).

Effect	Power, efficacy only	E[N]	Power, with P_max futility	E[N]	P(futility stop)
0.00	0.026	316	0.026	150	0.902
0.05	0.124	305	0.121	189	0.691
0.10	0.384	277	0.371	213	0.401
0.15	0.715	232	0.694	206	0.164
0.20	0.926	182	0.911	173	0.047

Under the null the predictive futility rule stops 90 percent of trials before the end and cuts the expected sample size from 316 to 150, more than half. It does this at a price in power of up to 2.2 percentage points, largest at the design alternative 0.15 (0.715 falls to 0.694), where a trial with a true moderate benefit has a real chance of looking hopeless at 40 or 80 per arm. That price is the cost of the futility rule, and the table shows it. A sponsor who chooses F by looking at this table is choosing a point on the frontier between expected sample size under the null and power against modest effects, which is the decision that the Goldilocks name refers to.

The Goldilocks sketch above omits the part that needs a delayed-outcome model (the stop-accrual rule and the completion of follow-up), and the design in a real FACTS project would add arms, longitudinal models for early outcomes, and a different success rule. DAWN's analysis plan is a trial-sized version of this construction, with predictive-probability stopping thresholds that fall with sample size and a success threshold that rises when the population is enriched (Section 14.11).

The same practice has built platform trials: I-SPY 2, a breast cancer platform enrolling since 2010 that shares controls, drops arms early for futility or efficacy, reinvests freed resources in new agents and uses longitudinal imaging to predict its pathologic response endpoint; GBM

AGILE, a glioblastoma platform; and the critical care and pandemic platforms including REMAP-CAP and the ACTIV family. Basket designs include a nine-cancer study of a combination in rare mutation-defined tumors whose primary analysis borrows across the nine groups, and which supported a pan-tumor approval in 2022 (Polson and Sokolov 2026b). These are the trials in which simulation is the only available route, and they are the subject of Chapter 20 and Chapter 19 (the I-SPY 2 and GBM AGILE decision rules are in Section 15.10). One caution in the same source: the public material contains no measured efficiency claims. The benefits asserted (smaller samples, fewer patients on placebo, lower cost and shorter timelines) have no quantity attached to them, which is why the numbers in this chapter are computed, not quoted.

14.8. What the simulation report must contain

The FDA's adaptive designs guidance, the Bayesian draft and ICH E20 differ in detail and agree in substance. Section VI of the 2019 guidance treats simulation as the way to estimate the operating characteristics of designs that have no analytic solution and says what a submission should document: the simulation plan, the software or code, the number of replicates and the results, across a range of plausible values that includes the null space (U.S. Food and Drug Administration 2019a). The Bayesian draft separates designs calibrated to type I error control (for which the key characteristic is the type I error, controlled at a given level, typically 0.025 one-sided) from designs that are not (where the sponsor reports the probability of a positive conclusion across effect sizes, along with the bias and variability of the estimates), and asks for extensive documentation of simulations, code and software, and for the effect of the prior on the operating characteristics (U.S. Food and Drug Administration 2026g). ICH E20 defines a simulation as "the repeated execution of a large number of hypothetical clinical trials" in Section 5.2 and, in Section 5.3, says that Bayesian adaptive designs must show targeted operating characteristics and maintain type I error control (International Council for Harmonisation 2025a). The three documents are drafts or guidance, not law, and a sponsor can propose another approach; but a submission that omits what they list is unlikely to survive review without a good reason.

A structure that satisfies all three is the ADEMP framework of Morris, White and Crowther: aims, data-generating mechanisms, estimands, methods and performance measures (Morris, White, and Crowther 2019). Table 14.12 maps the elements of a simulation report to it and to the source of each requirement.

Table 14.12.: Contents of a simulation report.

Element	What the report states	Where it comes from
Aims	The decisions the design supports; the target operating characteristics (α, power, maximum N)	FDA 2019 Section III; E20 Section 5.2
Design as simulated	Every rule: priors, success criterion (a, c), looks (by information, events or calendar), futility, allocation, selection; binding or non-binding futility	FDA Bayesian IV.A; E20 Section 5.2

Element	What the report states	Where it comes from
Calibration	How c (and a) were chosen; the null scenario used; the worst-case search over nuisance parameters; the grid	FDA Bayesian IV.A.1
Scenarios	All data-generating parameters of each scenario, and why chosen; null space, misspecification, operational	FDA 2019 Section VI.A
Estimand	The primary estimand per hypothesis and how intercurrent events are simulated	ICH E9(R1)
Performance measures	Type I error (per hypothesis and familywise); power by scenario; expected and maximum N (enrolled, not only completers); probability of stopping at each look and why; bias, variability, coverage of the estimates at stopping	FDA 2019 Section III; FDA Bayesian
Monte Carlo error	Number of replicates and the standard error of every reported rate	FDA 2019 Section VI
Prior	The prior, its justification and a sensitivity analysis of the operating characteristics to it	FDA Bayesian draft, Section V
Software and reproducibility	Program, version, random number generator, seeds, MCMC settings if any, the code itself	FDA 2019 Section VI; FDA Bayesian
Example trials	A handful of individual simulated trials plotted, so that a reader can see how the rules behave	Practice (Berry Consultants), E20
Traceability	The simulated design is the protocol and SAP design, line by line, with a change log	E20 Section 5.2

Three items in this table are where reports most often fall short. The first is the enrolled-versus-completed distinction. A trial with a 24-week endpoint that stops when 80 patients have completed has probably enrolled more: the analytic expected sample size of a group sequential design counts completers, while a simulation that models accrual counts everyone enrolled, which is larger because patients still in the pipeline count when the trial stops. The second is binding and non-binding futility: if the data monitoring committee can ignore a

futility recommendation, the type I error must be calibrated with the futility rule ignored. The third is traceability: the code that was simulated and the rules in the protocol must be the same rules, and the easiest way to ensure this is to generate both from one specification.

Two practice sources complement the regulatory documents. Mayer et al. (2019) survey simulation practices for adaptive designs in drug and device development, the practitioner counterpart of Table 14.12. And when the trial itself is reported, the CONSORT extension for adaptive designs (ACE) of Dimairo et al. (2020) is the checklist that journals ask for: its items on the pre-specified adaptations and on how they can affect the estimates correspond to the rows "Design as simulated" and "Performance measures" above, so the simulation report supports ACE reporting of design properties and estimation methods. ACE additionally calls for reporting the adaptations actually made, departures from planned rules, participant flow and arrangements protecting trial integrity.

In the protocol: language for the simulation section

[Template: bracketed values, Table X and Appendix Y are placeholders to be completed by the sponsor.] "The success criterion is $\Pr(p_1 - p_0 > 0 \mid \text{data}) > c$ at any of four looks (40, 80, 120 and 160 patients per arm), with independent Beta(1,1) priors. The threshold c was calibrated by simulation so that the familywise one-sided type I error rate is at most 0.025 across a grid of control rates (0.1 to 0.7 in steps of 0.1, and 0.9) and under all null hypotheses $p_1 \leq p_0$, using 10^5 simulated trials per scenario; the resulting c is [value], and the type I error was verified on an independent run of 4×10^5 trials with a different seed (maximum observed error [value], Monte Carlo SE 0.00025). Operating characteristics for the scenarios in Table X are in the simulation report (version, date, SHA of the code), which is part of this protocol. The simulation code, random seeds and software versions are provided in Appendix Y."

14.9. Software, reproducibility and what to submit

Four kinds of software appear in submissions. Commercial design packages include FACTS (Berry Consultants), which has core engines for continuous, dichotomous and time-to-event endpoints, dose-finding and staged designs, and a platform-trial engine, and East (Cytel), the industry standard for group sequential and frequentist adaptive designs. Analytic open-source packages such as rpact and gsDesign in R compute group sequential boundaries, power and sample size reassessment without simulation, and are the right tool for the traditional half of a design (Wassmer and Brannath 2016; Jennison and Turnbull 2000). Open-source simulation packages for Bayesian adaptive designs exist in R (adaptr, and the `bayestrials` package used in this book, for example) and are validated by their users. Custom code in R, Python or Stan is needed for bespoke platforms or borrowing, and with custom code the sponsor owns the verification. The regulator asks for the code.

Whichever is used, verification is the sponsor's job, and there are three cheap tests. Reproduce a known analytic result: with one look and $c = 0.975$ the simulator's type I error should be close to the exact enumeration of the fixed design (0.0252 above; the check below gives a little under that, because the Bayesian rule and the pooled z test differ slightly at discrete counts), and a Bayesian rule with a boundary borrowed from a group sequential design should reproduce the analytic rejection probability, so the code is checked at the points where the answer is known. Check invariances: the type I error should not depend on the

label of the arms, and a design with $K = 1$ should be the fixed design. Run the simulation twice with different seeds and compare: the difference should be of the size of the Monte Carlo error. Never leave debug settings in a final run (one burn-in sample to speed up a Markov chain Monte Carlo fit makes every Bayesian output meaningless while the frequentist outputs look fine).

```
one <- oc_simulate(0.3, 0.3, 160, nsim = 4e5, seed = 3)      # K = 1 must
 ↪  reproduce the fixed design
w1 <- unlist(oc_summarize(one, 0.975))[c("reject", "se_reject")]
knitr::kable(data.frame(Quantity = c("Type I error", "Monte Carlo SE"),
 ↪  Value = round(unname(w1), 5)))
```

Quantity	Value
Type I error	0.02456
Monte Carlo SE	0.00024

Reproducibility means fixing the random number generator and the seed, recording the software and package versions, and archiving the code and outputs with the report. In this book every simulation passes `seed =` to `oc_simulate`, which sets the generator before drawing. Parallel runs should use independent streams, for example L'Ecuyer-CMRG streams from `parallel::nextRNGStream`, not the same seed on every worker. A one-line record of the environment goes in the report:

```
knitr::kable(data.frame(Item = c("R", "bayestrials", "RNG"),
  Version = c(R.version.string,
   ↪  as.character(packageVersion("bayestrials")), paste(RNGkind(),
   ↪  collapse = "/"))))
```

Item	Version
R	R version 4.5.2 (2025-10-31)
bayestrials	0.1.0
RNG	Mersenne-Twister/Inversion/Rejection

The final run that supports the submission should be a clean execution of the frozen code from a frozen specification, not the last of a sequence of exploratory edits.

In submitting, the FDA's Complex Innovative Trial Design paired meeting program (Chapter 3) expects a simulation report ahead of the meeting; the report is then the object under discussion, and the agency's comments are on its scenarios and calibration. The practice for a late-phase confirmatory trial is the same: send the simulation plan before the trial starts, send the code if asked, and expect questions about the null space and the prior (Chapter 23). A simulation done after the data are in is not a prespecified operating characteristic, however carefully it is done.

14.10. Traditional and Bayesian side by side

	Traditional (fixed or group sequential)	Bayesian adaptive, calibrated
Decision rule	$Z_k > b_k$ at look k	$\Pr(\delta > a \mid \text{data}) > c$ at look k
Type I error	Computed analytically from the joint normal distribution of the Z_k	Computed by simulation; c (and a) chosen so that it is 0.025 one-sided
Power and expected N	Analytic for stated alternatives	Simulated across a scenario grid
What the SAP states	Sample size, boundaries or spending function, test	The above, plus the prior, the calibration and the simulation report
What changes under complexity	Closed form fails; simulation is added	Simulation is usually the method (exact recursion for some rules)
Agreement	With a flat prior and a normal likelihood the posterior probability of benefit is $1 - p$ for the one-sided p-value at a single look	The same, for a single look; calibration targets the overall probability of any false efficacy conclusion across the looks, and the distribution of that error across looks need not match the traditional design
What the regulator asks	Is the boundary prespecified and the multiplicity handled?	Is the simulation complete, reproducible, and is the null space covered?

Where the traditional design is the better choice, it is for reasons that this chapter makes concrete. When the design is a single look or a standard group sequential design with a known boundary, the analytic computation is exact, cheaper to verify and carries no Monte Carlo error: a reviewer can reproduce it in a minute. A sponsor that cannot commit to the prespecification and the code that a simulation-based design requires should not choose one.

14.11. In practice

Four submissions show what a calibrated posterior threshold looks like in a protocol, a statistical analysis plan or a device summary.

14.11.1. BNT162b2: thresholds chosen for a 2.5 percent overall error

The pivotal phase 2/3 trial of the Pfizer and BioNTech vaccine (protocol C4591001) is described, with its model and clinical result, in Section 21.5, and its posterior computation

in Chapter 6. For this chapter only the calibration matters. The analysis plan on ClinicalTrials.gov sets the posterior threshold at 0.995 at an interim analysis and 0.986 at the final analysis (164 first primary cases), chosen so that the overall one-sided type I error is 2.5 percent (Pfizer Inc. 2022). The futility rule of the plan is described in Section 15.10. The lesson for this chapter is the shape of the calibrated rule: a single posterior probability does not control the error across looks, so the threshold is stricter at the interim looks, which spend little of the error, than at the final analysis (0.995 early, 0.986 at the end). The plan posted on ClinicalTrials.gov is one of several amended versions.

14.11.2. REBYOTA: two thresholds, adjusted for the information fraction

The phase 3 trial of RBX2660 for recurrent *C. difficile* infection (the full case is in Section 19.17.1) chose two posterior thresholds as analogues of one-sided type I error rates of 0.00125 and 0.025 without borrowing, and at study end adjusted them for the information fraction actually reached (0.9993275 and 0.9750338) (U.S. Food and Drug Administration, Center for Biologics Evaluation and Research 2022). The result, a posterior probability of superiority of 0.991, met the second threshold but not the first. The lesson is the one the Lan-DeMets method teaches for z boundaries: a threshold calibrated for a planned information fraction is recomputed for the realized one, and the rule for doing so belongs in the plan before the data. A posterior of 0.991 is also close to the $c = 0.9907$ of this chapter's example, which 0.991 only just clears: a result well above the single-look 0.975 can sit at the edge of a calibrated multi-look bar.

14.11.3. DAWN: a success threshold that moves with the enrichment

The DAWN trial of thrombectomy 6 to 24 hours after stroke used a Bayesian adaptive enrichment design. Its analysis plan takes a baseline success threshold of 0.986 on the posterior probability of superiority, chosen by simulation to control the type I error, and raises it if enrichment occurs: the threshold is $\Phi(\sqrt{1 + N_2/(N_1 + N_3)}\ \Phi^{-1}(0.986))$, where N_1 patients are in the enriched population at the decision, N_2 outside it and N_3 enrolled after it (Stryker Neurovascular 2017). As an illustration, if the patients outside the enriched population were half of the rest the threshold would be 0.9964. The predictive-probability stopping schedule of the plan is in Section 15.10. The trial stopped at the first efficacy look with 206 patients and 49 percent against 13 percent functional independence (Nogueira et al. 2018). The lesson is that calibration extends to adaptations: every branch of the design (enrich or not, stop or not) needs its threshold computed so that the error is controlled on all of them.

14.11.4. Superion: a device threshold of 0.958

The Superion interspinous spacer trial (PMA P140004) randomized against a comparator device with a Bayesian adaptive sample size of 250 to 350 evaluable patients, a non-inferiority margin of 10 percent, flat Beta(1,1) priors and success if the posterior probability of non-inferiority was at least 0.958, a value chosen to control the type I error below 0.05; the stated power was 86.3 percent (U.S. Food and Drug Administration, CDRH 2015). The final modified intention-to-treat sample was 190 against 201 patients, and the posterior probability of non-inferiority was 0.9927 (0.9944 per protocol), above the threshold, with approval on 20 May 2015. The lesson is that 0.958 is not a conventional number: it is what a simulation

of this particular margin, prior, error budget and adaptive sample size returned, and the device summary reports it as such. The practice of calibrating a Bayesian threshold to a frequentist error rate by simulation is described in the FDA's 2010 guidance on Bayesian statistics in medical device clinical trials (U.S. Food and Drug Administration 2010), and in the accounts of reviewers who handled such submissions (Pennello and Thompson 2008b; Campbell 2011).

Try it in the lab

Open the adaptive Bayesian module at https://vsokolov.org/bayesian_trials_lab/09-adaptive-bayesian.html. Before moving the "Bar" slider, predict what happens to the false-positive rate when the trial is allowed to stop at any interim look at the same posterior threshold, then check. Then open the Bayesian trial simulator, module 14, at https://vsokolov.org/bayesian_trials_lab/14-simulator.html: set the null, vary the number of looks and the threshold, and find the threshold that holds the false-positive rate at 2.5 percent, as in Table 14.4. Its group sequential comparison uses Lan-DeMets O'Brien-Fleming-type spending by default (4.333, 2.963, 2.359, 2.014), with the classical boundaries (4.049, 2.863, 2.337, 2.024) as a toggle. The full open simulator is at https://vsokolov.org/bayesian_trials_lab/14-studio.html.

Key points

- A traditional SAP states analytic operating characteristics: the type I error follows from the sampling distribution of the test statistic, and power is computed at stated alternatives. Complex adaptations often require simulation, although some posterior and predictive stopping rules permit analytic calculation or exact recursive enumeration (FDA 2019 Section VI.A describes simulation as a major use).
- The Bayesian success rule $\Pr(\delta > a \mid \text{data}) > c$ has no type I error until the whole procedure (priors, looks, a, c) is fixed. At $c = 0.975$ with four looks the error in the example is 6.3 percent, and with 32 looks 12.8 percent; the working paper behind this book reports 12.0 percent for continuous monitoring in a SELECT-based setting.
- Calibration chooses c (given a) as the $(1 - \alpha)$ quantile of the maximum posterior probability under the null. In the example $c = 0.9907$ gives 2.5 percent for four looks, and the worst case over control rates needs about 0.991. Real plans state such calibrated thresholds: 0.995 and 0.986 for BNT162b2, 0.9993275 and 0.9750338 for REBYOTA, 0.958 for Superion.
- The type I error must hold over the null space, not at one point: report a grid over the nuisance parameters, the location of the maximum, and the boundary $p_1 < p_0$.
- Report power, expected sample size, probability of stopping early, and the bias at stopping. In the example the four-look design stops early for success 57 percent of the time at the design alternative, saves about 28 percent of the sample, loses about 8 points of power against the fixed design, and overstates the effect by about 0.08 among the trials that stop early.
- Monte Carlo error is binomial: 0.0016 at 10,000 trials for a rate of 0.025, 0.0005 at 100,000. Calibrate on at least 10^5 trials, and validate on a fresh seed.
- The Goldilocks design lets the sample size be an output, bounded by $N_{\max}$, governed by predictive probabilities of success; its thresholds are calibrated by simulation, not derived.

- A simulation report states aims, scenarios, every rule, the calibration, performance measures with Monte Carlo errors, the prior sensitivity, software, seeds and code, and it must match the protocol line by line (FDA Bayesian draft IV.A.1; FDA 2019 Section VI; ICH E20 Sections 5.2 and 5.3).

Discussion questions

1. The calibrated threshold for four looks is $c = 0.9907$, which is more demanding than the 0.975 of a one-sided test. A sponsor argues that this makes the Bayesian design "less efficient" than a fixed trial. Using the table of operating characteristics, say what the design buys in return, and under which scenarios it does not pay for itself.
2. A reviewer finds that the type I error is 2.59 percent at a control rate of 0.60, though the design was calibrated at 0.30. Is this a violation, a Monte Carlo artifact, or both? What would you do to decide, and how many simulated trials would you need?
3. Choose between $(a, c) = (0, 0.9907)$ and $(0.10, 0.849)$, both of which give 2.5 percent type I error. What clinical and regulatory arguments favor each, and what does the second pair do to power at small effects?
4. The estimate at stopping is biased upwards among trials that stop early for success. Should the protocol report the naive estimate, a bias-adjusted estimate, or both? What would a regulator accept?
5. A platform design has no closed-form operating characteristics. List five scenarios you would insist on seeing in the simulation report, including at least one operational and one misspecification scenario.

15. Predictive Probability, Interim Monitoring, and Stopping

Every interim look asks the same question in different words: given what we have seen, what will happen if we carry on? A data monitoring committee (DMC) that sees a borderline interim result wants to know whether the trial can still succeed, how many more patients it would take, and whether the money and the patients' time are better spent elsewhere. The traditional trial answers with conditional power, a number computed from the interim data and an assumed effect. The Bayesian trial answers with the predictive probability of success, the same calculation averaged over what the data say about the effect. This chapter builds both, shows that they are close relatives, and then follows the idea to its logical end: a stopping rule that is optimal for a stated loss, found by working backward from the end of the trial.

The chapter uses three tools that appear in the earlier chapters. The beta-binomial posterior is from Chapter 6. The group sequential boundaries and the error-spending logic are from Chapter 11. The simulation report that every design must carry is from Chapter 14.

> **!** Regulatory anchor
>
> The documents below govern interim monitoring and stopping. Chapter 3 describes each one in full and gives its status.
>
> - **ICH E9, Statistical Principles for Clinical Trials (1998), section 4.5, interim analysis and early stopping.** Defines an interim analysis as "any analysis intended to compare treatment arms with respect to efficacy or safety at any time prior to formal completion of a trial", and requires that "all interim analyses should be carefully planned in advance and described in the protocol" (PDF p23). The statistical methods "should be fully specified in advance of the availability of information on treatment outcomes" (section 3.4, PDF p20).
> - **FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics (final guidance, November 2019).** Treats sample size adaptation and futility stopping as adaptations that must be prospectively planned, with type I error control and bias addressed. Notes that adaptive features "can inflate the Type I error probability of a trial" (section III.A, PDF p11), and asks that interim results be walled off from those running the trial (sections III and VII).
> - **ICH E20, Adaptive Designs for Clinical Trials (Step 2 draft, June 2025).** Recommends that adaptations be "pre-specified in the clinical trial protocol prior to initiation of the trial" (section 1, PDF p6), warns that "the Type I error probability can be more than doubled" under analysis methods that ignore the adaptation (section 2, PDF p7), and expects simulation of operating characteristics such as "power, expected sample size, reliability of adaptation decisions" (section 2, PDF p8). Section 5.3 says a Bayesian adaptive design must

still show targeted operating characteristics "while maintaining Type I error probability control" (PDF p29). A draft, not binding.

- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft guidance, January 2026).** The success criterion is of the form $\Pr(\delta > a \mid \text{data}) > c$, and, for trials with an overall frequentist statistical analysis plan, familywise type I error is "almost always" held at 0.025 one-sided (section IV.A, PDF p10). The draft also describes direct interpretation of posterior probabilities and decision-theoretic criteria (section IV.A.3), and calibration to error control is this book's default, not the only route. In this book's reading (not the draft's wording), a predictive probability rule is a design component of the same kind: it is acceptable to the extent that its simulated operating characteristics are reported and its thresholds are fixed before the trial. A draft, not binding.

15.1. The traditional interim: conditional power and futility

15.1.1. Conditional power

A trial is designed for a fixed sample size with one-sided $\alpha = 0.025$ and power $1 - \beta$ at a design alternative ξ_a (Chapter 9). Partway through, at information fraction t (the share of planned patients or events observed), the DMC has a standardized statistic Z_t. The question "could this trial still win?" has a precise form. *Conditional power* is the probability that the final statistic Z_1 exceeds the critical value z_α, given the interim data and an assumed value of the effect.

The calculation uses the independent-increments structure of the test statistic. Write the *B-value* (Lan and Wittes 1988) $B_t = Z_t\sqrt{t}$, the cumulative sum of the standardized information. Let ξ denote the *drift*, the expected value of the final Z_1, so that under the design alternative $\xi = z_\alpha + z_\beta$. The symbol θ is reserved for the treatment effect; $\xi = \theta\sqrt{I_{\max}}$ is its standardized final drift. Then the future increment $B_1 - B_t$ is normal with mean $\xi(1-t)$ and variance $1 - t$, whatever the past was, and

$$\mathrm{CP}(\xi) = \Pr(Z_1 > z_\alpha \mid Z_t) = 1 - \Phi\left(\frac{z_\alpha - Z_t\sqrt{t} - \xi(1-t)}{\sqrt{1-t}}\right). \tag{15.1}$$

Everything depends on the choice of ξ, and three choices are in use. Under the *design alternative*, $\xi = z_\alpha + z_\beta$: this asks whether the trial can still succeed if the sponsor's original belief was right. Under the *current trend*, $\xi = Z_t/\sqrt{t}$: this asks what happens if the effect seen so far is the true effect. Under the *null*, $\xi = 0$: this asks whether the trial can win by luck alone. The first is optimistic after a poor interim, the second is pessimistic after a poor interim and optimistic after a good one, and the third is the floor.

Conditional power is the basis of *stochastic curtailment*, introduced for clinical trials by Lan, Simon, and Halperin (1982), with the companion conditional-power monitoring tool of Halperin et al. (1982): stop when the outcome is nearly decided one way or the other. Proschan, Lan, and Wittes (2006) give the modern treatment.

```
t <- 0.5; xi_a <- qnorm(0.975) + qnorm(0.90)   # design drift, 90% power at
 ↪  one-sided 0.025
z <- c(0, 0.5, 1, 1.5, 2)
cp <- data.frame(Zt = z, CP_design = cond_power(z, t, xi_a),
                 CP_trend = cond_power(z, t, z / sqrt(t)), CP_null =
                  ↪  cond_power(z, t, 0))
knitr::kable(cp, digits = 3, col.names = c("Interim z", "Design
 ↪  alternative", "Current trend", "Null"))
```

Table 15.1.: Conditional power at half the information (t = 0.5) of a trial designed with 90 percent power at one-sided 0.025 (design drift 3.24), under the design alternative, the current trend and the null, for five interim z statistics.

Interim z	Design alternative	Current trend	Null
0.0	0.316	0.003	0.003
0.5	0.508	0.038	0.012
1.0	0.699	0.220	0.038
1.5	0.846	0.590	0.102
2.0	0.936	0.890	0.220

Table 15.1 shows what the three choices do at the halfway point of a trial designed with 90% power. A trial that is exactly on the null path ($Z_t = 0$) still has conditional power 0.316 under the design alternative, because the design assumes that the effect, not the data so far, is the truth and half the information is still to come. Under the current trend its conditional power is 0.003. The two numbers describe different beliefs, not different data, and a DMC that reports only one of them has made a modeling choice without saying so.

Two further facts about conditional power matter in practice. First, it is a statement about the *frequentist* test: it treats ξ as fixed and known, so it gives no credit for uncertainty about ξ. At $Z_t = 1.0$, the current-trend value is 0.220 and the design-alternative value is 0.699, and the truth is probably between them; there is no principled way to pick a point in between without a prior on ξ. Second, it is a *decision tool*, not an inferential one. It does not change the final test, and under the usual non-binding rules the type I error of the final test is unaffected by whether the DMC acts on it, a point we return to below.

15.1.2. Futility rules

A *futility rule* stops the trial when the chance of success is too low to justify continuing. It can be written on any of several scales: a conditional power threshold (stop if $\mathrm{CP}(\xi_a) < 0.10$ or 0.20 are common choices), a lower bound on the interim Z or p-value, or a boundary from a group sequential design (the β-spending function of Chapter 11). Futility stopping can only reduce the probability of rejecting the null, so it never inflates the type I error. If the futility boundary is *non-binding*, meaning that the final test is conducted at the full α even when the trial continues past a crossing, the type I error is controlled whether or not the DMC follows the rule. The cost is a small loss of power (Lachin (2005) reviews futility rules based on conditional power and what they cost in power), and the benefit is that the sponsor can ignore a futility recommendation for ethical or scientific reasons without invalidating the trial. Non-binding futility boundaries are the usual choice for exactly this reason.

Stopping for efficacy is different. It spends type I error, and the boundary must be set by an alpha spending function or by simulation calibration (Chapter 11, Chapter 14). This chapter concentrates on futility and sample size, where the choice of rule is a design decision rather than a multiplicity problem.

15.1.3. Sample size re-estimation and the promising zone

The sample size of a trial is usually computed from a planning value of the effect and of the variance. When either is wrong the trial is under- or over-powered. There are two traditional responses.

The first is *blinded sample size re-estimation*: use the pooled (blinded) variance at an interim to update the nuisance parameter, leaving the effect assumption alone. It requires no unblinding and has negligible effect on type I error, and the adaptive designs guidance treats it as the least risky form of adaptation.

The second is *unblinded sample size re-estimation* based on the interim effect. This is where the danger lies. Suppose the interim effect is smaller than planned and the sponsor increases n so as to restore the conditional power to 90% under the current trend, then analyses the final data with the ordinary test. The sample size now depends on the data, and the type I error can exceed the nominal level, by more than a factor of two in extreme cases (E20 section 2 makes the same warning in general terms). Two families of remedies exist, both pre-specified. *Combination tests* compute a p-value from each stage and combine them with weights fixed in advance (P. Bauer and Kohne 1994; Cui, Hung, and Wang 1999; Lehmacher and Wassmer 1999). The *conditional error* principle (Proschan and Hunsberger 1995), generalized by H.-H. Müller and Schäfer (2001), permits any redesign of the second stage whose conditional type I error, given the interim data, is no larger than the original design's. In both, the price is that patients enrolled after the increase may count for less than the original ones, so the extra patients buy less power than a fixed trial of the same size.

The *promising zone* design of Mehta and Pocock (2011), which names the zone and gives the practical guide to an idea of Y.-H. J. Chen, DeMets, and Lan (2004) and Gao, Ware, and Mehta (2008) (raise the sample size when conditional power is intermediate and leave the final test unadjusted), restricts the increase to the interim results where it is most useful. If conditional power at the current estimate lies below a *futility* limit, the trial continues unchanged (or stops); if it lies above a *favorable* limit, the trial continues unchanged because it is already on track; if it lies in between, in the promising zone, the sample size is increased by a pre-specified rule up to a pre-specified cap, for example, to restore a target conditional power with the cap at twice the original size. The attraction is that the increase is made where the data suggest it will most likely turn a near-miss into a success. Comparative work (Jennison and Turnbull 2015) shows that a group sequential design with a futility boundary, planned for the larger sample size from the start, can match or beat a promising zone design in power and average sample size. The same conclusion follows from Tsiatis and Mehta (2003), who showed that the efficiency of an adaptive sample size rule can be matched or exceeded by a group sequential design. The adaptive design buys insurance against a misjudged effect, not free efficiency. Whether the unadjusted final test remains valid depends on how the zone is defined and capped; that has to be established for the design in question, by theory or by simulation, and not assumed.

A protocol that says "the DMC may recommend an increase of up to 50%" without a formula and a cap has not specified an adaptation in the sense of E20. One that gives the zone limits, the rule and the cap, and shows by simulation that the type I error is controlled, has.

15.2. Predictive probability of success

15.2.1. From conditional power to predictive probability

Conditional power fixes ξ. The Bayesian replaces the fixed value by the *posterior distribution* of ξ given the interim data and averages. The result is the *predictive probability of success*:

$$\mathrm{PP} = \int \mathrm{CP}(\xi)\, \pi(\xi \mid y_{1:n})\, d\xi = \Pr(\text{final success criterion met} \mid y_{1:n}), \tag{15.2}$$

where $\pi(\xi \mid y_{1:n})$ is the posterior at the interim (Chapter 6) and the second equality is the definition: the probability, over the posterior predictive distribution of the data still to come, that the trial's final success criterion is met. Predictive probability is conditional power averaged over what we have learned about the effect. Spiegelhalter, Freedman, and Blackburn (1986) compared the two quantities for monitoring, and called the Bayesian one predictive power; Dmitrienko and Wang (2006) use the same name, and the phase 2 literature below says predictive probability of success. Predictive probability accounts for uncertainty in the effect and conditional power does not. Platform trials use it as their routine decision quantity: I-SPY 2 graduates an arm on predictive probability, and GBM AGILE stops an arm for limited efficacy on a predictive power below 25% (Section 15.10).

The *final success criterion* in the definition is whatever the protocol says it is: a frequentist test rejecting at 0.025, or a Bayesian rule such as $\Pr(\delta > a \mid \text{data}) > c$ at the final analysis, as in the FDA draft. The predictive probability depends on that criterion, and the two worlds have different answers, so the criterion must be named when the number is reported.

15.2.2. The normal case

For the normal-drift model of Equation 15.1 with a normal prior on the drift, $\xi \sim N(\mu_0, \sigma_0^2)$, the posterior is normal and the integral has a closed form. After seeing B_t the posterior precision on ξ is $t + 1/\sigma_0^2$, the final statistic is normal given the data, and

$$\mathrm{PP} = 1 - \Phi\left(\frac{z_\alpha - \mu_t}{\sigma_t}\right),$$

with μ_t and σ_t^2 the predictive mean and variance of Z_1 (`pp_normal` in the package computes it). For a flat prior the result simplifies to $\mathrm{PP} = \Phi((Z_t/\sqrt{t} - z_\alpha)/\sqrt{(1-t)/t})$. The predictive distribution of Z_1 is wider than the conditional distribution, because it carries both the sampling variation of the future data and the posterior uncertainty about ξ.

Figure 15.1 and Table 15.1 show the pattern. At $Z_t = 1.0$ the flat-prior predictive probability is 0.293, between the two conditional power values (0.220 and 0.699, but nearer the trend) and a good deal more cautious than the design-alternative version. With a prior centered on the design alternative and unit variance (a prior worth as much information as the whole

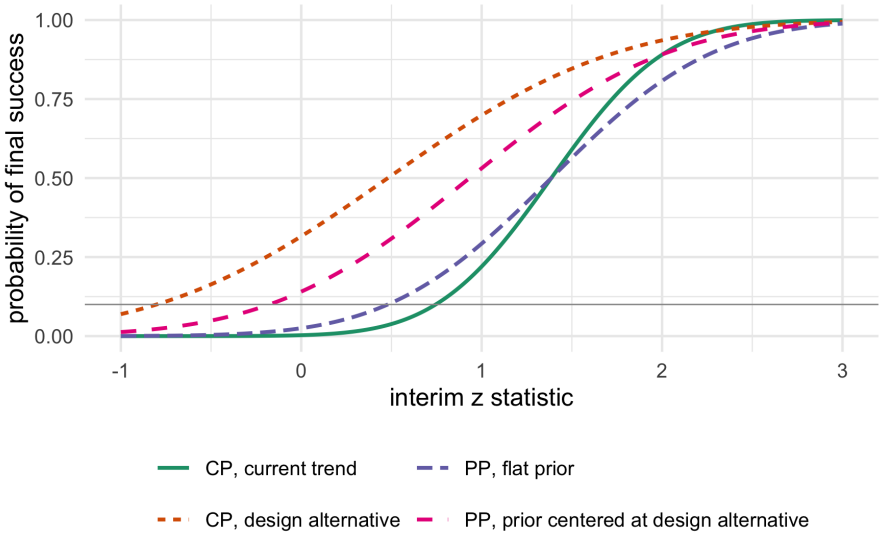


Figure 15.1.: Conditional power under the design alternative, under the current trend, and the predictive probability of success under a flat prior and under a design-alternative prior N(3.24, 1), as a function of the interim z statistic at half the information (t = 0.5, one-sided alpha 0.025, design power 90%).

trial, which is a strong prior), the interim data at $Z_t = 1.0$ give 0.531: the prior pulls the effect estimate up and the predictive probability with it. Neither number is the "right" one. They answer the question under different priors, which is why the prior must be pre-specified (Chapter 8) and why the sensitivity of the stopping decision to the prior belongs in the simulation report.

> **i** Technical note: the predictive distribution of the final statistic
>
> Let $B_t = Z_t\sqrt{t}$ be the interim B-value, with $B_1 - B_t \mid \xi \sim N(\xi(1-t), 1-t)$ independent of B_t. With prior $\xi \sim N(\mu_0, \sigma_0^2)$ and $B_t \mid \xi \sim N(\xi t, t)$, the posterior is $\xi \mid B_t \sim N(\mu_p, v_p)$ with $v_p = (t + 1/\sigma_0^2)^{-1}$ and $\mu_p = v_p(B_t + \mu_0/\sigma_0^2)$. Then $Z_1 = B_t + (B_1 - B_t)$ has predictive law $N(B_t + (1-t)\mu_p,\ (1-t) + (1-t)^2 v_p)$. A flat prior ($\sigma_0^2 = \infty$) gives $v_p = 1/t$, $\mu_p = B_t/t$, and predictive mean $B_t/t = Z_t/\sqrt{t}$ with variance $(1-t)/t$. The flat-prior predictive probability exceeds the current-trend conditional power when Z_t is below $z_\alpha\sqrt{t}$ and falls short of it above, because the added variance helps when the trend is below the target and hurts when it is above.

15.2.3. The exact beta-binomial computation

For a binary endpoint no normal approximation is needed. Suppose n of the planned N patients are observed, with s responders, and the prior on the response probability is $\text{Beta}(a_0, b_0)$. The posterior is $\text{Beta}(a_0 + s, b_0 + n - s)$, and the number of responders X among the $m = N - n$ remaining patients has the *beta-binomial* predictive distribution

$$\Pr(X = x \mid s, n) = \binom{m}{x} \frac{B(a_0 + s + x,\ b_0 + n - s + m - x)}{B(a_0 + s,\ b_0 + n - s)}, \qquad x = 0, \ldots, m, \tag{15.3}$$

with $B(\cdot, \cdot)$ the beta function. Each value of x completes a data set of N patients, to which the final success criterion can be applied directly, so the predictive probability is the sum of the beta-binomial probabilities of the x for which the criterion is met:

$$\mathrm{PP}(s, n) = \sum_{x=0}^{m} \Pr(X = x \mid s, n)\ \mathbf{1}\{\Pr(p > p_0 \mid s + x,\ N) > \eta\}. \tag{15.4}$$

There is no integration and no simulation. The same construction for two arms takes a double sum over the future counts in each arm, with the posterior probability of superiority computed for each pair by adaptive numerical integration on the rank scale, cached across states (`pp_two_arm` in `bayestrials`); the predictive sum itself is exact.

A single-arm phase 2 example, with a fixed design first. The control response rate is $p_0 = 0.20$, the target is $p_1 = 0.40$, and the trial enrolls $N = 40$ patients. With a Beta$(1, 1)$ prior, the final criterion is $\Pr(p > 0.20 \mid \text{data}) > 0.95$, which here works out to at least 13 responders of 40.

```
N <- 40; p0 <- 0.20; eta <- 0.95
succ <- pbeta(p0, 1 + 0:N, 1 + N - 0:N, lower.tail = FALSE) > eta   #
 ↪  succ[x + 1]: success with x responders
ps <- c(0.20, 0.30, 0.40)
knitr::kable(data.frame(p = ps, P = sapply(ps, function(p) sum(dbinom(0:N,
 ↪  N, p)[succ]))), digits = 3,
             col.names = c("True response rate", "P(success)"))
```

Table 15.2.: Probability of success of the fixed single-arm design (N = 40, criterion Pr(p > 0.20 | data) > 0.95, Beta(1,1) prior) at three true response rates.

True response rate	P(success)
0.2	0.043
0.3	0.423
0.4	0.871

The fixed design (Table 15.2) has type I error 0.043 and 87% power at $p = 0.40$, which is a conventional phase 2 operating profile. The *predictive probability at the interim* answers a different question: after, say, 20 patients with 6 responders, how likely is the criterion to be met at 40?

```
ints <- data.frame(n = c(10, 10, 20, 20, 20, 30, 30), s = c(3, 5, 5, 6, 7,
 ↪  8, 10))
ints$PP <- mapply(pp_one_arm, ints$s, ints$n, MoreArgs = list(N = N, p0 =
 ↪  p0, eta = eta))
knitr::kable(ints, digits = 3, col.names = c("Patients observed",
 ↪  "Responders", "Predictive probability"))
```

Table 15.3.: Predictive probability of meeting the final criterion (at least 13 responders of N = 40) for selected interim outcomes of the single-arm example.

Patients observed	Responders	Predictive probability
10	3	0.513
10	5	0.934
20	5	0.219
20	6	0.457
20	7	0.713
30	8	0.150
30	10	0.692

In Table 15.3, twenty patients with six responders (30%) gives a predictive probability of 0.457: a trial that is on course for an observed rate of 30% needs 13 of 40, and the remaining 20 patients have to deliver 7 more. Twenty patients with 5 responders gives 0.219; with 7 it gives 0.713. The predictive probability moves fast near the middle, and a single responder changes the picture at an interim of 20.

15.2.4. A time-to-event example

The normal-case formulas apply directly to an event-driven trial, where total information is $D/4$ for D events (so the information fraction is $D/D_{\max}$) and the log hazard ratio has variance about $4/D$. The worked example, with the predictive probability and the conditional power under the design effect and under the current trend side by side for four interim states, is in Section 18.4. The three summaries condition on different things (a fixed design effect, a fixed current estimate, and a posterior that averages over the effect), and which is more useful depends on how much the interim estimate is trusted against the design assumption; none is wrong, and the table there shows where they differ.

The same logic applies when the interim is combined with an earlier trial. In a continuous-endpoint program (an invented example), a skeptical posterior from phase 2 of mean 0.20 and standard deviation 0.115 is combined with a half-information interim of a pivotal trial (300 patients per arm at the interim, 600 per arm at the final analysis) that reports $\hat{\theta}_1 = 0.15$ with standard error $\sqrt{2/300}$.

```
w <- c(1 / 0.115^2, 300 / 2)                # precisions: phase 2
 ↪  posterior, interim data
m2 <- sum(w * c(0.20, 0.15)) / sum(w); s2 <- 1 / sqrt(sum(w))
# final estimate = (theta1 + theta2)/2, need z >= 2.02 at SE sqrt(2/600)
thr <- 2 * 2.02 * sqrt(2 / 600) - 0.15      # required value of the
 ↪  second-half estimate
pp_prog <- pnorm((m2 - thr) / sqrt(s2^2 + 2 / 300))
```

The combined posterior has mean 0.167 and standard deviation 0.067, and the predictive probability of crossing the final boundary of 2.02 is 0.79, high enough to continue. An interim estimate near zero would have started a futility discussion, and the predictive probability, not the conditional power at the design effect, would have been the quantity to discuss. The regulator analyzes the pivotal trial on its own data, so the type I error is controlled by that

analysis; the sponsor deciding whether to continue may use everything it knows. These are different decisions, and they can use different priors.

15.3. The traditional phase 2 design, and where Bayesian monitoring started

The single-arm phase 2 trial is a widely used early efficacy design in oncology, and its traditional form is older than the predictive probability designs that follow. A response rate p is judged against an uninteresting rate p_0 and a target rate p_1. Gehan (1961) proposed a first stage that stops when there are no responses, and Fleming (1982) a one-sample multiple testing procedure with early stopping. The design most protocols use is the two-stage design of Simon (1989): enroll n_1 patients and stop for futility if at most r_1 respond; otherwise enroll to a total of n and declare the drug promising if more than r respond. Among the pairs of stages that meet the stated type I error and power, the optimal design is the one with the smallest expected sample size under p_0. Both error rates are exact binomial sums, so the design is computed, not simulated.

```
s1 <- simon_optimal(0.2, 0.4, 0.05, 0.80)
tb3 <- lee_liu_table(N, p0, c(10, 20, 30), th_fut = 0.05, th_eff = 0.95,
 ↪  eta = eta)
ll0 <- lee_liu_oc(N, 0.2, tb3, p0, eta); ll1 <- lee_liu_oc(N, 0.4, tb3, p0,
 ↪  eta)
s2 <- simon_optimal(0.2, 0.4, ll0[[1]], ll1[[1]])
oc_row <- function(s) c(simon_oc(s$n1, s$r1, s$n, s$r, 0.2)[[1]],
 ↪  simon_oc(s$n1, s$r1, s$n, s$r, 0.4)[[1]], s$EN, simon_oc(s$n1, s$r1,
 ↪  s$n, s$r, 0.4)[[2]], s$n)
ph2 <- rbind(c(sum(dbinom(0:N, N, 0.2)[succ]), sum(dbinom(0:N, N,
 ↪  0.4)[succ]), N, N, N),
           oc_row(s1), oc_row(s2), c(ll0[[1]], ll1[[1]], ll0[[2]],
            ↪  ll1[[2]], N))
knitr::kable(data.frame(design = c(paste0("Fixed, ", which(succ)[1] - 1, "
 ↪  or more of ", N), "Simon optimal, alpha 0.05, power 0.80", "Simon
 ↪  optimal, matched to the three-look design", "Predictive probability,
 ↪  three looks (next section)"), ph2),
             digits = c(0, 3, 3, 1, 1, 0), col.names = c("Design", "Type I
              ↪  error", "Power", "Expected N at p0", "Expected N at p1",
              ↪  "Maximum N"))
```

Table 15.4.: Single-arm phase 2 designs for p0 = 0.20 and p1 = 0.40, with exact type I error (at p0), power (at p1) and expected sample size at p0 and at p1. Simon's optimal two-stage design is shown at the conventional error rates (0.05, 0.80) and at the error rates of the three-look predictive probability design of the next section.

Design	Type I error	Power	Expected N at p0	Expected N at p1	Maximum N
Fixed, 13 or more of 40	0.043	0.871	40.0	40.0	40
Simon optimal, alpha 0.05, power 0.80	0.050	0.800	20.6	37.9	43
Simon optimal, matched to the three-look design	0.048	0.870	26.8	43.6	46
Predictive probability, three looks (next section)	0.048	0.867	26.6	27.2	40

At the conventional error rates the optimal Simon design stops after 13 patients if at most 3 respond and otherwise continues to 43, declaring the drug promising if more than 12 respond; its type I error is 0.050, its power 0.800, and its expected sample size under p_0 is 20.6 (Table 15.4), which reproduces the published table of Simon (1989). It has a single look and stops for futility only. Matched to the error rates of the three-look predictive probability design of Section 15.4, the optimal Simon design is 5/21 and 13/46, with expected sample size 26.8 under p_0, against 26.6 for the three-look design: the two are close. Under p_1 the expected sample size is 27.2 for the three-look design against 43.6 for the matched Simon design, and that is the main gain: the possibility of stopping for efficacy. The other gain is a smaller maximum sample size (40 against 46). Under the null the reduction in the expected sample size is small.

The Bayesian single-arm designs came in two steps. Herson (1979) proposed stopping on the predictive probability of a positive final result. P. F. Thall and Simon (1994) monitored on the posterior probability that the response rate exceeds a target and chose the thresholds by their frequentist operating characteristics; P. F. Thall, Simon, and Estey (1995) extended the idea to several outcomes, and Heitjan (1997), Tan and Machin (2002) and the BOP2 design of H. Zhou, Lee, and Yuan (2017) are later versions. The next section is the predictive probability form of the same idea.

15.4. Predictive probability monitoring for phase 2

J. J. Lee and Liu (2008) turned the predictive probability of success into a two-threshold monitoring design for phase 2 with a printed stopping table. Related treatments include Johns and Andersen (1999) (phase 2 and 3), Dmitrienko and Wang (2006) (normal and time-to-event endpoints, with the link to conditional power) and Benjamin R. Saville et al. (2014) (practical interim use). The idea is the same as the predictive probability futility rule of Section 15.2, written in the units a phase 2 clinician uses: after each cohort, compute the predictive probability that the trial will end in success. Stop for futility if it is below a lower threshold η_L, stop for efficacy if it is above an upper threshold η_U, and otherwise continue. At the maximum sample size the trial declares success if the posterior probability criterion

$\Pr(p > p_0 \mid \text{data}) > \eta$ is met. Because the rule is a function of (n, s) only, the decision for every possible interim outcome can be tabulated before the trial begins and printed in the protocol. Real trials use the same logic: the BNT162b2 trial planned to stop for futility if the predictive probability of success fell to 5% or less, and the DAWN stroke trial used predictive probability for both futility and expected success (Section 15.10).

For the example above with $\eta_L = 0.05$, $\eta_U = 0.95$, and final threshold $\eta = 0.95$, and with looks after 10, 20 and 30 patients:

```
tab <- lee_liu_table(N, p0, c(10, 20, 30), th_fut = 0.05, th_eff = 0.95,
 ↪  eta = eta)
knitr::kable(tab, col.names = c("Patients observed", "Stop for futility if
 ↪  responders at most", "Stop for efficacy if responders at least"))
```

Table 15.5.: Stopping table of the three-look predictive probability design (N = 40, p0 = 0.20): stop for futility at or below the first number of responders, and for efficacy at or above the second.

Patients observed	Stop for futility if responders at most	Stop for efficacy if responders at least
10	0	6
20	3	9
30	7	12

```
tab7 <- lee_liu_table(N, p0, seq(5, 35, 5), 0.05, 0.95, eta)
oc_ll <- do.call(rbind, lapply(c(0.20, 0.40), function(p) {
  rbind(data.frame(p = p, looks = "Three", t(lee_liu_oc(N, p, tab, p0,
   ↪  eta))),
        data.frame(p = p, looks = "Seven", t(lee_liu_oc(N, p, tab7, p0,
         ↪  eta))))
}))
knitr::kable(oc_ll, digits = c(2, 0, 3, 1), col.names = c("True response
 ↪  rate", "Looks", "P(success)", "Expected N"))
```

Table 15.6.: Exact operating characteristics of the three-look design and of the same design with looks every five patients: probability of declaring success and expected sample size.

True response rate	Looks	P(success)	Expected N
0.2	Three	0.048	26.6
0.2	Seven	0.057	22.6
0.4	Three	0.867	27.2
0.4	Seven	0.862	23.2

This is the table a protocol prints. The three-look design stops for futility after 10 patients only if there are no responders, after 20 if there are three or fewer, and after 30 if there are seven or fewer; it stops for success after 10 with six or more responders, after 20 with nine

or more, and after 30 with twelve or more. The operating characteristics follow from exact enumeration of all paths, not from simulation: the probability of declaring success is 0.048 under the null rate (compared with 0.043 for the fixed design) and 0.867 at $p = 0.40$ (0.871 for the fixed design), with an expected sample size of about 27 patients under either, against 40 for the fixed design.

Three observations. First, power is essentially unchanged and the sample size falls by roughly a third. Second, the expected sample size is almost the same under the null and the alternative (26.6 and 27.2), because efficacy stops under the alternative offset futility stops under the null. Third, the type I error is *not* guaranteed to stay at the fixed-design value once efficacy stopping is allowed: with looks every five patients the null success probability rises to 0.057 (power 0.862, expected N 22.6 to 23.2), above 0.043, so the thresholds have to be recalibrated by exact enumeration or simulation (Chapter 14). Phase 2 trials often accept a type I error of 0.05 to 0.10 by design; a confirmatory trial targets 0.025 one-sided.

15.4.1. What the predictive probability design is, and is not

Predictive probability monitoring is a Bayesian way of *computing* a stopping boundary and a frequentist way of *judging* it. The decisions are made on a Bayesian quantity. The design is accepted or rejected on its operating characteristics under simulated truth. For a trial that adapts only by stopping, the realized design is a group sequential design with an unusual, data-driven boundary shape, and its probability of success under any true p is exact. This is how the FDA Bayesian draft expects a Bayesian design to be defended, and it is why the draft's demand for pre-specified thresholds and simulation is no obstacle: the table above is the pre-specification.

The approximation of Marion et al. (2025) shows that the exact calculation can be avoided in normal-approximation settings. If q_n is the current posterior probability of superiority, r the information fraction and η the final threshold, then

$$\mathrm{PP} \approx \Phi\left(\frac{\Phi^{-1}(q_n) - \Phi^{-1}(\eta)\sqrt{r}}{\sqrt{1-r}} \right), \tag{15.5}$$

which needs only the current posterior probability and the information fraction. The approximation assumes a Gaussian posterior and test statistic, and with a flat prior r is the share of the final information accrued, I_n/I_N. With an informative normal prior of precision J_0 the fraction for the posterior-threshold formula is $(J_0 + I_n)/(J_0 + I_N)$ rather than I_n/I_N, and with longitudinal outcomes r is the fraction of final information carried by the observed data, not the fraction of patients enrolled (Marion et al. 2025). In the binary example of the working paper by Polson and Sokolov (cited below as SIM) (Polson and Sokolov 2026a), with ten looks, thresholds 0.95 and 0.05 and final threshold 0.975, the approximation and the exact computation differ by 0.013 in mean absolute error across all interim states, and at most 9.0% of states receive different monitoring actions. Run through the full trial, the two give null success probabilities of 0.042 and 0.041, powers of 0.932 and 0.928, and expected sample sizes of 48.2 and 47.6 per arm. The approximation is good enough for planning, but the numbers also show that it is a design component to validate rather than an identity: where the final decision depends on a fine threshold, use the exact calculation.

15.4.2. The economic threshold PP^*

The thresholds 0.05 and 0.95 are conventions. A decision-theoretic reading ties the futility threshold to money. Suppose a false go costs c_3, the expected cost of a failed phase 3, and a missed active drug forfeits V, its expected net value if developed. Continuing from the interim costs the remaining patients, mc_2, and ρ is the probability that phase 3 succeeds given a phase 2 success. The myopic rule continues if $\text{PP}\,[\rho V - (1-\rho)c_3] > mc_2$, so the continuation threshold is

$$\text{PP}^* = \frac{mc_2}{\rho V - (1-\rho)c_3}, \tag{15.6}$$

which is small when the drug is valuable and the remaining phase 2 is cheap. The numbers below are invented for teaching, not taken from a real program.

```
pps <- rbind(p2_pp_star(V = 500, rho = 0.5, c3 = 150, mc2 = 2),
             p2_pp_star(V = 100, rho = 0.5, c3 = 150, mc2 = 2))
knitr::kable(pps, digits = 3, col.names = c("V", "rho", "c3", "m c2", "rho
 ↪  V - (1 - rho) c3", "PP*", "Viable"))
```

Table 15.7.: Economic futility threshold PP* for two invented drugs (values in millions of dollars): a valuable drug and a drug in a crowded indication. A negative denominator means a phase 2 success does not justify going on to phase 3.

V	rho	c3	m c2	rho V - (1 - rho) c3	PP*	Viable
500	0.5	150	2	175	0.011	TRUE
100	0.5	150	2	-25	-0.080	FALSE

For the valuable drug, $\text{PP}^* = 0.011$, so the conventional 0.05 is more aggressive than the economics justify: a sponsor following expected value would keep enrolling at an interim where the predictive probability is 0.02. For the crowded indication the denominator is negative, -25, which means a phase 2 success leads to a phase 3 of negative expected value and the trial should not be running at all. The lesson is not that 0.011 is the right number but that the futility threshold is a consequence of the economics and should be derived from them, then judged by operating characteristics like any other threshold (Chapter 14). The toxicity counterpart, a continuous Bayesian stopping rule for excess toxicity, is in Chapter 13.

15.5. The Goldilocks design: two predictive probabilities

Phase 2 monitoring asks one question: will the trial succeed at its maximum size? A confirmatory trial with a long follow-up asks two, and the *Goldilocks* sample size design of K. R. Broglio, Connor, and Berry (2014) separates them (Section 14.7 introduces it as a design chosen by simulation; this section gives the construction). The trial has a maximum sample size $N_{\max}$ per arm and enrolls continuously. Because the primary endpoint is observed after a delay (say at week 12), at an interim look some enrolled patients have outcomes and some are still *pending*. At each look the design computes two predictive probabilities, both for the final success criterion, say $\Pr(p_1 - p_0 > 0 \mid \text{data}) > 0.975$:

1. PP_{stop}: the probability of success if *enrollment stops now* and the pending patients are followed to complete their outcomes. Only the pending outcomes are unknown.
2. PP_{max}: the probability of success if the trial continues to the maximum sample size, where every patient not yet observed (pending or not yet enrolled) is unknown.

The decisions are: stop *enrolling for expected success* if PP_{stop} is high (the trial has already answered the question, and the pending patients will confirm it); stop *for futility* if PP_{max} is low (even the full sample is unlikely to succeed); otherwise, keep enrolling. The "not too big, not too small" name refers to the sample size that the data pick, between the two limits. The two probabilities are different questions with different thresholds, and the design adjusts the sample size to both. The full design of Broglio and colleagues, as summarized in Section 14.7, also has an immediate-success rule that declares success without following the pending patients when PP_{stop} passes a higher threshold; this chapter includes only the stop-enrollment rule above.

```
# Two arms, Nmax = 100 per arm. At the look: 60 enrolled per arm, 50 with
 ↪  outcomes, 10 pending.
sc <- data.frame(label = c("treatment ahead", "modest lead", "narrow lead",
 ↪  "no difference"),
                 s1 = c(30, 24, 22, 18), s0 = c(15, 15, 19, 18))
sc$post <- with(sc, prob_superior(1 + s1, 51 - s1, 1 + s0, 51 - s0))
sc$PP_stop <- with(sc, mapply(pp_two_arm, s1, 50, s0, 50, 10, 10, 0.975))
 ↪  # only the 10 pending per arm unknown
sc$PP_max <- with(sc, mapply(pp_two_arm, s1, 50, s0, 50, 50, 50, 0.975))
 ↪  # 10 pending + 40 unenrolled unknown
sc$action <- with(sc, ifelse(PP_stop > 0.95, "stop enrolling",
 ↪  ifelse(PP_max < 0.05, "stop for futility", "continue")))
knitr::kable(sc, digits = 3, col.names = c("Scenario", "Treatment
 ↪  responders", "Control responders", "Posterior P(superior)", "PP stop",
 ↪  "PP max", "Action"))
```

Table 15.8.: Goldilocks decision at the look (60 enrolled per arm, 50 with outcomes, maximum 100 per arm) for four interim scenarios: responders in each arm, posterior probability of superiority, the two predictive probabilities and the resulting action.

Scenario	Treatment responders	Control responders	Posterior P(superior)	PP stop	PP max	Action
treatment ahead	30	15	0.999	0.999	0.990	stop enrolling
modest lead	24	15	0.966	0.545	0.735	continue
narrow lead	22	19	0.727	0.001	0.133	continue
no difference	18	18	0.500	0.000	0.022	stop for futility

The four rows of Table 15.8 show the logic. With 30 of 50 against 15 of 50 the posterior probability of superiority is already 0.999, PP_{stop} is 0.999, and the design stops enrolling;

completion is predicted to meet the success criterion with probability about 0.999, although unfavorable pending outcomes could reverse the decision. The modest lead (24 against 15, posterior probability 0.966) is not yet there: $\mathrm{PP}_{\mathrm{stop}}$ is 0.545, $\mathrm{PP}_{\max}$ is 0.735, and the design keeps enrolling. The narrow lead of 22 against 19 gives $\mathrm{PP}_{\max} = 0.133$, above the futility floor of 0.05, so the trial continues but is in trouble. Equal rates at 18 of 50 give $\mathrm{PP}_{\max} = 0.022$ and the design stops for futility.

Two cautions, both of the kind a regulator raises. The thresholds (0.95 and 0.05 here are illustrative) must be calibrated by simulation so that the overall type I error is controlled (Section 14.4 shows this for the four-look example of Chapter 14, where the flat posterior threshold of 0.975 becomes $c = 0.9907$), and the look frequency and the delay length change the calibration. And stopping enrollment when the predictive probability of success is high is an *efficacy* decision: it spends type I error just as a group sequential efficacy boundary does (Chapter 14).

The same two questions run through the DAWN stroke trial (Section 15.10); I-SPY 2 uses a single predictive probability, that of success in a follow-on phase 3 trial. The Goldilocks design replaces three separate traditional elements (an interim efficacy boundary, a futility boundary, and a sample size re-estimation) with one pre-specified rule, and its operating characteristics are what justify it. It also has a price: the sample size is random, so budgets and supply have to be planned for the maximum and the expected sample size reported alongside it.

15.6. Backward induction: the Bayes-optimal stopping rule

Predictive probability monitoring is a rule with thresholds that the designer chooses and calibrates. A natural question is whether a *best* rule exists, for a stated loss. For a sequential decision problem with a finite horizon the answer is yes, and it can be found by working backward from the end. The method is *backward induction*, or dynamic programming (Bellman 1957). Sequential analysis itself goes back to Wald (1947), and Bayes sequential solutions by backward induction to Arrow, Blackwell, and Girshick (1949). Anscombe (1963b) and Colton (1963b) cast the trial as a Bayesian decision problem over a patient horizon, and D. A. Berry and Ho (1988a) derived clinical-trial stopping boundaries by backward induction; Lewis and Berry (1994b) evaluated such designs by their frequentist operating characteristics, the stance of this book. Later work includes Carlin, Kadane, and Gelfand (1998), Cheng and Shen (2005) and P. Müller et al. (2007) for sequential trials with a sufficient statistic; Brockwell and Kadane (2003) gave a gridding method for the normal model and F. Jiang, Lee, and Müller (2013) combined it with response-adaptive randomization. The presentation below follows SIM (Polson and Sokolov 2026a).

15.6.1. The decision problem

A two-arm trial with binary outcomes enrolls one pair of patients at a time, one per arm, up to a maximum of T pairs. After k pairs, the trial can *stop and declare* a winner, or *pay a cost* κ *for one more pair*. A stopping loss records the consequences of the terminal decision, and the continuation cost records the consequences of waiting. Two stopping losses are useful.

The *utility* version stops for treatment when the posterior mean effect $\hat{\theta}_k$ is positive and for control otherwise, with stopping value $h_k = -\max(0, \hat{\theta}_k)$, the negative of the expected

benefit of the chosen arm. It has no separate cost for a false positive, so it is a decision rule for patient benefit, not an evidence rule, and it is deliberately aggressive under the null.

The *classification* version fixes the losses L_{FP} for declaring superiority when it is absent and L_{FN} for failing to declare it when it is present. With $q_k = \Pr(p_1 > p_0 \mid \mathrm{data}_k)$, $\gamma = L_{\mathrm{FP}}/(L_{\mathrm{FP}} + L_{\mathrm{FN}})$ and κ the cost per pair normalized by $L_{\mathrm{FP}} + L_{\mathrm{FN}}$, the Bayes stopping loss and action are

$$h_k = \min\{\gamma(1-q_k),\ (1-\gamma)q_k\}, \qquad \text{declare superiority if } q_k > \gamma. \tag{15.7}$$

A loss ratio of $L_{\mathrm{FP}} : L_{\mathrm{FN}} = 39 : 1$ gives $\gamma = 0.975$, which is the Bayesian reading of one-sided 0.025: a false positive is 39 times worse than a false negative. This ratio is illustrative; the elicitation of the real ratio is the part of the design that belongs to clinicians, sponsors and regulators, not to the statistician.

15.6.2. The sufficient-statistic reduction

The data history up to pair k is a sequence of 4^k possible outcome patterns. The posterior depends on the history only through the success counts (s_1, s_0) and k, because the beta-binomial model has sufficient statistics. With the prior $\mathrm{Beta}(a_{0i}, b_{0i})$ in arm i, the predictive probability of the next outcome in that arm is

$$\tilde{p}_i(s_i, k) = \frac{a_{0i} + s_i}{a_{0i} + b_{0i} + k}, \tag{15.8}$$

so the future depends on the past only through (s_1, s_0, k). This is the *sufficient-statistic reduction*: the decision tree, which grows exponentially, collapses to a lattice with $(k+1)^2$ states at stage k, and the trial's whole future is a function of where on the lattice it stands.

With $V_k(s_1, s_0)$ the minimum expected loss from state (s_1, s_0) at stage k onward, the *Bellman equation* is

$$V_k(s_1, s_0) = \min\Big\{h_k(s_1, s_0),\ \kappa + \sum_{y_1, y_0 \in \{0,1\}} \Pr(y_1, y_0 \mid s_1, s_0, k)\, V_{k+1}(s_1 + y_1,\ s_0 + y_0)\Big\}, \tag{15.9}$$

for $0 \le k < T$, with $V_T = h_T$ at the horizon (the trial is forced to decide). The first term is the loss from stopping now. The second is the cost of one more pair plus the *expected* loss at the next stage, averaged over the four outcomes of the next pair with the predictive probabilities of Equation 15.8. The minimum of the two is the best that can be done from here, and the argument that achieves it is the optimal action. One computes V_T first, then V_{T-1} from it, and so on back to V_0: *backward* induction.

> **i** Technical note: why the optimal policy is a function of the lattice state, and its cost
>
> **Proposition** (SIM, Appendix). For fixed prior parameters and horizon T, the optimal policy at stage k is a function of (s_1, s_0, k) only, and the recursion in Equation 15.9 evaluates $\sum_{k=0}^{T}(k+1)^2 = (T+1)(T+2)(2T+3)/6 = O(T^3)$ states, each in constant time.

Proof sketch. By conjugacy the posterior at stage k is Beta$(a_{0i} + s_i, b_{0i} + k - s_i)$ in each arm, so (s_1, s_0, k) is a Markov sufficient state for the future. The one-step predictive probabilities of Equation 15.8 depend on no other feature of the history, so the continuation value is a function of the state, and so is the minimum over the two actions. Each update is the arithmetic of four lookups of V_{k+1}. Summing the number of states over stages gives the cubic count. For $T = 200$ pairs, about 2.7 million states. □

For normal data with known variance the same argument holds with the posterior mean as the scalar state, and the lattice is replaced by a grid of the posterior mean; Brockwell and Kadane (2003) used 12,000 grid points on $[-6, 6]$.

The result is *exact* for the binary endpoint: no simulation, no approximation, no quadrature. The only approximation in the classification version is numerical, the computation of q_k for each state, which is a one-dimensional integral that `bayestrials` evaluates by the same adaptive rank-scale integration, with caching. The classification-loss version is implemented in `backward_induction` of the `bayestrials` package, together with an exact forward propagation (`forward_oc`) that returns the probability of declaring superiority and the expected sample size under any true pair (p_1, p_0).

```
nT <- 60; gamma <- 0.975
scen <- list(null = c(0.30, 0.30), target = c(0.55, 0.30), small = c(0.45,
 ↪  0.30))
policies <- lapply(c(`0.0001` = 1e-4, `2e-05` = 2e-5), backward_induction,
 ↪  T = nT, loss = "class", gamma = gamma)
fx <- sapply(scen, function(p) fixed_oc(nT, gamma, p[1], p[2]))
bi <- do.call(rbind, lapply(names(policies), function(nm) {
  o <- sapply(scen, function(p) forward_oc(policies[[nm]], p[1], p[2]))
  data.frame(design = paste("Bayes-optimal, kappa =", nm), quantity =
   ↪  c("P(declare treatment)", "E[N per arm]"), o, check.names = FALSE)
}))
tabbi <- rbind(data.frame(design = "Fixed, 60 per arm", quantity =
 ↪  "P(declare treatment)", t(fx), row.names = NULL), bi)
knitr::kable(tabbi, digits = 3, row.names = FALSE, col.names = c("Design",
 ↪  "Quantity", "Null", "Target", "Small effect"))
```

Table 15.9.: Operating characteristics of the fixed 60-per-arm design and of two Bayes-optimal classification-loss policies (gamma = 0.975, cost kappa per pair) under the null (0.30, 0.30), the target (0.55, 0.30) and a small effect (0.45, 0.30): probability of declaring treatment better and expected sample size per arm.

Design	Quantity	Null	Target	Small effect
Fixed, 60 per arm	P(declare treatment)	0.024	0.806	0.393
Bayes-optimal, kappa = 0.0001	P(declare treatment)	0.023	0.708	0.335
Bayes-optimal, kappa = 0.0001	E[N per arm]	19.388	34.996	33.370
Bayes-optimal, kappa = 2e-05	P(declare treatment)	0.024	0.790	0.384
Bayes-optimal, kappa = 2e-05	E[N per arm]	30.237	45.880	46.147

In Table 15.9, at a cost of 2×10^{-5} per pair, the classification-loss policy declares superiority with probability 0.024 under the null (the fixed design: 0.024), 0.790 at (0.55, 0.30) (fixed:

0.806), and 0.384 at (0.45, 0.30) (fixed: 0.393). Its expected sample size under the target alternative is 45.9 pairs, against 60 for the fixed design. At the target alternative the Bayes policy gives power 0.790 against 0.806 for the fixed design, at 24% fewer patients on average: a small loss of power for a substantial saving, not equal operating characteristics. Raising the cost to 10^{-4} makes continuation more expensive: power at the target falls to 0.708 and expected sample size to 35.0 pairs. This is the *power frontier*, the trade-off between the expected number of patients and the probability of a correct declaration, indexed by the cost κ. A designer calibrates κ so that the frontier passes through the operating characteristics the protocol promises: that paper does this for a maximum of 100 per arm and finds the cost at which the null declaration probability is held below 0.05 over a grid of 99 equal-rate scenarios and the power at (0.55, 0.30) is at least 0.90 (Polson and Sokolov 2026a). That calibrated policy has a largest null declaration probability of 0.026 (slightly above 0.025, because the SIM calibration holds it below 0.05 over the grid), target power 0.903 and expected sample size 54.3 pairs, against 100 for the fixed design with 100 per arm.

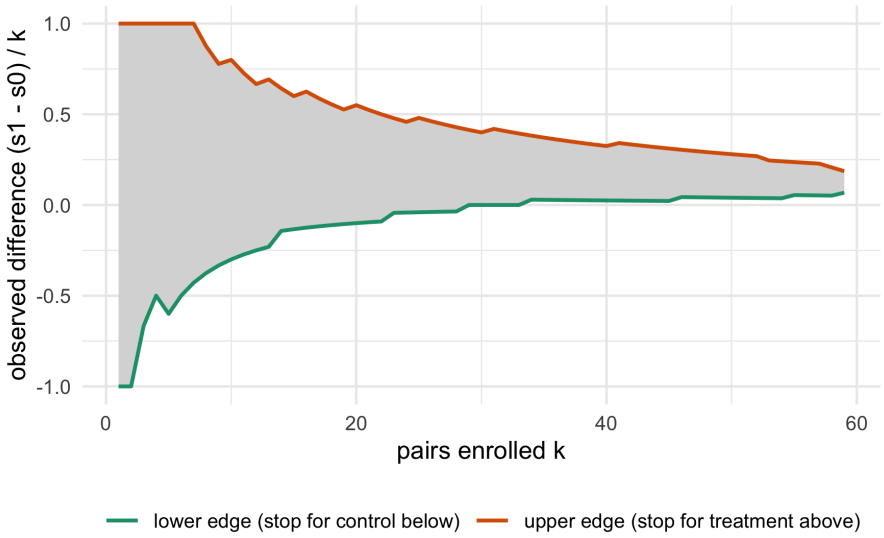


Figure 15.2.: Continuation region of the Bayes-optimal classification-loss policy (T = 60 pairs, gamma = 0.975, cost 2e-5 per pair), shown as the range of the observed difference in success proportions among continuation states at each stage (a projection: the full policy is indexed by the pair of success counts, and states with the same difference can have different actions). Continuation states lie within the shaded envelope between the two edges; the action at a particular point requires both success counts. Above the upper edge the trial stops for treatment, below the lower edge for control.

Figure 15.2 shows the familiar funnel. Early in the trial a large observed difference is needed to stop for treatment (0.8 after 10 pairs, 0.55 after 20, 0.4 after 30, 0.28 after 50), because a few patients say little; the trial stops for control once the observed difference falls to about 0 after 30 pairs. The region closes as k approaches the horizon, because the trial is forced to decide at T whatever happens. The optimal rule has the shape of a group sequential boundary, with efficacy and futility stopping, but it is derived from a loss and a cost, not from an error-spending function. The projection onto the observed difference is only a summary: the full policy is indexed by (s_1, s_0, k), and states with the same difference can have different

decisions.

15.6.3. What backward induction does and does not do

Three honest limits. First, a Bayes-optimal rule is optimal for the *stated loss under the stated prior*, and the loss ratio and cost are inputs somebody has to choose. A decision-theoretic submission should justify its loss function, prior, success criterion, and operating characteristics. For the error-calibrated approach used in this book, the route from one to the other is the calibration above: choose γ and κ, compute the policy, simulate (or enumerate) its operating characteristics under the full range of scenarios, and adjust until they meet the protocol targets. The decision-theoretic framing is a way of *generating* a good design; the frequentist operating characteristics are how it is *judged* (Chapter 14).

Second, the utility-only version shows what happens when a loss does not penalize false positives. In SIM's calculation, with $\kappa = 0.0005$ and a maximum of 100 per arm, utility-only backward induction declares treatment better with probability 0.513 under the null (Monte Carlo standard error 0.005) and 0.98 at the large effect, with an expected sample size of only 14.4 per arm. It is a good rule for choosing the arm that gives patients the best outcomes, and a useless rule for a confirmatory claim: its "null error" is a treatment-choice probability, not a type I error. The classification version, with $\gamma = 0.975$, has a null declaration probability of 0.026, and its expected sample size at the large effect is 54.3 per arm with power 0.903. The loss function is the design.

Third, the lattice assumes that outcomes are observed before the next decision, balanced allocation, and no pending patients. Delayed outcomes enlarge the state to include the pending counts, which is the setting of the Goldilocks design above, and response-adaptive allocation adds the allocation decision to the action set (Chapter 16). Those extensions are what make exact dynamic programming harder; the sufficient-statistic reduction is what makes it feasible at all.

15.6.4. A single-arm example

The two-arm lattice above has a one-arm counterpart that can be printed as a table. A single-arm trial enrolls patients one at a time, up to 60, with a Beta(3, 7) prior on the response rate p, centered at a standard-of-care rate $p_0 = 0.3$. Each patient costs $c = 0.002$ in utility units. When the trial stops at n patients with x responders it adopts the drug if the posterior mean exceeds p_0, earning $M(\hat{p} - p_0)$ with $M = 10$ standing for the population of future patients, and abandons it otherwise. With $\hat{p}_n = (3 + x)/(10 + n)$, the value $V_n(x)$ of state (n, x) satisfies

$$V_n(x) = \max\Big\{\max(M(\hat{p}_n - p_0),\, 0),\;\; -c + \hat{p}_n V_{n+1}(x+1) + (1-\hat{p}_n)V_{n+1}(x)\Big\}, \qquad (15.10)$$

solved backward from V_{60}. This is an invented teaching example with no regulatory standing: it has no type I error target.

```
st <- decide_stop_single_arm(p = c(0.2, 0.3, 0.45))
knitr::kable(st$table, col.names = c("Patients n", "Continue from x =",
 ↪  "Continue to x =", "Adopt if x at least"))
```

Table 15.10.: Optimal stopping of the single-arm problem (Beta(3, 7) prior, horizon 60, M = 10, cost 0.002 per patient): the trial continues while the number of responders lies in the given range, and adopts the drug on stopping at or above the last column.

Patients n	Continue from x =	Continue to x =	Adopt if x at least
5	0	4	2
10	1	5	4
20	4	8	7
30	7	11	10
40	10	14	13
50	14	16	16
55	15	17	17

```
knitr::kable(st$oc, digits = c(2, 3, 1), col.names = c("True response
 ↪  rate", "P(adopt)", "Expected N"))
```

Table 15.11.: Exact operating characteristics of the single-arm optimal stopping rule: probability of adopting the drug and expected number of patients at three true response rates.

True response rate	P(adopt)	Expected N
0.20	0.053	24.8
0.30	0.496	32.3
0.45	0.976	19.1

The continuation region of Table 15.10 is a band around the adoption boundary that narrows as the trial proceeds, because each additional patient moves the posterior less and so is worth less. Outside the band the decision is clear enough that more data cannot pay for themselves. The operating characteristics, computed exactly by forward propagation, are small where the stakes are large and large only where the two actions are nearly equivalent: at $p = 0.2$ the drug is adopted with probability 0.053 after 24.8 patients on average, at the boundary $p = 0.3$ with probability 0.50 after 32.3, and at $p = 0.45$ with probability 0.976 after 19.1. This is the general shape of Bayes sequential designs. The same caveat applies as above: the loss and the cost are inputs, and a regulator reviews the operating characteristics (or, under the other routes of the FDA draft, the posterior or benefit-risk criterion), not the loss.

15.7. Comparing the designs

SIM puts five specifications on a common footing, a two-arm trial with a maximum of 100 patients per arm, $p_0 = 0.30$ and a large effect of $\theta = 0.25$, and reports power and expected sample size per arm (Polson and Sokolov 2026a). The rows are different *specifications*, with different objectives and monitoring schedules, so the table is a description of each and not a ranking of methods.

Table 15.12.: Power and expected sample size at $\theta = 0.25$ (SIM, maximum 100 per arm; all rows are 10,000 simulated trials per scenario except the classification-loss row, which is exact enumeration, where SIM's simulation reports power 0.911. Realized null declaration probabilities: 0.033 for the O'Brien-Fleming-shaped rule (not recalibrated for discrete data), 0.028 for the fixed Wald test, 0.042 for predictive probability monitoring and 0.026 for classification-loss backward induction, but 0.513 for the utility-only rule, which is a treatment-choice probability).

Design	Power at $\theta = 0.25$	Expected N per arm
Fixed sample, one-sided Wald test	0.96	100
O'Brien-Fleming-shaped rule, five looks (efficacy only)	0.96	64.8
Predictive probability monitoring, ten looks (thresholds 0.95 and 0.05, final 0.975)	0.93	48.2
Classification-loss backward induction, $\gamma = 0.975$	0.903	54.3
Utility-only backward induction	0.98	14.4

Three lessons. The traditional group sequential design, fixed at one sample size and efficacy-only, cuts the expected sample size from 100 to 64.8 per arm at this large effect with simulated power unchanged at 0.96: the Bayesian predictive designs are not the first to stop early. That comparator is an approximate O'Brien-Fleming-shaped rule with boundary $\Phi^{-1}(0.975)\sqrt{5/j}$, not recalibrated for discrete data, and its realized null rejection probability is 0.033 against 0.028 for the fixed Wald test, so the comparison is not calibrated and no claim of a costless saving in power follows. The predictive probability and classification-loss designs go further because they also stop for futility, at a cost in power (0.93 and 0.90 against 0.96) that the calibration can trade back. And the utility-only rule shows that the objective changes everything: a rule that stops after 14 patients per arm and is right 98% of the time at the large effect is also wrong half the time under the null.

The honest reading of Table 15.12 is that backward induction does not "beat" group sequential monitoring. It chooses a rule for a stated loss, and the rule, once calibrated to the same operating characteristics, is in the family of the others. What the decision framing adds is the explicit price: the cost per patient and the loss per error are design parameters, where the group sequential design hides them in the choice of spending function.

15.8. Side by side: what changes in the protocol

Table 15.13.: Interim monitoring and stopping, traditional and Bayesian predictive.

	Traditional	Bayesian predictive
Interim quantity	Conditional power at the design alternative or the current trend; interim Z against a boundary	Predictive probability of final success (posterior-averaged), or posterior probability
Treatment of uncertainty in ξ	None: ξ is fixed at a chosen value	Averaged over the posterior; requires a prior
Futility	Non-binding boundary or conditional power threshold; adding it to an unchanged success rule cannot inflate the type I error	Threshold on PP (for example 0.05 to 0.10); the same holds when the success rule is unchanged, and the efficacy threshold is calibrated with the futility rule in place
Efficacy stopping	Alpha spending function (O'Brien-Fleming, Lan-DeMets)	Threshold on PP or posterior probability; calibrated by simulation
Sample size change	Blinded re-estimation; combination or conditional-error tests; promising zone	Goldilocks stopping of accrual; sample size is data-driven within a pre-specified maximum
Optimal rule	Group sequential designs optimized for expected sample size	Backward induction for a stated loss and cost
What the regulator asks	Spending function, boundaries, type I error	Thresholds, prior, simulated type I error and power, expected N, sensitivity to prior and delay
What stays the same	The final test criterion, the type I error target (0.025 one-sided), the pre-specification, the walled-off DMC	

The two approaches agree numerically in simple cases: with a flat prior and a large trial, the predictive probability of success at an interim is a function of the same Z_t and t as conditional power under the current trend, and the flat-prior posterior probability of benefit at the end is approximately one minus the one-sided p-value (Chapter 7). What the Bayesian design adds is the language, the prior, the forward-looking quantity that clinicians understand, and the framework for optimization. What it does not add is a lower bar.

When is the traditional design the better choice? A single interim for futility with a non-binding rule is simple, well understood by every DMC and regulator, and needs no prior and no simulation beyond the standard group sequential software. If the DMC will take its decision on clinical grounds, the extra machinery of a predictive probability rule has no one to use it. Where the decision is not a single stop-or-go but a sequence (several looks, a pending-outcome structure, a platform with many arms), the predictive design earns its complexity.

In the protocol

A predictive monitoring section should state, before any outcome data are seen:

- The *final success criterion* in full (for example, $\Pr(p_1 - p_0 > 0 \mid \text{data}) > 0.975$ at $N_{\max}$) and the prior used to compute the predictive probability, including its justification.
- The *look schedule* (information fractions or patient numbers) and the quantity used at each look.
- The *thresholds* for futility and for efficacy, and whether the futility threshold is binding. State that the DMC may recommend continuation against a non-binding futility threshold without affecting the final analysis.
- The *method of computation* (exact beta-binomial, or the normal approximation of Equation 15.5, with its validation).
- The *handling of pending outcomes*, if any: whether PP_{stop} and PP_{max} are both used, and how patients enrolled but not yet observed enter the computation.
- A pointer to the *simulation report*: scenarios (null, design alternative, several in-between and adverse), type I error, power, expected and maximum sample size, probability of each stopping reason, and sensitivity to the prior and to delays.

15.9. Regulatory reading of the chapter's designs

The position on each tool follows from two requirements: pre-specification and operating characteristics. A futility rule is the lowest-risk adaptation, provided it is pre-specified and interim access to comparative data is controlled (ICH E9 4.5; FDA 2019), and non-binding is the safe default; the new element in a Bayesian version is the prior inside the predictive probability, which has to be defended. Sample size re-estimation divides into blinded (low risk) and unblinded (needs type I error control by design and a prespecified algorithm); E20 says "the number and complexity of adaptations should generally be limited" in confirmatory trials (section 3.2, PDF p9) and requires a "clear and compelling justification" (PDF p8), which a promising zone or Goldilocks design in a pivotal trial must supply. A predictive probability rule inside a design under the January 2026 FDA draft is a stopping rule and needs the report of any adaptive decision rule: simulated type I error and power, expected sample size, prior sensitivity. No guidance mentions backward induction. A decision-theoretic submission should justify its loss function and prior (the FDA draft discusses expected-loss criteria in Section IV.A.3), and document the resulting policy as a table of boundaries for every (s_1, s_0, k) with its operating characteristics. A Bayes-optimal rule is submitted as a pre-specified table and justified by those characteristics, and a novel rule in a pivotal trial is a subject for the CID paired meeting program (Chapter 3, Chapter 23).

15.10. In practice

Five trials show the chapter's quantities at work: one that stopped on a posterior probability of futility, I-SPY 2 with two arms judged on predictive probability, one stroke trial and one vaccine trial with predictive stopping rules written into the analysis plan, and one platform arm that stopped for limited efficacy.

15.10.1. CALGB 49907: futility stopping in practice

CALGB 49907 compared capecitabine with standard adjuvant chemotherapy in women aged 65 or older with early-stage breast cancer (Muss et al. 2009). The design planned up to 1,800 patients, with a Bayesian monitoring rule on the noninferiority comparison and noninformative priors. At the first interim, after 600 patients, the posterior probability that capecitabine would fail the noninferiority criterion was 96%, over the pre-specified 80% futility boundary. Enrollment stopped at 633 patients, 65% fewer than the planned maximum. After further follow-up the hazard ratio was 2.09 (95% CI 1.38 to 3.17), confirming the superiority of standard chemotherapy (Muss et al. 2009). D. A. Berry (2025) describes it as, to his knowledge, the first fully Bayesian randomized trial published in the *New England Journal of Medicine*.

The case illustrates the value of futility stopping where the new drug is *worse*, not just no better: the trial spent 633 patients, not 1,800, to learn that capecitabine was inferior in this population. It also illustrates something regulators look for: the criterion was a probability statement with a pre-specified boundary, and the boundary was crossed by the data, not chosen after the fact. Note that the quantity reported at the interim is a posterior probability of failing noninferiority, and the design as described in the primary report did not stop on a conditional power value.

15.10.2. I-SPY 2: predictive probability as the graduation criterion

The original I-SPY 2 trial in neoadjuvant breast cancer (Barker et al. 2009; D. A. Berry 2015) tested multiple agents against a shared control. Its decision criteria are written in predictive probability. An agent *graduated* in a biomarker signature when its Bayesian predictive probability of success in an equally randomized 300-patient phase 3 trial reached at least 85%; it was dropped for futility when that probability fell below 10% in every one of the ten signatures (D. A. Berry 2025; J. W. Park et al. 2016). This is the Goldilocks logic applied to a platform: the question at each interim is not whether the data already show benefit, but whether a *specific follow-on trial* would succeed, and the follow-on trial is part of the definition.

Two arms show how the rule behaves. Neratinib (127 randomized, 115 evaluable, against 78 evaluable concurrent controls) graduated only in the HER2-positive, hormone-receptor-negative signature, with an estimated pathologic complete response rate of 56% against 33% for the control and a final predictive probability of phase 3 success of 79%, below the 85% threshold (J. W. Park et al. 2016). The drop was anticipated: graduation is declared on interim predictions that use imaging, and the probability is recomputed once all pathology is in, which is why the authors set the threshold high at 85%. Pembrolizumab with chemotherapy (69 patients against 181 controls) reached the threshold in all three signatures it was tested in and randomization was halted; the final predictive probabilities were 98.5% (HER2-negative), 99.6% (hormone-receptor-positive, HER2-negative) and 83.4% (triple negative) (Nanda et al. 2020). The triple-negative value is also under 85%, and the arm still counts as graduated.

The lesson is that a predictive probability threshold is a decision rule on *partially observed* outcomes, and its realized value at the end can fall below the nominal one. A protocol that sets the threshold should say which data enter the calculation and how the final recomputation is treated. The platform structure, the shared control and the arm-level multiplicity are treated in Chapter 20, and the allocation rule in Section 16.9.

15.10.3. DAWN: predictive probability for expected success and futility

DAWN tested thrombectomy with the Trevo device in acute ischemic stroke 6 to 24 hours after onset in patients with a mismatch between clinical deficit and infarct core (Jovin et al. 2017). Its statistical analysis plan (Stryker Neurovascular 2017) makes predictive probability the stopping quantity. From 200 randomized patients, with at least 100 since the last enrichment, the trial stops for expected success if the predictive probability of success exceeds 95% at 200 and 250 patients, 90% at 300 and 350, 85% at 400, and 80% at 450 and 500. It stops for futility if the predictive probability of success, under any enrichment possibility, at the maximum sample size is under 10%. The thresholds fall as the sample grows, so an early stop needs more certainty than a late one. The final success criterion and its adjustment for enrichment are in Section 14.11.

This is the Goldilocks structure of Section 15.5 with a schedule of thresholds, and it shows what a regulator reads in such a design: not the decision theory but the table of looks and thresholds and the simulation that calibrates the final criterion.

15.10.4. BNT162b2: a futility rule in a vaccine efficacy trial

The pivotal phase 2/3 trial of the Pfizer and BioNTech vaccine declared efficacy on a posterior probability with calibrated thresholds (the model and prior are in Chapter 6, the thresholds and their calibration in Section 14.11), and added a futility rule that is predictive: stop if the posterior predictive probability of success at the final analysis is 5% or less, with non-binding boundaries (for example, 27.8% observed efficacy at 62 cases and 38.6% at 92 cases). The first interim analysis planned at 32 cases was not performed, the thresholds were unchanged and the type I error stayed at 2.5% (Pfizer Inc. 2022).

The lesson is the one of the section on futility rules: a predictive futility boundary that is non-binding costs nothing in type I error, and a sponsor can pre-specify it as a table of case splits that a monitoring committee can read without a computer.

15.10.5. GBM AGILE: predictive power as the futility quantity

GBM AGILE is a Bayesian adaptive platform for glioblastoma with a screening stage and a confirmatory stage (Alexander et al. 2018). The first arm to finish was regorafenib, equally randomized against control in newly diagnosed MGMT-unmethylated (temozolomide plus radiotherapy control) and recurrent (lomustine control) disease. Efficacy would be declared at a Bayesian probability of benefit (hazard ratio below 1) of at least 98%. Analyses ran monthly, and the arm stopped enrolling for limited efficacy when the Bayesian predictive power was below 25% in all signatures (Wen et al. 2026). It did: the median hazard ratios were 1.05, 1.07 and 1.07 (newly diagnosed, recurrent, all) with probabilities of benefit of 0.421, 0.312 and 0.296, and the drug had more toxicity than control (Wen et al. 2026). An erratum to the regorafenib report has been published (Wen, Berry, et al. 2026).

The stopping rule needed no unblinded judgment: the threshold was in the design and the data crossed it. The lesson for this chapter is the choice of threshold. A 25% limit on predictive power is far more aggressive than the 5% or 10% limits above, and the loss in power that it causes has to be paid for in the simulation report.

15.11. Summary

Conditional power, predictive probability, the Lee-Liu design, the Goldilocks design and the backward induction policy are one idea at increasing ambition: conditional power fixes the effect, predictive probability averages over it, and backward induction derives the rule from a loss. All are judged by the operating characteristics of Chapter 14.

> Try it in the lab
>
> Open module 09 (Adaptive Bayesian) at https://vsokolov.org/bayesian_trials_lab/09-adaptive-bayesian.html. Before moving the sampling-cost control, predict what happens to the expected sample size and to the probability that the rule says "better" when the drug does nothing; then check. The module's CALGB 49907 example (a planned 1,800 patients, stopped at 633) is the case of this chapter. Then open the simulator at https://vsokolov.org/bayesian_trials_lab/14-simulator.html (the full version is `14-studio.html`) and set a futility threshold on predictive probability: predict how much the expected sample size falls under the null, then run it.

Key points

- Conditional power depends on an assumed effect: under the design alternative it is optimistic after a poor interim, under the current trend it is pessimistic. Report which, and do not mix them.
- Predictive probability of success is conditional power averaged over the posterior of the effect. It needs a prior, and the prior has to be pre-specified and stress-tested.
- For binary endpoints the predictive probability is exact: a beta-binomial sum over the future responder counts, with no integration or simulation.
- Futility stopping cannot inflate the type I error, and a non-binding rule keeps the final test valid even if the DMC does not follow it. Efficacy stopping spends error and requires calibration.
- Lee and Liu's predictive probability design tabulates, before the trial, the responder counts that stop for futility or success at each look, and its operating characteristics can be computed exactly. In the worked example, three looks cut the expected sample size from 40 to about 27 with power 0.867 against 0.871.
- The Goldilocks design uses two predictive probabilities: success if enrollment stops now ($\mathrm{PP}_{\mathrm{stop}}$) and success at the maximum size ($\mathrm{PP}_{\mathrm{max}}$). Stopping accrual for expected success spends type I error.
- Backward induction gives the optimal stopping rule for a stated loss and cost per patient. The sufficient-statistic reduction makes it exact and cheap for binary endpoints: $O(T^3)$ states.
- The optimal rule is only as good as its loss. A utility-only rule chooses the better arm quickly and says "better" half the time under the null; it is not a confirmatory rule. In the 60-pair example, at the target alternative the classification version uses about 24% fewer patients on average, with power 0.790 rather than 0.806; the null rejection probabilities both round to 0.024.
- Regulators judge all of these by simulated or exact operating characteristics under a pre-specified design (E20 section 5, FDA Bayesian draft), not by the elegance of the decision theory.

Discussion questions

1. A DMC meets at 50% information and the interim Z is 1.0. The sponsor reports conditional power of 0.70, the DMC statistician reports 0.22. Which assumptions produce each number, and what additional information would you ask for before recommending that the trial continue?
2. A futility rule is declared non-binding. The DMC recommends stopping, the sponsor continues, and the final p-value is 0.04 two-sided. Is the trial positive? What does your answer depend on in the protocol and the SAP?
3. In the Lee-Liu example, seven looks every five patients raised the null success probability from 0.043 to 0.057 at the same thresholds. List the ways to restore it to 0.043 or below, and say what each costs in power and expected sample size.
4. A Goldilocks design stops enrollment when $\mathrm{PP}_{\mathrm{stop}} > 0.95$, with ten patients per arm pending. Why can this be an efficacy decision in disguise, and what would you check in the simulation report to show that it is controlled?
5. Backward induction with a loss ratio of 39 to 1 and a cost per pair produces a stopping table. A regulator asks you to justify the loss ratio. What do you say, and what do you submit instead?

16. Response-Adaptive Randomization and Bandits

In 1933 William Thompson asked a question that every clinician has felt at the bedside: if the first patients treated with a new drug seem to do better than those on the old one, why should the next patient be assigned by a coin flip? His answer was to let the odds shift toward the treatment that is looking better, and his motive was ethical, to spare patients from being "sacrificed" to the inferior treatment. The idea is called response-adaptive randomization (RAR), and in the language of operations research it is a multi-armed bandit problem. It is the most intuitive of the adaptive features in this book and the most contested.

It is contested because it asks a randomized trial to do two jobs at once. A trial has to produce a comparison that a regulator will accept as substantial evidence, and that comparison is cleanest when allocation is fixed and balanced. A bandit has to maximize the benefit to the patients inside the study, and that is best done by abandoning the losing arm as soon as it is identifiable. The two objectives pull in opposite directions, and the history of RAR, from the ECMO trial of 1985 to the critiques of the 2010s, is a history of finding out how far.

This chapter follows the book's rule and starts with the traditional design: fixed allocation, usually 1:1, and the reasons it is the default. It then develops the bandit view, including the surprising fact (due to Polson and Sokolov (Polson and Sokolov 2026b)) that the logarithm that governs the Bayes sequential boundary also governs the regret of an optimal bandit policy. It tells the ECMO story, the one case every trialist knows. It then gives the critiques their due, simulates the failure that matters most in practice (drift in the patient population over calendar time), and shows what fixes it. It ends with the multi-arm setting, where RAR earns its place, and with what the regulators say.

! Regulatory anchor

The documents below govern changes of the allocation ratio on the basis of accumulating outcome data. Chapter 3 has the full map and the status of each document.

- **FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics (final, November 2019).** Treats changes to allocation based on accumulating outcome data as adaptive-design features and holds them to the four principles of its Section III: "the chance of erroneous conclusions should be adequately controlled, estimation of treatment effects should be sufficiently reliable, details of the design should be completely prespecified, and trial integrity should be appropriately maintained." Expects simulation of operating characteristics ("A major use of simulations in adaptive trial design is to estimate trial operating characteristics"). Final guidance: FDA's current thinking, not legally binding.
- **ICH E20, Adaptive Designs for Clinical Trials (Step 2 draft, June 2025).** "A proposed adaptive design requires a clear and compelling justification" and "the number and complexity of adaptations should generally be limited" in

confirmatory trials. A draft, not final.

- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft, January 2026).** For trials with an overall frequentist analysis plan, success criteria are "almost always" chosen so that the familywise type I error is "no greater than 0.025, one-sided". The draft also describes direct interpretation of posterior probabilities and decision-theoretic criteria. Calibration is the approach this chapter uses, and it applies to a Bayesian design with RAR as much as to any other. A draft.
- **FDA, Master Protocols for Drug and Biological Product Development (revised draft, June 2026).** With a shared control, a design that "allocates more participants to the control arm than each individual drug arm" is described as appropriate, and comparisons should use "only concurrently eligible control participants". Ratio changes in its examples are fixed schemes, not driven by outcome data. A draft.
- **EMA, reflection paper on confirmatory platform trials (draft, October 2026).** Recommends fixed allocation ratios for each comparison of interest and concludes that response-adaptive randomization is not supported in confirmatory platform trials. A draft, and the strictest position of these documents.
- **ICH E9 (1998)**, Section 2.3.2: randomization "introduces a deliberate element of chance into the assignment of treatments", the property that RAR modifies and that the analysis must still respect.

16.1. The traditional default: fixed allocation

A traditional two-arm trial fixes the allocation ratio in the protocol and keeps it for the whole study. Almost always the ratio is 1:1, implemented by permuted blocks, perhaps stratified by site or a prognostic factor (Chapter 5). The sample size comes from the power formula of Chapter 9, and the analysis is the test of Chapter 4. This is the design that RAR must beat, so it is worth saying exactly why it is hard to beat.

16.1.1. Why 1:1

For a comparison of two means or two proportions with the same variance in each arm, the standard error of the estimated difference is proportional to $\sqrt{1/n_1 + 1/n_0}$. For a fixed total $n = n_0 + n_1$ this is smallest when $n_0 = n_1$. If the ratio is $r = n_1/n_0$, the information about the treatment effect is proportional to $4r/(1+r)^2$ of its 1:1 value, and the sample size needed for the same power is inflated by the reciprocal. Table 16.1 gives the numbers.

Table 16.1.: Cost of unequal fixed allocation. Efficiency is the information relative to 1:1 for the same total n. Inflation is the factor by which the total sample size must grow to keep the power of a 1:1 design.

Ratio	Efficiency	Inflation of n	Share on the larger arm
1:1	1.000	1.000	50%
2:1	0.889	1.125	67%
3:1	0.750	1.333	75%

Table 16.1.: Cost of unequal fixed allocation. Efficiency is the information relative to 1:1 for the same total n. Inflation is the factor by which the total sample size must grow to keep the power of a 1:1 design.

Ratio	Efficiency	Inflation of n	Share on the larger arm
4:1	0.640	1.562	80%
9:1	0.360	2.778	90%

A 2:1 ratio costs 12.5 percent more patients. A 3:1 ratio costs a third more, and 9:1 costs almost three times as many. These are not small prices, and they are paid whatever the data show. When the outcome is binary the variances differ between the arms, and the optimal (Neyman) allocation puts more patients on the arm with the larger variance. For a control response of 0.30 and a treatment response of 0.45 the optimal share on the treatment arm is only 52 percent, so the textbook gain from Neyman allocation is negligible in the range where trials are run. Equal allocation is close to optimal for the questions trials usually ask.

Three other reasons keep 1:1 the default, and none is about the variance formula.

The first is simplicity of the protocol and the analysis. Allocation is a pre-specified schedule independent of the outcomes, so every patient's assignment is, in the words of ICH E9, a deliberate element of chance, and the randomization distribution of the test statistic is exactly known. The type I error of the test of Chapter 4 is a property of the design, not of a model for how patients enter.

The second is balance in calendar time. With permuted blocks, each arm receives, within one block, the same number of patients, so any change in the patient mix, the standard of care, the assay or the site mix during enrollment hits both arms equally. Time is the covariate nobody records and everybody has. Fixed allocation makes it harmless by construction. This property is the one RAR gives up, and the simulations below show what that costs.

The third is the absence of a statistical contract with the data. A fixed design says nothing about what the interim results may be. The data monitoring committee can look at them (Chapter 11), but allocation does not react, so the trial cannot amplify a lucky early streak.

16.1.2. Unequal fixed allocation and when it is justified

There are good reasons to depart from 1:1, and regulators accept them when stated in advance. A sponsor may want more patients on the experimental arm to collect safety exposure, a rare-disease trial may find enrollment on the experimental arm easier when the odds of getting it are 2:1, and a shared control in a multi-arm trial should be larger than each experimental arm. For k experimental arms and one shared control, Dunnett's classical result (Dunnett 1955) is that the control arm should be about $\sqrt{k}$ times as large as each treatment arm, to minimize the variance of the k comparisons with control. Table 16.2 shows how far this is from equal allocation.

Table 16.2.: Share of patients on the control arm under equal allocation and under Dunnett's square root rule, for k experimental arms.

k	Equal allocation	Dunnett sqrt(k) rule
2	33.3%	41.4%
3	25.0%	36.6%
4	20.0%	33.3%
5	16.7%	30.9%
8	11.1%	26.1%

With three experimental arms, equal allocation puts 25 percent of the patients on control and Dunnett's rule puts 36.6 percent. The FDA master protocols draft makes the same point in its own words (control allocated more than each drug arm), and the EMA draft acknowledges the sqrt(k) rule as more efficient for fixed designs while noting that it is not always suitable when arms enter and leave. These are fixed ratios. A fixed control share protects control accrual but does not establish regulatory acceptability of adaptive experimental allocation, because the ratio of each experimental arm to control still moves when the experimental allocation does. The question the rest of the chapter asks is what, if anything, is gained by letting the ratios among the experimental arms move with the data.

16.2. Allocation as a bandit problem

16.2.1. Exploring and exploiting

Suppose two treatments have unknown success probabilities p_0 and p_1 and patients arrive one at a time. Each assignment does two things. It treats the patient (the reward, a success or a failure), and it teaches us about the arm (information that improves later assignments). Assigning the arm that currently looks better exploits what we know. Assigning the other arm explores. A trial that explores too little may lock onto the wrong arm. A trial that explores too much wastes patients on the arm it already suspects is worse. This is the multi-armed bandit problem, named for a row of slot machines with unknown payout rates, and it was posed in medicine before it was named anywhere.

Thompson (Thompson 1933) proposed that the next patient be assigned to arm 1 with probability equal to the current posterior probability that arm 1 is better, $\Pr(p_1 > p_0 \mid \text{data})$. Robbins (Robbins 1952) formulated the sequential allocation problem for two populations with unknown distributions, together with its reward-shortfall criterion (regret, the shortfall of the reward actually obtained against the reward of always giving the best arm); Lai and Robbins (T. L. Lai and Robbins 1985) established the logarithmic lower bound (below). Clinical-trial versions of the problem are traced in Section 1.9 and Section 15.6. Gittins (J. C. Gittins and Jones 1974; J. C. Gittins 1979) showed that when future patients are discounted geometrically, the optimal policy is an index policy, and Berry and Fristedt (D. A. Berry and Fristedt 1985) gave the definitive Bayesian account. Villar, Bowden and Wason (Villar, Bowden, and Wason 2015) review the use of these models in trial design and are frank about the obstacles, which the simulations of this chapter quantify.

The allocation rules in the clinical literature came by three routes. The first is the urn: Zelen (1969) proposed the play-the-winner rule, and randomized versions followed (Wei

and Durham (1978); the ECMO trial in Section 16.3). The second is target allocation: Rosenberger and Lachin (1993) reviewed response-adaptive designs, Rosenberger et al. (2001) derived allocation proportions that minimize the expected number of treatment failures for a fixed power, and the doubly adaptive biased coin design (Eisele 1994; Hu and Zhang 2004; Hu and Rosenberger 2006) is the usual frequentist way to reach a stated allocation target with controlled variability. The Bayesian bandit rules of this section (Thompson sampling and the Gittins index, with the forward-looking version of Villar, Wason, and Bowden (2015) for several arms) are the third family.

16.2.2. Thompson sampling in four lines

With a Beta prior on each success probability, the posterior after s successes and f failures is $\text{Beta}(1+s, 1+f)$ (Chapter 6). Thompson sampling draws one value from each arm's posterior and assigns the patient to the arm with the largest draw. For two arms, the probability that arm 1 is chosen is exactly $\Pr(p_1 > p_0 \mid \text{data})$.

```
set.seed(1)
prob_better <- function(s1, f1, s0, f0, M = 2e5) mean(rbeta(M, s1 + 1, f1 +
 ↪  1) > rbeta(M, s0 + 1, f0 + 1))  # uniform priors
# after 20 patients: 7 of 10 respond on the new arm, 4 of 10 on control
early <- prob_better(7, 3, 4, 6)
# after 200 patients with the same observed rates, the posterior is far
 ↪  more concentrated
late <- prob_better(70, 30, 40, 60)
kable(data.frame(Stage = c("After 20 patients", "After 200 patients"),
 ↪  `P(new arm better)` = round(c(early, late), 3), check.names = FALSE))
```

Stage	P(new arm better)
After 20 patients	0.901
After 200 patients	1.000

Early in a trial the allocation probability is close to one half and the design behaves like equal randomization. As evidence accumulates it moves toward 0 or 1. That is the whole algorithm. Nothing in it asks for a threshold, a boundary or a stopping rule, and that fact is the subject of the "same logarithm" argument below.

16.2.3. The Bellman equation

A trial is a reinforcement-learning problem in the sense that its state is the posterior. Let π_t be the posterior over the arms' success probabilities p_k after t patients and m the number of patients still to be treated. The best policy solves

$$V(\pi_t, m) = \max\Big\{ \underbrace{m \max_k \mathrm{E}_{\pi_t}[p_k]}_{\text{stop and treat}},\ \max_k \Big(\mathrm{E}_{\pi_t}[p_k] + \mathrm{E}\big[V(\pi_{t+1}, m-1) \mid \pi_t, \text{arm } k\big] \Big) \Big\}, \quad (16.1)$$

where the first term stops the trial and gives every remaining patient the arm that currently looks best, and the second assigns the next patient to arm k, collects the expected response, and moves to the updated posterior by Bayes' theorem. Reinforcement learning calls this a belief-state Markov decision process, and statisticians have called it a Bayesian sequential design since the 1950s (Chapter 11). The two literatures solve the same equation. It is solvable exactly only for small problems, which is why the rest of this section looks at its two famous approximations: the Gittins index for the discounted infinite horizon, and Thompson sampling.

16.2.4. The Gittins index

The Gittins index (J. C. Gittins and Jones 1974) is the exact solution to a cleaner problem. Each arm has a state (for a binary outcome, the counts of successes and failures). Future rewards are discounted by a factor $\gamma < 1$ per patient. The index of an arm in state s is the highest achievable ratio of expected discounted reward to expected discounted time, over the stopping times τ at which one might abandon the arm,

$$G(s) = \sup_{\tau > 0} \frac{\mathrm{E}\left[\sum_{t=0}^{\tau-1} \gamma^t r_t \mid s\right]}{\mathrm{E}\left[\sum_{t=0}^{\tau-1} \gamma^t \mid s\right]}, \tag{16.2}$$

or, equivalently, the constant payoff per patient at which a decision maker would be exactly indifferent between continuing with that arm and retiring to a sure thing paying that constant forever (Whittle 1980). The optimal policy is to give each patient the arm with the highest index. The remarkable part is that each arm's index depends only on that arm's own history, so a problem with many arms is solved by tabulating one-arm indices. Table 16.4 shows a few values for $\gamma = 0.9$ and 0.99.

Table 16.4.: Gittins index for a Bernoulli arm with a Beta(a, b) posterior (that is, a - 1 successes and b - 1 failures counted from a uniform prior), discount factors 0.9 and 0.99. The index exceeds the posterior mean when the arm is uncertain, and the excess is the value of learning.

Posterior	Posterior mean	Index, 0.9	Index, 0.99
Beta(1, 1)	0.500	0.7029	0.8699
Beta(2, 1)	0.667	0.8001	0.9102
Beta(1, 2)	0.333	0.5001	0.7005
Beta(3, 3)	0.500	0.6010	0.7308
Beta(5, 5)	0.500	0.5676	0.6697
Beta(10, 10)	0.500	0.5373	0.6045
Beta(7, 3)	0.700	0.7573	0.8350
Beta(30, 70)	0.300	0.3076	0.3261

An arm we know nothing about (the uniform, Beta(1, 1)) has posterior mean 0.5 but index 0.703. A well-measured arm with the posterior Beta(30, 70) has mean 0.3 and index 0.308, almost no premium. The premium on uncertainty is what makes an optimal policy try a new arm even when its estimate is mediocre. A clinical reader can think of it as the value of finding out. Three arms with the same posterior mean of one half, Beta(1, 1), Beta(5, 5)

and Beta(10, 10), have indices 0.7, 0.57 and 0.54, falling as the arm becomes better known. A longer horizon raises the premium: with $\gamma = 0.99$ the index of the untried arm is 0.87, because the more future there is in which to exploit what is learned, the more an untried arm is worth.

The index is a beautiful result with two limits for trials. It is exact for an infinite, discounted horizon, whereas a trial has a fixed number of patients and a finite horizon problem has no index solution. Practical designs therefore use approximations, with Thompson sampling the most common. And the objective is the reward to patients in the trial, which, as we see below, is not the objective of a confirmatory comparison.

16.2.5. Regret and the Lai-Robbins bound

Regret measures how much a policy loses against an oracle that knows the best arm. For a two-arm trial the regret after T patients is the expected number of patients on the inferior arm times the gap in response probabilities. Equal randomization puts half the patients on the inferior arm, so its regret grows linearly in T. The question is how much better any policy can do.

Lai and Robbins (T. L. Lai and Robbins 1985) answered it. For any policy that is uniformly good in the sense of T. L. Lai and Robbins (1985) (expected regret of order $o(T^a)$ for every $a > 0$ and every parameter value; the bound holds for single-parameter families under regularity conditions), the expected number of patients on an inferior arm i satisfies

$$\liminf_{T \to \infty} \frac{\mathbb{E}[N_i(T)]}{\log T} \;\geq\; \frac{1}{\mathrm{KL}(p_i, p^*)}, \tag{16.3}$$

where $\mathrm{KL}(p_i, p^*)$ is the Kullback-Leibler divergence between the response distribution of the inferior arm and that of the best arm. In words: a good policy cannot avoid giving the inferior arm a number of patients that grows like the logarithm of the horizon, and the constant is the reciprocal of how distinguishable the arms are. The regret of the best possible policy is logarithmic, not linear. For response probabilities 0.30 and 0.45 the divergence is 0.047 and the bound is about 21 patients per unit of $\log T$.

Figure 16.1 simulates the number of patients on the inferior arm for equal allocation, Thompson sampling and the UCB1 rule of Auer, Cesa-Bianchi and Fischer (Auer, Cesa-Bianchi, and Fischer 2002), which adds to each arm's observed mean a bonus of $\sqrt{2 \log t / N_i(t)}$ and assigns the arm with the largest total.

At 100 patients, Thompson sampling has placed about 27 on the inferior arm against 50 for equal allocation. By 5,000 patients the figures are about 90 and 2,502, with UCB1 at about 387. These are the figures of an idealized setting: a very long horizon, an immediate binary outcome and one decision, whether to treat. A phase 3 trial is usually a few hundred patients and its outcome takes months to observe, and its main product is a statement about the drug for people outside the trial. The rest of the chapter is about what that does. The Lai-Robbins bound is asymptotic (a liminf as T grows), so finite-horizon counts can sit below the plotted slope, as Thompson sampling does at $t = 5{,}000$ (about 90 patients against about 181 from $\log T / \mathrm{KL}$).

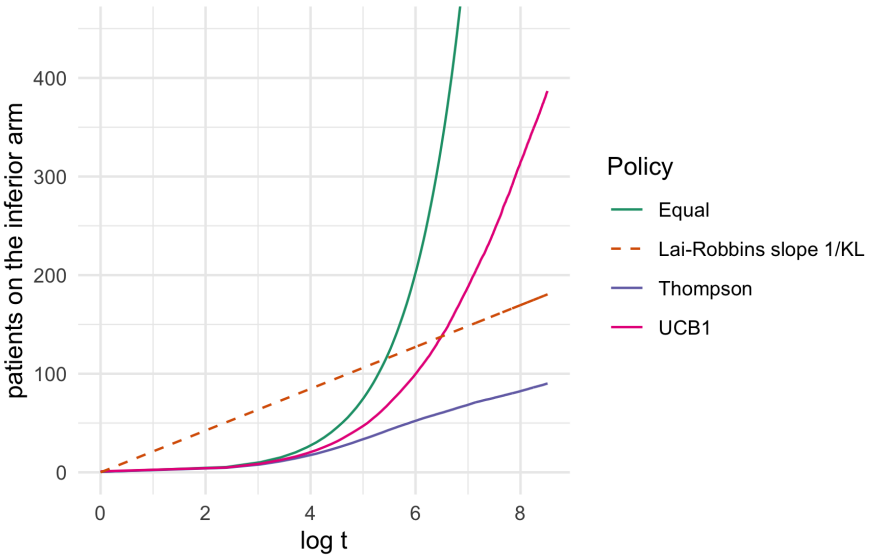


Figure 16.1.: Expected number of patients on the inferior arm (response 0.30 against 0.45) after t patients, 300 simulated trials per policy, plotted against log t. Equal allocation grows linearly in t, so it is exponential on this axis. Thompson sampling and UCB1 grow like log t.

Technical note: why the bound is logarithmic

To tell an inferior arm from the best arm at error probability ϵ, a policy needs about $\log(1/\epsilon)/\mathrm{KL}$ observations on that arm, because KL is the rate at which log likelihood ratio evidence accumulates per observation. A policy that must be uniformly efficient for every possible truth cannot let the error probability on this question fall slower than polynomially in the horizon T, since a mistaken commitment costs a number of patients linear in T. Setting $\epsilon \approx 1/T$ gives $\log T/\mathrm{KL}$ observations on the inferior arm, which is Equation 16.3. The logarithm is the price of a bar that tightens with the horizon, and the argument of the next section is that the same price appears in sequential testing.

16.2.6. The same logarithm

Polson and Sokolov (Polson and Sokolov 2026b) observe that the scale in Equation 16.3 is the scale of the Bayes sequential boundary of Chapter 11 and the moderate deviation sample size equation of Chapter 9. We give the argument as they make it and mark which parts are exact and which are an analogy.

For contiguous alternatives (a small standardized effect θ) the Kullback-Leibler divergence is $\mathrm{KL} \approx \theta^2/2$. The moderate deviation relation for a test that balances the two errors is $n\theta^2 \asymp \log n$, which in terms of divergence is $n\,\mathrm{KL} \asymp \frac{1}{2}\log n$. For asymptotically optimal policies, the Lai-Robbins bound is attained in order, $\mathbb{E}[N_i(T)]\,\mathrm{KL}_i \sim \log T$; the theorem itself supplies only the lower bound. These are the same relation with the horizon T in place of the sample size n, up to a factor of two. The number of patients a bandit policy must

waste on an inferior arm is the number of observations the Bayes rule needs to distinguish that arm at the moderate deviation scale.

The frequentist bandit algorithm makes the same point from the other side. UCB1 adds the bonus $\sqrt{2\log t/N_i}$, which comes from a Hoeffding bound for rewards in $[0,1]$ whose error probability shrinks polynomially in t (as t^{-4} in the analysis of Auer and coauthors). So UCB's confidence level also decays with the horizon, and its bonus grows like $\sqrt{\log t}$, the same order as the Bayes bar $\sqrt{\log T}$. The constants are not comparable in general, because the UCB radius is on the response scale and a testing boundary is on the standardized scale, so they depend on the variance proxy. Table 16.5 prints the two expressions for a unit variance proxy to show the shared growth, not a constant-by-constant comparison.

Table 16.5.: The UCB exploration expression and the moderate deviation (Bayes) bar, both on a standardized scale with unit variance proxy (an illustration of shared order, not a comparison of constants). Both grow like the square root of the logarithm of the horizon, in contrast with the fixed bar z = 1.96 of a one-look level 0.025 test.

Horizon	log T	UCB bar sqrt(2 log T)	Bayes bar sqrt(log T)
100	4.61	3.03	2.15
1,000	6.91	3.72	2.63
10,000	9.21	4.29	3.03
1,000,000	13.82	5.26	3.72

An algorithm that holds its confidence level fixed, which is the bandit analogue of an unadjusted fixed-cutoff test, has a constant probability of locking onto the inferior arm forever and suffers regret linear in the horizon. Sublinear regret needs a level that decays with the horizon. That is the lesson of Chapter 11 from a literature that was not trying to teach it: repeated marginal tests at an unchanged cutoff generally fail to control the overall error, whereas sequentially valid procedures (alpha spending, the e-process threshold $1/\alpha$ of Chapter 12) can retain a fixed overall α.

Thompson sampling has no explicit bar. It draws from the posterior and plays the maximizer, so its exploration is whatever the posterior's spread dictates, and yet Kaufmann, Korda and Munos (Kaufmann, Korda, and Munos 2012) and Agrawal and Goyal (Agrawal and Goyal 2013) showed that for Bernoulli arms it attains the Lai-Robbins bound asymptotically. The logarithm appears without being imposed. In testing the same thing happens: the Bayes factor carries an Ockham penalty of order $\log n$ (Chapter 7) that arrives as a consequence of integrating out the parameter and not as a threshold chosen by the analyst. In both problems the Bayesian procedure gets the logarithmic scale by construction (its type I error still has to be calibrated, Chapter 14), and in both the frequentist procedure has to be told its level, as a function of the horizon, by a separate calculation. Table 16.6 sets the two problems beside each other.

Table 16.6.: The two problems share a deviation scale. An unadjusted fixed cutoff fails in both; sequentially valid procedures retain a fixed overall error target.

	Testing	Allocation
Wrong calibration	Repeated marginal tests at an unchanged cutoff z_α	A confidence level that does not decay with the horizon, linear regret
Pointwise tail probability for the illustrative boundary	Bar $\sqrt{\log n}$, pointwise normal-tail probability of order $(n \log n)^{-1/2}$ at each n (not the probability of ever crossing)	Bar $\sqrt{2 \log t}$, regret of order $\log T$
Imposed by a threshold	Level chosen as a function of n	UCB
Supplied by the posterior	The Bayes factor's $\log n$ penalty	Thompson sampling

Russo and Van Roy (Russo and Van Roy 2016) give a third reading. They bound the regret of Thompson sampling through an information ratio, the squared expected reward shortfall divided by the information gained about which arm is best, and the bound involves the entropy of the prior over which arm is best. Regret becomes a quantity of information to be acquired divided by a rate at which it arrives, the same accounting as the Wald approximation of Chapter 11.

16.2.7. An analogy that does not transfer as design advice

Polson and Sokolov are explicit that the analogy is exact at the level of the deviation scale and misleading at the level of the objective, and that a recommendation of Thompson sampling for confirmatory trials on the strength of the first would be wrong. Regret is a loss in the trial. Power is a property of the final decision. Optimizing the first damages the second, and the next sections show it. As mathematics, the analogy shows that this particular normal-mixture construction shares a logarithmic scale with optimal bandit regret. As design advice it does not transfer to a trial whose conclusion will govern treatment for a population far larger than the trial. The place where it does transfer is early-phase work, where the population inheriting the decision is the next trial, and that is where the utility-based designs of Chapter 17 put it.

i If it works for Amazon: Thompson sampling on the web

Hill and colleagues (D. N. Hill et al. 2017) describe how Amazon used Thompson sampling to choose among the very many layouts of a web page, searching a combinatorial space of page elements in real time. They report that after a single week of online optimization the chosen layout lifted purchases by about 21 percent over the median layout. The comparison with a trial is instructive and the transfer is limited. A click is observed in seconds, the experiment never really ends, and the visitors after the experiment matter as much as the visitors in it, so in-experiment reward is close to the right objective. A confirmatory trial has delayed outcomes, a population of future

patients that may outnumber those enrolled a thousand to one, and a regulator who must trust the final estimate. That is the argument of the analogy section above and of Section 16.4.

16.3. ECMO: the case every trialist knows

16.3.1. What was done

Extracorporeal membrane oxygenation (ECMO) is a form of heart-lung bypass. In the early 1980s, Bartlett and colleagues were convinced that it saved newborns with persistent pulmonary hypertension and a very high predicted mortality, and that randomizing such infants to conventional medical therapy (CMT) was close to unethical. Their trial (R. H. Bartlett et al. 1985) used a randomized play-the-winner rule (Wei and Durham 1978), the randomized form of Zelen's play-the-winner rule (Zelen 1969). In its simplest form the rule is an urn: start with one ball for each treatment, draw a ball to assign each patient, and after each outcome add a ball for the treatment that succeeded (or for the other treatment after a failure). As the urn drifted toward ECMO, one infant was assigned to CMT and died, and the other eleven were assigned to ECMO and all survived. The final tally was 11 of 11 on ECMO and 0 of 1 on CMT, with intracerebral hemorrhage in 1 of the 11 ECMO infants (R. H. Bartlett et al. 1985). A Fisher exact test of 11 of 11 against 0 of 1 gives a one-sided $p = 1/12 = 0.083$, and Ware and Epstein (Ware and Epstein 1985) commented on the trial in the same journal.

16.3.2. The controversy

Ware (Ware 1989) used the trial as a case study of the difficulties of investigating therapies of potentially great benefit. The difficulty is specific. The conclusion that ECMO works rests on twelve infants, and the comparison with control rests on one death. A reader who accepts the trial does so largely because they already believe ECMO works and because the historical mortality of such infants was dreadful, and a reader who does not believe it can reasonably say that the trial did not provide the evidence. The design that was meant to reconcile ethics with science had allocated the information away from the comparison. The ethics of randomizing in such a case is treated as a problem of statistical inference by Royall (1991), who describes the randomized trial as a scientific and ethical dilemma and notes that statisticians have tended to emphasize only one side of it.

The Bayesian reading, as discussed by Kass and Greenhouse in their commentary (Kass and Greenhouse 1989b) and by Berry on the ethics (D. A. Berry 1989), is more forgiving in one respect: the posterior depends on the data observed and not on the rule that produced them. With uniform priors the posteriors are $\text{Beta}(12, 1)$ for ECMO and $\text{Beta}(1, 2)$ for CMT, and the posterior probability that ECMO is better is $90/91 \approx 0.989$, which the chunk below confirms by integration. With a $\text{Beta}(4, 16)$ prior for CMT encoding a historical survival of about 20 percent, the probability is above 0.999. That agreement of the two priors is, of course, a statement about the data being extreme and not about the design being good.

```
f  <- function(x) dbeta(x, 12, 1) * pbeta(x, 1, 2)     # uniform priors
f2 <- function(x) dbeta(x, 12, 1) * pbeta(x, 4, 17)    # Beta(4,16) prior
 ↪  for CMT, 0/1 observed
kable(data.frame(Quantity = c("Uniform priors", "Exact 90/91", "Beta(4,16)
 ↪  prior for CMT"), `Pr(ECMO better)` = round(c(integrate(f, 0, 1)$value,
 ↪  90 / 91, integrate(f2, 0, 1)$value), 4), check.names = FALSE))
```

Quantity	Pr(ECMO better)
Uniform priors	0.989
Exact 90/91	0.989
Beta(4,16) prior for CMT	1.000

16.3.3. The Harvard two-stage trial

The comment on the Michigan trial was that it could not settle the question, and the Harvard group (Boston Children's Hospital) designed a second trial with that in mind (O'Rourke et al. 1989). In the first phase infants were randomized between ECMO and CMT until a fourth death had occurred in one arm. That happened on the CMT arm: 9 of 9 ECMO infants survived against 6 of 10 on CMT. In the second phase the next 20 infants all received ECMO and 19 survived, so that overall 28 of 29 ECMO infants survived against 6 of 10 on CMT. The trial report is O'Rourke et al. (O'Rourke et al. 1989); the analysis that made it a teaching case is Ware's (Ware 1989). The first phase alone has a one-sided Fisher p of 0.054, just short of the conventional bar, and the data from the second phase cannot be used in a randomized comparison because they were not randomized.

16.3.4. The UK collaborative trial

The question was settled by a conventional randomized trial in the United Kingdom (UK Collaborative ECMO Trial Group 1996). It enrolled 185 infants in 55 hospitals from 1993 to 1995 and compared referral to one of five ECMO centers with continued conventional care. It was stopped early on the advice of the data monitoring committee. Thirty of 93 infants allocated to ECMO died against 54 of 92 allocated to conventional care, a relative risk of 0.55 (95% CI 0.39 to 0.77, $p = 0.0005$), or about one extra survivor for every three to four infants allocated to ECMO. (Only 84% of those allocated to ECMO received it, so this is an intention-to-treat comparison of a referral policy.)

16.3.5. How much the prior matters

Table 16.8 collects the four data sets and two summaries computed from them: the one-sided Fisher exact p (a nominal calculation that ignores the allocation rule and the stopping design of the adaptive trials, and is omitted for the pooled Harvard rows, where the second-phase ECMO infants were not randomized) and the posterior probability that ECMO has the higher survival under independent uniform priors.

Table 16.8.: The ECMO trials: survivors out of infants, the one-sided Fisher exact p (nominal: it ignores the allocation and stopping design of the adaptive trials, and is not shown for the pooled Harvard phases I and II, which combine nonrandomized ECMO infants with earlier controls), and the posterior probability that ECMO survival exceeds control survival under independent uniform priors. The Harvard rows take 28 of 29 on ECMO as given, which assumes the second phase adds only ECMO infants. The UK row counts survivors (63 of 93 and 38 of 92 are the complements of the reported deaths).

Trial	ECMO	Control	Fisher p	Pr(ECMO better)
Michigan (Bartlett 1985)	11/11	0/1	0.083	0.9890
Harvard, phase I	9/9	6/10	0.054	0.9773
Harvard, phases I and II	28/29	6/10	n/a	0.9968
UK collaborative (1996)	63/93	38/92	0.00025	0.9998

Kass and Greenhouse (Kass and Greenhouse 1989b) showed in their comment on Ware's paper how a Bayesian should handle the disagreement: not by hiding the prior but by displaying how the conclusion depends on it. Table 16.9 is our reconstruction in that spirit, not a reproduction of their priors or numbers (we did not extract those). Let η be the control log odds of survival and θ the log odds ratio for ECMO. The effect prior is $\theta \sim N(0, \sigma_\theta^2)$, centered on no effect, with $\sigma_\theta = 0.5$ a skeptic (odds ratios beyond about 2.7 are two standard deviations out) and $\sigma_\theta = 5$ open to almost anything. The control prior is either vague, $\eta \sim N(0, 2^2)$, or historical, $\eta \sim N(\text{logit}\, 0.4, 0.5^2)$, centered at 40 percent survival.

Table 16.9.: Posterior probability that ECMO is better, Pr(theta > 0 | data), by the prior SD of the log odds ratio. Our reconstruction in the spirit of Kass and Greenhouse (1989), not their numbers. Control prior vague N(0, 2^2) on the log odds of survival, or historical N(logit 0.4, 0.5^2) where stated.

	SD 0.5	SD 1	SD 2	SD 5
Michigan, vague control prior	0.753	0.894	0.975	0.997
Harvard phase I, vague control prior	0.825	0.939	0.984	0.997
Harvard phase I, historical control prior	0.934	0.989	0.998	1.000
UK collaborative, vague control prior	0.999	1.000	1.000	1.000

Three things follow. To a skeptic ($\sigma_\theta = 0.5$) the Michigan trial is weak evidence (0.75), because almost everything it says about control survival comes from one infant. The Harvard first phase is stronger (0.83), and to anyone open to a large effect it is convincing. The historical control prior, whose information the unaugmented Fisher analysis shown here excludes, moves the skeptic to 0.93, and this is exactly where the disagreement in the 1989 discussion lives: whether that prior is well founded is the question, and the table shows how much hangs on it. By the UK trial the data swamp any of these priors (0.999 even for the skeptic), and evidence that the earlier trials only suggested was confirmed.

16.3.6. What the rule did to the information

The design question is cleaner than the argument about whether ECMO works. Under a randomized play-the-winner rule, how many patients end up on the control arm, and how

much power does the trial have? The simulation below runs the urn with 12 infants and true survival 0.80 on ECMO and 0.20 on CMT, the extreme scenario the SIM paper uses as a stress test (Polson and Sokolov 2026a), and compares it with a fixed 6:6 trial analyzed by a one-sided Fisher exact test at level 0.025.

Play-the-winner puts on average about 3.7 of the 12 infants on CMT, and in about 11 percent of trials at most one, which is the situation in which a comparison is essentially impossible. Its power is 0.15, against 0.29 for a fixed 6:6 split, and neither is a trial anyone would call adequate (twelve patients is not a sample size at which any test can detect even a large effect with confidence, and the true advantage assumed here is huge). The point of the numbers is the direction. The rule gave more infants the treatment the investigators believed in, and it did so at the cost of the information the trial was meant to produce. That is the pattern of every RAR design: the benefit is in-trial reward, the price is terminal information.

16.3.7. The aftermath and what it means for design

A frequentist re-expression of the same logic appears in the later EOLIA trial of ECMO for adult ARDS (Combes et al. 2018), whose primary analysis gave $p = 0.09$. A Bayesian reanalysis reported 88 to 99 percent posterior probability that ECMO reduces mortality, depending on the prior (Goligher et al. 2018). As the SIM paper puts it, this is a model-dependent re-expression of evidence, not a resolution. The SIM paper also treats the ECMO data as a prior-sensitivity stress test for a backward-induction stopping design (Chapter 15), with a $\text{Beta}(4, 16)$ prior for CMT and a $\text{Beta}(1, 1)$ prior for ECMO. That analysis is explicitly not a reconstruction of the play-the-winner trial and not a proposed design: it asks what a pre-specified Bayesian rule would have concluded and how fast, and it finds that with an extreme generating effect and an asymmetric historical prior the rule stops after a handful of patients per arm. Its warning is the one to carry forward. Extreme early data and an informative prior can make a rule stop almost immediately, and a protocol must show, by simulation, that this is the behavior it wants (Chapter 14).

The lessons of ECMO for RAR are three. A rule that reacts to a few early outcomes can move allocation sharply on very little evidence. The treatment that looks best early is not always best, and an urn that has drifted toward it will take a long time to correct the error. And the argument over whether the evidence was adequate is an argument about the comparison, which was thin because the design starved it.

16.4. The critiques of response-adaptive randomization

16.4.1. The case against, in two arms

By the 2010s the experience with Bayesian adaptive randomization in oncology produced a set of published criticisms that the practitioners of RAR now regard as the fair statement of its costs. Korn and Freidlin (Korn and Freidlin 2011) argued that in two-arm trials outcome-adaptive randomization offers little over equal allocation, while its costs in efficiency and logistics are real. Thall, Fox and Wathen (P. Thall, Fox, and Wathen 2015) collected the scientific and ethical objections, among them lower power for a given sample size, highly variable arm sizes, a non-trivial chance of putting more patients on the inferior arm than equal allocation would have, and sensitivity to time trends, and they questioned whether the ethical gain that motivates RAR survives these defects. Hey and Kimmelman (2015)

argue that the ethical advantage claimed for outcome-adaptive allocation does not survive scrutiny in two-armed or early-phase studies. The summaries here follow the papers' main arguments and are not exhaustive. Wathen and Thall (Wathen and Thall 2017) followed with a simulation study of multi-arm trials, and Proschan and Evans (Proschan and Evans 2020) titled their review as advice: resist the temptation. The balanced review of Robertson and coauthors (Robertson, Lee, et al. 2023) separates the myths from the practical considerations and arrives at guidance that this chapter follows: use a burn-in, restrict extreme allocation, model time trends, and justify the ethical claim explicitly.

16.4.2. The case for, and the replies

The critiques have replies, and a fair account gives them. D. A. Berry and Eick (1995) analyzed adaptive assignment against balanced randomization as a decision problem whose objective is to treat as many patients, in and out of the trial, as effectively as possible. Randomization is satisfactory when the disease is at least moderately common, and adaptive procedures can play a role when a substantial proportion of all patients with the disease are in the trial. That is the premise the later simulations test, and, on our reading, the two-arm trials of the critique are mostly settings with a large population outside the trial. Villar, Robertson, and Rosenberger (2020) is a reply to the review of Proschan and Evans, and Villar, Bowden, and Wason (2018) propose response-adaptive designs for binary responses that keep patient benefit while being robust to time trends. J. J. Lee, Chen, and Yin (2012) revisited the usefulness of outcome-adaptive randomization, and Wason and Trippa (2014) compared Bayesian adaptive randomization with multi-stage designs for multi-arm trials, which puts the cost of RAR against alternatives that also drop arms early. On our reading the case is weakest in a two-arm confirmatory trial with a large outside population, and strongest with several arms and a small population (Section 16.6), provided the ethical claim is made explicitly and the design guards against drift (Section 16.5). This is not a point of agreement: P. Thall, Fox, and Wathen (2015) and Wathen and Thall (2017) argue that the problems persist with several arms.

16.4.3. Reproducing the critique

We simulate the two-arm trial of the Polson and Sokolov example: 400 patients, response 0.30 on control and 0.45 on the experimental arm, binary outcome observed immediately, outcomes analyzed by a one-sided pooled z test of two proportions at $\alpha = 0.025$. Three allocation rules are compared. Equal 1:1 is the traditional design, here simple randomization with probability one half (permuted blocks would give exact balance within periods). Thompson uses the posterior probability that the experimental arm is better, recomputed after every batch of 20 patients. Capped Thompson adds a burn-in of the first 25 percent of patients at 1:1 and then bounds the allocation probability between 0.10 and 0.90 (the bounds are enforced exactly, with the probabilities still summing to one). The simulator is `rar_simulate()` in the package, wrapped by `rar_two_arm()`. We report the number of responders in the trial, the share of patients on the better arm, the number of patients on the inferior arm, the standard deviation of the number of patients on the better arm (how variable the design is from trial to trial), and power.

Table 16.10.: Two-arm binary trial, n = 400, response 0.30 (control) and 0.45 (experimental), 5,000 simulated trials per rule, no drift. Power is one-sided at level 0.025; the last column is the power of the analysis stratified by enrollment period (Section 16.5).

Rule	Re-spon-ders	Share on better arm	Patients on inferior arm	SD of n on better arm	Trials with fewer than half on the better arm	Power, strati-Powerfied
Equal 1:1	150.0	50%	200	10	47.1%	87.6%87.3%
Thomp-son	172.3	87%	53	41	1.2%	61.1%49.0%
Capped Thomp-son	166.6	78%	89	22	0.7%	73.2%64.0%

The results confirm the critique and the original Polson and Sokolov table to within simulation error. Thompson sampling puts about 87% of patients on the better arm and adds about 22 responders to the trial, but power falls from 88% to 61%. The number of patients on the better arm has a standard deviation of about 41 from trial to trial, against 10 for equal allocation, which is the variability Thall and colleagues describe. The critique that RAR often puts more patients on the inferior arm than equal allocation would is weaker here. Fewer than half of the patients end up on the better arm in about 1% of Thompson trials (the equal-allocation figure of 47% is only sampling noise around a count centered at 200). In this scenario the effect is large, the outcome is immediate and the posterior converges quickly, and the costs that remain are power and variability.

Capped Thompson recovers part of the power (73%) and keeps 78% of patients on the better arm, at a cost of about 6 responders relative to uncapped. In both cases, the trade is what Polson and Sokolov describe: for a 400-patient trial whose conclusion will govern treatment of many more patients, the loss of 14 to 26 points of power is not a trade a sponsor should accept in a confirmatory setting.

i Technical note: why power falls

For a binary outcome the variance of the estimated difference is approximately $p_1(1-p_1)/n_1 + p_0(1-p_0)/n_0$, where n_1 and n_0 are the realized arm sizes. When allocation is driven by the data, n_0 falls quickly once arm 1 looks better and the term $p_0(1-p_0)/n_0$ dominates, so the standard error is much larger than that of the 1:1 design at the same total n. With 87 percent of 400 patients on the better arm, $n_0 \approx 52$ and the standard error is about 1.4 times the 1:1 value. In addition n_0 is random and correlated with the outcomes, which is why the nominal z test is no longer exactly standard normal under the null: the simulation below finds a type I error of about 0.049 for Thompson sampling, almost twice the nominal 0.025, even with no drift.

16.5. Time trends: the failure that matters most

16.5.1. The mechanism

RAR reacts to accumulating data, which means it reacts to chance. Under the null hypothesis, if the experimental arm happens to look better in the early patients, RAR sends it more of the later patients. Outcome-adaptive allocation can invalidate the naive analysis even without drift (the no-drift Thompson analysis above has type I error near 0.049 against a nominal 0.025). Changes in the patient population add confounding with calendar time. Suppose the background response rate rises during enrollment, because the standard of care improves, the site mix changes, a referral pattern shifts, or a variant changes. The arm that benefited from early luck now receives more of the later patients, who respond better on any treatment, and it looks better still. The bias runs through allocation alone. Nothing about the treatment has changed, and the comparison of response rates is confounded with calendar time.

This is the central practical objection to RAR in confirmatory trials, and it is the property that fixed allocation with permuted blocks was designed to defeat. The FDA draft on master protocols and the EMA draft both refer to the same mechanism: changes in the allocation probability over time introduce time-varying selection bias, so comparisons should use concurrent controls and the analysis can be hard to adjust.

16.5.2. Simulation

We simulate a null trial (response 0.30 in both arms) in which the response probability of every patient rises linearly from $0.30 - d/2$ at the first patient to $0.30 + d/2$ at the last, with drift d from 0 to 0.30. In the larger values this is a very large change in the patient mix across enrollment; the range is chosen to show the shape of the effect and not to describe a typical trial. Two analyses are applied to each trial. The naive analysis is the pooled z test. The stratified analysis divides enrollment into four consecutive periods, forms the difference in response rates within each period, and combines them with Mantel-Haenszel-type weights. Type I error is the probability that the experimental arm is declared better at one-sided level 0.025.

Table 16.11.: Type I error under the null (0.30 versus 0.30), n = 400, for drift d = 0 and d = 0.20. Nominal level 0.025 one-sided.

Rule	d = 0 naive	d = 0 stratified	d = 0.2 naive	d = 0.2 stratified
Equal 1:1	0.025	0.025	0.026	0.026
Thompson	0.049	0.053	0.196	0.052
Capped Thompson	0.029	0.032	0.138	0.031

The pattern is clear. Under equal allocation, drift is harmless: the type I error stays at its nominal level even for a 30-point drift, because allocation does not depend on the outcomes, so each period contributes equally to both arms in expectation. Under Thompson sampling, the naive test has a type I error that climbs with the drift, from 0.049 with none to 0.196 at $d = 0.2$ and 0.295 at $d = 0.3$, many times the nominal level. Capped Thompson reduces the

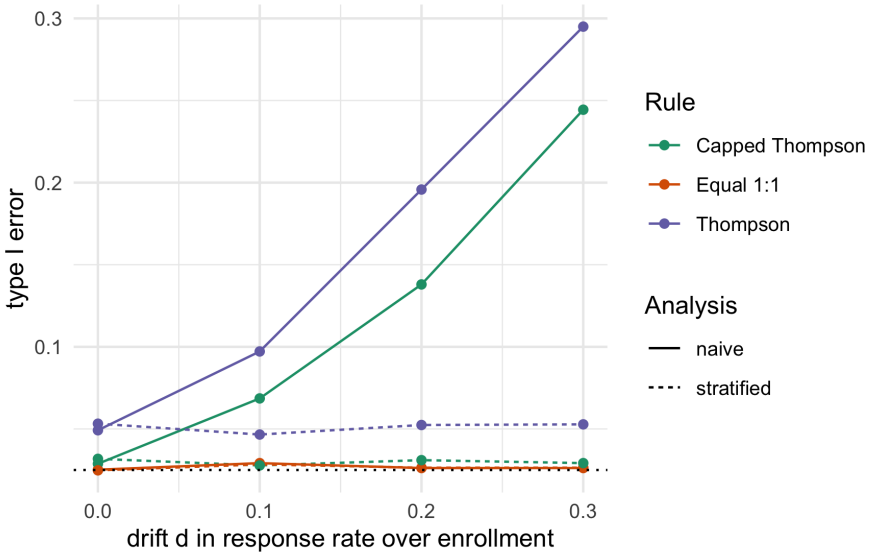


Figure 16.2.: Type I error (one-sided, nominal 0.025) of the experimental-versus-control comparison under the null, as the response rate drifts upward by d over the course of enrollment. Solid lines: naive pooled analysis. Dashed lines: analysis stratified by enrollment period. 5,000 simulated trials per point, n = 400.

problem (0.029 with no drift, 0.138 at $d = 0.2$) but does not remove it. A trial that drifts by 20 points has roughly a one-in-seven chance of declaring a useless drug effective under capped Thompson, and roughly one in five under plain Thompson.

16.5.3. The fixes

Three fixes are standard, and they should be pre-specified as a package.

Stratify the analysis by enrollment period. If allocation probabilities are fixed within each period and the analysis compares arms only within a period, then the allocation within a period is independent of the outcomes in that period and each stratum is a valid comparison. The type I error stops growing with the drift, as the dashed lines show: 0.047 to 0.053 for Thompson (about twice nominal), 0.028 to 0.032 for capped Thompson. The remaining excess for plain Thompson (about 0.05, not 0.025) comes from allocation updating within a period and from small-count behavior of the z test, and is removed by calibrating the critical value by simulation (Chapter 14). The price is power. In the no-drift table of the previous section, the stratified analysis has power 49% under Thompson and 64% under capped Thompson (the corresponding naive figures are 61% and 73%), because comparing within periods discards the information in between-period contrasts when allocation is unbalanced across them. Pre-specified adjustment for calendar time as a covariate in a regression model is the continuous counterpart, with the same trade.

Burn-in and a cap. Start with a period of fixed 1:1 allocation, and keep the allocation probability away from 0 and 1. A burn-in prevents the first handful of outcomes from steering the whole trial, as the urn in ECMO did. A cap guarantees that every arm keeps contributing concurrent data, which is what makes later calendar-time adjustment possible.

The cap reduced the type I error from 0.196 to 0.138 at $d = 0.2$ in the table above, which is helpful but not sufficient alone.

Keep the control allocation fixed. Hold the control arm at a pre-specified share, and let only the experimental arms trade patients. The comparison of each experimental arm with the control then rests on a control stream that is balanced in time. This keeps the control stream balanced in time and is the fix with the best operating characteristics in the multi-arm simulation below. It does not by itself make the design acceptable to regulators: the FDA master protocols draft describes a larger fixed control allocation as appropriate, but the ratio of each experimental arm to control still changes with the data, and the EMA draft recommends fixed ratios for every comparison of interest. In a two-arm trial there is nothing left to adapt once the control is fixed, which is the honest reason that two-arm RAR is hard to defend.

A fourth route is to model the time trend instead of designing around it. The platform-trial literature smooths the control rate across calendar periods and borrows from non-concurrent controls with discounting, the *Bayesian time machine* of Benjamin R. Saville et al. (2022), which is developed in Chapter 20. For multi-arm binary trials, Villar, Bowden, and Wason (2018) studied protected control allocation and found approximately nominal error under the time trends examined. They also assessed randomization-based inference and regression adjustment. The protocol should prespecify the allocation rule and analysis method, and simulations should assess plausible drift and any time-model misspecification.

In the protocol: pre-specifying a response-adaptive randomization rule

If a protocol includes RAR, the statistical analysis plan should state, before the first patient is enrolled:

- the allocation rule in full (for Thompson-type rules, the formula, the tuning exponent if any, and the update frequency);
- the burn-in size and the lower and upper bound on each arm's allocation probability;
- the share of patients on the control arm and whether it is fixed;
- the primary analysis, including how enrollment period or calendar time enters it, and the estimand (ICH E9(R1));
- the simulation report: type I error under the null and under pre-specified drift scenarios (a monotone drift of at least 10 and up to 20 percentage points in the control response is a reasonable starting range), power, expected patients per arm, and the distribution of the allocation, not only its mean;
- how the allocation algorithm is shielded: who runs it, who sees the allocation probabilities, and how the sponsor's blinding is preserved (the DMC charter of Chapter 23).

16.6. When response-adaptive randomization helps: many arms

16.6.1. The multi-arm case

The strongest argument for RAR is not the two-arm trial but the trial with several experimental arms (doses, regimens, or drugs) and a shared control. Two things differ. First, there are many inferior arms, so equal allocation wastes a larger share of the trial (75 percent of patients are on arms other than the best one in a four-arm trial with one best). Second, the comparison that matters, best arm against control, can be protected by fixing the control share while the experimental arms compete for the rest. This is the setting of I-SPY 2 (Barker et al. 2009) and REMAP-CAP (Angus et al. 2020) (both in Section 16.9 below, with BATTLE) and the one in which the review of Robertson and coauthors (Robertson, Lee, et al. 2023) finds the best case for RAR. Wathen and Thall (Wathen and Thall 2017) study the multi-arm setting by simulation.

We simulate a four-arm trial: a control at 0.30 and three experimental arms at 0.30, 0.40 and 0.50, 600 patients, with the same batch size of 20. The four rules are equal allocation, Thompson sampling with allocation proportional to the posterior probability that each arm is best (the multi-arm form of Thompson's rule, which generalizes to arbitrary exponents as in Thall and Wathen (P. F. Thall and Wathen 2007)), capped Thompson with a burn-in and a floor of 2.5 percent on each arm, and the same capped rule with the control share held fixed at 1/4 and the remaining three quarters divided among the three experimental arms by their posterior probabilities of being best among the experimental arms (control excluded). Power is for the best arm (0.50) against control, with a Bonferroni level of 0.025/3 for each experimental arm. Power and type I error are reported for the same prespecified analysis, first the naive pooled test and then the period-stratified one. Bonferroni gives familywise control only when each component test is valid, and the naive test is not valid under drift, so no claim of strong familywise control is made here: that would need calibration across nuisance values and partial-null configurations.

Table 16.12.: Four-arm trial (control 0.30, experimental 0.30, 0.40, 0.50), n = 600, 3,000 simulated trials per rule. Power is for the best arm against control, one-sided, Bonferroni level 0.025/3, for the naive and the period-stratified analyses. The last two columns are the type I error (probability of declaring any experimental arm better) of the same two analyses under the global null with a drift of 0.20 in the response rate over enrollment.

Rule	Responders	Patients not on best arm	Share on control	Power, naive	Power, stratified	Type I, naive, d=0.2	Type I, stratified, d=0.2
Equal	225	450	25%	87.2%	87.2%	1.6%	1.5%
Thompson	276	164	6%	49.5%	34.8%	16.9%	1.0%
Capped Thompson	267	215	9%	68.1%	49.7%	13.2%	1.7%
Capped, control fixed at 1/4	250	298	25%	93.3%	92.9%	3.4%	1.4%

The first three rows repeat the two-arm lesson with more force. Uncapped Thompson sampling puts 73% of patients on the best arm and adds about 51 responders to the 225 of equal allocation, but because it also reduces the control group to about 6% of the trial, power for the comparison with control falls from 87% to about 50%, and a drifting response rate pushes the familywise type I error of the naive analysis to 17%. Capping helps (power 68%) but is not sufficient (type I error 13% under drift).

The fourth row is the result that justifies RAR. With the control fixed at one quarter of the trial, a burn-in and a floor, the design places about half of the 600 patients on the best arm (against a quarter under equal allocation), delivers about 24 more responders than equal allocation, and has higher power than equal allocation for the best arm versus control (93% against 87% for the period-stratified analysis, 93% against 87% for the naive one), because the best arm now has about 300 patients to compare with the control's 150. The two analyses must be read each against its own error. For the stratified analysis the type I error under a 20-point drift is 1.4%, against 1.5% for equal allocation, so the power of the stratified analysis (93%) is the one to quote. For the naive analysis it is 3.4%, above the 2.5 percent target by about three Monte Carlo standard errors (0.3 points at 3,000 trials), so the naive power is not a power one can claim. Both error rates are under the global null only; partial nulls and other nuisance values would have to be checked before the procedure is called error-controlled. The gain is the observation that in a multi-arm trial much of the sample would be spent on arms that are visibly not going to win, and a design that moves that sample to the contender, while the control comparison stays balanced in time, spends it on the question being asked.

Two caveats keep the result honest. The power reported is for the best arm. The 0.40 arm gets less data under the adaptive rule than under equal allocation and its comparison with control has lower power, so if the question is to estimate every dose, equal allocation is the better design. And the result depends on the effect being large enough, and the endpoint fast enough, for the allocation to converge in the trial. With a delayed endpoint, the allocation moves on information that is months old, and the gain shrinks. A simulation that mimics the real lag between enrollment and outcome is a required part of the design report.

16.6.2. Dunnett-type and fixed control allocation

The fixed control share in the fourth row is 1/4, the share equal allocation would give. The Dunnett sqrt(k) share for three experimental arms is 36.6 percent (Table 16.2). A protocol that wants the best control comparison in a fixed design should use the latter, and a protocol with RAR among the experimental arms can choose any fixed share it can defend. The regulatory text asks for the choice to be justified and the ratio to be held fixed, and says nothing that requires either number. The extra control patients increase the power for every comparison with control and cost experimental-arm allocation, a trade the simulation report should show.

16.6.3. Where RAR fits

The conditions under which a case for RAR can be made are those the simulations suggest and the literature endorses (Robertson, Lee, et al. 2023). There are several experimental arms, so there is a lot to be saved from visibly inferior arms. The primary endpoint is observed quickly relative to the enrollment period, so allocation has something to learn from. The trial is large enough for the allocation to converge, and small enough in population

that the in-trial benefit matters (a rare serious disease, an early-phase screening trial). The control share is fixed and concurrent controls are used. And the trial's regulatory role is exploratory or seamless-with-confirmation, not the single confirmatory comparison of record. The platform designs of Chapter 20 and the dose-finding designs of Chapter 17 are where these conditions are most often met. Where they are not (two arms, a slow endpoint, a confirmatory objective) the traditional design wins, and a sponsor that chooses RAR anyway owes the regulator a simulation that shows what it buys.

16.7. What the regulators say

The regulatory texts agree on the principle and differ on strictness. The principle is that RAR is an adaptive design feature and so falls under the general requirements for adaptive designs: prospective planning, control of the false positive rate, reliable estimation, and integrity of the trial. The differences are in how much the text expects a sponsor to justify it.

FDA, Adaptive Designs (2019). Changes to allocation fall within its definition of an adaptive design (anchor above; Chapter 3). The drift simulation above illustrates its warning that adaptive features can inflate the type I error probability, and it expects the analysis, not only the allocation, to be pre-specified, since an analysis that ignores how patients were assigned can be biased. It does not forbid RAR. It places the burden of proof on the sponsor.

ICH E20 (draft, June 2025). For RAR in confirmatory trials the "clear and compelling justification" is the argument of Section 16.6: why the adaptive design is better than a fixed one for this question, shown by simulation under realistic drift and delay. For a two-arm confirmatory trial, Table 16.10 and Table 16.11 are what a reviewer would produce to ask for the justification.

FDA Bayesian draft (January 2026). Nothing in the Bayesian label relaxes the above. This chapter takes calibration by simulation as the default (anchor above). A Bayesian design with RAR is a design with a Bayesian analysis and a data-dependent allocation, and the calibration by simulation (Chapter 14) must include the allocation rule and the drift scenarios. A Bayesian analysis does not by itself solve the confounding of calendar time with arm in a drifting trial. The posterior is conditional on the data, which is why a Bayesian can say that the stopping rule is ignorable, but the model for those data must contain the drift for the posterior to be right.

FDA master protocols (revised draft, June 2026). Its worked examples of ratio changes are fixed schemes in which the ratio changes as drugs enter or leave a platform, not changes driven by outcome data, so it does not address outcome-driven allocation there. It asks that the primary analysis use only concurrently eligible controls.

EMA, draft reflection paper on confirmatory platform trials (October 2026). The EMA text is the strictest: it reasons that changes in allocation probability over time can cause time-varying bias that is difficult to adjust for, and concludes that response-adaptive randomization is not supported in confirmatory platform trials. It is a draft. A sponsor with a global program should assume that RAR in the confirmatory trial of record will be queried in Europe. The draft's conclusion concerns confirmatory platform trials; exploratory and seamless-stage uses still need case-specific justification and agency discussion.

Table 16.13.: Fixed allocation and response-adaptive randomization, side by side.

	Traditional fixed allocation	Bayesian RAR
Allocation	Fixed ratio, usually 1:1, by blocks	Probability updated from accumulating outcomes
Basis in regulation	Default of ICH E9, 21 CFR 314.126	Adaptive design under FDA 2019 and E20 draft, with justification
Type I error	Exact from the design (randomization or normal theory)	By simulation, including drift and delay scenarios
Time trends	Balanced by blocks	Must be handled by stratified or model-based analysis
Power for a fixed n	Highest in the two-arm case	Lower in two-arm trials, can be higher in multi-arm trials with fixed control
Patients on inferior arms	Equal share	Fewer, with higher variability
Where regulators are comfortable	Confirmatory trials	Exploratory, multi-arm, early phase; a fixed control share alone does not establish acceptability, and the EMA draft does not support RAR in confirmatory platform trials

What stays the same is the objective: a comparison that meets the standard of substantial evidence (Chapter 3). What changes is where the burden lies. Some designs, conventional or adaptive, permit analytic or exact recursive evaluation of their operating characteristics; designs with data-driven allocation generally require simulation. Both conventional and Bayesian designs require justification of the complete procedure, including the analysis, the multiplicity handling, the estimation and the operations. For the adaptive trial of this chapter the operating characteristics are known from the simulation report, so the sponsor must make the case, scenario by scenario.

16.8. Practical guidance

For a confirmatory two-arm trial, use fixed allocation. The simulations of this chapter and the critiques of Korn and Freidlin and of Thall and colleagues point the same way: at best a small gain to in-trial patients, a loss of power, and a new route to type I error inflation. If ethics requires protecting patients inside the trial, use a group sequential design with a futility and efficacy boundary (Chapter 11 and Chapter 15), which achieves much of the saving without touching allocation. SHINE (Section 16.9) paired a group sequential design with limited adaptive allocation.

For a multi-arm trial or a platform, decide whether to adapt among experimental arms, and if so, fix the control share, include a burn-in, cap the allocation probabilities, analyze with

concurrent controls, adjust for calendar time, and report type I error under drift, together with the power for each experimental arm and not only the best. Chapter 20 develops the platform setting, including how arms enter and leave. Chapter 17 develops the early-phase use, where Thall's utility-based designs are the strongest case for RAR because the objective there is to choose a dose for the next trial.

For a rare disease, where equal allocation means half of a handful of patients on an arm that may be inferior, a Bayesian design with RAR can be the least bad option and may be the only one the community will accept. The same safeguards apply, and so does the warning from ECMO: the more extreme the allocation, the less information the trial produces about the comparison that regulators will have to judge.

16.9. In practice: four trials

The trials below use adaptive allocation, or sit next to it, in settings where the chapter's conditions are partly met. The facts are from the primary reports cited in each subsection.

16.9.1. I-SPY 2

I-SPY 2 is a neoadjuvant breast cancer platform with ten prospectively defined biomarker signatures (all patients, HR+, HR-, HER2+, HER2-, MammaPrint high-2, and the four HER2 by HR combinations). Patients are classified into eight mutually exclusive subtypes defined by HR, HER2 and MammaPrint status for adaptive randomization, and efficacy and graduation are assessed in the ten overlapping signatures. The allocation probability of each experimental arm follows its current Bayesian probability of success in the subtype, so arms that look better receive more patients, while a fixed share of patients goes to control (J. W. Park et al. 2016; Nanda et al. 2020). No experimental arm exceeds 120 patients, and the model is a covariate-adjusted logistic regression on pathologic complete response (pCR) (J. W. Park et al. 2016). The graduation and futility rules, which are written in predictive probability, and the neratinib and pembrolizumab results are in Section 15.10. The lesson for this chapter is the multi-arm one of Section 16.6. Adaptation among experimental arms, with a concurrent control, serves a screening purpose, and the platform pays for it with a conservative graduation bar and a later confirmatory trial.

16.9.2. REMAP-CAP

REMAP-CAP (Angus et al. 2020) is a platform in which several domains of treatment are randomized at once, and its protocol allows response-adaptive updating of allocation within a domain. What is documented are the Bayesian stopping triggers in the interleukin-6 receptor antagonist domain (REMAP-CAP Investigators, Gordon, et al. 2021): efficacy if the posterior probability of an odds ratio above 1 against control exceeds 99 percent, futility if the probability of an odds ratio above 1.2 is below 5 percent, with neutral priors and a hierarchical model that nests tocilizumab and sarilumab under an overall IL-6 effect. At the interim analysis of 28 October 2020, tocilizumab met the efficacy criterion (posterior probability 99.75 percent, odds ratio 1.87, 95 percent credible interval 1.20 to 2.76), and assignment to control closed on 19 November. The final adjusted odds ratio on organ-support-free days was 1.64 (1.25 to 2.14) for tocilizumab. Here the adaptation is in the stopping

rule, a posterior probability, and the report was made while the platform kept running. The lesson is the one the drift simulation teaches: the analysis adjusts for 2-week time epochs, site and age, which is the calendar-time adjustment that adaptive allocation requires.

16.9.3. BATTLE

BATTLE (E. S. Kim et al. 2011; the design is in X. Zhou et al. 2008) was a biopsy-mandated trial of pretreated advanced non-small cell lung cancer with about 255 patients and four treatments (erlotinib, vandetanib, erlotinib plus bexarotene, sorafenib). After an initial period of equal randomization, a Bayesian probit model moved allocation toward the treatments performing better within biomarker groups. It is described as the first completed biomarker-based, adaptively randomized trial in this disease. An earlier Bayesian adaptive randomization design for recurrent glioblastoma, evaluated retrospectively on four phase 2 trials, is Trippa et al. (2012). The lesson is the caution of Section 16.6 and Section 16.5: allocation among treatments inside small biomarker strata moves on little data, so the burn-in, the cap and an analysis that respects calendar time matter more, not less.

16.9.4. SHINE and its Bayesian re-analysis

SHINE (NCT01369069) compared intensive with standard glucose control for 72 hours in acute ischemic stroke with hyperglycemia. Its design combined group sequential stopping with response-adaptive randomization, and it was not a Bayesian trial. It found no significant difference in favorable 90-day outcome, and severe hypoglycemia occurred in 15 of 581 patients (2.6 percent) on intensive treatment (Johnston et al. 2019). A later paper compared the Bayesian adaptive design developed in the ADAPT-IT project with the frequentist design that was used, in a retrospective comparison (K. Broglio et al. 2022). The lesson is regulatory in tone: a confirmatory-scale trial can combine a classical error-controlled stopping rule with limited adaptation of allocation, which is the conservative end of what Section 16.7 describes, and a Bayesian alternative is judged against it by simulation and not by assertion.

> Try it in the lab
>
> Open Module 9, https://vsokolov.org/bayesian_trials_lab/09-adaptive-bayesian.html, and go to “Send more patients to the winner” (control 30 percent, new drug 50 percent). Before running a trial, predict how many more successes Thompson sampling will give than the 80 expected from an equal split of 200 patients, then run ten more trials and check. Then open the Studio, https://vsokolov.org/bayesian_trials_lab/14-studio.html, switch the allocation from fixed to adaptive and add a drifting scenario: read the operating-characteristics table for power and type I error. The lab shows the gain in successes, and the book and the Studio show the price.

Key points

- Fixed 1:1 allocation is the default because it maximizes power for a given total, makes the randomization distribution exact, and balances calendar time within blocks. Unequal fixed ratios cost 12.5 percent more patients at 2:1 and a third more at 3:1.

- Allocation is a bandit problem. Thompson sampling assigns each patient to an arm with probability equal to its posterior probability of being best. The Gittins index solves the discounted infinite-horizon problem exactly, and the Lai-Robbins bound says the best possible regret grows like $\log T$.
- The logarithm that governs bandit regret is the same logarithm that governs the Bayes sequential boundary: $n\,\mathrm{KL} \asymp \frac{1}{2}\log n$ against $\mathbb{E}[N_i]\,\mathrm{KL}_i \sim \log T$ for asymptotically optimal policies (Polson and Sokolov). Repeated marginal tests at an unchanged cutoff fail in both. The analogy is exact for the scale and does not transfer as design advice, because regret is a loss in the trial and confirmatory power is a property of the final decision.
- In a two-arm trial of 400 patients (0.30 against 0.45), Thompson sampling adds about 22 responders and cuts power from about 88% to about 61%, with much larger variation in the arm sizes.
- Drift in the response rate over enrollment breaks RAR. With a 20-point drift the type I error of the naive test is about 0.196 under Thompson sampling and 0.138 under capped Thompson against 0.025 nominal, and it is unaffected under equal allocation. Stratifying the analysis by enrollment period removes the effect of drift (capped Thompson about 0.031; plain Thompson stays near 0.05 until the critical value is calibrated by simulation), at a cost in power.
- In a multi-arm trial with a fixed control share, a burn-in and a cap, RAR gave about 24 more responders, about half of the patients on the best arm, and higher power for the best arm than equal allocation, reported for the period-stratified analysis (93% against 87%) with its type I error under drift (1.4%); the naive analysis of the same design has type I error 3.4% and its power is not claimable.
- ECMO illustrates the cost of aggressive allocation: play-the-winner moved nearly all infants to one arm and starved the comparison that the trial existed to make.
- Regulators accept RAR as an adaptive feature subject to FDA 2019 and E20 (clear and compelling justification, limited adaptations), expect simulation including time trends, and describe a larger fixed control share as appropriate (which protects control accrual but does not by itself establish acceptability of adaptive allocation among experimental arms). The EMA draft of October 2026 recommends fixed allocation ratios and concludes that response-adaptive randomization is not supported in confirmatory platform trials.

Discussion questions

1. A sponsor proposes 2:1 allocation to the experimental arm in a two-arm confirmatory trial because "patients prefer it". What does the 2:1 design cost in sample size for the same power, and is that cost the same as the cost of a data-driven shift to 2:1 halfway through enrollment? Which of the two does the FDA 2019 guidance treat as an adaptive design?
2. In the drift simulation, equal allocation has the right type I error at any drift and Thompson sampling does not. Explain in words why blocking in calendar time protects the first and why data-dependent allocation exposes the second. What would you add to the SAP to defend a Thompson design against this objection?
3. In the four-arm simulation, the adaptive design with a fixed control has higher power for the best arm than equal allocation, while in the two-arm simulation every adaptive rule has lower power. What is different? For which arm of the four-arm trial does the adaptive design have lower power than equal allocation,

and does that matter for the objective of the trial?

4. Polson and Sokolov say the bandit analogy is "exact at the level of the deviation scale and misleading at the level of the objective". Restate this for a clinician. Where in the development program (phase 1b, phase 2, phase 3, platform) is the in-trial reward the right objective?
5. The Bartlett ECMO trial put 11 infants on ECMO and 1 on conventional therapy. Posterior probabilities were over 0.98 under both priors we tried. Should the regulator have accepted it as substantial evidence? What does the answer depend on besides the posterior probability?

17. Dose Finding and Dose Optimization

Every drug that reaches a confirmatory trial has already been given a dose, and that dose was chosen on very little data. A first-in-human trial may enroll twelve to thirty patients, and the dose that comes out of it is carried into trials of hundreds or thousands. If the dose is too high, patients are harmed and the product may be approved at a dose that later has to be reduced. If it is too low, an effective drug is lost. The statistical problem is small in sample size and large in consequence, which is where a Bayesian design has its clearest case: the data are scarce, the decisions are sequential, and the regulator has long expected, and in its August 2024 oncology guidance restated, that the dose be chosen on more than the highest tolerable level.

This chapter starts with the design most clinicians have actually run, the 3+3, and simulates it so that its operating characteristics are on the table rather than in folklore. It then introduces the model-based and model-assisted designs one at a time (the continual reassessment method, the Bayesian optimal interval design, escalation with overdose control), each as a change to the same trial and judged on the same scenarios. It moves from toxicity-only designs to designs that weigh efficacy against toxicity, which is where the regulatory conversation has gone, and it ends with seamless designs that carry the dose decision into the confirmatory stage, a comparison of the designs, and five real development programs. Chapter 14 supplies the general simulation discipline used here, and Chapter 16 treats response-adaptive randomization, which dose-finding designs resemble but do not copy.

! Regulatory anchor

- **FDA, Optimizing the Dosage of Human Prescription Drugs and Biological Products for the Treatment of Oncologic Diseases (final guidance, August 2024)** (U.S. Food and Drug Administration 2024d). Recommends that sponsors of oncology products compare multiple dosages, using pharmacokinetic, pharmacodynamic, safety and efficacy information, rather than defaulting to the maximum tolerated dose. This is the guidance behind FDA's Project Optimus (U.S. Food and Drug Administration 2021d). It is a guidance: it states FDA's current thinking and recommends, it does not bind.
- **ICH E4, Dose-Response Information to Support Drug Registration (1994)** (U.S. Food and Drug Administration 1994). The long-standing statement that dose-response information should be collected during development, and that the dose-response relationship is a property to be characterized, not assumed.
- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft guidance, January 2026)** (U.S. Food and Drug Administration 2026g). Recommends the Bayesian operating characteristics, priors and simulation report that a sponsor should supply when a design, including a model-based dose-finding design, rests on a Bayesian primary analysis. A draft, not binding.
- **ICH M15, General Principles for Model-Informed Drug Development**

(Step 4, January 2026; FDA version printed June 2026) (International Council for Harmonisation 2026a). The framework for exposure-response and other models used to support dose selection: define the question of interest and context of use, then assess model influence and the consequence of a wrong decision.

- **FDA Drug Development Tools Fit-for-Purpose Initiative** (U.S. Food and Drug Administration 2025a). Lists dose-finding methods that FDA has reviewed for a stated context of use. The list as printed (current as of 10 January 2025) includes the Bayesian optimal interval (BOIN) design, determination issued 10 December 2021, and MCP-Mod, issued 26 May 2016.
- **FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics (final, November 2019)** (U.S. Food and Drug Administration 2019a). Covers the seamless and dose-selecting designs of the later sections. See Chapter 3 for the map of these documents.

17.1. The traditional approach: the 3+3 design

17.1.1. What a phase 1 trial in oncology is for

A phase 1 trial of a cytotoxic cancer drug answers a narrow question. Doses of the drug are fixed in advance on an ascending ladder, say six levels. Patients are enrolled in small cohorts. At each dose, a patient either does or does not experience a *dose-limiting toxicity* (DLT), a prespecified adverse event within a prespecified window, usually the first cycle. The goal is to find the *maximum tolerated dose* (MTD), the highest dose whose DLT probability is acceptable. The acceptable level is a *target toxicity rate*, commonly 0.20 to 0.33. This chapter uses $\Gamma = 0.25$ for the target and writes p_d for the true DLT probability at dose level d. The reasoning behind the aim is old. For cytotoxic drugs, both efficacy and toxicity were expected to increase with dose, so the highest tolerable dose was expected to be the most active one.

17.1.2. The rule

The 3+3 design is the rule-based method that grew out of this setting, and it is still the most commonly used phase 1 design (Le Tourneau, Lee, and Siu 2009). It has no model. A dose-escalation rule, with no probability anywhere in it, tells the investigator what to do.

1. Treat three patients at the current dose.
2. If none of the three has a DLT, move to the next higher dose.
3. If one of the three has a DLT, treat three more at the same dose. If at most one of the six has a DLT, move up. If two or more of the six have a DLT, stop.
4. If two or more of the first three have a DLT, stop.
5. When the trial stops at dose d, the MTD is declared to be the dose below, $d - 1$. If it stops at the lowest dose, no dose is declared tolerable. If the highest dose is cleared by rule 2 or 3, it is recommended.

This is the variant simulated here. It has no step in which three or more additional patients are treated at the dose below the stopping dose to confirm it, and variants that add one (to have at least six patients at the declared MTD) will select somewhat differently.

The up-and-down idea behind the rule goes back to Dixon and Mood (1948), and Storer (1989) formalized and analyzed the traditional algorithm-based designs and proposed alternatives. The accelerated titration designs of Simon et al. (1997) initially used single-patient cohorts, reverting to conventional cohorts after one DLT or two grade 2 toxicities, with the assessment period depending on the variant. Their simulations showed fewer patients treated at low doses. The 3+3 design is attractive for good reasons. It is easy to explain to a clinical team and an ethics committee. Nobody needs software. It never escalates after seeing two DLTs in a cohort.

17.1.3. Operating characteristics, by simulation

The rule has no stated target toxicity rate. We can, however, ask what target it implies and what it does in practice. Operating characteristics are the currency of this book, and for the 3+3 they are most easily found by simulation (exact enumeration of the rule is also possible (Reiner, Paoletti, and O'Quigley 1999; Y. Lin and Shih 2001)), which is a few lines of code (the package functions `dose_three_plus_three()` and `dose_simulate()` do the work, and the rule itself is a short function).

We use the three scenarios with six dose levels shown in Table 17.1. In each, the true DLT probabilities are fixed, and the true MTD is the highest dose with $p_d \leq 0.25$. In scenario A the MTD is dose 3. In scenario B toxicity is low until late and the MTD is dose 5. In scenario C the MTD is dose 2. We run 2,000 simulated trials per design and scenario (the 3+3 check below uses 5,000 trials in scenario A alone).

Table 17.1.: Three true toxicity scenarios used in this chapter (target DLT rate 0.25). The true MTD is the highest dose with DLT probability at most 0.25.

Scenario	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Dose 6	MTD
A	0.05	0.10	0.25	0.40	0.55	0.70	3
B	0.02	0.05	0.08	0.12	0.25	0.45	5
C	0.10	0.25	0.45	0.60	0.70	0.80	2

```
set.seed(15)
runs <- lapply(1:5000, function(i) dose_three_plus_three(S$A))
sel <- tabulate(sapply(runs, `[[`, "selected") + 1, 7) / 5000
res <- c(setNames(sel, c("No dose", paste("Dose", 1:6))), `Mean patients` =
 ↪  mean(sapply(runs, function(r) sum(r$n))))
knitr::kable(data.frame(Outcome = names(res), Value = round(res, 2)),
 ↪  row.names = FALSE, caption = "Scenario A, 3+3: probability of selecting
 ↪  each dose, and mean sample size.")
```

Table 17.2.: Scenario A, 3+3: probability of selecting each dose, and mean sample size.

Outcome	Value
No dose	0.03
Dose 1	0.09
Dose 2	0.35
Dose 3	0.37
Dose 4	0.14
Dose 5	0.02
Dose 6	0.00
Mean patients	13.75

Table 17.3 sets the 3+3 beside two Bayesian designs (CRM and BOIN) introduced below, on the same scenarios. The columns are the probability of selecting the true MTD, the probability of selecting a dose above it (overdose selection), the mean number of patients, the mean number treated at doses whose true DLT rate exceeds 0.35, and the mean number of DLTs observed.

Table 17.3.: Operating characteristics of three designs over 2,000 simulated trials per cell. 3+3 stops by its own rule (mean sample size shown); CRM and BOIN use 30 patients in cohorts of 3. 'Toxic' means patients treated at doses with true DLT probability above 0.35.

Scen.	Design	P(correct MTD)	P(overdose)	Mean n	n toxic	Mean DLTs
A	3+3	0.38	0.15	13.82	2.98	2.80
A	CRM (n=30)	0.66	0.19	30.00	4.97	6.37
A	BOIN (n=30)	0.58	0.18	30.00	4.87	5.89
B	3+3	0.36	0.11	18.98	2.03	2.71
B	CRM (n=30)	0.52	0.08	30.00	1.73	4.31
B	BOIN (n=30)	0.52	0.11	30.00	2.53	4.32
C	3+3	0.41	0.13	10.35	2.78	2.69
C	CRM (n=30)	0.71	0.14	29.92	5.87	7.49
C	BOIN (n=30)	0.69	0.11	29.93	5.21	7.01

Two things stand out. First, the 3+3 finds the true MTD in only about 36 to 41 percent of trials across the three scenarios, and in roughly half of trials it recommends a dose below the MTD (for example 47 percent in scenario A, 53 percent in scenario B). The rule is conservative by construction: it stops as soon as a cohort of six shows two DLTs, and two DLTs in six is far from impossible at doses whose true DLT rate is well below the target. Second, its mean sample size is small, between about 10 and 19 patients. That is its virtue

and its limitation: few patients are exposed at toxic doses (about three or fewer, on average, in every scenario), and little information is gathered at the recommended dose.

Figure 17.1 shows the distribution of the selected dose in scenario A for the three designs.

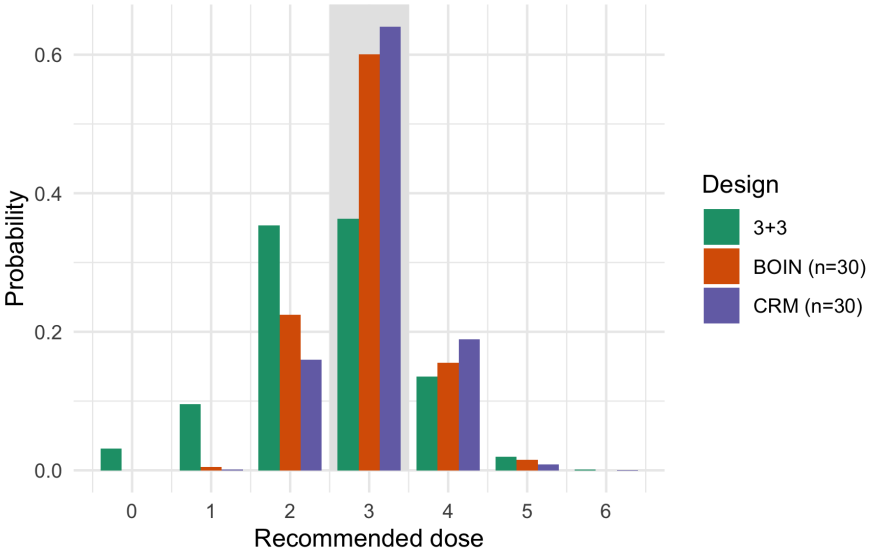


Figure 17.1.: Distribution of the dose recommended as the MTD in scenario A (true MTD is dose 3, shaded). 0 means no dose was judged tolerable.

17.1.4. What six patients can tell you

Most of the cost of the 3+3 is not in the rule but in the information it leaves behind. A trial that ends with one DLT in six patients at the recommended dose is described in the protocol as showing a "tolerable" dose. The thinness of what a 3+3 trial learns has been shown by exact enumeration of the rule (Reiner, Paoletti, and O'Quigley 1999; Y. Lin and Shih 2001) and in comparison with the CRM (Iasonos et al. 2008), and is discussed in our working paper (Polson and Sokolov 2026b); the arithmetic below is a Bayesian version of the same point. Take the prior on the DLT probability to be Beta(0.25, 0.75) (a weakly informative prior with prior mean 0.25 and the weight of one prior patient). One DLT in six patients gives a Beta(1.25, 5.75) posterior.

```
a <- 0.25 + 1; b <- 0.75 + 5
ci <- qbeta(c(0.025, 0.975), a, b)
post <- c(`Posterior mean` = a / (a + b), `Lower 95%` = ci[1], `Upper 95%`
 ↪  = ci[2],
  `P(DLT rate > 0.33)` = pbeta(0.33, a, b, lower.tail = FALSE))
knitr::kable(data.frame(Quantity = names(post), Value = round(post, 3)),
 ↪  row.names = FALSE)
```

Quantity	Value
Posterior mean	0.179
Lower 95%	0.010
Upper 95%	0.511
P(DLT rate $>$ 0.33)	0.141

The 95 percent posterior interval runs from 0.01 to 0.51. The same trial that is reported as having "found" a dose with a one-in-six observed DLT rate is compatible, at that level, with a DLT rate that is half. A frequentist exact (Clopper-Pearson) interval for one in six is wider still (roughly 0.004 to 0.64), so the problem is the sample size and not the Bayesian treatment of it. The designs below do not repair this by magic. They do three things: they use every patient treated at every dose (the 3+3 forgets all previous doses when it moves), they state the target toxicity rate, and they let the sample size be chosen deliberately instead of by the stopping rule.

17.1.5. What stays the same

A fair summary of the 3+3: it is simple, transparent and safe in the sense that few patients are exposed to high doses, at the price of a low chance of finding the MTD and a poor estimate of the DLT rate at the dose it recommends. For a trial of a drug with a well-understood toxicity profile and a small number of doses, a team may reasonably choose it. Simulation makes the consequences of choosing the 3+3 explicit and supports the design justification, because the simulation above is exactly the evidence on which it should be judged.

17.2. The Bayesian version: model-based and model-assisted designs

Chapter 6 updated a belief about a probability with data. A dose-finding design is that same update, repeated after every cohort and used to choose the next dose. The Bayesian designs differ from the 3+3 in two ways. They name a target toxicity rate Γ. And they assume that the DLT probability increases with dose, so information at one dose informs the neighboring ones. They differ from one another in how much of that structure they use.

17.2.1. The continual reassessment method

The continual reassessment method (CRM) of O'Quigley, Pepe and Fisher (O'Quigley, Pepe, and Fisher 1990) was the first Bayesian design in phase 1 and remains the reference model-based design. In its simplest form it posits a one-parameter curve through K doses,

$$p_d(a) = \pi_d^{\exp(a)}, \qquad d = 1, \dots, K,$$

where $\pi_1 < \cdots < \pi_K$ is a prespecified *skeleton* (a set of guessed DLT probabilities, one per dose) and a is an unknown parameter with a prior, here $a \sim N(0, 1.34)$. After each cohort, the posterior of a is computed from the data at all doses, and the next cohort is given the dose whose posterior mean DLT probability is closest to Γ. Two safety rules are standard:

no skipping a dose when escalating, and a stopping rule if the lowest dose is judged too toxic. The original proposal treats one patient at a time; the version in use adds cohorts, no skipping of untried doses and a start at the lowest dose (S. N. Goodman, Zahurak, and Piantadosi 1995). The likelihood version (O'Quigley and Shen 1996) and the book of Cheung (2011) popularized the power working model used here, and Garrett-Mayer (2006) is a tutorial.

The consequence for the protocol is that the sponsor now writes down a skeleton and a prior, which the 3+3 never needed. The skeleton is not the truth; it is a shape. The design uses the data to move the whole curve up or down. In our simulation the skeleton (0.05, 0.12, 0.25, 0.40, 0.55, 0.70) is close to the truth in scenario A (it differs only at dose 2) and wrong in scenarios B and C; a real simulation report should include scenarios in which the skeleton is badly wrong.

> **i** Technical note: the CRM posterior
>
> With y_d DLTs among n_d patients at dose d, the likelihood of a is $\prod_d p_d(a)^{y_d}\{1 - p_d(a)\}^{n_d - y_d}$. The posterior $\pi(a \mid y) \propto \pi(a)\prod_d p_d(a)^{y_d}\{1 - p_d(a)\}^{n_d - y_d}$ is one-dimensional, so it is computed on a grid of a few hundred points (as `dose_crm()` does) and the posterior mean of each $p_d(a)$ is a weighted sum over the grid. No MCMC is needed. The one-parameter form is deliberately rigid: it is well-behaved with few patients because it cannot fit anything but a monotone curve. Richer two-parameter logistic models, such as the Bayesian logistic regression model used with escalation with overdose control (Neuenschwander, Branson, and Gsponer 2008), estimate the slope too and need informative priors to behave.

The chapter's code uses `dose_crm()` from the book's package `bayestrials`, which is written for teaching and for the simulations here. For production work the maintained CRAN packages are the tools to use: `dfcrm` and `trialr` for CRM and related Bayesian designs, and `BOIN` for the Bayesian optimal interval designs below (the boundaries computed by `dose_boin_bounds()` agree with `BOIN::get.boundary`). A sponsor's statistical analysis plan should name the software and version that generated the simulation report.

17.2.2. The Bayesian optimal interval design

The Bayesian optimal interval (BOIN) design of Liu and Yuan (Liu and Yuan 2015) is a *model-assisted* design. It is derived from a Bayesian argument, but at the bedside it is a table. For each dose, the observed DLT rate $\hat{p}_d = y_d/n_d$ is compared with two fixed boundaries. If $\hat{p}_d \le \lambda_e$, escalate; if $\hat{p}_d \ge \lambda_d$, de-escalate; otherwise stay. The boundaries are chosen to minimize the probability of a wrong decision between three simple hypotheses: the dose is at the target, it is below the target toxicity rate, at $0.6\,\Gamma$, or it is overly toxic at $1.4\,\Gamma$. The boundaries do not depend on the dose or on the number of patients treated.

```
bd <- dose_boin_bounds(0.25)
n <- c(3, 6, 9)
knitr::kable(data.frame(Patients = n, `Escalate if DLTs at most` =
 ↪  floor(bd[["escalate"]] * n + 1e-9),
  `De-escalate if DLTs at least` = ceiling(bd[["deescalate"]] * n - 1e-9),
   ↪  check.names = FALSE), align = "ccc")
```

Patients	Escalate if DLTs at most	De-escalate if DLTs at least
3	0	1
6	1	2
9	1	3

With cohorts of three, this rule says escalate on zero DLTs and de-escalate on one or more, which is more cautious than the 3+3 in the first cohort. A dose is eliminated, together with all higher ones, if the posterior probability that its DLT rate exceeds Γ exceeds 0.95 (with a uniform prior and at least three patients treated). At the end, the MTD is chosen as the dose whose isotonically smoothed DLT rate is closest to Γ. The smoothing follows the reference implementation (`BOIN::select.mtd`): isotonic regression with inverse posterior-variance weights, and, when two doses are equally close with rates below the target, the higher one.

The practical appeal is that the protocol contains the same kind of table as the 3+3: a grid of numbers of DLTs and numbers of patients. A clinician needs no software on the day. The statistician has the operating characteristics, by simulation, in the design report. The FDA's Drug Development Tools Fit-for-Purpose Initiative lists the BOIN design for dose finding, with a determination issued on 10 December 2021 (U.S. Food and Drug Administration 2025a). That is a statement about a context of use (phase 1 dose finding with a toxicity endpoint) and does not exempt a sponsor from providing simulations for its own trial. A practical overview for oncology trialists is in Yuan et al. (2016), and an extension to drug combinations in R. Lin and Yin (2017).

17.2.3. The other interval designs: mTPI, mTPI-2 and Keyboard

BOIN is not the only design whose decisions are printed as a table. The toxicity probability interval designs divide the DLT-rate axis into an underdosing interval, an equivalence interval around the target and an overdosing interval. The first version is Ji, Li, and Bekele (2007), and its modified form (mTPI) (Ji et al. 2010) computes the posterior probability mass of each interval divided by the interval's width and moves toward the interval with the largest value. The mTPI-2 design (Guo et al. 2017) changes how the intervals are formed, in response to a weakness of mTPI that its authors frame as an Ockham's razor problem. The Keyboard design (Yan, Mandrekar, and Yuan 2017) lays equal-width keys, each as wide as the target interval, over the DLT-rate axis, and recommends the move toward the key with the highest posterior probability. As with BOIN, the protocol carries a table of decisions by number of DLTs and number treated, and the logic can be checked by hand.

Which to choose is an empirical question, and the evidence is simulation. Horton, Wages, and Conaway (2017) compared the CRM, mTPI and BOIN on the percentage of correct selection of the true MTD, patient allocation near it and an accuracy index, and H. Zhou, Yuan, and Nie (2018) compared several newer phase 1 designs on accuracy, safety and reliability. Those are comparisons in stated scenarios at stated sample sizes, like the comparison of this chapter, and the lesson is the same as for CRM and BOIN: the protocol should report the chosen design's own operating characteristics in scenarios that include a badly placed target.

17.2.4. Comparing the designs

Back to Table 17.3. With 30 patients in cohorts of three, CRM selects the true MTD in about 52 to 71 percent of trials and BOIN in about 52 to 69 percent, against 36 to 41 percent for the 3+3. CRM and BOIN also make fewer under-selection errors, which is where the 3+3 loses most. But the table is honest in the other direction too. CRM and BOIN treat 30 patients, the 3+3 about 10 to 19. They therefore treat more patients at toxic doses in absolute terms (about 5 to 6 for CRM and BOIN against about 3 for the 3+3 in scenarios A and C) and see more DLTs. (Section 17.8 reports how rarely model-based designs have been used, and what happened when published 3+3 trials were re-run with them.)

Table 17.6.: CRM and BOIN at a small sample size (n = 15, cohorts of 3), compared with the 3+3, scenarios A and B, 2,000 simulated trials per cell.

Scen.	Design	P(correct MTD)	Mean n	n toxic	Mean DLTs
A	3+3	0.38	13.82	2.98	2.80
A	CRM (n=15)	0.51	15.00	1.68	2.53
A	BOIN (n=15)	0.46	15.00	1.11	2.25
B	3+3	0.36	18.98	2.03	2.71
B	CRM (n=15)	0.28	15.00	0.00	1.25
B	BOIN (n=15)	0.35	15.00	0.00	1.18

The comparison is not like for like, since a model-based design spends more patients to find the dose more reliably. Cutting the sample size to 15 (Table 17.6), a size comparable with the 3+3, the probability of selecting the MTD falls in scenario A to about 51 percent for CRM and 46 percent for BOIN (and in scenario B to 28 and 35 percent), against 38 and 36 percent for the 3+3. Both are then better than the 3+3 in scenario A. In scenario B CRM is worse and BOIN about level with the 3+3, with fewer patients at toxic doses in both. The result is a design-time choice that the protocol should make explicitly: how many patients is the MTD worth?

These are results for particular scenarios, a particular skeleton, a particular prior and a particular cohort size. They do not rank the designs in general, and the literature shows both CRM and BOIN performing well when properly calibrated. The point for the protocol is that a design can be judged only by running it, and that the simulation must include scenarios in which the skeleton is wrong, the MTD is at the top or bottom of the range, and every dose is too toxic.

17.2.5. Escalation with overdose control

Escalation with overdose control (EWOC) (Babb, Rogatko, and Zacks 1998) changes the criterion for the next dose. Instead of the dose whose estimated DLT rate is closest to Γ, EWOC chooses the highest dose for which the posterior probability of overdosing (exceeding the MTD) does not exceed a feasibility bound η (several doses may satisfy the constraint), often starting at 0.25 and raised as the trial proceeds. A dose that has some chance of being too toxic is therefore not the one most likely to be chosen. This builds a patient-safety constraint into the decision itself, which is one reason EWOC was an early attempt to give ethical control a formal expression. The Bayesian logistic regression model (BLRM) of Neuenschwander, Branson, and Gsponer (2008) applies the same overdose-control idea to

interval probabilities of the DLT rate under a two-parameter logistic model with informative priors, and with either design the dose-escalation decisions then rest on a posterior that depends strongly on the prior, which the FDA draft on Bayesian methodology says should be justified and subjected to sensitivity analysis (see Chapter 8).

17.2.6. A worked escalation decision with the logistic model

To see what the posterior looks like at a decision point, take doses of 1, 2, 4, 8 and 16 mg with reference dose 8 mg, and the two-parameter logistic model $\operatorname{logit} p(d) = \log \alpha + \beta \log(d/8)$ with $\beta = e^{\log \beta}$. The prior is $\log \alpha \sim N(\operatorname{logit} 0.25, 2^2)$, which centers the toxicity rate at the reference dose on the target of 0.25 with wide uncertainty, and $\log \beta \sim N(0, 1)$. Four cohorts have been treated: 0 of 3 DLTs at 1 mg, 0 of 3 at 2 mg, 1 of 3 at 4 mg and 1 of 6 at 8 mg. Each dose is classified by its true DLT rate: underdosing below 0.16, target from 0.16 to 0.33, overdosing at 0.33 or above. Table 17.7 gives the posterior probability of each interval, computed on a grid.

Table 17.7.: Posterior mean DLT rate and posterior probability that the DLT rate lies in the underdosing (below 0.16), target (0.16 to 0.33) or overdosing (0.33 or above) interval, after four cohorts. A dose is admissible if the probability of overdosing is at most 0.25.

Dose (mg)	Mean DLT rate	Under	Target	Over	Admissible
1	0.049	0.95	0.05	0.00	yes
2	0.074	0.90	0.10	0.00	yes
4	0.123	0.73	0.25	0.02	yes
8	0.223	0.36	0.45	0.19	yes
16	0.375	0.17	0.32	0.51	no

The 16 mg dose is excluded, since its probability of overdosing is 0.51, above the bound of 0.25. Among the admissible doses, 8 mg has the highest probability of lying in the target interval (0.45, with 0.36 under and 0.19 over), so the next cohort receives 8 mg. The single DLT at 4 mg did not stop escalation: the model weighs it against the absence of toxicity at the lower doses and one in six at 8 mg, which is the use of all the data that the 3+3 rule forgoes. The result depends on the prior, as the text above says: rerun the table with a different `prior_sd` before trusting it.

17.2.7. Protocol language

In the protocol: a model-based dose-escalation design

State the target toxicity rate Γ and the DLT definition and window. State the dose ladder, the starting dose, the cohort size and the maximum sample size. For CRM, give the skeleton and the prior on its parameter, with the reason for each. For BOIN, give the escalation and de-escalation boundaries and the elimination rule. State the no-skipping rule and the stopping rule for an overly toxic lowest dose. Attach a simulation report with at least five scenarios (including a flat-toxic scenario and one where the

MTD is the top dose) with the probability of correct selection, the probability of overdose selection, the number of patients at doses above the target, and the sensitivity of each to the skeleton or prior. Specify who can override the design's recommendation (usually a safety review committee) and in what circumstances, for example when late toxicities, or toxicities after the DLT window, appear. See Chapter 23 for the general protocol table.

17.3. Beyond the MTD: efficacy, toxicity and dose optimization

17.3.1. Why the MTD is a poor target for modern agents

The MTD paradigm assumes that more is better until it hurts. That was a reasonable working belief for cytotoxic chemotherapy, and it was written into the designs above, which look only at toxicity. It is a poor assumption for targeted therapies, immunotherapies and antibody-drug conjugates. For such agents the dose-response curve for efficacy may reach a plateau well below the MTD, a drug may be taken for months or years rather than for a few cycles, and chronic low-grade toxicity may lead to dose reductions and discontinuations that cannot be seen in a short DLT window. A dose chosen as the MTD can then be both more toxic and no more effective than a lower one. The FDA's stated concern is that choosing the dose this way can leave patients on doses with more toxicity and no more benefit than a lower dose (U.S. Food and Drug Administration 2024d). The rationale for the initiative is set out by FDA authors in Shah et al. (2021), and the statistical principles, with a review of FDA initial approvals of small molecules and antibody-drug conjugates in 2019 to 2021, in Fourie Zirkelbach et al. (2022).

17.3.2. What the FDA guidance asks

The final guidance of August 2024 (docket FDA-2022-D-2827, Oncology Center of Excellence with CDER and CBER) has a stated scope. The guidance does not address first-in-human starting-dose selection. It does not specifically address dosage optimization for radiopharmaceuticals, cellular and gene therapies, oncolytic viruses, microbiota or cancer vaccines, or pediatric development, although some recommendations may apply to these settings. Developed through Project Optimus, launched in 2021 (U.S. Food and Drug Administration 2021d, 2024d), the guidance recommends that sponsors of oncology products take a dose selection approach different from that of the 3+3. In paraphrase, and the guidance should be consulted for the precise wording, it recommends the following.

- Doses to be taken to registrational trials should be chosen on a broader basis than the MTD alone: pharmacokinetics, pharmacodynamics, exposure-response for efficacy and safety, and clinical safety, tolerability and activity, including late and low-grade toxicities.
- Dose-ranging in the clinic should compare several dosages, and where feasible with randomization, rather than rely on a single-arm escalation to one dose. The comparison is intended to characterize the dose-response and dose-toxicity relationships, not to

establish superiority or noninferiority of one dose over another at registrational error rates.
- Model-based and model-informed methods, including Bayesian ones, are expected to be useful in this process, and sponsors should discuss the design with FDA early. The guidance does not require a specific design, and neither does it exclude the 3+3.
- Dose selection should be explained in the context of the intended population and the treatment's schedule.

ICH E4 (U.S. Food and Drug Administration 1994) already established these principles: the dose-response relationship should be characterized during development, several dose-response designs (including Bayesian and pharmacometric methods) are discussed, prospective randomized multidose studies are recommended, and the highest tolerated dose need not be optimal even in life-threatening disease. The August 2024 guidance gives them a focused application to contemporary oncology dosage optimization. For the exposure-response models that support the choice, ICH M15 (International Council for Harmonisation 2026a) is the framework. A sponsor who uses a model to select between two doses has to state the question of interest and the context of use, and assess how much weight the model carries in the decision and how costly a wrong decision would be.

17.3.3. A Bayesian formulation: utility

A Bayesian can state the dose problem directly. Let $\pi_E(x)$ be the probability of a favorable efficacy outcome at dose x and $\pi_T(x)$ the probability of a toxicity. Elicit a *utility* for each of the four joint outcomes, as in the utility-based designs described in our working paper (Polson and Sokolov 2026a): efficacy without toxicity is the best, no efficacy and toxicity the worst, with efficacy and toxicity together somewhere in between. The *expected utility* of a dose is the average of these utilities under the posterior. A design chooses the dose that maximizes it, among those that are *acceptable*: meeting a minimum efficacy and a maximum toxicity bound at a stated posterior probability.

In Figure 17.2 the dose with the largest expected utility is dose 4, one level below the MTD, because the efficacy gain from dose 4 to dose 5 is small and the toxicity cost is not. The utilities are the price of the method. They are an elicited, subjective choice, and the dose that wins depends on them. The protocol should record who elicited them, from whom, on what basis, and how the result changes if they change (see the sensitivity analysis in Chapter 8 for the same point about priors). The numbers 100, 40, 0 and -60 above are illustrative, not clinical.

17.3.4. EffTox and phase 1/2 designs

The EffTox design of Thall and Cook (P. F. Thall and Cook 2004) formalizes this. It models efficacy and toxicity probabilities as functions of dose, with a quadratic logistic form for efficacy so that the dose-efficacy curve can turn down, a monotone logistic form for toxicity, and a copula for the association between the two outcomes in the same patient (Polson and Sokolov 2026a). Rather than ask for utilities for four outcomes, the original EffTox design elicits a *trade-off contour* in the plane of (π_T, π_E): the investigator names several pairs of efficacy and toxicity probabilities that are equally desirable. A dose is *acceptable* if its posterior probabilities satisfy the minimum efficacy and maximum toxicity constraints, and the next cohort receives the acceptable dose with the highest desirability. Early phase

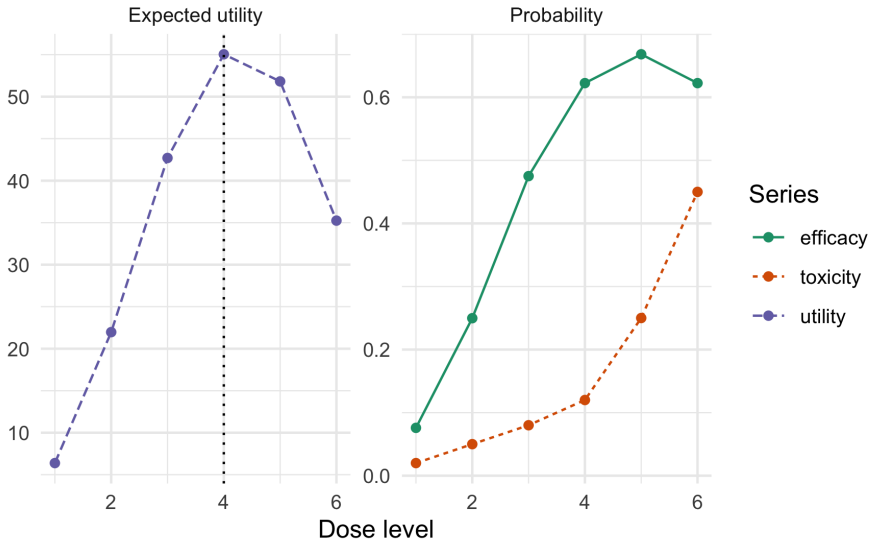


Figure 17.2.: Illustrative true curves. Toxicity (scenario B) reaches 0.25 at dose 5, so the MTD is dose 5. Efficacy is non-monotone and plateaus at dose 4 to 5. The dose that maximizes expected utility is dose 4, below the MTD.

1/2 designs of this kind are described in P. F. Thall and Russell (1998). Later work extends the utility approach to ordinal efficacy and toxicity outcomes and combinations of agents, including the utility-based combination design of Houede and colleagues (Houédé et al. 2010) and an adaptive-randomization version for bivariate ordinal outcomes (P. F. Thall et al. 2015). Book-length treatments are Yuan, Nguyen, and Thall (2016) and P. F. Thall (2024).

Three things are different from the toxicity-only designs. The trial needs a *probabilistic model for efficacy*, so it needs an efficacy endpoint that can be observed quickly enough to be used during the trial (a response, a biomarker, a PD measure). Early efficacy readouts that are only surrogate for the clinical endpoint bring a surrogacy question into the dose decision. The designs need larger samples, usually 40 to a few hundred. And the elicitation of the contour or utility is an extra step that must be defended. In return the trial can state what it says about the dose-response, and the dose that it picks is accompanied by a distribution of efficacy and toxicity at that dose.

17.3.5. Late-onset toxicity and utility-based interval designs

The phase 1/2 designs above need the outcome of each cohort before the next one is dosed, which is the logistic problem of late-onset toxicity. The time-to-event CRM of Cheung and Chappell (2000) incorporates incomplete follow-up from patients without an observed DLT, commonly using the fraction of the observation window completed, while observed DLTs contribute full event information, so that enrollment need not pause while earlier patients are followed. Yuan et al. (2018) built the same idea into the interval design: TITE-BOIN allows real-time dose assignment for new patients while some enrolled patients' toxicity data are still pending, and so addresses both late-onset toxicity, which is common for targeted agents and immunotherapy, and rapid accrual. For the efficacy-toxicity trade-off, BOIN12 of R. Lin et al. (2020) targets the optimal biologic dose that optimizes the risk-benefit trade-off,

for agents whose efficacy does not necessarily increase with dose, and is built to be simple enough that its decision rules can be tabulated in the protocol as BOIN's are. In the setting of Figure 17.2, where efficacy plateaus below the MTD, for the illustrative true curves the MTD is dose 5 and expected utility is highest at dose 4. Yuan, Zhou, and Liu (2024) review the design strategies for dose-optimization trials and the practical considerations in planning them, including the randomized comparison the FDA guidance recommends.

17.3.6. Randomized dose comparison, and what a regulator will ask

The guidance recommends comparing dosages directly, and a randomized comparison of two or three candidate doses after the escalation phase is the common implementation. The trial need not be powered for formal superiority or noninferiority between doses at registrational error rates. Its size should support sufficiently precise assessments of safety and antitumor activity at each dosage, and the doses are not expected to differ by much in efficacy, which is why the evidence is expected to include PK/PD, safety and tolerability, and modeling. A Bayesian formulation is to compute the posterior probability that each dose's efficacy is within a prespecified margin of the best dose and the posterior probability that its tolerability is better, and to select the lowest dose that meets both, with simulated operating characteristics. The chapter on simulation (Chapter 14) describes the report. For a design of this kind a reviewer is likely to ask the following.

1. What are the operating characteristics for choosing a dose that is not the one that maximizes efficacy: the probability of selecting a dose that is too low or too toxic under each scenario?
2. How were the utility, trade-off contour or margins elicited, and what changes if they are altered?
3. What priors were used for the efficacy and toxicity models, and how much do they affect the dose choice?
4. How are late toxicities and dose reductions after cycle 1 handled in the dose decision?
5. What is the role of the exposure-response model (Section 17.3) and how was its risk assessed under M15?

17.4. Dose-response estimation outside oncology: MCP-Mod

Most of this chapter concerns oncology, where the dose question is how much toxicity to accept. Most other programs run a phase 2 dose-ranging study whose question is the shape of the dose-response curve and which dose to take to phase 3. The traditional method that regulators know best here is MCP-Mod, which combines a multiple comparison step with a modeling step (Bretz, Pinheiro, and Branson 2005; Pinheiro et al. 2014). It is on FDA's fit-for-purpose list with a determination of 26 May 2016 (U.S. Food and Drug Administration 2025a), and it is the dose-finding method on that list that belongs to the frequentist tradition.

The method has two steps. First, before the data, the protocol names a candidate set of dose-response shapes (for example linear, Emax, exponential and umbrella) with guessed parameters, and from each shape computes the contrast of the dose means that has the most power if that shape is true. The test of "no dose-response" is the maximum of these contrast statistics, and its critical value comes from their joint (multivariate normal or t) distribution, so the familywise type I error over the candidate shapes is held at one-sided 0.025 when

there is no dose-response at all. Second, if the test finds a signal, the models that passed are fitted, and the dose-response curve and a target dose (for example the smallest dose with a clinically relevant effect) are estimated, by selecting a model or by averaging. The chunk below shows the first step for five doses and 40 patients per dose with a known standard deviation of 1 and candidate shapes that never fall below placebo, and compares it with two traditional analyses of the same data: the top dose against placebo, and the best of four pairwise comparisons with a Bonferroni correction.

```
doses <- c(0, 10, 25, 50, 100); n <- 40
shapes <- cbind(linear = doses / 100, emax = doses / (doses + 15),
 ↪  exponential = (exp(doses / 50) - 1) / (exp(2) - 1),
                umbrella = 1 - (doses / 50 - 1)^2)          # candidate
                 ↪  shapes, none below placebo
mcp <- mcp_contrasts(shapes)                                 # optimal
 ↪  contrasts and the critical value
truth <- list(Flat = rep(0, 5), `Emax with plateau` = 0.5 * doses / (doses
 ↪  + 10), Linear = 0.5 * doses / 100,
              Umbrella = 0.5 * (1 - (doses / 50 - 1)^2), `Harmful, linear
               ↪  decline` = -0.5 * doses / 100)
mcp_tab <- t(sapply(truth, mcp_power, contrasts = mcp$contrasts, crit =
 ↪  mcp$crit, n = n, seed = 1))
knitr::kable(mcp_tab, digits = 3, col.names = c("MCP-Mod contrast test",
 ↪  "Top dose vs placebo", "Bonferroni, four doses"))
```

Table 17.8.: Probability of a one-sided 0.025 dose-response signal under five true dose-response shapes (flat, an Emax shape with a plateau, a linear shape, an umbrella shape that returns to the placebo mean at the top dose, and a harmful linear decline) for the MCP-Mod multiple contrast test, the top dose against placebo, and the best of four Bonferroni-adjusted pairwise comparisons. Candidate shapes: linear, Emax, exponential and umbrella, all at or above placebo. Five doses, 40 patients per dose, standard deviation 1, 20,000 simulated trials per cell.

	MCP-Mod contrast test	Top dose vs placebo	Bonferroni, four doses
Flat	0.024	0.025	0.022
Emax with plateau	0.523	0.527	0.493
Linear	0.617	0.609	0.424
Umbrella	0.686	0.025	0.471
Harmful, linear decline	0.015	0.000	0.004

With four candidate shapes the critical value of the maximum contrast statistic is 2.37, against 1.96 for one test and 2.50 for a Bonferroni correction over four comparisons, because the contrasts are positively correlated for the monotone shapes. The type I error is held at the flat truth (0.024), which is where the familywise guarantee applies, and the power of the contrast test is similar to the best traditional test when the true shape is monotone (Table 17.8). Its advantage is visible when the true curve is not monotone: when the top dose returns to the placebo mean the top-dose comparison has only its type I error rate (0.025), the Bonferroni comparison has 0.47, and the contrast test with an umbrella among

the candidates has 0.69. The test detects a non-flat dose-response in the directions of the candidates, not benefit at some dose. A declining truth can still trigger a contrast when the candidates allow declines, which is why the candidates are chosen as benefit shapes (Bretz, Pinheiro, and Branson 2005); with the candidates here a harmful linear decline signals in only 0.015 of trials. The price is that the candidate set is a pre-specified list, and a truth outside it loses power; this is the dose-response counterpart of the skeleton choice for the CRM.

The Bayesian designs in this chapter allocate and decide adaptively: ASTIN (Grieve and Krams 2005; Krams et al. 2003) and the Bayesian dose selection in the lecanemab and AWARD-5 examples (Section 17.8). The operating characteristics of MCP-Mod, like those of any other design, are found by simulation in the scenarios the protocol names.

17.5. Adaptive enrichment and subgroup-specific doses

When benefit, or the best dose, differs by a biomarker-defined subgroup, broad enrollment can dilute the effect estimates, and a single dose may be wrong for part of the population. An enrichment design starts with broad enrollment and restricts to responsive subgroups as data accumulate (Polson and Sokolov 2026a). At each interim analysis, subgroup-specific posterior probabilities of benefit, $\Pr(\theta_g > 0 \mid \text{data})$ for subgroup g, map to one of three actions: continue broadly, restrict enrollment to the promising subgroups, or stop. The thresholds that govern these actions are calibrated by simulation, which is exactly the procedure the FDA draft Bayesian guidance asks for (U.S. Food and Drug Administration 2026g). The multiplicity cost is real: subgroup-by-interim looks multiply the number of opportunities for a chance finding, and Chapter 10 describes the shrinkage and error-control tools that apply. Hierarchical models that borrow strength across subgroups (Chapter 19) are the Bayesian answer to the same problem. For a dose-finding trial the usual practice is to treat the dose-selection and the enrichment decisions separately unless the trial is built for both.

17.6. Seamless designs: carrying the dose decision forward

17.6.1. Phase 2/3

A conventional program has a phase 2 dose-finding trial, a gap in which the sponsor analyzes the data, writes a new protocol and opens new sites, and a phase 3 trial. A *seamless* phase 2/3 design merges the two. A first stage compares several doses or regimens with a control and selects one or two on interim data, and a second stage continues into confirmation under a prespecified rule. The statistical machinery is older than the guidance: P. F. Thall, Simon, and Ellenberg (1988) proposed two-stage designs that select among treatments and then test, Stallard and Todd (2003) gave sequential phase III designs that select the most promising treatment at the first interim analysis, and Bretz, König, et al. (2009) review confirmatory adaptive designs, including combined phase II/III studies with treatment selection at an interim. The common structure is selection at the interim followed by a pre-specified final test that accounts for the selection. In an *inferentially seamless* design, data from the first stage for the selected dose are used in the final test, so the analysis has to account for the fact that the dose was selected because it looked good. In an *operationally seamless* design, the stages run under a single protocol but the final analysis uses second-stage data only. The

motivation is the elimination of the "dead time" between phases, and the use of the trial's own data to choose a dose rather than committing to a weak one. The price is complexity in the design, a need for a pre-specified selection rule, and the inferential adjustment. FDA's adaptive design guidance (U.S. Food and Drug Administration 2019a) expects the selection rule, the control of the type I error probability, and the firewall that protects the unblinded interim data to be written down in advance, and treats seamless designs under the same principles as other adaptive designs.

A simulation study of the inferentially seamless case in which one arm is selected at the end of the first stage found that pooling the two stages and using an adjusted final alpha, as in a group sequential design, does better than analyzing the second stage alone at full alpha, and slightly better than a combination test. It also found that putting roughly 30 to 50 percent of the patients in the first stage is close to optimal across a range of dose-response shapes (L. R. Berry et al. 2024). Two cautions apply to sizing. The power calculated for the arm that is eventually selected overstates the power of the design, because the first stage sometimes advances the wrong arm, so the true power must be found by simulating the selection. And when the first-stage endpoint is delayed, the patients enrolled while it matures belong to the cost of the first stage and the planned split should count them.

The two best-documented cases of Bayesian dose finding that continues into later stages, dulaglutide (AWARD-5) and lecanemab, are described in Section 17.8.

17.7. Side by side: what changes in the protocol

	3+3	CRM / BOIN (toxicity-only)	EffTox, utility, randomized dose comparison
Target	None stated (implies a DLT rate near 0.2 or below)	Target DLT rate Γ	Dose with best benefit-risk trade-off, subject to acceptability bounds
Uses data from other doses	No	Yes (model, or isotonic smoothing)	Yes
What goes in the protocol	The rule	Target, skeleton and prior (CRM) or boundaries (BOIN), cohort size, maximum n, stopping rules	Models for efficacy and toxicity, priors, utilities or contour, acceptability bounds, selection rule
Typical n	10 to 25 (rule-determined)	20 to 40, fixed	40 to a few hundred
Evidence the regulator expects	Rationale for the DLT definition and doses	Simulated operating characteristics in several scenarios	Same, plus elicitation record, sensitivity to utilities and priors, PK/PD and exposure-response support

	3+3	CRM / BOIN (toxicity-only)	EffTox, utility, randomized dose comparison
Where it is the better choice	Familiar agent, few doses, no model support available	Most phase 1 trials with a toxicity endpoint	Targeted, immunotherapy and ADC agents where the MTD is not the target (see the FDA 2024 guidance)
Main risk	Under-dosing, poor estimate at the recommended dose	More patients, a model-dependent answer if the skeleton or prior is poor	Utility elicitation, surrogate efficacy endpoint, larger sample

What stays the same is that a dose-escalation design is a decision rule fixed before the first patient and judged by its operating characteristics, and that safety oversight by a committee with the power to override stays in place for every design. What changes is that the rule can use all the data, that the question can be about the benefit-risk trade-off, and that the evidence offered to the regulator is a simulation report and a model assessment, not an appeal to convention.

17.8. In practice

17.8.1. How often are model-based designs used, and does it matter?

The 3+3 survives mostly out of habit. Model-based designs were used in 1.6 percent of phase 1 trials published from 1991 to 2006 and in 6.4 percent of those published from 2012 to 2014, according to the figures reported by Love et al. (2017) (who cite an earlier survey for them). Their review and survey of investigators named the barriers: a lack of training, investigators' preference for the 3+3, and limited design resources available before a trial is funded. Rogatko et al. (2007) is the earlier paper that documented the gap between published methods and the designs that trials used. The other side of the record is reassuring on safety: a review of 53 completed phase 1 cancer trials published from 2003 to 2013 that used the CRM or a variant concluded that the model-guided escalation was safe and generalizable (Iasonos and O'Quigley 2014).

The most direct evidence of what the 3+3 costs is a re-analysis. Silva et al. (2021) sampled 60 published dose-finding trials, took the 22 that had used the 3+3, and re-applied the CRM and BOIN to the published data, at target toxicity rates of 0.25 and 0.30 and without the expansion cohorts. The model-based designs recommended a different dose from the published MTD in a number of trials, treated fewer patients at suboptimal doses and escalated faster. *Lesson.* The reanalysis shows that alternative designs can recommend different doses. Because the true MTDs were unknown and some outcomes were simulated, it does not establish their real-world correct-selection probabilities. The obstacle to adoption is not statistical but organizational, and a protocol that offers a one-page table (BOIN) removes part of it.

17.8.2. BOIN and FDA's fit-for-purpose determination

The BOIN design of Liu and Yuan (2015) is the one model-assisted design for which FDA has issued a determination: the Drug Development Tools Fit-for-Purpose list records a determination for BOIN dose finding dated 10 December 2021 (U.S. Food and Drug Administration 2025a). What was reviewed was the method for a stated context of use (phase 1 dose finding with a toxicity endpoint). *Lesson.* The regulator's acceptance attaches to a method and a context of use, and a sponsor still has to supply its own simulations and a protocol-specific justification. Chapter 3 places the fit-for-purpose initiative among FDA's other tools.

17.8.3. Sotorasib and the dose that was not optimized

Sotorasib, a KRAS G12C inhibitor for non-small-cell lung cancer, is the case most often cited for Project Optimus. The pivotal single-arm trial (Skoulidis et al. 2021) used the 960 mg dose, and the accelerated approval of May 2021 came with a postmarketing requirement to compare that dose with a lower one. The randomized comparison of 960 mg and 240 mg was run only after approval. *Lesson.* This is the situation the August 2024 guidance (U.S. Food and Drug Administration 2024d) was written for: a dose chosen without a randomized comparison of lower doses, with the evidence for a lower dose collected only after approval. The cost of the dose question is then borne by patients and by a postmarketing study, rather than by a randomized comparison in phase 1/2.

17.8.4. Dulaglutide (AWARD-5): seamless dose finding

The best-documented example of a Bayesian seamless dose-finding design is the AWARD-5 trial of dulaglutide in type 2 diabetes, which started in 2008 with seven doses and two controls (Geiger et al. 2012; D. A. Berry 2025). Stage 1 used response-adaptive randomization with interim analyses every two weeks, scored by a composite *clinical utility index* combining HbA1c, weight, heart rate and blood pressure. Longitudinal modeling predicted the 12-month outcome from early data, so that doses could be chosen before anyone reached the primary endpoint. The algorithm selected two non-adjacent doses (0.75 and 1.50 mg) at the earliest permitted time and the trial moved to stage 2, fixed randomization against the controls. The dose-finding results are reported in Skrivanek et al. (2014), and the design in Geiger et al. (2012). The data monitoring board could depart from the algorithm only on safety, and did so: in April 2009 randomization to the 3.0 mg dose was stopped on its recommendation after increases in heart rate and concerns about pancreatic safety (European Medicines Agency 2014). Both doses were approved by FDA in 2014. By Berry's estimate the seamless design saved at least a year compared with separate phase 2 and phase 3 trials (D. A. Berry 2025).

Lesson. This is a utility-based design in the sense of the previous section, and it shows the pattern of regulation that has to be in place: a prespecified algorithm, an agreed role for the DMC, and a simulated operating-characteristics report. A sponsor of such a design today would seek agreement with the regulator in advance, by the routes described in Chapter 3.

17.8.5. Lecanemab: Bayesian dose finding in phase 2b

Study 201 of lecanemab in early Alzheimer's disease was a Bayesian phase 2b trial with five dose regimens and placebo, with response-adaptive allocation and interim analyses beginning after 196 patients and every 50 patients thereafter (Swanson et al. 2021; D. A. Berry 2025). It identified 10 mg/kg biweekly as the estimated 90 percent effective dose (the dose achieving 90 percent of the modeled maximum effect). The trial did not meet its prespecified 12-month primary success criterion: the posterior probability of at least a 25 percent reduction in decline was 0.64, short of the required 0.80 (D. A. Berry et al. 2023). The prespecified 18-month clinical and biomarker results were part of the broader evidence package, and the selected regimen was carried into the subsequent phase 3 program. *Lesson.* The case shows two things relevant to a dose-optimization protocol. A design can fail its primary criterion and still produce the right dose, which is why the protocol should say in advance how the dose-selection evidence will be read in that event. And a prespecified Bayesian criterion that is not met is a result, to be reported as one.

17.8.6. ASTIN: a Bayesian adaptive dose-response trial in stroke

ASTIN (Krams et al. 2003)

The Acute Stroke Therapy by Inhibition of Neutrophils (ASTIN) trial was a Bayesian adaptive dose-response study of a neutrophil inhibitor in acute ischemic stroke (Krams et al. 2003). Rather than assign patients to a few fixed doses, it modeled the dose-response curve over a wide dose range with a smoothing model, so that the response at each dose borrowed strength from its neighbors, and used the accumulating posterior to choose the doses of later patients, to target the dose giving close to the maximum effect, and to stop early for efficacy or futility under prespecified rules. The trial stopped for futility at an interim analysis. The design paper is Grieve and Krams (2005). The point for this chapter is that the dose-finding model, the allocation rule and the stopping rule were all specified in advance and evaluated by simulation, as the 2026 FDA draft expects for a Bayesian design. We keep the description general: the exact dynamic linear model and allocation criterion should be taken from the original papers, not from this summary.

Try it in the lab

The open simulator at https://vsokolov.org/bayesian_trials_lab/14-studio.html lets you set doses, adaptive allocation and scenarios and read off operating characteristics tables. Before opening it, predict how a design that updates after every cohort should behave when the true dose-response is flat across the top two doses, then change the true curves in the scenarios of this chapter (`dose_scenarios()`) and check your prediction.

Key points

- The 3+3 is simple and exposes few patients to toxic doses, but in our simulations it selects the true MTD in only about 36 to 41 percent of trials, and under-selection is its most common selection error in these scenarios. Its operating characteristics can be obtained by exact enumeration or simulation, and they should be reported.
- One DLT in six patients with a Beta(0.25, 0.75) prior leaves a 95 percent posterior interval for the DLT probability of 0.01 to 0.51. Small phase 1 trials identify a dose much less precisely than their rule suggests (see Reiner, Paoletti, and O'Quigley (1999) and Polson and Sokolov (2026b)).
- CRM and BOIN name a target toxicity rate and use the data from all doses. In our scenarios, with 30 patients, they select the true MTD more often than the 3+3, at the price of more patients, and a design must be compared at a stated sample size. BOIN is listed in FDA's fit-for-purpose initiative (determination of 10 December 2021) for a defined context of use (U.S. Food and Drug Administration 2025a).
- The MTD is the wrong target for many modern agents. The August 2024 FDA guidance recommends choosing the dosage on pharmacokinetic, pharmacodynamic, safety and efficacy evidence, preferably with a randomized comparison of several dosages (U.S. Food and Drug Administration 2024d).
- Utility-based and EffTox designs make the benefit-risk trade-off explicit. The price is an elicited utility or contour, an efficacy model and endpoint, and a larger sample, all of which a regulator will ask about.
- Seamless phase 2/3 designs remove the gap between dose selection and confirmation, and require a prespecified selection rule, type I error control that accounts for the selection, and simulated operating characteristics. The AWARD-5 trial (Section 17.8) is the documented precedent.
- Exposure-response models that support a dose choice fall under ICH M15: state the context of use and assess model influence and the consequence of a wrong decision (International Council for Harmonisation 2026a).

Discussion questions

1. A sponsor proposes a 3+3 design for a first-in-human trial of a targeted agent with six doses and a target DLT rate of 0.25. Using the simulation in this chapter, what would you ask them to add to the protocol, and what would you accept as a reason to keep the 3+3?
2. Table 17.3 shows the model-based designs treating more patients at toxic doses than the 3+3. Is that a defect of the designs, of the comparison, or both? How would you set the sample size for a CRM or BOIN trial?
3. For an antibody-drug conjugate with a plateau in response above dose 3 of 6, and a rising rate of dose reductions, how would you use the 2024 FDA guidance to design the dose-selection stage? What would be the primary decision rule, and what would the simulation report show?
4. A utility-based design selects a dose below the MTD. Who should provide the utilities, how should the trial record them, and what sensitivity analysis would convince a reviewer?
5. In a seamless phase 2/3 design with 3 candidate doses, what are the arguments for and against letting the first-stage data count toward the final test?

18. Time-to-Event Endpoints

Most confirmatory trials in cardiology and oncology are event driven. The primary endpoint is the time to death, progression or a major adverse cardiovascular event, the analysis is a log-rank test or a Cox model, and the trial ends not at a fixed sample size but when a target number of events has accrued. SELECT and REWIND (Chapter 21) are both of this kind: SELECT planned 1,225 events and observed 1,270, REWIND observed 1,257. A trial of 17,604 patients is, statistically, a trial of about 1,270 events.

The Bayesian treatment of these trials is simpler than its reputation. For the decisions that matter (whether to stop at an interim analysis, how confident to be in a hazard ratio, whether a promising interim justifies more events) almost everything follows from two facts: the information about a log hazard ratio is proportional to the number of events, and the information about a single hazard is a gamma posterior. The chapter starts with the traditional analysis (log-rank test, Cox model, Schoenfeld's event formula), then works through both facts with numbers, then turns to the case that breaks the first fact, non-proportional hazards.

! Regulatory anchor

No FDA or ICH document is devoted to the Bayesian analysis of time-to-event data, and, as far as we are aware, none is devoted to non-proportional hazards or to restricted mean survival time. The following documents set the frame; Chapter 3 gives each one's status.

- **ICH E9(R1), Addendum on Estimands and Sensitivity Analysis** (Step 4, November 2019) (International Council for Harmonisation 2019). For a time-to-event endpoint the estimand must name the population, the variable (time to first event, or a composite), the handling of intercurrent events such as treatment discontinuation, rescue therapy or death before the non-fatal event, and the summary measure. A hazard ratio is one summary measure; the difference in restricted mean survival time or in survival probability at a landmark time are others. The choice is made before the trial, not after the curves are seen.
- **ICH E9, Statistical Principles for Clinical Trials** (1998) (International Council for Harmonisation 1998). Section 3.5 on sample size and section 4.5 on interim analysis apply unchanged to event-driven trials: the number of events, and the assumptions behind it, must be stated in the protocol, and the interim plan must be fixed in advance.
- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products** (draft guidance, January 2026) (U.S. Food and Drug Administration 2026g). A draft, which recommends and does not require. Its success criteria are posterior probabilities of the form $\Pr(\delta > a \mid \text{data}) > c$, and nothing in it is specific to time-to-event data. For a trial with a frequentist statistical analysis plan, the operating characteristics of a Bayesian rule on the hazard ratio are calibrated as in Chapter 14.

- **FDA, Clinical Trial Endpoints for the Approval of Cancer Drugs and Biologics** (final guidance, December 2018) (U.S. Food and Drug Administration 2018a). Discusses overall survival, progression-free survival and related endpoints and the biases of each. It is the document sponsors most often cite for oncology time-to-event endpoints; it does not prescribe a method for non-proportional hazards.

18.1. The traditional analysis

Let T be the event time and $S(t) = \Pr(T > t)$ the survival function. The hazard is $h(t) = \lim_{\delta \to 0} \Pr(t \le T < t + \delta \mid T \ge t)/\delta$, and the cumulative hazard is $H(t) = \int_0^t h(u)\,du$, so that $S(t) = e^{-H(t)}$. Patients still event-free at the data cutoff, or lost to follow-up, are censored: their event time is only known to exceed the last contact. The Kaplan-Meier estimate of $S(t)$ (Kaplan and Meier 1958) handles censoring without a model, and it is the curve every trial report draws.

The treatment comparison has two standard forms. The log-rank test (Mantel 1966; Richard Peto and Peto 1972) compares, at each event time, the observed number of events in the treated arm with the number expected if the two arms shared one hazard, and sums the differences. It is the score test of a proportional hazards model. The Cox model (Cox 1972) writes $h(t \mid z) = h_0(t)e^{-\theta z}$ for a treatment indicator z, leaves the baseline hazard $h_0(t)$ unspecified, and estimates $\theta = -\log \mathrm{HR}$ (positive when treatment reduces the hazard, the book's benefit-positive convention) from the partial likelihood (Cox 1975). Under 1:1 allocation the variance of $\hat{\theta}$ is approximately

$$\mathrm{Var}(\hat{\theta}) \approx \frac{4}{D}, \tag{18.1}$$

where D is the total number of events (D. Schoenfeld 1981; D. A. Schoenfeld 1983). The approximation is accurate when the hazard ratio is not far from one. It carries the whole of the design: the number of patients enters only through the events they produce, which is why an event-driven trial fixes D and lets the sample size, the accrual period and the follow-up be chosen to deliver it.

When randomization was stratified, the log-rank test and the Cox model should be stratified on the same factors (a stratified log-rank test, or a Cox model with a baseline hazard per stratum), and prognostic baseline covariates can be added for precision, with the caveat that the hazard ratio is non-collapsible and so changes meaning when covariates are added; Section 5.10 sets out the conditional and marginal estimands and the regulatory guidance.

Inverting Equation 18.1 gives the sample size formula of Chapter 9. To detect a hazard ratio HR with one-sided level α and power $1 - \beta$, the trial needs

$$D = \frac{4\,(z_\alpha + z_\beta)^2}{(\log \mathrm{HR})^2} \tag{18.2}$$

events. With $\theta = -\log \mathrm{HR}$ the information per event is $I = 1/4$, so Equation 18.2 is Equation 9.2 with n replaced by D. Freedman's tables give the same requirement with the factor $(1 + \mathrm{HR})^2/(1 - \mathrm{HR})^2$ in place of $4/(\log \mathrm{HR})^2$ (Laurence S. Freedman 1982); the two agree when the hazard ratio is near one. Table 18.1 shows how fast the requirement grows as the target effect shrinks.

```
d_req <- function(hr, power) ceiling(4 * (qnorm(0.975) + qnorm(power))^2 /
 ↪  log(hr)^2)
hrs <- c(0.70, 0.80, 0.83, 0.90)
tab <- data.frame(`Design HR` = hrs, `Events, 80% power` = d_req(hrs,
 ↪  0.80),
                  `Events, 90% power` = d_req(hrs, 0.90), check.names =
                   ↪  FALSE)
knitr::kable(tab, format.args = list(big.mark = ","))
```

Table 18.1.: Events required by the Schoenfeld formula, one-sided 0.025, for four design hazard ratios and two levels of power.

Design HR	Events, 80% power	Events, 90% power
0.70	247	331
0.80	631	845
0.83	905	1,211
0.90	2,829	3,787

SELECT was designed for a hazard ratio of 0.83 with 90 percent power and 1,225 events, and the formula gives 1,211, close to the plan (the trial's group sequential design accounts for the difference). A cardiovascular outcomes trial hoping for a hazard ratio of 0.90 needs 3,787 events, and so tens of thousands of patients followed for years. Oncology trials, with larger effects and higher event rates, need hundreds.

Two cautions belong with the traditional analysis, and a Bayesian analysis inherits both. First, a single hazard ratio is a clean summary under proportional hazards. When the hazard ratio changes over time, the ordinary Cox estimate is a weighted average whose weights depend on the censoring pattern and on follow-up (Struthers and Kalbfleisch 1986), so that the same drug can yield different hazard ratios in different trials (Section 18.5). Time-specific or explicitly defined average hazard ratios remain possible, but they need careful definition and interpretation. Second, the information in Equation 18.1 is the number of events of the primary endpoint as defined, so any change to the endpoint definition or its intercurrent-event handling changes the information.

18.2. Events are information

Combine Equation 18.1 with the normal posterior of Chapter 6. With a flat prior, $\theta \mid \text{data} \sim N(\hat{\theta}, 4/D)$, and an observed hazard ratio of 0.80 gives the posterior probabilities of Table 18.2.

```
ev <- surv_events_table(c(100, 200, 400, 800), hr = 0.8, hr_mcid = 0.9)
tab <- data.frame(Events = ev$D, `SD of log HR` = round(ev$se, 3),
                  `Pr(HR < 1)` = round(ev$p_benefit, 3), `Pr(HR < 0.9)` =
                   ↪  round(ev$p_mcid, 3),
                  check.names = FALSE)
knitr::kable(tab)
```

Table 18.2.: Posterior probabilities for an observed hazard ratio of 0.80 under a flat prior on the log hazard ratio, by number of events.

Events	SD of log HR	Pr(HR $<$ 1)	Pr(HR $<$ 0.9)
100	0.200	0.868	0.722
200	0.141	0.943	0.798
400	0.100	0.987	0.881
800	0.071	0.999	0.952

The table is a reminder of how much information a convincing hazard ratio requires. To be 95 percent sure that a 20 percent reduction in hazard is at least a 10 percent reduction, a trial needs about 800 events: with 800 events the probability is 0.95, with 400 it is 0.88.

The same calculation run backwards translates a prior into events. To make the posterior standard deviation of θ equal to s the trial needs $D = 4/s^2$ events, and a prior of standard deviation s_0 on the log hazard ratio contributes $4/s_0^2$ effective events. A prior with $s_0 = 0.2$ is worth 100 events. This exchange rate is the cleanest way to explain prior borrowing (Chapter 19) to a clinical team: "the historical trials contribute as much as 100 events in this trial" is a statement a clinician can accept or reject, where "a normal prior with standard deviation 0.2" is not.

18.3. The gamma posterior for a hazard

For a single arm with constant hazard λ, observing d events in total follow-up time T gives the likelihood $\lambda^d e^{-\lambda T}$, whatever the censoring pattern, provided censoring is noninformative. With a Gamma(a, b) prior (shape a, rate b) the posterior is Gamma$(a + d, b + T)$. The prior is worth a events and b years of exposure.

Suppose a control arm records 60 events in 400 patient-years. With a vague Gamma$(0.01, 0.01)$ prior:

```
g <- surv_gamma_hazard(d = 60, T = 400, a = 0.01, b = 0.01)
tab <- data.frame(Quantity = c("Hazard (per year)", "Median survival
 ↪  (years)"),
                  `Posterior mean` = c(g$hazard["mean"], g$median["mean"]),
                  `2.5%` = c(g$hazard["lo"], g$median["lo"]),
                  `97.5%` = c(g$hazard["hi"], g$median["hi"]), check.names
                   ↪  = FALSE)
knitr::kable(tab, digits = 3, row.names = FALSE)
```

Table 18.3.: Posterior for a constant hazard and for the exponential median survival, 60 events in 400 patient-years, Gamma(0.01, 0.01) prior.

Quantity	Posterior mean	2.5%	97.5%
Hazard (per year)	0.150	0.114	0.190
Median survival (years)	4.699	3.643	6.054

The posterior mean hazard is 0.15 per year, with 95 percent interval 0.114 to 0.19. The median survival under the exponential model is $\log 2/\lambda$. Its posterior mean is 4.7 years, not the plug-in $\log 2/0.15 = 4.62$: the median is a reciprocal of the hazard, so averaging it is not the same as transforming the average. The interval, 3.6 to 6.1 years, is asymmetric for the same reason. The Bayesian computation transforms the posterior directly and so gets the asymmetry right without the delta method.

A constant hazard is rarely realistic over years of follow-up. Dividing time into intervals $I_1, \ldots, I_K$ with constant hazard λ_k in each, the likelihood factors into independent gamma terms, one per interval, with d_k events and T_k exposure. A hierarchical prior that ties neighboring hazards together, for example a random walk on $\log \lambda_k$, smooths the hazard and lets sparse late intervals borrow from earlier ones. The piecewise exponential model (Holford 1976; Breslow 1974), with a gamma-process prior for the cumulative hazard as the nonparametric Bayes extension (Kalbfleisch 1978), underlies most Bayesian analyses of time-to-event data (see Ibrahim, Chen, and Sinha (2001)), and with a proportional hazards term $\lambda_k e^{-\theta z}$ it approaches the Cox model as the intervals become finer.

Two further Bayesian survival models are worth knowing. The power prior reviewed in Ibrahim et al. (2015) is not limited to binary data: its discounting of the historical likelihood (Section 19.4) extends to a survival likelihood. When the historical data may not be exchangeable with the current trial, Murray and colleagues give a semiparametric commensurate model for time-to-event data from two possibly non-exchangeable sources, designed to attenuate bias and increase efficiency, and illustrated with post-market surveillance of stents in dialysis patients (Murray et al. 2014). Chen and colleagues develop a Bayesian promotion-time cure model, distinct from the mixture model illustrated in Section 18.5.

18.4. Predictive probability in an event-driven trial

Chapter 15 developed conditional power and predictive probability for a generic standardized statistic. For an event-driven trial the information scale is the event count, and the formulas become explicit. Suppose a trial is designed for $D_{\max} = 600$ events and succeeds if the final one-sided z statistic for $\theta > 0$ exceeds 1.96. At an interim analysis with D_1 events the estimate is $\hat{\theta}_1$. The remaining $D_2 = D_{\max} - D_1$ events contribute an independent estimate $\hat{\theta}_2 \sim N(\theta, 4/D_2)$, and the final estimate is the information-weighted average $\hat{\theta} = (D_1\hat{\theta}_1 + D_2\hat{\theta}_2)/D_{\max}$. Under a flat prior the posterior of θ is $N(\hat{\theta}_1, 4/D_1)$, so the predictive distribution of $\hat{\theta}_2$ is $N(\hat{\theta}_1, 4/D_2 + 4/D_1)$ and the final estimate is normal with mean $\hat{\theta}_1$ and variance $(D_2/D_{\max})^2(4/D_2 + 4/D_1)$. The predictive probability of success is a single normal probability. The calculation has a history in time-to-event monitoring. Spiegelhalter, Freedman and Parmar proposed skeptical and enthusiastic priors on the log hazard ratio for exactly this purpose (Spiegelhalter, Freedman, and Parmar 1994), and the CHART radiotherapy trials were the worked example of Bayesian design and monitoring in practice (Mahesh K. B. Parmar et al. 1994).

Two frequentist summaries are often reported instead. Conditional power under the design assumption plugs in the hazard ratio the trial was designed for, here 0.8. Conditional power under the current trend plugs in $\hat{\theta}_1$. Table 18.4 compares the three.

```
pp <- surv_pp_cp(D1 = c(300, 300, 300, 450), hr_obs = c(0.75, 0.85, 0.95,
 ↪  0.88),
```

```
                  Dmax = 600, hr_design = 0.8)
tab <- data.frame(`Events D1` = pp$D1, `Observed HR` = pp$hr_obs, `Interim
 ↪  z` = round(pp$z, 2),
                  PP = round(pp$pp, 2), `CP, design` = round(pp$cp_design,
                   ↪  2),
                  `CP, trend` = round(pp$cp_trend, 2), check.names = FALSE)
knitr::kable(tab)
```

Table 18.4.: Interim summaries in a trial designed for 600 events, success at final z > 1.96, design hazard ratio 0.80, flat prior for the predictive probability (PP). CP is conditional power.

Events D1	Observed HR	Interim z	PP	CP, design	CP, trend
300	0.75	2.49	0.94	0.95	0.99
300	0.85	1.41	0.51	0.72	0.52
300	0.95	0.44	0.09	0.35	0.03
450	0.88	1.36	0.25	0.42	0.22

The three numbers can tell different stories. In the third row, conditional power under the design assumption is 0.35, which might persuade a data monitoring committee to continue. But halfway through, the observed hazard ratio is 0.95, whose approximate 95 percent interval (0.76 to 1.19) includes the design value of 0.8 as well as no effect, so the data neither confirm nor exclude the design assumption. Conditional power under the trend, 0.03, conditions on the interim estimate as if it were the true effect, although half the events are still to come. The predictive probability, 0.09, averages over the posterior of the hazard ratio, so it reflects both the interim estimate and its uncertainty. The three differ in what they condition on, and a committee should see all three. (A predictive probability under a flat prior is itself pessimistic when the design effect was supported by earlier trials; the flat prior is the conservative end, and Chapter 15 shows the effect of a design-alternative prior.)

The second row, a promising but uncertain interim, is the situation in which adaptive designs propose to increase $D_{\max}$. The predictive probability makes the trade-off visible. Increasing $D_{\max}$ to 900 raises the predictive probability of success from 0.51 to 0.63 at a cost of 300 more events. Whether that is worth it depends on the value of approval and on what the extra events cost, the question of the value of the extra information, and not on a rule that conditional power lies in a "promising zone". Any such change must be pre-specified and its type I error consequences controlled, as the FDA adaptive design guidance (U.S. Food and Drug Administration 2019a) (final, November 2019) recommends; the calculation above is a decision aid, not a license to adapt.

18.5. Non-proportional hazards

Immunotherapy trials have taught oncologists that hazards need not be proportional (Alexander, Schoenfeld, and Trippa 2018). A treatment that engages the immune system may do nothing for months and then produce durable responses in a minority of patients. The survival curves overlap or cross early and separate late, and a single hazard ratio misdescribes both phases. The consequences for the traditional analysis are concrete: the log-rank test

loses power when the effect is delayed, because the early events, which carry weight in the test, are uninformative about the treatment; and the Cox hazard ratio becomes a quantity that depends on how long the trial runs (Hernán 2010).

Figure 18.1 shows three scenarios built from the same control arm, a constant hazard of 0.15 per year (median survival 4.6 years). Treatment either cuts the hazard by 40 percent from the start (proportional), cuts it by 40 percent only after a delay of one year, or cures 20 percent of patients and leaves the rest at the control hazard.

```
tt <- seq(0, 8, by = 0.05)
S_ctl <- surv_pwexp_S(tt, 0.15)
dd <- rbind(
  data.frame(t = tt, S = S_ctl, arm = "Control", scenario = "Control"),
  data.frame(t = tt, S = surv_pwexp_S(tt, 0.15 * 0.6), arm = "Treated",
    ↪  scenario = "Proportional"),
  data.frame(t = tt, S = surv_pwexp_S(tt, c(0.15, 0.09), c(0, 1)), arm =
    ↪  "Treated", scenario = "Delayed effect"),
  data.frame(t = tt, S = 0.2 + 0.8 * S_ctl, arm = "Treated", scenario =
    ↪  "Cure fraction 0.2"))
ggplot(dd, aes(t, S, color = scenario, linetype = arm)) + geom_line() +
  labs(x = "Years", y = "Survival probability", color = NULL, linetype =
    ↪  NULL) + theme_minimal()
```

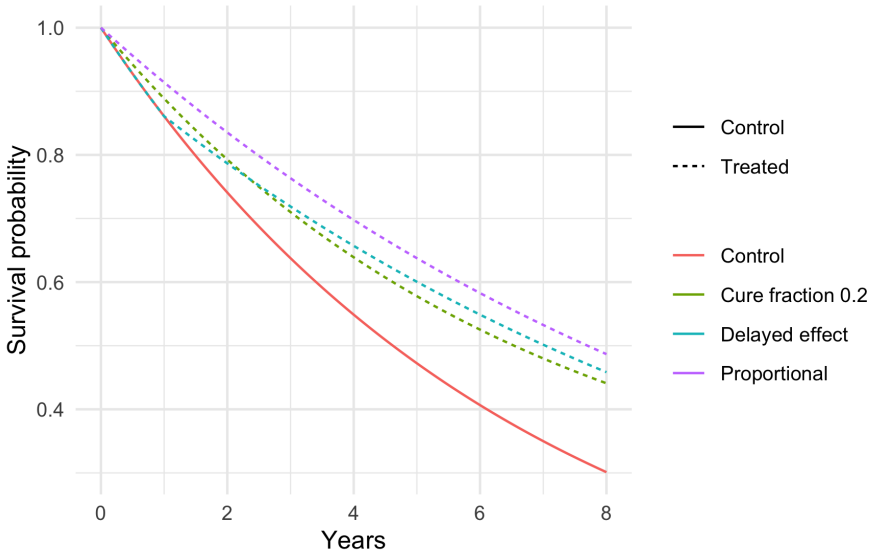


Figure 18.1.: Survival curves for a control arm with hazard 0.15 per year and three treatment scenarios: proportional hazards with hazard ratio 0.6, a delay of one year before the same hazard ratio takes effect, and a cure fraction of 0.2.

Three Bayesian responses are in common use.

The first is a piecewise treatment effect, $\theta_k = -\log \mathrm{HR}_k$ in interval k, with a hierarchical prior that lets the effect change over time but shrinks it toward a common value. It is the

piecewise exponential model of Section 18.3 with a treatment effect in each interval, and its posterior answers the clinical question "when does the benefit begin?" directly. The second is a mixture cure model (Boag 1949; Berkson and Gage 1952), in which a fraction π of treated patients are effectively cured and the rest follow the control hazard, $S_1(t) = \pi + (1-\pi)S_0(t)$ (in the Berkson-Gage model the uncured patients have a survival function of their own, and following the control hazard is a special case; M.-H. Chen, Ibrahim, and Sinha (1999) give a Bayesian promotion-time (non-mixture) cure model with a surviving fraction). The posterior for π is the quantity a patient cares about, but it is identifiable only if follow-up is long enough for the curve to plateau; with immature data it is dominated by the prior, which has to be stated. The third is to change the estimand to the restricted mean survival time up to a horizon τ,

$$\mathrm{RMST}(\tau) = \int_0^{\tau} S(t)\,dt, \tag{18.3}$$

whose posterior is computed from draws of the survival curve. The difference in RMST is the average event-free time gained within τ, in months or years, which is the unit in which patients and payers think, and it does not require proportional hazards (Royston and Parmar 2011, 2013; Uno et al. 2014). The idea of a restricted mean life is older than these papers (Irwin 1949; Karrison 1987). It does require a horizon, and the choice of τ is part of the estimand and must be fixed in the protocol.

```
nph <- surv_nph_example(tau = 5, h0 = 0.15, delay = 1, hr = 0.6, cure =
 ↪  0.2)
tab <- data.frame(Scenario = nph$scenario, `RMST treated` =
 ↪  nph$rmst_treated,
                  `RMST difference` = nph$rmst_diff, `Cumulative HR` =
                   ↪  nph$cumhaz_ratio, check.names = FALSE)
knitr::kable(tab, digits = 2)
```

Table 18.5.: RMST difference over 5 years and cumulative hazard ratio at 5 years for the three scenarios of the figure (control hazard 0.15 per year, hazard ratio 0.6, delay 1 year, cure fraction 0.2). The control arm has RMST 3.52 years.

Scenario	RMST treated	RMST difference	Cumulative HR
proportional	4.03	0.51	0.60
delayed effect	3.82	0.30	0.68
cure model	3.81	0.30	0.73

The scenarios give different RMST gains, 0.51, 0.3 and 0.3 years over five years, and a cumulative hazard ratio at five years of 0.6, 0.68 and 0.73: the hazard ratio is not a constant, which is the point.

18.5.1. Power loss under a delayed effect

How much does the log-rank test lose? The simulation below compares a trial of 300 patients per arm followed for five years, with the control hazard 0.15 per year, under a constant hazard ratio of 0.75 and under a delayed effect with no benefit for two years and a hazard ratio of 0.583 afterward. The two scenarios are matched so that the treated survival at five years is the same, 0.57, so they describe the same final benefit reached by different paths.

```
set.seed(1507)
rpwexp <- function(n, rates, breaks) {      # inversion for a piecewise
 ↪ exponential
  e <- rexp(n); H <- c(0, cumsum(rates[-length(rates)] * diff(breaks)))
  k <- findInterval(e, H); breaks[k] + (e - H[k]) / rates[k]
}
sim_power <- function(rates1, breaks1, n = 300, tau = 5, R = 2000) {
  rej <- numeric(R); ev <- numeric(R)
  for (r in seq_len(R)) {
    t0 <- rexp(n, 0.15); t1 <- rpwexp(n, rates1, breaks1)
    time <- pmin(c(t0, t1), tau); status <- as.integer(c(t0, t1) <= tau)
    grp <- rep(0:1, each = n)
    lr <- survival::survdiff(survival::Surv(time, status) ~ grp)
    z <- sign(lr$obs[1] - lr$exp[1]) * sqrt(lr$chisq)    # positive: more
 ↪ events in control
    rej[r] <- z > qnorm(0.975); ev[r] <- sum(status)
  }
  c(power = mean(rej), events = mean(ev))
}
p_ph  <- sim_power(0.15 * 0.75, 0)
p_del <- sim_power(c(0.15, 0.0875), c(0, 2))
tab <- data.frame(Scenario = c("Proportional, HR 0.75", "Delayed 2 years,
 ↪ then HR 0.583"),
                  Power = c(p_ph["power"], p_del["power"]), `Mean events` =
                   ↪ c(p_ph["events"], p_del["events"]))
knitr::kable(tab, digits = c(0, 2, 0))
```

Table 18.6.: Simulated power of the one-sided log-rank test (alpha 0.025), 300 patients per arm, administrative censoring at 5 years, 2,000 replications, with the mean number of events. Treated survival at 5 years is 0.57 in both scenarios.

Scenario	Power	Mean.events
Proportional, HR 0.75	0.68	287
Delayed 2 years, then HR 0.583	0.55	287

The simulated power is 0.68 under the constant hazard ratio and 0.55 under the delayed effect, with a similar number of events, although the five-year survival difference is the same. The Schoenfeld calculation of Table 18.1, which assumes a single hazard ratio, would not have warned about the difference. Two responses are used in practice: a weighted log-rank test (Fleming-Harrington weights (Harrington and Fleming 1982; Fleming and Harrington 1991)) that gives more weight to late events, and a design that states the delayed-effect model, with its simulation, in the protocol. Either choice is made before the trial; choosing the weights after seeing the curves invalidates the type I error. The next subsection shows what the weighted tests recover.

18.5.2. Tests built for a delayed effect

The Fleming-Harrington weighted log-rank test multiplies the contribution of each event time by a weight $S(t)^{\rho}\{1-S(t)\}^{\gamma}$, where S is the pooled Kaplan-Meier estimate (Harrington and Fleming 1982; Fleming and Harrington 1991). The choice $\rho = \gamma = 0$ is the ordinary log-rank test, and $(\rho, \gamma) = (0, 1)$ puts the weight on late events, where a delayed effect shows. A test that is best for one shape of effect can be poor for another, so the combination test takes the largest of several standardized weighted statistics, usually the four with $\rho, \gamma \in \{0, 1\}$, and refers it to the multivariate normal distribution implied by their correlation (S.-H. Lee 2007; R. S. Lin et al. 2020). This is the MaxCombo test. Table 18.7 repeats the simulation of Table 18.6 with the weighted test and the MaxCombo test, and adds a null scenario to check the level.

Table 18.7.: Simulated rejection rate (one-sided 0.025) of the log-rank test, the Fleming-Harrington (0, 1) test and the MaxCombo test of four Fleming-Harrington statistics, 300 patients per arm, administrative censoring at 5 years, 2,000 replications. The first row is the null of identical survival curves.

Scenario	Log-rank	FH(0,1)	MaxCombo
Null, identical curves	0.02	0.02	0.02
Proportional, HR 0.75	0.67	0.55	0.65
Delayed 2 years, then HR 0.583	0.55	0.80	0.74

Under the delayed effect the weighted test has power 0.8 and the MaxCombo test 0.74, against 0.55 for the log-rank test. Under proportional hazards the order reverses (this table is a fresh simulation, so its log-rank figure differs by Monte Carlo error from the 0.68 of Table 18.6): the log-rank test has 0.67, the MaxCombo test gives up a little (0.65) and the late-weighted test gives up more (0.55). That is the trade. A test chosen for one shape wins when the shape is right and loses when it is wrong, and the MaxCombo test pays a smaller premium for not knowing which shape it will meet, although, as the first caution below says, it too can reject when the experimental arm is worse. All three hold the level under the null of identical curves: the rejection rates are 0.024, 0.022 and 0.021 against the nominal 0.025.

Two cautions go with the table. First, these tests are tests of identical survival curves, and the weighted tests can reject when early harm and late benefit cancel in the average, so that a significant result does not by itself mean that patients on average did better; the modestly weighted tests of Magirr and Burman were proposed to control the risk of declaring a uniformly inferior drug superior (Magirr and Burman 2019). Second, the choice among them is a design decision with regulatory weight: the cross-pharma working group's straw-man guidance (Roychoudhury et al. 2021) treats the choice of test and the plausible non-proportional scenarios as something to settle, with simulation, at the design stage. Restricted mean survival time, the other main alternative, has an empirical record: in 54 oncology trials whose individual patient data were reconstructed from the published curves, the hazard ratio gave on average a larger treatment effect than the ratio of restricted mean survival times (average ratio 1.11, with substantial variation between trials), and the authors recommend that restricted mean survival time be reported routinely (Trinquart et al. 2016). The rule of this chapter is the same for every alternative: the primary test, its weights or horizon, and the simulation under the delayed-effect scenario go in the protocol before the curves are seen.

18.6. Regulation and practice

Four points follow from the guidance landscape.

First, the estimand comes first. Under ICH E9(R1) (International Council for Harmonisation 2019) the hazard ratio, the RMST difference and the survival difference at a landmark time are different estimands, and a Bayesian analysis reports the posterior of the one the protocol names. The five strategies for intercurrent events of Chapter 5 apply: for time to a non-fatal event in the presence of death, the usual choice is a composite (death counts as an event) or a competing-risks treatment, and each leads to a different model.

Second, what exists on non-proportional hazards is thin. We are not aware of an FDA or ICH guidance that sets a required method for non-proportional hazards or for RMST; the closest thing to a consensus document is the straw-man guidance of the cross-pharma working group discussed above (Roychoudhury et al. 2021), which is industry and academic guidance and not regulatory text. The oncology endpoints guidance (U.S. Food and Drug Administration 2018a) discusses the endpoints and their biases, and the practice of FDA reviewers, which can be read in public review documents, is to accept pre-specified alternatives to the log-rank test as sensitivity analyses and to scrutinize post hoc ones. The summary of a drug's effect on the product label is still usually a hazard ratio. This is an area where agreement with the agency at a pre-submission meeting, on the primary analysis and on the sensitivity analyses, matters more than the choice among statistical methods.

Third, the Bayesian elements map onto the existing framework. An informative prior on the log hazard ratio is judged as in Chapter 8 and Chapter 19, and its worth in events is stated as in Section 18.2. The book's default for a confirmatory adaptive design without borrowing is a Bayesian success criterion calibrated to the type I error by simulation, with the event-time process (accrual, censoring, the NPH model) inside the simulation. The FDA draft Bayesian guidance (U.S. Food and Drug Administration 2026g) describes this as one of three routes, the others being direct interpretation of posterior probabilities (for example with borrowing, by agreement with FDA) and decision-theoretic or benefit-risk criteria. The predictive probability of Section 18.4 is a monitoring aid and enters the formal rules only as a pre-specified futility or sample-size criterion.

Fourth, the event count itself is a design element. The final analysis is triggered by the event count, so the data monitoring committee and the sponsor must agree on how events are adjudicated and counted, and the timing rule for an interim analysis must be prespecified, whether by a number of events or by calendar date. If the analysis occurs at an information fraction other than the target, or events overrun the plan, the actual information fraction has to be handled by a prespecified method (for example an alpha-spending function evaluated at the observed fraction).

Key points

- The information about a log hazard ratio is one quarter of the number of events, $\mathrm{Var}(\widehat{\theta}) \approx 4/D$ under 1:1 allocation. Event-driven trials fix D, and the Schoenfeld formula $D = 4(z_\alpha + z_\beta)^2/(\log \mathrm{HR})^2$ is the sample size formula of Chapter 9 in event units.
- A prior with standard deviation s_0 on the log hazard ratio is worth $4/s_0^2$ events. A prior of standard deviation 0.2 is worth 100 events.

- To be 95 percent sure that an observed hazard ratio of 0.8 is below 0.9, a trial needs about 800 events (probability 0.95).
- For a constant hazard the posterior is gamma, and posterior summaries of the median survival come from transforming draws. In the worked example the posterior mean median is 4.7 years, not the plug-in 4.62.
- Conditional power under the design assumption and under the trend condition on different fixed effects; predictive probability averages over the posterior of the effect: in the worked row with an observed hazard ratio of 0.95 at half the events the three values are 0.35, 0.03 and 0.09.
- Under non-proportional hazards a single Cox hazard ratio can depend on censoring and follow-up, so the estimand needs explicit definition. The log-rank test loses power under a delayed effect (simulated 0.55 against 0.68 for a constant hazard ratio), and the RMST difference and the cure fraction are alternatives, each requiring pre-specification.
- No FDA or ICH document, to our knowledge, prescribes a method for non-proportional hazards or RMST. ICH E9(R1) requires that whatever summary measure is chosen be defined as an estimand before the trial.

Exercises

1. Derive Equation 18.1 from the score test for the Cox model with one binary covariate under 1:1 allocation and a hazard ratio near one.
2. With a Gamma$(2, 10)$ prior on the control hazard, recompute the posterior mean and interval for the control arm with 60 events in 400 patient-years. How many events is the prior worth, and how much does it move the median survival?
3. Reproduce Table 18.4. For the second row, compute the predictive probability if $D_{\max}$ is increased to 900, and the conditional power under the design assumption.
4. Under a flat prior, investigate when the predictive probability lies between the two conditional powers. Is it always so when the design hazard ratio is more favorable than the observed one?
5. Simulate the cure scenario (cure fraction 0.2, control hazard 0.15 per year). Over follow-up of 3, 5 and 8 years, compute the Cox hazard ratio and the RMST difference. Which is more stable as follow-up lengthens?
6. A protocol specifies a primary analysis by the log-rank test and a "supportive" RMST analysis chosen after the blinded review of the pooled curve. Write the objection a regulator would raise, and the pre-specification that would meet it.

19. Borrowing Information

A sponsor planning a trial of a new drug in a rare disease already knows something about the control arm. Earlier trials enrolled similar patients, a registry holds their outcomes, and a pediatric program can lean on what is known in adults. Using that knowledge looks like plain common sense, and there is a cost argument too: every patient who does not have to be randomized to control is a patient who can be treated, or a trial that can finish sooner. This chapter is about how to use such information, how regulators judge it, and what it costs when the borrowed information turns out to be wrong.

Of the four mechanisms by which Bayesian designs save patients (monitoring, multiple arms, shrinkage, and prior information), borrowing is the one with the sharpest trade. Done well, it recovers precision that a small trial cannot supply. Done badly, it can introduce substantial bias into a small current trial, whose data may be insufficient to reveal the mismatch, and it does so under the sponsor's own prior. The central message of the chapter is that borrowing is not free, that the price is measurable by simulation, and that the regulator will ask to see that price before agreeing to the design.

! Regulatory anchor

- **FDA draft guidance, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (January 2026, draft)** (U.S. Food and Drug Administration 2026g). Informative priors "will generally need a greater amount of justification". Discusses static versus dynamic discounting, effective sample size (ESS), prior-data conflict and sensitivity analyses. Recommends against measuring the amount of borrowing by type I error inflation. Case examples include REBYOTA, pediatric extrapolation, platform-trial controls and basket trials. Details in Chapter 3.
- **ICH E10, Choice of Control Group (Step 4, July 2000)** (International Council for Harmonisation 2000). Names the externally (historically) controlled trial as one of the five types of control and states its main weakness, the inability to control bias.
- **FDA draft guidance, Considerations for the Design and Conduct of Externally Controlled Trials (February 2023, draft)** (U.S. Food and Drug Administration 2023a). Recommendations for building external control arms from real-world data, with emphasis on pre-specification and comparability.
- **FDA, Considerations for the Use of Real-World Data and Real-World Evidence to Support Regulatory Decision-Making for Drug and Biological Products (final guidance, August 2023)** (U.S. Food and Drug Administration 2023b). FDA's current thinking on how real-world data and evidence support new indications and postapproval studies. It recommends and does not require, and for external control arms it refers to the externally controlled trials draft above.
- **FDA, Rare Diseases: Considerations for the Development of Drugs**

and Biological Products (final guidance, December 2023) (U.S. Food and Drug Administration 2023c). Finalizes the draft "Rare Diseases: Common Issues in Drug Development" of February 1, 2019. It says that external control designs are usually reserved for specific circumstances, such as a well-understood and characterized natural history, high and predictable mortality or progressive and predictable morbidity, or a drug effect that is large and self-evident, that each such design warrants a case-by-case assessment, and that early discussion with the review division is recommended. It also says that sponsors may be able to use Bayesian methods to maximize the use of information from early-phase or natural history studies, and that Bayesian methods may inform pediatric investigations through incorporation of adult data.

- **ICH E11A, Pediatric Extrapolation (Step 4, August 2024)** (International Council for Harmonisation 2024). A framework for deciding how much pediatric efficacy can be inferred from adult or other source data, and a place for Bayesian methods in the quantitative part.
- **FDA, Guidance for the Use of Bayesian Statistics in Medical Device Clinical Trials (final, February 2010)** (U.S. Food and Drug Administration 2010). The device guidance that has allowed borrowing from prior studies for more than fifteen years, with simulated operating characteristics.
- **MHRA draft guideline on external control arms from real-world data (2025)** (Medicines and Healthcare products Regulatory Agency 2025) and **EMA concept paper on external controls (draft of July 2025, adopted May 2026)** (European Medicines Agency 2026b). Both stress pre-specified protocols and fitness for purpose of the external data.
- **EMA CHMP qualification opinion on prognostic covariate adjustment, PROCOVA (September 2022)** (European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022). Qualifies a covariate-adjustment method that uses external data to build a prognostic score, without borrowing outcomes.

19.1. What the traditional design does with prior data

A traditional trial uses prior data heavily, but only before the trial starts. Phase 2 results and earlier studies of the control treatment set the design alternative (the effect size the trial is powered to detect), the assumed control response rate or event rate, the variance, the dropout rate, and therefore the sample size. In the notation of Chapter 9, the sample size is a function of θ and of nuisance parameters, and all of them come from earlier data. A sponsor who assumed a 30% control response rate because three earlier trials averaged 30% has used those trials fully.

Then the data are set aside. In a conventional standalone randomized-trial analysis, the analysis of the new trial uses only the new trial. The earlier studies appear in the introduction of the protocol and in the clinical overview, and nowhere in the p-value or the confidence interval. This is the usual convention, and it has a reason. The frequentist guarantee, a one-sided type I error of 0.025 at every parameter value in the null hypothesis, is a statement about repeated use of the same procedure on the same kind of data. If last year's data entered this year's test statistic, the guarantee would depend on how last year's data happen to relate to this year's patients, and nothing in the protocol controls that. Frequentist methods

can incorporate historical information under explicit assumptions (the margin synthesis later in this chapter is one, and FDA's 2016 noninferiority guidance describes a frequentist synthesis method that combines the historical and current estimates and their uncertainties (U.S. Food and Drug Administration 2016)), but the guarantee then holds only under those assumptions.

The rule has a cost that is easy to see in a small trial. If the control arm must be as large as the treatment arm, half of a scarce population is randomized to a comparator whose behavior is, to a good approximation, already known. In a pediatric rare-disease program the number of eligible children may not support a trial of conventional size at all. The traditional design answers this with a smaller trial and a looser level of evidence, a larger effect size in the alternative, or a single-arm design. The single-arm design is where the question of external information becomes unavoidable.

19.2. External and historical controls: the traditional objections

In a single-arm trial all patients receive the new treatment, and the effect is judged by comparing the observed outcome with what is expected without it. The expectation comes from somewhere: a published control arm, a registry, a set of earlier trials, a cohort of patients treated in routine care. ICH E10 calls this the externally controlled trial, including the historical control, and lists it among the five kinds of control group (International Council for Harmonisation 2000). The guideline's stance is cautious. Because treatment assignment between the trial and the external control is not randomized, and the external controls may also be nonconcurrent, differences between the groups can arise from sources other than the treatment, and the design is regarded as suitable only where the effect of treatment is large relative to the variability of the natural course of the disease. FDA's substantial-evidence reasoning is the same: "Because of the lack of randomized treatment assignment", differences between groups are a concern (see Chapter 3 for the sources).

Pocock (1976) gave the classic list of conditions under which a historical control group can be combined with a randomized one (Pocock 1976). In paraphrase, the historical controls should have: received exactly the treatment now given to the concurrent controls; been part of a recent study with the same protocol, so that eligibility and the evaluation of outcome are the same; been treated at the same institutions or by the same investigators, so far as possible; had the same distribution of important prognostic factors; and shown no unexplained differences in outcome from the concurrent controls. These criteria are still the working checklist. They are demanding, because the situations in which a sponsor wants historical data are exactly the ones in which standard of care, diagnostic criteria, supportive care and patient selection have drifted since the older study.

Three mechanisms make an external control misleading. First, *selection*: patients who enter a trial are not the same as patients in a registry, because trial entry requires being well enough to travel, to consent and to meet eligibility, and investigators choose whom to offer a new drug. Second, *time*: outcomes of untreated and standard-treated patients move over time as care improves and as diagnosis happens earlier, so a control from five years ago may be worse than a control today. Third, *measurement*: endpoints assessed in routine care (a response recorded in a chart) are not assessed as in a trial (a central radiology read at scheduled visits). Each of these gives a bias, not a variance, so a larger external sample does not help.

The regulatory response has been to give external controls a structure rather than to ban them. FDA's draft guidance on externally controlled trials was issued in February 2023 (U.S. Food and Drug Administration 2023a). It asks for a pre-specified protocol and analysis plan, a justification of why a concurrent randomized control is not feasible, careful choice of the data source, and an assessment of whether the external group is comparable on the prognostic factors and on how outcomes were measured. In 2025 the UK MHRA published a draft guideline on external control arms built from real-world data, and the EMA published a concept paper (draft in July 2025, adopted in May 2026) announcing a reflection paper on the same topic (Medicines and Healthcare products Regulatory Agency 2025; European Medicines Agency 2026b). Both are drafts or plans, not binding, and both repeat the same two planning considerations: pre-specification before the external data are examined, and demonstration that the data are fit for the purpose. None of these documents describes a statistical method as the answer. They describe a design discipline in which the statistical method sits.

Two final guidances of 2023 sit beside the draft and are summarized in the regulatory anchor at the start of the chapter: the real-world data and evidence guidance points to the externally controlled trials draft for external-control designs (U.S. Food and Drug Administration 2023b), and the rare-diseases guidance reserves external controls for specific circumstances (U.S. Food and Drug Administration 2023c). Both recommend and do not require.

The DIA working group's roadmap for historical controls gives a checklist for judging the quality and comparability of a historical source (Ghadessi et al. 2020). Propensity-score methods add a second step: the propensity-score-integrated power prior and composite likelihood use the overlap in propensity scores between external and current patients to decide how much of the external data enters, so that external patients unlike the trial's patients count less (C. Wang et al. 2019; W.-C. Chen et al. 2020; J. Lin, Gamalo-Siebers, and Tiwari 2019). They adjust for the covariates that were measured and cannot repair the ones that were not, which is the lesson of the NSCLC case study below (Section 19.14).

The Bayesian methods in the rest of this chapter should be read with that background. They are a way of deciding *how much* to rely on external data once a regulator has accepted that the data are admissible. They do not make inadmissible data admissible.

19.3. Bayesian borrowing: the idea

Let p_c be the response rate on control in the current trial, and suppose historical data y_h from n_h control patients are available. Treating the two sources as the same population and pooling gives a posterior that is a Beta distribution with the historical successes and failures added to the current ones. In Chapter 8 the prior was measured in patients: a $\text{Beta}(a, b)$ prior has an effective sample size of $a + b$. Borrowing the whole historical control arm is a prior with effective sample size n_h. If the historical data are right, the current control arm is effectively enlarged by n_h patients. If they are wrong, the posterior is anchored on the wrong rate.

The methods below differ in what they assume about the relationship between p_c and the historical rate p_h:

1. $p_c = p_h$ exactly (full pooling).
2. p_c is close to p_h, and a single number says how close (power prior with a fixed discount, static borrowing).

3. p_c and p_h are draws from a common distribution whose spread is estimated (hierarchical models, meta-analytic-predictive priors).
4. p_c is either like p_h or unrelated to it, and the data decide which (mixture priors, robust MAP).
5. How close p_c is to p_h is itself a parameter that is updated by comparing the two datasets (commensurate priors, normalized power priors).

Methods 1 and 2 are *static*: the amount of borrowing is fixed before the trial. Methods 3 to 5 are *dynamic*: the amount of borrowing depends on how well the current data agree with the historical data. FDA's draft guidance uses exactly this split and says that dynamic methods "borrow less when the data are less similar and more when the data are more similar" (U.S. Food and Drug Administration 2026g). The distinction between exchangeable and non-exchangeable sources is the Bayesian analogue of Pocock's checklist. Real programs illustrate both kinds: the OPTIMIZER device approval used a fixed 30% weight on an earlier study (Section 19.17.3), and the REBYOTA phase 3 trial used a hierarchical model with dynamic borrowing from a phase 2b study (Section 19.17.1).

Borrowing methods have a lineage, and these approaches make different assumptions about historical comparability. Table 19.1 lists it so that the names used in the rest of the chapter can be placed.

Table 19.1.: A short lineage of borrowing methods.

Year	Contribution	What it changed
1976	Pocock's comparability criteria and random-bias model (Pocock 1976)	Models disagreement between historical and concurrent controls through a normal bias distribution
1994, 2004	Skeptical and enthusiastic priors, and priors expressed as equivalent patients (Spiegelhalter, Freedman, and Parmar 1994; Spiegelhalter, Abrams, and Myles 2004)	Prior information discounted explicitly, but by a fixed judgment
2000	The power prior (Ibrahim and Chen 2000)	A single parameter a_0 down-weights the historical likelihood
2006, 2009	The normalized power prior (Duan, Ye, and Smith 2006; Neuenschwander, Branson, and Spiegelhalter 2009)	Retains the a_0-dependent normalizing constant, so the specified prior on a_0 is its marginal prior
2010	The meta-analytic-predictive prior (Neuenschwander et al. 2010)	Several historical trials, with between-trial heterogeneity
2011	The commensurate prior (Hobbs et al. 2011)	Agreement between the sources becomes a parameter
2014	The robust MAP prior (Schmidli et al. 2014); a simulation comparison of pooling, power priors and hierarchical borrowing (Viele et al. 2014)	A vague component against conflict; the cost of drift put into numbers

Year	Contribution	What it changed
2020	Limits of borrowing under strict conditional type I error control (Kopp-Schneider, Calderazzo, and Wiesenfarth 2020)	A statement of what dynamic borrowing cannot deliver

19.3.1. Exchangeability and hierarchical models

Two groups of patients are *exchangeable* if, before seeing any data, we would not be surprised by relabeling them: nothing we know distinguishes the response rate of one group from that of the other. Exchangeability does not say the rates are equal. It says their joint prior is symmetric. We model the group rates as conditionally independent draws from a common distribution with unknown parameters, which is the hierarchical model. This is a modeling assumption; de Finetti's representation (Chapter 6) motivates it when exchangeability extends to an infinite sequence.

For K groups with response rates $p_1, \dots, p_K$, a standard hierarchical model for binary data is

$$y_k \mid p_k \sim \text{Binomial}(n_k, p_k), \qquad \operatorname{logit} p_k \mid \mu, \tau_b \sim N(\mu, \tau_b^2), \qquad (\mu, \tau_b) \sim \pi(\mu, \tau_b). \tag{19.1}$$

The between-group standard deviation τ_b controls borrowing. When τ_b is near zero the groups are close to identical and each group's estimate is pulled towards the common mean. When τ_b is large the groups are nearly independent and each keeps its own data. The key feature is that τ_b is estimated from the data: if the groups disagree, the posterior for τ_b moves up and borrowing falls. For a normal outcome with group effects θ_k, the posterior mean of group k is, approximately, a weighted average,

$$\mathbb{E}[\theta_k \mid y] \approx \lambda_k \bar{\theta} + (1 - \lambda_k)\hat{\theta}_k, \qquad \lambda_k = \frac{\sigma_k^2}{\sigma_k^2 + \tau_b^2}, \tag{19.2}$$

where σ_k^2 is the sampling variance of the group's own estimate $\hat{\theta}_k$ and $\bar{\theta}$ is the precision-weighted overall mean. A small group has large σ_k^2 and so is shrunk more. This is the same shrinkage that Chapter 10 used for several endpoints and Stein's normal-means result (1956) shows that carefully chosen shrinkage estimators can reduce total squared error under its assumptions (Stein 1956). Whether shrinkage improves estimation in the hierarchical binomial model above has to be assessed under plausible heterogeneity, because the theorem does not cover estimated heterogeneity or arbitrary shrinkage rules.

The same structure is used in multicenter trials, where the groups are centers. Polson and Sokolov (2026a) fit a hierarchical logistic-normal model to the eight-center topical-cream trial of Skene and Wakefield, with a bivariate normal on the pair of center-specific log odds for the two arms, so that centers borrow strength on both arms and on their difference (Skene and Wakefield 1990). Individual centers had 5 to 37 patients per arm, so single-center estimates were imprecise, and the model improves them without discarding center-level variation.

Two warnings. First, shrinkage lowers *total* error only under conditions, and not the error of every group: under the normal model with known variances and three or more means, Stein's result is that shrinkage lowers the total squared-error risk, but an individual group can be estimated worse, and a group that really is different usually is, compared with its own estimate. This is the price of borrowing in one sentence. Second, with few groups

τ_b is poorly estimated, and the answer depends on its prior. With four historical trials, a half-normal prior with scale 0.5 on τ_b and one with scale 1.0 can give quite different degrees of borrowing, and the sensitivity analysis should show it. The FDA draft's insistence on sensitivity analyses over "a range of alternative reasonable choices for the prior" is aimed at precisely this point (U.S. Food and Drug Administration 2026g).

19.4. Power priors and their normalized version

The power prior of Ibrahim and Chen (2000) is the most direct way to down-weight historical data (Ibrahim and Chen 2000; Ibrahim et al. 2015). Given a vague initial prior $\pi_0(\theta)$ and historical data D_h with likelihood $L(\theta \mid D_h)$, the power prior is

$$\pi(\theta \mid D_h, a_0) \propto L(\theta \mid D_h)^{a_0}\, \pi_0(\theta), \qquad 0 \le a_0 \le 1. \tag{19.3}$$

Setting $a_0 = 0$ discards the historical data and $a_0 = 1$ pools them fully. For a binomial response with y_h successes in n_h patients and a Beta$(1, 1)$ initial prior, the power prior is Beta$(1 + a_0 y_h,\ 1 + a_0(n_h - y_h))$, so its effective sample size is $a_0 n_h + 2$ and one can read a_0 as the fraction of the historical patients that count. With $n_h = 100$ and $a_0 = 0.3$, the historical arm counts as about 30 patients.

If a_0 is fixed in advance, the method is static. It is easy to describe and easy to simulate, and the regulator can see the borrowing in one number. It also has a feature that surprises people: the power prior discounts the *precision* of the historical data but does not move their *center*. Historical data that say 30% are still a prior centered near 30%, only a weaker one. With a Beta$(1, 1)$ initial prior and $a_0 = 0.3$ the prior is Beta$(10, 22)$, centered at 0.3125, with effective sample size 32. If the truth is 40%, it pulls the estimate from the data towards that center with weight $32/(32 + n_c)$ and nothing in the method notices the conflict.

Figure 19.1 reads Bayes' rule as a product of bars. With a vague prior the posterior is the likelihood, with a standard deviation of 0.085. The power prior at $a_0 = 0.3$ adds the equivalent of 32 patients to the 30 current controls (ESS $a_0 n_h + 2$), and the posterior narrows to a standard deviation of 0.06 and moves from 0.41 towards the historical 0.30, to 0.35. The narrowing is the gain, and the shift is the risk: the current data say 40% and the posterior is pulled down by an amount the trial cannot verify. The larger the gap between the historical and the current rate, the more the borrowed bars mislead, and Table 19.3 (and the drift scenarios in Section 19.8) put numbers on what that costs in type I error.

A natural improvement is to treat a_0 as unknown with its own prior, so that the data choose it. This runs into a technical difficulty: the prior in Equation 19.3 as written has a normalizing constant that depends on a_0, and ignoring it makes the posterior favor small a_0 in a way that has nothing to do with agreement between the datasets. The *normalized power prior* divides by the constant,

$$\pi(\theta, a_0 \mid D_h) = \frac{L(\theta \mid D_h)^{a_0}\, \pi_0(\theta)}{\int L(\theta \mid D_h)^{a_0}\, \pi_0(\theta)\, d\theta}\ \pi(a_0), \tag{19.4}$$

and then the marginal posterior of a_0 increases when the current data resemble the historical data and decreases when they do not. The normalized power prior is due to Duan, Ye, and Smith (2006) and Neuenschwander, Branson, and Spiegelhalter (2009), and M.-H. Chen and Ibrahim (2006) relate it to hierarchical models. For FDA a fixed a_0 is a static method, and a random a_0 updated by the data is a dynamic one (the draft also lists supervised power priors, a term we do not use again). The cost of dynamic methods is more parameters to specify, including the prior on a_0, and an operating characteristic that has to be simulated.

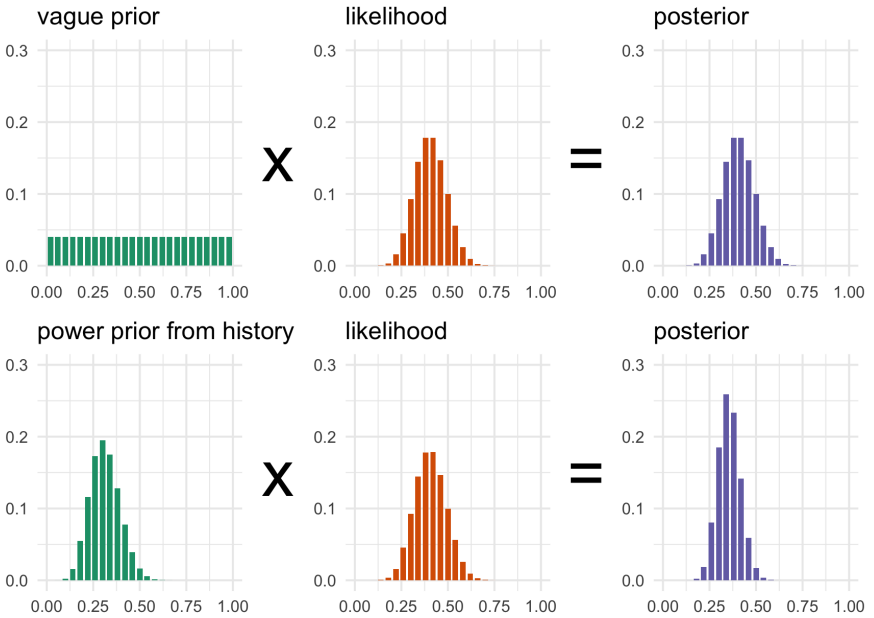


Figure 19.1.: Bayes' rule on a grid of control response rates. Top row: vague Beta(1,1) prior times the likelihood of 12 responders in 30 current controls. Bottom row: power prior from the historical control arm (30/100, weight 0.3) times the same likelihood. Each panel is scaled to sum to one.

19.4.1. A rare-disease trial: choosing a_0 by the marginal likelihood

A fixed a_0 is a judgment. The data can say something about it, and the simplest way to see how is a small example. Consider a rare metabolic disease. A natural-history study of 40 patients on standard care found 12 responders by a functional criterion (30%). The new trial randomizes 25 patients to the treatment and 15 to standard care, a 5:3 allocation (hypothetical, like the rest of this example) chosen to limit the number of patients on a comparator of doubtful benefit, and observes 14 of 25 responses on treatment and 6 of 15 on control. With uniform priors and no borrowing, the posterior probability that the treatment rate exceeds the control rate is 0.83, an inconclusive result by the 0.975 standard. Table 19.2 shows what the power prior adds.

Table 19.2.: Rare-disease trial (14/25 treated, 6/15 control) with a power prior from 12/40 historical controls. The control size, excluding the reference prior, is 15 + 40 a0. The second probability is that the treatment rate exceeds the control rate by more than 0.10.

a0	Control size (excluding reference prior)	Pr(pT > pC)	Pr(pT - pC > 0.1)
0.00	15	0.830	0.619
0.25	25	0.918	0.743
0.50	35	0.951	0.803
1.00	55	0.974	0.859

At $a_0 = 0.5$ the control sample is more than doubled and the probability of benefit reaches 0.95. Whether that borrowing is justified is a question about the current controls, and the data speak to it through the marginal likelihood of the current control data as a function of a_0,

$$m(a_0) = \frac{B(1 + a_0 y_h + y_c,\ 1 + a_0(n_h - y_h) + n_c - y_c)}{B(1 + a_0 y_h,\ 1 + a_0(n_h - y_h))}, \tag{19.5}$$

with B the beta function. It measures how well the historical controls predicted the current ones.

Here the current control rate is 6/15, or 40%, close to the historical 30%, and m is largest at $a_0 = 1$: full borrowing is supported, with $m(1)/m(0) = 2.3$. Had the current controls responded 11 times in 15 (73%), the maximum would move to $a_0 = 0$ and the trial would stand on its own data. This is an empirical Bayes choice and it is crude, because fifteen patients are asked to judge a disagreement, and the maximizer $a_0 = 1$ at a boundary says "no evidence against full borrowing", not "the populations are the same". It captures the logic of the commensurate and robust-mixture priors of Section 19.5 and Section 19.6: borrow when the sources agree and stop when they conflict. It does not remove the need for the drift simulation of Section 19.8, because a regulator will ask what the design concludes if the true control rate is 45%, not 30%. Real instances of borrowing from natural-history and earlier-trial controls are told in Section 19.17. Before relying on a source like the 12 of 40 above, check it against the DIA roadmap's criteria for the quality and comparability of historical controls (Ghadessi et al. 2020), and expect the review division to read it against the rare-disease guidance, which reserves external controls for specific circumstances (U.S. Food and Drug Administration 2023c).

19.5. Commensurate priors

Hobbs, Carlin, Mandrekar and Sargent (2011) put the question of agreement into a parameter of its own (Hobbs et al. 2011) (the paper treats Gaussian data, and the extension to generalized linear models, the case the binary examples of this chapter need, is in Hobbs, Sargent, and Carlin (2012)). Write η_c for the current parameter and η_h for the historical one (for binary data, the logits $\eta_c = \operatorname{logit} p_c$ and $\eta_h = \operatorname{logit} p_h$, so that the normal prior is placed on the logit scale). The *commensurate prior* says

$$\eta_c \mid \eta_h, \tau_c \sim N(\eta_h,\ 1/\tau_c), \tag{19.6}$$

where the *commensurability parameter* τ_c is a precision, with its own prior, typically a spike-and-slab or a discrete mixture that allows a large value (near agreement) and a small value (near independence). Large posterior τ_c means the current data support the claim that the two parameters are close, and the historical data count heavily. Small posterior τ_c means they do not, and the historical data are down-weighted. In this form the method looks like a hierarchical model with two groups, but there is an important difference in how the historical group is used. In the two-stage version the historical posterior is held *fixed* and the direction of borrowing is one way, which is the way a regulator thinks about an earlier trial that has already been reported. In the fully Bayesian joint model, information flows both ways through the commensurability parameter.

Hobbs and colleagues also give a version in which the commensurate model sits on a power prior, so that the two ideas, discounting and agreement, are combined. In applications there is a practical issue the original paper discusses: a method that learns agreement from a small current control arm has little to learn from, and so in a small trial the commensurate prior behaves much like a static one with weights fixed by the prior on τ_c.

19.6. Meta-analytic-predictive priors and robust mixtures

Often there is not one historical control arm but several, from different trials. The *meta-analytic-predictive* (MAP) prior of Neuenschwander and colleagues (2010) does a random-effects meta-analysis of the historical control rates, using the hierarchical model of Equation 19.1, and then takes the *predictive distribution* of the control parameter in a new trial as the prior (Neuenschwander et al. 2010). The predictive distribution is wider than the meta-analytic mean because it includes the between-trial variation τ_b: even if the historical trials were estimated without error, a new trial's control rate would differ from their average by about τ_b. This is the logic of an exchangeable new trial, and it is the reason a MAP prior does not become more informative than the heterogeneity warrants however many historical patients there are.

Figure 19.2 shows this with four illustrative historical control arms (80, 100, 70 and 150 patients, 400 in total). The calculation is a normal-normal random-effects model on the logit scale, with a half-normal prior with scale 0.5 on τ_b, and the predictive distribution is converted to a Beta prior by matching mean and variance.

The MAP prior has median 0.326 and 95% interval (0.212, 0.471), with posterior mean τ_b of 0.18, and its Beta approximation is Beta(18.8, 38.2). The MAP prior is worth about 57 patients, although 400 patients contributed. The pooled analysis, which ignores between-trial variation, would claim a prior effective sample size of 402 (the 400 historical patients plus

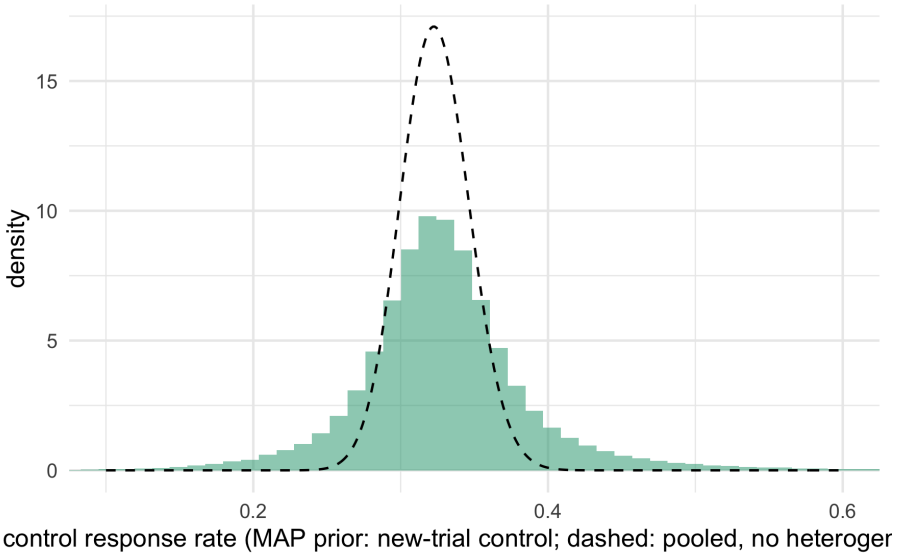


Figure 19.2.: Meta-analytic-predictive prior for the control response rate in a new trial (histogram) compared with the pooled-historical posterior that ignores between-trial variation (dashed). Four illustrative historical arms.

the 2 patients of the Beta(1, 1) reference prior). The difference is the between-trial variation, and it is the most important number a regulator can ask for, since it quantifies how much a *new* trial's control arm can be expected to differ from the old ones.

A MAP prior is still an exchangeability assumption: it assumes that the new control rate is another draw from the same distribution as the historical ones. If the new trial differs systematically, because standard of care changed or the population is different, the MAP prior is confidently wrong. Schmidli and colleagues (2014) proposed protecting against this with a *robust MAP* prior, a mixture of the MAP prior and a vague prior (Schmidli et al. 2014):

$$\pi_{\text{rob}}(\theta) = w\,\pi_{\text{MAP}}(\theta) + (1 - w)\,\pi_{\text{vague}}(\theta). \tag{19.7}$$

The conjugate update of a mixture is a mixture of conjugate updates, with the weights multiplied by the marginal likelihood of the current data under each component. If the current control data look like the MAP prior, the weight stays on the informative component. If they do not, the weight moves to the vague component and the posterior reverts to the current data (Schmidli et al. 2014). The weight $1 - w$ on the vague component is a choice made at the design stage; values of 0.1 to 0.5 are our working range, not a figure taken from the literature on robust priors. The mixture weight after the data are in plays the role of the commensurability parameter in the previous section, and it is the quantity to report: it says how much of the final answer comes from outside the trial. For production use, the R package RBesT (on CRAN) fits MAP priors by MCMC, approximates them by mixtures of conjugate distributions, and computes robust versions and operating characteristics (Weber et al. 2021); the chunks in this chapter use a simpler normal-normal approximation from `bayestrials` so that every step is visible. The belimumab pediatric analysis (Section 19.17.2) is the clearest regulatory example of a robust mixture prior.

i Technical note: why robust mixtures learn agreement slowly

Let the informative component be Beta(a, b) and the vague one Beta$(1, 1)$, with prior weights w and $1 - w$. After y successes in n current control patients, the posterior weight on the informative component is proportional to $w\,B(a + y, b + n - y)/B(a, b)$, against $(1 - w)\,B(1 + y, 1 + n - y)$. The ratio of the two marginal likelihoods is a Bayes factor between a sharp and a diffuse prior. For small n it takes a large discrepancy to change the Bayes factor by much, because both components assign comparable marginal likelihood to a wide range of y. The weight starts to move only when $\hat{\theta}$ is more than a couple of standard errors, $\sqrt{\theta(1-\theta)/n}$, from the informative component's mean. In a control arm of 30 patients the standard error is about 0.08, so a true rate 0.10 above the historical rate is within two standard errors and the weight barely moves. This is not a defect of the method. It is the statement that 30 patients cannot tell 30% from 40%, and therefore that dynamic borrowing in a small trial offers weak protection against moderate drift. The simulation below shows the consequence.

19.7. How much does the borrowed information weigh? Effective sample size and conflict

Chapter 8 defined the effective sample size of a prior, and its best known definition is that of Morita, Thall and Mueller (Morita, Thall, and Müller 2008). Definitions differ and give different numbers for the same prior, so a request for the "ESS" should be answered with the definition stated. For borrowing it is the central summary, and FDA asks for it: the draft guidance defines ESS as "a measure of the information in a probability distribution in terms of the equivalent number of patients in the target population", asks that it be reported across the plausible range of outcomes, with its maximum and its mean, and asks for the ratio of prior ESS to trial size (U.S. Food and Drug Administration 2026g). For a static prior the ESS is a number. For a dynamic prior it is a function of the observed current data: the ESS is high when the current control agrees with the historical one and falls when it does not.

For a robust mixture there is no unique patient-equivalent ESS. A simple convention, used here, is the component-weighted ESS: the posterior weight on each component times that component's ESS (102 for the informative Beta$(31, 71)$, 2 for the vague Beta$(1, 1)$), summed over the two components. Other conventions give different numbers. Figure 19.3 shows it for the example used in the rest of the chapter: a historical control arm of 100 patients with 30 responders, and a current control arm of 30 patients, with a robust prior that puts weight 0.8 on the historical-data Beta$(31, 71)$ and 0.2 on Beta$(1, 1)$.

At the historical rate (9 of 30) the weight on the historical component is 0.94 and the borrowed ESS is about 96. At 12 of 30, which is an observed rate ten percentage points above the historical one, the weight is still 0.91. At 15 of 30 it is 0.69, and at 18 of 30 it has fallen to 0.18. The method protects against a gross conflict and does very little against a moderate one.

Prior-data conflict is the general name for this problem: the current data are unlikely under the prior. Two practical diagnostics are used. The first is a prior predictive check: compute the probability, under the prior, of data at least as extreme as the observed control data (a Box p-value), and report it. The second is the weight on the vague component or the

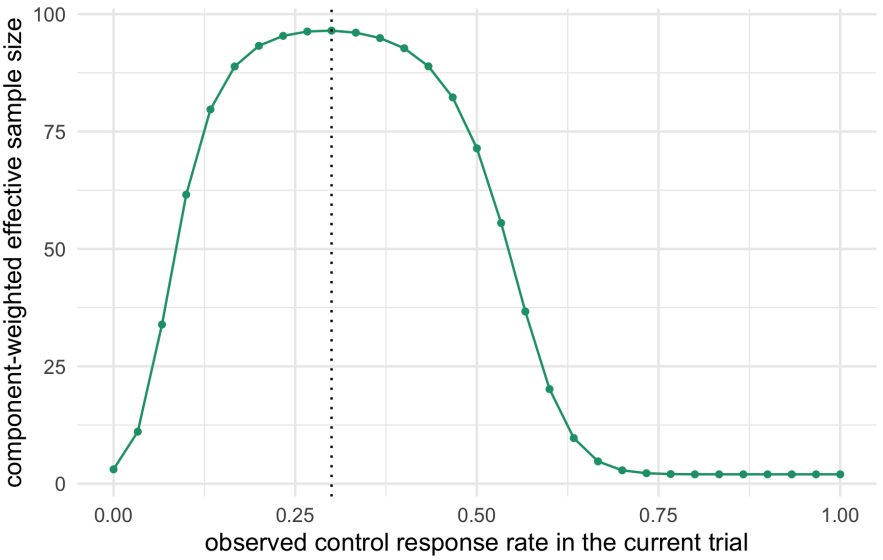


Figure 19.3.: Dynamic (component-weighted) effective sample size of the robust prior as a function of the number of responders in the current control arm of 30. The historical control is 30/100.

posterior of a_0 or τ_c. Neither detects conflict that the current data are too small to see. The FDA draft asks sponsors to explore prior-data conflict in the design simulations, including a scenario with no effect in the target population (U.S. Food and Drug Administration 2026g).

19.8. What borrowing costs: type I error and power under drift

Now the central calculation. Take a two-arm trial with a binary endpoint, with 60 patients on the new treatment and 30 on control (2:1 randomization, so that a smaller concurrent control arm is accepted in return for borrowing). The historical control arm has 100 patients and 30 responders. The decision rule is the one the FDA draft uses, $\Pr(\delta > 0 \mid \text{data}) > c$ with $\delta = p_1 - p_0$ and $c = 0.975$, with a Beta$(1, 1)$ prior on the treatment arm. The control arm gets one of four priors:

- *No borrowing*: Beta$(1, 1)$.
- *Full pooling*: Beta$(31, 71)$, the historical arm added as if it were current.
- *Power prior*, $a_0 = 0.3$: Beta$(10, 22)$.
- *Robust MAP-type mixture*: 0.8 Beta$(31, 71)$ + 0.2 Beta$(1, 1)$, with weights updated by the data.

We then ask what happens when the true control rate in the current trial is not the historical 30% but $0.30 + d$, for *drift* d between -0.15 and $+0.20$. For type I error the true treatment rate equals the true control rate. For power the true treatment effect is 0.20 above the control rate. The calculation is exact: the number of responders in each arm is binomial, the

decision is a function of the two counts, and the rejection probability is a double sum with no Monte Carlo error.

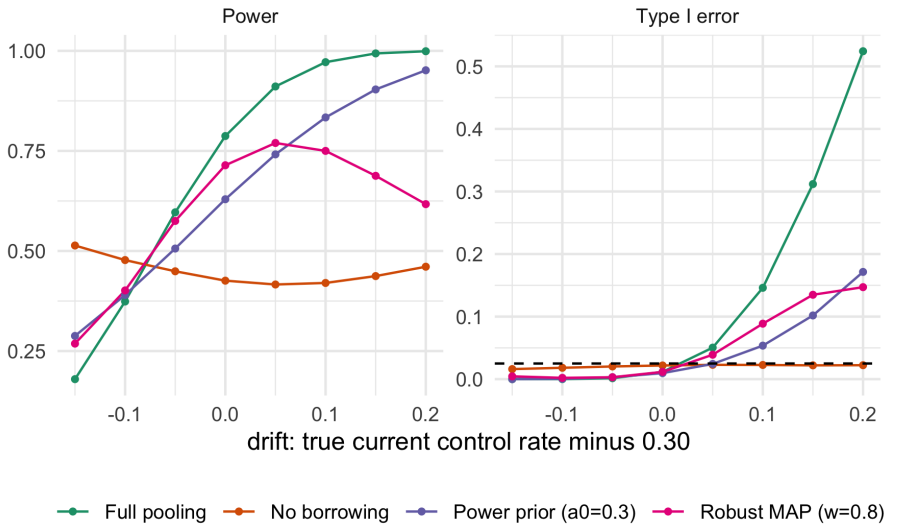


Figure 19.4.: Type I error (left) and power (right) of four ways of handling a historical control arm (30/100), as the true current control rate drifts from the historical 30%. Trial: 60 treated, 30 control, rule Pr(p1 > p0 | data) > 0.975. Dashed line: 0.025.

Figure 19.4 plots the whole range, and Table 19.3 reads the numbers that matter off the simulation above.

Table 19.3.: Selected operating characteristics from the simulation above (exact enumeration).

True control rate	No borrowing	Full pooling	Power prior	Robust MAP
0.25 (drift −0.05): power	0.449	0.597	0.506	0.575
0.30 (no drift): type I error	0.022	0.012	0.010	0.012
0.30 (no drift): power	0.426	0.787	0.629	0.714
0.40 (drift +0.10): type I error	0.023	0.146	0.054	0.089
0.40 (drift +0.10): power	0.420	0.972	0.834	0.750
0.50 (drift +0.20): type I error	0.022	0.524	0.171	0.147

Four features of this table are the content of the chapter.

Without drift, borrowing buys power. At the historical rate, full pooling raises power from 0.43 to 0.79 with the same 90 patients, and the type I error is below 0.025 (it is conservative because the decision threshold has not been recalibrated for the borrowed information). The saving is real: it would take a no-borrowing trial with roughly 170 control patients, rather than 30, to match this power (computed with the same exact enumeration and 60 treated patients). The power prior at $a_0 = 0.3$ and the robust prior sit in between.

With drift upward, borrowing inflates the type I error, and the inflation is large. If the true control rate is 40% instead of 30%, the Bayesian rule that was conservative at $d = 0$ rejects a true null 14.6% of the time under full pooling, 5.4% under the power prior and 8.9% under the robust prior. At a 20 point drift, full pooling rejects 52% of the time. Nothing is wrong with the arithmetic: the rule "is the treatment rate above the posterior control rate" is correctly computed, and the posterior control rate is anchored at the wrong value. The false positive arises because the historical control appears worse than the treatment, which is exactly the signature of a successful drug.

Without borrowing the type I error does not move. The first column is nearly flat, between 0.016 and 0.023 for every drift. This is the property the traditional design buys by discarding prior data: validity does not depend on any assumption about the past.

The same drift that inflates type I error gives spurious power. The columns for full pooling and the power prior at $d = +0.10$ show power of 0.97 and 0.83, but this is a mixture of a real effect and an artifact, and a sponsor cannot tell which from the trial. Conversely, at $d = -0.15$ (the current control rate is 15 points lower than history) full pooling loses power, falling from 0.79 at no drift to 0.18, against 0.51 without borrowing: the posterior control rate is pulled up towards the old 30%, which hides a real treatment effect. Drift in one direction costs type I error and drift in the other costs power. Neither is controlled by the trial.

The robust prior is not a cure. At a ten point drift it still has a type I error of 0.089, higher than the power prior's 0.054, and only at a twenty point drift (0.147 against 0.171) does it start to protect.

Viele and colleagues compared pooling, power priors and hierarchical borrowing in simulations of this kind and found, as here, that the gain from borrowing is lost or reversed when the current control rate drifts, and that dynamic methods protect only in part (Viele et al. 2014). Galwey argues that, unless a strong assumption is made about the degree of discounting, dynamic borrowing may give little or no gain over using the current data alone, and recommends comparing the consequences of a range of assumptions (Galwey 2017); for a control arm of 30 this agrees with the technical note above. When there is a single historical study, as in the example here, Röver and Friede treat the discount as shrinkage estimation in a Bayesian random-effects meta-analysis of two studies, and illustrate it with a trial and a clinical registry in Creutzfeldt-Jakob disease (Röver and Friede 2020).

19.8.1. What the regulator does with this

In the settings studied by Kopp-Schneider and colleagues, borrowing cannot improve power over an optimal current-data test while preserving strict conditional type I error control given the historical data (Kopp-Schneider, Calderazzo, and Wiesenfarth 2020). Recalibration can restore control, but may remove the gain. So the claim "type I error at most 0.025 over every drift" is available only for a recalibrated rule, and for the uncalibrated rules above inflation at some drift is unavoidable: a rule that is conservative when the historical rate is right gives up something when it is wrong. The FDA draft recognizes this. For designs with borrowing, "this default success criterion may not be applicable or appropriate", and the sponsor should agree an alternative with the Agency (U.S. Food and Drug Administration 2026g). The draft says that it does *not* recommend using type I error inflation as the *measure of how much has been borrowed*, because it regards that as philosophically inconsistent with

having chosen to borrow, and observes that dynamic methods already reduce borrowing near the null (U.S. Food and Drug Administration 2026g). What it asks for instead:

- the ESS of the prior and the ratio of prior ESS to the trial's size, over the range of plausible data;
- simulations of prior-data conflict, including a scenario with no effect in the target population (that is the type I error row of the table above, for a chosen set of drift values);
- sensitivity analyses that vary the strength of borrowing;
- justification that the source and target are relevant: same estimand, comparable standard of care and population, patient-level data where possible, and a pre-specified rule for including sources so that the sponsor cannot pick favorable studies.

The table above is the shape of the simulation report. The one number to add to it is the *range of drift that the sponsor believes is plausible and the type I error at the edge of that range*. A sponsor who can say "we think the control rate in 2027 lies between 0.25 and 0.35, and at 0.35 our type I error is 0.04" has stated what is being bought and what it costs, and the regulator can negotiate on that basis. A sponsor who reports only the no-drift column has not.

Two protective devices are also in use. A *cap* on borrowing limits the ESS from the historical source to a fraction of the current trial, such as 30% to 50% (an illustrative range, not a regulatory figure), so that the current data always dominate. A *recalibrated threshold* raises c until the type I error at no drift, or at a pre-agreed drift, is at most 0.025, which gives up part of the power gain. Both are choices to put to the Agency early, as FDA recommends for any informative prior.

In the protocol: borrowing from a historical control

State, before the historical data are examined by the sponsor's statisticians where possible: (1) the source trials and the pre-specified criteria that selected them; (2) the estimand in the historical and current trials and any differences in the definition of response or event; (3) the discounting method, the parameters (for example a_0, w, or the prior on τ_b) and the reason for each; (4) the prior ESS and its ratio to the planned control arm; (5) the simulation scenarios, with drift values stated as ranges, and the type I error and power at each; (6) a primary analysis with borrowing and a sensitivity analysis without it, and what happens if they disagree; (7) who sees the current control data when, so that the borrowing parameters are not tuned after the data are seen.

19.9. Traditional versus Bayesian: side by side

	Conventional standalone design	Bayesian borrowing
Role of prior data	Sets the design alternative, control rate and n; absent from the analysis	Enters the analysis through a prior on the control parameter

	Conventional standalone design	Bayesian borrowing
Control arm	Concurrent and randomized, full size	Concurrent, possibly smaller (for example 2:1), supplemented by external information
Validity claim	Type I error at most 0.025 at every null value, whatever the past	No general 0.025 guarantee for the uncalibrated rules: report error rates over the justified drift range, and calibrate over that range if a 0.025 guarantee is claimed (recalibration can remove the power gain)
Main risk	A larger or longer trial than the history suggests is necessary	Bias from a mismatched source, undetectable from a small control arm
Report	p-value, confidence interval	Posterior probability, prior ESS, simulations over conflict
What the regulator asks	The pre-specified analysis, multiplicity	Source relevance, estimand match, pre-specification of discounting, conflict simulations, sensitivity analysis
Where it is the better choice	A common disease with a feasible randomized trial, where external controls are not needed	A rare disease or pediatric population where a full concurrent control is infeasible, with strong evidence that the source applies

Where the two agree: with $a_0 = 0$, or a vague prior, the Bayesian design is the traditional one in a different notation and (up to the tail calculation) gives the same decision. In a trial large enough that the borrowed ESS is small relative to the trial, the two also agree in practice. Where traditional is the better choice: when a randomized comparison of full size is feasible and the question is the effect of the drug now, the borrowing buys little and risks something.

19.10. Pediatric extrapolation and ICH E11A

Pediatric trials are the setting where borrowing is most often justified, because enrollment is small and withholding treatment from children is difficult ethically. The ICH E11A guideline on pediatric extrapolation, adopted at Step 4 in August 2024, sets out a general approach (International Council for Harmonisation 2024). It asks for a *pediatric extrapolation plan* with three parts: an assessment of how similar the disease, the drug's pharmacology and the response to treatment are in the source (adult) and target (pediatric) populations,

using whatever is known about the mechanism; a description of the gaps and uncertainties; and a plan for the data to be generated in children, which can range from PK and safety only (full extrapolation of efficacy) to a reduced efficacy trial to a full one. The more similar the populations, the less pediatric data are needed. Bayesian borrowing is one of the quantitative tools that the guideline allows for the middle case, in which adult data are used to inform the pediatric efficacy analysis, and the guideline emphasizes that the same evidence-based judgment of similarity should come first and the statistics second. The intravenous belimumab program in pediatric lupus, analyzed with a robust mixture prior and a tipping-point analysis over the borrowing weight, is described in Section 19.17.2. The FDA framework for extrapolating adult data to children is described by Dunne et al. (2011), and Bayesian modeling for it by Gamalo-Siebers et al. (2017). For rare diseases generally, FDA's final guidance of December 2023 says that Bayesian methods may inform pediatric investigations through incorporation of adult data (U.S. Food and Drug Administration 2023c).

The FDA Bayesian draft gives pediatric extrapolation as a core use of an informative prior and names empagliflozin and linagliptin in pediatric type 2 diabetes as examples where Bayesian analyses were supportive (U.S. Food and Drug Administration 2026g). It draws two practical points. First, undiscounted adult data would often overwhelm a small pediatric sample, so a prior centered on the adult effect with an inflated variance, representing the possibility that children respond differently, is often more reasonable than pooling. Second, the sources can be related in several ways: equal relevance (adolescents enrolled with adults), exchangeability (a hierarchical model across age strata after adjusting for known effect modifiers), a bias parameter (the pediatric effect equals the adult effect plus an unknown shift), and functional dependence (a pharmacometric model predicting the pediatric effect from adult data and exposure). The bias-parameter form is the one that fits the discount most naturally: if the pediatric effect is $\theta_p = \theta_a + b$ with $b \sim N(0, \omega^2)$, then the adult data enter the pediatric prior with variance $\sigma_a^2 + \omega^2$, and the sponsor's judgment of pediatric similarity is the number ω. With a large ω the adult data contribute little. With $\omega = 0$ they are pooled. The protocol, in effect, writes down ω and defends it with the pharmacology.

The design consequence is the same as in the simulation above. Drift in this setting is "the pediatric effect is not the adult effect", and its plausible range is a clinical judgment about physiology, not a statistical estimate. The simulation of conflict should therefore cover scenarios in which children respond less well than adults, including no effect, and say what the pediatric trial would conclude.

A numerical example shows the bias-parameter form with a mixture prior. Let θ_A be the adult log odds ratio, estimated as -0.5 with standard error 0.08, and let θ_P be the pediatric effect. A pediatric trial of modest size gives $\hat{\theta}_P = -0.35$ with standard error 0.30. The prior is

$$\theta_P \sim w\, N(\hat{\theta}_A,\, 0.08^2 + \omega^2) + (1 - w)\, N(0, 1), \tag{19.8}$$

where ω is the allowed adult-pediatric difference and w the prior probability that extrapolation is valid.

With $w = 0.7$ and $\omega = 0.15$, the posterior weight on the informative component rises to 0.87, because the pediatric estimate is consistent with the adult one, and the posterior probability of benefit, $\Pr(\theta_P < 0 \mid \text{data})$, is 0.98. The pediatric trial alone gives 0.88. The mixture is a weak safeguard in a trial this small, and the example shows it: if the pediatric estimate had been -0.05 instead, the weight on the informative component would fall only to 0.75, and the probability of benefit would still be 0.89 against 0.57 for the trial alone. With a standard error of 0.30 the pediatric data cannot tell the two components apart, so the answer

is mostly the prior, which is why the simulation of conflict above must include scenarios in which children respond less well. The prior weight w is the honest summary of the clinical judgment about similarity in the E11A extrapolation plan, and writing it down is what lets a review committee debate it.

19.11. Noninferiority with borrowed information

A noninferiority trial is a borrowing problem from the start. Its margin comes from the control's effect in earlier placebo-controlled trials, and its validity rests on the constancy assumption that this effect still holds (Section 4.9; FDA 2016 (U.S. Food and Drug Administration 2016); ICH E10 (International Council for Harmonisation 2000)). The traditional analysis uses the historical data once to fix a margin and then treats the margin as known. The Bayesian analysis can state the same thing as a model.

Take a difference in event rates $\theta = p_N - p_C$ (new minus control, events bad), noninferior if $\theta < M$. Suppose a trial estimates $\widehat{\theta} = 0.01$ with standard error 0.015 and the margin is $M = 0.04$. The upper limit of the two-sided 95% interval is $0.01 + 1.96 \times 0.015 = 0.0394$, just under the margin, so the traditional test succeeds. Table 19.5 gives the posterior probability of noninferiority under three priors.

Table 19.5.: Posterior probability of noninferiority, estimate 0.01, standard error 0.015, margin M = 0.04. Skeptical prior: N(M, 0.02^2). Enthusiastic prior: N(0, 0.01^2). Flat: improper uniform.

Prior	Posterior mean	Pr(theta < M)
Flat	0.010	0.977
Skeptical, N(M, 0.02^2)	0.021	0.945
Enthusiastic, N(0, 0.01^2)	0.003	>0.999

Under the flat prior the posterior probability is one minus the one-sided p-value, 0.977, as it must be (Section 4.10). The skeptical prior, centered at the margin, is the natural regulatory prior for noninferiority: it represents a reviewer who thinks the new drug is as likely to be unacceptably worse as acceptably close. Under it the probability is 0.945, below the 0.975 that the traditional test implicitly demands. The prior is doing visible work, and the reviewer and the sponsor can argue about its standard deviation instead of about a p-value.

19.11.1. Margin synthesis

Regulatory practice preserves a fraction f of the control's effect against placebo, δ_{CP}, estimated from the historical trials as $N(d, s_{CP}^2)$. The fixed-margin approach takes the lower 95% limit of δ_{CP}, multiplies by $1 - f$ (the loss of effect that is allowed), and uses the result as the margin M_2. The synthesis alternative asks directly for the posterior probability that the new drug retains at least the fraction f,

$$\Pr(\theta_{NC} < (1 - f)\,\delta_{CP} \mid \text{data}), \tag{19.9}$$

with θ_{NC} the new-minus-control difference (events bad). The network meta-analytic-predictive approach of Schmidli, Wandel and Neuenschwander is the hierarchical-model

counterpart: it combines the historical trials with data on the active control and on placebo, together with the noninferiority trial, in a network meta-analysis, and predicts the effect of the test treatment against placebo in the noninferiority trial while carrying the between-trial variability of all parameters (Schmidli, Wandel, and Neuenschwander 2013). Prior-data conflict can be handled by a robust mixture, as in Section 19.6. The synthesis method is usually credited to Rothmann et al. (2003), with an earlier method for comparison to placebo in active-control trials in Hasselblad and Kong (2001), its preservation-of-effect interpretation and error properties are in Snapinn and Jiang (2008), a Bayesian formulation is in Simon (1999), and the constancy assumption is discussed by Fleming (2008). The fixed-margin approach reduces the historical estimate to a single conservative number, the lower confidence limit, and so builds in a deliberate conservatism; the synthesis carries the full uncertainty of the historical estimate into the comparison. Both rest on the constancy assumption, and the synthesis makes it visible: if care has improved since the historical trials and the control's effect has shrunk, discounting by a power prior with $a_0 < 1$ would only widen the distribution of δ_{CP} and leave its center where it was. A shrunken effect has to be modeled directly, for example by shifting or scaling the center of δ_{CP} by an attenuation or bias parameter with its own prior, and the retained fraction should be assessed over a range of plausible downward shifts.

An illustration, with hypothetical numbers: let the historical control effect be a 10-point event reduction with standard error 0.02, and let the sponsor aim to preserve half of it. The fixed margin is half the lower 95% limit, 0.03, tighter than the 0.04 above, and the trial with $\widehat{\theta} = 0.01$ and standard error 0.015 (upper limit 0.039) does not meet it. The synthesis probability of retaining at least half of the effect is 0.987. The two answers differ because the fixed margin spends all of the historical uncertainty at its worst limit and then asks the current trial to clear that limit at 0.025, which is two conservative operations stacked. The synthesis is not automatically more lenient: for a trial with $\widehat{\theta} = 0.04$ the probability of retaining half the effect is only 0.71. Whether a regulator will accept a synthesis rule in place of a fixed margin, and with what threshold, is a question for the protocol and for early discussion with the agency; the 2016 guidance describes both approaches.

19.12. Indirect comparison through a common comparator: the tirzepatide chain

The question a guideline committee asks of SURPASS-CVOT is how much tirzepatide lowers cardiovascular risk compared with no incretin therapy at all. No trial answers it directly: SURPASS-CVOT compared tirzepatide with dulaglutide, and REWIND compared dulaglutide with placebo (Nicholls et al. 2025; Gerstein et al. 2019). The answer has to be assembled, and on the log scale the effects add,

$$\log \mathrm{HR}_{T:P} = \log \mathrm{HR}_{T:D} + \log \mathrm{HR}_{D:P}, \tag{19.10}$$

with variances adding if the two estimates are independent. SURPASS-CVOT gave a hazard ratio of 0.92 (95.3% CI 0.83 to 1.01) and REWIND 0.88 (95% CI 0.79 to 0.99).

Table 19.6.: Tirzepatide versus placebo, chained through dulaglutide: SURPASS-CVOT (HR 0.92) times REWIND (HR 0.88), with the REWIND variance divided by the power-prior weight a0. Standard errors from the published intervals: 0.0496 and 0.0577.

a0	HR	95% interval	Pr(HR < 1)
1.00	0.81	0.70 to 0.94	0.997
0.50	0.81	0.67 to 0.98	0.987
0.25	0.81	0.63 to 1.04	0.954
0.10	0.81	0.56 to 1.17	0.868

At $a_0 = 1$ the chain gives a hazard ratio of 0.81 with interval 0.70 to 0.94. The calculation assumes the constancy assumption again: the dulaglutide effect in REWIND is the dulaglutide effect in the SURPASS-CVOT population. The populations differ. REWIND enrolled patients aged 50 or older, with and without established cardiovascular disease, in 2011 to 2013, whereas SURPASS-CVOT enrolled only patients with established disease, a decade later and on different background therapy. A power prior discounts the REWIND evidence: dividing its variance by a_0 widens the interval and leaves the estimate where it was. At $a_0 = 0.5$ the interval is 0.67 to 0.98, and at $a_0 = 0.25$ it is 0.63 to 1.04, so the conclusion of a benefit against placebo survives a halving of the weight on REWIND but not a quartering. Reporting the table lets a reader pick the weight, and shows where the conclusion changes.

As in Section 19.4, the discount widens the interval but does not move the center. If the REWIND effect in the SURPASS-CVOT population differs from 0.88, the discount is the wrong correction. The published indirect comparison of Sattar and colleagues addresses exactly this (Sattar et al. 2026). It was prespecified, and it used propensity-score adjustment to select and weight 2055 of the 9901 REWIND participants who resemble the SURPASS-CVOT population, then multiplied the SURPASS-CVOT hazard ratio by the REWIND hazard ratio in that subset. Its primary result is a hazard ratio of 0.72 (95% CI 0.55 to 0.94) for the primary composite against an imputed placebo. That is lower than the simple chain's 0.81, which is consistent with a larger dulaglutide effect in the REWIND subset with established disease than the overall 0.88 (the paper itself reports a hazard ratio of 0.78, 95% CI 0.61 to 1.01, for the selected REWIND cohort, which is a different quantity from the propensity-weighted indirect analysis). The two answers rest on different assumptions about exchangeability, and the Bayesian formulation lets both be reported: the unadjusted chain with its discount table, and the adjusted analysis with its own uncertainty. Neither is a randomized comparison with placebo. For a regulator, this is the evidence that supports a label discussion or a guideline recommendation, not a primary efficacy claim.

19.13. Borrowing across subgroups: basket trials

A basket trial tests one therapy in several tumor types or biomarker-defined subgroups that share a molecular target. Each basket is small, and the question is which baskets respond. The traditional design runs a separate single-arm test in each basket, with a multiplicity adjustment across them (Chapter 10). The Bayesian design puts the baskets in a hierarchical model, Equation 19.1, so that responses in one basket inform the others. Thall and colleagues (2003) proposed the hierarchical model for phase II trials in diseases with multiple subtypes, and Berry and colleagues (2013) applied it to oncology subpopulations, reporting designs that

need fewer patients per subgroup than independent analyses at the same type I error under their scenarios (P. F. Thall et al. 2003; Scott M. Berry et al. 2013). In Polson and Sokolov (2026a) the quoted figure is 4 to 7 patients per group fewer than independent analyses. It comes from the scenarios that Berry and colleagues chose themselves, so it shows what the method can achieve and not what it will achieve in a given trial.

The next chunk shows the mechanism with ten baskets of 30 patients each. In the first scenario nine baskets have 12 responders (40%) and one has 3 (10%). In the second all ten have 12.

In the all-equal scenario the model estimates a small τ_b and every basket's estimate stays at its observed 40%. The estimate is close to the observed rate because nothing disagrees, and the posterior interval for each basket is narrower than it would be alone. In the outlier scenario, the observed heterogeneity pushes τ_b up, so that the nine typical baskets are *pulled down* from 40% to about 38% by the outlier and the outlier is pulled up from 10% to about 29%. The outlier is moved a lot because it has only 30 patients and the model, informed by nine similar baskets, regards a rate of 10% as probably a low draw. If the outlier is in fact a non-responding tumor type, the model has overstated its response rate by almost three-fold, which is the "estimated worse" warning of the earlier section. The treatment effect in a basket that really does not respond is overestimated, and a decision rule such as $\Pr(p_k > 0.2 \mid y) > c$ will, for some true rates, reject more often than it would with independent analyses. The inflation is again the price, and the same remedy applies: calibrate the threshold by simulation, over scenarios in which some baskets respond and others do not, and report the false positive rate for the non-responding baskets in each. FDA's draft guidance lists basket trials under borrowing across similar diseases or subtypes, and recommends that the sponsor justify exchangeability and to model known effect modifiers, such as biomarker status, explicitly (U.S. Food and Drug Administration 2026g). Chapter 20 returns to the master-protocol setting, and Chapter 10 treats the shrinkage idea for endpoints.

Several variants are worth knowing. The calibrated Bayesian hierarchical model (CBHM) of Chu and Yuan (2018a) sets the amount of shrinkage as a function of the heterogeneity seen in the data, so that baskets that agree borrow more. The Bayesian latent subgroup design (BLAST) of Chu and Yuan (2018b) assigns each cancer type to a latent sensitive or insensitive subgroup and jointly models the binary response and a longitudinal biomarker, so that a non-responding tumor type is not averaged with responding ones. The exchangeability-nonexchangeability model (EXNEX) of Neuenschwander et al. (2016) gives each basket a prior probability of belonging to the common group, much as the robust prior does for a single historical control. Other designs monitor exchangeability as data accrue (Hobbs and Landin 2018) or estimate which baskets are exchangeable (Kaizer, Koopmeiners, and Hobbs 2018), and Simon et al. (2016) give a Bayesian basket design. All fit the same template: the amount of borrowing is a parameter, and the operating characteristics under heterogeneity, above all the false positive rate when some baskets do not respond, have to be simulated.

19.14. When borrowing failed: the NSCLC case study

A sponsor deciding whether a fixed-weight power prior is enough can look at a case study that tested it. Struebing and colleagues (2024) built a matched real-world external control arm in first-line non-small cell lung cancer, augmented it with historical-trial data through a fixed-weight power prior, and compared the result with the survival result of the randomized

trial whose control arm it replaced (Struebing et al. 2024). The result was a negative one: the augmented external control did not reproduce the randomized result.

The lesson is the one in the simulation, from a real dataset. A power prior with a fixed discount lowers the weight of the historical data but does not change their center. Discounting does not remove differences between the sources, and if real-world patients differ from trial patients in prognostic factors that matching did not capture, the augmented control stays biased. The study is one case and should not be generalized beyond what it shows, but it is the right example to put in front of a team that proposes to replace a concurrent control with a matched external one: matching on measured covariates removes only the measured bias.

19.15. Prognostic covariate adjustment and digital twins

There is a way of using external data that does not borrow outcomes of the control arm and therefore carries no bias risk of this kind. *Prognostic covariate adjustment* (PROCOVA) builds a prognostic model of the control-arm outcome from external data, such as historical trial data, and then uses each trial participant's predicted outcome (their "digital twin" prediction under control) as a baseline covariate in the analysis of the randomized trial. The treatment effect is still estimated by comparing the randomized arms, so the analysis remains valid, and the covariate absorbs variation in the outcome, so the variance falls. The EMA's Committee for Medicinal Products for Human Use adopted a qualification opinion for the method in September 2022, for phase 2 and 3 trials with continuous outcomes (European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022). The key regulatory feature, as with the covariate adjustment guidance of FDA, is that the model is fixed before the trial starts: with an independently developed, pre-specified score and appropriate analysis and variance estimation, PROCOVA can preserve asymptotic type I error control despite poor prediction, whereas small trials need suitable finite-sample inference (U.S. Food and Drug Administration 2023d; European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP) 2022). Precision gains depend on prognostic performance. What the sponsor gains is variance reduction, which can be converted into a smaller sample size, up to the limit the qualification opinion sets; no assumption about the stability of the control arm over time is made. The contrast to take away is that PROCOVA uses external data to sharpen a randomized comparison, whereas borrowing uses them to replace part of it (Ross et al. 2024).

This is why the method is often the first one to propose to a regulator who is wary of external controls. It gives up less (the randomized control stays) and so buys less than a full borrowing design, but what it buys carries no assumption about drift. It is a special case of the covariate adjustment of Section 5.10, with a single prognostic score as the covariate, and the guidance of FDA and EMA on prespecification and the estimand applies to it.

19.16. Putting it together: a decision guide

The chapter's methods fall on a line from no use of external information to full reliance on it. A practical ordering for a team choosing among them:

1. **Is a concurrent randomized control feasible?** If yes, and the question is the effect of the drug in the current population, use it at full size, or use PROCOVA-type adjustment to raise precision without borrowing. This is the traditional design and it is hard to beat.
2. **If the concurrent control must be small, is there a randomized source for the same control treatment in the same population?** If so, a pre-specified dynamic borrowing design (robust MAP or commensurate) is defensible, as in REBYOTA (Section 19.17.1). Report the ESS, the conflict simulations and a sensitivity analysis without borrowing.
3. **If the only sources are non-randomized (registries, real-world data, historical single-arm series)?** Follow the FDA external-controls draft: pre-specify, demonstrate comparability, quantify what is borrowed, and treat the result as supportive unless the effect is large and the natural history is predictable (ICH E10). Expect the NSCLC lesson to apply.
4. **Pediatric?** Begin with the E11A extrapolation plan. Write the similarity judgment as a number (ω, a_0, w), defend it, and simulate the case in which it is wrong.
5. **Subgroups of a disease or baskets?** A hierarchical model is natural. Justify exchangeability across baskets, include known effect modifiers, and simulate scenarios with some baskets not responding.

19.17. In practice

Three regulatory cases show what borrowing buys and what it risks. Two more appear earlier in the chapter: the NSCLC external-control study (Section 19.14), where a matched real-world control with a fixed-weight power prior failed to reproduce a randomized result, and the EMA qualification of PROCOVA (Section 19.15), which uses external data to sharpen a randomized comparison without borrowing outcomes.

19.17.1. REBYOTA: dynamic borrowing from a phase 2b study

REBYOTA (RBX2660, fecal microbiota, live-jslm) prevents recurrence of *Clostridioides difficile* infection after antibiotics, and was approved on 30 November 2022. The sponsor first planned two phase 3 trials of about 300 patients each. After FDA's July 2013 enforcement discretion on fecal microbiota transplantation made enrollment difficult, it proposed a single phase 3 trial (study 2017-01, PUNCH CD3) plus a Bayesian integration of the earlier phase 2b study 2014-01. FDA wrote that the studies "are generally exchangeable" but not identical, and accepted a Bayesian hierarchical model with dynamic borrowing, in which the extent of borrowing depends on how similar the effect is in both the active and placebo arms (U.S. Food and Drug Administration, Center for Biologics Evaluation and Research 2022). The January 2026 FDA draft guidance cites this as its example of borrowing from a previous trial of the same drug (U.S. Food and Drug Administration 2026g).

The success criteria are the interesting part. The sponsor set two thresholds on the posterior probability of superiority as analogues of a one-sided type I error of 0.025 (one adequate and well-controlled trial) and one-sided 0.00125 (a substitute for two trials), each without borrowing. After adjusting for the information fraction actually reached, they were 0.9750338 and 0.9993275. The modified intention-to-treat result was an estimated success rate of 0.71 on RBX2660 against 0.57 on placebo (difference 0.13, computed from unrounded estimates, 95%

credible interval 0.02 to 0.24), with posterior probability of superiority 0.991. That met the one-trial threshold and missed the two-trial threshold (U.S. Food and Drug Administration, Center for Biologics Evaluation and Research 2022).

The lesson is about data alignment, not about the model. During the review FDA requested a refined analysis that aligned the success definition, the assessment period (9 weeks in 2014-01 against 8 weeks in 2017-01) and the analysis populations of the old study to those of the new one, so as to improve exchangeability. The sponsor's primary analysis had also used non-final 2014-01 data, from which six randomized but undosed subjects had been wrongly excluded. Both the specified analysis and the refined one reached the same conclusion. The briefing document gives no borrowing weight or ESS, so the amount borrowed cannot be read from it. For a sponsor, the checklist of Section 19.16 applies: same product, randomized source, pre-specified model, and alignment of definitions before the data are analyzed.

19.17.2. Belimumab in pediatric lupus: a robust mixture and a tipping point

The pediatric trial of intravenous belimumab in systemic lupus erythematosus (C1109, PLUTO) randomized 93 children (92 in the analysis below). At week 52 the SRI-4 response was 28 of 53 (52.8%) on belimumab and 17 of 39 (43.6%) on placebo, an observed difference of 9.2 points with an odds ratio of 1.5 (95% CI 0.6 to 3.5) (Brunner et al. 2020; U.S. Food and Drug Administration, Center for Drug Evaluation and Research 2026). That interval is far too wide to stand alone. FDA recommended a post hoc Bayesian analysis that borrowed from two adult trials (C1056 and C1057) with the same SRI-4 endpoint and the same 10 mg/kg dose (U.S. Food and Drug Administration, Center for Drug Evaluation and Research 2026).

The prior was a robust mixture of the form of Equation 19.7: an informative component built from the adult placebo and 10 mg/kg arms, and a minimally informative component, with borrowing weight w. Instead of fixing one value of w, the analysis was a tipping-point analysis. The posterior probability of efficacy reached at least 97.5% whenever w was 0.55 or more, and the review division judged such a weight reasonable given a similar disease course, similar pharmacokinetics and similar baseline characteristics (U.S. Food and Drug Administration, Center for Drug Evaluation and Research 2026). The case study stresses that the analysis was post hoc and that future programs should pre-specify it, with the discussion starting before the IND.

The tipping-point analysis is the practical answer to the question "what weight should we use?". It reports the smallest amount of borrowing that supports the conclusion and leaves the judgment of its plausibility to the clinical and pharmacological evidence, which is the role of ω in the pediatric section above.

19.17.3. OPTIMIZER Smart: what a fixed 30% weight buys

The OPTIMIZER Smart System (cardiac contractility modulation, Impulse Dynamics, PMA P180036) was reviewed by an FDA device panel on 4 December 2018 and approved on 21 March 2019 for heart failure patients in NYHA class III with ejection fraction 25 to 45%. The pivotal trial FIX-HF-5C randomized 160 patients 1:1 and was analyzed with a Bayesian model that included "30% fixed borrowing" of data from the 229 patients with ejection fraction above 25% in the earlier FIX-HF-5 study. The primary endpoint was the change in

peak VO2 at 24 weeks, and success required a posterior probability above 0.975 (U.S. Food and Drug Administration, Center for Devices and Radiological Health 2019).

Table 13 of the summary of safety and effectiveness gives the result under four ways of handling missing data. With borrowing the posterior probability was between 0.977 and 0.989 in all four, so the success criterion was met. FIX-HF-5C alone gave 0.957 and 0.960 under two of the imputation models, 0.916 for the complete-case analysis and 0.911 with last observation carried forward, below 0.975 in each (U.S. Food and Drug Administration, Center for Devices and Radiological Health 2019). The success of the device therefore depended on the borrowed 30%, and this is the clearest illustration in the book of both sides of borrowing. The borrowed information was worth a decision, which is the gain of the simulation above. It also carries the risk of the same simulation: if the earlier study overstated the effect, the conclusion rests on that overstatement. The FDA device guidance expects the sponsor to show simulated operating characteristics under such discrepancies (U.S. Food and Drug Administration 2010), and the fixed weight is the device analogue of the power prior with $a_0 = 0.3$ used in our example.

Key points

- A conventional standalone randomized-trial analysis uses prior data to choose the alternative, the control rate and the sample size, and then ignores them in the analysis. This is why its type I error does not depend on the past.
- External and historical controls are regarded with caution by ICH E10 and by FDA, MHRA and EMA because of selection, time trends and measurement differences. Pocock's 1976 criteria are still the checklist. FDA's externally controlled trials draft (February 2023), the MHRA draft and the EMA concept paper (2025, adopted May 2026) ask for pre-specification and fitness for purpose.
- Bayesian methods choose how much to borrow: fixed discount (power prior), estimated heterogeneity (hierarchical model, MAP prior), estimated agreement (commensurate, normalized power prior), or a mixture with a vague component (robust MAP). Static methods fix the borrowing in advance; dynamic ones let the data adjust it.
- Effective sample size is the summary of how much of the answer comes from outside the trial. A MAP prior built from 400 historical patients can be worth about 57 patients once between-trial variation is counted.
- Borrowing buys power when the historical rate is right (0.43 to 0.79 in the example) and inflates type I error when it is wrong (0.023 without borrowing, 0.146 with full pooling, 0.054 with the power prior and 0.089 with the robust prior at a ten point drift). A control arm of 30 cannot detect a ten point drift well enough for a dynamic method to protect against it.
- FDA's January 2026 draft asks for the prior ESS, simulations of prior-data conflict (including no effect), and sensitivity analyses. It does not recommend using type I error inflation to measure the amount of borrowing.
- REBYOTA is the template of a good case: a randomized phase 2b source for the same product, a pre-specified hierarchical model with dynamic borrowing, a posterior of 0.991 that met the one-trial threshold (0.975) but not the two-trial one (0.99933), and a data-alignment fix from FDA. In OPTIMIZER the success criterion (0.975) was met with a fixed 30% borrowing (0.977 to 0.989) and missed without it (0.911 to 0.960). Belimumab in children was supported by a robust mixture prior with a tipping point

at borrowing weight 0.55. ICH E11A (August 2024) puts the similarity judgment first in pediatric extrapolation, and Bayesian borrowing second.

- Basket trials use hierarchical models to borrow across tumor types, at the price of over-estimating the response in baskets that do not respond. In an NSCLC case study, a matched real-world control augmented with a fixed-weight power prior did not reproduce the randomized result.
- PROCOVA uses external data without borrowing outcomes, so the randomized comparison and its type I error are unchanged, and the gain is variance reduction.

Discussion questions

1. A sponsor proposes a 2:1 trial with a control arm of 30, borrowing from a historical control of 100 patients with a robust prior. What drift would you want the simulation to cover, who should decide the range, and what type I error at the edge would you accept for a serious but non-life-threatening condition? For a condition with no approved therapy?
2. The FDA draft recommends against using type I error inflation as the measure of how much a design borrows, yet asks for conflict simulations that include a no-effect scenario. How are these two positions consistent? Where would you draw the line between a legitimate price of borrowing and an unacceptable one?
3. A MAP prior from four trials has an ESS of 57 while the pooled analysis would give 400. Which of those two numbers would you put in the protocol, and what does the difference tell a clinician about how much a new control arm should be expected to differ from the old ones?
4. In the pediatric setting, the similarity judgment ω is a clinical, not a statistical, number. Who should supply it, when should it be fixed, and what should the sponsor do if the pediatric interim data suggest it was too optimistic?
5. In a basket trial, nine tumor types respond and one does not. The hierarchical model reports the non-responding type at 29% against an observed 10%. What decision rule and what calibration would you propose so that a regulator would not label the basket "responding" on the strength of the other nine?

20. Master Protocols and Platform Trials

A conventional program builds a trial, answers one question, and takes the trial down. The site contracts, the database, the randomization system, the monitoring board and the regulatory file are all rebuilt for the next drug. A master protocol keeps that machinery standing and lets questions pass through it. The idea is not new, and it is not Bayesian. The UK prostate cancer platform STAMPEDE opened in 2005 with frequentist stopping rules and is still the clearest demonstration that a shared control arm and a standing infrastructure pay for themselves. What the Bayesian designs added is a different toolkit for the arms that nobody had planned yet: a decision rule per arm, calibrated by simulation, and hierarchical models that share information. Both frameworks can accommodate new arms and share information. They differ in their inferential machinery and decision rules, and both need a justified hypothesis family and demonstrated operating characteristics.

This chapter follows the order the rest of the book uses. It starts with the traditional answer to "we have five drugs to test", which is five separate two-arm trials, and measures what is wasted. It then describes the frequentist multi-arm multi-stage platform (STAMPEDE), the shared-control arithmetic that all platforms inherit, the problem of non-concurrent controls and calendar drift, and the multiplicity rules that differ between FDA and EMA. Only then does it turn to the Bayesian platforms (I-SPY 2, GBM AGILE, REMAP-CAP and the COVID anticoagulation multiplatform), to governance, and to the one famous platform that never enrolled an arm.

! Regulatory anchor

All documents below are described in Chapter 3. Status as of November 2026.

- **FDA, Master Protocols for Drug and Biological Product Development, revised draft guidance (June 2026)** (U.S. Food and Drug Administration 2026f). Revises the draft of December 2023. Defines master protocol, basket, umbrella and platform; recommends concurrent controls, a larger control allocation than any single arm, and generally no multiplicity adjustment across unrelated drugs; asks for amendments that substantively affect safety, quality or scope to be submitted at least 30 days before they start. A draft, so it recommends and does not bind.
- FDA, Master Protocols: Efficient Clinical Trial Design Strategies to Expedite Development of Oncology Drugs and Biologics, final guidance (March 2022) (U.S. Food and Drug Administration 2022a). The final text for cancer drugs and biologics (docket FDA-2018-D-3292), and FDA's final oncology-specific master protocol guidance: the 2026 revision above is a draft. For master protocols evaluating multiple drugs in one disease, FDA generally recommends comparisons with a common standard-of-care control. Experimental-arm comparisons may be proposed prospectively when the IND sponsor has the necessary right of reference. When changed standard of care makes the previous control unethical, FDA recommends suspending enrollment until the protocol and informed consent

incorporate the new control, and updating the SAP before analysis. For adaptive and Bayesian master protocols it asks that the statistical analysis plan give the details of the method and the futility analyses. It recommends and does not bind.

- **EMA, reflection paper on platform trials (draft, October 2026)** (European Medicines Agency 2026a). States a strong preference for concurrent controls, treats non-concurrent controls as generally unacceptable for the primary analysis of the primary endpoint, discusses control allocation (Dunnett's square root rule) and the error rate across arms, and, for confirmatory platforms, does not support response-adaptive randomization (Section 3.2.1). A draft.
- **EC, EMA and HMA, Complex clinical trials: Questions and answers (2022)** (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). The operative EU document: definitions, the sub-protocol structure, substantial modifications when arms are added, and the roles of sponsor and investigators.
- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products, draft (January 2026)** (U.S. Food and Drug Administration 2026g). Names the Bayesian "time machine" for non-concurrent controls in platforms (it cites GBM AGILE and Precision Promise) and asks that borrowing across substudies justify exchangeability and include known effect modifiers.
- **ICH E20, Adaptive Designs for Clinical Trials, Step 2 draft (June 2025)** (International Council for Harmonisation 2025a). Pre-specification and simulation-based operating characteristics for adaptations such as dropping arms. See Chapter 14.
- **FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics, final (November 2019)** (U.S. Food and Drug Administration 2019a). The governing text for arm dropping, sample-size re-estimation and the control of type I error in adaptive designs.

20.1. Separate trials, and what they cost

Suppose a sponsor, or a network, has k experimental treatments for the same disease. The textbook plan runs k two-arm trials. Each has its own protocol and its own control group of m patients, so the program randomizes $2km$ patients, half of them to a control arm that answers one question each. The plan is correct, easy to audit, and it is what regulators have approved drugs on for decades. Each trial has its own type I error rate and its own power, and nothing in one trial can contaminate another.

Its cost is the control arm. A trial of five drugs at 80 patients per arm randomizes 800 patients, 400 of them to the same standard of care measured five times. In the separate-trials plan, 400 participants receive standard of care, measured five times, so that five separate questions could each have an isolated comparison. The sponsor pays for it in time as well, since each trial must be started, sited and closed separately.

A platform answers the same k questions with one control group of size m. Each drug-versus-control comparison has the same precision as before, because its variance depends only on the two arms it uses, and the program enrolls $(k+1)m$ patients rather than $2km$. The saving is

$$1 - \frac{k+1}{2k} = \frac{k-1}{2k}, \tag{20.1}$$

which is 25% for two drugs, 37.5% for four and 40% for five, approaching one half as k grows (Polson and Sokolov 2026b). Saville and Berry simulated whole development programs and found the same direction on a larger scale: platform designs found beneficial treatments with fewer patients, fewer patient failures, less time and a greater probability of success than a traditional two-arm strategy (Benjamin R. Saville and Berry 2016). These are simulations under stated assumptions, not trial results. The saving falls on control assignments. Whether that reduces exposure to an inferior treatment depends on the experimental treatments' actual effects.

```
tab <- platform_shared_control_table(2:6, m = 80)
knitr::kable(tab, digits = 3, align = "rrrr",
             col.names = c("Drugs tested", "Patients, separate trials",
              ↪  "Patients, shared control", "Fraction saved"))
```

Table 20.1.: Patients needed by k separate two-arm trials against one platform with a shared control arm, with m = 80 patients per arm, and the fraction saved.

Drugs tested	Patients, separate trials	Patients, shared control	Fraction saved
2	320	240	0.250
3	480	320	0.333
4	640	400	0.375
5	800	480	0.400
6	960	560	0.417

The five-drug row of Table 20.1 is the example of module 10 of the companion lab: 800 patients become 480. Two things in that arithmetic need care before it is believed. First, it assumes all k arms run at the same time and that each comparison is run on the same data as in a separate trial. Real platforms add arms later, which creates the control problem discussed below. Second, it treats the k questions as separate. Whether the k results then need a joint error rate is a regulatory question as much as a statistical one, and it is treated in the multiplicity section.

20.2. Definitions: master protocol, basket, umbrella, platform

The terms and the taxonomy were set out by Woodcock and LaVange (2017), and the platform trial was presented to clinicians in Scott M. Berry, Connor, and Lewis (2015). FDA's June 2026 draft defines the terms as follows. A master protocol is "a protocol designed with multiple substudies, which may have different objectives and involve coordinated efforts to evaluate one or more drugs" (Section I, printed pages 1 to 2). The three kinds it names are:

- **Umbrella.** "a trial designed to evaluate multiple drugs concurrently for a single disease or condition."
- **Basket.** "a trial designed to evaluate a drug for multiple diseases, conditions, or disease subtypes."
- **Platform.** "evaluate multiple drugs for one or more diseases, conditions, or disease subtypes in an ongoing manner, with drugs entering or leaving the platform."

The three differ in what is held fixed. A basket fixes the drug and varies the population, as in the tissue-agnostic approvals built on a molecular marker. An umbrella fixes the disease and varies the drug, often assigning patients to arms by biomarker. A platform fixes the disease and the infrastructure and lets the drugs change over time. In practice the terms overlap: I-SPY 2 is a platform with biomarker-defined signatures, and many baskets are platforms for one drug family. The Adaptive Platform Trials Coalition uses *adaptive platform trial* for a platform whose adaptations (dropping or graduating arms, response-adaptive randomization, sample-size changes) follow decision rules fixed in advance (Coalition et al. 2019). The EC/EMA/HMA question-and-answer document defines a platform trial as a shared framework in which several investigational products enter and leave at different times under pre-specified decision rules, possibly in different diseases (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). Early reviews of how often the three designs were used, and with what mix of features, are J. J. H. Park et al. (2019) and Meyer et al. (2020). Basket designs and the borrowing that makes them efficient belong to Chapter 19 and are not treated further here.

The treatment-specific part of a platform sits in a module attached to the master protocol. It is called an intervention-specific appendix, sub-protocol, substudy appendix or, in REMAP-CAP, a domain-specific appendix. The master protocol holds what is common: eligibility, endpoints, schedule of assessments, the randomization system, the statistical framework and the governance. The module holds what is specific to one treatment: its dose, its safety monitoring, its exclusion criteria, and its decision thresholds (Woodcock and LaVange 2017).

20.3. The frequentist platform: multi-arm multi-stage designs

The platform with the longest record is frequentist. STAMPEDE (Systemic Therapy in Advancing or Metastatic Prostate Cancer: Evaluation of Drug Efficacy) was designed at the MRC Clinical Trials Unit at UCL as a multi-arm multi-stage (MAMS) trial. The first patient was randomized in October 2005 to one of six arms, a control on standard hormone therapy and five research arms, with allocation 2:1:1:1:1:1, so that the control received twice the share of each research arm (Mahesh K. B. Parmar et al. 2017). The MAMS design for time-to-event outcomes, with stopping for lack of benefit on an intermediate outcome, was introduced by Royston, Parmar, and Qian (2003), with stopping guidelines in Royston et al. (2011); Stallard and Todd (2003) gave a sequential treatment-selection design and Wason and Jaki (2012) optimal designs. The design follows the logic of a group sequential trial (Chapter 11) with one change: at each interim look every research arm is compared with the control, and arms that fail a pre-specified bar are dropped while the others continue (Sydes et al. 2009).

Three features of the design carry over to every platform.

Intermediate outcome first. The final outcome in STAMPEDE is overall survival, which takes years. The early stages use failure-free survival as an intermediate outcome, with a hazard-ratio cut-off at each stage that is lenient early and tighter later, so that arms with no sign of benefit leave the trial at a fraction of the cost. In STAMPEDE the early stages are therefore futility gates on an intermediate outcome, failure-free survival, ahead of the definitive outcome, overall survival. The intermediate outcome need not be a validated surrogate, and a MAMS design can instead use the same endpoint at successive stages (Sydes et al. 2009). General MAMS methods calibrate the stopping boundaries through the joint distribution of the test statistics across stages and arms, so as to control a chosen error rate such as the pairwise or familywise rate for the comparisons in a family (Magirr, Jaki, and Whitehead 2012). Which error family is required depends on the claims the trial intends to make. STAMPEDE's own strategy was comparison-level testing: each research arm is compared with the control at its own level, and its investigators report a familywise error rate of 6.75% for the three original comparisons reaching their targets (Mahesh K. B. Parmar et al. 2017). These hazard-ratio cutoffs were derived under proportional hazards (Royston et al. 2011). For non-proportional hazards, define an appropriate summary measure (Section 18.5) and include delayed-effect scenarios in the design simulations.

Concurrent controls only. A new comparison added after the start is compared with controls randomized from the time it opened, using the same intermediate-outcome rules shifted in time. STAMPEDE kept its control arm in step with the standard of care: when the standard of care changed, the control regimen changed with it, and an arm opened after such a change would have a different control population from one opened before it (Nicholas D. James et al. 2016).

New arms by amendment. Over its life STAMPEDE added comparisons as earlier ones reported. New arms were substantial amendments, each requiring ethics and regulatory review before the arm could open. The abiraterone comparison joined later and, in men starting long-term hormone therapy, reduced the hazard of death (hazard ratio 0.63) (Nicholas D. James et al. 2016; Nicholas D. James et al. 2017). The project reports a correlation of 0.12 between the test statistic of the abiraterone comparison and those of the original arms (Mahesh K. B. Parmar et al. 2017).

STAMPEDE is a standing reminder that the Bayesian machinery is optional for a platform. A common control, pre-specified stopping rules, intermediate outcomes and a standing infrastructure were enough to answer several questions in the time that a sequence of separate trials would need (Mahesh K. B. Parmar, Carpenter, and Sydes 2014). Frequentist platforms can also let the set of arms grow: STAMPEDE itself added comparisons by amendment, an added arm can be treated as a new family with its own level (Mahesh K. B. Parmar et al. 2017), and online error-rate control methods exist for hypotheses that arrive over time (Robertson, Wason, et al. 2023). What a MAMS error calculation for a fixed family assumes is that the family is known, so for added arms the choice of family has to be made and justified. The Bayesian platforms below start from a per-arm decision rule calibrated by simulation, with optional hierarchical sharing of information.

20.4. Shared-control arithmetic

Suppose every experimental arm $j = 1, \dots, k$ has n patients, the control has $n_0 = rn$ patients, and outcomes have unit variance. The estimate of the effect of arm j is the difference of arm mean and control mean. Its variance is

$$\mathrm{Var}(\hat{\theta}_j) = \frac{1}{n} + \frac{1}{n_0} = \frac{1}{n}\left(1 + \frac{1}{r}\right), \tag{20.2}$$

and it depends on r and on n, with $N = (k + r)n$ the total enrollment. Fixing N and minimizing the variance over the split gives Dunnett's rule: the best allocation puts $\sqrt{k}$ patients on the control for every patient on an experimental arm, $r = \sqrt{k}$ (Dunnett 1955). Equal allocation $r = 1$ is optimal only for $k = 1$.

The test statistics for two arms are correlated, because both subtract the same noisy control mean. With $n_0 = rn$ the correlation between the z statistics of two arms is $1/(1 + r)$: 0.5 under equal allocation, and $1/(1 + \sqrt{k})$ under the square-root rule, which is 0.33 for four arms.

```
k <- c(1, 2, 4, 6)
v1 <- mapply(platform_contrast_variance, k, 480, 1)
vs <- mapply(platform_contrast_variance, k, 480, sqrt(k))
tab <- data.frame(k, var_r1 = v1, var_sqrt = vs, gain = 1 - vs / v1,
                  corr_r1 = platform_arm_correlation(1), corr_sqrt =
                   ↪ platform_arm_correlation(sqrt(k)))
knitr::kable(tab, digits = c(0, 4, 4, 3, 2, 2), align = "rrrrrr",
             col.names = c("Experimental arms", "Variance, equal
              ↪ allocation", "Variance, square-root rule", "Gain",
              ↪ "Correlation, equal allocation", "Correlation, square-root
              ↪ rule"))
```

Table 20.2.: Variance of each experimental-versus-control contrast at 480 total patients, for equal allocation (r = 1) and the square-root rule, with the resulting gain and the correlation between two arms' test statistics.

Experimental arms	Variance, equal allocation	Variance, square-root rule	Gain	Correlation, equal allocation	Correlation, square-root rule
1	0.0083	0.0083	0.000	0.5	0.50
2	0.0125	0.0121	0.029	0.5	0.41
4	0.0208	0.0188	0.100	0.5	0.33
6	0.0292	0.0248	0.150	0.5	0.29

At 480 total patients (Table 20.2) the square-root rule lowers the variance of each contrast by 10% with four arms and 15% with six. Those are gains in precision at a fixed total sample size, which equals the same power with about 10 to 15% fewer patients. The control arm is larger than each experimental arm but is still a smaller share of the whole than in k separate trials. With four arms and $r = 2$, 2 patients in 6 are on control, compared with 1 in 2 in a two-arm trial, so the chance that a given participant gets the control falls even though the control arm is the largest.

FDA's draft asks for the same thing in words: with a shared control, the master protocol should consider allocating "more participants to the control arm than each individual drug arm" (PDF p. 9), and the EMA draft gives the square-root rule as one example (U.S. Food and Drug Administration 2026f; European Medicines Agency 2026a). Both texts are drafts.

The statistical content is old and exact, and the protocol should state the allocation ratio and the reason for it.

In practice the ratio is a design parameter that moves. When arms enter and leave, the number of arms open at a given time changes, so a platform fixes either a constant ratio per open arm or a constant control fraction. I-SPY 2 and GBM AGILE hold a fixed fraction of patients on control per subtype (20% in both (Nanda et al. 2020; Alexander et al. 2018)), and let the rest be allocated according to performance. For confirmatory platforms, the October 2026 EMA draft recommends maintaining each treatment-to-control allocation ratio and does not support response-adaptive randomization (Section 3.2.1). Sponsors proposing it should address this position explicitly (European Medicines Agency 2026a).

20.4.1. The number of false positives

Every analysis of k arms against one control makes k tests. If each test is run at one-sided level 0.025 and all arms are null, the expected number of false positives is $0.025k$, regardless of the correlation between tests. The correlation changes the *distribution* of that count. With a shared control, a control sample that happens to be unlucky low will make several arms look good together, and a lucky control will make none look good. In the four-arm illustration below, the probability of at least one false positive falls below the independent-trials value and the probability of two or more rises above it. That comparison of count tails is specific to the illustration, and the general statement is only the $k\alpha$ expectation.

```
k <- 4; a <- 0.025
pm <- rbind(independent = dbinom(0:k, k, a),
            shared_r1 = platform_false_positive_counts(k, 1, seed = 1),
            shared_r2 = platform_false_positive_counts(k, sqrt(k), seed =
             ↪  2))
tab <- data.frame(design = c("Independent trials", "Shared control, equal
 ↪  allocation", "Shared control, square-root allocation"),
                  P0 = pm[, 1], P_ge1 = 1 - pm[, 1], P_ge2 = rowSums(pm[,
                   ↪  -(1:2)]), mean = as.vector(pm %*% 0:k))
pct <- function(x) sprintf("%.1f", 100 * x)
knitr::kable(tab, digits = 3, align = "lrrrr",
             col.names = c("Design", "No false positive", "At least one",
              ↪  "At least two", "Mean count"))
```

Table 20.3.: Distribution of the number of false positives among four null arms tested at one-sided 0.025: independent trials, shared control with equal allocation, and shared control with square-root allocation. Shared-control rows are Monte Carlo.

	Design	No false positive	At least one	At least two	Mean count
independent	Independent trials	0.904	0.096	0.004	0.1
shared_r1	Shared control, equal allocation	0.922	0.078	0.017	0.1
shared_r2	Shared control, square-root allocation	0.914	0.086	0.012	0.1

Under four independent trials the chance of at least one false positive is 9.6%, and the chance of two or more is 0.4%. With a shared control and equal allocation, at least one drops to 7.8% and two or more rises to 1.7%. The mean count is 0.10 in every case. The square-root allocation, whose correlation is lower, gives 8.6% for at least one and 1.2% for two or more (shared-control rows are Monte Carlo, 400,000 draws; see Table 20.3).

The result is a lesson for how a platform is judged. A design that reports only "the probability of at least one false positive is below the independent-trial value" is correct and leaves out the thing a sponsor with five programs would worry about, which is that when the control is unusually good or bad, several programs are misjudged together. FDA's draft makes this point when it says that a larger control allocation reduces the risk that a poorly or unusually well-performing control arm yields several correlated erroneous findings, and EMA notes that bringing the probability of multiple false positives down by tightening each arm's level would require thresholds more stringent than Bonferroni, so more control participants is the more efficient remedy (U.S. Food and Drug Administration 2026f; European Medicines Agency 2026a). The simulation report required by Chapter 14 should therefore show the distribution of the number of errors across arms and not only the familywise rate.

i Technical note: why the mean is unchanged

Each test statistic Z_j is marginally standard normal under the null, whatever the allocation, so $\mathbb{P}(Z_j > z_\alpha) = \alpha$ and by linearity of expectation $\mathbb{E}[\#\text{false positives}] = k\alpha$. The correlation $\rho = 1/(1+r)$ affects only the joint tail probabilities. For two arms, $\mathbb{P}(Z_1 > z_\alpha, Z_2 > z_\alpha)$ is larger than α^2 for any $\rho > 0$, which is why the count of two or more rises in the four-arm illustration. The expectation calculation holds for any number of arms and any correlation, but the shape of the count distribution does not, so the distribution should be computed for the design at hand. The code simulates the arm means directly: a common control mean with variance $1/n_0$, and k independent arm means with variance $1/n$.

20.5. Concurrent and non-concurrent controls

A platform that runs for years faces a question that separate trials never face: which control patients may be compared with a given arm? *Concurrent* controls are those randomized while the arm was open. *Non-concurrent* controls were randomized before it opened. They are tempting because they are free patients, and they are dangerous because of drift. Over a platform's life the standard of care, the mix of patients, the sites and the conduct of the trial change, so an arm that opens late is compared with controls treated under different conditions (Dodd, Freidlin, and Korn 2021).

20.5.1. Regulatory position

FDA's June 2026 draft is the stricter of the two. It asks that the primary comparison for a drug use only concurrently eligible controls: those who met that drug's eligibility criteria and could have been randomized to it but were randomized to control instead (PDF p. 11, "only concurrently eligible control participants"). Controls enrolled earlier, and controls enrolled at the same time who could not have received the drug (a kidney-toxic drug excludes some patients, or a site offers only some arms), are left out of the primary analysis. The draft

says that using non-concurrent data can bias the estimate and change the type I and type II error probabilities even when the analysis tries to model a time trend, and that the shared control keeps most of its efficiency gain when only concurrent controls are used. A sponsor who wants to use non-concurrent data should justify this in advance, weigh the amount of data against the calendar gap, pre-specify the method, and run sensitivity analyses that give the non-concurrent data less weight or none (U.S. Food and Drug Administration 2026f).

The EMA draft reflection paper defines concurrent controls as those with a positive probability of being randomized to the arm of interest, states a strong preference for them, and says that non-concurrent controls are usually not acceptable for the primary analysis of the primary endpoint, though they may support supplementary analyses or rarely observed secondary endpoints (European Medicines Agency 2026a). Both documents are drafts. They agree on the default and differ in how much room they leave for model-based exceptions.

20.5.2. How big is the bias? A simulation

Module 10 of the book's companion online lab (https://vsokolov.org/bayesian_trials_lab/10-platform-trials.html) draws a platform in which success on standard care rises 2 percentage points a year and shows a drug that does nothing and appears to help. The simulation below makes the arithmetic explicit. The platform runs for four years, enrolling 300 patients per year in equal proportions between control, arm A and (after two years have elapsed) arm B. Arm B has no effect in the first scenario, and in the second a true absolute improvement of 10 percentage points. Control success starts at 30% and rises by the drift. We analyze arm B three ways: against concurrent controls (those randomized in years 3 and 4), against all controls, and against all controls with a linear calendar-time term.

```
rows <- list("no drift, B null" = list(drift = 0, effect = 0),
             "drift 2 pts/yr, B null" = list(drift = 0.02, effect = 0),
             "drift 2 pts/yr, B +10 pts" = list(drift = 0.02, effect =
              ↪  0.10))
tab <- do.call(rbind, lapply(names(rows), function(n)
  cbind(scenario = n, do.call(platform_drift_simulation, c(rows[[n]], nsim
   ↪  = 4000, seed = 17)))))
tab$mean_estimate <- 100 * tab$mean_estimate; tab$rejection <- 100 *
 ↪  tab$rejection
drift_tab <- tab
dt <- function(s, m, col) drift_tab[[col]][drift_tab$scenario == s &
 ↪  drift_tab$method == m]
knitr::kable(tab, digits = 1, align = "llrr", row.names = FALSE,
             col.names = c("Scenario", "Method", "Mean estimate (points)",
              ↪  "Rejection (%)"))
```

Table 20.4.: Analysis of arm B against concurrent controls, all controls, and all controls with a linear time term, in 4,000 simulated platforms. Mean estimate is the average estimated effect in percentage points, and rejection is the one-sided rejection rate in percent.

Scenario	Method	Mean estimate (points)	Rejection (%)
no drift, B null	concurrent	0.1	2.5
no drift, B null	all controls	0.1	2.3
no drift, B null	time-adjusted	0.1	2.5
drift 2 pts/yr, B null	concurrent	0.0	2.9
drift 2 pts/yr, B null	all controls	2.4	8.1
drift 2 pts/yr, B null	time-adjusted	0.0	2.8
drift 2 pts/yr, B +10 pts	concurrent	10.1	53.0
drift 2 pts/yr, B +10 pts	all controls	12.4	85.8
drift 2 pts/yr, B +10 pts	time-adjusted	10.1	59.4

With no drift (Table 20.4), pooling all controls costs nothing and the three methods agree. With a 2-point-per-year drift and no true effect, the concurrent comparison is unbiased and rejects at about the nominal 2.5% (2.9% in 4,000 simulations), while pooling all controls produces a bias of 2.4 percentage points and a one-sided type I error of 8.1%, more than three times the nominal level. The platform's own control arm turns into a source of false positives. When arm B has a true effect of 10 points, pooling "improves" power from 53% to 86%, and the improvement is partly spurious: the estimate is 12.4 points rather than 10.

The calendar-time adjustment does what it was designed to do under this smooth, linear drift: no bias, nominal type I error, and slightly higher power than the concurrent-only analysis (59% against 53%), because the non-concurrent controls are used to estimate the slope. That result depends on the model being right. FDA's draft warns that modeling does not guarantee error control, and the time-machine simulation below shows a related failure when history breaks.

20.5.3. The Bayesian time machine

A less rigid alternative to a linear term lets the control rate drift as a random walk across calendar periods and estimates the control rate in the current period by borrowing from the neighboring periods, with a smoothing parameter σ_s that controls how fast the rate is allowed to change. This is the idea behind the *time machine* of Saville and colleagues, used in I-SPY 2, proposed for GBM AGILE and Precision Promise, and cited as such in FDA's Bayesian draft (Benjamin R. Saville et al. 2022; U.S. Food and Drug Administration 2026g). Small σ_s pools almost everything, large σ_s is nearly the concurrent-only estimate. The code below is an illustrative first-order Gaussian smoother over period-level control rates, fitted from the simulated data only (the period variances are estimated from the observed rates, not taken from the truth). It is not the published second-order model, so its results illustrate the trade-off and do not establish the robustness of the published method. It is run two ways: with σ_s learned from the data by restricted maximum likelihood (REML), and with σ_s fixed at three values for illustration.

The simulation (Table 20.5) estimates the control success rate in the current period under three histories of the true rate: constant, linear drift of 2 points per period, and a step of 8

points in the last period (a change in standard of care). There are four periods, each with 100 control patients. We report bias and root mean squared error, and the mean smoothing parameter.

```
scen <- list(constant = c(0, 0, 0, 0), `linear drift` = c(0, .02, .04,
 ↪  .06), `step in last period` = c(0, 0, 0, .08))
tab <- do.call(rbind, lapply(names(scen), function(n) cbind(scenario = n,
 ↪  platform_time_machine(scen[[n]], seed = 17))))
tab$bias <- 100 * tab$bias; tab$rmse <- 100 * tab$rmse
tm <- function(h, m, col) with(tab, get(col)[startsWith(scenario, h) &
 ↪  startsWith(method, m)])
f1 <- function(x) sprintf("%.1f", x)
knitr::kable(tab, digits = c(0, 0, 1, 1, 3), align = "llrrr", row.names =
 ↪  FALSE,
             col.names = c("History", "Method", "Bias (points)", "RMSE
              ↪  (points)", "Mean smoother scale"))
```

Table 20.5.: Bias and root mean squared error, in percentage points, of estimates of the current control success rate under three histories, for the current period alone, pooling all periods, the smoother with σ_s learned by REML (mean estimated σ_s shown), and the smoother at three fixed smoothing values.

History	Method	Bias (points)	RMSE (points)	Mean smoother scale
constant	current only	0.1	4.5	NA
constant	pool all	0.0	2.3	NA
constant	learned smoothing (REML tau)	-0.1	3.0	0.015
constant	fixed smoothing, tau=0.01	-0.3	2.3	0.010
constant	fixed smoothing, tau=0.03	-0.2	2.7	0.030
constant	fixed smoothing, tau=0.1	0.0	3.9	0.100
linear drift	current only	0.0	4.8	NA
linear drift	pool all	-3.0	3.8	NA
linear drift	learned smoothing (REML tau)	-1.7	3.9	0.026
linear drift	fixed smoothing, tau=0.01	-3.1	4.0	0.010
linear drift	fixed smoothing, tau=0.03	-2.0	3.4	0.030
linear drift	fixed smoothing, tau=0.1	-0.5	4.1	0.100
step in last period	current only	0.0	4.8	NA
step in last period	pool all	-6.0	6.4	NA

Table 20.5.: Bias and root mean squared error, in percentage points, of estimates of the current control success rate under three histories, for the current period alone, pooling all periods, the smoother with σ_s learned by REML (mean estimated σ_s shown), and the smoother at three fixed smoothing values.

History	Method	Bias (points)	RMSE (points)	Mean smoother scale
step in last period	learned smoothing (REML tau)	-3.3	5.4	0.037
step in last period	fixed smoothing, tau=0.01	-6.1	6.6	0.010
step in last period	fixed smoothing, tau=0.03	-4.5	5.3	0.030
step in last period	fixed smoothing, tau=0.1	-1.4	4.3	0.100

When nothing changes, borrowing is nearly free: pooling cuts the root mean squared error from 4.5 points (current period alone) to 2.3, the smoother with learned σ_s to 3.0 (mean estimated σ_s 0.015), and the smallest fixed σ_s to 2.3. Under linear drift, pooling is biased by 3.0 points and the error is 3.8. The learned smoother has bias 1.7 and error 3.9, against 4.8 for the current period alone; the fixed $\sigma_s = 0.03$ has the lowest rmse of the methods shown (3.4 points), and the loosest fixed smoothing, $\sigma_s = 0.10$, has little bias (0.5 points). When the rate jumps at the end, pooling is biased by 6.0 points with an rmse of 6.4, worse than using the current period alone (4.8). The learned smoother, which has to infer the break from four periods of 100 patients, is also worse than the current period alone (bias 3.3, rmse 5.4), and only the loosest fixed smoothing improves on concurrent-only data, with an rmse of 4.3 and bias of 1.4 points. That value of σ_s was fixed with knowledge of the scenarios, which a real protocol does not have.

Two points follow for the protocol. First, a time machine involves a smoothing choice, and whether it is learned from a few periods or fixed in advance, it depends on how the standard of care can change, which is a clinical judgment to be made before the data. Second, in all three illustrated histories the loosest fixed smoother ($\sigma_s = 0.10$) has a lower rmse than the concurrent-only analysis (3.9, 4.1 and 4.3 against 4.5, 4.8 and 4.8 points). Three scenarios do not establish that it is better for other histories (an abrupt drift, for example), the value was fixed with knowledge of the scenarios, and smoothing introduces bias that has to be checked: borrowing gains precision when history is stable and gives it back when history breaks. The simulation report of Chapter 14 therefore has to include scenarios with a step change, which is what FDA means by asking that the method be pre-specified and accompanied by sensitivity analyses. The simulation studies and reviews of non-concurrent controls that examine this trade-off are K. M. Lee and Wason (2020), Roig et al. (2022), Roig et al. (2023) and Z. Jiang et al. (2023), and the R package ncc implements the analyses (Krotka et al. 2023). FDA's Bayesian draft adds that the relevance of borrowed data depends on recency and on changes in standard of care, which is the same point in different words (see Chapter 19 and Chapter 16).

20.6. Multiplicity across arms

A traditional trial with several treatment arms and one control usually controls the familywise error rate at 0.025, through Bonferroni, Holm, or Dunnett's step-down procedure, because a regulator who approves a drug on one of several comparisons must know how many chances the sponsor had (Chapter 10). A platform raises the question of what the family is. Five unrelated drugs from five sponsors that happen to share a control group would never be put in a common family in separate trials, and their arrival in the same protocol does not change their clinical independence.

20.6.1. FDA

The FDA draft says so directly: FDA "generally does not recommend the use of multiplicity adjustments to strongly control the probability of making at least one type I error across the multiple comparisons of different drugs" (Section H, PDF pp. 18 to 19). The reason it gives is the one of the previous section: "the expected total number of type I errors is the same regardless of whether these comparisons are made in independent trials or in a single trial" (PDF p. 19). The unit of control is the drug, with its own level, not the platform. Collignon and colleagues similarly argue that sharing a control arm does not itself call for an additional multiplicity adjustment across substudies investigating different drugs, while recognizing the possibility of multiple false positive regulatory decisions (Collignon et al. 2020).

The draft keeps the adjustment where the comparisons answer one question: several doses or formulations of one drug, the components of a combination, or several endpoints for one drug. In those cases the sponsor should define the family arm by arm and defend the definition before the data are seen. Doses of one drug compared with a shared control, for example, are one family with d comparisons, and the usual tools of Chapter 10 apply: Bonferroni, Holm, a graphical procedure, or the Dunnett critical value, which uses the correlation of the shared-control statistics and is less conservative than Bonferroni.

20.6.2. EMA

The EMA draft is more cautious about the same fact. It notes that the shared control makes errors correlated, that the probability of at least one false positive may fall while the probability of several rises, and that tightening each arm's level to hold down the chance of multiple false positives would require thresholds stricter than Bonferroni (European Medicines Agency 2026a). It treats the choice of family as a design decision to be made in advance and argued, and it offers alternatives such as larger control groups. The comparison with FDA is subtle. Both reject a blanket adjustment across unrelated arms, and both want the sponsor to define the families. EMA's draft puts more weight on the arms that share a regulatory submission or a sponsor, and on the distribution of the number of errors, which is the quantity the earlier simulation computed. For a program whose arms are all developed by one sponsor for one eventual label, a regulator in either jurisdiction may regard the arms as a single family; this needs a conversation, and Chapter 23 describes how to have it.

20.6.3. Bayesian error control across arms

A Bayesian platform does not remove the question, it moves it. Each arm has a success criterion of the form $\Pr(\delta > a \mid \text{data}) > c$ (Chapter 7). For confirmatory adaptive comparisons without borrowing, this book uses simulation-calibrated criteria, and the same calibration can be extended to a familywise rate over a defined family if the regulator asks for one. FDA's draft also discusses direct posterior interpretation and decision-theoretic or benefit-risk criteria; the proposed route should be discussed with FDA. FDA's Bayesian draft specifies that the sponsor describe how the success criteria map to error control, and the simulation report shows it by arm and by family (U.S. Food and Drug Administration 2026g). The Bayesian framework also offers a natural route to sharing information across arms, shrinkage, which frequentist analyses can reach as well through hierarchical or empirical-Bayes estimation with its own calibration.

If arms in the same platform are modeled as exchangeable, with each arm's effect drawn from a common distribution with between-arm variance τ_b^2 and sampling variance v, each estimate keeps weight $w = \tau_b^2/(\tau_b^2 + v)$ on its own data and is pulled toward the common mean with weight $1 - w$, and its posterior variance falls from v to vw, an information multiple of $1/w$ (Polson and Sokolov 2026b). The reverse is also a cost: the multiple is an upper bound that treats the common mean as known, and an arm whose effect is far from its neighbors is shrunk toward them.

```
tau <- 0.264; v <- 0.09   # between-arm sd and sampling variance (log-odds
 ↪  scale, illustrative)
w <- tau^2 / (tau^2 + v)
knitr::kable(data.frame(quantity = c("Weight on own data", "Information
 ↪  multiple", "Posterior sd before", "Posterior sd after"),
                         value = c(w, 1 / w, sqrt(v), sqrt(v * w))), digits
                          ↪  = 2, align = "lr", col.names = c("Quantity",
                          ↪  "Value"))
```

Table 20.6.: Shrinkage across arms with between-arm standard deviation 0.264 and sampling variance 0.09 on the log-odds scale (illustrative): weight on own data, information multiple, and posterior standard deviation before and after.

Quantity	Value
Weight on own data	0.44
Information multiple	2.29
Posterior sd before	0.30
Posterior sd after	0.20

With these values (Table 20.6; the between-arm spread from the ulcer example of Chapter 10) each arm's posterior standard deviation falls from 0.30 to 0.20. That information is paid for by an assumption: that the arms are exchangeable. Between unrelated drugs that assumption is hard to defend, and FDA's Bayesian draft asks that a model borrowing across substudies justify exchangeability and include known effect modifiers, such as biomarker status, as covariates rather than treating the groups as interchangeable (U.S. Food and Drug Administration 2026g). Shrinkage across *arms* is therefore most defensible for arms from one drug class, doses of one drug, or the same drug in related subgroups, which is the territory

of Chapter 19. It is not a substitute for the multiplicity decision across unrelated drugs, and the protocol should not present it as one.

20.7. Bayesian platforms

The Bayesian platforms differ from STAMPEDE in what they do at each interim look and in how they treat information shared across arms.

20.7.1. I-SPY 2: graduation by predictive probability

I-SPY 2 is a platform for neoadjuvant breast cancer that has enrolled since 2010 and was later reconfigured as I-SPY 2.2 (J. W. Park et al. 2016; D. A. Berry 2025). The primary endpoint of each arm was pathologic complete response (pCR) at surgery, with tumor volume measured by MRI during treatment used to predict pCR for patients still in follow-up. Patients were classified into biomarker subtypes by hormone receptor status, HER2 status and a genomic profile, and each therapy was evaluated in each of up to ten *signatures*, the subtype combinations that might become a regulatory indication (Barker et al. 2009; H. Wang and Yee 2019).

An arm *graduated* in a signature when its Bayesian predictive probability of success in a hypothetical 300-patient, equally randomized phase 3 trial against the same control, with pCR as endpoint, reached 85%; it was dropped for futility if that probability fell below 10% in all ten signatures, and no arm took more than 120 patients (J. W. Park et al. 2016). The decision is made from phase 2 data, so it answers the question a sponsor asks at a go/no-go meeting. It differs from a posterior probability of benefit, which asks whether the effect is positive and says nothing about the size and power of the trial that would follow. The simulation below (Table 20.7) shows the difference with four phase 2 results, 50 patients per arm in the signature, and a phase 3 with 150 per arm tested at one-sided 0.025.

```
cases <- list("clear win, 24/50 vs 10/50" = c(24, 50, 10, 50), "good, 20/50
 ↪  vs 10/50" = c(20, 50, 10, 50),
              "modest, 18/50 vs 13/50" = c(18, 50, 13, 50), "none, 15/50 vs
               ↪  15/50" = c(15, 50, 15, 50))
res <- t(sapply(cases, function(x) do.call(platform_predictive_graduation,
 ↪  c(as.list(x), seed = 5))))
tab <- data.frame(phase2 = names(cases), res, graduates = ifelse(res[,
 ↪  "pp"] >= 0.85, "yes", "no"))
knitr::kable(tab, digits = 3, align = "lrrc", row.names = FALSE,
             col.names = c("Phase 2 result", "Predictive probability",
              ↪  "Posterior probability", "Graduates"))
```

Table 20.7.: Predictive probability of phase 3 success and posterior probability of benefit for four phase 2 results (50 patients per arm), with the 85 percent graduation rule applied.

Phase 2 result	Predictive probability	Posterior probability	Graduates
clear win, 24/50 vs 10/50	0.942	0.999	yes

Table 20.7.: Predictive probability of phase 3 success and posterior probability of benefit for four phase 2 results (50 patients per arm), with the 85 percent graduation rule applied.

Phase 2 result	Predictive probability	Posterior probability	Graduates
good, 20/50 vs 10/50	0.814	0.985	no
modest, 18/50 vs 13/50	0.470	0.857	no
none, 15/50 vs 15/50	0.161	0.499	no

The second row has a 98.5% posterior probability that the new arm beats the control, and a predictive probability of phase 3 success of 81.4%, below the threshold. The posterior probability is a statement about the present, and the predictive probability includes the chance that a true effect of ordinary size is missed by a finite phase 3. Setting the graduation threshold on the predictive scale therefore ties the platform's decision to the confirmatory trial it feeds, and a sponsor can see at the start what phase 2 result would be enough. Predictive probabilities are developed in Chapter 15.

Three further features of I-SPY 2 are relevant. Randomization was response adaptive within a signature, with a fixed share held on control (20% (Nanda et al. 2020)), the remainder going to arms that looked better (J. W. Park et al. 2016). A control cohort for each signature accumulated over the years, with a time machine used to smooth the control response over time. And the platform also tested combination regimens as experimental arms against the shared control. The graduation results of the first two graduates, neratinib and pembrolizumab, and why their final predictive probabilities fell after all pathology was in, are told in Section 15.10. Because allocation adapts toward responsive subtypes, the raw difference should not be read as the treatment effect, and a design that allocates by subtype responsiveness needs a model that adjusts for it; Chapter 16 describes the mechanism and the critiques.

I-SPY 2 is a phase 2 screening platform. A graduation is a decision to run a phase 3, and it carries a false positive rate that was calibrated by simulation to be acceptable for a screening stage (Chapter 14). It does not replace a confirmatory trial, and a graduate is not an approval.

20.7.2. GBM AGILE: a registration platform

GBM AGILE (Adaptive Global Innovative Learning Environment, NCT03970447) is a Bayesian response-adaptive platform in glioblastoma designed with registration in mind, which I-SPY 2 was not (Alexander et al. 2018). An antecedent is the Bayesian adaptive randomized design for recurrent glioblastoma of Trippa et al. (2012). It uses a seamless phase 2/3 structure: in stage 1 each arm is randomized adaptively against a shared control (temozolomide in newly diagnosed disease, lomustine in recurrent), with allocation in proportion to the posterior probability of a hazard ratio below a benchmark and arms evaluated monthly; an arm that clears the stage 1 bar graduates to stage 2 with fixed randomization to confirm the finding for registration (Alexander et al. 2018). In the first completed arm, regorafenib, efficacy required a Bayesian probability of benefit (hazard ratio below 1) of at least 98%, and enrollment stopped for limited efficacy when the Bayesian predictive power fell below 25% in all signatures (Wen et al. 2026). The published report has since been corrected by an erratum (Wen, Berry, et al. 2026). Patients are classified into signatures

based on newly diagnosed versus recurrent disease and MGMT methylation status, and 20% of each subtype's patients are assigned to control (Alexander et al. 2018).

The 98% is the kind of number the FDA Bayesian draft discusses under calibration to type I error: the threshold is set by simulation so that the false positive rate over the design's scenarios is acceptable, not by a view about the prior, and the simulation report is part of the submission (U.S. Food and Drug Administration 2026g). GBM AGILE is also the example FDA's Bayesian draft names for the time machine: the shared control over the years has to be used carefully when standard of care changes. The platform's own registry record lists 17 arm entries (ClinicalTrials.gov 2026f), including regorafenib, paxalisib and VAL-083, with several arms (VT1021, troriluzole, ADI-PEG 20, tinostamustine) running dose-finding and enhanced safety management phases before the main arm. That reflects an amendment pattern worth copying: a safety lead-in as part of the arm's appendix, with its own go/no-go rule, ahead of the efficacy comparison.

20.7.3. REMAP-CAP: domains, response-adaptive randomization, and an embedded design

REMAP-CAP (Randomized, Embedded, Multifactorial Adaptive Platform trial for Community-Acquired Pneumonia) is a platform designed before the pandemic it became known for (Angus et al. 2020). After the 2009 H1N1 pandemic, an EU-funded network built the infrastructure, which began enrolling in April 2016 according to its ClinicalTrials.gov record (NCT02735707) and carried a dormant pandemic stratum that was activated for COVID-19. Its design has four features that distinguish it from the oncology platforms.

Multifactorial. The platform is organized by *domains*, each a set of alternative interventions for one aspect of care: corticosteroids, antivirals, immune modulators, anticoagulation, antibiotic strategy, ventilation. A patient is randomized in every domain for which he or she is eligible, so a single patient can contribute to several comparisons, and the domains are analyzed together in a model that includes the interactions that were prespecified. Each domain has its own appendix and its own statistical analysis plan, which means the master protocol is stable while the domains come and go.

Embedded. The trial runs inside routine care and uses routinely collected data where it can, which keeps the burden per patient low enough for ICUs under pandemic load.

Ordinal primary endpoint. For the pandemic stratum the primary endpoint was organ-support-free days to day 21, an ordinal scale with death as the worst outcome, analyzed with a Bayesian proportional-odds model (REMAP-CAP Investigators, Gordon, et al. 2021).

Response-adaptive randomization and decision rules. Allocation within a domain is updated at pre-specified intervals to favor arms with a higher posterior probability of being best, and the domain stops when a pre-specified trigger is crossed. In the IL-6 domain the triggers were: superiority if the posterior probability of being best exceeded 99%; efficacy if the probability that the odds ratio against control exceeded 1 was above 99%; futility if the probability that the odds ratio exceeded 1.2 was below 5%; and equivalence of two non-control arms if the probability of an odds ratio between 0.83 and 1.2 exceeded 90% (REMAP-CAP Investigators, Gordon, et al. 2021; Angus et al. 2020). The platform declared that interleukin-6 receptor antagonists (tocilizumab and sarilumab) improved outcomes, including survival, in critically ill patients with COVID-19 (REMAP-CAP Investigators, Gordon, et al. 2021). Note that a platform with this structure can declare a domain complete and move on without waiting for the others, which is the property the sponsors of separate trials have to give up.

The domains share a hierarchical structure in one sense: the pandemic and non-pandemic strata, and subgroups within a domain, can be modeled with partial pooling. This is borrowing in the sense of Chapter 19, and the conditions under which it is acceptable are the same.

20.7.4. The COVID anticoagulation multiplatform

Three independently funded Bayesian platforms (REMAP-CAP, ATTACC, an international platform led from Canada, and ACTIV-4a in the United States) asked the same question early in the pandemic: therapeutic-dose against prophylactic-dose anticoagulation in COVID-19. Separate reports would have invited conflicting results and a post hoc meta-analysis. The three teams instead harmonized their protocols and statistical analysis plans before the data were in, and combined patient-level data in a Bayesian hierarchical model with dynamic borrowing across severity strata (ATTACC Investigators et al. 2021; REMAP-CAP Investigators, ACTIV-4a Investigators, et al. 2021). The stopping rule was superiority at a posterior probability above 99% or futility at above 95% for an odds ratio below 1.2, applied separately in each severity cohort. In critically ill patients enrollment stopped for futility (adjusted odds ratio 0.83, 95% CrI 0.67 to 1.03, futility probability 99.9%); in non-critically ill patients therapeutic dosing increased organ-support-free days (odds ratio 1.27, 95% CrI 1.03 to 1.58, probability of benefit 98.6%) (REMAP-CAP Investigators, ACTIV-4a Investigators, et al. 2021; ATTACC Investigators et al. 2021). The severity strata had been specified in advance, so the finding was not a post hoc subgroup.

The anticoagulation project is an unusual use of the platform idea. The shared object was a pre-agreed analysis, not a shared control group, and the platforms are the data sources for a single multi-site analysis. It shows what is possible when the sponsors agree on the estimand in advance, and it is the nearest thing the field has to a meta-analysis designed before the data exist.

20.7.5. Pragmatic pandemic platforms: RECOVERY and PANORAMIC

Not every pandemic platform was an elaborate Bayesian adaptive design. RECOVERY (Randomized Evaluation of COVID-19 Therapy) was a large, pragmatic and open-label platform in hospitalized patients, analyzed with frequentist methods, with deliberately little data collected beyond what the health service already records and 28-day mortality as the primary outcome. Its first major result was that dexamethasone reduced 28-day mortality overall (rate ratio 0.83, a reduction of about a sixth), by more among patients on invasive mechanical ventilation (0.64), and not at all among those who needed no oxygen (RECOVERY Collaborative Group et al. 2021). PANORAMIC took the pragmatic approach into UK primary care and analyzed its primary outcome with Bayesian logistic regression using weakly informative Cauchy priors. Its molnupiravir comparison randomized 25,783 participants between December 2021 and April 2022 to molnupiravir plus usual care or usual care alone (the platform's total of 26,411 includes other comparisons), and hospitalization or death within 28 days in about 1% of each group (adjusted odds ratio 1.06), a clear negative result from a platform that could enroll that many community patients in under five months (Butler et al. 2023). The two trials show that simplicity and volume can do the work that a more elaborate adaptive design does elsewhere, and that a platform's value comes first from its infrastructure and its speed.

20.7.6. Traditional against Bayesian platforms: what differs

Table 20.8 sets the frequentist MAMS and Bayesian platform side by side, as the book does for each design.

Table 20.8.: Frequentist and Bayesian platforms compared

	Frequentist MAMS (STAMPEDE)	Bayesian platform (I-SPY 2, GBM AGILE, REMAP-CAP)
Interim decision	hazard-ratio cut-off on an intermediate outcome, tightening by stage	posterior or predictive probability threshold, often monthly
Error control	error rates for the chosen comparisons, calibrated by the joint distribution of statistics (STAMPEDE: comparison-level testing, 6.75% familywise error reported for the three original comparisons)	per-arm or per-family type I error calibrated by simulation (Chapter 14)
Control allocation	fixed ratio (2:1:1:1:1:1 at the start), concurrent controls only	fixed control fraction, often response-adaptive for the rest (not supported for confirmatory platforms by the EMA draft), time machine optional
Arms nobody planned	added by amendment, as a new family with its own level or by online error allocation	added with a pre-specified decision rule; the family and its error rate still have to be defined
Information sharing	the shared control; hierarchical or empirical-Bayes estimation across arms is possible in principle	optional hierarchical borrowing across arms or strata with a justified exchangeability argument (Chapter 19)
What the regulator asks	the MAMS calculation and the stopping boundaries	the simulation report, prior and borrowing justification, time-trend handling
Where they agree	for a normal likelihood with known standard error, a flat prior and a single final look, the posterior probability of benefit equals one minus the one-sided p-value, and the corresponding thresholds agree	the same (for binary outcomes the correspondence is approximate)

The traditional design is the better choice when the arms are known, the intermediate outcome is trusted, and the sponsor wants the simplest possible regulatory conversation. A

Bayesian platform can earn its complexity when the outcome can be modeled longitudinally, or when information can be shared across strata with a defensible exchangeability argument. Frequentist platforms also accommodate added arms, and for confirmatory use the EMA draft does not support response-adaptive randomization, so the allocation rule is a point to settle with the regulator.

20.8. Governance, firewalls and the monitoring board

A platform survives its first arm only if the governance is built for arms that are not yet known. Typically a single data monitoring committee (DMC) oversees all arms, an independent statistical center runs the unblinded interim analyses and feeds the allocation algorithm, and a separate committee approves applications for new arms. The arrangement serves one aim: no sponsor sees how a competitor's arm is performing. Under a Bayesian response-adaptive design, in which allocation probabilities depend on accumulating results, the allocation proportions themselves leak information, so the sponsor-facing report should show enrollment counts only when they cannot be inverted into outcomes. A burn-in period of equal allocation before adaptation begins, with the statistical center and not the sponsor holding the algorithm, is the usual safeguard (U.S. Food and Drug Administration 2026f; European Commission and European Medicines Agency and Heads of Medicines Agencies 2022).

Several other elements of governance have regulatory consequences.

Who is the sponsor. A master protocol needs a sponsor responsible for the platform (the "master protocol sponsor" of FDA's draft) and, for each arm, an arm sponsor who owns the drug and its safety reporting. In the US, when all substudies sit under one master IND, FDA's draft recommends that a new substudy be submitted at least 30 days before it begins and that the regulatory project manager is notified at least 48 hours before the amendment is submitted (U.S. Food and Drug Administration 2026f). In the EU, a new arm is a substantial modification of the authorization, and since a change to one sub-protocol can force changes to others, each sub-protocol states how far it depends on the rest (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). STAMPEDE's experience was that each new comparison was a substantial amendment, and that drug supply could set the pace.

Simulating before opening. Analytical error-rate and power calculations are available for some platform designs with added arms. Platforms combining staggered entry, allocation changes and adaptive decisions generally need simulation to assess their full operating characteristics (Choodari-Oskooei et al. 2020; Meyer et al. 2025), and the simulation report of Chapter 14 is the vehicle. Meyer and colleagues draw lessons on why and how platform trials should be simulated from the EU-PEARL consortium.

What the firewall protects. The sponsor of one arm should not see another arm's data, the control arm's results, or the platform's allocation algorithm in a way that would let it infer them. The DMC sees everything. The master protocol's governance section should state who sees what, and the confidentiality agreements for arm sponsors should say so.

Rules fixed before data. The DMC charter should list the decision rules for each arm (stopping for efficacy, futility, graduation) and state that the DMC can override them for safety but not for convenience. The decision rules and the allocation algorithm are changed only by protocol amendment, with the simulation report updated accordingly. Operating characteristics must

be re-established when anything that affects them changes (Chapter 14); a late-arriving arm changes the correlation structure and the allocation ratio, and the previous simulation may no longer apply.

In the protocol

A master protocol should state, in the master document or an appendix: (1) the families of hypotheses, arm by arm, and the multiplicity method for each family; (2) the control allocation rule and when it changes as arms enter or leave; (3) the definition of a concurrent control, including eligibility; (4) the primary analysis population for each arm, and any use of non-concurrent data with its method, weight and sensitivity analyses; (5) the simulation report scenarios, including a step change in standard of care; (6) the decision rules and thresholds for each arm, and who may change them; (7) who sees which results, and the information given to arm sponsors; (8) the process and timing for adding an arm, with the 30-day regulatory submission that FDA's draft recommends and the ethics review.

20.9. EPAD: a platform that never opened an arm

The European Prevention of Alzheimer's Dementia program (EPAD) is the standard example of a platform that did not work. A 64-million-euro, five-year IMI-funded project, it built a 2,000-participant readiness cohort with detailed pre-characterization (amyloid status, cognitive baselines and longitudinal follow-up) to reduce the very high screen-failure rates of Alzheimer's prevention trials (Ritchie et al. 2016; Saunders et al. 2022). It never enrolled a single treatment arm. In the project's own post-mortem, two obstacles were decisive: owners of candidate interventions were reluctant to hand sponsorship of a critical asset to a university-run platform, and the stakeholders could not agree on a legal framework that suited every party (Saunders et al. 2022). When the five-year term ended in 2020, the treatment platform had never opened and the pandemic had made funding harder. The cohort data and biobank outlived the project.

The lesson is organizational, not statistical: nothing in EPAD's statistical design was at fault, and the sample-size, borrowing and decision rules were never tested. Platforms need three things before a first arm. A legal framework that arm sponsors will sign, because they are handing control of an asset to a shared structure. A funding model that does not depend on the commercial pipeline the platform is meant to serve; the Healey ALS Platform Trial is the contrast (Quintana et al. 2023). And a governance structure that answers the question "who decides?" before the first arm's sponsor asks it.

20.10. What changes in the protocol

A traditional two-arm trial and a platform differ less in the statistics of one comparison than in what surrounds it, as Table 20.9 shows.

Table 20.9.: Separate trials and a platform compared

	k separate two-arm trials	Platform with shared control
Patients ($k = 5$, 80 per arm)	800	480
Control allocation	1:1 in each trial	$r > 1$ in favor of the control, often $\sqrt{k}$
Control data used	each trial's own	concurrent only for primary; non-concurrent by pre-specified method
Multiplicity	none across trials	none across unrelated drugs (FDA); family defined per arm; distribution of the number of errors reported
New question	new trial, new submission	amendment; FDA draft recommends submission at least 30 days before
Simulation report	one design	all scenarios, including arms entering late and a step change in control
Consequences of failure	one trial	a comparison, or several comparisons if shared controls, analyses, or governance fail

What stays the same: randomization within the platform defends each comparison against confounding exactly as in a two-arm trial, the estimand for each arm is still defined in the protocol table of Chapter 23, and the legal standard of substantial evidence of effectiveness is not lowered. A master protocol supports a marketing application for each drug on the evidence of that drug's comparison. What the regulator will ask about is what is new: the controls, the multiplicity rule, the governance, and the simulation.

Try it in the lab

Open module 10 of the companion tutorial, https://vsokolov.org/bayesian_trials_lab/10-platform-trials.html. Before you press the button, predict how many patients five separate trials need compared with one platform, then drop a drug into the platform in year 2 and check which control group the lab regards as fair when success on standard care is rising 2 points a year.

Key points

- A platform shares control patients: k comparisons need $(k + 1)m$ patients rather than $2km$, a saving of $(k - 1)/(2k)$, with the same precision per comparison. Five drugs at 80 per arm need 480 patients, not 800.
- The best allocation puts $\sqrt{k}$ patients on the control for each patient on an arm (Dunnett). FDA recommends more patients on the control than on any single arm, and the EMA draft cites the square-root rule.

- A shared control keeps the expected number of false positives at $k\alpha$ but makes them correlated: in the four-arm illustration the chance of at least one falls and the chance of several rises. Report the distribution of the number of errors.
- Non-concurrent controls carry drift. In the simulation, pooling them under a 2-point-per-year drift raised the one-sided type I error from about 2.5% to 8.1%. FDA asks for concurrently eligible controls for the primary analysis, and EMA generally rejects non-concurrent controls for the primary endpoint.
- Time-trend models and time-machine smoothers recover power when history is stable and lose it, or bias, when it breaks. In our three illustrated histories the loosest fixed smoother had a lower rmse than the concurrent-only analysis, but three scenarios do not show it is better for other histories (an abrupt drift, for example), and the smoothing bias must be checked.
- FDA generally does not recommend multiplicity adjustment across unrelated drugs in a master protocol, and expects the sponsor to define families (doses, combinations, endpoints of one drug) in advance. Hierarchical shrinkage across arms needs a defensible exchangeability argument.
- STAMPEDE shows a frequentist MAMS platform works. I-SPY 2 graduates arms by predictive probability, GBM AGILE adds a seamless registration stage, and REMAP-CAP combines domains, ordinal outcomes and adaptive allocation. The anticoagulation multiplatform pre-agreed one analysis across three platforms. For confirmatory platforms the EMA draft does not support response-adaptive randomization.
- Governance decides whether a platform opens: EPAD had the cohort and the design and never enrolled an arm.

Discussion questions

1. A sponsor has four drugs of its own for the same disease and proposes a platform with a shared control. Should the four comparisons share one familywise error rate under FDA's draft, and what would change your answer if two of the drugs were doses of the same compound?
2. In the drift simulation the time-adjusted analysis beat the concurrent analysis on power. Write the scenario in which it would not, and say what you would put in the simulation report to show it.
3. An arm opens in year 4 of a platform in which the standard of care changed in year 3. What would you pre-specify about the use of controls from years 1 and 2?
4. I-SPY 2 graduates an arm at a predictive probability of 85% for a later phase 3. A sponsor argues that the graduation should be at a posterior probability of 95%. What does each criterion say, and what trial does each assume?
5. List what EPAD had and did not have that Healey ALS had, and say which of those a statistician could have fixed.

Part V.

From Design to Approval

21. Case Studies: Cardiovascular, Alzheimer's and Vaccine Trials

The earlier chapters argued in general terms: that a Bayesian rule monitored continuously can reach the same conclusion as a fixed design with fewer events, that the saving comes from the monitoring and not from a lower bar, and that every gain has a price. This chapter tests the argument on five trials that were actually run, and it keeps the argument honest by choosing them to differ. SELECT is the favorable case: a large effect, cleanly estimated, in a trial big enough for the asymptotics to apply. REWIND is the unfavorable case: an effect half as large on the log scale, which barely cleared its threshold and in which the scheduled design wins. SURPASS-CVOT is an active-comparator case with a noninferiority margin, where the lesson is what the margin and a transportability assumption allow a Bayesian reading to say. TRAILBLAZER-ALZ 2 is a different kind of case: a trial with two prespecified design features, a biomarker-guided treatment duration and baseline tau strata, that a Bayesian analysis could be proposed to handle differently. BNT162b2, the last case, is a trial that was Bayesian in its confirmatory analysis as well as in its design, and that led to an emergency use authorization and a biologics license.

Each case follows the same order, in line with the rule of this book that the traditional design comes first. We say what the trial did and how it was monitored; we then reconstruct the Bayesian alternative on the trial's own published numbers; and we close with what a data monitoring committee (DMC) and a regulator would have had to accept in order to act on it. Everything in the SELECT and REWIND reconstructions and in the stratum illustration for TRAILBLAZER-ALZ 2 is a counterfactual, and we say so each time it matters. BNT162b2 is the exception: its Bayesian analysis was the trial's actual analysis. No sponsor ran these trials the way we simulate them, and nothing here is a claim that any of them should have stopped early. The reconstructions are a way of pricing the design choices in units that a sponsor, a payer and a regulator can all read: events, patients, years, quality-adjusted life years (QALYs) and money.

> **!** Regulatory anchor
>
> The governing texts for this chapter are the ones that decide whether a monitoring rule is acceptable, not the ones that decide whether a drug is. Status as of November 2026.
>
> - **ICH E9, Statistical Principles for Clinical Trials** (Step 4, 5 February 1998), Section 4.5: "all interim analyses should be carefully planned in advance and described in the protocol." A rule that looks at the data repeatedly is an interim-analysis plan, Specify the monitoring plan prospectively. E9 permits justified amendments before unblinded access to comparative data, with their statistical consequences addressed. Section 3.4 names group sequential designs as the usual way to look repeatedly.
> - **FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics** (final,

November 2019), Section III: the chance of erroneous conclusions should be adequately controlled, treatment-effect estimates should be sufficiently reliable, design details should be completely prespecified, and trial integrity should be maintained. Section VI.A: simulations are the way to estimate operating characteristics. These four principles are the test every reconstruction below has to pass.

- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products** (draft, January 2026), Section IV.A: the draft describes three routes for Bayesian success criteria. One is calibration to a type I error rate: for trials with an overall frequentist statistical analysis plan the draft says criteria are "almost always" chosen so that the familywise type I error is no greater than 0.025 one-sided, and a criterion of the form $\Pr(\delta > a \mid \text{data}) > c$ can have a and c chosen so that this holds (Section IV.A.1 uses $a = 0$ for calibrated superiority designs). The others are direct interpretation of posterior probabilities, for example with borrowing and by agreement with FDA, and decision-theoretic or benefit-risk criteria (Section IV.A.3). Calibration is this book's default for confirmatory adaptive designs without borrowing, and the route this chapter follows; it is one of FDA's routes and not its universal requirement, and a draft recommends and does not require. The prior must be prespecified (Sections IV.B and V).
- **FDA, Demonstrating Substantial Evidence of Effectiveness** (revised draft, June 2026): the statutory standard of adequate and well-controlled investigations, and the conditions under which one trial with confirmatory evidence can suffice. Cardiovascular outcomes trials are usually the only trial behind the cardiovascular claim, which makes this the standard in play.
- **FDA, Diabetes Mellitus: Evaluating Cardiovascular Risk in New Antidiabetic Therapies to Treat Type 2 Diabetes** (final guidance, December 2008; withdrawn in March 2020, when FDA issued the draft guidance "Type 2 Diabetes Mellitus: Evaluating the Safety of New Drugs for Improving Glycemic Control", which remains a draft), which asked sponsors of new type 2 diabetes drugs for cardiovascular safety evidence, in practice a comparison of the upper bound of the 95 percent confidence interval for the cardiovascular risk ratio against margins of 1.8 (to support approval) and 1.3 (to be shown, if the premarketing bound lies between the two, by a postmarketing trial) (U.S. Food and Drug Administration 2008). This guidance is the setting of REWIND. SELECT enrolled people without diabetes and is not governed by them.

None of these documents requires or forbids what is reconstructed here. The chapter's position is that the reconstructions would have to satisfy these texts, and it checks them one by one at the end of each case. See Chapter 3 for the full map, Chapter 11 for group sequential boundaries, Chapter 12 for e-values and Chapter 14 for the simulation report.

21.1. SELECT: semaglutide and cardiovascular events

21.1.1. The trial as conducted

SELECT randomized 17,604 adults with overweight or obesity and established cardiovascular disease, without diabetes, to weekly semaglutide 2.4 mg or placebo, at 804 sites in 41 countries, enrolling from 2018 to 2021 (Lincoff et al. 2023). The primary endpoint was a composite of cardiovascular death, nonfatal myocardial infarction and nonfatal stroke, analyzed as time to first event. It occurred in 569 of 8,803 patients on semaglutide and 701 of 8,801 on placebo: 1,270 events in all, hazard ratio 0.80 with a 95 percent confidence interval from 0.72 to 0.90, and $p < 0.001$. Mean follow-up was 39.8 months.

This is the traditional trial in its standard form: an event-driven design with a single primary hypothesis, a Cox model for the hazard ratio, an independent DMC, and a final analysis at which the type I error is spent. SELECT was designed for a hazard ratio of 0.83 (a 17 percent reduction) with 90 percent power at one-sided 0.025, which calls for 1,225 planned events, and it planned one superiority interim analysis at two thirds of the planned events (about 817) followed by the final analysis, with the group sequential design accounted for in the final test (D. H. Ryan et al. 2020). The textbook formula at the design hazard ratio gives about 1,211 events, close to the plan. The observed hazard ratio of 0.80 was more favorable than the design alternative. The four-look O'Brien-Fleming design below is a separate hypothetical comparator, and SELECT's actual two-look design is carried through every comparison as a fourth rule. The reconstruction asks a different question from the one the sponsor faced: if the true hazard ratio were 0.80 and the trial's information as observed, what would each monitoring scheme, each held to the same one-sided error of 2.5 percent, have needed?

In information time the trial is a clean object. With $d = 1{,}270$ events, the standard error of the log hazard ratio is approximately $2/\sqrt{d}$, because each event carries about $1/4$ of a unit of Fisher information under 1:1 allocation (this is the standard approximation for a log-rank or Cox analysis, and it is why $I_n = d/4$ throughout this chapter). The log hazard ratio is $\theta = -\log(0.80) = 0.2231$ in the book's sign convention, so that benefit is positive, and the final standardized score is $Z = 3.98$.

```
d_tot <- select_design$d_tot; HR <- select_design$HR; tau <-
 ↪  select_design$tau
theta <- -log(HR); se <- 2 / sqrt(d_tot)
z_final <- theta / se
```

The published interval implies a standard error of 0.0569 against our 0.0561, so this parametrization reproduces the trial to two digits. That is all the model has to do: it is a summary of how information accrued, not a model of the disease.

21.1.2. Three monitoring schemes and SELECT's own design, one error rate

A speed comparison means nothing unless the rules carry the same type I error under the monitoring each actually uses, and this is where naive comparisons fail. We compare three rules and add SELECT's actual design as a comparator.

1. **Group sequential, O'Brien-Fleming.** Four equally spaced looks at 25, 50, 75 and 100 percent of the events, with the O'Brien-Fleming boundaries on the Z scale of 4.049, 2.863, 2.337 and 2.024 (see Chapter 11; the working paper printed 4.05, 2.87, 2.34 and 2.03 from a rounded constant; the reconstruction uses the exact values above). It can act only at a scheduled look. This is the design most event-driven outcomes trials use today, and it is the one a regulator will expect to see first.
2. **Bayesian posterior probability, monitored on a fine event grid.** Stop for benefit when $\Pr(\theta > 0 \mid \text{data}) \geq p^*$, with a normal prior on the log hazard ratio centered at zero with scale $\tau = 0.25$ (a prior 95 percent interval for the hazard ratio of 0.61 to 1.63, a skeptical prior in the sense of Chapter 8), and p^* chosen so that the one-sided type I error is 2.5 percent when the rule is evaluated at every second event from 4 to 1,270. The simulator monitors on this grid and to this horizon, not literally continuously, and "continuous" below means this grid. This is the criterion $\Pr(\delta > a \mid \text{data}) > c$ of the FDA draft, with $a = 0$ and $c = p^*$.
3. **E-value mixture, monitored on the same grid.** Stop for benefit when the mixture e-value reaches $1/\alpha = 40$ (see Chapter 12). By Ville's inequality, this cannot have a type I error above 2.5 percent however often it is looked at, and in practice it realizes less.
4. **SELECT's actual design (comparator).** One superiority interim at two thirds of the 1,225 planned events (817) and a final analysis at 1,225, with Lan-DeMets O'Brien-Fleming-type spending of the one-sided 0.025 at the planned interim, as in the SELECT statistical analysis plan (section 2.1, Table 2-1: nominal one-sided levels of about 0.00605 at the interim and 0.02314 at the final). Its boundaries on the Z scale are reported with the results below. We do not calibrate it; we report its realized error. Crossing the boundary is necessary but not sufficient for stopping: at the actual interim, at 831 events, the efficacy boundary was crossed and the data monitoring committee recommended continuing (FDA clinical review of NDA 215256 s011, pp. 65-66).

The Bayesian and e-value rules use the same normal prior with the same scale and are functions of the same score; the Bayesian rule thresholds the posterior probability of benefit, the e-value rule the mixture likelihood ratio. Because the Bayesian rule's type I error is calibrated by simulation, it is acceptable under the calibration route of the FDA draft: the posterior is the way the decision is expressed, and the frequentist operating characteristics are the way it is justified.

The simulation below uses `select_simulate()` from the companion package, which ports the code of the working paper. It generates 40,000 event-time score paths on the grid of every second event (plus 817, 1,225 and 1,270) with the hazard ratio held at the observed 0.80 for the power calculation and at 1.0 for the type I error, evaluates each rule on the same paths, and calibrates p^* as the empirical $1 - \alpha$ quantile of the maximum of the posterior probability along each null path.

```
res <- select_simulate(nsim = 40000, seed = 31415)
```

```
# Independent check of the calibrated threshold: 40,000 fresh null score
 ↪  paths (a different seed) on the same event grid.
set.seed(2718)
I_g <- res$dg / 4; A_g <- I_g + 1 / tau^2; dI <- diff(c(0, I_g)); K <-
 ↪  length(I_g)
hit <- 0; nval <- 40000
```

```
for (b in 1:10) {
  S <- t(apply(matrix(rnorm(4000 * K, sd = rep(sqrt(dI), each = 4000)),
 ↪  4000), 1, cumsum))
  hit <- hit + sum(apply(pnorm(sweep(S, 2, sqrt(A_g), "/")), 1, max) >=
 ↪  res$pstar)
}
val_typeI <- hit / nval; se_mc <- sqrt(0.025 * 0.975 / nval)
```

A posterior probability rule at the conventional threshold of 0.975, monitored on the same grid, has a type I error of about 11.8 percent rather than 2.5. That is the phenomenon of a repeated-significance test (Chapter 11) in a different notation. Any claim that a Bayesian rule decides sooner at the conventional threshold is an artifact unless the error is reported with it. The calibrated threshold is 0.9952. Selecting an empirical null quantile fixes the rejection proportion in the calibration sample at 2.5 percent; it does not make the population error exactly 2.5 percent. With 40,000 paths the binomial standard error near 0.025 is about 8×10^{-4}. On 40,000 fresh null paths with a different seed the calibrated threshold rejects in 2.68 percent, which is 0.18 points from 2.5 percent, or 1.6 standard errors of the difference between two independent estimates of that precision. The realized error of the calibrated rule is therefore known only to within a few tenths of a point, which is why the calibration and its validation should use more paths than a protocol-stage sketch needs. A protocol would repeat the validation, and the calibration, across its nuisance scenarios (event rate, accrual, nonproportional hazards). The e-value rule realizes well under its 2.5 percent budget, in our run 0.58 percent, because Ville's inequality protects against every stopping time and this trial uses only a few of them; Polson and Sokolov report 0.55 percent in their run of the same design, and the two differ by less than two Monte Carlo standard errors (Polson and Sokolov 2026b). The four-look O'Brien-Fleming design realizes about 2.55 percent because its looks are discrete, and SELECT's actual design 2.54 percent; both agree with 2.5 percent within Monte Carlo error. The comparator's boundaries are 2.51 at 817 events and 1.99 at 1,225 on the Z scale.

Table 21.1.: SELECT reconstruction, hazard ratio held at 0.80, prior scale 0.25, 40,000 simulated paths. The first events column is conditional: the median among simulated trials that declare efficacy. The all-trial columns count a trial that never crosses as ending at its maximum (1,270 events, or the planned 1,225 for SELECT's actual design). Power is the probability of crossing. The Bayesian rule is calibrated to 2.5 percent one-sided on the event grid; the realized one-sided type I error of every rule is shown (Monte Carlo standard error about 0.0008).

rule	median events, trials that declare efficacy	median events, all trials	mean events, all trials	power	type I error
O'Brien-Fleming, four scheduled looks, alpha = 0.025	952	952	835	97.7%	2.55%
SELECT actual design: interim at 817 of 1225 events, then final	817	817	917	97.4%	2.54%
E-value mixture, continuous, E >= 40	612	714	743	83.1%	0.58%

Table 21.1.: SELECT reconstruction, hazard ratio held at 0.80, prior scale 0.25, 40,000 simulated paths. The first events column is conditional: the median among simulated trials that declare efficacy. The all-trial columns count a trial that never crosses as ending at its maximum (1,270 events, or the planned 1,225 for SELECT's actual design). Power is the probability of crossing. The Bayesian rule is calibrated to 2.5 percent one-sided on the event grid; the realized one-sided type I error of every rule is shown (Monte Carlo standard error about 0.0008).

rule	median events, trials that declare efficacy	median events, all trials	mean events, all trials	power	type I error
Bayes posterior, continuous, $p^* = 0.995$	464	486	553	94.7%	2.50%
The trial as conducted	1270	1270			

The headline numbers are in Table 21.1, and every stopping time is stated with its conditioning. Among simulated trials that declare efficacy, the median stopping point is 464 events for the Bayesian rule, 612 for the e-value rule and 952 for the four-look O'Brien-Fleming design. Over all simulated trials, counting a trial that never crosses as ending at its maximum, the medians are 486, 714 and 952. The all-trial figures are the fairer basis for comparing rules of different power: the e-value rule reaches its bar in only 83 percent of replications even at full information, so its conditional median describes only the trials that succeed, and its all-trial median is 47 percent above the Bayesian rule's (32 percent on the conditional scale). Power is 94.7 percent for the Bayesian rule, 83.1 for the e-value rule and 97.7 for the four-look design. SELECT's actual two-look design stops at a median of 817 events (it ends at 1,225 if it never crosses), with power 97.4 percent and realized type I error 2.54 percent. Against that actual design the Bayesian rule's all-trial median is 41 percent lower, with 2.7 points less power. The four-look design, at 952, is later than SELECT's actual design for two reasons: it cannot act between looks, and its early boundaries are very high. Crossing the first interim boundary of 4.05 would need the equivalent of about 1,300 events at this effect size, more than the trial ever accrued, so the first interim analysis of such a design is, on the mean path, arithmetically incapable of stopping this trial (about 2 percent of simulated paths cross 4.05 at that look). The conditional medians (464, 612, 952) are close to the working paper's 464, 608 and 952 (Polson and Sokolov 2026b); the R random stream differs from the Python stream used there, and every simulated number in this book is the R one.

The two group sequential designs have the highest power in the table (97.7 and 97.4 percent), and the reason is instructive: they spend their error budget where the information is, and do not need to protect against stopping times they never use. The Bayesian rule is faster and has a little less power (94.7 percent). That trade is the price of monitoring on a fine grid, so the table is a trade and not a ranking. A sponsor who will in fact look only a few times has no reason to buy protection for looking at every second event.

Figure 21.1 shows the same story on the scale a DMC works on. On the mean path the statistic Z grows like $\sqrt{d}$, and each rule is a curve (or four points) that the statistic has to cross.

The figure also shows why the conditional medians in the table are not the mean-path crossings (about 600 events for the Bayesian bar and 870 for the e-value bar, mean-path approximations and not stopping times): a rule that can act at any moment stops early on

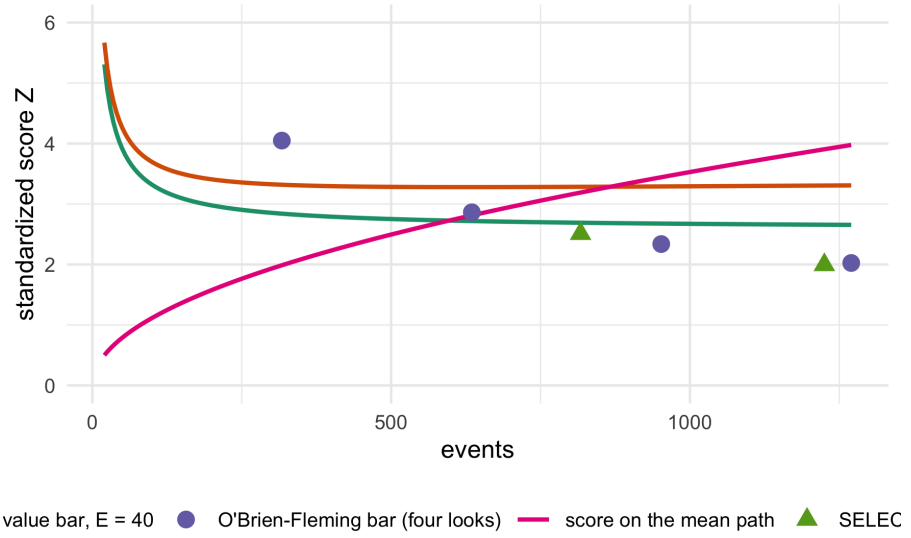


Figure 21.1.: SELECT on the mean path (hazard ratio 0.80). The standardized score grows with the square root of the events. The Bayesian bar (calibrated posterior threshold) and the e-value bar are curves; the O'Brien-Fleming bar exists only at the four scheduled looks, and SELECT's actual design has two (817 and 1,225 events). Each rule acts where the score first crosses its bar. The mean path is an approximation, not a trial.

the paths where noise has helped, and the median over paths that succeed sits below the crossing of the average path. The group sequential rules have far fewer moments at which noise can help them.

The ordering of the Bayesian and e-value rules is robust to the prior scale. Over $\tau \in \{0.15, 0.25, 0.40, 0.60\}$, whose prior intervals for the hazard ratio run from tight to very loose, the calibrated Bayesian rule stops earlier than the e-value rule, on both the conditional and the all-trial scale (Table 21.2):

```
taus <- c(0.15, 0.25, 0.40, 0.60)
sens <- do.call(rbind, lapply(taus, function(tau) {
  r <- select_simulate(d_tot, HR, nsim = 20000, tau = tau, seed = 5000 +
 ↪  round(100 * tau))$rows
  data.frame(tau = tau, prior_HR = sprintf("%.2f to %.2f", exp(-1.96 *
   ↪  tau), exp(1.96 * tau)),
             bs = r$median_events_success[3], ba = r$median_events_all[3],
              ↪  pb = r$power[3],
             es = r$median_events_success[2], ea = r$median_events_all[2],
              ↪  pe = r$power[2])
}))
knitr::kable(data.frame(sens$tau, sens$prior_HR, sprintf("%.0f / %.0f
 ↪  (%.1f%%)", sens$bs, sens$ba, 100 * sens$pb),
                         sprintf("%.0f / %.0f (%.1f%%)", sens$es, sens$ea,
                          ↪  100 * sens$pe)),
```

```
            col.names = c("tau", "prior 95% HR interval", "Bayes
             ↪  conditional / all-trial median (power)", "e-value
             ↪  conditional / all-trial median (power)"), align = "rlrr")
```

Table 21.2.: Sensitivity of the SELECT reconstruction to the prior scale tau: median events among trials that declare efficacy, median events over all trials, and power in parentheses, for the calibrated Bayesian and e-value rules, 20,000 simulated paths per row.

tau	prior 95% HR interval	Bayes conditional / all-trial median (power)	e-value conditional / all-trial median (power)
0.15	0.75 to 1.34	494 / 510 (95.4%)	674 / 762 (83.1%)
0.25	0.61 to 1.63	476 / 502 (93.9%)	612 / 710 (82.9%)
0.40	0.46 to 2.19	462 / 494 (93.1%)	594 / 715 (81.0%)
0.60	0.31 to 3.24	466 / 506 (92.5%)	618 / 766 (78.1%)

In our run the Bayesian conditional median ranges from 462 to 494 events and the all-trial median from 494 to 510, and the e-value conditional median from 594 to 674 and its all-trial median from 710 to 766, with power as shown. The ranges differ slightly from those in the working paper, which are for a different random stream; the ordering does not. All of it is conditional on a hazard ratio held at 0.80, which is the point estimate of a trial that succeeded. The point estimate of a trial that succeeded is on average somewhat optimistic, so a trial run at the true effect would, to that extent, need more events than these medians, and this reconstruction is not a power calculation made before the trial. It is a description of what each rule needs to see if the world is as the trial found it.

21.1.3. What the saving would be worth in calendar time and exposure

From here to the end of the chapter every conversion of events into time, exposure, QALYs or money is a modeled counterfactual: it combines the simulated stopping distribution with an accrual model and with assumptions about what would have happened to patients after a decision. Translating events into calendar time needs an accrual model. We take enrollment linear over 36 months, the trial running to 60 months, and events accruing in proportion to cumulative person time, which reproduces a mean follow-up of 42 months against the reported 39.8.

```
m <- select_month_of_events(c(Bayes = mda_b, `e-value` = mda_e,
 ↪  `O'Brien-Fleming` = mda_o, `SELECT actual` = mda_a), d_tot)
m_b <- m[["Bayes"]]
```

The all-trial median decision points fall at month 34 for the Bayesian rule, month 41.6 for the e-value rule, month 49.5 for the four-look group sequential design and month 45 for SELECT's actual design, against a read-out at month 60. The Bayesian median falls about 8 months before the e-value median, 15 months before the four-look design's, 11 months before SELECT's actual design's, and 2.2 years before the trial's read-out.

The quantity that matters clinically is exposure. Total exposure in SELECT was about 58,400 patient-years (enrollment times the reported mean follow-up of 39.8 months), of which about 29,200 were on placebo. The accounting below (Table 21.9) converts the expected stopping point of the Bayesian rule, averaged over all simulated trials with the trials that never cross counted at their maximum, into the placebo patient-years and placebo events that would come after it. Three assumptions carry it: events are uniform in person time, placebo participants would have been offered active treatment at the decision so that the events they would otherwise have had are prevented at the trial's observed relative reduction, and the decision would have been acted on. The figures are an order of magnitude and not a count. For REWIND the median follow-up of 5.4 years stands in for the mean, and the two are not the same.

21.1.4. What the monitoring committee would have seen

The medians in Table 21.1 are a summary. The question a DMC faces is concrete: at this look, with these data, does the rule permit a recommendation? Table 21.3 answers it on the mean path at the four looks a group sequential design would have scheduled.

Table 21.3.: SELECT look by look on the mean path with the hazard ratio at 0.80. The e-value and Bayesian rules are not restricted to these looks; they are shown there for comparison. The e-value bar is log 40 = 3.69. The Bayesian bar is the calibrated threshold p*.

events	Z	OBF bar	OBF stops	log E	e-value stops	Pr(benefit)	Bayes stops
318	1.99	4.05	no	0.75	no	0.9651	no
635	2.81	2.86	no	2.40	no	0.9963	yes
952	3.44	2.34	yes	4.17	yes	0.9996	yes
1270	3.98	2.02	yes	6.01	yes	0.9999	yes

Two readings of the table should be separated. At the first interim, the group sequential committee is shown a standardized score of 1.99 against a boundary of 4.05 and must continue. The Bayesian committee is shown a posterior probability of benefit of 96.5 percent and must also continue, since its calibrated threshold is 99.5 percent. The two agree. At the second interim they part. The score is 2.81 against a boundary of 2.86, so the four-look design continues by a margin of 0.05; the e-value stands at $\log E = 2.4$ against the 3.69 it needs; and the posterior probability of benefit is 99.63 percent, which clears the calibrated threshold, so the Bayesian rule stops. On the mean path, the four-look design stops at the third look, another 317 events after the Bayesian decision at the second look.

The near miss at the second look is a reminder of how much turns on the spacing of looks, which a group sequential design fixes in advance. The calibrated Bayesian rule may act at any count on its prespecified monitoring grid, within the 1,270-event horizon. It is also a reminder of why a mean path is not a trial: a single real path would sit above or below 2.81 at that look, and the four-look rule would have stopped on a substantial fraction of paths there. SELECT's actual interim, at 817 events, would show a mean-path score of 3.19 against a boundary of 2.51, but the trial's actual interim came at 831 events, where the efficacy boundary was crossed and the data monitoring committee recommended continuing (FDA clinical review of NDA 215256 s011, pp. 65-66). Crossing is necessary but not sufficient

for stopping, which is one more reason to read the reconstruction through the all-trial table and not through the mean path.

> **i** Technical note: why the e-value rule is slow here
>
> The e-value rule uses the same normal prior scale τ and the same score S as the Bayesian rule, with $I_n = d/4$ events-based information and $A = I_n + 1/\tau^2$. The simulated Bayesian rule acts when $\Pr(\theta > 0 \mid \text{data}) = \Phi(S/\sqrt{A}) \geq p^*$, which on the scale $Z = S/\sqrt{I_n}$ is $Z \geq \sqrt{1 + 1/(I_n\tau^2)}\, z_{p^*}$ with $z_{p^*} = \Phi^{-1}(p^*)$, or $Z^2 \geq (1 + 1/(I_n\tau^2))\, z_{p^*}^2$ with $Z > 0$. The e-value rule acts when the mixture likelihood ratio E_n reaches $1/\alpha$, which on the same scale is $Z^2 \geq (1 + 1/(I_n\tau^2))\,[\log(1 + I_n\tau^2) + 2\log(1/\alpha)]$ with $Z > 0$. At $\tau = 0.25$ and $p^* = 0.9952$ the Bayes bar is $Z^2 \geq 7.6$ at 464 events and 7.0 at 1,270, and the e-value bar is 10.8 and 10.9, so the e-value rule asks for 3.2 more units of Z^2 at 464 events and 3.9 more at 1,270. The constant $2\log(1/\alpha) = 7.38$ at $\alpha = 0.025$ is, for this normal-mixture construction and in the large-information approximation, the surcharge of the e-value bar over the uncalibrated Bayes-factor rule at posterior odds 1, $Z^2 \geq \log(1 + I_n\tau^2)$, of Chapter 11 and Chapter 12; the exact surcharge is $7.38\,(1 + 1/(I_n\tau^2))$. It is not a universal cost of anytime validity, since other e-processes behave differently. It is also not the surcharge over the calibrated posterior-probability rule simulated here, whose bar p^* was itself raised by simulation to pay for monitoring on the grid. The e-value rule's realized type I error of about 0.6 percent against a budget of 2.5 percent is the visible form of the remaining difference: it is paying for protection at stopping times this trial never used. In this construction the surcharge is a constant in the large-information approximation, and its relative cost falls only as $1/\log I_n$; see Chapter 12.

21.1.5. What a DMC and a regulator would have had to accept

The reconstruction is only useful if the Bayesian rule could have been proposed, accepted and run. Walk through the four principles of the FDA adaptive-designs guidance (U.S. Food and Drug Administration 2019a) and the success-criterion language of the Bayesian draft (U.S. Food and Drug Administration 2026g).

Error control. This is met by construction in the reconstruction, on the event grid and to the 1,270-event horizon, because p^* was found by simulation so that the one-sided familywise type I error under the monitoring plan is 2.5 percent. That fixes the rejection proportion in the calibration sample, not the population error, so a protocol would report the Monte Carlo uncertainty (about 0.0008 at 40,000 paths), validate the threshold on independent null simulations as above, and repeat both across nuisance scenarios. The draft describes calibration as one of three routes, and for a design with a frequentist analysis plan and no borrowing it is the usual one. The weakness is that the calibration holds under a specific monitoring plan. A rule calibrated to a fine grid is conservative for a DMC that meets twice a year, because any subset of the grid inherits the bound. A rule calibrated to meetings twice a year is anti-conservative if the committee also looks ad hoc, for example after a safety signal. A DMC that cannot commit to a schedule can calibrate to a dense grid at the cost of a higher threshold, but monitoring off the grid or beyond the horizon needs recalibration, and a trial that cannot bound its looks should use a fixed design, a spending function or an anytime-valid rule.

Reliable estimation. A trial that stops at the conditional median of 464 events yields a hazard ratio estimate that is biased away from the null on average, as is true of trials that stop early for efficacy, and the confidence interval is wider than the interval from 1,270 events by about $\sqrt{1270/464} = 1.65$. The Bayesian posterior shrinks the estimate towards the prior, which is a correction in the right direction but not a free one. The trial's other jobs would also suffer: the key secondary endpoints, the safety database for an agent intended for long-term use, and the subgroup results a label depends on all need events and patient-years, not just a primary endpoint that crossed a bar. For a drug intended for chronic use by people without diabetes, a regulator may well prefer the longer exposure. This is a real cost and the reason the calendar saving in this chapter should not be read as a recommendation.

Prespecification. The prior scale, the threshold, the monitoring schedule and the stopping rule would all have to be written into the protocol and the statistical analysis plan, prospectively. E9 requires interim analyses to be planned and described in the protocol (Section 4.5) and permits justified amendments before unblinded access to comparative data, with their statistical consequences addressed (International Council for Harmonisation 1998). The simulation report that supports them is the operating-characteristics evidence the FDA asks for (Chapter 14).

Trial integrity. The DMC would have been shown a posterior probability and a threshold instead of a boundary. The sponsor would remain blind to the interim comparisons, as it does under a group sequential plan. Nothing about the Bayesian rule changes the firewall; what changes is the number the committee reads.

The regulator's question about the prior is the one a group sequential design never faces. Here it is answered the way the draft asks: the prior is skeptical, it is prespecified, and its influence on the decision is bounded by the calibration, since a different τ changes p^* and leaves the simulated type I error unchanged (see Chapter 8). A regulator who accepts that the final decision rule is "reject when the calibrated posterior probability exceeds p^*" is accepting a test with simulated size and power, which is also what it accepts from an O'Brien-Fleming design. In the language of the substantial-evidence draft (U.S. Food and Drug Administration 2026a), the evidence in either case is a single well-controlled trial whose persuasiveness comes from its effect size and its error control, and a hazard ratio of 0.80 with a final score near 4 is persuasive under any of the rules.

21.2. REWIND: where the scheduled design wins

21.2.1. The trial as conducted

REWIND randomized 9,901 patients with type 2 diabetes to weekly dulaglutide or placebo, enrolling from August 2011 to August 2013 with a median follow-up of 5.4 years, and reported in 2019 (Gerstein et al. 2019). The primary composite cardiovascular outcome occurred in 594 of 4,949 on dulaglutide and 663 of 4,952 on placebo, so 1,257 events, with a hazard ratio of 0.88, a 95 percent interval from 0.79 to 0.99, and $p = 0.026$.

REWIND ran in the regulatory setting of the 2008 diabetes guidance (U.S. Food and Drug Administration 2008), which asked sponsors of new glucose-lowering drugs to show that the drug did not raise cardiovascular risk beyond a prespecified margin on the hazard ratio. That is the historical context, but REWIND itself was designed to demonstrate cardiovascular superiority, with 90 percent power, and not merely to exclude the 2008 margin (Gerstein et

al. 2019). A reconstruction that asks about stopping for benefit is therefore as natural for REWIND as for SELECT, and the contrast below is about the size of the effect and not about the purpose of the trial.

The information content is almost identical to SELECT, 1,257 events against 1,270. The effect is half the size on the log scale, 0.128 against 0.223, so the final standardized score implied by the rounded hazard ratio is 2.27 rather than 3.98 (2.23 is implied by the published two-sided p value of 0.026; the gap is rounding in the hazard ratio). This is a trial that barely cleared its threshold, and it is exactly the case in which the advantages of continuous monitoring should be expected to shrink.

21.2.2. The Bayesian reconstruction

Table 21.4.: REWIND reconstruction, hazard ratio held at 0.88, 40,000 simulated paths, the three rules of the SELECT table (SELECT's own two-look design is not shown, since it was built for SELECT). The conditional median is among simulated trials that declare efficacy; the all-trial median counts a trial that never crosses as ending at the trial's 1,257 events. Power is the probability of crossing by 1,257 events. Compare the SELECT table.

rule	median events, trials that declare efficacy	median events, all trials	mean events, all trials	type I power error
O'Brien-Fleming, four scheduled looks, alpha = 0.025	942	1257	1112	60.7%2.57%
E-value mixture, continuous, E >= 40	766	1257	1123	25.7%0.60%
Bayes posterior, continuous, p* = 0.995	680	1257	973	49.1%2.50%
The trial as conducted	1257	1257		

Three things change from SELECT, and all three come from the weaker effect.

The speed advantage shrinks, and the conditioning matters more. Among trials that declare efficacy the Bayesian rule acts at a median of about 680 events, 54 percent of the trial's total, against 37 percent in SELECT. But it declares efficacy in only 49 percent of the trials, so over all trials the median ending point is the maximum, 1,257 events, and the mean is 973 (77 percent of the total, against 44 percent in SELECT). A conditional median here describes only the trials that happened to succeed. A weaker effect takes longer to establish at any bar, and the relative gain from lowering the bar falls accordingly.

Power falls for every rule and the gap between the rules widens. In SELECT the calibrated Bayesian rule had power of 94.7 percent against 97.7 percent for the four-look design; here the four-look design again has the highest power, about 60.7 percent against the Bayesian rule's roughly 49 percent. A design that concentrates its error budget at the final analysis is better placed to detect an effect that only ever reaches $Z = 2.27$ than a rule that must hold its level against monitoring on a fine grid.

The e-value rule fails outright. Its power is about 26 percent, so about 74 percent of the time it would not have reached its bar by the end of a trial of this size at this effect. Its realized type I error is about 0.6 percent against a budget of 2.5. This is the surcharge at its most damaging: on the mean path the trial's total evidence is $Z^2 = 5.1$, against an exact e-value bar at 1,257 events of 10.9, of which the large-information approximation of the surcharge over the uncalibrated Bayes-factor bar is 7.38 (the exact surcharge is $7.38(1 + 1/q)$, about 7.8 at $q = 19.6$). A bar that high cannot be met on the mean path. Sokolova and Sokolov report the same phenomenon in a mortality trial with a small absolute effect, where the e-value rule needed several times the patients the trial enrolled (A. O. Sokolova and Sokolov 2026).

```
lt_rw <- select_look_table(1257, 0.88)
knitr::kable(round(lt_rw, 3), row.names = FALSE, align = "r")
```

Table 21.5.: REWIND look by look on the mean path (hazard ratio 0.88, 1,257 events): standardized score, O'Brien-Fleming bar, log e-value and posterior probability of benefit at each scheduled look.

events	z	obf_bound	log_bf	post_prob
314	1.133	4.049	-0.355	0.849
628	1.602	2.863	-0.026	0.937
943	1.963	2.337	0.426	0.971
1257	2.266	2.024	0.930	0.986

The look-by-look table (Table 21.5) tells the same story as the medians. On the mean path the four-look design does not stop at the first three looks and crosses at the last, where the score is 2.27 against a bar of 2.02. The Bayesian posterior probability of benefit reaches 0.986 at the end, and the calibrated threshold of 0.995 is not reached on the mean path at any of the four looks. The Bayesian rule that stops at a conditional median of 680 events does so on the paths where chance has helped, and fails to stop on about 51 percent of paths.

21.2.3. What a DMC and a regulator would have had to accept

The honest conclusion for REWIND is that the scheduled design was the right choice, and a Bayesian statistician advising the sponsor would have said so if the expected effect was a hazard ratio near 0.88. Four things follow for a protocol.

First, the reconstruction does not rescue the argument for continuous monitoring in a marginal trial, and it should not be offered to a regulator as if it did. The regulator's concern with a glucose-lowering drug would also include the one stated in the 2008 guidance: precision at the upper bound of the interval, which is a property of the full trial and cannot be got from half of it.

Second, the same Bayesian machinery is useful for a different purpose in such a trial. A posterior distribution for the hazard ratio from the final data, with a skeptical prior, answers the question the 2008 guidance asks (is the upper bound of the effect on cardiovascular risk below the margin?) in the units of the margin. This does not change when the trial can stop; it changes how the result is reported (Chapter 6 and Chapter 7).

Third, the contrast between SELECT and REWIND is itself the finding. The saving from continuous Bayesian monitoring grows with the size of the effect relative to the precision, which a sponsor knows only roughly in advance. The assurance calculation of Chapter 9 is the tool for deciding which design to pick before the trial: the average of the probability of success over a prior on the effect. A prior that puts most of its mass on hazard ratios near 0.80 favors continuous monitoring; a prior centered at 0.90 does not.

Fourth, the reconstruction conditions on the observed effect. For REWIND this is a hazard ratio of 0.88 that barely cleared the bar. Had the true effect been closer to 0.90 the trial would more likely have failed, and the reconstruction would show fewer simulated trials declaring efficacy at a true hazard ratio of 0.90. Assessing early futility would require an additional prespecified rule and a separate operating-characteristics calculation (Chapter 15).

21.2.4. What the final data say: REWIND under skeptical priors

The reconstruction asks when a decision could have been taken. A separate question is what the final data say, and there the normal approximation is enough. The log hazard ratio is estimated at $\log 0.88 = -0.128$ with a standard error of about 0.058, read from the published interval. Table 21.6 gives the posterior under a flat prior and under three skeptical priors centered on no effect, with prior standard deviations of 0.2, 0.1 and 0.05 on the log hazard ratio (a standard deviation of 0.05 says that a hazard ratio below 0.90 or above 1.11 is a two-standard-deviation event before the trial).

Table 21.6.: REWIND, normal approximation to the log hazard ratio (estimate log 0.88, standard error 0.058 from the published interval). Each prior is normal on the log hazard ratio with mean zero and the standard deviation shown, or flat. Posterior hazard ratio, 95 percent credible interval, probability of any benefit and probability of a reduction of at least 10 percent.

prior on log HR	posterior HR	95% interval	Pr(HR < 1)	Pr(HR < 0.9)
flat	0.880	0.79 to 0.99	0.987	0.65
N(0, 0.2^2)	0.889	0.80 to 0.99	0.984	0.59
N(0, 0.1^2)	0.908	0.82 to 1.00	0.973	0.43
N(0, 0.05^2)	0.947	0.88 to 1.02	0.927	0.09

The direction of the effect is robust to any reasonable skepticism, and its size is not. A clinician asking whether dulaglutide lowers cardiovascular risk can be fairly confident that it does (the probability is 0.93 even under the strictest prior). A payer asking whether it lowers risk by 10 percent or more should be much less confident: that probability falls from 0.65 under a flat prior to 0.09 under the strictest, and the answer depends on how skeptical the payer's prior is. This is the same lesson as the reconstruction, seen from the end of the trial.

21.3. SURPASS-CVOT: an active comparator and a noninferiority margin

The preceding two cases compared a drug with placebo. SURPASS-CVOT compared two active drugs, which changes the question (Nicholls et al. 2025). It randomized 13,299 adults with type 2 diabetes and established atherosclerotic cardiovascular disease, double blind and 1:1, to weekly tirzepatide (up to 15 mg) or dulaglutide 1.5 mg, the dose that REWIND had shown to reduce cardiovascular events. After 134 exclusions for ineligibility the modified intention-to-treat population was 13,165 (6,586 on tirzepatide and 6,579 on dulaglutide). The primary outcome was the first of cardiovascular death, myocardial infarction or stroke, and the primary hypothesis was noninferiority: the upper limit of the 95.3 percent confidence interval for the hazard ratio had to lie below 1.05, and superiority could be claimed if the upper limit was below 1.00. The traditional logic of the noninferiority test, a margin fixed in advance and a one-sided bound on the interval, is the one set out in Chapter 4.

The primary outcome occurred in 801 patients (12.2 percent) on tirzepatide and 862 (13.1 percent) on dulaglutide, a hazard ratio of 0.92 with a 95.3 percent confidence interval from 0.83 to 1.01. Noninferiority was established ($p = 0.003$) and superiority was not ($p = 0.09$). We omit the secondary and component hazard ratios, which we have not checked against the full report.

Table 21.7.: SURPASS-CVOT, normal approximation to the log hazard ratio (estimate log 0.92, standard error from the 95.3 percent interval 0.83 to 1.01, flat prior). The first three rows are posterior probabilities for tirzepatide against dulaglutide. The last row is the indirect hazard ratio against placebo, the product of the SURPASS-CVOT and REWIND hazard ratios with standard errors added in quadrature (a flat prior on each, no discount for the difference in populations).

quantity	value
Pr(HR $<$ 1.05), noninferiority	0.996
Pr(HR $<$ 1), superiority	0.954
Pr(HR $<$ 0.9), a reduction of 10% or more	0.33
Indirect HR against placebo (95% interval)	0.81 (0.70 to 0.94)

The Bayesian reading (Table 21.7) answers the questions a prescriber asks, and it does so on a continuous scale. The posterior probability that tirzepatide is noninferior, that is, that the hazard ratio is below 1.05, is 0.996, and the probability that it is superior is 0.954. The trial "failed" to show superiority because the upper limit of 1.01 sat just above 1.00, yet the probability that the hazard ratio is below one is 0.95. These are not competing statements: the frequentist $p = 0.09$ is the chance, if the hazard ratio were exactly one, of data at least this extreme in either direction, and the posterior probability is a statement about the hazard ratio given the data and a flat prior. Neither is wrong, and a decision-maker needs the second to weigh a benefit against a cost. The probability of a reduction of at least 10 percent is only 0.33, which is the honest size of the uncertainty about how much better tirzepatide is. As with REWIND, the posterior under a skeptical prior would be pulled toward one, and for a trial of this size the flat-prior numbers are the upper end of what the data support.

Tirzepatide was never tested against placebo for cardiovascular outcomes in this trial, and the missing comparison can be filled by an indirect chain through dulaglutide, shown in the

last row of Table 21.7 (0.81, interval 0.70 to 0.94). The assumption of transportability from REWIND is the whole of the argument, and it is exactly what a Bayesian borrowing model makes explicit and discounts. The calculation, the discounted versions and the comparison with the published indirect analysis of Sattar et al. (2026) are in Section 19.12; the clinical point is that two numbers for the same question differ in what they adjust for, and an indirect comparison rests on its assumptions.

For Bayesian reanalyses of other landmark trials, with the priors and the arithmetic laid out, see Chapter 22.

21.4. TRAILBLAZER-ALZ 2: a prespecified biomarker strategy and baseline tau strata

21.4.1. The trial as conducted

TRAILBLAZER-ALZ 2 randomized 1,736 participants with early symptomatic Alzheimer's disease to donanemab or placebo, with efficacy measured at 18 months (76 weeks), and reported a 35 percent slowing of decline on its integrated Alzheimer's Disease Rating Scale (iADRS) in the prespecified low and medium tau population of 1,182 participants, and a 22 percent slowing in the combined population of 1,736 (Sims et al. 2023). The primary analysis was prespecified for two populations, the tau-defined one and the combined one, and the trial therefore had to choose how to spend its type I error between them. The order of testing is given in the publication and its supplement (Sims et al. 2023).

This is a traditional design in the sense of this book: a fixed duration, a prespecified primary analysis with a multiplicity procedure across populations, and a single final analysis. It differs from a textbook parallel-group trial in two respects. Both are prespecified design features, and neither is Bayesian as conducted; a Bayesian analysis could be proposed for the second.

21.4.2. Two prespecified design features

The first is that treatment duration was guided by a biomarker. Participants completed their course of donanemab once they reached a prespecified level of amyloid plaque clearance, and the report gives 52 percent of the low and medium tau participants as having reached the clearance criterion by one year, among those with a PET assessment at that visit (Sims et al. 2023). This is a stopping rule applied to each participant rather than to the trial, driven by an intermediate biomarker. It is a prespecified biomarker-guided treatment strategy, and it does not require Bayesian inference. It is related to individualized dosing in the patient-specific dose-finding designs reviewed in Chapter 17, and because it depends only on each participant's own biomarker and not on accumulating comparative results, it is a treatment strategy and not a trial adaptation in the sense of the FDA guidance. It is also a rule that a fixed-duration comparison cannot evaluate on its own: the trial estimates the effect of the strategy "stop when cleared" against placebo, not the effect of a fixed dose and duration, and the estimand has to say so (ICH E9(R1), Chapter 5).

The second is baseline tau stratification. The trial reports a 35 percent effect in one stratum and a 22 percent effect in the combined population. Tau status is measured at baseline, and the strata do not change in response to accumulating results, so this is baseline stratification

and not an adaptive feature. The standard apparatus requires that the populations and the order of testing be designated before the data arrive, and the designation then governs the claim. A prespecified hierarchical model (Chapter 19, Chapter 10) could estimate effects in both disjoint strata with borrowing between them. Population claims and their success criteria would still require prespecification and an evaluation of false-positive risks. The sizes of the two strata, 1,182 participants in the low and medium tau stratum and 552 in the high tau stratum as reported, do not by themselves establish that borrowing would be large; that depends on the between-stratum variance, as the illustration below shows.

The arithmetic of that shrinkage is simple enough to show. Suppose the two strata have estimated effects with standard errors proportional to $1/\sqrt{n}$, and the between-stratum standard deviation τ_b of the true effect is the quantity the data cannot tell us much about. The weight on the stratum's own estimate is $\tau_b^2/(\tau_b^2 + s^2)$, where s is its standard error and τ_b is the between-stratum standard deviation. The smaller the stratum and the smaller the between-stratum variance, the more the estimate is pulled to the pooled one.

```
# Illustration only: two strata with effect estimates on a common scale,
 ↪  standard errors scaling as 1/sqrt(n).
# These are NOT the trial's estimates; they show how the shrinkage weight
 ↪  depends on stratum size and tau.
n <- c(1182, 552)
s <- 10 / sqrt(n)                  # arbitrary common scale, so the weights
 ↪  are what matter
tau_b <- c(0.1, 0.2, 0.4)           # between-stratum sd, same scale
W <- t(sapply(tau_b, function(t) t^2 / (t^2 + s^2)))
knitr::kable(data.frame(tau_b, W), digits = 2, align = "rrr",
             col.names = c("Between-stratum sd", "Weight, large stratum",
              ↪  "Weight, small stratum"))
```

Table 21.8.: Illustrative shrinkage weights on a stratum's own estimate for the large (1,182) and small (552) strata, by between-stratum standard deviation. The values are not the trial's estimates.

Between-stratum sd	Weight, large stratum	Weight, small stratum
0.1	0.11	0.05
0.2	0.32	0.18
0.4	0.65	0.47

The numbers in Table 21.8 are illustrative and not the trial's, which we do not have in the form the calculation needs.

21.4.3. What a DMC and a regulator would have had to accept

Which analysis a regulator prefers is not a question we can settle. The trial as conducted had to choose a primary population and a multiplicity procedure, and it did, in a way the FDA could review against the substantial-evidence standard. A hierarchical Bayesian analysis of the two strata would be a proposed alternative, not a description of what the trial did, and would need three things the draft guidance asks of any borrowing: a prespecified model, a

stated rationale for exchangeability of the strata, and a simulation of its type I error across the scenarios in which the effect is present in one stratum only (Chapter 19). Those scenarios are the ones that matter, because a hierarchical model that shrinks a null stratum towards a positive pooled effect will raise the false-positive rate for that stratum. The regulatory cost is a longer simulation report and a more demanding conversation about the prior on the between-stratum variance.

For the biomarker-guided duration, the regulator's concern is the estimand and the integrity of blinding (the blind has to survive the switch when a participant stops active treatment after clearance, which is a matter of masking procedure and not of the analysis). Neither is Bayesian or frequentist. They are the design properties that made a prespecified biomarker-guided feature acceptable to the FDA in this case.

The lesson is different from SELECT's. In SELECT, a Bayesian monitoring rule would change when the trial stops. In TRAILBLAZER-ALZ 2, the traditional apparatus handled two prespecified features, and the trial shows that regulators accept them when they are specified in advance. A hierarchical Bayesian analysis of the strata would not make the trial more or less adaptive. It would express the stratum analysis through one model in place of a testing hierarchy, at the price of the prespecification and simulation described above.

21.5. BNT162b2: a Bayesian confirmatory trial that led to an authorization

21.5.1. The trial as conducted

The phase 2/3 trial of the BNT162b2 mRNA vaccine (Protocol C4591001, Pfizer and BioNTech) began in July 2020 and reported the efficacy of the vaccine in preventing COVID-19 (Polack et al. 2020). Its statistical analysis plan, posted on ClinicalTrials.gov, makes the primary analysis Bayesian (Pfizer Inc. 2022). This case differs from the other four in kind: the Bayesian rule is not a counterfactual reconstruction but the decision rule the trial itself used, and it supported a regulatory action. The beta-binomial model of Chapter 6 is the whole of the machinery.

21.5.2. The Bayesian element

The conjugate beta-binomial model, with prior Beta(0.700102, 1) on the vaccine share of cases, the transformation to VE and the posterior arithmetic, is worked in full in Section 6.12. The plan calls the prior minimally informative and says it can be considered pessimistic. The case-specific points are the success criterion, $\Pr(\text{VE} \geq 30\% \mid \text{data})$, which must exceed 0.995 at an interim analysis and 0.986 at the final analysis, with the thresholds calibrated so that the overall one-sided type I error is 2.5 percent (Section 7.1 of the plan; the calibration is the kind of simulation of Chapter 14). Two primary objectives, VE in participants with no evidence of prior infection and VE in all participants, were evaluated in sequence. The final analysis was planned at 164 first primary cases, with interim looks and a futility rule based on the predictive probability of success, nonbinding, of 5 percent or less.

Two features set it apart from SELECT's reconstruction. The bar is on VE of 30 percent, a margin of clinical relevance, and not on zero, so the criterion is the $\Pr(\delta > a \mid \text{data}) > c$ of the FDA draft with a different from zero. And the model is conjugate, so each look is

a closed-form calculation with no simulation of the posterior itself, only of the operating characteristics.

```
a0 <- 0.700102; pi30 <- 0.7 / 1.7              # VE = 30% corresponds to
 ↪  pi_v = 0.7/1.7
pr_ve30 <- function(v, p) pbeta(pi30, a0 + v, 1 + p)   # v vaccine cases, p
 ↪  placebo cases
th <- qbeta(c(0.975, 0.5, 0.025), a0 + 8, 1 + 162)
ve <- round(100 * (1 - (2.222 / 2.214) * th / (1 - th)), 1)   # posterior
 ↪  2.5%, 50%, 97.5% for VE; surveillance-time ratio 2.222/2.214 (FDA
 ↪  briefing, Table 6)
```

21.5.3. What happened and what the regulator did

The published analysis counted 8 cases in the vaccine group and 162 in the placebo group, a vaccine efficacy of 95.0 percent with a 95 percent credible interval of 90.3 to 97.6 percent (Polack et al. 2020). Our calculation (Section 6.12), with the surveillance-time ratio of 2.222 to 2.214 reported in the FDA briefing document, gives 90.3 to 97.6 percent against the published 90.3 to 97.6 (the counts of 8 and 162 are exact), and a posterior probability of VE above 30 percent that is indistinguishable from one (1), far above 0.986. FDA issued an emergency use authorization on 11 December 2020 for individuals 16 years of age and older (U.S. Food and Drug Administration 2021a), and approved the biologics license application (Comirnaty) on 23 August 2021, again for those 16 and older (U.S. Food and Drug Administration 2021b).

21.5.4. The lesson for this chapter

Where the SELECT and REWIND reconstructions are counterfactuals, BNT162b2 shows the regulatory path already traveled: a minimally informative prior, a margin, calibration to 0.025 one-sided, and a protocol that fixed the thresholds before the first case. One caution. The plan on ClinicalTrials.gov is a sponsor document with several amendments. The efficacy boundaries tabulated in the plan, written as case splits, reproduce from the stated prior: at the final analysis of 164 cases, 53 vaccine cases is the largest count with $\Pr(\text{VE} > 30\% \mid \text{data}) > 0.986$ (53 gives 0.990, 54 gives 0.985). The backing for a rule of this kind comes from Chapter 6 for the model and Chapter 14 for the calibration.

21.6. What the saving is worth

A claim that a design decides at 486 events (the all-trial median; 464 among trials that declare efficacy) rather than 1,270 is not yet an argument that anyone outside the field will weigh. This section converts it into five currencies. Every conversion is a modeled counterfactual, and the conversions differ in how many assumptions they stack. They use the expected stopping point over all simulated trials, so that trials which never cross, and therefore save nothing, are counted.

21.6.1. Inside the trial: events and QALYs

These figures are modeled counterfactuals: arithmetic on the trial's own counts and the simulated all-trial stopping distribution, under the assumptions stated above (events uniform in person time, placebo participants switched at the decision, the decision acted on).

```
x <- data.frame(trial = c("SELECT", "REWIND"), d_dec = c(mna_b, mna_brw),
 ↪  d_tot = c(1270, 1257),
                n = c(17604, 9901), yrs = c(39.8 / 12, 5.4), ev_placebo =
                 ↪  c(701, 663), ev_active = c(569, 594))
x$share <- x$d_dec / x$d_tot                          # expected share of
 ↪  events accrued at the Bayesian decision
x$placebo_py <- (1 - x$share) * x$n * x$yrs / 2    # placebo patient-years
 ↪  afterwards
x$placebo_events <- (1 - x$share) * x$ev_placebo
x$excess <- (1 - x$share) * (x$ev_placebo - x$ev_active)
sel <- x[1, ]; rew <- x[2, ]
knitr::kable(transform(x[, c("trial", "share", "placebo_py",
 ↪  "placebo_events", "excess")], share = round(share, 3),
  placebo_py = round(placebo_py), placebo_events = round(placebo_events),
   ↪  excess = round(excess)), row.names = FALSE,
  align = "lrrrr", col.names = c("Trial", "Expected share of events at
   ↪  decision", "Placebo patient-years after", "Placebo events after",
   ↪  "Excess placebo events after"))
```

Table 21.9.: Modeled exposure after the Bayesian decision point in SELECT and REWIND: the expected share of events accrued at the decision (mean over all simulated trials, a trial that never crosses counted at its maximum), and the placebo patient-years, placebo events and placebo events in excess of the active arm that would follow it. Uniform event rates in person time and a switch of placebo participants at the decision are assumed. REWIND person-time uses the median follow-up as a stand-in for the mean.

Trial	Expected share of events at decision	Placebo patient-years after	Placebo events after	Excess placebo events after
SELECT	0.435	16490	396	75
REWIND	0.774	6037	150	16

Averaged over all simulated trials, the Bayesian rule had accrued 43.5 percent of SELECT's events when it stopped (a trial that never crosses counted at 1,270), so on average 56.5 percent of the events and of the exposure would have come afterwards. Under the modeled assumptions that is about 16,500 placebo patient-years and 396 of the 701 placebo primary events. The excess over the semaglutide arm is about 75 events: the major adverse cardiovascular events that the treatment would have prevented in a placebo arm switched to treatment at the decision, if the trial's relative effect held afterwards. This is a model-based expectation and not an observed count. It is also smaller than the figure the conditional median gives (63 percent of events), which describes only the trials that succeed.

Converting to QALYs requires a value per event averted. The composite mixes events worth very different amounts: a nonfatal event in a patient of about sixty carries a utility decrement of roughly 0.05 to 0.12 a year over a remaining life of ten to fifteen years, so of order 0.5 to 1.5 QALYs, while a cardiovascular death averted is worth the discounted remainder of that life, of order six to nine. These are our assumptions, and we report a range and not a figure.

Table 21.10.: QALYs inside the trial: placebo events in excess of the active arm after the Bayesian decision point, times an assumed QALY value per event averted. The QALY values are assumptions.

	1.5 QALY per event	2 QALY per event	3 QALY per event
SELECT	112	149	224
REWIND	23	31	47

For REWIND, the expected Bayesian stopping point is at about 77 percent of events, so about 6,000 placebo patient-years and 150 of the 663 placebo events come afterwards, of which about 16 are the excess over the dulaglutide arm, under the same assumptions. The modeled in-trial saving is of order a hundred QALYs for SELECT and a small fraction of that for REWIND, not thousands.

21.6.2. Trial cost

Moore and colleagues reviewed the pivotal trials of 59 novel agents approved by the FDA in 2015 and 2016 (T. J. Moore et al. 2018). The median estimated cost of a pivotal trial was $19.0 million, the costliest single trial in the review was about $347 million, and the mean estimated cost of the five cardiovascular pivotal trials was $157.2 million (their Table 2). SELECT, with 17,604 patients at 804 sites in 41 countries and about five years of follow-up, is far larger than the typical trial in that review, and we have no published cost for it. We therefore take the cited cardiovascular mean of $157 million as the base case and $350 million and $600 million as scenarios for a larger trial, not as estimates of the cost of any particular trial.

The saving is not proportional to the whole budget. Variable costs, which are site payments, patient visits, monitoring, laboratory work and drug supply, scale with person-time; fixed costs, which are protocol development, startup, regulatory work and the final analysis, do not. With 56.5 percent of the person-time avoided on average, the modeled saving is that share of the variable share.

Table 21.11.: Modeled expected trial cost saving at the SELECT Bayesian decision point, by the variable share of cost. The variable share is an assumption: the split between fixed and variable cost is proprietary. The $157 million is the cardiovascular mean of Moore et al. (2018); the other totals are scenarios.

	50% variable	60% variable	70% variable
saving as a share of total cost	28%	34%	40%
on a $157 million trial	$44 million	$53 million	$62 million
on a $350 million trial	$99 million	$119 million	$138 million

Table 21.11.: Modeled expected trial cost saving at the SELECT Bayesian decision point, by the variable share of cost. The variable share is an assumption: the split between fixed and variable cost is proprietary. The $157 million is the cardiovascular mean of Moore et al. (2018); the other totals are scenarios.

	50% variable	60% variable	70% variable
on a $600 million trial	$169 million	$203 million	$237 million

For a trial of SELECT's size the order of magnitude is therefore tens to a couple of hundred million dollars, and that is the number a sponsor will find most persuasive. It leaves out the cost of the longer exposure that a regulator might require to build the safety database, which would reduce the saving.

21.6.3. Timeline and the acceleration bound

Under the accrual model above, the all-trial median Bayesian decision falls at month 34 against a read-out at month 60, so 26 months or 2.2 years earlier. For REWIND the all-trial median decision is the end of the trial, so there is no median calendar saving, and the expected saving, with the decision at about 77 percent of events on average, is much smaller than SELECT's, since the Bayesian rule reaches its bar in only about half of the replications.

Multiplying the acceleration by the treated population gives a very large number. We give it, and then explain why it is a bound and not a forecast. Suppose approval moved earlier by 2.2 years (the all-trial median saving) and N patients received the treatment during that window, at the trial's placebo event rate of about 2.4 percent a year and its relative reduction of 20 percent.

Table 21.12.: Acceleration bound. Events averted if approval moved earlier by the SELECT Bayesian saving and N patients were treated during that window, at a placebo event rate of 2.4% a year and a 20% relative reduction. This is an upper bound under assumptions that did not hold.

patients treated in the window	events averted	QALYs at 2 per event
100,000	1,039	2,079
500,000	5,196	10,393
1,000,000	10,393	20,786

Two orders of magnitude separate this from the in-trial figures, and that asymmetry is the real economics of trial design: the value of a trial is overwhelmingly in the decision it enables for the population outside it, which is the post-trial population size M of the loss function in the working paper (Polson and Sokolov 2026b). But the calculation assumes that approval, reimbursement, manufacturing and clinical uptake would all have moved forward by the full 2.2 years, and none of those would. Manufacturing capacity for these agents was the binding constraint on access during this period, so the realistic acceleration is a fraction of the statistical one.

21.6.4. Label revenue brought forward

The largest sum in the accounting is commercial, and it rests on the least. Patent expiry is fixed, so a cardiovascular indication reached earlier adds the same number of years of protected, label-driven sales. Novo Nordisk reported global Wegovy sales of DKK 58.2 billion for 2024 (Novo Nordisk A/S 2025). The share of sales that the cardiovascular label drives is not public, and we assume 10 to 25 percent. At an assumed 6.9 DKK per dollar the 2024 sales are about $8.4 billion, and brought forward by the all-trial median saving of 2.2 years the label-driven revenue is $1.8 billion to $4.6 billion. This is a modeled counterfactual of revenue, not profit, and an upper bound: approval, reimbursement and uptake would not all have moved forward by the full statistical acceleration, for the reasons given for the population bound above, and the sales level, the label share and the exchange rate are assumptions.

REWIND gives the same arithmetic only conditionally. Trulicity (dulaglutide) had revenue of $4.13 billion in 2019 (Eli Lilly and Company 2022), and the FDA approved its cardiovascular indication on 21 February 2020 (U.S. Food and Drug Administration 2020b). In the reconstruction the Bayesian rule acts in only 49 percent of simulated REWIND trials, so the all-trial median saving is zero, because fewer than half of the simulated trials cross. The unconditional expected saving can still be positive: the mean decision falls at 973 of 1,257 events, so on average 23 percent of the events and person time would come after it. The revenue figures below describe the conditional median among trials that cross, and multiplying the crossing probability by a conditional median would not give an expected revenue (that needs the calendar saving averaged over all paths, which the summary output does not keep). Conditional on its acting, the median decision falls at 680 of 1,257 events. Under an accrual model with an assumed 24 months of linear enrollment and the 85 months between the registered start and primary completion dates (ClinicalTrials.gov 2026k), that is month 51, or 2.8 years before the read-out. With an assumed 10 to 20 percent of 2019 sales attributable to the label, the revenue brought forward would be $1.2 billion to $2.3 billion, but only in the trials in which the rule acts, and the scheduled design acts more often there.

Two caveats from the earlier sections apply to every currency in this section. Among trials stopping early for efficacy, naive estimates tend to exaggerate benefit. Skeptical priors may reduce this bias; bias and interval coverage should be evaluated for the specified estimator and stopping rule. And a regulator would want safety follow-up to continue after an efficacy decision, which recovers only part of the cost saving.

21.6.5. Summary of the accounting

Table 21.13.: The accounting for SELECT, by the strength of its basis.

currency	SELECT	basis
events averted in trial	about 75	modeled: expected share of events plus a placebo switch assumption
placebo patient-years avoided	about 16,500	modeled: expected share of events plus uniform events in person time

currency	SELECT	basis
QALYs inside the trial	112 to 224	plus a QALY value per event
calendar time	26 months before read-out (11 before SELECT's actual design, at the all-trial medians)	plus an accrual model
trial cost	28 to 40 percent	plus a fixed and variable cost split
label revenue brought forward	\$1.8 to \$4.6 billion of Wegovy revenue	plus a label share, an exchange rate and full acceleration; revenue, not profit
population QALYs	thousands	plus an access counterfactual

Every row is a modeled counterfactual computed from the all-trial distribution of simulated stopping points, and none is an observed fact about the trial. The first two rows rest on the stopping distribution and on assumptions about treatment switching and the uniformity of events. The rest add assumptions we have stated and a reader can vary. We would defend the first two as modeling exercises with few assumptions, the third and fourth as modeling exercises, the fifth as an order of magnitude, the sixth as an upper bound on revenue that rests on an assumed label share, and the seventh only as a bound.

21.7. Side by side: what changes in the protocol

Table 21.14.: The traditional and Bayesian designs on the same decision.

	Traditional (what the trials did)	Bayesian reconstruction	What the regulator will ask
Monitoring	Event-driven; DMC reviews at scheduled looks, boundaries from a spending function (SELECT: one interim at two thirds of the planned events)	Monitoring of a posterior probability on a fine event grid	The monitoring plan, written in advance (E9 4.5)
Decision rule	Test statistic crosses a boundary	$\Pr(\delta > a \mid \text{data}) > p^*$	Calibration of p^* to 0.025 one-sided by simulation, with Monte Carlo error
Type I error	Spent by alpha spending	Calibrated by simulation under the plan	Simulation report, scenarios, Monte Carlo error
Prior	None	Skeptical normal on the log hazard ratio	Prespecified, with sensitivity to τ

	Traditional (what the trials did)	Bayesian reconstruction	What the regulator will ask
Where it wins	Marginal effects, fixed looks (REWIND)	Large effects, DMC that can commit to a plan (SELECT)	Evidence the choice was made before the data
What does not change	Randomization, blinding, endpoint, firewall, intention-to-treat estimand	Same	Same

Where the two agree numerically: with a flat prior, the posterior probability of benefit equals one minus the one-sided p value, so a single-look Bayesian rule with threshold c is the fixed one-sided test at level $1 - c$.

21.8. Trials running now

The cases above are trials that have reported. The ones below are running, or have just reported, and each sits next to relevant evidence from a sibling or rival trial. We list them to show where the ideas of this book would apply now, not to predict their results. Design facts are from the ClinicalTrials.gov records (retrieved 10 November 2026, with each record's last-update date in the references), and the status of a trial is its status on that date; dates marked "est." are registry estimates and move. We give no result that we could not read in a registry record or a sponsor announcement.

Table 21.15.: Nine registered trials, their designs as registered, their status on 10 November 2026, and what a Bayesian design or analysis would change.

Trial (drug, sponsor)	Design as registered	Status on 10 November 2026	What a Bayesian design or analysis would change
PREVAIL (obicetrapib, NewAmsterdam) (ClinicalTrials.gov 2026a)	Randomized, quadruple-masked, placebo-controlled phase 3; 9,541 participants; primary endpoint major adverse cardiovascular events	Active, not recruiting; primary completion est. November 2026 (the record was last updated June 2024)	A discounted prior from the sponsor's earlier phase 3 trial of the same drug, and an interim decision by the predictive probability of success at completion (Chapter 15)

Trial (drug, sponsor)	Design as registered	Status on 10 November 2026	What a Bayesian design or analysis would change
AHEAD 3-45 (lecanemab, Eisai and Biogen) (ClinicalTrials.gov 2026g)	Randomized, quadruple-masked phase 3 in preclinical Alzheimer's disease; est. 1,400 participants; two primary outcomes, change in the PACC5 composite (A45) and in amyloid PET (A3) at week 216	Active, not recruiting; primary completion est. December 2028	The two prevention trials as estimates of a common class effect, so that the first readout updates the second, with biomarker change modeled jointly with cognition
TRAILBLAZER-ALZ 3 (donanemab, Lilly) (ClinicalTrials.gov 2026b)	Randomized, double-masked phase 3 in preclinical Alzheimer's disease; est. 2,996 participants; primary endpoint time to clinical progression on the Clinical Dementia Rating	Active, not recruiting; primary completion est. November 2027	As above; the endpoints differ (a time to progression and a change score), so exchangeability would have to be argued on a common scale
SURMOUNT-MMO (tirzepatide, Lilly) (ClinicalTrials.gov 2026d)	Randomized, double-masked phase 3; 15,374 participants; primary endpoint time to first of all-cause death, nonfatal myocardial infarction, nonfatal stroke, coronary revascularization or heart failure event, up to 5 years	Active, not recruiting; primary completion est. October 2027	A prior from SELECT and SURPASS-CVOT (Section 21.3), discounted for a five-component composite and a different drug, and calibrated monitoring as in the SELECT reconstruction
REDEFINE 3 (CagriSema, Novo Nordisk) (ClinicalTrials.gov 2026j)	Randomized, quadruple-masked phase 3; 7,101 participants aged 55 or older with established cardiovascular disease; primary endpoint time to first 3-point MACE	Active, not recruiting; primary completion est. September 2027	A power prior for the semaglutide component from SELECT, discounted for the added uncertainty about cagrilintide (Chapter 19)

Trial (drug, sponsor)	Design as registered	Status on 10 November 2026	What a Bayesian design or analysis would change
TRIUMPH-3 (retatrutide, Lilly) (ClinicalTrials.gov 2026c)	Randomized, double-masked phase 3, 80 weeks, in obesity with established cardiovascular disease; 1,946 participants in the registry; registered primary outcome percent change in body weight	Completed; topline announced 23 July 2026 (Eli Lilly and Company 2026)	A class prior turns an imprecise cardiovascular interval into a posterior (below)
Lp(a)HORIZON (pelacarsen, Novartis) (ClinicalTrials.gov 2026h)	Randomized, double-masked phase 3; 8,323 participants; two primary outcomes, expanded MACE in patients with Lp(a) of at least 70 mg/dL and of at least 90 mg/dL	Completed; Novartis announced on 4 September 2026 that the primary endpoint was not met (Novartis 2026); full results published 7 November 2026 (Cho, Nicholls, et al. 2026)	A posterior for the sizes of benefit the published hazard ratio (1.02, 95 percent interval 0.90 to 1.15) rules out (below)
OCEAN(a)-Outcomes (olpasiran, Amgen) (ClinicalTrials.gov 2026i)	Randomized, double-masked phase 3; 7,297 participants; primary endpoint time to coronary heart disease death, myocardial infarction or urgent coronary revascularization	Active, not recruiting; primary completion est. March 2028	A hierarchical model across the Lp(a) outcomes trials with the depth of Lp(a) lowering as a covariate, and a futility rule on the predictive probability of a clinically meaningful effect
ACCLAIM-Lp(a) (lepodisiran, Lilly) (ClinicalTrials.gov 2026e)	Randomized, double-masked phase 3; est. 17,300 participants; primary endpoint time to first MACE-4	Active, not recruiting; primary completion est. March 2029	A futility rule and a final sample size set from OCEAN(a) as it reads out

The traditional designs here are sound for what they are asked to do. All are randomized and masked. Several cardiovascular outcomes trials are event driven; AHEAD and TRIUMPH-3 assess their primary outcomes at specified follow-up times. Each is built to stand alone as substantial evidence, and a sibling trial's result does not automatically change the analysis plan. Justified amendments based solely on external information may be acceptable while

comparative interim results remain protected (U.S. Food and Drug Administration 2019b). For the proposed Bayesian designs, the borrowing and updating procedures, including any incorporation of future sibling results, should be prespecified. The Bayesian changes in the last column are therefore proposals with a price: each borrowed prior must be fixed before the data, discounted and justified, and each design with it simulated for its operating characteristics under a plan the regulator has seen (Chapter 19, Chapter 14). Where we write that a Bayesian rule could stop sooner, the size of the gain depends on an effect size that is not yet known, and the SELECT reconstruction above is not a forecast for any of these trials.

21.8.1. Retatrutide in TRIUMPH-3

Lilly's announcement reports 44 five-component events (all-cause death, myocardial infarction, stroke, heart failure event or coronary revascularization) on pooled retatrutide against 52 on placebo, a hazard ratio of 0.82 with a 95 percent interval from 0.55 to 1.22, in prespecified analyses of a trial whose registered primary outcome was weight (Eli Lilly and Company 2026). The three-component MACE hazard ratio in the same announcement was 1.12 (0.64 to 1.96). A frequentist reading of the five-component interval says that benefit is not established, and that is a correct statement about the interval. A Bayesian reading adds what the interval does not say, and shows how much of it depends on the prior. The interval gives the log hazard ratio $\log 0.82$ with standard error $(\log 1.22 - \log 0.55)/3.92$. For a class prior we use SELECT, with a hazard ratio of 0.80 and interval 0.72 to 0.90 (Lincoff et al. 2023), discounted by inflating its variance (doubling it for half weight, quadrupling it for quarter weight), and for contrast a skeptical normal prior with standard deviation 0.15 and a flat prior (Section 22.1).

```
est <- log(0.82); se_t <- (log(1.22) - log(0.55)) / 3.92
se_s <- (log(0.90) - log(0.72)) / 3.92
pri <- list(`Flat` = c(0, Inf), `SELECT, half weight` = c(log(0.80),
 ↪  sqrt(2) * se_s),
            `SELECT, quarter weight` = c(log(0.80), 2 * se_s), `Skeptical,
             ↪  N(0, 0.15^2)` = c(0, 0.15))
tri <- t(sapply(pri, function(p) {
  u <- normal_update(p[1], p[2], est, se_t)
  c(exp(u$mean), pnorm(0, u$mean, u$sd), pnorm(log(0.9), u$mean, u$sd),
   ↪  pnorm(log(0.8), u$mean, u$sd))
}))
knitr::kable(tri, digits = c(2, 3, 3, 3), align = "rrrr",
  col.names = c("Posterior hazard ratio", "Pr(HR < 1)", "Pr(HR < 0.9)",
   ↪  "Pr(HR < 0.8)"))
```

Table 21.16.: Posterior for the five-component cardiovascular hazard ratio of pooled retatrutide against placebo in TRIUMPH-3 (estimate 0.82, 95 percent interval 0.55 to 1.22, from the sponsor announcement of 23 July 2026), by prior. The SELECT prior is the semaglutide hazard ratio 0.80 (0.72 to 0.90) for a three-component outcome in a different population, with its variance doubled or quadrupled; the rows differ by how much that borrowing is believed.

	Posterior hazard ratio	Pr(HR < 1)	Pr(HR < 0.9)	Pr(HR < 0.8)
Flat	0.82	0.836	0.677	0.452
SELECT, half weight	0.80	0.998	0.937	0.482
SELECT, quarter weight	0.80	0.986	0.870	0.476
Skeptical, N(0, 0.15^2)	0.93	0.719	0.385	0.102

With no prior the posterior probability of some benefit is 0.84, and of a reduction of at least 10 percent 0.68. With SELECT at quarter weight the probability of some benefit is 0.99 and of at least a 10 percent reduction 0.87, while the skeptical prior gives 0.72 and 0.38. The disagreement among the rows is the scientific question, whether a class effect measured with a three-component outcome in semaglutide carries over to a triple agonist and a five-component outcome, and the analysis puts it on the table instead of leaving it behind "not significant". The three-component hazard ratio above 1 is a reason to give the class prior a low weight, not to hide it. With 96 events in all, none of these is a confirmatory result.

21.8.2. What a Lp(a)HORIZON result does to OCEAN(a)

When an analyst moves the probability that a related trial succeeds from p_0 to p_1 after a sibling reads out, the analyst has implicitly applied a Bayes factor, the ratio of the posterior odds to the prior odds:

$$B = \frac{p_1/(1-p_1)}{p_0/(1-p_0)}.$$

We could not confirm a published analyst estimate of the probability of success of OCEAN(a)-Outcomes after the Lp(a)HORIZON announcement of 4 September 2026, so we use an illustrative move from 50 to 20 percent, which is not a finding about any analyst.

On that illustration $B = 0.25$: the analyst treats the Lp(a)HORIZON result as 4 times as likely if olpasiran fails as if it succeeds. A fall from 50 percent to 40, 30 or 10 percent implies factors of 0.67, 0.43 and 0.11. Whether the factor is reasonable depends on how much the two drugs share, which the announcement does not settle: it reports that Lp(a) was lowered and that the primary endpoint was not met, with no hazard ratio and no comparison with the depth of lowering of other agents (Novartis 2026). That is the information set of the September announcement. The full results of 7 November 2026 reported a hazard ratio of 1.02 (95 percent interval 0.90 to 1.15) overall and 1.00 (0.87 to 1.15) among patients with Lp(a) of at least 90 mg/dL, with Lp(a) at 12 months of 64.4 nmol/L on pelacarsen against 233.4 on placebo, a difference of 72 percent (Cho, Nicholls, et al. 2026). On the flat-prior

normal approximation to the log hazard ratio, the posterior probability that the hazard ratio is below 0.90 is 0.023, below 0.85 0.002 and below 0.80 0.0001, so a 10 percent or larger benefit is largely ruled out for pelacarsen at its depth of lowering. These are the sizes of benefit compatible with HORIZON, and they inform OCEAN(a) only to the extent that olpasiran's benefit is expected to track that of pelacarsen. If the benefit of Lp(a) lowering is proportional to the absolute reduction, a drug that lowers it further should be judged against a different benchmark and the factor should be closer to 1. A hierarchical model with the depth of lowering as a covariate makes that assumption explicit and lets the data weigh it, which one revised probability does not.

21.9. Limits of the reconstruction

The numbers above are order-of-magnitude arguments, and a reader should know exactly where they would break.

- **The counterfactual is not the trial.** The Bayesian rule is evaluated on simulated paths with the observed hazard ratio. Nobody observed a SELECT that stopped at 486 events, and the conditional medians describe only the paths that crossed, which are the lucky ones.
- **Calibration is under a known grid and horizon.** A threshold of 0.9952 is calibrated for evaluation at every second event up to 1,270 events, not for literally continuous monitoring. Any subset of that grid, including unscheduled meetings at event counts on it, inherits the error bound, because the maximum over a subset cannot exceed the maximum over the grid. Monitoring off the grid or beyond the horizon needs additional justification and a recalibration, and the calibration is a property of the plan and not of the number.
- **The error rates are estimates.** A type I error read from 40,000 paths has a Monte Carlo standard error of about 0.0008, and the calibrated threshold should be validated on independent null simulations and across nuisance scenarios.
- **The information model is approximate.** The calculation assumes a Gaussian score process with $I_n = d/4$, proportional hazards and 1:1 allocation, which is accurate to two digits for the final analysis and approximate for early looks.
- **Early stopping costs something not priced here.** The secondary endpoints, safety exposure, subgroup results and the label depend on events and time, and a regulator weighs them.
- **The economic figures are scenarios.** The QALY value per event, the variable cost share and the access counterfactual are assumptions that a reader can vary and should.
- **Selection.** SELECT and REWIND are trials that were published. The fixed hazard ratio is the successful trial's own estimate.

In the protocol

Language a sponsor could use if it chose the Bayesian monitoring rule for an outcomes trial: "The DMC will review unblinded efficacy data at each scheduled meeting, at most twice a year. At each review the trial may be stopped for efficacy if the posterior probability that the hazard ratio is below 1 exceeds 0.9952 under a normal prior on the log hazard ratio with mean zero and standard deviation 0.25. The threshold was calibrated by simulation (Appendix X) on a grid of event counts, every second

event up to 1,270 events, so that the one-sided familywise type I error is estimated at 0.025 (Monte Carlo standard error 0.0008) and was confirmed on independent null simulations. A review at any event count on the grid, including a meeting not on the schedule above, is covered by that calibration. A review beyond 1,270 events or at an event count off the grid is not covered, requires a protocol amendment and a recalibration, and the simulation report gives the operating characteristics for hazard ratios from 0.70 to 1.00 and a range of event rates." The values are those of this chapter's reconstruction, not a recommendation for any particular trial.

Key points

- Using a common nominal one-sided bound of 2.5 percent (the Bayesian rule is calibrated to it by simulation, the four-look design by its boundaries, and the e-value rule is controlled by its bound through Ville's inequality), a Bayesian rule monitored on a fine event grid stops, among simulated trials that declare efficacy, at a median of 464 events on SELECT's data, against 612 for an e-value rule and 952 for a four-look O'Brien-Fleming design. Over all trials, counting those that never cross at the 1,270-event read-out, the medians are 486, 714 and 952, with power 95, 83 and 98 percent and realized type I errors of 2.5, 0.58 and 2.55 percent. SELECT's actual two-look design (one interim at two thirds of 1,225 planned events) stops at a median of 817 with power 97 percent and realized type I error 2.5 percent; the trial reported at 1,270. This is a reconstruction that uses the trial's observed hazard ratio and is not a power calculation.
- An uncalibrated posterior threshold of 0.975 on the same grid has a type I error of about 12 percent. Any claim of speed has to come with the error it was calibrated to, and with the power.
- On REWIND, where the hazard ratio is 0.88 and the final score is 2.27, the advantage shrinks: the Bayesian rule declares efficacy in only 49 percent of runs, so its all-trial median is the maximum of 1,257 events (680 among the trials that succeed), and the four-look design has the highest power. Monitoring on a fine grid is expensive when the effect is marginal.
- SURPASS-CVOT shows what a Bayesian reading adds to a noninferiority trial: posterior probabilities of noninferiority, of superiority and of a clinically meaningful reduction on a continuous scale, and an indirect comparison with placebo whose transportability assumption has to be stated and can be discounted.
- TRAILBLAZER-ALZ 2 shows that regulators accept a prespecified biomarker-guided treatment strategy and a prespecified two-population analysis with baseline tau strata when the rules are written in advance. A hierarchical Bayesian analysis of the strata is a proposed alternative that would itself need prespecification and a simulation of its false-positive risk.
- Unlike the counterfactual reconstructions above, BNT162b2 is a Bayesian confirmatory trial whose actual analysis supported an EUA and a BLA: a prespecified beta-binomial posterior, a bar set on vaccine efficacy of 30 percent and not on zero, and thresholds calibrated to a one-sided type I error of 2.5 percent.
- The in-trial value of the SELECT saving is, as a modeled counterfactual, of order a hundred QALYs (112 to 224 under stated assumptions) and tens to hundreds of millions of dollars, computed from the all-trial stopping distribution; the population-level value is orders of magnitude larger but is a bound, not a forecast.

- The label-driven revenue that an earlier SELECT decision would have brought forward is, as a modeled counterfactual, of order a few billion dollars of Wegovy revenue ($1.8 billion to $4.6 billion under an assumed label share of 10 to 25 percent), an upper bound and not profit. The REWIND analogue holds only conditionally on the Bayesian rule acting.
- Trials still running show where the same ideas apply now: a discounted prior from a sibling trial, a predictive interim, and a hierarchical model across a drug class. The TRIUMPH-3 posterior depends on the prior, with the probability of some benefit between 0.72 and 1.00 across the priors considered, and none of the rows is a confirmatory result.
- A calibration is a property of the monitoring plan, its event grid and its horizon, and it is estimated with Monte Carlo error. The reconstruction is a counterfactual, and no part of it argues that these trials should have stopped early.

Discussion questions

1. A DMC for an outcomes trial meets twice a year and sometimes convenes extra meetings after safety signals. What monitoring plan would you write into the protocol if you wanted to use a calibrated posterior threshold, and how would you describe the extra meetings in the simulation report?
2. In REWIND the scheduled design had the highest power. What would you need to believe about the effect size before choosing continuous monitoring for a trial like it, and how would you show a regulator that belief was reasonable? (Think of the assurance calculation of Chapter 9.)
3. The Bayesian rule in SELECT would, on average over all simulated trials, have reached its decision with about 44 percent of the events. List what a regulator would still need from the remaining follow-up before approving a drug for chronic use, and say which of those needs could be met by a longer extension study.
4. In TRAILBLAZER-ALZ 2, what is the estimand when treatment stops at amyloid clearance? Does a hierarchical analysis of the two tau strata change the estimand, the multiplicity procedure, or both?
5. The acceleration bound in the chapter assumes instant approval and uptake. Which of the assumptions would you relax first to make it a forecast, and what data would you need?

Try it in the lab

Module 12 (Decisions: what Type I control costs) at https://vsokolov.org/bayesian_trials_lab/12-decisions.html lets you tighten the bar for a "yes" and watch the cost in power, the same trade as the calibrated Bayesian threshold and the REWIND table. Before you click, predict how much power a stricter bar costs when the true effect is small, then check.

22. Bayesian Reanalyses of Landmark Trials

A published trial gives a likelihood, not a conclusion. The conclusion depends on the prior the reader brings and on the loss attached to acting wrongly, and a trial that ends with $p = 0.09$ or $p = 0.06$ can support action, inaction, or a new trial depending on both. This chapter reanalyses four trials from their published summaries. In each case the same three steps recur. First, reduce the report to an approximately normal likelihood for a log relative risk, a log hazard ratio or a risk difference. Second, combine it with a small family of priors chosen to represent defensible positions. Third, report posterior probabilities of effects large enough to matter, not only the probability that the effect is nonzero.

Each trial reported a prespecified primary analysis at a prespecified level, and it did its job: GUSTO, EOLIA and ANDROMEDA-SHOCK each answered the question the protocol asked. The designs differ: EOLIA, for example, was a group sequential trial with a triangular boundary, which matters for the futility discussion below. A reanalysis asks a different question, and the last section of this chapter says what it can and cannot claim. Every number below is computed from the counts or intervals quoted, with the `reanalysis_` functions of the bayestrials package (one per trial, plus the general `reanalysis_normal()` and the hierarchical `reanalysis_hier()`). Where published Bayesian reanalyses exist, we compare with them. Our numbers differ from theirs because we use normal approximations on the stated effect scale for each trial and because our priors are simple stand-ins for theirs.

! Regulatory anchor

- **FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft guidance, January 2026)** (U.S. Food and Drug Administration 2026g). The draft recommends that the prior be prespecified and justified in the protocol, that sensitivity to the prior be examined, and describes a posterior probability criterion as one way to define success; skeptical priors are one option for particular circumstances, not a requirement (Sections IV.A, V.A, V.C and V.F). All of this concerns a trial designed to be analyzed this way. It says nothing that turns a post hoc reanalysis of a finished trial into evidence of effectiveness. A draft, not binding.
- **ICH E9 (1998), sections 3.4 and 5.1.** Analysis methods are fixed before the data are unblinded, and the primary analysis is the prespecified one. A reanalysis is a secondary or exploratory analysis in this sense. See Chapter 3.

22.1. The normal approximation and its posterior

Let $\hat{\theta}$ be the estimate on the log scale and s its standard error. A reported interval (ℓ, u) at 95 percent gives

$$s = \frac{\log u - \log \ell}{2 \times 1.96}. \tag{22.1}$$

With a normal prior $\theta \sim N(m_0, s_0^2)$, the posterior is normal with precision equal to the sum of the precisions and mean equal to the precision-weighted average,

$$\frac{1}{s_1^2} = \frac{1}{s^2} + \frac{1}{s_0^2}, \qquad m_1 = s_1^2 \left(\frac{\hat{\theta}}{s^2} + \frac{m_0}{s_0^2} \right). \tag{22.2}$$

Write $\theta = -\log RR$ (or minus the log hazard ratio), so that positive means benefit, as in the code. For a benefit threshold θ^* (for example $\theta^* = -\log 0.8$ for a twenty percent relative reduction), $\Pr(\theta > \theta^* \mid \text{data}) = \Phi((m_1 - \theta^*)/s_1)$. The prior is worth s^2/s_0^2 trials of the present size: a prior with $s_0 = s$ counts as much as the trial itself (this is the effective sample size of Chapter 8). Ratios are reported back on their natural scale.

Spiegelhalter, Freedman, and Parmar (1994) proposed a standard family of priors for this exercise, and we use versions of them. Elicited skeptical priors for stopping rules have an earlier statement in L. S. Freedman and Spiegelhalter (1983), and the CHART trials were the worked example (Mahesh K. B. Parmar et al. 1994). A skeptical prior is centered at no effect with enough spread that a large benefit is unlikely a priori. An enthusiastic prior is centered at the effect the trial was designed to detect. A reference prior is flat. A positive trial is convincing when it persuades the skeptic, and a negative trial is convincing when it discourages the enthusiast. Section 8.3 discusses the three families and Section 8.8 the sensitivity analysis over the prior scale.

22.2. GUSTO: when prior trials disagree with the new one

22.2.1. The trial and its traditional analysis

GUSTO (GUSTO Investigators 1993) randomized 41,021 patients with acute myocardial infarction to four strategies: streptokinase with subcutaneous heparin, streptokinase with intravenous heparin, accelerated tissue plasminogen activator (tPA) with intravenous heparin, and a combination. Thirty-day mortality was 6.3 percent on accelerated tPA and 7.2 and 7.4 percent on the two streptokinase-only arms, about 7.3 percent pooled. The prespecified comparison of tPA with the two streptokinase arms gave a 14 percent relative reduction in mortality (95 percent interval 5.9 to 21.3 percent), $p = 0.001$. The frequentist conclusion was that tPA was superior, with an absolute benefit of one percentage point, or ten lives per thousand treated, at a substantially higher drug cost.

22.2.2. The Bayesian question

Brophy and Joseph (1995) asked what a cardiologist should have believed given the two earlier megatrials that compared the agents, GISSI-2 and ISIS-3, with 2,411 deaths among 24,118 patients on tPA and 2,384 among 24,176 on streptokinase between them (their Table 1). They gave the earlier trials weights of 100, 50 and 10 percent, took one percentage point of absolute mortality as the smallest clinically superior difference, and concluded that the clinical superiority of tPA "remains uncertain". They put the prior on each arm's mortality rate, a beta distribution built from that arm's earlier deaths and survivors, and with GUSTO's data the probability that tPA lowers mortality was 0.17, 0.44 and 0.98 at the three weights and 0.9995 with GUSTO alone (their Table 2).

In the counts Brophy and Joseph used, GUSTO had 652 deaths among 10,343 patients on accelerated tPA and 1,473 among 20,173 on the two streptokinase arms combined. The risk difference (streptokinase minus tPA) is $\hat{\theta} = 0.0100$ with standard error

$$s = \sqrt{\frac{\hat{p}_S(1-\hat{p}_S)}{20173} + \frac{\hat{p}_T(1-\hat{p}_T)}{10343}} = 0.0030.$$

We put the prior on the risk difference itself. Pooled, the earlier trials give a difference of -0.0014 (tPA very slightly worse) with standard error 0.0027, and the prior is normal with that mean and that standard error divided by $\sqrt{w}$, so $w = 0.1$ is a prior worth a tenth of its nominal information. Table 22.1 shows this model next to Brophy and Joseph's, which the `reanalysis_gusto()` function reproduces.

Table 22.1.: GUSTO risk difference (streptokinase minus tPA) with the earlier trials GISSI-2 and ISIS-3 at weight w. Difference model: normal prior on the risk difference. Arm model (Brophy and Joseph): beta priors on each arm's mortality; it reproduces their Table 2.

Prior	Posterior mean	Pr(effect > 0)	Pr(effect > 0.01)	Arm model: Pr(effect > 0)	Arm model: Pr(effect > 0.01)
None (GUSTO only)	0.0100	1.000	0.497	1.000	0.492
w = 0.1	0.0087	0.999	0.329	0.977	0.067
w = 0.5	0.0057	0.992	0.034	0.447	0.000
w = 1	0.0037	0.968	0.001	0.173	0.000

The two models agree on GUSTO alone and part company as the earlier trials gain weight. Within each earlier trial the agents had nearly the same mortality (9.6 percent on tPA and 8.9 on streptokinase in GISSI-2, 10.3 and 10.6 in ISIS-3), so on the scale of the difference the earlier evidence says "close to zero" and little more. Pooling each arm's rate across trials does something else. GUSTO's patients died at about 7 percent and the earlier trials' at about 10, and GUSTO put twice as many patients on streptokinase as on tPA, so the pooled streptokinase rate leans toward GUSTO's low mortality and the pooled tPA rate toward the earlier trials' high mortality. The comparison then absorbs the difference between trials into the difference between agents, the mechanism of Simpson's paradox, and at full weight the arm model makes tPA probably worse although no trial found it so. For these trials, borrowing within-trial treatment contrasts avoids pooling dissimilar baseline mortality rates.

An arm-level model would need trial-specific baseline risks and justified borrowing between trials (Chapter 19).

The table still separates two questions: whether there is any benefit, and whether it is large enough to matter. A test against zero addresses the first, and an interval or a test against the one-point threshold addresses the second; the posterior probabilities read both directly, conditional on the prior. On the difference scale, that tPA lowers mortality at all is very probable under every weight: the probability is at least 0.968. That it lowers mortality by a full percentage point, enough to justify its price, is a coin toss (0.50) with no prior information, 0.33 at weight 0.1, and unlikely once the earlier trials count at half weight or more. Brophy and Joseph's conclusion, that clinical superiority remains uncertain, holds in both models; their low probabilities of any benefit come from the arm-wise pooling. The decision to pay for tPA turns on the second question and on the price of a life year.

22.3. EOLIA: probable benefit after a nonsignificant primary analysis

22.3.1. The trial and its traditional analysis

EOLIA (Combes et al. 2018) randomized 249 patients with very severe acute respiratory distress syndrome to early venovenous extracorporeal membrane oxygenation (ECMO) or to conventional lung-protective ventilation, with rescue ECMO allowed in the control arm. Sixty-day mortality was 35 percent (44 of 124) with ECMO and 46 percent (57 of 125) in the control arm, a relative risk of 0.76 with 95 percent interval 0.55 to 1.04 and $p = 0.09$. Thirty-five control patients (28 percent) crossed over to ECMO, and 20 of them died. The trial was widely reported as negative. The design was group sequential with a two-triangle boundary, with analyses after every 60 randomized participants and a maximum of 331, and the trial stopped for futility at the fourth interim analysis (Combes et al. 2018, Methods).[1]

22.3.2. The Bayesian analysis

Goligher et al. (2018) carried out the published Bayesian reanalysis. Under a minimally informative prior they reported a posterior relative risk of 0.78 (95 percent credible interval 0.56 to 1.04), a posterior probability of any mortality reduction of 96 percent, of a relative risk below 0.67 of 18 percent, and of an absolute risk reduction of at least 2 percentage points of 92 percent. Across their other priors (moderately enthusiastic, strongly skeptical, and one built from earlier studies) the probability of benefit ranged from 88 to 99 percent.

Our own calculation uses the counts, so $\log \widehat{RR} = \log(0.355/0.456) =$ -0.251 with $s^2 = 1/44 - 1/124 + 1/57 - 1/125 = 0.0242$, $s = 0.156$. The counts give a relative risk of 0.778, a little different from the published 0.76, which we take as printed (it equals the ratio of the rounded percentages, 35 to 46). The priors are normal on the log relative risk and every one is stated in Table 22.2.

[1]The sample size assumed 60-day mortality of 60 percent with conventional care and 40 percent with ECMO, with 80 percent power at a two-sided level of 0.05. That is a relative risk of 0.67, the center of the enthusiastic prior below.

Table 22.2.: EOLIA priors on the log relative risk, with the prior mean and SD on that scale and the prior probability of a relative risk below 0.8.

Prior	Mean of log RR	SD of log RR	Median RR	Prior Pr(RR < 0.8)
Flat	none	infinite	none	none
Skeptical, s0 = 0.25	0	0.25	1	0.19
Skeptical, s0 = 0.15	0	0.15	1	0.07
Enthusiastic	-0.400	0.25	0.67	0.76

Table 22.3.: EOLIA: posterior relative risk of death at sixty days under four priors (our calculation from the counts 44/124 and 57/125, normal approximation).

Prior	Posterior RR	Pr(RR < 1)	Pr(RR < 0.8)	Pr(RR < 0.67)
Flat	0.78	0.95	0.57	0.168
Skeptical, s0 = 0.25	0.83	0.91	0.37	0.048
Skeptical, s0 = 0.15	0.89	0.87	0.17	0.005
Enthusiastic	0.75	0.99	0.70	0.207

On the absolute scale, with independent uniform priors on the two mortality rates (a calculation of ours, not Goligher's), the posterior probability that ECMO reduces mortality by at least two percentage points is 0.90, and by at least ten points it is 0.50. Goligher's own figure for two points, under their minimally informative prior, is 92 percent. The two agree closely even though the models differ, which is itself a robustness statement.

The lesson is about what the trial ruled out. EOLIA did not rule out the effect it was designed to find: it made that effect less likely (0.17 under the flat prior), but not unlikely. What it established is that some benefit is probable under the four priors examined (the probability of benefit runs from 0.87 to 0.99). A hospital deciding whether to maintain an ECMO program for severe ARDS faces a decision with a large upside and a known cost, and for that decision a probability of about 0.9 that the treatment helps is a strong argument. A nonsignificant result is nonrejection of the null, not evidence of no effect, and the interval from 0.55 to 1.04 already said as much. The posterior adds probabilities at thresholds of interest, which a test of the same threshold or an interval would also address. It does not make EOLIA a positive trial: the primary analysis missed its threshold, and a reanalysis cannot change that (see Section 22.7). The earlier randomized ECMO trials are told in Section 16.3.

22.4. ANDROMEDA-SHOCK: the same pattern with a hazard ratio

ANDROMEDA-SHOCK (Hernández et al. 2019) randomized 424 adults with septic shock in 28 intensive care units to an 8-hour resuscitation targeting normalization of capillary refill time or targeting serum lactate. Twenty-eight-day mortality was 34.9 percent (74 of 212) in the capillary refill arm and 43.4 percent (92 of 212) in the lactate arm, a hazard ratio of 0.75 with 95 percent interval 0.55 to 1.02 and $p = 0.06$.

The reported interval gives log $\widehat{HR}$ = -0.288 and $s = 0.158$. Zampieri et al. (2020) published the Bayesian reanalysis with four priors (optimistic, neutral, null and pessimistic) and found a posterior probability of benefit at 28 days above 90 percent under all of them. Their model was on the odds-ratio scale, with priors on the odds ratio. Ours is on the hazard-ratio scale, so the rows below illustrate the same logic and are not a replication of their table.

Table 22.4.: ANDROMEDA-SHOCK: posterior hazard ratio for death under five normal priors on the log hazard ratio (our hazard-ratio calculation, not Zampieri et al.'s odds-ratio analysis). Prior means and SDs: flat, none; neutral, mean 0, SD 0.35; skeptical, mean 0, SD 0.15; pessimistic, mean log 1.1, SD 0.15; optimistic, mean log 0.8, SD 0.20.

Prior	Posterior HR	Pr(HR < 1)	Pr(HR < 0.8)
Flat	0.75	0.97	0.66
Neutral, s0 = 0.35	0.79	0.95	0.54
Skeptical, s0 = 0.15	0.87	0.90	0.21
Pessimistic, centered at HR = 1.1	0.92	0.79	0.10
Optimistic, centered at HR = 0.8	0.77	0.98	0.63

Zampieri's pessimistic prior still left a probability of benefit above 0.90, while our pessimistic row gives 0.79. The difference is not a contradiction: the priors, the scale and the model differ. The pessimistic row is one sensitivity analysis and not the only way to discount the result. It centers the prior at a hazard ratio of 1.1, an instantaneous hazard ten percent higher, which is not a ten percent higher probability of death, and its posterior probability of benefit is 0.79. A sufficiently concentrated prior centered at no effect also lowers the posterior probability of benefit, so the table shows how much prior skepticism the result tolerates.

There is also a decision argument that no p value captures. Capillary refill time is measured with a finger and a stopwatch, and lactate needs a laboratory. If the two strategies were equally effective, the cheaper and faster one would be preferred. A strategy that is cheaper, at least as safe, and probably better need not clear $p < 0.05$ to be adopted. Whether it should be adopted is a question of expected net benefit, and we do not compute it here: that needs a utility function, a measured cost difference and a mapping from the hazard ratio to absolute mortality, none of which the posterior probabilities in Table 22.4 supply. The argument is conditional: if the strategies are equally safe and capillary refill is cheaper, adoption is favored to the extent that the posterior makes a loss in survival improbable, and Chapter 1 shows how to put numbers on that.

22.5. RECOVERY: subgroups and shrinkage

The dexamethasone comparison in RECOVERY (RECOVERY Collaborative Group et al. 2021) randomized hospitalized patients with Covid-19 to dexamethasone 6 mg once daily for up to ten days or usual care. Twenty-eight-day mortality was 22.9 percent (482 of 2104) on dexamethasone and 25.7 percent (1110 of 4321) on usual care, an age-adjusted rate ratio of 0.83 with 95 percent interval 0.75 to 0.93. The subgroup results by respiratory support at randomization were as important as the overall result. All rate ratios here are age-adjusted.

Table 22.5.: RECOVERY dexamethasone by respiratory support: published age-adjusted rate ratios and the hierarchical posterior (flat prior on the common mean, uniform prior on tau over (0, 1]).

Stratum	Mortality	Published RR	Shrunk RR	Pr(RR < 1), shrunk	Pr(RR < 1), unshrunk
Invasive ventilation	29.3% vs 41.4%	0.64 (0.51 to 0.81)	0.66	1.000	1.000
Oxygen only	23.3% vs 26.2%	0.82 (0.72 to 0.94)	0.82	0.998	0.998
No oxygen	17.8% vs 14.0%	1.19 (0.92 to 1.55)	1.13	0.187	0.096

Subgroup analyses are a familiar source of false findings, and the frequentist instinct is to distrust them unless an interaction test is significant (Section 10.3). The Bayesian instinct is to shrink them toward a common mean by an amount the data choose (Section 10.4). Let $\theta_j = -\log RR_j$ be the benefit-positive effect in stratum j, with $\hat{\theta}_j \sim N(\theta_j, s_j^2)$ and $\theta_j \sim N(\mu, \tau_b^2)$. With a flat prior on μ and a prior on τ_b, the marginal likelihood of τ_b is

$$p(\tau_b \mid \hat{\theta}) \propto p(\tau_b)\, V_\mu(\tau_b)^{1/2} \prod_j (s_j^2 + \tau_b^2)^{-1/2} \exp\left\{ -\frac{(\hat{\theta}_j - \hat{\mu}(\tau_b))^2}{2(s_j^2 + \tau_b^2)} \right\}, \tag{22.3}$$

where $\hat{\mu}(\tau_b)$ is the precision-weighted mean and $V_\mu(\tau_b)$ its variance. The conditional posterior mean of each θ_j is $(1 - B_j)\hat{\theta}_j + B_j \hat{\mu}$ with $B_j = s_j^2/(s_j^2 + \tau_b^2)$.

With only three strata, the data say little about τ_b and its posterior is sensitive to the prior placed on it. We use a uniform prior on τ_b over $(0, 1]$ on the log scale. Under that prior the posterior median of τ_b is 0.41 and the posterior mean is 0.45. The mean is the less stable summary: moving the upper bound from 1 to 3 changes the median to 0.52 and the mean to 0.73, because with three groups the posterior has a heavy right tail that the truncation point controls. The shrinkage factors are small in either case, and the shrunk estimates in Table 22.5 barely move (0.64 to 0.66, 1.19 to 1.13). That is the right answer. Shrinkage is not a rule that every subgroup be pulled to the mean. It is a rule that the data decide how much to pull, and here the pattern of benefit rising with severity is both large and biologically expected, because the inflammatory phase of Covid-19 is where a corticosteroid should help.

The decision implication is the one clinicians drew: treat patients on oxygen or ventilation, and do not treat patients without oxygen. In our model the posterior probability of harm (rate ratio above 1) in the no-oxygen stratum is 0.81, and even if one discounts the point estimate, no benefit is plausible enough to justify exposure. This probability, like the shrunk estimates, comes from our model, not from RECOVERY. Published Bayesian reanalyses of other RECOVERY comparisons exist for comparison, for example of tocilizumab (a preprint) (Albuquerque et al. 2021).

22.6. What the four reanalyses share

Four patterns recur.

Existence and magnitude are separate questions. In GUSTO some benefit was very probable and its magnitude was the open question. In EOLIA and ANDROMEDA-SHOCK the primary test did not reject the null, which is nonrejection and not evidence of absence, and the intervals (0.55 to 1.04 and 0.55 to 1.02) already showed that sizeable benefits were compatible with the data. Frequentist intervals describe the uncertainty about magnitude, and tests can target a clinically meaningful threshold instead of zero. A posterior adds direct probability statements at chosen thresholds, conditional on the prior, and the comparison with the estimate, interval and threshold test is the useful one.

The priors are arguments, not confessions. Each prior in the tables represents a position someone in the field actually held or could defend. Displaying the posterior under each is how a Bayesian report shows readers where they stand, and it is far more informative than a single interval.

A stopping rule can stop a trial despite a high probability of some benefit. EOLIA stopped for futility at its fourth analysis, after 240 patients, when the lower boundary of its triangular design was crossed (Combes et al. 2018).[2] A boundary rule and a predictive-probability rule can both do this, depending on the final success criterion each is built around, because each asks whether that criterion is likely to be met, which is not the same as whether the treatment works.

Decisions need utilities. In each case the action depends on more than the posterior: on the cost of tPA, the burden of ECMO, the convenience of capillary refill and the harm of steroids in early disease. The reanalyses supply the probabilities, and a utility turns them into choices (Chapter 1).

22.7. What a reanalysis can and cannot claim

22.7.1. Prespecification

The trials above were designed, powered and analyzed as frequentist trials. Their prespecified analysis is the one the protocol and statistical analysis plan fixed, and for EOLIA, ANDROMEDA-SHOCK and GUSTO that analysis was fixed before the data were seen. A Bayesian reanalysis is done after the results are known, by people who know them. Even with the best intentions, the choice of priors, thresholds and scale is made in the light of the answer. That does not make it wrong, and Goligher and Zampieri both chose their priors in a documented way. It does mean the reanalysis is exploratory in ICH E9's sense, and that the honest report lists every prior tried and not only the favorable ones. The tables above follow that rule: each row is shown, including the pessimistic prior that gives the weakest result.

22.7.2. Prior choice

The posterior is as defensible as the prior. A flat prior on the log scale is not neutral on the absolute scale (Section 8.4), a skeptical prior centered at no effect is a belief and not an absence of belief, and a prior built from earlier trials, as Brophy and Joseph did for GUSTO,

[2]The lower boundary of a triangular test is a futility boundary on the efficacy statistic, and the design is explained in Chapter 11. A predictive-probability futility rule asks a related but different question, how likely the final success criterion is to be met given the data so far (Chapter 15).

depends on how much weight they get and on whether the populations are exchangeable (Chapter 19). The recommended practice is the one the FDA draft asks of a prospective design: state each prior's mean and standard deviation, show the posterior across a range of them, and say which readers each prior represents. The weight table for GUSTO and the four priors for EOLIA do that. A reanalysis that reports one prior, chosen after the data, does not.

22.7.3. The regulator's view

The FDA Bayesian draft (U.S. Food and Drug Administration 2026g) is written for a trial that is designed to be Bayesian. It asks for a prespecified prior, a success criterion of the form $\Pr(\delta > a \mid \text{data}) > c$, simulation of operating characteristics, and, for trials with an overall frequentist analysis plan, familywise type I error control at 0.025 one-sided. It allows direct interpretation of the posterior probability and decision-theoretic criteria. It does not describe a post hoc reanalysis as substantial evidence, and nothing in it says a posterior probability of 0.9 computed after a negative primary analysis converts that analysis into a positive one. In this book's reading (not the draft's words), a reanalysis is useful input to a sponsor planning the next trial, to a guideline committee weighing a decision, and to a hospital choosing a practice, and it is not a substitute for the prespecified analysis in a regulatory submission. It is a draft, not binding, and the substantial-evidence standard is in the statute (Chapter 3).

22.7.4. A reanalysis is not a new trial

A reanalysis uses the data once more. It cannot add patients, cannot repair a design flaw such as open-label assessment or crossover, and cannot change the probability that the original trial would have rejected a null effect. What it can do is change the question from "did the trial cross its threshold?" to "what should we believe now, and what should we do?". The honest consequence is that a strongly positive posterior after a negative trial is a reason to run a confirmatory trial designed around the effect sizes the reanalysis found plausible, with the priors written in the protocol. EOLIA's reanalysis and the later ECMO literature are a reminder of this: the posterior pointed to probable benefit, and the decision to adopt rested on clinical judgment and further evidence, not on the posterior alone. RECOVERY shows the other side: a trial so large that its prespecified analysis was decisive, where shrinkage of subgroups changes the numbers but not the practice.

22.7.5. What critics say

Reanalyzing a trial after the result is known has a literature of its own, with supporters and critics, and the cases above sit inside it. An early instance is the debate over extracorporeal membrane oxygenation in newborns, in which Ware described the trials and Kass and Greenhouse commented from a Bayesian perspective (Ware 1989; Kass and Greenhouse 1989a). EOLIA put the same question to adults (Section 16.3), and its Bayesian reanalysis (Goligher et al. 2018) was published beside an editorial asking whether clinicians should embrace their inner Bayesian (Lewis and Angus 2018). Systematic programs followed. Wijeysundera and colleagues reanalyzed 88 randomized trials from four general medical journals and concluded that Bayesian and frequentist analyses complement each other and that trial reports should include both (Wijeysundera et al. 2009). Diamond and Kaul used

Bayesian approaches to interpret clinical megatrials (Diamond and Kaul 2004), Quintana, Viele and Lewis describe how prior information is used to interpret trial results (Quintana, Viele, and Lewis 2017), and Ryan and colleagues showed what a simple Bayesian analysis of the OPTIMISE trial of hemodynamic therapy in major gastrointestinal surgery adds to the traditional one (E. G. Ryan et al. 2019). The case for these reanalyses is that they state probabilities at clinically meaningful thresholds, and that showing the posterior under several priors makes the dependence on the prior visible, in the spirit of the acknowledgment of multiple perspectives that Gelman and Hennig propose as a virtue of statistical work (Gelman and Hennig 2017).

The criticism we cite is a commentary by de Grooth and Cremer, "Bayes and the Evidence Base: Reanalyzing Trials Using Many Priors Does Not Contribute to Consensus" (Grooth and Cremer 2024). It recommends testing whether a result would persuade a reasonable adversary and assessing bias and generalizability separately. That connects to the skeptical-prior analyses above, in which the skeptical prior stands in for the adversary, and to a point that we make in our own words, that a table of many priors invites each reader to pick the row that agrees with what he or she already believed, so the table can sharpen a disagreement instead of resolving it. Two further objections follow from the structure of a reanalysis and are not attributed to any source. First, the priors are chosen after the result is known. Second, which trials are reanalyzed is itself a choice that nobody prespecified. A cardiology example shows both the power and the risk. Brophy reanalyzed the EXCEL trial of stenting against bypass surgery for left main disease: EXCEL's prespecified three-year analysis established noninferiority against a 4.2 percentage point margin (Stone et al. 2016). Applying that original margin to the five-year results, Brophy observed that noninferiority was no longer established by this comparison, and his Bayesian reanalysis estimated a 95 percent probability of more composite events with stenting, so the headline moved with the question asked. Adding three other trials found by a stated search as informative priors lowered the probability of any excess mortality with stenting from 99 to 85 percent, which shows how much depends on which trials enter, and it is the stated search rule that makes that choice auditable (Brophy 2020).

Our reading is that each side is right about something different. A reanalysis is informative as a way of seeing what the data can and cannot support, and weak as evidence when the priors and the trials were chosen after the fact. The rule that follows extends the one stated above: write the reanalysis down before looking, with the priors fixed and the trials chosen by a stated rule, report every prior tried, and treat the result as input to the next trial and not as a verdict on the last one.

Key points

- A published trial yields a likelihood, and a conclusion needs a prior and a loss. Use an approximately normal likelihood on the stated effect scale, such as a risk difference or a log ratio, then show the posterior under a family of stated priors.
- GUSTO: with the prior on the risk difference, that tPA helps is very probable under every weight, but whether it helps by a full percentage point depends on how much weight the earlier trials get. Brophy and Joseph's priors on each arm's rate pool baseline risks across trials and make any benefit look doubtful. Borrowing within-trial treatment contrasts avoids that, and an arm-level model would need trial-specific baseline risks and justified borrowing between trials.
- EOLIA and ANDROMEDA-SHOCK: a missed threshold at $p = 0.09$ or 0.06 left

a high posterior probability of benefit under most priors. Our hazard-ratio rows for ANDROMEDA-SHOCK are not a replication of Zampieri et al.'s odds-ratio analysis.

- RECOVERY: hierarchical shrinkage leaves strongly supported subgroup effects nearly unchanged and pulls a weak one only a little. The posterior of the between-stratum SD depends on its prior, so state the prior and report the median.
- A reanalysis is exploratory, prior choices are made after the data, and it cannot replace the prespecified analysis or serve as a new trial. Report every prior tried, and use the result to design the next trial.

22.8. Exercises

1. Derive Equation 22.2 by completing the square, and show that a prior with $s_0 = s$ contributes as much information as the trial.
2. For EOLIA, find the prior standard deviation s_0, centered at no effect, for which $\Pr(RR < 1 \mid \text{data}) = 0.8$. Would anyone in the field have held a prior that skeptical?
3. Repeat the GUSTO analysis on the relative risk scale. Does the conclusion about the one percentage point threshold change?
4. For ANDROMEDA-SHOCK, take 28-day mortality of 0.434 under lactate-guided resuscitation and treat the hazard ratio as a risk ratio applied to it (a stated approximation). Suppose capillary refill monitoring saves a laboratory cost worth the equivalent of 0.002 deaths per patient. Under the skeptical prior, compute the expected net benefit of capillary refill in deaths per patient, using the posterior mean of the hazard ratio, and say how the answer changes with the approximation.
5. In RECOVERY, put a half-normal prior with scale 0.1 on τ_b instead of the uniform prior. How much do the shrunk estimates move, and what does that say about the prior's influence with three strata?
6. A trial reports $HR = 0.85$ with interval 0.70 to 1.03. Construct the skeptical prior that makes the posterior probability of $HR < 0.9$ exactly one half.

23. Writing the Bayesian Protocol and Working With Regulators

A design is only as good as the document that fixes it. The earlier chapters argued that a trial is a decision, that a regulator approves on evidence meeting a legal standard, and that Bayesian adaptive designs reach that standard sooner by monitoring, by handling several arms together, and by using prior information, not by lowering the bar. All of that is worth nothing to a sponsor until it has been turned into sentences in a protocol, a statistical analysis plan, a simulation report and a data monitoring committee charter, and until a regulator has read those sentences and not objected.

The practical question is which sentences change. A sponsor who has written a dozen group sequential protocols knows what the document looks like. A Bayesian protocol shares most of its sections with that document and differs in a handful, and the handful is where reviewers will spend their time. The chapter walks through the protocol section by section, traditional version first, then the Bayesian version, with model sentences in "In the protocol" callouts that a sponsor can adapt. It then covers the statistical analysis plan and the simulation report, the structured protocol formats now in use, the data monitoring committee and its firewalls, the sequence of meetings with FDA and the parallel routes at EMA, the standard of substantial evidence and where Bayesian evidence fits in it, what the Bayesian design does not buy, and two checklists, one for sponsors and one for reviewers. Nothing here is new methodology. It assembles material from Chapter 3, Chapter 8, Chapter 11, Chapter 14, Chapter 15, Chapter 16, Chapter 19 and Chapter 20 into the form a submission takes.

! Regulatory anchor

FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products (draft guidance, January 2026, U.S. Food and Drug Administration (2026g)). Not binding. Section VIII asks that the prior, the success criteria, the operating characteristics and the computational details be documented "as early as possible" in the protocol, the statistical analysis plan (SAP) and a simulation report, and that complex designs be compared with simpler alternatives and discussed with FDA before the trial starts. FDA's draft recommends prespecifying and justifying the prior in the protocol; this book treats prespecification as essential design practice.
FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics (final guidance, November 2019, U.S. Food and Drug Administration (2019a)). Four principles: the chance of erroneous conclusions should be adequately controlled, estimation of treatment effects should be sufficiently reliable, details of the design should be completely prespecified, and trial integrity should be appropriately maintained. Simulations are the accepted route to operating characteristics for complex designs.
ICH E20, Adaptive Designs for Clinical Trials (Step 2 draft, endorsed June 2025, International Council for Harmonisation (2025a)). Adaptations are "prospectively planned" only if prespecified in the protocol before the trial starts, a "clear and

compelling justification" is required, and Section 5 sets out what a simulation report should contain. Draft, not final.
ICH E9 (1998, International Council for Harmonisation (1998)) and E9(R1) (Step 4, November 2019, International Council for Harmonisation (2019)). Interim analyses must be planned in advance and described in the protocol. The estimand, the precise description of the treatment effect, must be defined before the trial.
FDA, Demonstrating Substantial Evidence of Effectiveness (revised draft guidance, June 2026, U.S. Food and Drug Administration (2026a)). When one adequate and well-controlled trial plus confirmatory evidence can carry an approval, and the advice to discuss the plan with FDA before the trial starts.
FDA, Formal Meetings Between FDA and Sponsors of PDUFA Products (guidance, August 2026, U.S. Food and Drug Administration (2026c)); **FDA, Interacting with the FDA on Complex Innovative Trial Designs** (final guidance, December 2020, U.S. Food and Drug Administration (2020a)); **ICH M11** structured protocol (Step 4, November 2025, International Council for Harmonisation (2025c)); **SPIRIT 2025** (Chan et al. (2025)). How to ask, when, and in what format.

23.1. The protocol, section by section

23.1.1. Traditional first

A traditional confirmatory protocol states a null hypothesis and a one-sided type I error of 0.025 (two-sided 0.05 divided by two, as everywhere in this book). It fixes a sample size, or for a time-to-event endpoint a number of events, from a power formula at a design alternative, and fixes the allocation, usually one to one. If there are interim analyses it schedules them, names the boundary family and the alpha spending function, and says that unscheduled looks are protocol deviations. It handles multiple comparisons by a stated procedure, Bonferroni, Holm, a gatekeeping sequence or a graphical scheme (Chapter 10). It lists safety stopping rules, often informally. It names the primary analysis population and, since E9(R1), the estimand. The final report carries a point estimate, a confidence interval and an adjusted p-value. Each of these has a regulatory basis: ICH E9 on prespecification of interim analyses and on multiplicity, E9(R1) on estimands, the 2019 adaptive designs guidance on anything that departs from a fixed design.

The Bayesian protocol keeps the skeleton and changes the parts that depend on how evidence is summarized and how the trial may change as data accumulate.

23.1.2. What changes

Table 23.1 sets the two designs side by side. It is a generalization of the table in the working paper that motivated this book (Polson and Sokolov 2026b), which was written for one cardiovascular outcomes trial (the SELECT reconstruction of Chapter 21). The entries in the last column are written for any Bayesian confirmatory design, and the numbers in the traditional column are the ones that example used where a number helps.

Table 23.1.: Protocol sections, traditional and Bayesian. Based on the protocol comparison in Polson and Sokolov (2026b), generalized beyond one trial, with a column for the review questions each row invites.

Protocol section	Traditional design	Bayesian design	What the reviewer will ask
Objective	Test H_0 of no effect at one-sided 0.025.	Decide whether the evidence meets a prespecified criterion, for example $\Pr(\delta > a \mid \text{data}) > c$; optionally, choose the action that maximizes expected utility.	Is the criterion the one that supports a labeling claim?
Estimand	Defined per E9(R1): population, treatment, variable, intercurrent event strategy, summary.	Identical. The estimand is not a statistical philosophy choice.	Does the posterior target the same estimand as the clinical question?
Prior	Not stated. The design alternative enters only the power calculation.	Stated in full, with its effective sample size (ESS), the source of any informative component, and a prespecified sensitivity analysis.	Why this prior? What happens under a skeptical or pessimistic one? What if the data conflict with it?
Sample size	Fixed (or number of events), from a power formula.	A maximum, with the distribution of the sample size at stopping reported from simulation.	What is the largest trial this protocol can produce, and is it feasible?
Interim looks	Prescheduled; alpha spending function; unscheduled looks are deviations.	Prescheduled in a confirmatory trial, with decision rules at each look. The posterior does not depend on when data were examined, but the operating characteristics do.	Is the schedule fixed before unblinding, and were the operating characteristics simulated on that schedule?

Protocol section	Traditional design	Bayesian design	What the reviewer will ask
Stopping rules	Boundary on Z_n (for example O'Brien-Fleming: 4.05, 2.86, 2.34, 2.02 across four looks).	Posterior or predictive probability thresholds for efficacy, futility and safety, calibrated by simulation (the working paper's SELECT example used 0.995 for efficacy).	How were a and c chosen, and is familywise type I error controlled at 0.025?
Multiple arms or endpoints	Bonferroni, closed testing, graphical procedures.	Hierarchical model across arms or subgroups when the effects are scientifically exchangeable, with the hypothesis family defined by the intended claims and the complete decision rule calibrated where strong familywise error control is needed; or the same traditional procedure applied to the posterior criterion.	Is familywise error still controlled where the guidance expects it?
Randomization	Fixed, usually 1:1.	May be response-adaptive, with a stated rule, floors on allocation, burn-in, and time-trend handling.	Is the rule prespecified, is bias from drift handled, and is the integrity risk managed?
Safety	Prespecified rules, often informal, outside the error budget.	A posterior statement, for example exclude an arm if $\Pr(\text{toxicity} > \text{limit} \mid \text{data}) > 0.90$, with its false-stop rate simulated.	What is the probability of stopping a safe arm, and of missing a harmful one?

Protocol section	Traditional design	Bayesian design	What the reviewer will ask
Operating characteristics	Type I error and power at the design alternative, analytically.	The full set by simulation over a grid of scenarios: error, power, expected sample size, bias, coverage, probability of correct decision.	Is the scenario set prespecified, comprehensive and plausible, and is the code provided?
Computation	Standard software, rarely discussed.	Model code, sampler settings, convergence diagnostics, seeds, software versions.	Can FDA reproduce the simulation and the analysis?
Final report	Estimate, confidence interval, adjusted p-value.	Posterior distribution of the effect, whether the criterion was met, sensitivity to the prior; a p-value and confidence interval alongside where FDA expects them.	Does the report show the effect of the prior?

Four things stay the same. The estimand is the same, because it is a statement of the clinical question and E9(R1) applies to any statistical framework. The randomization is, in the default case, the same, and the reasons for randomizing do not depend on the inferential paradigm (Chapter 5). The default bar in this book is the same: calibration to a familywise type I error of 0.025 one-sided. For a trial with an overall frequentist statistical analysis plan the Bayesian draft says that success criteria are "almost always" chosen that way, and that a and c may be chosen so that the overall error is controlled at that level. The draft offers two other routes, direct interpretation of posterior probabilities and decision-theoretic criteria, and as a draft it recommends and does not require. Calibration is this book's default for confirmatory adaptive designs without borrowing, not FDA's universal requirement. And the need to prespecify is the same. What moves is where the numbers come from, and in each moved row the sponsor owes the reviewer a simulation rather than a formula.

Three rows carry most of the practical difference. The interim looks row is where the speed comes from, and the one a data monitoring committee cares about. The multiple arms row is where robustness comes from, and matters for platform trials in which the number of contrasts is not known when the protocol is written (Chapter 20). The operating characteristics row is where the burden falls. A Bayesian protocol must simulate, and the simulation is the deliverable that replaces the closed form power calculation.

Where the two designs agree numerically, say so in the protocol. With a flat prior, a normal approximation and a single look, the posterior probability that the effect is positive is one minus the one-sided p-value, so a criterion $\Pr(\theta > 0 \mid \text{data}) > 0.975$ is the same test as one-sided $\alpha = 0.025$ (Chapter 7).

23.1.3. The prior

The prior is the section with no traditional counterpart, and the one FDA's draft recommends prespecifying and justifying in the protocol; this book treats prespecification as essential design practice. The Bayesian draft calls it "always critical" that a Bayesian analysis have a prespecified prior, asks that the full details be fixed and justified in the protocol, and, where the prior is informative or borrows external data, asks for strong justification, a description of each source including excluded ones, the discounting method, how prior-data conflict will be handled, and the ESS and other influence metrics (U.S. Food and Drug Administration 2026g). Chapter 8 and Chapter 19 give the methods. For the protocol, three habits matter. State the ESS, the number of patients the prior is worth, in the sense of Morita, Thall, and Müller (2008). A prior worth five patients in a trial of five hundred is a modeling choice, and one worth two hundred is an assumption that needs defending. Distinguish the analysis prior (used to compute the posterior that decides success) from design priors (used only in simulation to explore scenarios, including pessimistic ones). And plan a sensitivity analysis, naming the alternative priors in advance.

In the protocol: the prior and its ESS

"The analysis prior for the negative log hazard ratio $\theta = -\log \mathrm{HR}$ is $N(0, \sigma_0^2)$ with $\sigma_0 =$ [value], a skeptical prior centered at no effect. Its effective sample size is [value] events, which is [percentage] of the maximum [number] events. No external data are borrowed. The prior was chosen so that [justification, for example: a hazard ratio of [x] has prior probability [p]]. Prior sensitivity will be assessed by repeating the primary analysis with prior standard deviations of [values] and with a flat prior; the primary conclusion will be reported as robust only if the success criterion is met under [stated condition]. In simulation, design priors representing [pessimistic scenario] and [optimistic scenario] were used to assess performance when the analysis prior is wrong."

23.1.4. The success criterion and its calibration

The Bayesian draft describes the usual form of the criterion as based on the posterior probability that the true effect exceeds a threshold, $\Pr(\delta > a \mid \text{data}) > c$, where a is the minimum benefit and c the minimum probability. It asks the sponsor to justify both a and c, to give criteria for each primary and key secondary estimand, and, if the design is calibrated, to provide simulation evidence that the familywise error rate is controlled (U.S. Food and Drug Administration 2026g). The draft also recognizes that calibration to 0.025 "may not be applicable or appropriate" in some cases, for instance with borrowing, and describes two other routes, in which success rests on a direct interpretation of the posterior or on a decision-theoretic or benefit-risk criterion (Section IV.A.3) if sponsor and FDA agree in advance. A sponsor who wants that route has to ask for it, in a meeting, with a reason. The BNT162b2 statistical analysis plan is a worked example: posterior thresholds of 0.995 at the interim analyses and 0.986 at the final analysis, chosen so that the overall one-sided type I error is 2.5% (see "In practice" below).

In the protocol: success criterion and calibration

"The trial will be declared successful for the primary estimand if, at the final analysis or at an interim analysis specified in Section [x], $\Pr(\theta > a \mid \text{data}) > c$, with $a =$ [value, with clinical rationale] and $c =$ [value]. The threshold c at each analysis was calibrated by simulation so that the probability of declaring success under the null hypothesis $\theta = 0$, across all analyses in the schedule and under the stated enrollment and dropout assumptions, does not exceed 0.025 one-sided. Simulation details are in the simulation report (version [x], dated [date]); the code and seeds are provided. If the number or timing of analyses departs from the schedule, the thresholds will be recomputed as described in Section [y] only if the change is demonstrably independent of comparative interim results, follows a prespecified rule whose timing mechanism was simulated, or is handled by a prespecified conditional-error method."

23.1.5. Interim schedule and decision rules

E9 says that interim analyses "should be carefully planned in advance and described in the protocol", and E20 repeats the point for adaptations in general. A Bayesian posterior is formally unaffected by how many times it is computed, and that is a property of the posterior, not of the decision rule. A rule that stops when a posterior probability crosses a threshold has operating characteristics that do depend on when and how often it is evaluated (Chapter 11). The protocol therefore does not say "continuous monitoring". It says when looks occur, by information fraction or calendar rule, who sees them, what decisions are possible at each (stop for efficacy, stop for futility, continue, change allocation, drop an arm), and what happens if a look is missed or added. A look added after an external publication is a deviation from the simulated design, and the protocol should say what will be done about it, which is acceptable to handle by recalibration only if the change was demonstrably independent of comparative interim results, followed a prespecified rule whose timing mechanism was simulated, or is handled by a valid conditional-error approach; keeping the sponsor blinded does not by itself establish independence, and treating a schedule shaped by interim results as fixed ignores the selection mechanism (U.S. Food and Drug Administration 2019a). Chapter 12 describes the alternative of an anytime-valid rule, whose guarantee holds at any stopping time but at the price discussed there. The DAWN stroke trial wrote its whole predictive-probability stopping schedule into the SAP (see "In practice" below).

In the protocol: interim schedule and decision rules

"Interim analyses will take place when [number] participants (or events) have been observed on the primary endpoint, planned at [information fractions]. At each interim analysis the independent statistical center will compute (a) $\Pr(\theta > a \mid \text{data})$ and (b) the predictive probability of success at the maximum sample size. The DMC will be informed of the following rules, and the sponsor will not see unblinded comparisons: stop for efficacy if (a) exceeds $[c_k]$; stop for futility if (b) is below [value]; otherwise continue. The DMC may recommend stopping for safety or any other reason at any time under the charter, independent of these rules. No unscheduled efficacy analysis will be used to support an efficacy claim without a protocol amendment and recalculation of the operating characteristics; safety review and DMC action under the charter are unaffected."

23.1.6. Adaptive randomization

If the design uses response-adaptive randomization, the rule belongs in the protocol, and so do its limits. Chapter 16 covers why a reviewer's first questions concern time trends, bias in the estimate, and whether the control arm retains enough participants. EMA's October 2026 draft reflection paper on confirmatory platform trials does not support response-adaptive randomization in that setting (European Medicines Agency 2026a) (a draft, and its position on this point is a reason to ask EMA early). FDA's Adaptive Designs guidance does not forbid it but asks that the chance of erroneous conclusions be controlled and the estimates be reliable. Adaptive randomization has a long Bayesian pedigree in dose finding and elsewhere (P. F. Thall et al. 2015). Many Bayesian protocols therefore use a restricted rule: a fixed burn-in at equal allocation, a floor on each arm's probability, a cap on any change in allocation between updates, and analysis stratified by allocation period.

In the protocol: adaptive randomization rule

"After a burn-in of [number] participants randomized 1:1, allocation to each experimental arm will be updated every [number] participants in proportion to the posterior probability that the arm is best, raised to the power [value], subject to a minimum of [percentage] of participants per arm and a maximum change of [percentage points] between updates. The control arm will receive at least [percentage] of participants. The primary analysis will adjust for calendar period [method]. The operating characteristics of the rule, including bias of the effect estimate and the probability of allocating more than [x] participants to an inferior arm, are in Section [z] of the simulation report."

23.1.7. Safety rules

Traditional protocols often leave safety stopping to the DMC's judgment, with a guideline. A Bayesian protocol can state it as a posterior probability, which is more transparent and, more usefully, can be simulated: what is the probability that a safe arm is stopped, and that a harmful arm is not? The safety rule sits outside the efficacy error budget, so the protocol should say what its own false-stop rate is, and who may override it.

In the protocol: safety rules

"An arm will be suspended and referred to the DMC if, at any safety review, $\Pr([\text{event rate}] > [\text{limit}] \mid \text{data}) > 0.90$ under the prior in Section [x]. Reviews take place after every [number] participants per arm. By simulation, the probability of suspending an arm whose true event rate equals [acceptable rate] is [value], and the probability of suspending an arm whose true rate equals [unacceptable rate] within [number] participants is [value]. The rule is a trigger for DMC review, not an automatic stopping rule."

23.1.8. The estimand

Nothing about the estimand is different in a Bayesian trial, and that is the point. E9(R1) defines an estimand as "a precise description of the treatment effect reflecting the clinical

question posed by a given clinical trial objective" and requires that "the targets of estimation are to be defined in advance of a clinical trial" (International Council for Harmonisation 2019). It has five attributes: the treatment, the population, the variable, the strategy for each intercurrent event, and the population-level summary. The posterior is a summary of the data about a parameter, and the parameter is the estimand. The common error is to build a Bayesian model for a convenient analysis, a model on the observed outcome ignoring discontinuation, and then describe it afterwards in estimand language. The protocol should do it in the other order.

In the protocol: the estimand

"The primary estimand is the [summary, for example hazard ratio] of [variable] at [time] for participants with [condition], comparing [treatment] with [control], under a treatment policy strategy for [intercurrent event 1] and a [hypothetical, composite, while on treatment] strategy for [intercurrent event 2]. The Bayesian model in Section [x] targets this estimand: the parameter θ is [definition]; the likelihood handles [intercurrent event] by [method]. Sensitivity analyses address [list]."

23.2. The statistical analysis plan and the simulation report

The SAP is finalized before unblinding, and for a Bayesian design it carries the detail the protocol should not: the exact model, the likelihood, the parameterization of the prior, the computation (software, sampler, chains, burn-in, convergence diagnostics, seeds), the handling of missing data and intercurrent events in the model, the full set of success criteria at each analysis, and the plan for sensitivity analyses. The Bayesian draft asks for all of this "as early as possible" and says that FDA does not require any particular software (U.S. Food and Drug Administration 2026g). It asks that reproducibility be built in at the results stage as well: software versions, documented code and the seeds used to initialize chains. For the primary analysis it also fixes the baseline covariates for adjustment, the stratification factors that enter the model, the handling of missing covariate values and whether the estimand is conditional or marginal, as set out in Section 5.10.

The simulation report is a separate document, and for a Bayesian or complex adaptive design it is the central one. It is what replaces, for the reviewer, the closed form power calculation. Chapter 14 describes how to design the simulation and what to report; choosing a design by simulation has a longer history in Bayesian decision theory (P. Müller and Parmigiani 1995). For the protocol, what matters is the contents a regulator will look for, assembled from the Bayesian draft, the Adaptive Designs guidance and E20 Section 5:

1. The design as simulated, with every rule of the protocol, in enough detail to be reproduced, including enrollment rate, delay in observing the endpoint, dropout and any operational assumptions.
2. The scenarios: the null (global null where there are several arms), the design alternative, a range of plausible effects, effects smaller than planned, prior-data conflict scenarios, and, for several arms, configurations such as one effective arm or a non-monotone dose response. The draft asks that scenarios be "pre-specified, comprehensive, and plausible".
3. The operating characteristics under each scenario: probability of success (type I error under the null, power under the alternative), expected and maximum sample size,

probability of stopping at each look, bias, mean squared error, interval coverage, and for adaptive randomization the allocation distribution. For a non-calibrated design the draft asks for Bayesian power, expected bias, interval width and mean squared error.
4. Monte Carlo error: the number of simulated trials and the resulting precision on each reported probability. A type I error of 0.024 estimated from a thousand runs is not evidence that it is below 0.025.
5. Comparison with a simpler design. The draft asks for it for complex designs, and the question it answers is whether the complexity buys anything.
6. Sensitivity of the operating characteristics to the prior and to model misspecification.
7. The code, the seeds and the software environment.

The calibration that the SAP and the simulation report both reference can be a few lines of R. The chunk below simulates a binary-endpoint trial with four looks under the null (30% response in both arms), takes the success threshold c as the value for which the chance of crossing at any look is 2.5% in that calibration run, and then reports the operating characteristics at that threshold in a fresh null run with another seed and under a design alternative (30% against 45%). The report would give the same output for every scenario in the grid, with the seed and the package version.

```
sim0   <- oc_simulate(0.30, 0.30, looks = c(50, 100, 150, 200), nsim =
 ↪  20000, seed = 2026)
c_star <- oc_calibrate(sim0, alpha = 0.025)
sim1   <- oc_simulate(0.30, 0.45, looks = c(50, 100, 150, 200), nsim =
 ↪  20000, seed = 2027)
sim0v  <- oc_simulate(0.30, 0.30, looks = c(50, 100, 150, 200), nsim =
 ↪  20000, seed = 2028)
oc0 <- oc_summarize(sim0, c_star); oc0v <- oc_summarize(sim0v, c_star); oc1
 ↪  <- oc_summarize(sim1, c_star)
```

The calibrated threshold is $c = 0.9904$. The rejection proportion in the calibration run itself, 0.0249, is the calibration target and not an error assessment. On the fresh null run the type I error is 0.025 (Monte Carlo standard error 0.0011), and a protocol would repeat that validation across its prespecified nuisance grid (event rate, dropout, accrual). The power is 0.815, and the expected total sample size is 264 under the alternative against a maximum of 400. The same simulation gives the bias of the observed difference among trials that stop early for success, 0.062 under the alternative, which is the number the simulation report must show for the early-stopping reasons given in Chapter 21. The scenario grid, the futility rules (`oc_with_futility`) and the prior (the `prior` argument) are added the same way.

A sponsor that has done the calibration of Chapter 14 can write this report from its outputs. The report is versioned, and the protocol references a version by date, so that the design that was simulated is demonstrably the design that was run.

A final point about the SAP. The temptation in an adaptive design is to leave details to be fixed at an interim. E20 says the opposite: the number and complexity of adaptations in a confirmatory trial should generally be limited, and everything that drives a decision is prespecified. A Bayesian SAP that defers the prior or the success threshold to a later version has not prespecified the design.

23.3. Structured protocols: ICH M11 and SPIRIT 2025

Two standards shape the document itself, and neither is specific to Bayesian designs. ICH M11, the Clinical Electronic Structured Harmonised Protocol (CeSHarP), reached Step 4 in November 2025 and comes with a harmonized template and a technical specification, so that protocols present the same information in the same place and can be exchanged in a format acceptable to regulators (International Council for Harmonisation 2025c). SPIRIT 2025 is a checklist of minimum items a protocol of a randomized trial should address (Chan et al. 2025). The reporting counterpart for adaptive trials is the CONSORT 2010 adaptive-designs extension (ACE) (Dimairo et al. 2020).

The practical consequence is a mapping problem. M11 has sections where the design and statistics go, and a Bayesian design has components (prior, simulation-based calibration, predictive probability rules) that the template does not name. The sponsor should place them where a reviewer would look for the traditional counterpart: the prior with the statistical hypotheses and analysis methods, the decision rules with the interim analysis section, the operating characteristics with the sample size justification, and the detailed model and computation in the SAP, which the protocol references. The test of a good mapping is that a reviewer who has read only the protocol knows that the design is Bayesian, what the success criterion is, and where the simulation report is. A protocol that mentions "Bayesian methods" only in an appendix invites the first review question to be "where is the prior".

Several reporting guidelines sit around the protocol, and none of them is a regulatory requirement for a Bayesian trial. SPIRIT 2013 (Chan et al. 2013) is the antecedent of SPIRIT 2025, CONSORT 2025 (Hopewell et al. 2025) is the reporting guideline for the results of a randomized trial, ACE, which extends CONSORT 2010 and predates CONSORT 2025, adds items on the pre-specified adaptations, and a guideline on the content of statistical analysis plans lists what the SAP should contain (Gamble et al. 2017). Only a few guidelines are written for Bayesian analyses, and they are author guidelines: the seven-item ROBUST checklist for reporting a Bayesian analysis of a clinical study (Sung et al. 2005), the Bayesian Analysis Reporting Guidelines (BARG) (Kruschke 2021), and a consensus checklist for clinicians who read a trial analyzed by Bayesian methods (Ferreira et al. 2020). The items a Bayesian trial adds to the generic lists are the same ones this chapter has put in the protocol: the prior and its source, the success criterion with its threshold and calibration, the interim schedule and the simulation report, the software and seeds, and the sensitivity analyses to the prior. A report that gives each of them where the generic checklist asks for its traditional counterpart satisfies the intent of all of these lists.

23.4. The data monitoring committee, firewalls, and who sees what

Adaptive trials add a requirement that fixed trials mostly avoid: someone must see unblinded comparative data during the trial in order to apply the adaptation rules, and the sponsor must not. The Adaptive Designs guidance lists trial integrity as one of its four principles, and the integrity concern is that knowledge of interim results, or even of what decision was made, can change the behavior of investigators, patients and sponsor staff, and so bias the trial. The practice is by now standard, and the DMC guidance of 2006 (U.S. Food and Drug Administration 2006) remains the starting document (FDA issued a draft revision in

February 2024). What a Bayesian adaptive design adds is that the DMC charter must spell out the rules in more detail, because more of the design is executed at interim.

A workable arrangement has four parties.

- **The independent statistical center** (or an unblinded statistician walled off from the sponsor's team) holds the randomization code, runs the interim analyses, computes posterior and predictive probabilities, and reports to the DMC. It is also the place where an adaptive randomization update is computed and passed to the randomization system.
- **The DMC** reviews the unblinded reports, applies or overrides the rules, and makes recommendations to the sponsor. The charter states that the prespecified rules are guides and that the committee keeps its judgment on safety, so that nothing in the Bayesian machinery turns the DMC into a rule executor.
- **The sponsor's blinded team** receives from the DMC only what the charter allows: typically a recommendation to continue, stop or modify, and for a modification, the minimum information needed to implement it. For response-adaptive randomization, the allocation ratios themselves can reveal results, so the charter says who knows the current ratios and how site staff are kept from inferring a trend.
- **Regulators** may, under the charter, see more than the sponsor, and the sponsor should expect that a regulator reviewing a submission will want to see the interim reports, the minutes of the closed sessions, and the log of who had access.

In the protocol: DMC and firewall

"An independent DMC, whose membership and charter are described in Appendix [x], will review unblinded data at each analysis in Section [y]. The independent statistical center will compute all decision quantities; no member of the sponsor's study team, the investigators or the CRO's blinded team will have access to unblinded comparative results or to the current allocation ratios, except [named roles and what they see]. Access to the unblinded system is logged. The DMC charter specifies the prespecified rules, the DMC's authority to depart from them for safety or other reasons with documentation, the meeting schedule, the content of closed and open reports, and the communication path to the sponsor. A copy of the simulation report is available to the DMC."

What does a Bayesian design change about who sees what? Two things. First, the DMC should receive the simulation report, or a summary, so that its members know what the rules were designed to do and how often they are expected to trigger. A DMC that is asked to apply a posterior probability rule it does not understand will either follow it blindly or ignore it, and neither is acceptable. Second, firewalls in a platform trial are more complex, since arms enter and leave, and a sponsor of one arm should not see data of competing arms. Chapter 20 describes the structure, and one independent statistical center with one DMC over all arms is a possible arrangement. FDA permits an independent external DMC or another appropriate independent entity, and the EMA reflection paper discusses separate analysis or study teams for different interventions, so the proposed structure has to protect independence and confidentiality across arms (U.S. Food and Drug Administration 2026f; European Medicines Agency 2026a).

23.5. The interaction timeline with FDA

A regulator should never meet a Bayesian design for the first time in the submission. The Bayesian draft says that for complex designs the sponsor should request a meeting before the trial starts, and the substantial evidence draft says that a single-trial plan should be discussed with FDA before the single trial is started. The sequence below is the one a sponsor with a Bayesian confirmatory design should plan around. Meeting types and timelines are from the PDUFA formal meetings guidance of August 2026 (U.S. Food and Drug Administration 2026c); the program descriptions are from FDA's program pages and the CID guidance (U.S. Food and Drug Administration 2020a), which a sponsor should reread before applying, since program details change with each PDUFA cycle.

Table 23.2.: Where a Bayesian design is raised with FDA.

Stage	Meeting or program	What a Bayesian sponsor brings
Early development	INTERACT (for novel products before first-in-human); Critical Path Innovation Meeting (CPIM) for a method not tied to a product	Questions about a general approach, for instance a dose-finding design (Chapter 17) or a borrowing method.
Pre-IND	Type B meeting	The development plan and, where a Bayesian phase 1 or 2 design is proposed, the design and the safety rules.
Phase 2 design	Type C or Type B; CID or MIDD paired meeting for a complex design	The proposed adaptive or Bayesian design, with a simulation report.
End of phase 2	Type B (end of phase)	The phase 3 design: estimand, prior, success criterion and calibration, interim plan, evidence package.
Phase 3 protocol	Special Protocol Assessment (SPA) request (a written submission, optionally with a Type C meeting)	The final protocol and SAP; agreement on the design and analysis.
Before database lock	Type C	Changes to the SAP (but not the design, which cannot change after unblinding), and handling of deviations.
Pre-NDA or BLA	Type B	The planned submission, including the simulation report and the prior sensitivity analyses.

Meeting types. Type A is for a stalled program or an important safety issue, Type B covers milestone meetings including pre-IND and end-of-phase meetings, Type C is any other

meeting, Type D is a focused meeting on one or typically not more than two issues, and INTERACT is for novel products with unique early development challenges (U.S. Food and Drug Administration 2026c). In calendar days from receipt of the request, FDA's response time is 14 for Type A, 21 for Type B, 14 for an end-of-phase Type B, 21 for Type C, 14 for Type D and 21 for INTERACT. Scheduling the meeting, or delivering a written response only (WRO), is 30 days for Type A, 60 for Type B, 70 for an end-of-phase meeting, 75 for Type C and 50 for Type D (Tables 1 and 2 of the guidance; the guidance says when the meeting is scheduled relative to the request, and the times in force should be checked against the guidance in effect). A sponsor planning a protocol should count backwards: an end-of-phase meeting requested in January is a spring meeting, and the simulation report has to be ready before the briefing package is due, which is earlier. The meeting is an opportunity to get written FDA feedback on the design, and the format of a written response only is often enough for a narrow question such as whether the proposed prior is acceptable.

The CID paired meeting program. FDA's Complex Innovative Trial Design program was established under the PDUFA VI commitments, and continued under PDUFA VII, to give sponsors extra interaction on designs whose operating characteristics cannot be derived in closed form, which in practice means Bayesian designs and complex adaptive designs (U.S. Food and Drug Administration, n.d.c). The December 2020 guidance describes how to interact with FDA on such designs (U.S. Food and Drug Administration 2020a). As the program page describes it, the program now runs under PDUFA VII for fiscal years 2023 to 2027 and is run by CDER and CBER. Requests are due on the last day of each quarter (31 March, 30 June, 30 September, 31 December), FDA notifies sponsors within 45 days, and it selects one or two proposals per quarter, up to eight a year. A granted request receives an initial meeting and a follow-up meeting on the same proposal, the second approximately 90 days after FDA receives the briefing materials, which is when the simulation work is done and a reviewer can say something concrete. The sponsor needs a pre-IND or IND number, the trial cannot be first in human, and there must be enough clinical information to inform the proposed design. Before the first meeting the sponsor and FDA agree which information FDA may publish in a public case study, which is how the January 2026 draft can name examples. To apply, a sponsor submits a request that describes the product and the development program, the proposed design and why it needs the program (what is complex about it), the planned use in an application, the simulation plan, and why the design is better for this product than a conventional alternative. A selection is not an agreement on the design, and the program is not a substitute for the regular Type B and C meetings.

MIDD paired meetings. The Model-Informed Drug Development Paired Meeting Program offers a similar structure for modeling and simulation in a specific program (U.S. Food and Drug Administration, n.d.g). A Bayesian design with a pharmacometric component, such as exposure-response modeling to support dose selection or a prior built from a model, may fit here as well as at CID. The ICH M15 guideline on model-informed drug development (Step 4 in January 2026, International Council for Harmonisation (2026b)) frames the discussion in terms of question of interest, context of use, model influence, consequence of a wrong decision and model risk. A Bayesian trial whose prior or analysis depends on a model can use that vocabulary: the more the model decides, and the worse the consequences of a wrong decision, the more evaluation the regulator will want.

CPIM. Critical Path Innovation Meetings are nonbinding and not tied to a product, and are aimed at discussing emerging methods and technologies (U.S. Food and Drug Administration, n.d.e). A sponsor who has developed a method, for example a way to build a prior from historical data in a specific disease area, can seek FDA's informal views before any product depends on it. FDA describes CPIM discussions as non-regulatory, product-independent and

nonbinding on both FDA and the requester, and says they do not substitute for pre-IND, IND, NDA or BLA meetings.

Special Protocol Assessment. An SPA is a request for FDA's written agreement on the design, endpoints and statistical analysis of certain trials, including trials intended to form the primary basis of an efficacy claim, whatever their phase (U.S. Food and Drug Administration 2018b). The current guidance is the April 2018 revision. If FDA agrees, the agreement is documented, and FDA will not change its position on the agreed critical design elements unless a substantial scientific issue essential to determining safety or effectiveness is identified after the trial begins and the sponsor is notified (the statute also allows changes by agreement with the sponsor). An SPA resubmission after FDA comments is reviewed on a 45-day goal. It is worth considering for a Bayesian confirmatory trial, because it gets the prior, the criterion and the calibration agreed in writing before the data are seen. It is also demanding: FDA will not agree to what it has not seen, so the SAP and the simulation report must be complete, and the process will expose any design that cannot be defended. A sponsor should read the guidance on what happens if the sponsor changes the design afterwards. An SPA does not guarantee approval, or even that FDA will file the application: the agreement is about the design and not the result.

What to send. The Bayesian draft's Section VIII is a ready list of what FDA wants in a pre-trial package: the prior and its justification; for informative priors the sources, including those excluded; the success criteria and the justification of a and c; simulation evidence of familywise error control if calibrated, or Bayesian power and related metrics if not; the full simulation report with code; the computational details; and a comparison with simpler designs (U.S. Food and Drug Administration 2026g). A briefing package structured around that list makes the meeting short.

What to ask. Meetings are won by specific questions. "Is a Bayesian design acceptable?" gets a vague answer. "Does FDA agree that the success criterion $\Pr(\theta > 0.1 \mid \text{data}) > 0.985$, calibrated to a one-sided familywise error of 0.025 under the scenarios in Table [x], is acceptable for the primary estimand?" gets a yes, a no with reasons, or a request for more simulation. Good questions name the prior, the criterion, the scenario grid, the interim schedule, the borrowing (if any) and the evidence package.

23.6. EMA and other regulators

At EMA the corresponding route is scientific advice from the CHMP, with the Scientific Advice Working Party, and for novel methods a qualification procedure or the Innovation Task Force; in the EU the Complex Clinical Trials Q&A of the Commission, EMA and the Heads of Medicines Agencies discusses Bayesian methods and recommends early dialogue (European Commission and European Medicines Agency and Heads of Medicines Agencies 2022). The Q&A asks sponsors to justify the use of Bayesian methods and to document priors. The 2007 reflection paper on adaptive designs is more conservative than the FDA's guidance; ICH E20 is a Step 2 draft, and until it is final the 2019 guidance is FDA's current final guidance. The October 2026 draft reflection paper on confirmatory platform trials takes a cautious position on response-adaptive randomization and non-concurrent controls (European Medicines Agency 2026a). The practical advice is that a global development should plan for both agencies, request advice from EMA in parallel, and raise the points where the agencies may differ (adaptive randomization, borrowing from non-concurrent controls) in both. Where agencies differ, the protocol follows the stricter reading unless

advice says otherwise. The structured protocol format, ICH M11, is harmonized, which helps. Details of the programs, with their dates and links, are in Appendix B.

23.7. Substantial evidence: two trials, one trial, and where Bayesian evidence fits

The legal standard is in FD&C Act section 505(d), and 21 CFR 314.126 defines the adequate and well-controlled investigation. The statute speaks of "adequate and well-controlled investigations", and for decades the working interpretation was two trials, each significant at two-sided 0.05, which is one-sided 0.025 each. The book's companion online lab (module 11, https://vsokolov.org/bayesian_trials_lab/11-substantial-evidence.html) shows what that rule means arithmetically: a useless drug wins one trial in 40, and wins two independent trials about one time in 1,600 (one-sided 0.025 squared). Two trials are a strong filter, and a costly one.

FDA's revised draft guidance of June 2026 describes when one adequate and well-controlled trial plus confirmatory evidence can satisfy the standard (U.S. Food and Drug Administration 2026a). A draft, not final, and not binding. It says that such an approach "will involve a highly persuasive trial or a source of strong confirmatory evidence" (Section IV.A). Among the factors it weighs: the seriousness of the disease and the unmet need, and "whether it is ethical and practicable to conduct more than one adequate and well-controlled trial". Confirmatory evidence need not be another trial. The draft allows mechanistic evidence of the drug's treatment effect in a particular disease as confirmatory evidence in appropriate cases. And it says that the plan "should be discussed with FDA prior to initiating the single trial". The 2026 text revises the 2019 draft of the same name.

Where does Bayesian evidence fit? Three places, and they need to be kept apart.

First, a Bayesian trial can be one of the two trials, or the one trial. If its success criterion is calibrated to a familywise one-sided error of 0.025, it meets the default frequentist bar, and a reviewer treats it as a trial with a different analysis. This is the cleanest case, and nothing in the substantial evidence standard requires a p-value.

Second, a Bayesian analysis can combine the one trial with other evidence in a formal way. A prior built from earlier data, a borrowing model from an adult trial for a pediatric population, or a hierarchical model across related populations is a way to represent the confirmatory evidence. FDA's Bayesian draft discusses borrowing, and the REBYOTA program, where a prior phase 2 trial informed the phase 3 analysis, is the example it gives (Chapter 19; the "In practice" section below shows how its two success thresholds map onto one trial and two trials). The draft warns that a prior that is too informative "risks overwhelming the data collected in the target population", that dynamic discounting should borrow less when the data are less similar, and that type I error inflation is a poor measure of borrowing, so the sponsor must propose and justify other metrics such as the ESS. Here the Bayesian method does what the substantial evidence draft asks to be done informally. It does not remove the need for the persuasion: the judgment of whether the other evidence is relevant and strong remains with FDA.

Third, a Bayesian posterior probability can be offered as the measure of persuasiveness. A trial whose posterior probability of benefit is 0.9999 under a skeptical prior is "highly persuasive" in a way that a trial with $p = 0.024$ is not. The draft does not say that a particular posterior probability makes a trial highly persuasive, and a sponsor should not

claim so. The honest position is that the measure of how persuasive a single trial is can be stated on the posterior scale, as it can be stated on the p-value scale or as predictive probability of replication, and the reviewer decides. Module 11 of the companion lab makes this concrete: a trial that barely wins repeats only about half the time if its observed effect is the true effect, so a single marginal win is not persuasive on any scale. And the substantial evidence standard is not lowered by a fast review: expedited programs speed the review of serious conditions and do not lower the proof.

A final caution, from Chapter 21 and Chapter 7: among single trials that stop early for efficacy under a calibrated Bayesian rule, the observed effect estimate is biased away from the null on average, and the sponsor should show the bias in the simulation report and report estimates that account for it where possible. A highly persuasive single trial is persuasive about the existence of an effect, and its estimate may still need shrinkage.

23.8. In practice

Five documents show how sponsors wrote Bayesian rules into protocols and SAPs and what regulators did with them. Two are SAPs, one is an FDA briefing document, one an FDA case study, and one the FDA's own January 2026 list of design examples.

23.8.1. BNT162b2: thresholds calibrated to 2.5%

The vaccine trial's SAP wrote a beta-binomial success criterion with posterior thresholds of 0.995 at the interim analyses and 0.986 at the final analysis, calibrated so that the overall one-sided type I error was 2.5% (Pfizer Inc. 2022). The full case, with the prior, the arithmetic and the regulatory outcome, is in Section 21.5. FDA's announcements contain no statement on the Bayesian criterion, so the lesson is about the sponsor's side: a prior with its numbers stated, a calibrated threshold, a nonbinding futility rule and a changed look schedule (the first interim analysis was not performed) handled by saying what remained fixed. The binomial model assumes equal surveillance time in the two arms, an approximation that FDA's briefing document adjusts with the surveillance-time ratio.

23.8.2. DAWN: enrichment and a predictive-probability schedule

DAWN's SAP is the most complete Bayesian adaptive design among this chapter's sources: every threshold, every look, the enrichment rule and its irreversibility, and the predictive-probability stopping bounds were fixed before unblinding, with the type I error a simulation output (Stryker Neurovascular 2017; Jovin et al. 2017). The design is discussed in Chapter 15 and Chapter 14.

23.8.3. REBYOTA: the data alignment and the one-trial threshold

The protocol lesson is that success thresholds can be written in the language of the substantial evidence standard (analogues of a one-sided 0.025 for one trial and 0.00125 for a substitute for two), and that exchangeability is something the reviewer tests in the data alignment. The full case is in Section 19.17.1.

23.8.4. Belimumab in children: a post hoc Bayesian analysis requested by FDA

The protocol lesson is that when a borrowing weight cannot be justified in advance, a tipping-point table is a defensible fallback, and it is a post hoc one. FDA's case study recommends early consultation, ideally at a pre-IND meeting, and discussion of the detailed analysis plan before the clinical investigation begins (U.S. Food and Drug Administration, Center for Drug Evaluation and Research 2026). The full case is in Section 19.17.2.

23.8.5. The January 2026 draft's named CID examples

Two of the examples in FDA's Bayesian draft come from the CID program, whose case studies are published (U.S. Food and Drug Administration 2026g). A pediatric multiple sclerosis study is a randomized, double-blind, Bayesian group sequential non-inferiority trial against an approved active control, with one interim analysis for efficacy, using a robust meta-analytic-predictive prior from a completed pediatric trial and several adult trials, and FDA lists the first use of an active-controlled non-inferiority design in this setting, the borrowing, and model-based extrapolation from adults as the innovative features (U.S. Food and Drug Administration, n.d.b). An epilepsy with myoclonic-atonic seizures (EMAS) study is a placebo-controlled trial that borrows from related epilepsy populations through a hierarchical model, with sample size updated at unblinded interim analyses by Goldilocks predictive probabilities (U.S. Food and Drug Administration, n.d.a). Both are proposals discussed in the program, not approvals, and the case studies say so by calling them proposed. The lesson for this chapter is what the CID route produced: a published description of a design FDA found worth discussing, which a sponsor can cite in its own briefing package.

23.9. What the Bayesian design does not buy

A protocol chapter should say what the design cannot do, because sponsors who believe otherwise write the wrong protocol. There are four limits, taken from the working paper (Polson and Sokolov 2026b) and stated plainly.

The saving is not against a single unadjusted analysis at a fixed level. The Bayes-factor bar is below the unadjusted z_α^2 only at small information: it exceeds it once $I_n\tau^2$ passes about 41.6 for the exact normal-mixture boundary (for a unit-information prior, $n > 41.6$; Chapter 11). In the SELECT reconstruction an unadjusted single look crosses at 309 events on the mean path, earlier than the calibrated Bayesian rule's mean-path crossing near 600 events (conditional median stopping point 464). The unadjusted single look is not available to anyone who intends to look more than once, and the Bayesian gain is against the repeatedly monitored trial with the right error control, not against the fixed single look. Against a single analysis, the calibrated Bayesian bar is never lower unless an informative prior or borrowing is used.

The surcharge the Bayesian rule avoids buys a real guarantee. An e-value rule holds its level at every stopping time, including ones the protocol never contemplated: a look inserted after an external publication, an early stop ordered for reasons outside the trial, continuous unblinded inspection by a committee. A calibrated Bayesian rule holds 2.5 percent under the scheme to which it was calibrated, and under another scheme it holds

something else. A sponsor who cannot commit to a monitoring plan should read Chapter 12 before choosing.

The calibration requires knowing the monitoring plan and the horizon. That is exactly the knowledge the anytime valid construction is built to do without. The two paradigms make different promises. In protocol terms: the Bayesian design is credible only if the schedule is fixed, the horizon is stated, and deviations trigger a recalibration that is independent of interim results.

The power ordering is not universal. In the SELECT reconstruction the calibrated Bayesian rule had higher power than the e-value rule, 94.7 against 83.1 percent. In a two arm binary endpoint design reported by A. O. Sokolova and Sokolov (2026) the order is reversed, 68.8 against 72.3 percent, with the Bayesian rule stopping slightly earlier. The comparison depends on prior scale, endpoint and monitoring grid. The robust finding concerns stopping time, not power, and the direction of the power difference should be checked by simulation for the design at hand.

Two further statements belong with these, from the case studies. The traditional scheduled design wins in some settings: the REWIND comparison in Chapter 21 is one, a single look with a sponsor that has no interest in early stopping. And no design, Bayesian or otherwise, removes the need for an adequate and well-controlled trial. A Bayesian design with a poor control, a bad endpoint or an unblinded outcome assessment is still a poor trial.

23.10. Checklists

For the sponsor, before the end-of-phase 2 meeting and again before the protocol is final:

1. The estimand is written per E9(R1), with intercurrent event strategies, and the model targets it.
2. The prior is stated in full, with ESS, source of any external data (including excluded sources), design priors for simulation, and named sensitivity analyses.
3. The success criterion gives a and c with their clinical rationale, for each primary and key secondary estimand and at each interim analysis.
4. Calibration is by simulation to familywise one-sided error 0.025, or the sponsor has asked FDA in advance to accept a different standard and said why.
5. The interim schedule is fixed, with information fractions or calendar rules, and the operating characteristics were simulated on that schedule. The protocol says what happens if the schedule is missed.
6. The simulation report is complete (scenarios, Monte Carlo error, comparison with a simpler design, code, seeds, software) and the protocol cites its version.
7. Adaptive randomization, if used, has burn-in, floors, caps, time-trend handling and a bias assessment.
8. Safety rules are stated as posterior statements with simulated false-stop and missed-harm rates, and the DMC retains its judgment.
9. The DMC charter, firewall and access log are in place, and the DMC has seen the simulation summary.
10. For multiple arms or endpoints, familywise error control is shown where the guidance expects it, or the departure is explained.
11. The computation is documented: software, sampler settings, convergence diagnostics, seeds, non-convergence plan.

12. The evidence plan is clear: two trials, or one trial plus confirmatory evidence, and FDA has been asked about it before the trial starts.
13. The meeting plan is dated: which meeting, which request type, which package, which questions.
14. The protocol is mapped into the M11 template and SPIRIT items, with the Bayesian components placed where a reviewer would look.

For the reviewer, reading a Bayesian submission:

1. Is the prior prespecified, and could the data have changed it? Is the ESS reported and reasonable relative to the trial size?
2. Is each informative component justified by relevant, complete evidence, including skeptical evidence and excluded sources, and what happens under prior-data conflict?
3. What is the success criterion, and is it calibrated to the standard that applies? If not, was a different standard agreed?
4. Do the simulated scenarios include the null, plausible smaller effects, and prior-data conflict, and does the Monte Carlo error support the reported rates?
5. Was the simulated design the design that was run, in schedule, sample size, randomization and analysis?
6. Does the estimate at an early stop show the bias that stopping early implies, and does the report address it?
7. Were firewalls maintained, and does the access log show it?
8. How much does the conclusion depend on the prior, and does the report show a range of priors?
9. For one trial plus confirmatory evidence: is the trial highly persuasive on the stated scale, and is the other evidence relevant to the population and endpoint?
10. Is the whole analysis reproducible from the code and seeds supplied?

23.11. What to take away

Two pieces of advice are distinctive to this book and worth stating once. Clinicians and payers should ask what prior was used and how many patients it was worth, and should read a posterior probability as conditional on the model and the prior. Sponsors, regulators and statisticians should decide early whether the trial needs validity at stopping times the protocol cannot anticipate, and should report the realized type I error of whatever rule is used, with power beside it and stopping times as distributions. The key points below summarize the chapter.

23.11.1. Open questions

Several questions remain open. The e-value price is exact for mixture e-values built on the same statistic as the Bayes factor, as an extra squared-standardized gap of $2\log(1/\alpha)(1 + 1/q)$, whose constant large-information approximation is $2\log(1/\alpha) = 7.38$ at $\alpha = 0.025$ (Chapter 11). For betting e-processes with adaptively chosen bets the analogous quantity depends on the strategy and we have no general expression. The calibration of a posterior rule to a frequentist error under continuous monitoring, which this book did numerically, has no asymptotic characterization here. A posterior-probability cutoff cannot itself diverge, so any such result would concern a transformed boundary, such as the implied standardized boundary, over a stated monitoring horizon, and we do not state one. Finally, the utility-based designs

of P. F. Thall and Cook (2004) and Houédé et al. (2010) have operating characteristics that have not been compared with the conventional apparatus on the deviation-scale ledger of Chapter 11, and doing so would show whether the phase 1 and 2 savings have the same source as the confirmatory ones.

Try it in the lab

Module 11, Substantial evidence (https://vsokolov.org/bayesian_trials_lab/11-substantial-evidence.html): choose a first trial result and predict how often a second trial of the same drug would also win, then run it. Compare the answer for a marginal win with a very small p-value, and see why "highly persuasive" matters for the one-trial route.
Module 12, Decisions (https://vsokolov.org/bayesian_trials_lab/12-decisions.html): move the type I error limit and watch what it costs in patients. Predict whether tightening or loosening the limit changes the false alarms or the missed drugs more, then check.
Module 13, Regulators' map (https://vsokolov.org/bayesian_trials_lab/13-regulators-map.html): open the meeting clocks and find the days FDA takes to reply to a Type B end-of-phase request and the days to the meeting. Count back from a planned protocol date to the date by which the request must be filed.
Module 14, Studio (https://vsokolov.org/bayesian_trials_lab/14-studio.html): the full open simulator. Build the four-look design of the code chunk above, set the scenarios and read the operating characteristics table, the same table a simulation report carries. Predict first whether the calibrated threshold changes when the looks move, then check.

Key points

- A Bayesian protocol keeps the estimand, the randomization and the prespecification of a traditional one. It changes the objective, the prior, the sample size (a maximum), the interim rules, the multiplicity handling, the safety rules and, above all, how operating characteristics are shown.
- The prior should be prespecified and stated with its effective sample size, with named sensitivity analyses and design priors for simulation.
- The success criterion is $\Pr(\delta > a \mid \text{data}) > c$, calibrated by default in this book to one-sided familywise error 0.025. FDA's draft also describes direct interpretation and decision-theoretic routes; a different standard, for instance with borrowing, should be agreed with FDA in advance.
- The simulation report is the central deliverable: scenarios, operating characteristics, Monte Carlo error, comparison with a simpler design, code and seeds.
- The DMC charter and firewall are part of the design, and the DMC should see the simulation summary.
- Raise the design early: pre-IND, Type B end of phase 2, the CID and MIDD paired meetings for complex designs, CPIM for methods, SPA for agreement on the phase 3 protocol, and EMA scientific advice in parallel.
- One adequate and well-controlled trial plus confirmatory evidence can meet the standard under the June 2026 draft, if highly persuasive or strongly confirmed, and it should be discussed with FDA before the trial starts.
- The Bayesian design does not beat a single unadjusted analysis, does not give validity at unplanned stopping times, requires a known monitoring plan, and

does not always give more power than an anytime valid rule.

Discussion questions

1. A sponsor proposes a Bayesian phase 3 with four interim looks and a success criterion calibrated to 0.025 under a schedule of looks at 25, 50, 75 and 100 percent of information. Enrollment is faster than expected and the third look happens at 62 percent. What should the protocol have said, and what will the reviewer ask?
2. A prior is built from a phase 2 trial in the same program, with an ESS of 40 events in a phase 3 of 300 events. Write the three sentences you would want in the protocol, and the two additional simulations you would offer in the briefing package.
3. Should a sponsor request a CID paired meeting, a Type B end-of-phase meeting, or an SPA first? What does each buy, and what does each require the sponsor to have ready?
4. A rare disease trial can enroll 80 patients. The sponsor wants to rely on a single trial plus a Bayesian borrowing prior from natural history data. Using the June 2026 draft, list what FDA will weigh, and what the simulation report must show to make the borrowing defensible.
5. The authors list four limits of the Bayesian design. For a trial you know, which of the four bites hardest, and what would you put in the protocol in response?

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A. Notation

This appendix fixes the meaning of every symbol the book uses, defines the abbreviations, and gives short definitions of the key terms. Each symbol has one meaning throughout. Where a chapter needs a local symbol, it defines it on the spot and does not reuse one from this table.

A.1. Symbols

Symbol	Meaning
θ	Treatment effect (generic). For time-to-event outcomes $\theta = -\log \mathrm{HR}$, so that benefit is positive.
δ	Treatment effect in the FDA draft's success criterion $\Pr(\delta > a \mid \text{data}) > c$. Used only when quoting that criterion; elsewhere the book writes θ.
a, c	The minimum clinically relevant effect and the posterior probability threshold in that criterion.
b	Threshold on the Bayes factor (or on the e-value, $b = 1/\alpha$), distinct from the posterior probability threshold c.
p_j	Response probability in arm j. The index $j = 0$ is the control arm.
n, n_j	Sample size in patients or, for time-to-event outcomes, the number of events. n_j is the size of arm j.
I	Fisher information per patient (or per event). The total information is nI.
Z_n	Standardized test statistic after n observations.
y, $y_{1:n}$, Y	Observed data, the first n observations, and the corresponding random variable.
$\pi(\theta)$	Prior density of θ.
$\pi(\theta \mid y)$	Posterior density of θ given the data.
$p(y \mid \theta)$	Likelihood of the data, read as a function of θ.
π_0, π_a	Prior probabilities of the null and of the alternative hypothesis.
α	One-sided type I error rate, 0.025 unless stated otherwise.

Symbol	Meaning
β	Type II error rate. Power is $1 - \beta$.
z_α	Upper α quantile of the standard normal distribution.
BF, BF_{10}	Bayes factor, with BF_{10} in favor of the alternative over the null.
E_n	E-value, or test martingale, at stage n.
t	Information fraction in a group sequential design, the share of the planned information accrued so far.
$\mathrm{Beta}(a, b)$	Beta distribution with shape parameters a and b, the conjugate prior for a response probability. Here a is a shape parameter and not the threshold a of the success criterion above.
τ	Prior scale of the treatment effect: $\theta \sim N(0, \tau^2)$ in the SELECT reconstructions of Parts III to V, with $\tau = 0.25$.
τ_b	Between-stratum or between-arm standard deviation in the shrinkage examples of Section 21.4, Section 22.5 and Chapter 20. A local symbol, distinct from the prior scale τ.
p^*	Posterior probability threshold for continuous monitoring, calibrated by simulation so that the type I error is 0.025 (about 0.995 in the SELECT reconstruction).
d	Number of events in a time-to-event trial. It plays the role of n, with total information $I_n = d/4$ for the log hazard ratio (per-event information 1/4) under one to one allocation.
S	Score statistic, $S = \theta I_n$ on the mean path, with $Z = S/\sqrt{I_n}$.
L_n	Ockham penalty of a normal prior of scale τ, $L_n = \log(1 + I_n\tau^2)$; in the large-information approximation the Bayes-factor bar is $Z_n^2 \geq L_n + 2\log b$, with $Z_n > 0$ for a one-sided claim; the exact bar for the normal mixture multiplies the right side by $1 + 1/(I_n\tau^2)$ (Chapter 7, Chapter 11, Chapter 12).
I_n	Total information after n observations, nI.
$N(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2.
$X \sim F$	X is distributed as F.
$\mid$	Conditioning, as in $\pi(\theta \mid y)$.

$\mathbb{E}$, $\mathbb{P}$	Expectation and probability. The text says with respect to what: the prior, the posterior, or the sampling distribution.

A.2. Notation and package arguments

The companion package `bayestrials` (Appendix C) uses plain-text argument names. The table maps the book's symbols to them.

Table A.2.: Book symbols and `bayestrials` arguments.

Symbol	Argument	Meaning in the package
α	`alpha`	One-sided type I error, default 0.025.
θ	`theta`	Treatment effect; for time-to-event outcomes $\theta = -\log \mathrm{HR}$.
p_0, p_1	`p0`, `p1`	Response probabilities in the control and treatment arms.
n	`n`	Number of patients or events. In Chapter 4, n is the per-arm sample size with $n_0 = n_1 = n$, an explicit exception to the notation n_j for arm sizes used elsewhere. In `oc_simulate` the interim sample sizes are given per arm in `looks`.
number of simulated trials	`nsim`	Number of Monte Carlo replications.
random seed	`seed`	If not `NULL`, the function calls `set.seed(seed)` first.
a	`a`	Minimum clinically relevant effect in $\Pr(\delta > a \mid \text{data}) > c$.

A.3. Abbreviations

Abbreviation	Meaning
BLA	Biologics License Application.
BOIN	Bayesian optimal interval design for dose finding.
CID	Complex innovative trial design. Also the name of the FDA paired meeting program for such designs.

Abbreviation	Meaning
CPIM	Critical Path Innovation Meetings, an FDA program for nonbinding, product-independent discussion of methods.
CRM	Continual reassessment method, a model-based dose-finding design.
CSR	Clinical study report.
DDT	Drug development tool, as in the FDA qualification programs.
DMC	Data monitoring committee.
EMA	European Medicines Agency.
ESS	Effective sample size of a prior, the number of patients' worth of information it carries.
FDA	United States Food and Drug Administration.
FWER	Familywise error rate, the probability of at least one false positive among a family of hypotheses.
HMA	Heads of Medicines Agencies.
HR	Hazard ratio.
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use.
IND	Investigational New Drug application.
IRT	Interactive response technology, the system that randomizes patients and assigns drug supply.
ITT	Intention to treat, analysis of all randomized patients in the arms to which they were assigned.
MAMS	Multi-arm multi-stage design.
MAP	Meta-analytic-predictive prior, built from historical trials.
MHRA	Medicines and Healthcare products Regulatory Agency (United Kingdom).
MIDD	Model-informed drug development.
NDA	New Drug Application.
OC	Operating characteristics of a design, such as type I error, power and expected sample size.
PDUFA	Prescription Drug User Fee Act, the statute and commitment letters under which FDA programs such as CID are run.
QALY	Quality-adjusted life year.
RAR	Response-adaptive randomization.
SAP	Statistical analysis plan.
SPA	Special protocol assessment, an agreement with FDA on a protocol's design and analysis.

A.4. Key terms

Assurance. The probability of trial success averaged over a prior for the true effect, as opposed to power, which is computed at one assumed effect (Chapter 9).

Concurrent control. Control patients randomized during the same period as the patients on a given experimental arm. Control patients enrolled before that arm joined a platform trial are non-concurrent (Chapter 20).

Effective sample size. The number of patients whose data would carry the same information as the prior distribution (Chapter 8).

E-value. A nonnegative statistic whose expectation under the null is at most one. A running product of nonnegative factors forms a test supermartingale when each factor has conditional expectation at most one given the preceding data under every null distribution, and then, by Ville's inequality, it allows valid decisions at any stopping time (Chapter 12).

Estimand. The precise description of the treatment effect to be estimated: the treatment condition, the population, the variable, how intercurrent events are handled, and the population-level summary (ICH E9(R1), Chapter 5).

Intercurrent event. An event after treatment starts that affects the interpretation or existence of the measurements, such as discontinuation, rescue medication or death.

Operating characteristics. The frequentist behavior of a design under stated scenarios, usually by simulation: type I error, power, expected sample size, bias (Chapter 14).

Predictive probability. The probability, given the data so far, that a trial will meet its success criterion at the planned end, averaging over the unseen future data (Chapter 15).

Posterior probability of benefit. $\Pr(\delta > a \mid \text{data})$, the quantity compared with the threshold c in the FDA Bayesian draft (Chapter 7).

B. Regulatory Documents and Programs

This appendix lists, in one place, each regulatory document and program the book relies on. Status is as of November 2026. A draft guidance is not binding: it proposes recommendations that may change before they are finalized, and it does not represent finalized agency thinking. Final FDA guidances also generally contain nonbinding recommendations. Dates are those printed on the document. Where this book has not confirmed an official address, the entry says "URL: see issuer website". Chapter 3 is the map that explains how the documents fit together; the cross-references below say where each is used.

B.1. Documents

FDA, Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products. Draft guidance, January 2026. URL: https://www.fda.gov/media/190505/download. It sets out when Bayesian primary inference is acceptable, how priors and borrowing should be justified, and how success criteria such as $\Pr(\delta > a \mid \text{data}) > c$ may be calibrated to a type I error target or justified on other grounds. It expects a simulation report of operating characteristics. Used in Chapter 1 to Chapter 4, Chapter 6 to Chapter 9, Chapter 12, Chapter 14, Chapter 17 to Chapter 23.

ICH E20, Adaptive Designs for Clinical Trials. Step 2 draft, endorsed 25 June 2025. URL: https://www.fda.gov/media/188961/download (FDA-hosted). It gives principles for confirmatory adaptive designs: pre-specification, control of the type I error rate, reliable estimation, and simulation to show operating characteristics. Not final. Used in Chapter 3, Chapter 6, Chapter 8, Chapter 12, Chapter 14, Chapter 23 (cited by key; Chapter 11, Chapter 15 and Chapter 16 discuss it in text).

FDA, Adaptive Designs for Clinical Trials of Drugs and Biologics. Final guidance, November 2019. URL: https://www.fda.gov/media/78495/download. It states principles for adaptive designs and says that simulation may be needed to estimate error rates when analytic methods are unavailable. It also covers confidentiality of interim data. Used in Chapter 1, Chapter 3, Chapter 12, Chapter 14, Chapter 17, Chapter 20, Chapter 21, Chapter 23.

FDA, Master Protocols for Drug and Biological Product Development. Revised draft guidance, June 2026 (it replaces the draft of December 2023). URL: https://www.fda.gov/media/174976/download. It covers umbrella, basket and platform protocols, including shared controls and the handling of multiplicity across substudies. Used in Chapter 3, Chapter 20.

FDA, Demonstrating Substantial Evidence of Effectiveness for Human Drug and Biological Products. Revised draft guidance, June 2026 (it revises the draft of December 2019). URL: https://www.fda.gov/media/133660/download. It explains how the statutory standard can be met, including with one adequate and well-controlled trial plus confirmatory evidence. Used in Chapter 3, Chapter 21, Chapter 23.

ICH E8(R1), General Considerations for Clinical Studies. Final (Step 4), 6 October 2021. URL: https://database.ich.org/sites/default/files/E8-R1_Guideline_Step4_2021_1006.pdf. It describes quality by design, the role of randomization and control groups, and states that adaptive designs are acceptable when prospectively planned. Used in Chapter 1, Chapter 3.

ICH E9, Statistical Principles for Clinical Trials. Final (Step 4), 5 February 1998. URL: https://database.ich.org/sites/default/files/E9_Guideline.pdf. It covers randomization, blinding, the intention-to-treat principle, multiplicity and interim analysis. Used in Chapter 1 to Chapter 4, Chapter 6, Chapter 9, Chapter 10, Chapter 21, Chapter 23.

ICH E9(R1), Addendum on Estimands and Sensitivity Analysis. Final (Step 4), 20 November 2019. URL: https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf. It introduces the estimand framework and intercurrent events. Used in Chapter 1, Chapter 3, Chapter 9, Chapter 23.

ICH M15, General Principles for Model-Informed Drug Development. Final (Step 4), endorsed January 2026; the FDA edition is dated June 2026. URL: https://www.fda.gov/media/184747/download. It sets general principles for planning, documenting and assessing model-informed evidence. Used in Chapter 3, Chapter 17, Chapter 23.

FDA, Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products. Draft guidance, January 2025. URL: https://www.fda.gov/media/184830/download. It proposes a risk-based framework for credibility of AI models used in regulatory submissions. Used in Chapter 3.

FDA, General Considerations for the Use of New Approach Methodologies in Drug Development. Draft guidance, March 2026. URL: https://www.fda.gov/media/191589/download. It gives general considerations for non-animal and other new methods in drug development. Used in Chapter 3.

FDA, Formal Meetings Between the FDA and Sponsors or Applicants of PDUFA Products. Final procedural guidance, August 2026 (it withdraws the December 2017 guidance and a June 2018 guidance). URL: https://www.fda.gov/media/172311/download. It describes meeting types (A, B, C and others) and how to request them. Used in Chapter 3, Chapter 23.

ICH E10, Choice of Control Group and Related Issues in Clinical Trials. Final (Step 4), 20 July 2000. URL: https://database.ich.org/sites/default/files/E10_Guideline.pdf. It describes the types of control and when each is suitable, including the limits of external controls. Used in Chapter 19 (Chapter 5 does not cite it by key).

ICH E11A, Pediatric Extrapolation. Final (Step 4), adopted 21 August 2024. URL: https://database.ich.org/sites/default/files/ICH_E11A_Guideline_Step4_2024_0821.pdf. It sets out how evidence from adults or other sources may be extended to children, including the use of Bayesian methods. Used in Chapter 3, Chapter 19.

ICH E4, Dose-Response Information to Support Drug Registration. Final (Step 4), 10 March 1994. URL: https://database.ich.org/sites/default/files/E4_Guideline.pdf. It describes how dose-response data should be obtained and used. Used in Chapter 17.

FDA, Multiple Endpoints in Clinical Trials. Final guidance, October 2022. URL: https://www.fda.gov/media/162416/download. It sets out strategies for controlling the familywise error rate with several endpoints. Used in Chapter 3, Chapter 10.

FDA, Optimizing the Dosage of Human Prescription Drugs and Biological Products for the Treatment of Oncologic Diseases. Final guidance, August 2024. URL: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/optimizing-dosage-human-prescription-drugs-and-biological-products-treatment-oncologic-diseases. It recommends comparing multiple doses before pivotal trials rather than relying on the maximum tolerated dose. Used in Chapter 3, Chapter 17. Background on the initiative: https://www.fda.gov/about-fda/oncology-center-excellence/project-optimus.

FDA, Establishment and Operation of Clinical Trial Data Monitoring Committees. Final guidance, March 2006; a draft revision, Use of Data Monitoring Committees in Clinical Trials, was issued in February 2024. URL: https://www.fda.gov/media/75398/download (2006 text). It describes when a DMC is needed, its independence and the confidentiality of interim data. Used in Chapter 3, Chapter 23.

FDA, Considerations for the Design and Conduct of Externally Controlled Trials for Drug and Biological Products. Draft guidance, February 2023. URL: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/considerations-design-and-conduct-externally-controlled-trials-drug-and-biological-products. It gives recommendations on external control arms built from real-world or earlier trial data. Used in Chapter 3, Chapter 19.

FDA, Guidance for the Use of Bayesian Statistics in Medical Device Clinical Trials. Final guidance, February 2010. URL: https://www.fda.gov/media/71512/download. It is the long-standing precedent for Bayesian primary analysis in device trials, covering priors, borrowing and simulation of error rates. Used in Chapter 3, Chapter 8, Chapter 19.

EC, EMA and HMA, Complex Clinical Trials: Questions and Answers. Final, EMA/298712/2022, version of 23 May 2022, published June 2022. URL: https://health.ec.europa.eu/system/files/2022-06/medicinal_qa_complex_clinical-trials_en.pdf. It addresses the authorization and conduct of trials with complex designs in the European Union. Used in Chapter 3, Chapter 20, Chapter 23.

EMA, Reflection Paper on Methodological Aspects in Confirmatory Platform Trials. Draft, EMA/217509/2026, public consultation 19 October 2026 to 31 January 2027. URL: https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-methodological-aspects-confirmatory-platform-trials_en.pdf. It covers pre-specification, randomization, controls including non-concurrent controls, type I error control and estimands for platform trials. Used in Chapter 3, Chapter 20, Chapter 23.

MHRA, Draft Guideline on the Use of External Control Arms Based on Real-World Data to Support Regulatory Decisions. Draft, 2025. URL: https://www.gov.uk/government/consultations/mhra-draft-guideline-on-the-use-of-external-control-arms-based-on-real-world-data-to-support-regulatory-decisions. It states the United Kingdom regulator's expectations for external control arms. Used in Chapter 3, Chapter 19.

FDA, Diabetes Mellitus: Evaluating Cardiovascular Risk in New Antidiabetic Therapies to Treat Type 2 Diabetes. Final guidance, December 2008; withdrawn in March 2020 and replaced by the draft guidance "Type 2 Diabetes Mellitus: Evaluating the Safety of New Drugs for Improving Glycemic Control" (March 2020, draft; Federal Register 10 March 2020, https://www.govinfo.gov/content/pkg/FR-2020-03-10/html/2020-04877.htm). URL: https://www.federalregister.gov/documents/2008/12/19/E8-30086/guidance-for-industry-on-diabetes-mellitus-evaluating-cardiovascular-risk-in-new-antidiabetic (Federal Register availability notice). It asks sponsors to show that the upper bound of the 95 percent confidence interval for the cardiovascular risk ratio lies below 1.8 (and, where it lies between

1.3 and 1.8, to show it below 1.3 after approval). This is why cardiovascular outcomes trials such as REWIND are large. Used in Chapter 21.

FDA, Special Protocol Assessment. Guidance for industry, April 2018 (Revision 1). URL: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/special-protocol-assessment-guidance-industry. It describes the written agreement between FDA and a sponsor on the design, endpoints and statistical analysis of certain phase 3 protocols, and when FDA may change its position. Used in Chapter 23.

ICH M11, Clinical Electronic Structured Harmonised Protocol (CeSHarP). Final (Step 4), November 2025. URL: https://database.ich.org/sites/default/files/ICH_Step4_M11_Final_Guideline_2025_1119.pdf. It gives a harmonized protocol template and a technical specification so that protocols present the same information in the same place. Used in Chapter 23.

B.2. Products and trials named in FDA's January 2026 Bayesian draft guidance

Section III of the January 2026 draft guidance (printed pages 3 to 5) lists situations where Bayesian methods have been used and names the following products and trials. This list is of what the draft names, not an assessment of each case. Where the book has read a primary FDA document on a case, that source is given.

- **REBYOTA (RBX2660), phase 3 trial.** Borrowing from a previous phase 2 placebo-controlled study of the same product (Section III.A). The draft says the Bayesian analysis supported effectiveness and that REBYOTA was approved in 2022. The FDA briefing document for the advisory committee meeting of 22 September 2022 describes the hierarchical model with dynamic borrowing and its two success thresholds (U.S. Food and Drug Administration, Center for Biologics Evaluation and Research 2022).
- **GBM AGILE (NCT03970447) and Precision Promise (NCT04229004).** Oncology platform trials that augment the concurrent control with external or non-concurrent control data, using Bayesian models for temporal shifts (Section III.B).
- **A pediatric multiple sclerosis non-inferiority study** run through the CID program, with a prespecified Bayesian analysis that borrows historical active-comparator data (Section III.B, revisited in Section V). It is described in an FDA CID case study (https://www.fda.gov/media/172313/download).
- **Empagliflozin and linagliptin in pediatric type 2 diabetes.** Supportive Bayesian analyses in supplements, with borrowing from adult data (Section III.C).
- **An epilepsy trial in epilepsy with myoclonic-atonic seizures (EMAS)**, proposed through the CID program, with a Bayesian hierarchical model that borrows from trials in other epilepsy types (Section III.D; CID case study https://www.fda.gov/media/172312/download).
- **Liraglutide, LEADER trial.** A Bayesian hierarchical model for regional effects, and Bayesian hierarchical subgroup analyses in FDA Drug Trials Snapshots, with the Rinvoq snapshot as the example (Section III.E).

Oncology dose finding is discussed in Section III.F by method (CRM, BLRM, BOIN, mTPI) without a named product. Belimumab (Benlysta) in pediatric lupus is not named in the draft, but FDA's own case study on that approval shows a post hoc Bayesian analysis with a robust mixture prior borrowed from adult trials (U.S. Food and Drug Administration,

Center for Drug Evaluation and Research 2026), and it is used in Chapter 19 as a related example.

B.3. FDA programs

Complex Innovative Trial Design (CID) Meeting Program. Active, a PDUFA commitment. URL: https://www.fda.gov/drugs/development-resources/complex-innovative-trial-design-meeting-program. It offers sponsors additional meetings with FDA on novel designs, including Bayesian and simulation-heavy ones. The final guidance Interacting with the FDA on Complex Innovative Trial Designs (December 2020) describes the process. Used in Chapter 3, Chapter 14, Chapter 23.

Model-Informed Drug Development (MIDD) Paired Meeting Program. Active. URL: https://www.fda.gov/drugs/development-resources/model-informed-drug-development-paired-meeting-program. It offers paired meetings on modeling and simulation approaches for a specific development program. Used in Chapter 3, Chapter 23.

Critical Path Innovation Meetings (CPIM). Active. URL: https://www.fda.gov/drugs/novel-drug-approvals-fda/critical-path-innovation-meetings-cpim. These are nonbinding, product-independent discussions of methods and technologies. Used in Chapter 3, Chapter 23.

Drug Development Tool (DDT) Qualification Programs. Active. URL: https://www.fda.gov/drugs/development-approval-process-drugs/drug-development-tool-ddt-qualification-programs. They qualify a tool for a stated context of use that can then be used across development programs. Used in Chapter 3.

C. Software and Code

All numerical results in the book are produced by R code that the reader can run. Nearly all of the computation lives in a companion R package, `bayestrials`, and the chapters call it in short chunks and draw the figures with ggplot2. This appendix describes how to install the package, how it is organized, a worked example, how to reproduce the results, the companion online lab, and the established software that practitioners use for trial design.

C.1. Installation

The book was built with R 4.5.2. The package declares R 4.1 as its minimum version; it is version 0.1.0, it exports 203 functions, it passes `R CMD check`, and its tests run under testthat. The source is in the `bayestrials/` directory of the book repository. From a clone of the repository it installs with

```
R CMD INSTALL bayestrials
```

or, directly from its GitHub repository (https://github.com/VadimSokolov/bayestrials), with the line below.

```
remotes::install_github("VadimSokolov/bayestrials")
```

The chapters also use ggplot2 for figures, and the package imports only `stats` and `parallel`, with `mvtnorm` and the other packages it can use listed under Suggests. Quarto renders the book through knitr, and each chapter begins with a hidden setup chunk that loads `bayestrials` and `ggplot2`.

C.2. How the package is organized

Functions are named in snake_case, and the prefix tells the reader which part of the book the function belongs to. Arguments follow one convention throughout (see Section A.2): `alpha` is the one-sided type I error, `theta` an effect, `p0` and `p1` response rates, `n` a number of patients or events, `nsim` the number of simulated trials, and `seed` an optional random seed. Functions return numbers, named vectors, data frames or named lists. They never print or plot.

Table C.1.: Function families of `bayestrials`.

Family	What it does	Where used
Core posterior and `gs_`	Beta-binomial and normal posteriors (`beta_post`, `prob_superior`, `normal_update`), effective sample size (`ess_beta`, `ess_normal`), group sequential boundaries and crossing probabilities (`gs_boundary`, `gs_spending_boundary`, `gs_crossing_prob`, `gs_tolls`)	Chapter 6, Chapter 8, Chapter 11
Basics and `assign_`	Permutation test, binary z test, simple, block, stratified and minimization randomization (`assign_*`), covariate imbalance, small simulations of two-arm trials	Chapter 4, Chapter 5
Evidence	Bayes factors, minimum Bayes factor, false positive risk, the Bayesian bar	Chapter 7
`n_` and power	Sample size for binary, continuous and time-to-event endpoints (`n_fixed`, `n_binary`, `n_continuous`, `n_md`), power, assurance, number of events	Chapter 9
`oc_`	Simulation of Bayesian designs with interim looks, calibration of the threshold to a type I error target, operating characteristics tables (`oc_simulate`, `oc_calibrate`, `oc_summarize`)	Chapter 14
`pp_`	Predictive probability of success, conditional power, backward induction	Chapter 15
`rar_`	Response-adaptive randomization, Thompson sampling, allocation probabilities	Chapter 16
`dose_`	3+3, CRM, BOIN, isotonic regression, dose-finding simulation	Chapter 17

Family	What it does	Where used
`borrow_`	Power priors, meta-analytic-predictive and robust priors, basket posteriors, borrowing decision tables	Chapter 19
`platform_`	Shared control, correlation between arms, false positive counts, time trends	Chapter 20
`mult_` and `shrink_`	Bonferroni, Holm, Hochberg and graphical procedures, empirical Bayes shrinkage and posterior probabilities	Chapter 10
`ev_`	E-values, test martingales, mixtures, betting paths	Chapter 12
`select_`	The reconstruction of the SELECT trial: log Bayes factor, look table, event timing, simulation	Chapter 21

The package also contains four datasets: `ulcer` (the four-operation ulcer trial of Chapter 10), `select_design` (the design of the SELECT trial, Chapter 21), `ctgov_counts` and `ctgov_bayesian_records` (counts and records of registered trials that mention Bayesian methods, retrieved from ClinicalTrials.gov). Each function has a help page with an example that runs in under a second, and the files `bayestrials/data-raw/map-*.md` map the book's earlier Python functions to their R replacements.

C.3. A worked example

The example calibrates a two-arm Bayesian design with three looks, at 100, 200 and 300 patients per arm and a control response rate of 0.30. The success rule is that the posterior probability of superiority exceeds a threshold c at any look. `oc_calibrate` chooses the smallest c that keeps the type I error at 0.025 under the null, and then the same threshold is applied to the alternative of a response rate of 0.45.

```
library(bayestrials)
looks <- c(100, 200, 300)
sim0 <- oc_simulate(p0 = 0.3, p1 = 0.3,  looks = looks, nsim = 20000, seed
 ↪  = 1)
c_star <- oc_calibrate(sim0, alpha = 0.025)
sim0v <- oc_simulate(p0 = 0.3, p1 = 0.3,  looks = looks, nsim = 20000, seed
 ↪  = 3)
sim1 <- oc_simulate(p0 = 0.3, p1 = 0.45, looks = looks, nsim = 20000, seed
 ↪  = 2)
oc0 <- oc_summarize(sim0, c_star); oc0v <- oc_summarize(sim0v, c_star); oc1
 ↪  <- oc_summarize(sim1, c_star)
```

The script gives a calibrated threshold of $c = 0.989$. The rejection proportion in the calibration run, 0.0246, is the calibration target and not an error assessment. On a fresh null run with another seed the type I error is 0.0234 and the expected total sample size is 594 (the maximum is 600). Under the alternative the power is 0.943 and the expected total sample size is 346, because the trial stops early at the first look that crosses the threshold. These numbers are Monte Carlo estimates from 20,000 trials; the seed makes them repeatable and a different seed gives values that differ in the third decimal place. A full validation would repeat the fresh null run across the prespecified nuisance grid. Chapter 14 explains how to report the Monte Carlo error.

C.4. Reproducibility

Every simulation either calls `set.seed()` in the chunk or passes `seed =` to the package function, so the seed is visible next to the result. Functions draw random numbers from R's global generator after the seed is set, so a chunk is repeatable but its results depend on the order of the random calls inside it. Monte Carlo results are reported with their Monte Carlo error (Chapter 14). Quarto's freeze mechanism (`freeze: auto` in `_quarto.yml`) stores the computed output of each chapter, so a rebuild reuses stored results unless the source of that chapter has changed, and the slow simulation chunks are also knitr-cached with the cache keyed to a checksum of the installed package. To regenerate a result after the package changes, reinstall it, delete the chapter's directory under `_freeze/` and its knitr cache directory, and render the chapter again. The R version and the installed package versions can be recorded with `sessionInfo()`, and the authors recommend that a reader who needs exact reproduction record them for their own protocol.

C.5. The companion lab

An interactive tutorial accompanies the book at https://vsokolov.org/bayesian_trials_lab/. It runs in the browser and has modules that match the chapters, where the reader predicts an outcome and then checks it. The Bayesian trial simulator at https://vsokolov.org/bayesian_trials_lab/14-simulator.html lets the reader set a design (arms, prior, thresholds, interim looks) and see simulated operating characteristics, in the spirit of the simulation report discussed in Chapter 14. The full open simulator at https://vsokolov.org/bayesian_trials_lab/14-studio.html is a free tool in the style of commercial design software such as FACTS: it handles several doses, adaptive allocation, scenarios and operating characteristics tables, and is the place to explore a design before writing code. Chapters that have a matching module end with a "Try it in the lab" callout. An online encyclopedia built on this book, at https://vsokolov.org/bayesian_trials_lab/encyclopedia/, has entries for the methods, trials, regulatory documents and people in the book, each linked to the section that treats it.

C.6. Established trial-design software

The following tools are in wide use. The descriptions are brief and cover only the purpose of each tool; consult each tool's documentation for current features and versions. The R packages below are on CRAN, except bcrm, which CRAN archived on 8 May 2026. On the

machine where the book was built, rpact 4.4.0, gsDesign 3.11.0, BOIN 2.7.2, brms 2.23.0 and rstan 2.32.7 were installed; the other tools were not checked.

rpact (R package). Confirmatory adaptive design: group sequential designs, sample size reassessment and combination tests, with simulation.

gsDesign (R package). Group sequential design, including spending functions and boundaries, and power and sample size calculation. A reader can compare its boundaries with `gs_boundary` and `gs_spending_boundary` (Chapter 11).

RBesT (R package). Bayesian evidence synthesis tools: construction of meta-analytic-predictive priors, robustification, and operating characteristics for designs that use historical data. Relevant to Chapter 19.

SampleSizeProportions and SampleSizeMeans (R packages, by Joseph and Bélisle). Bayesian sample size criteria (ALC, ACC, WOC) for proportions and means. The lupus example of Section 9.2.5 uses SampleSizeProportions 1.1.3.

BOIN (R package). Design and simulation of Bayesian optimal interval phase I dose-finding trials. Relevant to Chapter 17.

trialr (R package). Bayesian clinical trial designs fitted with Stan, including dose-finding and phase II designs.

adaptr (R package). Simulation of Bayesian adaptive multi-arm trials, with response-adaptive randomization, early stopping and calibration of the stopping rules, summarized as operating characteristics. It is the closest open R counterpart to the simulation part of `bayestrials` for multi-arm designs.

bcrm (R package; archived on CRAN on 8 May 2026, so it no longer installs from CRAN by default). Bayesian continual reassessment method for dose finding. Relevant to Chapter 17.

Stan and brms. Stan is a probabilistic programming language for Bayesian inference by Hamiltonian Monte Carlo. brms is an R package that fits multilevel models through Stan with a formula interface. Both are used to fit analysis models when conjugate formulas are not available.

FACTS (Berry Consultants). Commercial software for designing and simulating Bayesian adaptive trials, including dose finding, response-adaptive randomization and platform designs. Simulation output of this kind is what regulators ask for in a CID meeting (Chapter 3).

East Horizon (Cytel). Commercial software for the design, simulation and monitoring of fixed, group sequential and adaptive trials. It belongs to the same family as the East product line from the same company.